

Wolfgang G. Stock, Mechtild Stock
Handbook of Information Science

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DE GRUYTER
SAUR

Translated from the German, in cooperation with the authors, by Paul Becker.

ISBN 978-3-11-023499-2
e-ISBN 978-3-11-023500-5

Library of Congress Cataloging-in-Publication Data

A CIP catalog record for this book has been applied for at the Library of Congress.

Bibliographic information published by the Deutsche Nationalbibliothek

The Deutsche Nationalbibliothek lists this publication in the Deutsche Nationalbibliografie; detailed bibliographic data are available in the Internet at <http://dnb.dnb.de>.

© 2013 Walter de Gruyter GmbH, Berlin/Boston
Typesetting: Michael Peschke, Berlin
Printing: Hubert & Co. GmbH & Co. KG, Göttingen
♾ Printed on acid free paper
Printed in Germany

www.degruyter.com

Preface

Dealing with information, knowledge and digital documents is one of the most important skills for people in the 21st century. In the knowledge societies of the 21st century, the use of information and communication technology (ICT), particularly the Internet, and the adoption of information services are essential for everyone—in the workplace, at school and in everyday life. ICT will be ubiquitous. Knowledge is available everywhere and at any time. People search for knowledge, and they produce and share it. The writing and reading of e-mails as well as text messages is taken for granted. We want to be well informed. We browse the World Wide Web, consult search engines and take advantage of library services. We inform our friends and colleagues about (some of) our insights via social networking, (micro-)blogging and sharing services.

Information, knowledge, documents and their users are research objects of Information Science, which is the science as well as technology dealing with the representation and retrieval of information—including the environment of said information (society, a certain enterprise, or an individual user, amongst others).

Information Science meets two main challenges. On the one hand, it analyzes the creation and representation of knowledge in information systems. On the other hand, it studies the retrieval, i.e. the searching and finding, of knowledge in such systems, e.g. search engines or library catalogs. Without information science research, there would be no elaborate search engines on the World Wide Web, no digital catalogs, no digital products offered on information markets, and no intelligent solutions in corporate knowledge management. And without information science, there would be no education in information literacy, which is the basic competence that guarantees an individual's success in the knowledge society.

This handbook focuses on the fundamental disciplines of information science, namely information retrieval, knowledge representation and informetrics. Whereas information retrieval is oriented on the search for and the retrieval of information, knowledge representation starts in the preliminary stages, during the indexing and summarization of documents. We will also discuss results from informetrics, in so far as they are relevant for information retrieval and knowledge representation. The subjects of informetrics are the measurement of information, the evaluation of information systems, as well as the users and usage of information services.

We do not discuss research into information markets and the knowledge society, as those topics are included in another current monograph (Linde & Stock, 2011).¹

Information science is both basic research and applied research. Our scientific discipline nearly always orients itself on the applicability and technological feasibility of its results. As a matter of principle, it incorporates both the users and usage of its tools, systems and services.

¹ Linde, F., & Stock, W.G. (2011). *Information Markets. A Strategic Guideline for the I-Commerce*. Berlin, New York, NY: De Gruyter Saur. (Knowledge & Information. Studies in Information Science.)

The individual chapters of our handbook all have the same structure. After the research areas and important research results have been discussed, there follows a conclusion, summarizing the most important insights gleaned, as well as a list of relevant publications that have been cited in the text.

The handbook is written from a systematic point of view. Additionally, however, we also let scholars have their say with regard to their research results. Hence, this book features many quotations from some of the most important works of information science. Quotations that were originally in the German language have been translated into English. We took care to not only portray the current state of information science, but also the historical developments that have led us there.

We endeavor to provide the reader with a well-structured, clear, and up-to-date handbook. Our goal is to give the demanding reader a compact overview of information retrieval, knowledge representation, and informetrics. The handbook addresses readers from all professions and scientific disciplines, but particularly scholars, practitioners, and students of

- Information Science,
- Library Science,
- Computer Science,
- Information Systems Research, and
- Economics and Business Administration (particularly Information Management and Knowledge Management).

Acknowledgments

Special thanks go to our translator *Paul Becker* for his fantastic co-operation in translating our texts from German into English. Many thanks, also, to the Information Science team at the *Heinrich-Heine-University Düsseldorf*, for their many thought-provoking impulses! And finally, we would like to thank the staff at *Walter de Gruyter* publishing house for their perfect collaboration in producing this handbook.

Mechtild Stock
Wolfgang G. Stock
May, 2013

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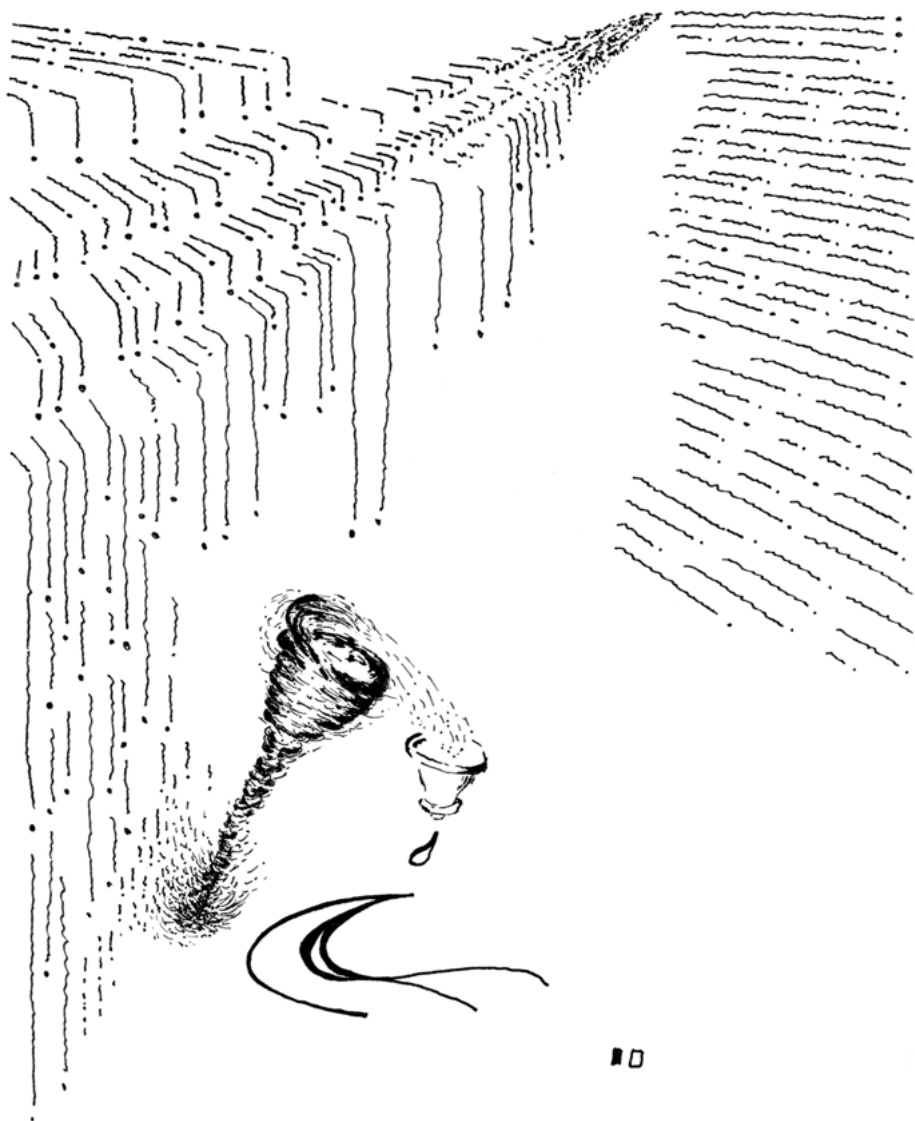
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Part A
Introduction to Information Science



A.1 What is Information Science?

Defining Information Science

What is information science? How can we define this budding scientific discipline? In order to describe and delineate information science, we will examine several approaches and pursue the following lines of questioning:

- What are the fundamental determinants and tasks of information science?
- What roles are played by knowledge, information and documents?
- How has our science developed? Do the branches of information science look back on different histories?
- To which other scientific disciplines is information science closely linked?

There is no generally accepted definition of information science (Bawden & Robinson, 2012; Belkin, 1978; Yan, 2011). This is due, first of all, to the fact that this science, at around fifty years of age, is still relatively young when compared to the established disciplines (such as mathematics or physics). Secondly, information science is strongly interrelated with other disciplines, e.g. information technology (IT) and economics, each of which place great emphasis on their own definitions. Hence, it is not surprising that information science today is used for the divergent purposes of foundational research on the one hand, and applied science on the other.

Let us begin with a working definition of “information science”!

Information Science studies the representation, storage and supply as well as the search for and retrieval of relevant (predominantly digital) documents and knowledge (including the environment of information).

Information science is a fundamental part of the development of the knowledge society (Webster, 1995) or the network society (Castells, 1996). In everyday life (Catts & Lau, 2008) and in professional life (Bruce, 1999), information science finds its expression in the skills of information literacy. “We already live in an era with information being the sign of our time; undoubtedly, a science about this is capable of attracting many people” (Yan, 2011, 510).

Information science is strongly related to both computer science (since digitally available knowledge is principally dependent upon information-processing machines) and to the economic sciences (knowledge is an economic good, which is traded or—sometimes—distributed for free on markets), as well as to other neighboring disciplines (such as library science, linguistics, pedagogy and science of science).

We would like to inspect the determinants of our definition of *information science* a little more closely.

- *Representation*: Knowledge contained in documents as well as the documents themselves (let us say, scientific articles, books, patents or company publications, but also websites or microblog posts) are both condensed via short content-

descriptive texts and labeled with important words or concepts with the purpose of information filtering.

- *Storage and supply*: Documents are to be processed in such a way that they are ideally structured, easily retrievable and readable and stored in digital locations, where they can be managed.
- *Search*: Information science observes users as they satisfy their information needs, it observes their query formulations in search tools and it observes the way they use the retrieved information.
- *Retrieval*: The focal points of information science are systems for researching knowledge; prominent examples include Internet search engines, but also library catalogs.
- *Relevance*: The objective is not to find “any old” information, but only the kind of knowledge that helps the user to satisfy his information needs.
- *Predominantly digital*: Since the advent of the Internet and of the commercial information industry, large areas of human knowledge are digitally available. Even though digital information represents a core theme of information science, there is also room left for non-digital information collections (e.g. in archives and libraries).
- *Documents*: Documents are texts and non-textual objects (e.g., images, music and videos, but also scientific facts, economic objects, objects in museums and galleries, real-time facts and people). And documents are both physical (e.g., printed books) and digital (e.g., blog posts).
- *Knowledge*: In information science, knowledge is regarded as something static, which is fixed in a document and stored on a memory. This storage is either digital (such as the World Wide Web), material (as on a library shelf) or psychical (like the brain of a company employee). Information, on the other hand, always contains a dynamic element; one informs (active) or is informed (passive). The production and the use of knowledge are deeply embedded in social and cultural processes; so information science has a strong cultural context (Buckland, 2012).

The possible applications of information science research results are manifold and can be encountered in a lot of different places on the information markets (Linde & Stock, 2011). By way of example, we will emphasize six types of application:

- Search engines on the Internet (a prominent example: Google),
- Web 2.0 services (such as YouTube for videos, Flickr for images, Last.fm for music or Delicious for bookmarks of websites),
- Digital library catalogs (e.g. the global project WorldCat, or catalogs relating to precisely one library, such as the catalog of the Library of Congress),
- Digital libraries (catalogs *and* their associated digital objects, such as the Perseus Digital Library for antique documents, or the ACM Digital Library for computer science literature),

- Digital information services (particularly specialist databases with a wide spectrum reaching from business and press information, via legal information, all the way to information from science, technology and medicine),
- Information services in corporate knowledge management (the counterparts of the above-mentioned five cases of Internet application, transposed to company Intranets).

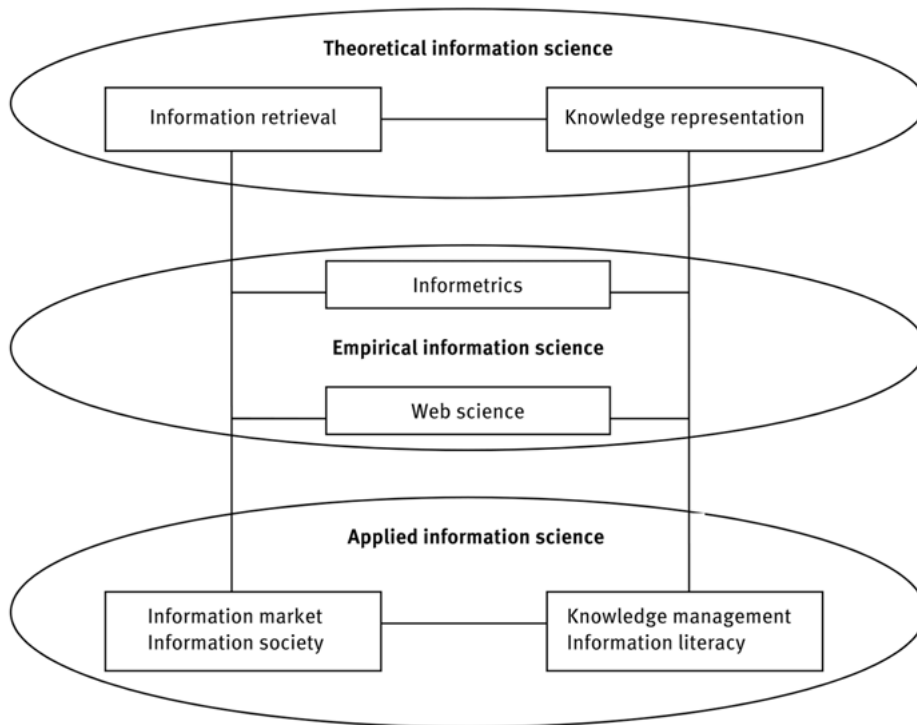


Figure A.1.1: Information Science and its Sub-Disciplines.

Information Science and Its Sub-Disciplines

Information science comprises a spectrum of five sub-disciplines:

- Information Retrieval,
- Knowledge Representation,
- Knowledge Management and Information Literacy,
- Research into the Information Society and Information Markets,
- Informetrics, including Web Science.

A central role is occupied by *Information Retrieval* (Baeza-Yates & Ribeiro-Neto, 2011; Manning, Raghavan, & Schütze, 2008; Stock, 2007), which is the science and engi-

neering of search engines (Croft, Metzler, & Strohman, 2010). Information retrieval investigates not only the technical systems, but also the information needs of the people (Cole, 2012) who use those systems (Ingwersen & Järvelin, 2005). The fundamental question of information retrieval is: how can one create technologically ideal retrieval systems in a user-optimized way?

Knowledge Representation addresses surrogates of knowledge and of documents in information systems. The theme is data describing documents and knowledge—so-called “metadata”—, such as a catalog card in a library (Taylor, 1999). Added to this are methods and tools for use during indexing and summarization (Chu, 2010; Lancaster, 2003; Stock & Stock, 2008). Essential tools are Knowledge Organization Systems (KOSs), which provide concepts and designations for indexing and retrieval. Here, the fundamental question is: how can knowledge in digital systems be represented, organized and condensed in such a way that it can be retrieved as easily as possible?

One subfield on the application side is *Knowledge Management* (Nonaka & Takeuchi, 1995), which focuses on the distribution and sharing of in-company knowledge as well as the integration of external knowledge into the corporate information systems. Another application of information science is research on *Information Literacy* (Eisenberg & Berkowitz, 1990), which means the use of (some) results of information science in everyday life, the workplace (Bruce, 1999), school (Chu, 2009) and university education (Johnston & Webber, 2003).

Research into the *Information Markets* (Linde & Stock, 2011) and the *Information Society* (Castells, 1996; Cronin, 2008; Webster, 1995) is particularly important with regard to the knowledge society. These research endeavors of information science have an extensive subject area, comprising everything from a country’s information infrastructure up to the network economics, by way of the industry of information service providers.

Information science proceeds empirically—where possible—and analyzes its objects via quantitative methods. *Informetric Research* (Weber & Stock, 2006; Wolfram, 2003) comprises webometrics (research into the World Wide Web; Thelwall, 2009), scientometrics (research into the information processes in science and medicine; van Raan, 1997; Haustein, 2012), “altmetrics” as an “alternative” metrics (informetrics with regard to services in Web 2.0; Priem & Hemminger, 2010) as well as patent informetrics (Schmitz, 2010). Empirical research is also performed during the evaluation of information systems (Voorhees, 2002) as well as during user and information needs analyses (Wilson, 2000). Additionally, informetrics attempts to detect the regularities or even laws of information processes (Egghe, 2005). One cross-section of informetrics, called *Web Science*, concentrates on research into the World Wide Web, its users and their information needs (Berners-Lee et al., 2006).

Information science has strong connections to a broad range of other scientific fields. It seems to be an interdisciplinary science (Buckland, 2012, 5). It combines, among others, methods and results from computer science (e.g., algorithms of

information retrieval and knowledge representation), the social sciences (e.g., user research), the humanities (e.g., linguistics and philosophy, analyzing concepts and words), economics (e.g., endeavors on the information markets), business administration (e.g., corporate knowledge management), education (e.g., applying information services in e-learning), and engineering (e.g., the construction of search engines for the World Wide Web) (Wilson, 1996). However, information science is by no means a “mixture” of other sciences, but a science on its own right. The central concern of information science—according to Buckland (2012, 6)—is “cultural engagement”, or, more precisely, “enabling people to become better informed (learning, become more knowledgeable)” (Buckland, 2012, 5). The application of information science in the normal course of life is to secure information literacy for all people. Information literate people are able to inform other people adequately (to represent, to store and to supply documents and knowledge) and they are always able to be well informed (to search for and to retrieve relevant documents and knowledge).

Information Science as Basic and Applied Research

We regard information science as both basic and applied research. In economic terms, it is an “importer” of ideas from other disciplines and an “exporter” of original results to other scientific branches (Cronin & Pearson, 1990). Information science is a basic discipline (and so an exporter of ideas) for some subfields of IT, of library science, of science policy and of the economic sciences. For instance, information science develops an innovative ranking algorithm for search engines, using user research and conceptual analyses. Computer science then takes up the suggestions and develops a workable system from them. On the other hand, information science also builds upon the results of computer science and other disciplines (e.g. linguistics) and imports their ideas. Endeavors toward the lemmatization of terms within texts, for instance, are doomed without an exact knowledge of linguistic results. The import/export-ratio of information science varies according to the partner discipline (Larivière, Sugimoto, & Cronin, 2012, 1011):

Although LIS’s (LIS: Library and Information Science, A/N) import dependency has been steadily decreasing since the mid-1990s—from 3.5 to about 1.3 in 2010 (inset)—it still has a negative balance of trade with most fields. The fields with which LIS has a positive balance of trade are from the natural, mostly medical sciences. Several of the fields with which LIS has a negative balance of trade are from the social sciences and the humanities.

Information science nearly always—even in questions of basic research—looks toward technological feasibility, incorporating user or usage as a matter of principle. It locates its object of research by investigating existing systems (such as search engines, library catalogs, Web 2.0 services and other information service providers’ systems) or creating experimental systems. Information science either analyzes and

evaluates such systems or does scientific groundwork for them. Designers of information services have to consider the understanding of the systems' users, their background knowledge, tradition and language. Thus information science is by no means only a technology, but also analyzes cognitive processes of the users and other stakeholders (Ingwersen & Järvelin, 2005).

Methods of knowledge representation are regarded independently of their historical provenance, converging into a single repertoire of methods: knowledge representation typically spans computer science aspects (ontologies; Gómez-Pérez, Fernández-López, & Corcho, 2004; Gruber, 1993; Staab & Studer, Eds., 2009), originally library-oriented or documentary endeavors (nomenclatures, classifications, thesauri; Lancaster, 2003) as well as ad hoc developments in the collaborative Web (folksonomies; Peters, 2009). Knowledge representation is an important building stone on the way to the "Semantic Web" (Berners-Lee, Hendler, & Lassila, 2001) or the "Social Semantic Web" (Weller, 2010), respectively.

Information Science as the Science of Information Content

The fixed point of information science is information itself, i.e. the structured information content which expresses knowledge. According to Buckland (1991), "information" has three aspects of meaning, all of which are objects of information science:

- information as a process (one informs / is informed),
- information as knowledge (information transports knowledge),
- information as a thing (information is fixed in "informative things", i.e. in documents).

Documents are both texts (Belkin & Robertson, 1976) and non-textual objects (Buckland, 1997) such as images, music and videos, but they are also objects in science and technology (e.g., chemical substances and compounds), economic objects (such as companies or markets), objects in museums and galleries, real-time facts (e.g., weather data) as well as people (Stock, Peters, & Weller, 2010). Besides documents which are created by machines (e.g., flight-tracking systems), the majority of documents include "human recorded information" (Bawden & Robinson, 2012, 4). Documents are very important for information science (Davis & Shaw, Eds., 2011, 15):

Before information science was termed *information science*, it was called *documentation*, and documents were considered the basic objects of study for the field.

Of less interest to information science are technological information processing (which is also an object of IT) and the organization of information activities toward the sale of content (these are subject to the economic sciences as well). Thus Rauch (1988, 26) defines our discipline via the concept of knowledge:

For *information science* ... information is *knowledge*. More precisely: knowledge that is needed in order to deal with problematic situations. Knowledge is thus *possible* information, in a way. Information is knowledge that becomes effective and relevant for actions.

Kuhlen (2004, 5) even discusses the term “knowledge science”, which in light of the sub-disciplines of “knowledge representation” and “knowledge management” would not be completely beside the point. Kuhlen, too, adheres to “information science”, but emphasizes its close connection to knowledge (Kuhlen, 2004, 6):

Information does not exist in itself. Information refers to knowledge. Information is generally understood to be a surrogate, a representation or manifestation of knowledge.

The specific content plays a subordinate role in information science; its objects are the structure and function of information and information processing.

(For information science) it is of no importance whether the object of discussion is a new insect, for example, or an advanced method of metal working,

Mikhailov, Cernyi and Gilyarevsky (1979, 45) write. One of the “traditional” definitions of information science comes from Borko (1968, 3):

Information science is that discipline that investigates the properties and behavior of information, the forces governing the flow of information, and the means of processing information for optimum accessibility and usability. It is concerned with that body of knowledge relating to the origination, collection, organization, storage, retrieval, interpretation, transmission, transformation, and utilization of information. This includes the investigation of information representation in both natural and artificial systems, the use of codes for efficient message transmission, and the study of information processing devices and techniques such as computers and their programming systems. ... It has a pure science component, which inquires into the subject without to its application, and an applied science component, which develops services and products.

“That definition has been quite stable and unvarying over at least the last 30 years,” is Bates’s (1999, 1044) comment on this concept definition. Borko’s “pure science” can be further subdivided—as we have already seen in Figure A.1.1—into a theoretical information science, which works out the fundamentals of information retrieval as well as of knowledge representation, and into an empirical information science that systematically studies information systems and users (who deal with information systems). Applied information science addresses the use of information in practice, i.e. on the information markets, within a company or administration in knowledge management and in everyday life as information literacy. One application of information science in information practice aims toward keeping all members of a company, college, city, society or humanity in general as ideally informed as possible: everyone, in the right place and at the right time, must be able to receive the right amount of

relevant knowledge, rendered comprehensibly, clearly structured and condensed to the fundamental quantity.

Whereas information science does have some broad theoretical research areas, as a whole the discipline is rather oriented towards application. Even though certain phenomena may not yet be entirely resolved on a theoretical basis, an information scientist will still go ahead and create workable systems. Search engines on the Internet provide an illustrative example. The fundamental elements of processing queries and documents are nowhere exhaustively discussed and resolved in the theory of information retrieval; even the concept of relevance, which ought to be decisive for the relevance ranking being used, is anything but clear. Yet, for all that, the search engines built on this basis operate very successfully. Looking back on the advent of information science, Henrichs (1997, 945) describes the initial sparks for the discipline, which almost exclusively stem from information practice:

The practice of modern specialist information ... undoubtedly has a significant chronological advantage over its theory. Hence, in the beginning—and there can be no doubting this fact in this country (Germany, A/N), at least—there was practice.

As early as 1931, Ranganathan formulated five “laws” of library science. For Ranganathan, these are rules describing the processing of books in libraries. For today’s information science, the information content—whether fixed in books, websites or anywhere else—is the crucial aspect:

1st Law: Information is for use.

2nd Law: Every user his or her information.

3rd Law: Every information its user.

4th Law: Save the time of the user!

5th Law: Information practice and information science are growing organisms.

The first four rules refer to the aspect of practice, which always comes to bear on information science. The fifth “Law” states that the development of information practice and information science is not over yet by a long shot, and is constantly evolving.

A Short History of Information Science and Its Sub-Disciplines

The history of searching and finding knowledge begins during the period in which human beings first began to systematically solve problems. “As we strive to better understand the world that surrounds us, and to control it, we have a voracious appetite for information,” Norton (2000, 4) points out. The history of information science as a scientific discipline, however, goes back to the 1950s at its earliest (Bawden & Robinson, 2012, 8). The term “information science” was coined by Jason Farradane in the 1950s (Shapiro, 1995). Information science finally succeeded in establishing itself,

particularly in the USA, the United Kingdom and the former socialist countries, at the end of the 1960s (Schrader, 1984).

Information science has deep historical roots accented with significant controversy and conflicting views. The concepts of this science may be at the heart of many disciplines, but the emergence of a specific discipline of information science has been limited to the twentieth century (Norton 2000, 3).

In its “prehistory” and history, we can separate five strands that each deal with partial aspects of information science.

Information Retrieval

The retrieval of information has been discussed ever since there have been libraries. Systematically organized libraries help their users to purposefully retrieve the information they desire. Let the reader imagine being in front of a well-arranged bookshelf, and discovering a perfectly relevant work at a certain place (either via the catalog or by browsing). Apart from this direct hit, there will be further relevant books to its left and right, the significance of which to the topic of interest will diminish the further the user moves away from the center. The terms “catalog” (or, today, “search engine”), “bookshelf” (or “hit list”), “browsing”, “good arrangement” and “relevance” give us the basic concepts of information retrieval. The basic technology of information retrieval did not change much until the days of World War II. The advent of computers, however, changed the situation dramatically. Entirely new ways of information retrieval opened up. Experiments were made using computers as knowledge stores and elaborate retrieval languages were developed (Mooers, 1952; Luhn, 1961; Salton & McGill, 1983). Experimental retrieval systems were created in the 1960s, while commercial systems with specialist information have existed since the 1970s. Retrieval research reached its apex with the Internet search engines of the 1990s.

Knowledge Representation

What is a “good arrangement” of information? How can I represent knowledge ideally, i.e. condense it and make it retrievable via knowledge organization systems? This strand of development, too, stems from the world of libraries; it is closely related to information retrieval. Here, organization systems are at the forefront of the debate. Early evidence of such a system includes the systematics of the old library of Alexandria. Research into knowledge representation was boosted by the Decimal Classification developed by Mevil Dewey in the last quarter of the 19th century (Dewey, 1876), which was then developed further by Paul Otlet and Henri La Fontaine, finally leading to the plan of a collection and representation of world knowledge (Otlet, 1934). Over the course of the 20th century, various universal and specialized classification systems

were developed. With the triumph of computers in information practice, information scientists created new methods of knowledge representation (such as the thesaurus) as well as technologies for automatically indexing documents.

Knowledge Management and Information Literacy

In economics and business administration, information has long been discussed in the context of entrepreneurial decisions. Information always proves imperfect, and so becomes a motor for innovative competition. With conceptions for learning organizations, and, later, knowledge management (Nonaka & Takeuchi, 1995), the subject area of industrial economics and of business administration on the topic of information has broadened significantly from around 1980 onward. The objective is to share and safeguard in-company knowledge (Probst, Raub, & Romhardt, 2000) and to integrate external knowledge into an organization (Stock, 2000).

Information literacy has two main threads, namely retrieval skills and skills of creation and representation of knowledge. Retrieval literacy has its roots in library instruction and was introduced by librarians in the 1970s and 1980s (Eisenberg & Berkowitz, 1990). Since 1989, there exists a standard definition of information literacy, which was formulated by the American Library Association (Presidential Committee on Information Literacy, 1989). With the advent of Web 2.0 services, a second information literacy thread came into life: the skills of creating documents (e.g., videos for publishing on YouTube) and of representing them thematically (e.g., with tags).

Information Markets and Information Society

With the “knowledge industry” (Machlup, 1962), the “information economy” (Porat, 1977), the “postindustrial society” (Bell, 1973), the “information society” (Webster, 1995) or the “network society” (Castells, 1996), the focus on knowledge has for around 50 years been shaped by a sociological and economic perspective. In economics, information asymmetries (Akerlof, 1970) are discovered and measures to counteract the unsatisfactory aspects of markets with information asymmetries are developed. Thus, the sellers of digital information know a lot more about the quality of their product than their customers do, for example.

Originally, the information market was conceived as a very broad approach that led to the construction of a fourth economic factor (besides labor, capital and land) as well as to a fourth economic sector (besides agriculture, industry and services). Subsequently, the research area of the information markets was reduced to the sphere of digital information, which is transmitted via networks—the Internet in particular, but also mobile telephone networks (Linde & Stock, 2011). Research into the information, knowledge or network society thus discusses a very large area. Its subjects reach from “informational cities” as prototypes of cities in the knowledge society (Stock, 2011)

up to the “digital divide”, which is the result of social inequalities in the knowledge society (van Dijk & Hacker, 2003).

Informetrics and Web Science

Starting in the second quarter of the 20th century, researchers discovered that the distribution of information follows certain regularities. Studies of ranking distributions—e.g. of authors in a discipline according to their numbers of publications—led to laws, of which those formulated by Samuel C. Bradford, Alfred J. Lotka or George K. Zipf have become classics (Egghe, 2005). Chronological distributions of information are described via the concept of “half-life”. Since the 1970s, informetrics has established itself as information science’s measurement method. Some much-noticed fields of empirical information science include scientometrics and patentometrics, since informetrics allows us to measure scientific and technological information. This leads to the possibility of making statements about the “quality” of research results, and furthermore about the performance and influence of their developers and discoverers.

From around 2005 onward, an interdisciplinary research field into the World Wide Web, called Web Science, has emerged (Hendler et al., 2008). Information science takes part in this endeavor via webometrics, altmetrics, user research as well as information needs research.

Information Science and Its Neighbors

In his classification of the position of information science, Saracevic (1999, 1052) emphasizes its relations to other disciplines. Among these, he assigns particular importance to the role of information technology and the information society in interdisciplinary work.

First, information science is interdisciplinary in nature; however, the relations with various disciplines are changing. The interdisciplinary evolution is far from over.

Second, information science is inexorably connected to information technology. A technological imperative is compelling and constraining the evolution of information science (...).

Third, information science is, with many other fields, an active participant in the evolution of the information society. Information science has a strong social and human dimension, above and beyond technology.

Information science actively communicates with other scientific disciplines. It is closely related to the following:

- Computer Science,
- Economics,
- Library Science,

- (Computational) Linguistics,
- Pedagogy,
- Science of Science.

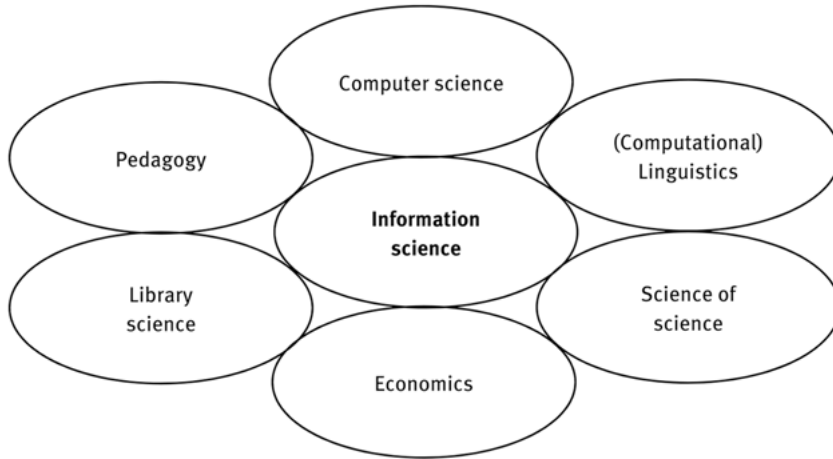


Figure A.1.2: Information Science and its Neighboring Disciplines.

Computer science provides the technological basis for information science applications. Without it, there would be no Internet and no commercial information industry. Some common research objects include information retrieval, aspects of knowledge representation (e.g., ontologies) and aspects of empirical information science, such as the evaluation of information systems. Computer science is more interested in the technology of information processing, information science in the processing of content. Unifying both perspectives suggests itself almost immediately. It has become common practice to refer to the combination between the two as Computer and Information Science—or “CIS”.

On the level of enterprises, the streams of information and communication between employees as well as with their environment are of extreme importance. Organizing both information technology and the usage thereof are tasks of information systems research and of information management. Organizing the in-company knowledge base as well as the sharing and communication of information fall within the domain of knowledge management. Information management, information systems research and knowledge management together form the corporate information economy. On the industry level, we regard information as a product. Electronic information services, search engines, Web portals etc. serve to locate a market for the product called “knowledge”. But this product has its idiosyncrasies, e.g. it can be copied at will, and many consumers regard it as a free common good. On the business level, lastly, network economics discusses the specifics of networks—of which the

Internet is one—and information economics asks about the buyer's and the seller's respective standards of knowledge, which already lays bare some considerable asymmetries.

The object of library science is the empirical and theoretical analysis of specific activities; among these are the collection, conservation, provision and evaluation of documents and the knowledge fixed therein. Its tools are elaborate systems for the formal and content-oriented processing of information. Topics like the creation of classification systems or information dissemination were common property of this discipline even before the term “information science” existed. This close link facilitates—especially in the United States—the development of approaches toward treating information science and library science as a single aggregate discipline, called “LIS” (Library and Information Science) (Milojević, Sugimoto, Yan, & Ding, 2011).

Search and retrieval are mainly performed via language; the knowledge—aside from some exceptions (such as images or videos)—is fixed linguistically. Computational linguistics and general linguistics are both so important for information science that relevant aspects from the different areas have been merged into a separate discipline, called information linguistics.

Pedagogy sometimes works with tools and services of Web 2.0, e.g. wikis or blogs. Additionally, in educational science learning management systems and e-portfolio systems are of great interest. Mainly in the aspects of e-learning and blended learning (Beutelspacher & Stock, 2011), there are overlaps with information science.

The analysis of scientific communication provided by empirical information science has made it possible to describe individual scientists, institutes, journals, even cities and countries. Such material is in strong demand in science of science and science policy. These results help sociology of science in researching for communities of scientists. History of science gets empirical historical source material. Science of science can check its hypotheses. And finally, science evaluation and science policy are provided with decision-relevant pointers toward the evaluation of scientific institutions.

Conclusion

- The object of information science is the representation, storage and supply as well as the search for and retrieval of relevant (predominantly digital) documents and knowledge.
- Information science consists of five main sub-disciplines: (1) information retrieval (science and engineering of search engines), (2) knowledge representation (science and engineering of the storage and representation of knowledge), (3) informetrics (including all metrics applied in information science), (4) endeavors on the information markets and the information society (researches in economics and sociology as parts of applied information science), and (5) efforts on knowledge management as well as on information literacy (researches in business administration and education).

- Some application cases of information science research results are search engines, Web 2.0 services, library catalogs, digital libraries, specialist information service suppliers as well as information systems in corporate knowledge management.
 - The application of information science in the normal course of life is to secure information literacy for all people.
 - Historically speaking, five main lines of development can be detected in information science: information retrieval, knowledge representation, informetrics, information markets / information society and knowledge management / information literacy. Information science has begun its existence as an independent discipline in the 1950s, and is deemed to have been firmly established since around 1970.
 - Information science works between disciplines and has significant intersections with computer science, with economics, with library science, with linguistics, with pedagogy and with science of science.
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A.2 Knowledge and Information

Signals and Data

“Information” and “Knowledge” are concepts that are frequently discussed in the context of many different scientific disciplines and in daily life (Bates, 2005; Capurro & Hjørland, 2003; Lenski, 2010). They are crucial basic concepts in information science.

The concept of information was decisively influenced by Shannon (1948), during the advent of computers and of information transmission following the end of the Second World War. Shannon’s interpretation of this concept was to become the foundation of telecommunications, and continues to influence all areas of computer and telecommunication technology. Wyner (2001, 56) emphasizes Shannon’s importance for science and technology:

Claude Shannon’s creation in the 1940’s of the subject of information theory is arguably one of the great intellectual achievements of the twentieth century. Information theory has had an important and significant influence on mathematics, particularly on probability theory and ergodic theory, and Shannon’s mathematics is in its own a considerable and profound contribution to pure mathematics. But Shannon did his work primarily in the context of communication engineering, and it is in this area that it stands as a unique monument. In his classical paper of 1948 and its sequels, he formulated a model of a communication system that is distinctive for its generality as well as for its amenability to mathematical interest.

Let us take a look at Shannon’s model (Figure A.2.1). In it, information is transmitted as a physical signal from an information source to a receiver via a channel.

Signals are physical entities, i.e. newsprint (on the paper you are reading), sound waves (from the traffic outside that keeps you from reading) or electromagnetic waves (from the television). The channel is prone to disruptions, and thus contains “noise” (e.g. when there is an error on one page).

An “encoder” is activated in the transmitter, between the source and the channel, transforming the digits into certain physical signals. Concurrently, a “decoder” that can interpret these signs is established between channel and receiver. Encoder and decoder must be attuned to each other in order for the signals to be successfully transmitted. As an illustrative example, let us consider a telephone conversation. Here, the information source is a person communicating a message via the microphone of their telephone. In the transmitter (the telephone), acoustic signals are transformed into electronical ones and then sent forth via telephone lines. At the other end of the line, a receiver transforms the electronical signals back into acoustic ones and sends them on to their destination (here: the person being called) via the second telephone.

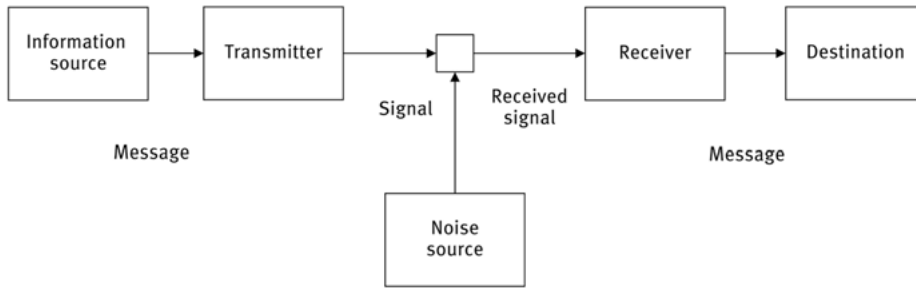


Figure A.2.1: Schema of Signal Transmission Following Shannon. Source: Shannon, 2001 [1948], 4.

The information content of a sign is calculated via this sign's probability of occurrence. Signs that occur less often receive high information content; frequently occurring ones are assigned a small value. When a sign z_i has the probability p_i of occurring, its information content I will be:

$$I(z_i) = \lg 1/p_i \text{ bit or (mathematically identical):}$$

$$I(z_i) = -\lg p_i \text{ bit.}$$

The logarithm dualis being used ($\lg$) and the measure “bit” (for ‘binary digit’) can both be explained by the fact that Shannon considers a sign's probability of occurrence to be the result of a sequence of binary decisions. The $\lg$ calculates the exponent x in the formula $2^x = a$ (e.g. $2^3 = 8$; hence, in the reverse function, $\lg 8 = 3$). Let us clarify the principle via an example: we would like to calculate the information content I for signs that have a probability of occurrence of 17% (like the letter e in the German language), of 10% (like n) and of 0.02% (like q). The sum of the probabilities of occurrence of all signs in the presupposed repertoire, which in this case is the German alphabet, must always add up to 100%. We apply the values to the formula and receive these results:

$$I(e) = -\lg 0.17 \text{ bit} = -(-2.6) \text{ bit} = 2.6 \text{ bit}$$

$$I(n) = -\lg 0.10 \text{ bit} = -(-3.4) \text{ bit} = 3.4 \text{ bit}$$

$$I(q) = -\lg 0.0002 \text{ bit} = -(-12.3) \text{ bit} = 12.3 \text{ bit.}$$

A q thus has far higher information content than an e , since it is used far less often by comparison.

The following observation by Shannon is of the utmost importance for information science (2001 [1948], 3):

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have *meaning*; that

is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem.

Here he is merely talking about a probability-theoretical observation of signal transmission processes; the meaning, i.e. the content of the transmitted signals, does not play any role at all (Ma, 2012, 717-719). It is the latter, however, which forms the object of information science. In so far, Shannon's information theory is of historical interest for us but it has only little significance in information science (Belkin, 1978, 66). However, it is important for us to note that any information is tied to a signal, i.e. a physical carrier, as a matter of principle.

Signals transmit signs. In semiotics (which is the science of signs), signs can be observed more closely from three points of view:

- in their relations to each other (syntax),
- in the relations between signs and the objects they describe (semantics),
- in the relations between signs and their users (pragmatics).

Shannon's information theory only takes into consideration the syntactic aspect. From an information science perspective, we will call this aspect data. Correspondingly, the processing of data concerns the syntactical level of signs, which are analyzed with regard to their type (e.g. alphanumerical signs) or their structure (e.g. entry in a field named "100"), among other aspects. This is only information science from a peripheral point of view, though. This sort of task falls much more under the purview of computer science, telecommunications and telematics (as a mixed discipline formed of telecommunications and informatics). Boisot and Canals (2004, 43) discuss the difference between "data" and "information" via the example of cryptography:

Effective cryptography protects information as it flows around the word. ... Thus while the data itself can be made "public" and hence freely available, only those in possession of the "key" are in a position to extract information from it (...). Cryptography, in effect, exploits the deep differences between data and information.

In information science, we analyze information as data *and* meaning (knowledge) in context. For Floridi (2005), information consists of data, which is well-formed as well as meaningful.

Information, Knowledge and Documents

The other two sub-disciplines of semiotics, semantics and pragmatics, investigate the meaning and usage of signs. Here the concepts of "knowledge" and "information" enter the scene. Following Hardy and Meier-Oeser (2004, 855), we understand knowledge to be:

partly a skill, i.e. the skill of conceiving of an object as it really is on the one hand, and that of successfully dealing with the objects of knowledge on the other. It is partly the epistemic state that a person occupies as a consequence of successfully performing his or her cognitive tasks, and partly the content that a cognitive person refers to when doing so, as well as the statement in which one gives linguistic expression to the result of a cognitive process.

Here, knowledge is assigned the trait of certainty from the perspective of a subject; independently of the subject, knowledge is accorded a claim for truth. We should re-analyze the above definition of knowledge and structure it simplistically. Knowledge falls into the two aspects of skill and state:

- (1) Knowledge as the skill
 - (1a) of correctly comprehending an object (“to know *that*”),
 - (1b) of correctly dealing with an object (“to know *how*”);
- (2) Knowledge as the state
 - (2a) of a person that knows (something),
 - (2b) that which is known itself, the content,
 - (2c) its linguistic expression.

Knowledge has two fixed points: the knowing subject and what is known, in which aspects (1a), (1b) and (2a) belong to subjective knowledge and (2b) and (2c) are objective knowledge—of course, (2c) can only be objective if the linguistic expression is permanently fixed on a physical carrier, a document.

The warrantor of the distinction between subjective and objective knowledge is Popper (1972; see also Brookes, 1980; Ma, 2012, 719-720). According to him, objective knowledge is knowledge that is fixed in documents, such as articles or books, whereas subjective knowledge is the knowledge that a person has fixed in their mind. This distinction is embedded within Popper’s theory of the three worlds:

- World 1: the physical world,
- World 2: the world of conscious experience,
- World 3: the world of content.

World 1 is material reality; this world contains the signals that are known as data or documents when bundled accordingly. Subjective knowledge (World 2) is clearly fixated on objective knowledge (World 3): practically all subjective knowledge (World 2) depends upon World 3. It is also clear, conversely, that objective knowledge develops from knowledge that was formerly subjective. There is a close interrelationship between the two: subjective knowledge orients itself upon objective knowledge, while objective knowledge results from the subjective. And without the physical world—World 1—“nothing goes”, as Brookes (1980, 127) emphasizes:

Though these three worlds are independent, they also interact. As human living on Earth, we are part of the physical world, dependent for our continued existence on heat and light from the Sun ... and so on. Though our mentalities we are also part of World 2. In reporting the ideas Popper has recorded in his book, I have been calling on the resources of World 3. Books and all other

artifacts are also physical entities, bits of World 1, shaped by humans to be exosomatic stores of knowledge which have an existence as physical things independent of those who created them.

Objective knowledge, or our point (2b) above, exists independently of people (in Popper's World 3). Semiotically speaking, we are in the area of semantics, which is known for disregarding the user of the sign, the subject. How does the transition from World 3 to World 2, or within World 2, take place? The (formless) content from World 3 must be "cast into a mould", put into a *form*, that can be understood in World 2. The same procedure is performed when attempting to transmit knowledge between two subjects in World 2. We are thus dealing with in*FORM*ation, if you'll excuse the pun. It is not possible to transmit knowledge "as such", we need a carrier (the "form") to set the knowledge in motion. "Information is a thing—knowledge is not," Jones (2010) declares.

Indeed the word information, etymologically speaking, has exactly this origin: according to Capurro (1978) and Lenski (2010, 80), the Latin language uses "informatio" and "informo" to mean education and formation, respectively. (Information is thus related to schooling; for a long time, the "informer" referred to a home tutor.) Information makes content move, thus also having a pragmatic component (Rauch, 2004). The groundbreaking work of economic research into the information market is "The Production and Distribution of Knowledge in the United States" (1962) by Machlup. Machlup was one of the first to formulate knowledge as static and information as dynamic. Knowledge is not transmitted: what is sent and received is always information. Machlup (1962, 15) defines:

to *inform* is an activity by which knowledge is conveyed; to *know* may be the result of having been informed. "Information" as the act of informing is designed to produce a state of knowing in someone's mind. "Information" as that which is being communicated becomes identical with "knowledge" in the sense of which is known. Thus, the difference lies not in the nouns when they refer to *what* one knows or is informed about; it lies in the nouns only when they are to refer to the *act* of informing and the *state* of knowing, respectively.

Kuhlen (1995, 34) proceeds similarly: "*Information is knowledge in action.*" Kuhlen continues:

Correspondingly, from an information science perspective we are interested in methods, procedures, systems, forms of organization and their respective underlying conditions. With the help of these, we can work out information for current problem solutions from socially produced knowledge, or produce new knowledge from information, respectively. The process of working out information does not leave knowledge in its *raw state*; rather, it should be regarded as a process of transformation or, with a certain valuation, of refinement ... This transformation of knowledge into information is called the creation of informational added value.

Knowledge has nothing to do with signals, since it exists independently of them, whereas information is necessarily tied to these physical carriers.

Bates (2005; 2006) distinguishes three levels of information and knowledge in regard to matter and energy (“information 1”; e.g., genetic information), to beings, especially to men (“information 2”) and to meaning and understanding (“knowledge”). These levels are similar to the three worlds of Popper. Bates (2006, 1042) defines

Information 1: The pattern of organization of matter and energy.

Information 2: Some patterns of organization of matter and energy by a living being (or its constituent parts).

Knowledge: Information given meaning and integrated with other contents of understanding.

If we put knowledge into carriers of Popper’s World 1 (e.g., into a book or a Web page), the meaning is lost. A reader of the book or the Web page has to create his understanding of the meaning on his own. Bates (2006) emphasizes:

Knowledge in inanimate objects, such as books, is really only information 1, a pattern of organization of matter and energy. When we die, our personal knowledge dies with us. When an entire civilization dies, then it may be impossible to make sense out of all the information 1 left behind; that is, to turn it into information 2 and then knowledge.

According to Buckland (1991b), information has four aspects, of which three fall under the auspices of information science and one (at least additionally) to computer science. Information is either tangible or intangible, it can either be physically grasped or not; in Popper’s sense, it is either tied to World 1 or it is not. Additionally, information is always either a process or a state. This two-fold distinction leads to the following four-field chart (modified following Buckland, 1991b, 352):

<i>Information</i>	<i>Intangible</i>	<i>Tangible</i>
<i>State</i>	Knowledge	Information as thing
<i>Process</i>	Information as process	Information processing

Information is tangible as a thing in so far as the things, i.e. the documents, are always tied to signals, and thus to World 1. They are to be regarded—at least over a certain period of time—as stable. Processes for dealing with information can also be physically described and must thus be allocated to World 1. Here Buckland locates the domain of information processing and data processing, and hence a discipline that is more a part of computer science. Knowledge is intangible; it falls either to World 2 (as subjective knowledge) or World 3 (as objective knowledge). In the process of informing or being informed (information as process), Buckland abstracts from the physical signal and only observes the transitions of knowledge from a sender to a receiver. Buckland is also aware that such transitions always occur in the physical world (Buckland, 1991b, 352):

Knowledge ... can be represented, just as an event can be filmed. However, the representation is no more knowledge than the film is the event. Any such representation is necessarily in tangible forms (sign, signal, data, text, film, etc.) and so *representations* of knowledge (and of events) are necessarily “information-as-thing”.

Here the great significance of documents (in Buckland’s words: information as thing) for information science becomes clear, since they are what shelters knowledge. In the documents, knowledge (which is not tangible as such) is embedded. For Mooers (1952, 572), the knowledge is a “message”, and the documents containing those messages are the “channels”:

(A) “channel” is the physical document left in storage which contains the message.

As knowledge is not directly tangible, how can it then be fixed and recognized? Apparently, this requires a subject to perform a knowledge act: knowledge is not simply given but is acquired on the basis of the respective subject’s foreknowledge.

How can knowledge be recorded and presented, and what are its exact characteristics? In the following sections, we will investigate the manifold aspects of knowledge. At first, we will concentrate on the question: can only texts contain knowledge, or are other document types, such as pictures, capable of this as well?

Non-Textually Fixed Knowledge

Knowledge fixed in documents does not exclusively have to be grounded in texts. Non-textual documents—images, videos, music—also contain knowledge. We will demonstrate this on the example of pictures and Panofsky’s theory (2006). The reader should call to mind Leonardo da Vinci’s famous painting *The Last Supper*. According to Panofsky, there exist three levels on which to draw knowledge from the painting. An Australian bushman—as an example for “everyman”—will look at the picture and describe its content as “13 persons either sitting at or standing behind one side of a long table.” For Panofsky, this semantic level is “pre-iconographical”; it only presupposes practical experiences and a familiarity with certain objects and events on the part of the viewer. On the second semantic level—the “iconographical” level—the viewer additionally requires foreknowledge concerning cultural traditions and certain literary sources, as well as familiarity with the thematic environment of the painting.

Armed with this foreknowledge, a first viewer will identify 13 men, one of which being Jesus Christ and the other 12 his disciples. Another viewer, with different foreknowledge (perhaps nourished by Dan Brown’s novel *The Da Vinci Code*), however, sees 12 men and a woman, by replacing the depiction of John—to Jesus’ left as seen from the viewer’s perspective—with that of Mary of Magdala. On this level, it also

depends upon who is looking at the picture, or more precisely, what background knowledge the viewer has of the picture. On the iconographical level, one recognizes the last supper of Christ with his disciples (or with 11 disciples and a woman).

The third, or “iconological” knowledge level can be reached via expert knowledge. Our exemplary image is now interpreted as a particularly successful work of Leonardo’s (due, among other aspects, to the precise rendering of perspective), as a textbook example of the culture of the Italian Renaissance, or as an expression of the Christian faith. Panofsky (1975, 50) summarizes the three levels in tabular form:

- I *Primary*, or *natural* subject—(A) factual, (B) expressive—that forms the world of *artistic motifs*.
- II *Secondary*, or *conventional* subject, which forms the world of *pictures, anecdotes and allegories*.
- III *Actual Meaning or content*, forming the world of “*symbolic*” values.

In correspondence to these three levels of knowledge, there are the three interpretative acts of (I) pre-iconographical description, (II) iconographical analysis and (III) iconological interpretation. The three levels of pictorial knowledge thus described should, analogously, be distinguishable in other media (films and music). In certain texts, too, particularly in fiction, it seems appropriate to divide access to knowledge in three. The story of the fox and the grapes (by Aesop) contains the primary knowledge that there is a fox who sees some grapes dangling above him, unreachable, and that the fox then rejects them, citing their sour taste. The secondary knowledge sees the fable as an example of the way people develop resentment out of impotence; and level three literary-historically places the text into the context of antique fables.

Knowing That and Knowing How

A simple initial approximation regards knowledge as instances of true propositions. The following definition of knowledge (Chisholm, 1977, 138) holds firm in some varieties of epistemology:

h is known by S =Df h is accepted by S; h is true; and h is nondefectively evident for S.

h is a proposition and *S* a subject; =*Df* means “equals by definition”. Hence, Chisholm demands that the subject accepts the proposition *h* (as true), which is in fact the case (objectively speaking) and that this is so not merely through a happy coincidence, but precisely “nondefectively evident”. Only when all three determinants (acceptance, truth, evidence) are present can knowledge be seen as well and truly established. In the absence of one of these aspects, such a statement can still be communicated—as information—but it would be an error (when truth and evidence are absent), a supposition (if acceptance and evidence are given, but the truth value is undecided) or a lie (when none of the three aspects applies).

Is knowledge always tied to statements, as Chisholm's epistemology suggests? To the contrary: the propositional view of knowledge falls short, as Ryle (1946, 8) points out:

It is a ruinous but popular mistake to suppose that intelligence operates only in the production and manipulation of propositions, i.e., that only in ratiocinating are we rational.

Propositions describe “knowing that” while neglecting “knowing how”. But knowledge is not exclusively expressed in propositions; and knowing how is most often (and not only for Ryle) the more important aspect of knowledge by far. Know-how is knowledge about how to do certain things. Ryle distinguishes between two forms of know-how: (a) purely physical knowledge, and (b) knowledge whose execution is guided by rules and principles (which can be reconstructed). Ryle (1946, 8) describes the first, physical variant of know-how as follows:

(a) When a person knows how to do things of a certain sort (e.g., make good jokes, conduct battles or behave at funerals), his knowledge is actualised or exercised in what he does. It is not exercised (...) in the propounding of propositions or in saying “Yes” to those propounded by others. His intelligence is exhibited by deeds, not by internal or external dicta.

In the second variant of know-how, knowledge is grounded in principles, which can be described at least in their essence (Ryle, 1946, 8):

(b) When a person knows how to do things of a certain sort (e.g., cook omelettes, design dresses or persuade juries), his performance is in some way governed by principles, rules, canons, standards or criteria. (...) It is always possible in principle, if not in practice, to explain why he tends to succeed, that is, to state the reasons for his actions.

According to Ryle, it is not possible to trace such “implicit” know-how (Ryle, 1946, 7) to know-that, and thus to propositions. In the know-how variant (b), it is not impossible to reconstruct and objectify the implicit knowledge via rules etc.; this is hardly the case in variant (a).

Subjective Implicit Knowledge

Polanyi enlarges upon Ryle's observations on know-how. According to Polanyi, people dispose of more knowledge than they are capable of directly and comprehensibly communicating to others. Polanyi's (1967, 4) famous formulation is this:

I shall consider human knowledge by starting from the fact that *we can know more than we can tell*. This fact seems obvious enough; but it is not easy to say exactly what it means.

Implicit, or tacit, knowledge consists of various facets (Polanyi, 1967, 29):

The things that we know in this way included problems and hunches, physiognomies and skills, the use of tools, probes, and denotative language.

This implicit knowledge is basically “embedded” within the body of the individual, and they use it in the same way that they normally use their body (Polanyi, 1967, X and XI):

(The structure of tacit knowing) shows that all thought contains components of which we are subsidiary aware in the focal content of our thinking, and that all thought dwells in its subsidiaries, as if they were parts of our body. ... (S)ubsidiaries are used as we use our body.

We must strictly differentiate between the meaning that the person might ascribe to the (implicit) knowledge, and the object bearing this meaning. Let the person carry an object in their hand, say: a tool; this object is thus close to the person (proximal). The meaning of this object, on the other hand, may be very far from the person (distal)—it does not even have to be known in the first place. The meaning of the object (unexpressed or inexpressible) arises from its usage. Polanyi (1967, 13) introduces the two aspects of implicit knowledge via an example:

This is so ... when we use a tool. We are attending to the meaning of its impact on our hands in terms of its effect on the things to which we are applying it. We may call this the *semantic aspect* of tacit knowledge. All meaning tends to be displayed *away from ourselves*, and that is in fact my justification for using the terms “proximal” and “distal” to describe the first and second terms of tacit knowledge.

Explicit knowledge can also contain implicit components. The author of a document has certain talents, is well acquainted with certain subjects and is steeped in traditions, all of which put together makes up his or her personal knowledge (Polanyi, 1958). Since these implicit factors are unknown to the reader, the communication of personal knowledge—even in written and thus explicit form—is always prone to errors (Polanyi, 1958, 207):

Though these ubiquitous tacit endorsements of our words may always turn out to be mistaken, we must accept this risk if we are ever to say anything.

Here Polanyi touches upon hermeneutical aspects (e.g. understanding and preunderstanding).

How can implicit knowledge be transmitted? In the case of distal implicit knowledge, transmission is nearly impossible, whereas proximal implicit knowledge, according to Polanyi, is communicated via two methods. On the one hand, it is possible to “physically” transmit knowledge via demonstration and imitation of the relevant activities (Polanyi, 1967, 30):

The performer co-ordinates his moves by dwelling in them as parts of his body, while the watcher tries to correlate these moves by seeking to dwell in them from outside. He dwells in these moves by interiorizing them. By such exploratory indwelling the pupil gets the feel of a master's skill and may learn to rival him.

The second option consists of intellectually appropriating the implicit knowledge of another person by imitation, or more precisely, by rethinking. Polanyi (1967, 30) uses the example of chess players:

Chess players enter into a master's spirit by rehearsing the games he played, to discover what he had in mind.

According to Polanyi, it is important to create “coherence” between the bearer of implicit knowledge and the person who wants to acquire it—be it of the physical or intellectual kind (Polanyi, 1967, 30):

In one case we meet a person skillfully using his body and, in the other, a person cleverly using his mind.

For implicit knowledge to be communicated without a hitch, it must be transformed into words, models, numbers etc. that can be understood by anyone. In other words, it must be “externalized” (ideally: stored in written documents). Externalized knowledge—and only this—is accessible to knowledge representation, and hence information science. In practice, according to Nonaka and Takeuchi (1995), this is not achieved via clearly defined concepts and statements that can be understood by anyone, but only via rather metaphorical and vague expressions. Should it prove impossible to comprehensibly externalize a person's implicit knowledge, we will be left with four more ways toward at least conserving some aspects of their knowledge:

- firstly, we regard the person themselves as a document which is then described via metadata (e.g. in an expert database),
- secondly, we draw upon those artifacts created by the person themselves and use them to approximate their initial knowledge (as in the chess player example above), or we attempt—as in Ryle's scenario (b)—to reconstruct the rules governing the lines of action,
- thirdly, we “apprentice” to the person (and physically appropriate their knowledge),
- fourthly, it is possible to compile information profiles of the person via their preferences (e.g. of reading certain documents), and to extrapolate their implicit knowledge on this basis (with the caveat of great insecurity).

Variant (1) amounts to “Yellow Pages”, which ultimately become useless once the person is no longer reachable (e.g. after leaving his company), variant (2) tries to make do with documenting the artifacts (via images, videos and descriptions) and hopes that will be possible for another person to reconstruct the original knowledge

on their basis. Variant (4) is, at best, useful as a complement to the expert database. The ideal path is probably variant (3), which involves being introduced to the subject area under the guidance of the individual in question. Here, Nonaka and Takeuchi (1995) talk about “socialization”. Memmi (2004, 876) finds some clear words on the subject:

In short, know-how and expertise are only accessible through contact with the appropriate individuals. Find the right people to talk to or to work with, and you can start acquiring their knowledge. Otherwise there is simply very little you could do (watching videos of expert behavior is a poor substitute indeed).

Socialization is an established method in knowledge management, but it has nothing to do with digital information and very little with information science (since in this instance, knowledge is *not* being represented at all, but directly communicated from person to person). From the purview of information science, then, implicit knowledge creates a problem. We can approach it in the context of knowledge representation (using methods 1, 2 and 4), but we cannot conclusively resolve it. Implicit knowledge already creates problems on the level of interpersonal relationships; if we additionally require an information system in knowledge representation, the problem will become even greater. Thus, Reeves and Shipman (1996, 24) emphasize:

Humans make excellent use of tacit knowledge. Anaphora, ellipses, unstated shared understanding are all used in the service of our collaborative relationships. But when human-human collaboration becomes human-computer-human collaboration, tacit knowledge becomes a problem.

Knowledge Management

In Nonaka’s and Takeuchi’s (1995) approach, we are confronted with four transitions from knowledge to knowledge:

- from the implicit knowledge of one person to that of another (socialization),
- from the implicit knowledge of a person to explicit knowledge (externalization)—which includes the serious problems mentioned above,
- from explicit knowledge to the implicit knowledge of a person (internalization, e.g. by learning),
- from explicit knowledge to explicit knowledge (combination).

Information science finds its domain in externalization (where possible) and combination. Explicit, and thus person-independent, knowledge is conserved and represented in its objective interconnections. For Nonaka, Toyama and Konno (2000), the ideal process of knowledge creation in organizations runs, in a spiral form, through all four forms of knowledge transmission and leads to the SECI Model (Socialization, Externalization, Combination, Internalization) (see Figure A.2.2). In order to exchange

knowledge, the actors must occupy the same space (Japanese: “*ba*”), comparable to the “World Horizon” of hermeneutics (Nonaka, Toyama, & Konno, 14):

ba is here defined as a shared context in which knowledge is shared, created and utilised. In knowledge creation, generation and regeneration of *ba* is the key, as *ba* provides the energy, quality and place to perform the individual conversions and to move along the knowledge spiral.

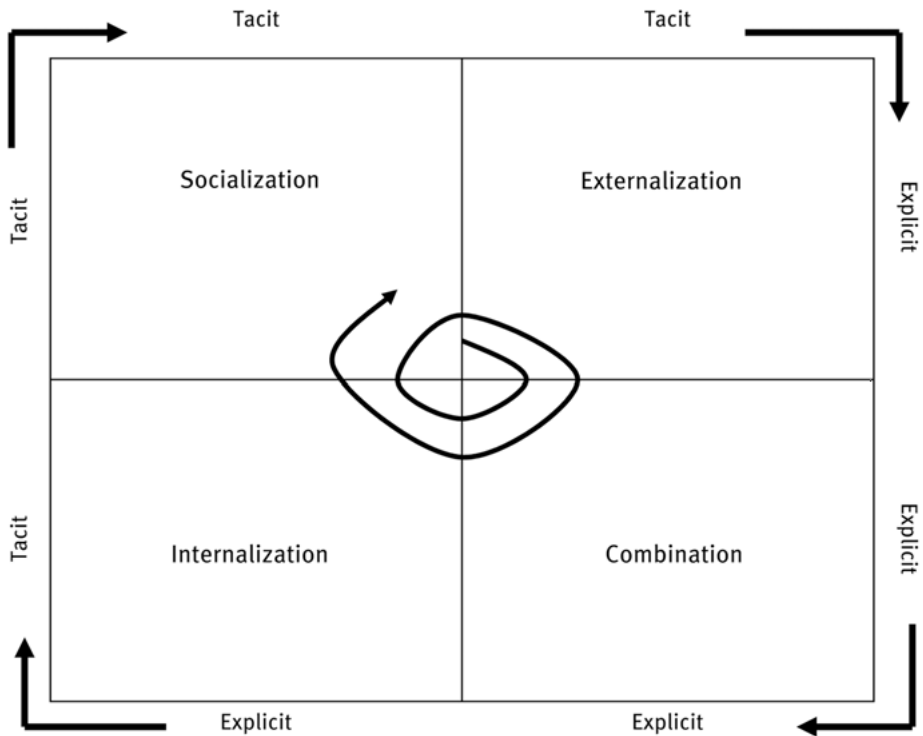


Figure A.2.2: Knowledge Spiral in the SECI Model. Source: Modified from Nonaka, Toyama & Konno, 2000, 12.

In enterprises, several building blocks are required in order to adequately perform knowledge management (Figure A.2.3). According to Probst, Raub and Romhardt (2000), we must distinguish between the work units (at the bottom) and the strategic level. The latter dictates which goals should be striven for in the first place. Then, the first order of business is to identify the knowledge required by the company. This can already be available internally (to be called up from knowledge conservation) or externally (on the Web, in a database, from an expert or wherever) and will then be acquired. If no knowledge is available, the objective will be to create it by one's self (e.g. via research and development). Knowledge is meant to reach its correct recipi-

ents. To accomplish this, the producers must first be ready to share their knowledge, and secondly, the system must be rendered capable of adequately addressing, i.e. distributing, said knowledge. Both knowledge compiled in enterprises and crucially important external knowledge must be kept easily accessible. The central aspect is the building block of Knowledge Use: the recipient translates the knowledge into actions, closes knowledge gaps, prepares for decisions or lets the system notify him—in the sense of an early-warning system—about unexpected developments. Finally, Knowledge Measurement evaluates the operative cycle and sets the results in relation to the set goals.

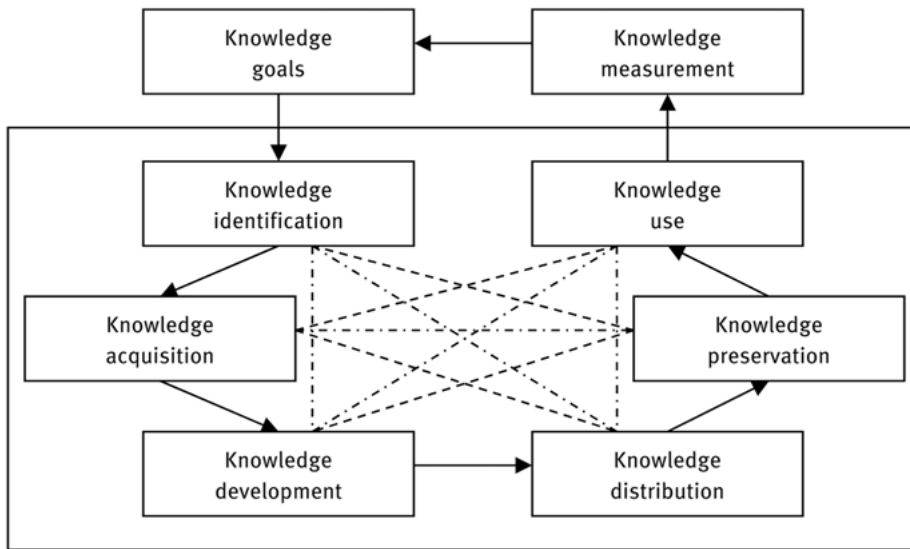


Figure A.2.3: Building Blocks of Knowledge Management Following Probst et al. Source: Probst, Raub, & Romhardt, 2000, 34.

Types of Knowledge

For Spinner (1994, 22), knowledge has become so important—we need only consider the “knowledge society”—that he places the knowledge order on the same level as the economic order and the legal order of a society:

The outstanding importance of these *three fundamental orders* for all areas of society results, ... in the case of the *knowledge order*, from the function of scientific-technological progress as the most important productive force, as well as from extra-scientific information as a mass medium of communication and control, i.e. as a means of entertainment and administration.

Such a knowledge order contains “knowledge of all kinds, in every conceivable quantity and quality” (Spinner, 1994, 24), which is granted by the triad of form, content and expression, as well as via the validity claim. Studying the form involves the logical form of statements and their area of application, where Spinner distinguishes dichotomously between general statements (statements in theories or laws, e.g. “All objects, given gravitation, fall downward”) and singular existential statements with reference to place and time (“My pencil falls downward in our office on December 24th, 2007”). Knowledge, according to Spinner, is devoid of content if it has only slight informational value (e.g. entertainment or advertisement). On the other hand, it is “informative” in the case of high informational value (e.g. in scientific statements or news broadcasts). The expression of knowledge aims for the distinction between implicit (tacit knowledge, physical know-how) and explicit (fixed in documents). The application of knowledge can also be considered as an epistemic “additional qualification” (Spinner, 2002, 21). An application can be apodictic if the knowledge concerned is presupposed to be true (as in dogmas), or hypothetical if its truth value is called into question.

Can all knowledge be equally well structured, or represented? Certainly not. A well-formulated scientific theory, published in an academic journal, should be more easily representable than a microblog on Twitter or a rather subliminal message in an advertising banner. Spinner (2000, 21) writes on this subject:

We can see that there are knowledge types, knowledge stores and knowledge characteristics that can be ordered and those that resist being ordered by looking at the species-rich knowledge landscape. This landscape reaches from implicit notions and unarticulated ideas all the way to fully formulated (‘coded’) legal texts and explicit (‘holy’) texts that are set down word for word.

Normal-Science Knowledge

Let us restrict our purview to scientific knowledge for a moment. According to Kuhn’s (1962) deliberations, science does not advance gradually but starts its research—all over again, as it were—in the wake of upheavals, or scientific revolutions. Kuhn describes the periods between these upheavals as “normal sciences”. We must understand these normal sciences as a type of research that is accepted by scientists of certain schools of thought as valid with regard to their own work. Research is based on scientific accomplishments of the past and serves as a basis for science; it represents commonly accepted theories which are applied via observation and experimentation. The scientific community is held together by a “paradigm”, which must be sensational in order to draw many advocates and adherents, but which also offers its qualified proponents the option of solving various problems posed by the paradigm itself. The researchers within a normal science form a community that orients itself on the common paradigm, and which disposes of a common terminology dictated by the

paradigm. If there is agreement within the community of experts as to the specialist language being used, it is easily possible—at least in theory—to represent the specialist language via exactly one system of a knowledge order. Kuhn (1962, 76) emphasizes the positive effects of such a procedure of not substituting established tools:

So long as the tools a paradigm supplies continue to prove capable of solving the problems it defines, science moves fastest and penetrates most deeply.

For the record: In normal sciences, specific knowledge orders can be worked out for the benefit of science. But there is more than just the normal sciences. Sometimes, anomalies arise—observations that do not fit the paradigm. If these anomalies pile up, the scientific community enters a critical situation in which the authority of the paradigm slackens. If the scientists encounter a new paradigm during such a crisis, a scientific revolution is to be expected—a paradigm shift, a change in perspectives, the formation of a new scientific community. The new paradigm must offer predictions that differ from those of the old (anomalous) one, otherwise it would be unattractive for the researchers. For Kuhn, however, this also means that the old paradigm and the new one are logically irreconcilable: the new one supplants the old. Both paradigms are “incommensurable”, i.e. they cannot be compared to each other. For the knowledge order of the old paradigm, this means that it has become as useless as the paradigm itself, and must be replaced.

There are sciences that have not (yet) found their way toward becoming normal sciences. These are in their so-called “pre-paradigmatic” phase, in which individual schools of thought, or “lone wolves”, attempt to solve the prevalent problems via their own respective terminologies. Since there is no binding terminology in such cases, it is impossible to build up a unified knowledge order.

Brier (2008, 422) emphasizes the importance of different languages in scientific fields for information practice:

(A) bibliographical system such as BIOSIS will only function well within a community of biologists. This means that both their producers and the users must be biologists—and so must be the indexers.

If for example a chemist is looking for biological information, he cannot use his chemical terminology but must speak the language of biology (Brier, 2008, 424):

(C)hemists must use the correct biological name for a plant in order to find articles about a chemical substance it produces.

Kuhn’s observations only claim to hold true for scientific paradigms. However, in spite of his caution, it seems that they can be generalized for all types of knowledge. It is only possible to create a binding knowledge order when all representatives of a community (apart from scientists, these may include all employees of a company, or

the users of a subject-specific Web service) speak a common language, which then makes up the basis of the knowledge order. If there is no common paradigm (e.g. in enterprises: the researchers have *their* paradigm, the marketing experts an alternative one), the construction of *one single* knowledge order from the perspective of exactly one group makes little sense. “Revolutionary changes” that lead to a paradigm shift require equally revolutionary changes in the knowledge order. It is a great challenge in information science “how to map semantic fields of concepts and their signifying contexts into our systems” (Brier, 2008, 424).

Information as Knowledge Put in Motion

Let us turn back to Figure A.2.1, in which we are discussing the process of signal transmission. If we want to put knowledge “into a form”, or in motion, we cannot do so by disregarding this physical process. Information is thus fundamentally a unit made up of two components: the document as signal, and the content as knowledge. For the purposes of information science, we must enhance Shannon’s schema by adding the knowledge component. The triad of sender—channel—receiver is preserved; added to it are, on the sender’s side, the knowledge he is referring to and wants to put in motion, and on the receiver’s side, the knowledge as he understands it. The problem: there is no direct contact between the knowledge that is meant by a sender and that which is understood by the receiver; the transmission always takes a detour via encoding, channel (including noise) and decoding. And this route, represented schematically in Figure A.2.4, is extremely prone to disruptions.

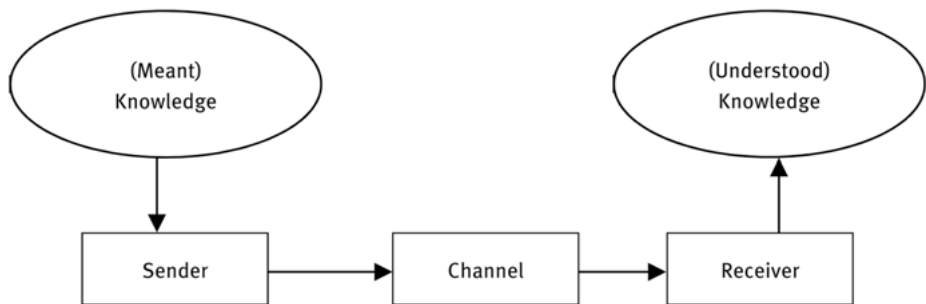


Figure A.2.4: Simple Information Transmission.

Apart from the physical problems of information transmission, several further bundles of problems must be successfully dealt with in order for the understood knowledge to at least approximately match the meant knowledge. First, we need to deal with the characters being used. Sender and receiver must dispose of the same character set, or employ a translator. (As a self-experiment, you could try speaking BAPHA out

loud. If this appears to yield no meaning at the first attempt, let us enlighten you that it includes Cyrillic characters.) Secondly, the language being used should be spoken fluently by both sender and receiver. A piece of “hot information”, written in Bulgarian, will be useless to someone who does not speak the language. Thirdly, natural languages are by no means unambiguous. (Another experiment: what are you thinking of when you hear or see the word JAVA? There are at least three possible interpretations: programming language, island and coffee.) Fourthly, there must be a common “world horizon”, i.e. a certain socio-cultural background. Imagine the problems of an Eskimo in explaining to an Arab from Bahrain the various forms of snow. When information transmission is successful, the knowledge (carried by the information) meets the pre-existing knowledge of the receiver and changes it.

According to Brookes (1980, 131), the connection between knowledge and transmitted information can be expressed via a (pseudo-mathematical) equation. Knowledge (K) is understood as a structure (S) of concepts and statements; the transmitted information (ΔI) carries a small excerpt from the world of knowledge. The equation goes:

$$K[S] + \Delta I = K[S + \Delta S].$$

The knowledge structure of the receiver is modified via ΔI . This effects a change to the structure itself, as signified by the ΔS . The special case of $\Delta S = 0$ is not ruled out. The same ΔI , received by different receivers with different $K[S]$ than each other, can effect different structural changes ΔS . The process of information differs “from person to person and from situation to situation” (Saab & Riss, 2011, 2245). So the “understanding of a person’s knowledge structure” (Cool & Belkin, 2011, 11) is essential for information science.

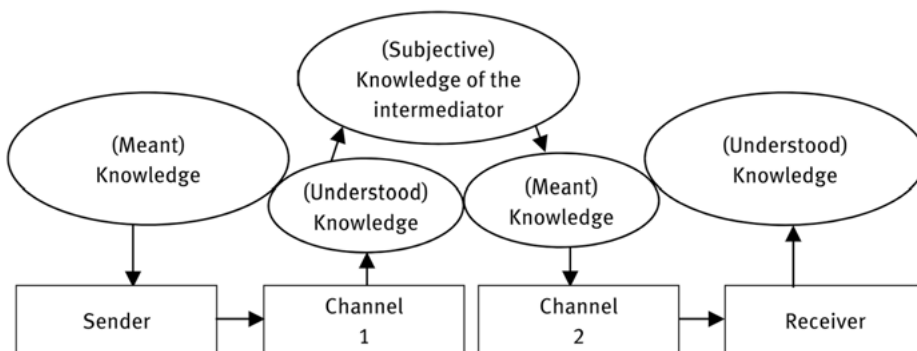


Figure A.2.5: Information Transmission with Human Intermediator.

The intermediation is set between instances of the chain of information transmission; this is where the specific informational added value provided by information science is being developed.

When interposing a human information intermediary, the blurry areas between the knowledge that is meant and that which is understood are widened due to the addition of the intermediary's subjective knowledge. Let us discuss the problems via an example: Over the course of this chapter, we will talk about Bacon's "Knowledge is Power". Bacon (Sender) has fixed his knowledge in several texts (Channel 1), from which we can hardly work out exhaustively what he "really" meant (meant knowledge). An information intermediary (in this case, we will assume this role) has read Bacon's works (Channel 1) and, ideally, understood them—on the basis of the existing knowledge base (this knowledge base also contains other sources, e.g. Schmidt's (1967) essay and further background knowledge). An opinion about Bacon's "Knowledge is Power" has been developed and fixed in this book (Channel 2). This is where you, the reader (Receiver), come into play. On the basis of your foreknowledge, you understand our interpretation of Bacon's "Knowledge is Power". How much of what Bacon actually meant would have reached you in its unadulterated form? To clarify: Information science attempts to minimize such blurry areas via suitable methods and tools of knowledge representation and information retrieval, and to provide for a (possibly) ideal information flow.

From Kuhlen we learned that the objective in information science is not only to put knowledge in motion, but also to refine it with informational added value. We now incorporate into our schema of information transmission a human intermediary with his or her own subjective knowledge (Figure A.2.5) on the one hand, and an automatic mechanical intermediary (Figure A.2.6) on the other.

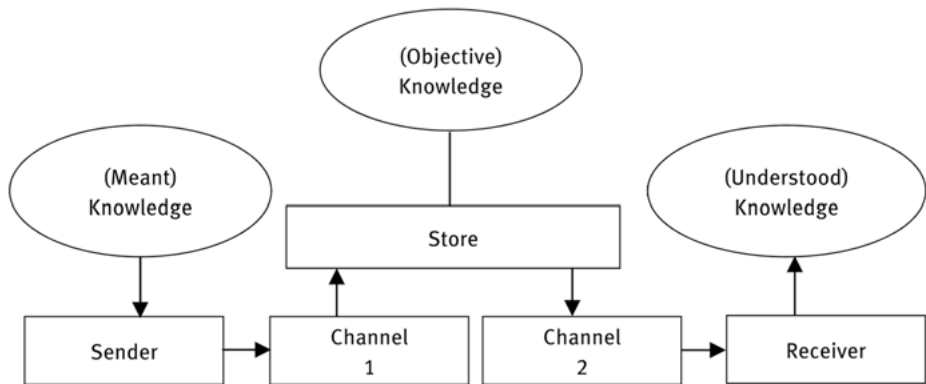


Figure A.2.6: Information Transmission with Mechanical Intermediation.

The situation is slightly different in the case of mechanical intermediation. Instead of a human, a machine is set between sender and receiver; say, a search engine on

the Internet. The problems posed by the human intermediary's (meant and understood) knowledge no longer exist: the machine stores—in Popper's sense—objective knowledge. The questions are now: How does it store this knowledge? And: How does it then yield it (i.e. in what order)? If the receiver wants to search comfortably and successfully, he must (at least to some extent) grasp the techniques used by search engines. Information science provides such techniques of Information Indexing and also makes sure users find it as easy as possible to search and retrieve knowledge. Is it possible to replace a subjective bearer of knowledge—say, an employee with valuable know-how or know-that—with an objective one in such a way that the knowledge is still accessible after the employee has left the company? The answer is: yes, in principle, only such a current employee must dispose of a knowledge structure that allows him to adequately understand the stored knowledge.

What are the characteristics of information? Does information have anything to do with truth? Is information always new? "Knowledge is power", Bacon says. Are there any relations between knowledge and power?

Information and Truth

Knowledge has a truth claim. Is this also the case for information, if information is what sets this knowledge in motion? Apart from knowledge, there are further, related forms of dealing with objects. Thus we speak of belief when we subjectively think something is true that cannot be objectively explained, of conjecture when we introduce something as new without (subjective or objective) proof, or of lies when something is clearly not true but is communicated regardless. If beliefs, conjectures or lies are put in motion, are they not information? For predominantly practical reasons, it would prove very difficult, if not impossible, for information science to check every piece of informational content for its truth value. "Information is not responsible for truth value," Kuhlen (1995, 41) points out (for a contrary opinion cf. Budd, 2011). Buckland (1991a, 50) remarks, "we are unable to say confidently of anything that it could not be information;" and Latham (2012, 51) adds, "even untrue, incorrect or unseen information is information." The task of checking the truth value of the knowledge, rather, must be delegated to the user (Receiver). He then decides whether the information retrieved represents knowledge, conjecture or untruth. At the same time, though, it must of course be required of the information's user to be competent enough to perform this task without any problems, i.e. he is well trained in information literacy.

Information and Novelty

Does information have anything to do with novelty? Must knowledge always be new to a receiver in order to become information? And conversely, is every new knowledge always information? Let us start off with the last question and imagine we are at the airport, planning to board a plane to Graz, Austria. There is a message over the PA system: “Last call for boarding Flight 123 to Rome!” This is new to us; we find it to be of no interest at all, though, and will probably hardly even notice it before immediately forgetting we heard it at all. Now another announcement: “Flight 456 to Graz will board half an hour late due to foggy weather.” This is also new, and it does interest us; we will modify our actions (e.g. by going for a bite to eat before boarding) accordingly. This time, the announcement is useful information to us.

We now come to the first question. Staying with the airport example (and sitting in the airport restaurant by now), the repeated announcement “Flight 456 to Graz will board half an hour late due to foggy weather” is not new to us, but it does affect our actions, as it allows us to continue eating in peace (after all, the fog could have lifted unexpectedly in the meantime). By now it should have become clear that novelty does not have to be an absolute requirement for information; it is still a relative requirement, though. By the fourth or fifth identical announcement, however, the knowledge being communicated loses a lot of its value for us. A certain quantum of repetition is acceptable—it is sometimes even desirable, as a confirmation of what is already known—but too much redundancy reduces the information’s relevance to our actions. In this context, von Weizsäcker (1974) identifies the vertices of first occurrence and repetition as the components of the pragmatic aspect of information: the “correct” degree of informedness lies between the two extremes.

“Knowledge is Power”

What does information yield its receiver? The knowledge that is transmitted via information meets the user’s pre-existing knowledge and leads to the user having more know-how or know-that than previously. Such knowledge enhancement is not in service of purely intellectual pursuits (at least not always); rather, it serves to better understand or be able to do something in order to gain advantages vis-à-vis others—e.g. one’s competition. Knowledge and having knowledge is thus related to power. The classical dictum “Knowledge is Power” has been ascribed to Bacon. In his “*Novum Organum*” (Bacon, 2000 [1620], 33), we read:

Human knowledge and human power come to the same thing, because ignorance of cause frustrates effect. For Nature is conquered only by obedience; and that which in thought is a cause, is like a rule in practice.

A further reference in the “New Organon” shows the significance of science and technology (“active tendency”) to power (Bacon, 2000 [1620], 103):

Although the road to human knowledge and the road to human power are very close and almost the same, yet because of the destructive and inveterate habit of losing oneself in abstraction, it is altogether safer to raise the sciences from the beginning on foundations which have an active tendency, and let the active tendency itself mark and set bounds to the contemplative part.

Bacon’s context is clear: he is talking about scientific knowledge, and about the power that allows us to reign over nature with the help of this (technological) knowledge. The aspect of knowledge as power over other people cannot be explicitly found in Bacon, but it is—according to Schmidt (1967, 484-485)—an implication hidden in the text.

The power that Bacon speaks of is the power of man to command nature; we must, however, pay attention to the dialectical nuance that man is not outside of nature, but a part of it. Every victory of man over nature is, by implication, a victory of man over himself. Power over other people can only be gained because every man is subject to natural conditions.

Uncertainty

What is the power that can be obtained through knowledge, and which is what we talk about in the context of information science? Ideally, a receiver will put the transmitted information to use. On the basis of the experienced “discrepancy between the information regarded as necessary and that which is actually present” (Wittmann, 1959, 28), uncertainty arises, and must be reduced. The information obtained—thus Wittmann’s (1959, 14) famous definition—is “purpose-oriented knowledge”. For Belkin (1980) such “anomalies” lead to “anomalous states of knowledge” (ASK), which are the bases of a person’s information need. Belkin, Oddy and Brooks (1982, 62) note:

The ASK hypothesis is that an information need arises from a recognized anomaly in the user’s state of knowledge concerning some topic or situation and that, in general, the user is unable to specify precisely what is needed to resolve that anomaly.

Wersig (1974, 70) also assumes a “problematic situation” that leads to “uncertainty”. Wersig (1974, 74) even defines the concept of information with reference to uncertainty:

Information (...) is the reduction of uncertainty due to processes of communication.

Kuhlthau (2004, 92) introduces the Uncertainty Principle:

Uncertainty is a cognitive state that commonly causes affective symptoms of anxiety and lack of confidence. Uncertainty and anxiety can be expected in the early stages of the information search process. The affective symptoms of uncertainty, confusion, and frustration are associated with vague unclear thoughts about a topic or question. As knowledge states shift to more clearly focused thoughts, a parallel shift occurs in feelings of increased confidence. Uncertainty due to a lack of understanding, a gap in meaning, or a limited construction initiates the process of information seeking.

What does “purpose-oriented knowledge” mean? What is “correct” information, information that fulfills its purpose and reduces uncertainty? Information helps us master “problematic situations”. Such situations include preparations for decision-making, knowledge gaps and early-warning scenarios.

In the first two aspects, the information deficit is known: decisions must be prepared with reference to the most comprehensive information available, knowledge gaps must be filled. Both situations are closely linked and, for the most part, coincide. In the third case, it is the information that alerts us to the critical situation in the first place. The early-warning system signals dangers (e.g. posed by the competition, such as the market entry of new competitors). The “correct” internal knowledge (including the “correct” bearers of knowledge, i.e. the employees), combined with the “correct” internally used and externally acquired knowledge shows up business opportunities for the company and at the same time warns of risks. In any case, it leads to conscious action (or forbearance). This is the practical goal of all users’ information activities: to translate knowledge into action via information. For a specific information user in a specific situation (Henrichs, 2004), information is action-relevant knowledge—beyond true or false, new or confirmed. Similarly, Kuhlen (2004, 15) states:

Corresponding to the pragmatic interpretation, information is that quantity of knowledge which is required in current action situations, but which the current actor generally does not personally dispose of or at the very least does not have a direct line to.

Once an actor has assembled the quantity of knowledge that he needs in order to perform his action, he is—in Kuhlen’s sense—“knowledge-autonomous”. Under the term “information autonomy”, Kuhlen summarizes the activities of gathering relevant knowledge (besides mastery of the relevant retrieval techniques) as well as assessing the truth and innovation values of the knowledge retrieved.

To put it in the sense of Bacon’s adage, we are granted the ability to act appropriately on the basis of knowledge if and only if we have informational autonomy. The totality of a person’s skills for acting information-autonomous (e.g. basic IT and smartphone skills, skills for mastering information retrieval and skills for controlling the uploading and indexing of documents) is called information literacy (see Ch. A.5).

The power of information literacy is constrained. Even information that is ideally “correct” is generally not enough to reduce any and all uncertainty during the decision-making process. The information basis for making decisions can only be smaller or greater. However, increasing the size of the information basis brings enormous competitive advantages in economic life, compared to the competition. Imperfect information and the respective relative uncertainty are key aspects of information practice: in an environment that as a matter of principle cannot be comprehensively recorded, and which is in constant flux to boot, it is important for individuals, enterprises, and even for entire regions and countries, to reduce the uncertainties resulting from this state of affairs as far as possible. However, complete information and thus the complete reduction of uncertainty can never be attained. Rather, information creates a state of creative uncertainty. The perception and assessment of uncertainty lead to information being used as a strategic weapon. By disposing of and refining knowledge, companies and individuals can stand to gain advantages over their competitors. The competitive advantage is given for as long as any party monopolizes certain information, or at least for as long as the information is asymmetrically distributed in favor of our person, our company or our country. The other competitors must necessarily follow suit, until there is once more an information equilibrium in a certain field. Then, information in other fields or new information in the original field is used to work out a new information advantage etc. Information thus takes center stage in the arena of innovative competition.

Production of Informational Added Value

Information activities process knowledge, present it in a user-friendly manner, represent it via condensation and the allocation of information filters, and prepare it for easy and comprehensive search and retrieval. All aspects that go beyond the original knowledge are informational added values, which can be found in private as well as public information goods. In addition to *knowing how* and *knowing that* the informational value added is *knowing about*. Information activities lead to knowledge *about* documents and *about* the knowledge fixed in the documents. For Buckland (2012), *knowing about* is more concerned with information science than *knowing how* and *knowing that*. Buckland (2012, 5) writes,

in everyday life we depend heavily and more and more on second-hand knowledge. We can determine little of what we need to know by ourselves, at first hand, from direct experience. We have to depend on others, largely through documents. ... In this flood of information, we have to select and have to decide what to trust. What we believe about a document influences our use of it, and more importantly, our use of documents influences what we believe.

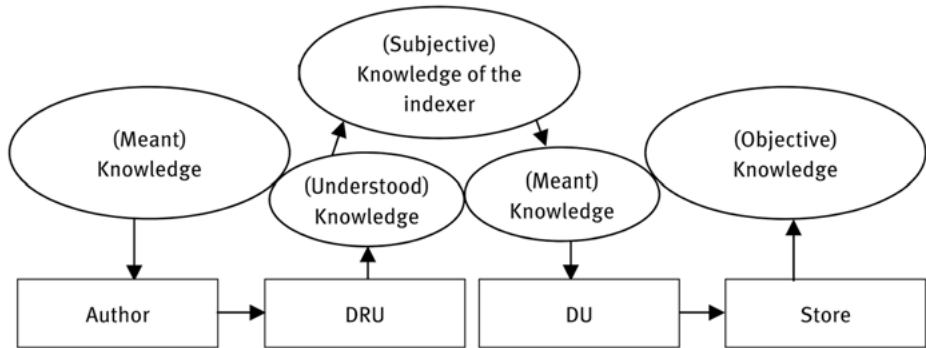


Figure A.2.7: Intermediation—Phase I: Information Indexing (here: with Human Indexer). DRU: Documentary Reference Unit, DU: Documentary Unit.

In an initial, rough approximation of theoretical information science, we distinguish between three phases of information transmission (Figures A.2.7 through A.2.9). In Phase 1, the objective is to represent knowledge fixed by an author in a document (e.g. a book, a research article, a patent document or a company-internal memo). This document is called a “documentary reference unit” (DRU). Then comes an information specialist or—in the case of automatic indexing—an information system, reads this unit and represents its content (via a short summary) and its main topics (via some keywords) in a so-called “documentary unit” (DU), also called “surrogate”. In the ideal scenario, this DU (as an initial informational added value) is saved in an information store together with the DRU. Warner (2010, 10) calls the activities of knowledge representation “description labor” as the labor involved “in transforming objects into searchable descriptions”. This process, sketched just above, proves to be far more complex than assumed and is what forms the central object of the second part of this book.

The DU may be sent to interested users by the respective information system itself, as a push service; until then, it remains in the knowledge store awaiting its users and usage. One day, a user will feel an information need and start searching, or let an information expert do his search for him. Via its pull service, the information system allows users to systematically search through the documentary units (with the informational added value contained within them) and also to simply browse through the system. The language of the searcher is then aligned with the languages of the DUs and DRUs, and the retrieval system arranges the list of DUs according to relevance. The searcher will select some relevant DUs and through them acquire the DRUs, i.e. the original documents, which he will then transmit to his client (of course client and searcher can be the same person). For Warner (2010, 10) the processes of information retrieval are connected with “search labor”, which is understood as “the human labor expended in searching systems”. Information retrieval is where the second amount

of informational added values is worked out. The entire first part of this book is dedicated to this subject.

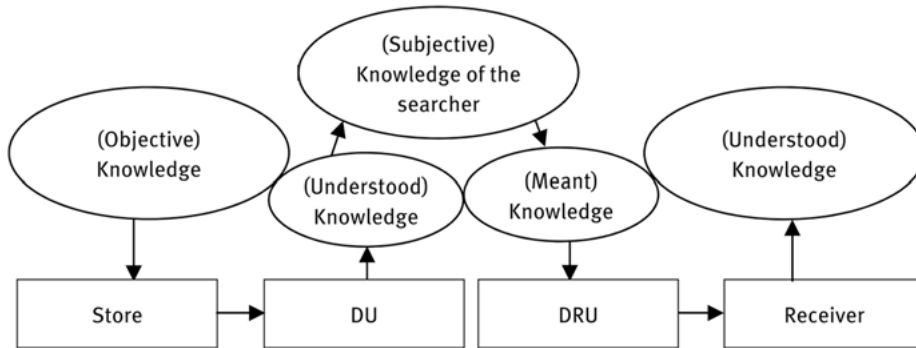


Figure A.2.8: Intermediation—Phase II: Information Retrieval.

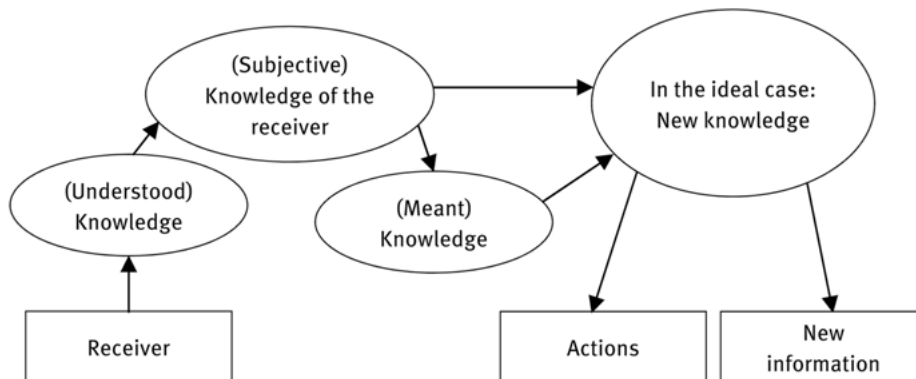


Figure A.2.9: Intermediation—Phase III: Further Processing of Retrieved Information.

Figure A.2.9 seamlessly picks up where Figure A.2.8 left off. The receiver has received the relevant documentary reference units containing his action-relevant knowledge. The transmitted information meets the receiver's state of knowledge and, in the ideal case, merges with it to create new knowledge. The new knowledge leads to the desired actions and—if the actor reports on it—new information.

Conclusion

- “Information” must be conceptually differentiated from the related terms “signal”, “data” and “knowledge”. The basic model of communication—developed by Shannon—takes into consideration sender, channel and receiver, where the channel is used to transmit physical signals. Sender and receiver respectively encode and decode the signals into signs.
- General semiotics distinguishes between relations of signs among each other (syntax), the meaning of signs (semantics) and the usage of signs (pragmatics). When we neglect semantics and pragmatics and concentrate on the syntax of transmitted signs, we are speaking of “data”; when we regard transmitted signs from a semantic and pragmatic perspective, we are looking at “knowledge” and “information”, respectively.
- Knowledge is subjective, and thus forms part of Popper’s World 2, when a user disposes of it (know-that and know-how). Knowledge is objective and thus an aspect of World 3 in so far as the content is stored user-independently (in books or digital databases).
- Knowledge is thus fixed either in a human consciousness (as subjective knowledge) or another store (as objective knowledge). Knowledge as such cannot be transmitted. To transmit it, a physical form is required, i.e. an inFORMation. Information is knowledge put in motion, where the concept of information unites the two aspects of signal and knowledge in itself.
- Knowledge is present not only in texts, but also in non-textual documents (images, videos, music). Panofsky distinguishes between three semantic levels of interpretation: the pre-iconographical, the iconographical and the iconological level.
- Knowledge—as one of the core concepts of knowledge representation—allows for different perspectives. Knowledge can be regarded as both (true, known) statements (know-that) and as the knowledge of how to do certain things (know-how).
- According to Polanyi, implicit knowledge is pretty much physically embedded in a person, and hence can be objectified (externalized) only with difficulty.
- Knowledge management means the way knowledge is dealt with in organizations. According to Nonaka and Takeuchi, the knowledge spiral consisting of socialization, externalization, combination, and internalization (SECI) leads to success, even if externalization (transforming implicit knowledge into explicit knowledge) can hardly be achieved comprehensively. Probst et al. use several building blocks of corporate knowledge management.
- According to Spinner’s knowledge theory, we are confronted by a multitude of different types of knowledge, amounts of knowledge and qualities of knowledge, which all require their own methods of knowledge representation.
- Only knowledge in normal sciences (according to Kuhn) can be put into a knowledge organization system. In normal sciences, there are discipline-specific special languages (e.g., the terminologies of chemists or of biologists).
- Information does not (in contrast to knowledge) have a truth claim, and neither does it claim to be absolutely new; ideal informedness is achieved between the extreme poles of novelty and repeated confirmation.
- The meant knowledge of a sender does not have to match the understood knowledge of a receiver. If we interpose further instances between sender and receiver (besides signal transmission via the channel), i.e. human and mechanical intermediators, the problems of adequately communicating knowledge via information will become worse.
- Information science allows knowledge to be put in motion, but it also facilitates the development of specific added values (knowledge about), which, at the very least, reduce problems during the transferral of knowledge.
- “Knowledge is Power” refers, in Bacon, to man’s power over nature. Thinking further, it can also be taken to refer to power over other people. The negative consequences of “Knowledge is Power”

can be counteracted by granting everybody the opportunity of requesting knowledge. The goal is the subjects' information autonomy, guaranteed by their information literacy.

- In problematic situations, when there is uncertainty, purpose-oriented knowledge will be researched in order to consolidate decisions, close knowledge gaps and detect early-warning signals. The goal of the information-practical value chain is to translate knowledge into concrete actions via information. Information, however, is never “perfect” and cannot reduce uncertainties to zero. What it can do is to create relative advantages over others.
 - Humans as well as institutions require information autonomy in order to be able to research and apply the most action-relevant knowledge possible.
 - Both information science and information practice work out informational added values (knowledge about). This is accomplished over the course of professional intermediation, both in the input area of storage (knowledge representation) and during the output process (information retrieval).
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A.3 Information and Understanding

Information Hermeneutics

Hermeneutics is the art of *hemeneuein*, i.e. of proclaiming, interpreting, explaining and expounding. <Hermes> is the name of the Greek messenger of the Gods, who passed the messages of Zeus and his fellow immortals on to mankind. His proclamations are obviously more than mere communication. In explaining the divine commands, he translates them into the language of the mortals, making them understandable. The virtue of H(ermeneutics) always fundamentally consists in transposing a complex of meaning from another “world” into one’s own. This also holds for the original meaning of *hermeneia*, which is “the statement of thought”. This concept of statement itself is ambivalent, comprising the facets of utterance, explanation, interpretation and translation (Gadamer, 1974, 1061-1062).

Without Hermes, there would have been no understanding between the two worlds of gods and men. Hermes is the mediator. Hermeneutics is the science of (correct) understanding. An important source on hermeneutics, for us, is Gadamer’s “Truth and Method” (1975), which is significantly influenced by the philosophy of Heidegger (1962 [1927]).

What does information science activity have in common with that of Hermes? Looking at the sheer unstructured amount of documents on the one hand, and contrasting this opaque mass with the specific information need of an individual, it is impossible to reach a satisfactory understanding without a mediator of some kind. Information science helps create a “bridge” between documents and information needs. The complex of meaning—hermeneutically speaking—is transposed from one “world” to the other; figuratively speaking, from a text to a query. The problems and perspectives of hermeneutics, cognitive sciences and other disciplines on human cognition are manifold. In the context of this book, we will only emphasize some fundamental points of discussion that concern the area of information.

If we neglect the world outside of information, the social periphery, we will run the danger of remaining in permanent tunnel vision (Brown & Duguid, 2000, 5). A linear, purely goal-oriented path of information science can easily lead into a dead end, since a stable knowledge organization system and stable bibliographical records, respectively, will one day cease to correspond to the requirements of the day. A holistic point of view, on the other hand, opens up one’s field of vision.

Information retrieval concentrates on the searching for and finding of information. In knowledge representation, the evaluation and presentation of information is at the foreground. Both areas of information science only serve as means to a particular end: their focus is the potential user who is supposed to find the processed information, and the knowledge behind it, useful. Even the “best” indexing of content is of no use to the user if he cannot himself interpret or understand and explain it. Information science, with all its methods and applications, interacts dynamically with its

agents, the users and authors. We must not forget, here, that all the technologies, methods, tools and tasks of information science being used are founded, once more, by human activity.

Interaction can only come about when one recognizes what the other means and wishes, and when one declares oneself ready to accommodate him or her. This is accomplished via language. Linguistic ability alone is not enough to make one understandable to another, however. What has influenced and continues to influence every single human being has been learned via interaction with his environment, culture, and past. Hence, language is not neutral but is always used from a certain socio-cultural viewpoint. Gadamer (1975) calls this the “horizon”. Language is not formed and used without understanding and pre-judgments. Otherwise, people would talk past one another. A stream of communication is thus based on a commonly understood language used within a culture. However, the language changes over time, due to individual usage, and man likewise changes via/alongside his use of language.

This circumstance represents a particular challenge for the analysis and usage of a language in time, space and interaction. Since information science deals with the form and content of different languages, in various times and places as well as different areas of knowledge and cultural spheres, it must do justice to this diversity and the constant change underlying it.

Heraclitus already recognized: “Panta rhei—Everything is in a state of flux” (Simplicius, 1954, citing Heraclitus). Plato writes, “Heraclitus, I think, says that everything moves and nothing rests” (Plato, 1998, *Cratylus* 402a). Except for natural laws and mathematical formulas, perhaps, everything in the world is in constant change. Language is a part of this world. Information (and its interpretations) streams like a river, like water endlessly spreading itself into countless directions, coming apart into separate streams before flowing back together. Even if the stream of information appears to run the same course for a long time, we will never see the same exact stream containing exactly the same information. So every knowledge organization system and every subject description in bibliographical records cannot be stable over the course of time, but are objects of possible change (Gust von Loh, Stock, & Stock, 2009). For Buckland (2012, 160)

(i)t is a good example of a problem that is endemic in indexes and categorization systems: Linguistic expressions are necessarily culturally grounded, and, for that reason, in conflict with the need to have stable, unambiguous marks to enable library systems to perform efficiently. A static, effective subject indexing vocabulary is a contradiction in terms.

Water constantly seeks a way to keep flowing. It never flows uphill, as this would contravene nature. Rather, it is in interaction with nature. It follows certain requirements. What are the hermeneutic and cognitive bases, respectively, that resonate throughout the entire information process, and which we do not think about explicitly and tend to ignore? They are fundamentals of complex build, which form the activities of both

information professionals and users. We are dealing with a holistic perspective on information science, where man is incorporated into the information process with all of his characteristics and activities.

Four large complexes of human characteristics combine in a feedback loop (Figure A.3.1):

- Horizons of the author / information professional / user,
- Understanding of the documents (interpretation),
- Readiness to cooperate (commitment),
- Practice and innovation.

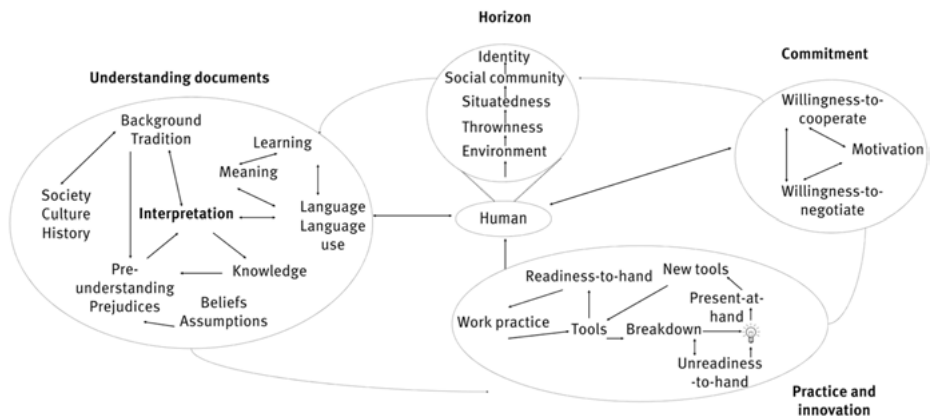


Figure A.3.1: Feedback Loop of Understanding Information.

Understanding

Man is born into the world, into an environment, without getting a say in the matter or the option of doing anything against it, a condition which Heidegger (1962 [1927]) calls “Thrownness”. There is no escaping from this environment. Man is not alone—this he would not survive—but he comes in contact with others. Accordingly, he must follow his own instincts while always being tied to the situation he finds himself in. His agency is a goal-oriented one, the goal being self-preservation. He learns to (continue to) exist by becoming a member of a social community. Embedded into this community, he develops his identity via his actions. The individual thus represents his standpoint from his own perspective.

Understanding, like the creation of knowledge, is never done independently of the respective context. In Japanese philosophy, this “shared context” is called “ba”. Ba is that place where information is understood as knowledge. Nonaka, Toyama and Konno (2000, 14) write:

Ba is a time-space nexus, or as Heidegger expressed it, a locationality that simultaneously includes space and time. It is a concept that unifies physical space such as an office space, virtual space such as e-mail, and mental space such as shared ideals.

The thoughts and feelings of individuals never exist outside of reality. For Heidegger, subject and existence (being-in-the-world) form a fundamental unit, which is ontologically grounded and cannot be further called into question. The world as such is explored by the individual via his understanding, which is present in human beings from the beginning. There is a multitude of possibilities in the world, and man carries the key—to put it metaphorically—that allows him to picture them. This key is understanding, or—seen from the cognitive standpoint—human cognition. Understanding, according to Heidegger, is what makes humans human. Accordingly, every activity in information science is irrevocably built on this fundamental understanding. Understanding is amenable to development, which relates it to interpretation. Interpretation means the act of processing the possibilities that have been developed through understanding. When interpreting something as something, that which is already understood is joined by something new and the whole is brought into a relation. Since the pre-opinion of the interpreter always forms a part of the text's interpretation, this process is not merely one of unconditionally registering the given data. In analogy to the way the interpreter approaches the text via his horizon, the text itself is founded by the horizon of the author. Pre-judgments and prejudices, without which no understanding would come about at all, are the respective bases of any interpretation. Interpretation is what grants a text its meaning.

History, culture and society form human background and tradition in the past and in the present. The *Dasein* of man is closely related to “man's historicity” (Day, 2010, 175). Knowledge—as the result of interpretation—depends upon earlier experience and on man's situatedness in a tradition. The constant learning process plays an important role in the usage and development of a language. In order for interpretation not to be influenced by arbitrary ideas, man—shaped as he is by pre-opinion and language use—directs his focus onto the thing itself.

A text is not read without understanding and pre-judgment (Gadamer, 1975). The draft or the expectation of meaning one approaches the text with bears witness to a certain openness, in the sense that one is prepared to accept the text in the first place. Understanding means knowing something and only then to sound another's opinion. The movement of understanding is not static but circular: from the whole to the part and back to the whole. Gadamer emphasizes that the hermeneutic circle is neither subjective nor objective or formal in nature—instead, it is determined via the commonality that links man to tradition, and hence to each other via the use of language. The circle (Gadamer, 1975, 293)

describes understanding as the interplay of the movement of tradition and the movement of the interpreter. The anticipation of meaning that governs our understanding of a text is not an act of

of subjectivity, but proceeds from the commonality that binds us to the tradition. But this commonality is constantly being formed in our relation to tradition.

The objective is to regard the interval as a positive and productive possibility, and not as a reproductive characteristic. Understanding as a process of effective history means to take into account one's own historicity. When we deal with communications, we do so from a certain standpoint, our "horizon". Having a horizon means (Gadamer, 1975, 301-302):

A person who has a horizon knows the relative significance of everything within this horizon, whether it is near or far, great or small. Similarly, working out the hermeneutical situation means acquiring the right horizon of inquiry for the questions evoked by the encounter with tradition.

We move dynamically in and with the horizon. The past is always involved in the horizon of the present. In understanding, finally, there is a "fusion of horizons"—an interpreter wants to understand the communication contained in the text, and to do so relates the text to hermeneutical experience. This communication is language in the purest meaning of the word, since it speaks for itself in the same way as an interlocutor (you) does. A *you*, however, is not an object but something that relates to another. Hence, one must admit the communication's claim to having something to say.

A forwarded text that becomes subject to interpretation asks a question of the interpreter. A text, with all the subjects addressed within it, can only be understood if one grasps the question answered by the text. This does not merely involve comprehending an outside opinion, but rather the necessity of setting the subjectivity of the text in relation to one's own thinking (fusion of horizons).

How does a user deal with a documentary unit that has been created with the help of one of the methods of knowledge representation (e.g. classification or thesaurus) and while using a specific tool (e.g. the International Patent Classification)? In the ideal scenario, he will try to understand and explain both the surrogate *and* the original document.

Understanding, as a form of cognition, means understanding something *as something*. Processing the documentary unit as well as reading the original document involves all hermeneutical aspects:

- the presence of a corresponding key,
- concentration on the forwarded text (i.e. the thing),
- here: reconstruction of the question answered by the text,
- understanding the parts necessitates understanding the whole, *and* understanding the whole necessitates understanding the parts (hermeneutic circle),
- text and reader are in different horizons, which are fused in the understanding,
- in this, the reader's own pre-understanding is to be estimated positively—as pre-judgment—as long as it contributes dynamically (in the hermeneutic circle) to the fusion of horizons.

Hermeneutics and Information Architecture

Of central importance to our subject is the conjunction of the results obtained by Winograd and Flores (1986) with regard to the architecture and design of information systems and, simultaneously, to the use of methods of knowledge representation and information retrieval. The focus is not on some computer programs—what is being analyzed, rather, are the decisive fundamentals influencing the works and innovations achieved while using computers. Here, the basis is formed by man's social background and his language as a tool of communication; computers only aid real communication in the form of a medium, or a tool.

Winograd and Flores argue as follows: tradition and interpretation influence one another, since man—as a social animal—is never independent of his current and historically influenced environment. Any individual that understands the world interprets it. These interpretations are based on pre-judgment and pre-understanding, respectively, which always contain the assumptions drawn upon by the individual in question. We live with language. Language, finally, is learned via the activities of interpretation. Language itself is changed by virtue of being used by individuals, and individuals change by using it. Hence, man is determined via his cultural background and can never be free of pre-judgments. Concerning the inevitability of the hermeneutic circle according to Gadamer, we read (Winograd & Flores, 1986, 30):

The meaning of an individual text is contextual, depending on the moment of interpretation and the horizon brought to it by the interpreter. But that horizon is itself the product of a history of interactions in language, interactions which themselves represent texts that had to be understood in the light of pre-understanding. What we understand is based on what we already know, and what we already know comes from being able to understand.

According to Heidegger, we deal with things in everyday life without thinking about the existence of the tool we are using. This tool is ready-to-hand (*Zuhandenheit* in Heidegger's original German text). We have been thrown into the world and are thus coerced into acting pragmatically. At the moment when the stream of action is interrupted by whatever circumstances, man begins to interpret. The tool has become useless. It is unready-to-hand (*Unzuhandenheit*). We search for solutions. Only now do the characteristics of the tool come to light, and the individual becomes aware of them. He realizes the tool's *Vorhandenheit*—it is now present-at-hand.

Such a “breakdown” with regard to language plays a fundamental role in all areas of life, including—Winograd and Flores maintain—in computer design and management and, related to our own problem area, in knowledge representation and information retrieval. A breakdown creates the framework about which something can be said, and language as a human act creates the space of our world (Winograd & Flores, 1986, 78):

It is only when a breakdown occurs that we become aware of the fact that ‘things’ in our world exist not as a result of individual acts of cognition but through our active participation in a domain of discourse and mutual concern.

Man gives meaning to the world in which he lives together with other men. However, we must not understand language as a series of statements about objective reality, but rather as human acts that rest on conventions regarding the background that informs the respective meaning. Background knowledge and expectations guide one’s interpretation. Information retrieval knows this problem all too well, as there exist, in many places, different words for one and the same concept (synonyms), or anaphora, i.e. expressions that point to other terms (e.g. pronouns to their nouns). Without interpretation, there can be no meaning. Interpretation, meaning and situation, respectively tradition, cannot be regarded in isolation here, either. According to Winograd and Flores, the phenomena of background and interpretation permeate our entire everyday life.

When understanding situations or sentences, man links them to suppositions or expectations similar to them. The basis of understanding as a process is memory. The origin of human language as an activity is not the ability to reflect on the world, but the ability to enter commitments. The process is an artistic development that goes far beyond a mere accumulation of facts.

Concerning the question whether computers also enter such connections and comprehend commands in this way, Winograd and Flores emphasize that computers and their programs are nothing other than actively structured media of communication that are “fed” by the programmer. Winograd and Flores (1986, 123) write:

Of course there is a commitment, but it is that of the programmer, not the program. If I write something and mail it to you, you are not tempted to see the paper as exhibiting language behavior. It is a medium through which you and I interact. If I write a complex computer program that responds to things you type, the situation is still the same—the program is still a medium through which my commitments to you are conveyed. ... Nonetheless, it must be stressed that we are engaging in a particularly dangerous form of blindness if we see the computer—rather than the people who program it—as doing the understanding.

Hence, we should not equate the technologies of artificial intelligence with the understanding of human thought or human language. The design of computer-based systems only facilitates human tasks and interaction. The act of designing itself is ascribed an ontological position: since man generally uses tools as part of his “man-ness”, in the same way the programmer uses his computer. The indexer uses the respective programs of the computer, the methods of indexing as well as the specific tools. By using tools, we ask and are told what it means to be human. The programmer designs the language of the world in which the user works; or, at least, he tries to create the world that regards or could regard the user. During this creative dynamic process, a fundamental role is played by breakdowns as conspicuous incidents,

imperfections and the like. They necessitate and sometimes cause a counteraction to fix the problem. True to the motto “mistakes are for learning”, breakdowns help analyze human activity and create new options or changes.

Analysis of Cognitive Work

The main characteristic of the analysis of cognitive work is that it observes human action in the midst of, and in dependence of, its environment. CWA (Cognitive Work Analysis) was developed in the early 1980s by Danish researchers at the Risø National Laboratory in Roskilde (Rasmussen, Pejtersen, & Goodstein, 1994; see also: Vicente, 1999). Where human work requires decisions to be made, it is designated as cognitive work. Fidel (2006) grounds the necessity of empirical work analyses in information science on the following: since information systems are created for the reason of supporting people in their activities, the architecture and design of these systems should consequently be based on those activities as performed by them. In CWA, the actors are referred to in this context as carriers of activities. Factors that necessarily shape these activities are called “constraints”. It often happens in the day-to-day that one and the same person performs different activities at different times in the public and private spheres, respectively, and correspondingly has different information needs (Fidel, 2006, 6):

This diversity introduces the idea that each type of activity may require its own information system.

CWA, with the help of work analyses of certain user groups, strives to answer the following questions: what are the constraints formed by the different activities, and how do these constraints affect the activities?

Proponents view the requirement of CWA as based in various different areas. For instance, Mai (2006) relates the approach of contextual analysis to knowledge representation—more specifically, to the development of controlled vocabularies (including classifications, semantic networks and up to complex thesauri). According to Mai, the developers of classification systems can use analysis to gain an insight into which factors (can) influence the work of the actor (Mai, 2006, 18):

The outcome is not a prescription of what actors should do (a normative approach) or a detailed description of what they actually do (a descriptive approach), but an analysis of constraints that shape the domain and context.

The structure of the analysis of cognitive work is displayed graphically in Figure A.3.2.

At the center of the dimensions is the actor with his respective education, technological experience, preference for certain information tools, knowledge about the

specific area of application and values he adheres to. Hence, six further dimensions that influence and shape the activities are analyzed.

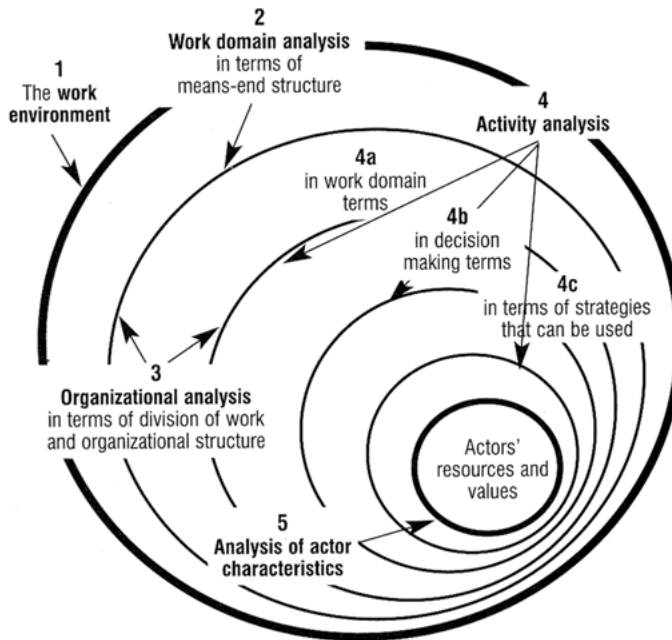


Figure A.3.2: Dimensions of the Analysis of Cognitive Work. Source: Pejtersen & Fidel, 1998.

Analyzing the work area generates the context in which the actor works. Actors may take part in the same work topic as others, while (due to their placement in a certain organization) setting different priorities and goals than others. Organizational analysis provides insight into the workplace, management or position of the actor in the organization. Activity analysis brings to light what actors do to fulfill their tasks. How, when and where does the actor require which type of information? Is the information present or not? Which information do decision makers require? Finally, the actor's strategies of publishing and searching information are observed. All of these empirical aspects of analysis, which serve to uncover the various interactions between people and information, first provide an insight into the actor's possible information need. After this, they are useful for the development of methods and tools of knowledge representation. The procedure following CWA is elaborate, but purposeful. It helps uncover the conditions for good knowledge representation and information retrieval to aid an actor's specific tasks (Mai, 2006, 19):

Each dimension contributes to the designer's understanding of the domain, the work and activities in the domain and the actors' resources and values. The analyses ensure that designers bring the relevant attributes, factors and variables to design work. While analysis of each dimension

does not directly result in design recommendations, these analyses rule out many design alternatives and offer a basis from which designers can create systems for particular domains. To complete the design, designers need expertise in the advantages and disadvantages of different types of indexing languages, the construction and evaluation of indexing languages and approaches to and methods of subject indexing.

Next up is evaluation, which, in analogy to the analysis of cognitive work, may provide insights into a potentially improved further processing (Pejtersen & Fidel, 1998). It is checked whether the implemented system corresponds to the previously defined goals and whether it serves the user's need. In the evaluation, too, the objective is not to focus on the system as a whole, but on the work-centric framework. The evaluation questions and requirements concern the individual dimensions, and a structure of analytical, respectively empirical, examination characteristics is compiled. Various facets and measurements are employed here, relative to the different dimensions of cognitive work. Pejtersen and Fidel demonstrate, via a study observing students and their working environment during Web searches, how the analysis and evaluation of cognitive work can uncover problem criteria for this user group.

Ingwersen and Järvelin (2005) create a model for information retrieval and knowledge representation that places the cognitive work of the actor at the foreground (Figure A.3.3). The actor (or a team of actors) is anchored in cultural, social and organizational horizons (context) (Relation 1). Actors are, for instance, authors, information architects, system designers, interface designers, creators of KOSs, indexers and users (seekers of information). Via a man-machine interface (2), the actor comes face to face with the information objects as well as with information technology (3). "Information objects" are the documentary units (surrogates) as well as the documents (where digitally available) that are accessible via methods and tools of knowledge representation and that have been indexed by form and content. "Information technology" summarizes the programming-related building blocks of an indexing and retrieval system (database, retrieval software, algorithms of Natural Language Processing) as well as retrieval models. Since the information objects can only be processed via information technology, there is another close link between these two aspects (4).

The actor always interacts with the system via the interface, but he requires additional knowledge about the characteristics of the information objects (Relation 5) as well as the information technology being used (7); additionally, information objects and information technology do not work independently of the social, cultural and organizational backgrounds (Relations 6 and 8). As opposed to the interactive relations 1 through 4, Relations 5 through 8 feature cognitive influences "in the background", which—as long as an information system is "running"—must always be taken into consideration, but cannot be interactively influenced.

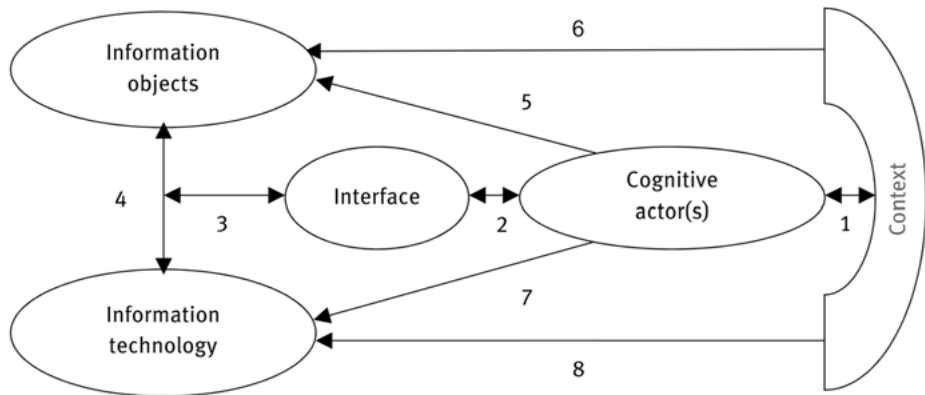


Figure A.3.3: The “Cognitive Actor” in Knowledge Representation and Information Retrieval. Source: Ingwersen & Järvelin, 2005, 261. Arrows: Cognitive Transformation and Influence; Double Arrows: Interactive Communication of Cognitive Structures.

Ingwersen and Järvelin (2005, 261-262) describe their model:

First, processes of social interaction (1) are found between the actor(s) and their past and present socio-cultural or organizational context. ... Secondly, information interaction also takes place between the cognitive actor(s) and the cognitive manifestations embedded in the IT and the existing information objects via interfaces (2/3). The latter two components interact vertically (4) and constitute the core of an information system. This interaction only takes place at the lowest linguistic sign level. ... Third, cognitive and emotional transformations and generation of potential information may occur as required by the individual actor (5/7) as well as from the social, cultural or organizational context towards the IT and information object components (6/8) *over time*. This implies a steady influence on the information behaviour of other actors—and hence on the cognitive-emotional structures representing them.

Conclusion

- Knowledge representation and information retrieval are influenced by hermeneutical aspects (understanding, background knowledge, interpretation, tradition, language, hermeneutic circle and horizon) and cognitive processes.
- Every knowledge organization system and every subject description of documents cannot be stable over the course of time, but are subjects to possible changes.
- If a forwarded text, which becomes subject to interpretation, is to be understood, the underlying subjectivity must—according to Gadamer—be set into a relation with one’s own thinking. The act of understanding effects a “fusion of horizons” between text and reader.
- Winograd and Flores establish a link between hermeneutics and information architecture. Tradition and language form the basis of human communication. In the same way that every man uses tools, the programmer uses his computer and the indexer his respective computer programs, methods of indexing and specific tools. “Breakdowns”, in the sense of errors in the human flow

of activity, are regarded as positive occurrences, since in requiring solutions they encourage creative activity.

- Human work that requires decisions is called “cognitive work”. Using an analysis of cognitive work, several dimensions that affect the entire working process within an environment can be uncovered. CWA (Cognitive Work Analysis) can contribute to the improvement of the further development of methods and tools of knowledge representation and information retrieval.
 - When processing and searching information, the cognitive actor always stands in relation to information objects, information systems and interfaces as well as to the peripheral environment (organization, social and cultural context).
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A.4 Documents

What is a Document?

The “document” is a crucially important concept for information science (Buckland, 1997; Frohmann, 2009; Lund, 2009). It is closely related to the concept of “resource”, which is why both terms are used synonymously. In cases where only the intellectual content of a document is being considered, we speak of a “work”. Floridi (2002, 46) even defines our entire scientific discipline via the concept of the document:

Library and information science ... is the discipline concerned with documents, their life cycles and the procedures, techniques and devices by which these are implemented, managed and regulated.

For a long time, an area of information practice used to be described as “documentation” (Buckland & Liu, 1995)—some of the leading information science journals were called “American Documentation” (in the U.S.) or “Nachrichten für Dokumentation” (in Germany). Even today, the “Journal of Documentation” remains one of the top publications of information science (Hjørland, 2000, 28). Documentation deals with documents, both conceptually and factually. Like “documentation”, “document” is etymologically made up of the Latin roots “doceo” and “mentum” (Lund, 2010, 743). “Doceo” refers to teaching and educating, pointing—like the etymology of “information”—to pedagogical contexts; “mentum” in Latin is a suffix used to turn verbs into nouns. Only from the 17th century onwards has “document” been understood as something written or otherwise fixed on a carrier (Lund, 2009, 400).

What is being stored in databases for the purposes of retrieval? We already know that distinctions must be made between documentary reference unit, documentary unit and original document—the documentary unit serving as the representation (surrogate) of the documentary reference unit. The latter is created through the database-appropriate analytical partition of documents into meaningful units. For instance: a book is a document, a single chapter within is the documentary reference unit and the data set containing the metadata (e.g. formal bibliographical description and content-depicting statements, perhaps complemented by the full text) is the documentary unit.

In a first approximation, we follow Briet in defining “document” as “any concrete or symbolic indexical sign [*indice*], preserved or recorded to the ends of representing, of reconstituting, or of providing a physical or intellectual phenomenon” (2006[1951], 10). It is intuitively clear that all manner of texts are documents. There is no restriction in terms of the documents’ length. An extreme example of short texts is a correspondence between Victor Hugo and his publisher (Yeo, 2008, 139). After the publication of his novel “Les Misérables”, Hugo sent a telegraph with only one character—“?”—apparently meaning to inquire about the sales numbers for his book. His publisher,

having understood the question, responded—as the novel had in fact been a great success—with “!” On the other hand, there are patent documents comprising more than 10,000 pages, and which still represent one single document.

Are there further document types besides texts? Buckland (1991, 586) claims that there is “information retrieval of more than text”. The idea of infusing non-textual documents with informational added value goes back to Briet (2006[1951]). The antelope she described has become the paradigm for the (additional) processing of non-textual documents in documentation (Maack, 2004). Briet asks: “Can an antelope in the wilderness of Africa be a document?”, to which the answer is a resounding “no”. However, if the animal is captured and kept in a zoo, it becomes a “document”, since it is now being consciously presented to the public with the clear intention of showing something. Buckland (1991, 587), discussing Briet’s antelope, observes:

Could a star in the sky be considered a “document”? Could a pebble in a babbling rock? Could an antelope in the wilds of Africa be a document? No, she (Briet, A/N) concludes. But a photograph of a star would be a document and a pebble, if removed from nature and mounted as a specimen in a mineralogical museum, would have become one. So also an antelope if it is captured, held in a zoo, and made an object of research and an educational exhibit would have become a “document”. ... (A)n article written about the antelope would be merely a secondary, derived document. The primary document was the antelope itself.

Objects are “documents” if they meet the following four criteria:

- materiality (they are physically—including the digital form—present),
- intentionality (they carry meaning),
- development (they are created),
- perception (they are described as a document).

In Buckland’s words (1997, 806), this reads as follows:

Briet’s rules for determining when an object has become a document are not made clear. We infer, however, from her discussion that:

1. there is materiality: Physical objects and physical signs only;
2. there is intentionality: It is intended that the object be treated as evidence;
3. the objects have to be processed: They have to be made into documents; and we think,
4. there is a phenomenological position: The object is perceived to be a document.

This definition is exceedingly broad. Following Martinez-Comeche (2000), it can be stated that 1st anything can be a document, but that 2nd nothing is a document unless it is explicitly considered as such. “A document is”, according to Lund (2010, 747), “any result of human documentation.” Documents are “knowledge artefacts that reflect the cultural milieu in which they arose” (Smiraglia, 2008, 29) and each document “is a product of its time and circumstances,” Smiraglia (2008, 35) adds. For Mooers (1952), a document is the channel between a creator (e.g., author) and a receiver (e.g., reader) which contains messages.

This comprehensive definition of “document” now includes textual and non-textual, digital and non-digital resources. Textual and factual indexing and retrieval are always oriented on documents, leading Hjørland (2000, 39) to suggest “documentation science” as an alternative name for the corresponding scientific discipline (instead of “information science”).

The Documents’ Audience

Documents are no ends in themselves, but act as means of asynchronous knowledge sharing for the benefit of an audience. This audience consists of the factual or—in future—the hypothetical users. Like the documents’ creators, the documents’ users have different intellectual backgrounds and speak different jargons. Where author and user share the same background, Østerlund and Crowston (2011) speak of “symmetric knowledge”, where they don’t, of “asymmetric knowledge”. The asymmetric knowledge of heterogeneous communities leads to the conception of “boundary objects” by Star and Griesemer (1989). “We can assume that documents typically shared among groups with asymmetric access to knowledge may take the form of boundary objects” (Østerlund & Crowston, 2011, 2). Such boundary documents “seem to explicate their own use in more detail” (Østerlund & Crowston, 2011, 7). It is possible for the document’s user to produce a new document, which is then addressed to the author of the first document. Now we see a document cycle (Østerlund & Boland, 2009)—e.g. in healthcare between medical records, request forms, lab work reports, orders, etc. Boundary documents call for polyrepresentation in the surrogates: the application of either different Knowledge Organization Systems (KOSs) for the heterogeneous user groups or of one KOS with terminological interfaces for the user groups.

Documents are produced in a certain place at a certain time. For Østerlund (2008, 201) a document can be a “portable place” that “helps the reader locate the meaning and the spatio-temporal order out of which it emerges.” In Østerlund’s healthcare case study doctors communicate via documents. Here is a typical example: “One doctor could be reporting on her relation to the patient to another physician who has no prior knowledge of that patient” (Østerlund, 2008, 202).

Documents have a time reference. They have been created at a certain time and are—at least sometimes—only valid for a certain duration. The date of creation can play a significant role under certain circumstances. E.g., patents, when they are granted, have a 20-year term, starting from the date of submission.

It is a truism, but all documents are created in a social context and against a cultural background. What has influenced and continues to influence every single human being has been learned via interaction with his environment, culture, and past. Hence, language and documents are not neutral but are always used from a certain socio-cultural viewpoint. In hermeneutics, this is called the document’s “horizon” (see Ch. A.3).

The surrogates of documents in information systems are documents as well. For Briet (2006[1951], 13), the creation of surrogates (called “documentation” following Otlet, 1934) is a “cultural technique”. So the indexer (or the librarian, the knowledge manager, the information systems engineer, etc.) is an active and important part in the process of maintaining the institutional memory. He decides whether a document should be stored or not, and which metadata are used to describe the document. Such decisions are of great importance for the preservation (Larsen, 1999) and retrievability of documents. For Briet, the indexer is embedded in the social context of an institution; she calls him a “team player” (2006[1951], 15). “Documentation is part of public spaces of production—social networks and cultural forms,” Day (2006, 56) comments on Briet’s thesis.

Digital Documents and Linked Data

Documents are either available in digital or non-digital form; indeed, it is possible for both forms to coexist side by side. One example for such a parallel form is an article in the printed edition of a scientific journal, whose counterpart is available digitally in the World Wide Web (as a facsimile). Texts can always be digitalized (via scanning), at least in principle, which is not always the case for other objects. A museum object, e.g. our pebble, can never be digitalized. Here, we are always forced to represent the document exclusively via the documentary unit. Other objects, such as images, films or sound recordings, are principally digitizable, like texts.

The complete versions of the documents can only be entered into information retrieval systems if they are digitally available. If digitization is impossible as a matter of principle or for practical reasons (such as Briet’s antelope) documentary reference units must be defined and documentary units created. In the case of digital documents, the original may under certain circumstances be entered into the system without any further processing. It is doubtful, however, whether it makes sense to forego the informational added value of a documentary unit. If both the documents and their surrogates are unified in a single system, we speak of “digital libraries”.

In contrast to typical documents written on paper, digital documents (Buckland, 1998; Liu, 2004) display two new characteristics: fluidity and intangibility (Liu, 2008, 1). They do not have a tangible carrier (such as paper), consisting merely of files on a computer; they can principally be altered at any point (one good example for this being Wikipedia entries). Phases of stability alternate with phases of change (Levy, 1994, 26), in which the very identity of a document may be lost. It is easily possible at any point to preserve a stable phase (and thus the identity of a digital document), at least from a technological standpoint (Levy, 1994, 30).

A French research group—RTP-DOC (Réseau Thématique Pluridisciplinaire—33: Documents and content: creating, indexing, browsing) at the Centre National de la Recherche Scientifique (CNRS)—has discussed the concept “document” under the

aspects of digitization and networking. In publications, “RTP-DOC” became the pseudonym “Roger T. Pédaque” (Pédaque, 2003; cf. also Francke, 2005; Gradmann & Meister, 2008; Lund, 2009, 420 et seq.). Analogously to syntax, semantics and pragmatics, R. T. Pédaque (2003, 3) studies documents from the points of view of form, sign and medium. On the level of form, digital documents are a whole made up of structure and data. The structure describes, for instance, fonts that are used (e.g. in Unicode) or markup languages for web pages (such as HTML or XML), whereas data refers to the character sequences occurring within the document. On the level of sign, digital documents present themselves as the connection between informed text and knowledge; insofar as the text bears knowledge that a competent reader is able to extrapolate. The object in discussing digital documents as a medium is the pairing of inscription and legitimacy. That which is digitally written down has different levels of legitimacy. Pédaque (2003, 18) emphasizes:

A document is not necessarily published. Many documents, for instance ones on private matters (...), or ones containing undisclosable confidential information can only be viewed by a very limited number of people. However, they have a social character in that they are written according to established rules, making them legitimate, are used in official relations and can be submitted as references in case of a dysfunction (dispute). Conversely, publication, general or limited, is a simple means of legitimization since once a text has been made public ... it becomes part of the common heritage. It can no longer easily be amended, and its value is appreciated collectively.

In the sense of legitimacy, we will distinguish between formally published, informally published and unpublished documents from here on out.

The fact that we can distinguish between structure and data on the syntactical level of digital documents means—at least in theory—that we are always capable of automatically extracting the data contained in a document and to save or interconnect them as individual factual documents. “In principle, any quantity of data carrying information can be understood as a document,” Voß (2009, 17) writes. In order for all documents—text documents, data documents or whatever—to be purposefully accessible (to men and machines), they must be clearly designated. This is accomplished by the Uniform Resource Identifiers (URI) or the Digital Object Identifiers (DOI; Linde & Stock, 2011, 235-236). Under these conditions, it stands to reason that sets of factual knowledge which complement each other should be brought together even if they originate in different documents. This is the basic idea of “Linked Data” (Bizer, Heath, & Berners-Lee, 2009, 2):

Linked Data is simply about using the Web to create typed links between data from different sources. ... Technically, Linked Data refers to data published on the Web in such a way that it is machine-readable, its meaning is explicitly defined, it is linked to other external data sets, and can in turn be linked to from external data sets.

In cases of automatic fact extraction in the context of Linked Data, the original text document fades into the background, in favor of the received data and their factual documents (Dudek, 2011, 28). (Of course it remains extant as a document.) It must be noted that the data must be defined in a machine-readable manner (with the help of standards like RDF, the Resource Description Framework). Additionally, we absolutely require a knowledge organization system (KOS) that contains the needed terms and sets them in relation to one another (Dudek, 2011, 27).

In addition to extracted facts there are data files which cover certain topical areas, e.g. economic data (say, employment figures) and other statistical materials (cf. Linde & Stock, 2011, 192-201), geographic data (used for example by Google Maps) as well as real-time data (such as current weather data, data on delays in mass traffic or tracking data of current flights). All these data can—inasmuch as it makes sense—be combined in so-called “mash-ups”. So it is possible to merge employment figures with geographical data resulting in a map which shows employment rates in different regions. Dadzie and Rowe (2011, 89) sum up:

The Web of Linked Data provides a large, distributed and interlinked network of information fragments contained within disparate datasets and provided by unique data publishers.

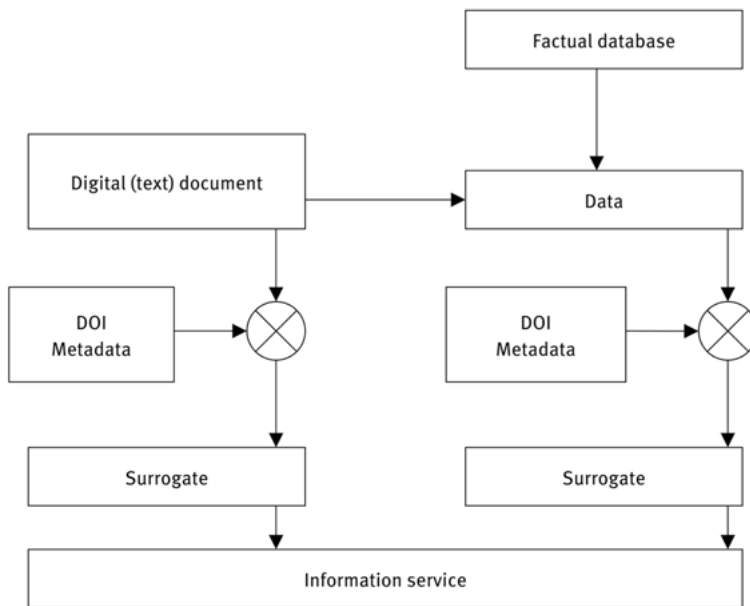


Figure A.4.1: Digital Text Documents, Data Documents and Their Surrogates in Information Services.

In Figure A.4.1, we attempt to graphically summarize the relations between digital text documents, the data contained therein and their representations as surrogates

in information services (such as the WWW, specialist databases or the so-called “Semantic Web”). In the information service, the linking of text documents (among one another and to the data) and data documents (also to one another and to the text documents) is facilitated via standards (like RDF) and the mediation of the KOS.

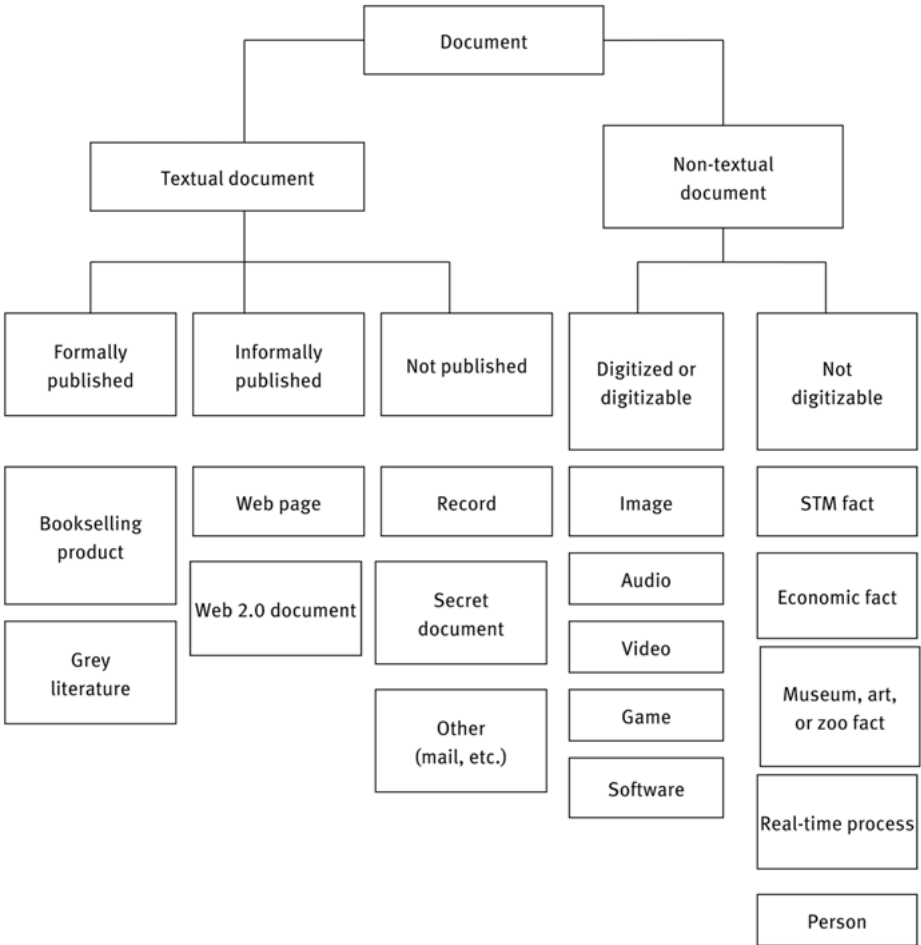


Figure A.4.2: A Rough Classification of Document Types.

An Overview of Document Types

In a first rough classification (Figure A.4.2), we distinguish between textual and non-textual documents. Text documents are either formally published (as a book, for instance), informally published (as a weblog) or not published (such as busi-

ness records). In the case of non-textual documents, we must draw lines between those that are available digitally or that can at least in principle be digitized (images, music, speech, video) and undigitizable documents. The latter include factual documents that have been extracted either intellectually, via factual indexing (resulting for example in a company dossier), or automatically from digital documents. In the context of information science, all documents are initially partitioned into units (dividing an anthology into its individual articles, for instance) and enriched via informational added values. In other words, they are formally described (e.g. by stating the author's name) and indexed for content (via methods of knowledge representation) (see Figure A.4.3). The resulting surrogates, or documentary units, carry metadata: text documents and digital (digitizable) non-textual documents contain bibliographical metadata (Ch. J.1), whereas undigitizable documents carry metadata about objects (Ch. J.2).

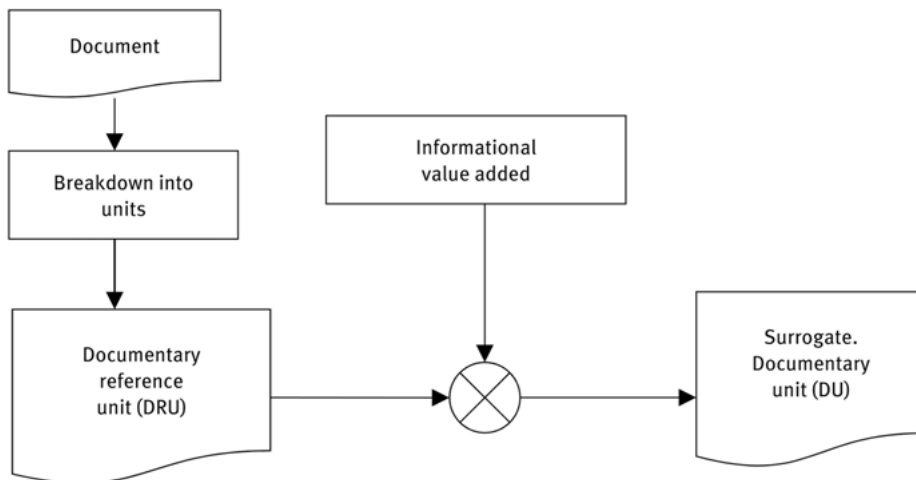


Figure A.4.3: Documents and Surrogates.

Formally Published Documents

There are textual documents that are published and there are those that are not. The former are divided into the two classes of (verified) formal and (generally unverified) informal texts. We define “text” in a way that includes documents that may partly contain elements other than writing (e.g. images).

Formal publications include all texts that have passed through a formal publishing process. This goes for all bookselling products: books, newspapers (including press reports from news agencies) and journals (including their articles), sheet music, maps etc. They also include all legal norms, such as laws or edicts as well as court

rulings, if they are published by a court. Furthermore, they include those texts that are the result of intellectual property rights processes, i.e. patents (applications and granted patents), utility models, designs and trademarks. A characteristic of formally published texts is that they have been verified prior to being published (by the editor of a journal, newspaper or publishing house, an employee of a patent office etc.). This “gatekeeper” function of the verifiers leads us to assume a degree of selection, and thus to ascribe a certain inherent quality to formally published documents.

From the perspective of content, we will divide formally verified documents into the following classes:

- Business, market and press documents (Linde & Stock, 2011, Ch. 7),
- Legal documents (Linde & Stock, 2011, Ch. 8),
- STM documents (science, technology and medicine) (Linde & Stock, 2011, Ch. 9),
- Fiction and art.

So-called “grey literature” (Luzi, 2000) is a somewhat problematic area, since there is no formal act of publication. It includes texts that are distributed outside the sphere of publishing and the book trade, e.g. university theses, series of “working papers” by scientific institutions or business documents by enterprises. In the case of university theses, the formal act of verification is self-evident; the situation is similar in the case of working papers, which, in general, present research results a lot more quickly than journals do. Business documents, such as company magazines or press reports, may contain texts about inventions that are not submitted for patents but whose priority must still be safeguarded. This alone is a reason for us to pay attention to these texts. For Artus (1984, 145),

there can be no doubt ... that grey literature is just as legitimate an object of documentation as formally published literature.

Where can we locate the indivisible entity of a formally published document (IFLA, 1998)? Bawden (2004, 243) asks:

(W)e ... have a problem deciding when two documents are “the same”. If I have a copy of a text-book, and you have a copy of the same edition of the same book, then these are clearly different objects. But in a library catalogue they will be treated as two examples of the same document. What if the book is translated, word for word, as precisely as possible, into another language: is it the same document, or different? At what point do an author’s ideas, on their way to becoming a published book, become a “document”?

We will answer Bawden’s question in Chapter J.1.

Informally Published Texts

We now come to informally published texts. These mainly include those texts that are published on the Internet, respectively on one of its services. Note: we say published, but not verified.

Documents on the World Wide Web are marked via an individual address (URL: Uniform Resource Locator). The basis of the collaboration between servers on the WWW is the Hypertext Transfer Protocol (HTTP). The most frequent data type is HTML (Hypertext Markup Language); it allows users to directly navigate to other documents via links. A web page is a singular document with precisely one URL. A connected area of web pages is called a website (in analogy to a camping site, on which the individual tents, i.e. web pages, are arranged in a certain fashion), with the entry page of a website being called the homepage. Websites are run by private individuals, companies, scientific institutions, other types of institutions as well as specific online service providers (e.g. search engines or content aggregators). Certain web pages contain hardly any content or even none at all, only serving purposes of navigation. Besides HTML files, websites can contain other document types (such as PDF or WORD, but also graphic or sound files).

In Web 2.0 services in particular, many people collaborate in the publication of digital documents (Linde & Stock, 2011, Ch. 11). Texts on message boards are of a rather private or semi-private character. It is here that people who are interested in a given subject meet online in order to “chat”. In contrast to chatrooms, the entries on message boards are saved.

Some sites on the WWW are continuously updated. They can be written in the form of a diary (like weblogs, or blogs in short) and contain either private or professional content, or they can filter information from other pages and incorporate them on their own URL. Short statements are called microblogs (like the “tweets” on Twitter or Weibo). Status updates in social networks (like Facebook) are posted frequently—depending on the user’s activity—while personal data remain rather static.

A position in between formally and informally published texts is occupied by documents in Wikis. Since anyone can create and correct entries at any point (as well as retract erroneous corrections), we can describe such Web pages as cooperatively verified publications. The criteria for such verification, however, are entirely undefined. Likewise, the question as to which documents should be accepted into a Wiki in the first place, and to what level of detail, is probably not formally defined to any degree of exhaustiveness (Linde & Stock, 2011, 271-273).

All of these Web 2.0 services confront information retrieval systems with the task of locating the documents, indexing them and making them available to their users incredibly quickly—as soon after their time of publication as possible.

Unpublished Texts

The last group of text documents are texts that are not published. These include all texts that accrue for private individuals, companies, administrations, etc., i.e. letters, bills, files, memos, internal reports, presentations etc. and additionally secret documents. This area also includes those documents from institutions which are released on their Intranet or Extranet (part of the Internet that is protected and can only be accessed by authorized personnel, e.g. clients and suppliers). Such texts can be of great importance for the institutions concerned; their objective is to store them and make them available via information retrieval. This opens up the large area of corporate knowledge management.

Equally unpublished are texts in certain areas of the Internet, such as e-mails and chats. Here, too, information retrieval is purposeful, e.g. to aid the individual user in searching his sent and received mails or—with great ethical reservations—for intelligence services to detect “suspicious” content.

Documents and Records

In a specification for records management, the European Commission defines “record” via the concept of “document” (European Commission, 2002, 11):

(Records are) (d)ocument(s) produced or received by a person or organisation in the course of business, and retained by that person or organisation. ... (A) record may incorporate one or several *documents* (e.g. when one document has attachments), and may be on any medium in any format. In addition to the content of the document(s), it should include contextual information and, if applicable, structural information (i.e. information which describes the components of the record). A key feature of a record is that it cannot be changed.

Are all files really documents (Yeo, 2011, 9)? Yeo (2008, 136) defines “records” with regard to activities:

Records (are) persistent representations of activities or other occurrences, created by participants or observers of those occurring by their proxies; or sets of such representations representing particular occurrences.

According to Yeo (2007; 2008), this relation to activities, or occurrences, is missing in the case of documents, so perhaps records cannot be defined as documents. However, we will construe records as a specific kind of documents, since, in the information-science sense, the same methods are used to represent them as for other documents. It is important to note, though, that a record (say, a personnel file) may contain various individual records (school reports, furloughs, sick leaves etc.). Depending on the choice of documentary reference unit, one can either display the personnel file as

a whole or each of the documents contained therein individually. When selecting the latter option, one must add a reference to the file as a whole. Otherwise, the connection is lost.

A similar division between superordinate document and its component parts (which are also documents) exists in archiving, specifically in the case of dossiers. In a press archive, for instance, there are ordered dossiers about people—generally in so-called press packs. These dossiers contain collections of press clippings with reports on the respective individual.

Non-Textual Image, Audio and Video Documents

In the case of non-textual documents, we distinguish between digital (or at least digitizable) and principally non-digital forms. Digitizable documents include visual media such as images (photographs, schematic representations etc.) and films (moving images), including individual film sequences, as well as audio media such as music, spoken word and sounds (Chu, 2010, 48 et seq.). Digital games (Linde & Stock, 2011, Ch. 13) as well as software applications (Linde & Stock, 2011, Ch. 14) also belong to this group. Such documents can be searched and retrieved via special systems of information retrieval—even without the crutch of descriptive texts—as in the example of searching images for their shapes and distributions of light and color (Ch. E.4).

Images, videos and pieces of music from Web 2.0 services (such as Flickr, YouTube and Last.fm) fall under the category of digital, informally published non-textual documents; they all have in common the fact that they can be intellectually indexed (via folksonomy tags).

Undigitizable Documents: Factual Documents

Undigitizable documents are represented in information retrieval systems via documentary units—the surrogates—only. We distinguish between five large groups:

- STM facts,
- Business and economic facts,
- Facts about museum artifacts and works of art,
- Persons,
- Real-time facts.

Of central importance for science, technology and medicine (STM) are facts, e.g. chemical compounds and their characteristics, chemical reactions, diseases and their symptoms, patients and their medical records as well as statistical and demographic data.

Non-textual business documents regard entire industries or single markets (i.e. with statements about products sold, revenue, foreign trade), companies (ownership, balance sheet ratio and solvency) up to individual products. The latter are important for information retrieval systems in e-commerce; the products must be retrievable via a search for their characteristics.

The third group of non-textual documents contains museum artifacts (Latham, 2012) and works of art. These include, among others, archaeological finds, historico-cultural objects, works of art in museums, galleries or private ownership as well as objects in zoos, collections etc. This is the documentary home of Briet's famous antelope.

A fourth group of undigitizable documents is formed by persons. They normally appear in the context of other undigitizable documents, e.g. as patients (STM), employees of a company (business) or artists (museum artifacts).

The last group of factual documents consists of data which are time-dependent and are collected real-time. There is a broad range of real-time data; examples are weather data or tracking data for ongoing flights.

In the intellectual compilation of metadata, the documentary reference units of undigitizable non-textual documents are described textually and, in addition, represented via audio files (e.g. an antelope's alarm call), photos (e.g. of a painting) or video clips (e.g. of a sculpture seen from different perspectives or in different lighting modes), where possible. In automatic fact extraction from digital documents, the respective data are either explicitly marked in the document or—with the caveat of great uncertainty—gleaned via special methods of information retrieval (Ch. G.6).

Conclusion

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- Apart from texts, there are non-textual (factual) documents. According to the general definition, objects are documents if and only if they meet the criteria of materiality (including digitality), intentionality, development, and perception as a document.
 - Digital documents display characteristics of fluidity and intangibility. Stable digital documents can be created and stored at any time.
 - The data contained in digital documents can be extracted and transformed into individual data documents. All digital documents (texts as well as data) require a clear designation (e.g., a DOI) and must be indexed via metadata. It is subsequently possible to link texts and data with one another ("linked data").
 - When discussing the legitimacy of documents, one must distinguish between formally published, informally published and unpublished documents. In the former, we can assume that the documents have been subject to a process of verification. This allows us to implicitly ascribe a certain measure of quality to these formal publications.
 - Formally published text documents include bookselling media, legal norms and rulings, texts of intellectual property rights as well as grey literature.
 - Informally published texts are available in large quantities on the World Wide Web. On the Web, we find interconnected individual web pages within a website, which has a firm entry page called

the homepage. Especially in the case of Wikis, there are web pages whose content is subject to frequent changes.

- Unpublished texts are all documents that accrue for companies, administrations, private individuals etc., such as letters, bills, memos etc. Important special forms are records, which refer to certain activities or other occurrences. These can be available non-digitally, but they can just as well be stored in an in-house Intranet or Extranet. These documents are objects of both records management and corporate knowledge management.
 - Non-textual documents are either digital (or at least digitizable), such as visual (images, films) and audio media (music, spoken word, sounds) or they are principally non-digital, such as facts from science, technology and medicine, economic objects (industries, companies, products), museum artifacts, persons as well as facts from ongoing processes (weather, delays, flights, etc.).
 - All documents without exception are, at least in principle, material for retrieval systems. Where possible, the systems should dispose of both the documentary units (with metadata as informational added value) and the complete digital forms of the documents (always including their DOI).
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A.5 Information Literacy

Information Science in Everyday Life, in the Workplace, and at School

According to UNESCO and the IFLA, information literacy is a basic human right in the digital world. In the “Alexandria Proclamation on Information Literacy and Lifelong Learning” (UNESCO & IFLA, 2005), we read that

Information Literacy ... empowers people in all walks of life to seek, evaluate, use and create information effectively to achieve their personal, social, occupational and educational goals.

If information literacy is understood as such a human right (Sturges & Gastinger, 2010), social inequalities in the knowledge society—the digital divide—can be counteracted and the individual’s participation in the knowledge society strengthened (Linde & Stock, 2011, 93-96). Information literacy refers to the usage of information science knowledge by laymen in their professional or private lives. It refers to a level of competence in searching and finding as well as in creating and representing information—a competence that does not require the user to learn any information science details. In the same way that laypeople are able to operate light switches without knowing the physics behind alternating current, they can use Web search engines or upload resources to sharing services and tag them without having studied information retrieval and knowledge representation. To put it very simply, information literacy comprises those contents of this book that are needed by everyone wishing to deal with the knowledge society. Information literacy plays a role in the following three areas:

- information literacy in the Everyday,
- information literacy in the Workplace,
- information literacy at School (Training of information literacy skills).

Competences are arranged vertically in a layer model (Catts & Lau, 2008, 18) (Figure A.5.1). Their basis is literacy in reading, writing and numeracy. ICT and smartphone skills as well as Media Literacy are absolutely essential in an information society (where information and communication technology are prevalent; Linde & Stock, 2011, 82). Everybody should be able to operate a computer and a smartphone, be literate in basic office software (text processing, spreadsheet and presentation programs), know the Internet and its important services (such as the WWW, e-mail or chat) and use the different media (such as print, radio, television and Internet) appropriately (Bawden, 2001, 225). All these competences mentioned above are preconditions of information literacy. The knowledge society is an information society where, in addition to the use of ICT, the information content—i.e., the knowledge—is available everywhere and at any time. Lifelong learning is essential for every single member of such a society (Linde & Stock, 2011, 83). Accordingly, important roles are played by people’s

information literacy and their ability to learn (Marcum, 2002, 21). Since a lot of information is processed digitally, a more accurate way of putting it would be to say that it is “digital information literacy” (Bawden, 2001, 246) which we are talking about.

In investigating information literacy, we pursue two threads. The first of these deals with practical competences for information retrieval. It starts with the recognition of an information need, proceeds via the search, retrieval and evaluation of information, and leads finally to the application of information deemed positive. The second thread summarizes practical competences for knowledge representation. Apart from the creation of information, it emphasizes their indexing and storage in digital information services as well as the ability to sufficiently heed any demands for privacy in one’s own information and others’. For both of these threads, it is of great use to possess basic knowledge of information law (Linde & Stock, 2011, 119-157) and information ethics (Linde & Stock, 2011, 159-180).

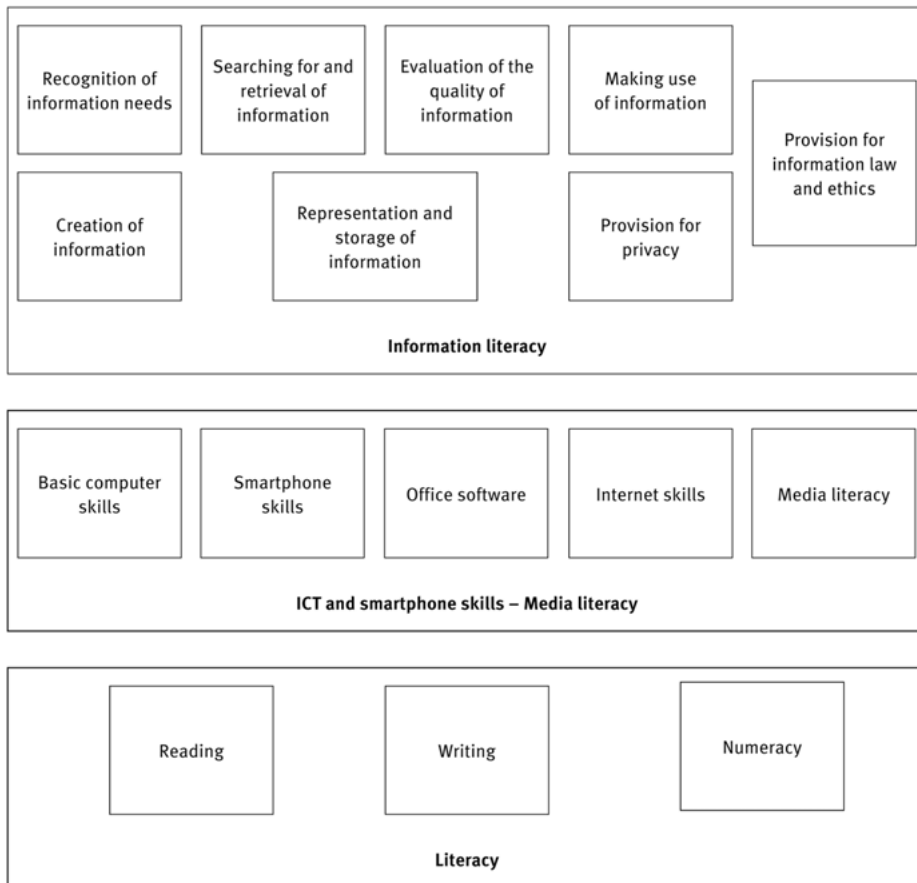


Figure A.5.1: Levels of Literacy.

Thread 1, which includes Information Retrieval Literacy, has been pursued for more than thirty years. It is rooted in bibliographic and library instruction and is practiced mainly by university libraries. A closely related approach is information science user research, e.g. the modelling of research steps in Kuhlthau (2004) (see Ch. H.3). Eisenberg and Berkowitz (1990) list “Six Big Skills” that make up information literacy: (1) task definition, (2) information seeking strategies, (3) location and access, (4) use of information, (5) synthesis, and (6) evaluation. The Association for College and Research Libraries (ACRL) of the American Library Association provides what has become a standard definition (Presidential Committee on Information Literacy, 1989):

To be information literate, a person must be able to recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.

The British Standing Conference of National and University Libraries (SCONUL) summarizes Retrieval Literacy into seven skills (SCONUL Advisory Committee on Information Literacy, 1999, 6):

- The ability to recognise a need for information
- The ability to distinguish ways in which the information ‘gap’ may be addressed
- The ability to construct strategies for locating information
- The ability to locate and access information
- The ability to compare and evaluate information obtained from different sources
- The ability to organise, apply and communicate information to others in ways appropriate to the situation
- The ability to synthesise and build upon existing information, contributing to the creation of new knowledge.

When confronted with technically demanding information needs, Grafstein (2002) emphasizes, the general retrieval skills must be complemented by a specialist component capable of dealing with the contents in question. Thus it would hardly be possible to perform and correctly evaluate an adequate research for, let us say, Arsenicin A (perhaps regarding its structural formula) without any specialist knowledge of chemistry. “(B)eing information literate crucially involves being literate *about something*” (Grafstein, 2002, 202).

In the year 2000, the ACRL added another building block to their information literacies (ACRL, 2000, 14):

The information literate student understands many of the economic, legal, and social issues surrounding the use of information and accesses and uses information ethically and legally.

Important subsections here are copyright as well as intellectual property rights.

The second thread, including Knowledge Representation Literacy, has become increasingly important with the advent of Web 2.0 (Huvila, 2011). Web users, who used to be able only to request information in a passive role, now become producers

of information. These roles—information consumers and information producers—are conflated in the terms “prosumer” (Toffler, 1980) and “produsage” (Bruns, 2008), respectively. Users create information, such as blog posts, wiki articles, images, videos, or personal status posts, and publish them digitally via WordPress, Wikipedia, Flickr, YouTube and Facebook. It is, of course, important that these resources be retrievable, and so their creators give them informative titles and index them with relevant tags in the context of folksonomies (Peters, 2009). Here, too, aspects of information law and ethics must be regarded (e.g. may I use someone else’s image on my Facebook page?). Additionally, the user is expected to have a keen sense for the level of privacy they are willing to surrender and the risks they will thus incur (Grimmelmann, 2009).

Information Literacy in the Workplace

In corporate knowledge management, an employee is expected to be able to acquire and gainfully apply the information needed to perform his tasks. Knowledge thus acquired in the workplace must be retrievable at any time. In knowledge-intensive companies in particular (e.g. consultants or high-tech companies), competences in information retrieval and knowledge representation are of central importance for almost all positions. With regard to Web 2.0 services, employees are expected not to reveal any secrets or negative opinions about their employer—be it via the company’s official website or via private channels. For Lloyd (2003) and Ferguson (2009), knowledge management and information literacy are closely related to one another. Bruce (1999, 46) describes information competences in the workplace:

Information literacy is about peoples’ ability to operate effectively in an information society. This involves critical thinking, an awareness of personal and professional ethics, information evaluation, conceptualising information needs, organising information, interacting with information professionals and making effective use of information in problem-solving, decision-making and research. It is these information based processes which are crucial to the character of learning organisations and which need to be supported by the organisation’s technology infrastructure.

Bruce (1997) lists seven aspects (“faces”) of information literacy, each communicating with organizational processes in the enterprise (Bruce, 1999, 43):

- Information literacy is experienced as using information technology for information awareness and communication (workplace process: environmental scanning).
- Information literacy is experienced as finding information from appropriate sources (workplace process: provision of inhouse and external information resources and services).

- Information literacy is experienced as executing a process (workplace process: information processing; packaging for internal/external consumption).
- Information literacy is experienced as controlling information (workplace process: information / records management, archiving).
- Information literacy is experienced as building up a personal knowledge base in a new area of interest (workplace process: corporate memory).
- Information literacy is experienced as working with knowledge and personal perspectives adopted in such a way that novel insights are gained (workplace process: research and development).
- Information literacy is experienced as using information wisely for the benefit of others (workplace process: professional ethics / codes of conduct).

The task of organizing the communication of information literacy in enterprises falls to the department of Knowledge Management or to the Company Library (Kirton & Barham, 2005, 368). Learning in the workplace, and thus learning information literacy, “is a form of social interaction in which people learn together” (Crawford & Irving, 2009, 34). In-company sources of information gathering also include digital and printed resources, but a role of ever-increasing importance is being played by one’s colleagues. Crawford and Irving (2009, 36) emphasize the significance of face-to-face information streams:

The traditional view of information as deriving from electronic and printed sources only is invalid in the workplace and must include people as a source of information. It is essential to recognize the key role of human relationships in the development of information literacy in the workplace.

Thus it turns out to be a task for corporate knowledge management to create and maintain spaces for information exchange between employees. Information literacy in the workplace requires company libraries to install information competence in employees, create digital libraries (to import relevant knowledge into the company and to preserve internal knowledge), as well as to operate a library as a physical space for employees’ face-to-face communication. Methods of knowledge management “ask, e.g., for knowledge cafés (...), space for story telling (...) and for corporate libraries as places” (Stock, Peters, & Weller, 2010, 143).

Information Literacy in Schools and Universities

According to Catts and Lau (2008, 17), the communication of information literacy begins at kindergarten, continues on through the various stages of primary education and leads up to university. The teaching and learning of information literacy is fundamentally “resource-based learning” (Breivik, 1998, 25), or—in our terminology (see Ch. A.4)—“document-based learning”. Retrieval literacy allows people to retrieve documents that satisfy their information needs. Knowledge representation grants people

the competence of creating, indexing and uploading documents to information services. The centers of attention are always documents, which thus become educational learning tools. The results of document-based learning can be easily documented, being “more tangible and varied than the writing of a term paper or the delivery of a class speech” (Breivik, 1998, 25). One thing students might achieve while learning information retrieval could be the compilation of a dossier on a scientific subject. To achieve this, they would have to browse WWW search engines, Deep Web search tools (such as Web of Science or Scopus), or print sources, thus retrieving, evaluating and processing relevant documents. Some achievements in the area of information creation and knowledge representation include creating wiki pages on a predefined topic, tweeting important topics discussed in lessons or lectures, or creating a Web presence for an enterprise (real or imaginary).

Document-based learning is suitable for learning in groups. Breivik (1998, 25) emphasizes:

Resource-based learning ... frequently emphasizes teamwork over individual performance. Not only does working in teams allow students to acquire people-related skills and learn how to make the most of their own strengths and weaknesses, but it also parallels the way in which they will need to live and work throughout their lives.

Certain educational contents easily lend themselves to teacher-centred teaching, e.g. communicating the basic functionality of Web of Science and Scopus, or learning HTML. However, the literature unanimously recognizes that the prevalent mode of teaching in communicating information literacy is project-based learning. In this approach, students are presented with a task, which they must then process and solve as a team. In addition, the introduction of methods of gamification to information literacy instruction seems to be very useful, especially for the “digital natives”, who have grown up with digital games (Knautz, 2013).

There are already detailed curricula (such as that of the River East Transcona School Division, 2005) leading from kindergarten up to grade K-12.

Chu et al. (2012), Chu (2009) and Chu, Chow and Tse (2011) report on their experiences teaching information literacy in primary schools. Their case studies involve schools in Hong Kong, which always provide for an ICT teacher as well as a school librarian (with teacher status). Their fourth-grade students were given projects in the area of “General Studies” (GS), which they had to complete in working groups. This inquiry project-based learning (PBL) proved to be fun for the children, while enhancing their ability to deal with information (i.e. using Boolean Operators or the Dewey Decimal Classification). Additionally, it improved their general ability to read (in this case: Chinese) (Figure A.5.2).

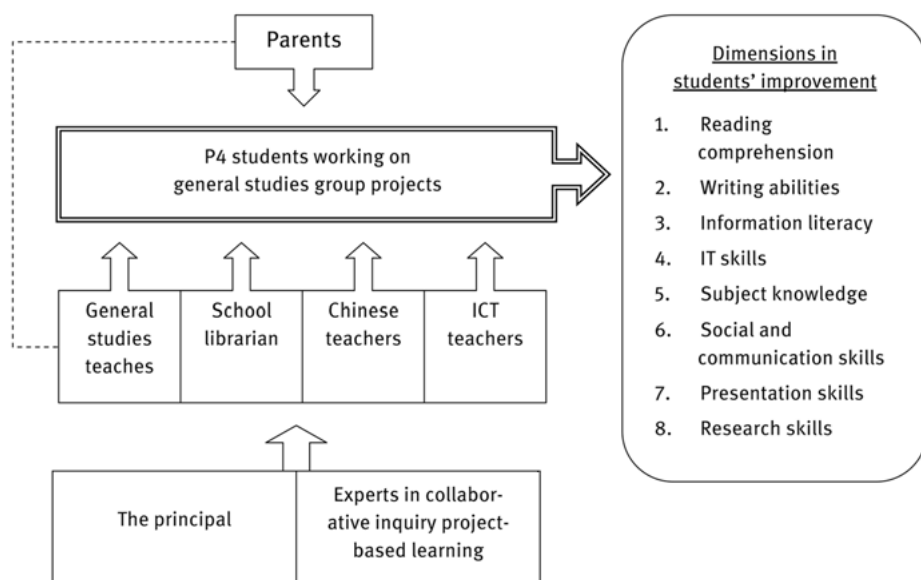


Figure A.5.2: Studying Information Literacy in Primary Schools. Source: Chu, 2009, 1672.

There are also research projects concerned with implementing the communication of information literacy in secondary schools. Virkus (2003) reports on experiences made in European countries. Among others, Herring (2011) studied the situation in Australia, Nielsen and Borlund (2011) in Denmark, Streatfield, Shaper, Markless and Rae-Scott (2011) in the United Kingdom, Asselin (2005) in Canada, Latham and Gross (2008) in the U.S., Abdullah (2008) in Malaysia, Mokhtar, Majid and Foo (2008) in Singapore and van Aalst et al. (2007) in Hong Kong. An additional object is always “learning how to learn”. In some countries, a key role in teaching information literacy is performed by teacher librarians (Tan, Gorman, & Singh, 2012). Mokhtar, Majid and Foo (2008, 100) emphasize the positive role of students’ levels of information literacy for their ability to study and for their future academic achievements.

Neither primary nor secondary schools currently feature information literacy as a separate subject in their curricula. In higher education, on the other hand (Johnston & Webber, 2003), there are manifold offers for students, mainly provided by university libraries. We can distinguish two separate approaches: either information literacy education takes its cue from the subjects that are taught, or information literacy forms a subject in its own right (Webber & Johnston, 2000). In the latter case, information literacy is taught as an interdisciplinary subject.

Conclusion

- Information literacy is the ability to apply information science results in the everyday and in the workplace. We distinguish between information retrieval literacy and literacy in creating and representing information.
 - Corporate knowledge management and information literacy are closely interrelated. Many businesses expect their employees to be able to acquire and create information in order to perform their tasks.
 - Information literacy is always acquired via document-based learning, and is often taught in groups as project-based learning. Sometimes methods of gamification are applied.
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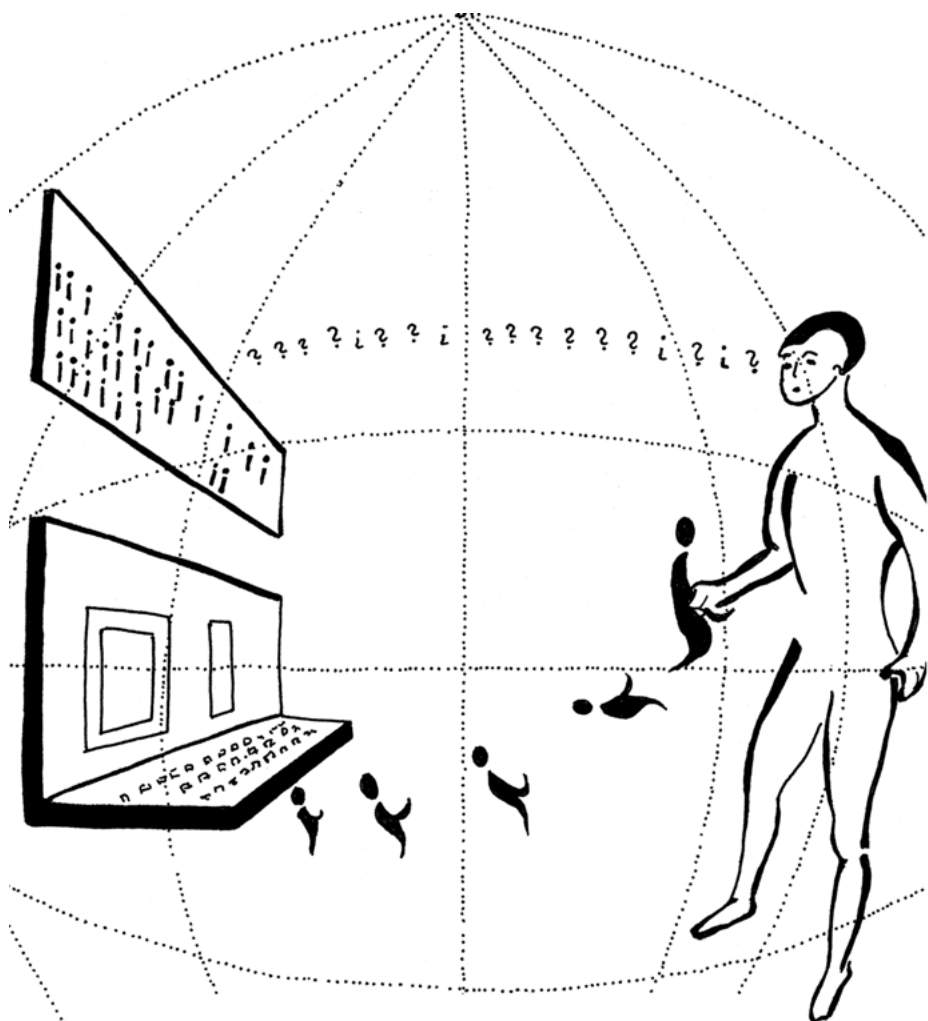
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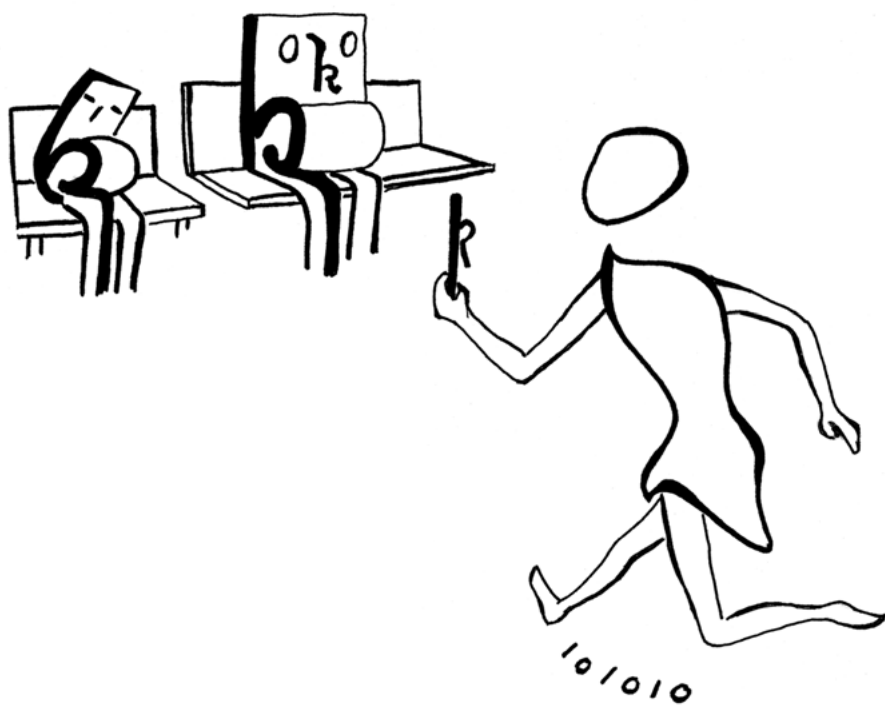
Information Retrieval



Information Retrieval

Part B

Propaedeutics of Information Retrieval



B.1 History of Information Retrieval

From Memex via the Sputnik Shock to the Weinberg Report

Information retrieval is that information science subdiscipline which deals with the searching and finding of stored information. One such task has been performed by libraries and archives since the day they were created. Information retrieval as a scientific discipline, though, is also inseparably connected with the use of computers. The discipline's history goes back to the early period of computer usage. The term "information retrieval" is first found in Mooers in the year 1950 (published 1952). He defines "information retrieval" (Mooers, 1952, 572):

The problem of directing a user to stored information, some of which may be unknown to him, is the problem of "information retrieval".

The article "As we may think" by the American electrical engineer and research organizer Bush (1890 – 1974) is regarded as one of the first (visionary) approaches to information retrieval (Veith, 2006). Bush's experiences—in the time during the second World War—stem from the area of militarily relevant research and development. His knowledge, acquired in war, is now available to the civilian user. Scientific knowledge is fixed in documents. For Bush, the important thing is not merely to create new knowledge, but to use that knowledge which already exists. In order to simplify this, one must use suitable storage media.

A record, if it is to be useful to science, must be continuously extended, it must be stored, and above all it must be consulted. Today we make the record conventionally by writing and photography, followed by printing; but we also record on film, on wax disks, and on magnetic wires (Bush, 1945, 102).

This involves not only the printing of articles or books, but also the mechanical provision of knowledge once acquired. The then-current methods used by libraries did not meet the new expectations:

The real heart of the matter of selection, however, goes deeper than a lag in the adoption of mechanisms by libraries, or a lack of the development of devices for their use. Our ineptitude in getting at the record is largely caused by the artificiality of systems of indexing. When data of any sort are placed in storage, they are filed alphabetically or numerically, and information is found (when it is) by tracing it down from subclass to subclass (ibid., 106).

Humans do not think in one-dimensional classification systems; rather, they think associatively. Machines that simulate such forms of multidimensional and even changing structures are far ahead of libraries' knowledge stores.

The human mind does not work that way. It operates by association. With one item in its grasp, it snaps instantly to the next that is suggested by the association of thoughts ... It has other characteristics, of course; trails that are not frequently followed are prone to fade, items are not fully permanent, memory is transitory. Yet the speed of action, the intricacy of trails, the detail of mental pictures, is awe-inspiring beyond all else in nature (*ibid.*).

When we think of the World Wide Web of today, we can see that a big part of Bush's vision has been realized: we can jump directly from one document to another via links, and the documents are (at least partially) updated all the time. One can browse the Web or make purposeful searches. In Bush, the system was called "Memex"; all sorts of knowledge carriers are incorporated within it:

Books of all sorts, pictures, current periodicals, newspapers, are thus obtained and dropped into place. Business correspondence takes the same path. And there is provision for direct entry (*ibid.*, 107).

Bush also kept in mind retrieval tools, today's search engines:

There is, of course, provision for consultation, of the record by the usual scheme of indexing. If the user wishes to consult a certain book, he taps its code on the keyboard, and the title page of the book promptly appears before him, projected onto one of his viewing positions. ... Moreover, he has supplemental levers. On deflecting one of these levers to the right he runs through the book before him, each page in turn being projected at a speed which just allows a recognizing glance at each. ...

A special button transfers him immediately to the first page of the index. ... As he has several projection positions, he can leave one item in position while he calls up another. He can add marginal notes and comments ... (*ibid.*).

New knowledge products will be created, e.g. new forms of encyclopedias, and information usage will grant experts completely new options of implementing their knowledge into action.

The lawyer has at his touch the associated opinions and decisions of his whole experience, and of the experience of friends and authorities. The patent attorney has on call the millions of issued patents, with familiar trails to every point of his client's interest. The physician, puzzled by its patient's reactions, strikes the trail established in studying an earlier similar case, and runs rapidly through analogous case histories, with side references to the classics for the pertinent anatomy and histology. ... (*ibid.*, 108).

The associative paths between the documents and within the documents are not (or not only) created by Memex, but by members of a new profession:

There is a new profession of trail blazers, those who find delight in the task of establishing useful trails through the enormous mass of the common record (*ibid.*).

“Trail blazers”, i.e. experts who mark (intellectual) paths and explore new vistas in the structure of knowledge, are challenged. What is needed are specialists in information retrieval and knowledge representation, in other words: information scientists.

The advent of the computer indeed sees work being done in order to implement Bush’s visions from the year 1945. These endeavors are boosted—in the United States at first—via a scientific-technological quantum leap made by the U.S.S.R.: the “conquest” of outer space in the year 1957 via the satellite Sputnik 1. Two shocks can be observed: a first and obvious shock is due to the technological lag of the U.S. in space research, which would then be successfully combated via large-scale programs, like the Apollo programs. The second shock regards information retrieval: Sputnik sends encrypted signals in certain intervals. Since one wanted to understand these, groups of scientists were charged with breaking the codes. Rauch (1988, 8-9) describes the second Sputnik shock:

American scientists needed approximately half a year to decrypt Sputnik’s signals. This would not have been that bad, had it not become known, shortly after, that the meaning of the signals and their encryptions had already been published in a Soviet physics journal two years prior. On top of that, it was a magazine whose articles were continuously being translated into English by an American translation agency. The corresponding journal article had been available in English in many American libraries long before the start of Sputnik.

It had never before become so clear that the scientific information system had stopped functioning properly; that the steady influx of information in libraries, archives and company files had not led to better knowledgeability—to the contrary.

The second Sputnik shock was also a deep one, and it led the President of the United States to commission a dossier about science, government and information. Alvin M. Weinberg submitted his report in 1963; John F. Kennedy writes, in the foreword (The President’s Science Advisory Committee, 1963, III):

One of the major opportunities for enhancing the effectiveness of our scientific and technological effort and the efficiency of the Government management for research and development lies in the improvement our own ability to communicate information about current research efforts and the results of past efforts.

At least since the Weinberg Report (see also Weinberg, 1989), it has been clear that we are confronted with an “information explosion”, and that information retrieval represents an effective means of not allowing the surplus of information to create an information deficit. The job of the information scientist becomes absolutely necessary (The President’s Science Advisory Committee, 1963, 2):

We shall cope with the information explosion, in the long run, only if some scientists and engineers are prepared to commit themselves deeply to the job of sifting, reviewing, and synthesizing

information; i.e., to handling information with sophistication and meaning, not merely mechanically. Such scientists must create new science, not just shuffle documents ...

This early period of information retrieval as a science was also the time of endeavors by Hans-Peter Luhn (1896 – 1964). Luhn developed foundations for the automatic summarizing and indexing of documents on the basis of text-statistical procedures: even in the 1950s, he was thinking of push services in order to safeguard knowledgeability in the context of individual information profiles (so-called SDI: Selective Dissemination of Information) as well as of the highlighting of search terms in retrieved documents (KWIC: Keyword in Context). It is the “machine talents” (Luhn, 1961, 1023) that must be discovered and developed for the purposes of information retrieval.

Another “classical” author of information retrieval is Gerard Salton (actually Gerhard Anton Sahlmann; 1927 – 1995). Like Luhn, he discovered the “talents” of the machine and implemented them in research systems: the counting of word frequencies and calculations on this basis. His Vector-Space Model registers documents and queries both as vectors in an n -dimensional space, where the dimensions represent words. The similarity between a query and the document (or of documents among one another) is calculated via the angle that exists between the single vectors. In the 1960s, the theory of the Vector-Space Model is founded and, at the same time, the experimental retrieval system SMART (System for the Mechanical Analysis and Retrieval of Text) is constructed (Salton & McGill, 1983).

Even in the 1950s, Eugene Garfield (born 1925) had the idea of securing the knowledgeability of scientists via copies of the contents of relevant specialist journals (Current Contents). Additionally, he had the idea of representing the paths of information transmission in the scientific journal literature via citation indices (Science Citation Index). His “Institute for Scientific Information” (ISI), founded in 1960, was one of the first private enterprises to run an information retrieval system with commercial goals (Cawell & Garfield, 2001).

In Germany, too, the 1960s saw the beginnings of information retrieval research. Worthy of note are the activities of Siemens’ groups of researchers, which resulted in both an operative retrieval system (GOLEM—*großspeicherorientierte, listenorganisierte Ermittlungsmethode*; mass storage-oriented, list-based derivation method) and an information-linguistic concept for automatically indexing German-language texts (PASSAT—*Programm zur automatischen Selektion von Stichwörtern aus Texten*; Program for the automatic selection of keywords from texts). PASSAT forms the document terms’ stems (excluding noise words) and compares them with a dictionary on file. In case of a match, the words are weighted and the highest-rated terms are allocated to the document as preferred terms (Hoffmann et al., 1971). GOLEM is used, for instance, in the (intellectual) documentation of philosophical literature. As early as 1967, Norbert Henrichs reported on the successful implementation of GOLEM as a retrieval system in the service of philosophy (Henrichs, 1967; see also Hauk & Stock, 2012).

Over the course of the 1950s and 60s, information science established itself as a scientific discipline; apart from information retrieval (Cool & Belkin, 2011), it concentrated on knowledge representation. There has long been a practical area of documentation, which mainly deals with the information activities of indexing and search. In the U.S.A., documentalists have been organized in the American Documentation Institute (ADI) since 1937. In the course of the success of the new discipline of information science, the ADI changed its name to American Society for Information Science (ASIS) in 1968. In her history of the ADI and ASIS “(f)rom documentation to information science”, Farkas-Conn (1990, 191) writes:

By the late 1950s, a movement started to change the name of the American Documentation Institute. The structure of the ADI was that of a society, and ‘*documentation*’ projected an old fashioned image: the people working in the field were by now concentrating on information and its retrieval and on representation of information. Their concern extended from the microactivity of symbol manipulation to the macroactivity of determining what information systems were important for the nation.

Apart from governmental actions for information retrieval, the Weinberg Report considers the private sector; it talks of an equilibrium between the initiative of the government and that of the private hand (The President’s Science Advisory Committee, 1963, 4). The question is now: what has the “private hand” accomplished for the development of information retrieval? Where are the origins of commercial information services?

Early Commercial Information Services: DIALOG, SDC Orbit and Lexis-Nexis

In the 1960s, developments were made in the U.S.A. in the area of commercial electronic information services. Using three examples, we will introduce some pioneering companies of the information market (cf. also Bourne & Hahn, 2003; Hall, 2011).

In 2002, the system DIALOG (Bjørner & Ardito, 2003b) already celebrated its 30th anniversary. Roger K. Summit (born 1930), the founder and former president and CEO of DIALOG, reminisces (Summit, 2002, 6):

Imagine a time when Dialog offered only two databases instead of the 573 it offers today. This time was long before PCs and searching was done on a dial-up terminal running at 300 baud (30cps). We called it ‘on-line’ and the term ‘end-user’ had not yet come into general use.

The actual flash-back does not begin in 1972, but much earlier, in 1960, when Summit, as a doctoral candidate, worked a summer job at Lockheed Missiles and Space Company. As Summit (2002) remembers it,

(a) common statement around Lockheed at the time was that it is usually easier, cheaper and faster to redo scientific research than to determine whether it's been done previously.

This circumstance led Summit to the conclusion that computer-based information retrieval could have a durably positive effect on research. This innovative idea was joined, four years later, by the propitious moment of a technological innovation: the IBM 360 computer, which was able to summarize large quantities of data as well as make them retrievable and accessible. During the development of a project plan as to how far the new technology could be applied to technical literature, the product's name was found: "Dialog" means an interactive system between man and machine (Summit, 2002).

The name for the system, 'Dialog', occurred to me in 1966. ... I was dictating a project plan, for what was to become Dialog, into a small, voice-activated tape recorder. ... The system was to be interactive between human and machine. Described that way, 'Dialog' was the obvious choice.

The creation of a (simple) command language was next. 1972 was the official birth date of Dialog, when two government databases (ERIC and NTIS) were made available for public usage. The world's first commercial online service was started. New products and database providers formed continuously from then on.

The story of Orbit (now Questel) goes back to the year 1962. Over the course of the 1960s, the retrieval system CIRC (Centralized Information Retrieval and Control) was developed by the SDC (System Development Corporation). This system was used to perform online test runs under the code name COLEX (CIRC On-Line Experiments). COLEX is the direct predecessor of Orbit. Carlos A. Cuadra (born 1925), who apart from his work for SDC gained prominence mainly through the "Annual Reviews of Information Science and Technology" and the "Cuadra Directory of Online Databases" he initiated, as well as through his "Star" software, was the foremost developer of COLEX (and Orbit). The name Orbit can be traced back to 1967, initially signifying "On-Line-Retrieval of Bibliographic Information Time-shared". While the early developments of the retrieval system were set in the context of research for the Air Force, scientific interest shifted, in the early 1970s, to medical information. A search system for the bibliographic medical database MEDLARS was constructed for the National Library of Medicine, which was offered, from 1974 onwards, under the name of "Medline" (MEDLARS On-Line). The launch of the SDC Search Service—including, among other services, MEDLARS—occurred in December of 1972, almost simultaneously to the start of DIALOG. From 1972 to 1978, the SDC Search Service was led by Cuadra. Whereas the early developments of Lockheed's DIALOG were rather economically-minded, the development of Orbit was more driven by research and development. Bourne and Hahn write, in their history of online information providers (2003, 226), that

(t)he two major organizations in the early online search service industry—SDC and Lockheed—had different philosophies and approaches to their products. SDC staff closely associated with the online search system were R&D-driven. With a history of published accomplishments, SDC hired staff with training, experience, and inclination toward solid R&D work. The online services were focal points around which staff could pursue their R&D interests. Building a business empire based on online searching did not seem to be a major goal for the SDC staff.

Orbit concentrated its content on databases in the area of natural sciences and patents (chemistry databases such as Chemical Abstracts were added very early on, with further databases like INSPEC for physics to follow).

Two problems accompanied Cuadra and the SDC Search Service. The mother company SDC demanded that Orbit use its in-house computers. Since SDC mainly worked for the Air Force, the technology it used was top of the line, with top-of-the-line prices. SDC's internal transfer prices were just as high as external prices, which put the SDC Search Service under enormous cost pressure. The modern computers guaranteed online users very short computing times. Since Orbit (just like DIALOG) priced connection time, this led to a second problem—not for the users, but for the company. Short computing times meant short connection times and thus less profit. Carlos Cuadra remembers (in Bjørner & Ardito, 2003a):

so while I was paying high-end retail at SDC, he (referring to DIALOG's Roger Summit) was getting almost free disk storage. Not only that, the Dialog response time was so slow that, with the connect hour charge, Roger was making two or three times as much money as we were, because our system was stunningly fast.

He left SDC in 1978.

We are back in the 1960s, zooming in on Ohio (Bjørner & Ardito, 2004a, 2004b). Due to the U.S.A.'s legal system, which is heavily case-based, attorneys were finding it ever harder to keep an overview of the entirety of legal literature, which includes many thousands of court rulings. The initiative of saving cases via the then innovative medium of computers was the American Bar Association of Ohio's. In 1965, the Ohio Bar Association founded OBAR (Ohio Bar Automated Research), with the goal of electronically managing legal texts (laws and mainly cases from Ohio). The commission for creating the retrieval software for OBAR was won by Data Corporation from Dayton, Ohio, which had hitherto specialized in photographic representation and multi-color inkjet printing, experimenting, at the time, with information retrieval. In 1968, Richard H. Giering, recently employed by Data Corporation, demonstrated the Data Central System, an online retrieval system for full-text searches. Apart from noise words, all words of the entire texts were entered into an inverted file. The system was ideally suited for the full-text search of Ohio verdicts; the company was local and, consequently, was charged with the OBAR project.

In 1968, Mead, a paper manufacturing company (also based in Dayton) acquired Data Corporation. The progressive printing technology of Data Corp. was the main

reason behind this step. At a time when it was feared that computers would replace paper, the company bet on modern, computer-assisted printing techniques. The OBAR project was adopted along with Data Corp. without the buyer knowing about it. Bourne and Hahn (2003, 246) write that

(t)hey (the Mead people) bought it for other reasons, and then they took inventory and found the OBAR activity.

The development proceeded speedily. The OBAR retrieval system had Boolean and Word Proximity operators as early as 1969, as well as the Focus Command (a search within a hit list), information-linguistic basic functionality (regular and irregular plural forms in the English language) and, from 1970 on, a highlighting of search terms in the hit list as KWIC (Keyword in Context; here—since it was marked in color—Keyword in Color). Keyword in Color could not be used commercially, though, since no customer at the time had color monitors.

In a company restructuring in 1970, all of Data Corp.'s OBAR activities were outsourced to a separate (but still Mead-owned) company: Mead Data Central (MDC), later renamed Mead Technology Laboratories. MDC concentrated on legal information, Data Corp. retained all non-legal activities.

In-house, the split was described as the 'legal' market and the 'illegal' market (Bourne & Hahn, 2003, 256).

Giering, who had been instrumental in developing full-text retrieval, stayed on at Data Corp. OBAR's full-text basis was expanded massively. In April of 1973, MDC entered the market with the online service LEXIS (LEX Information Service). LEXIS was sold on a subscription basis, its customers, apart from universities' legal departments, mainly larger companies.

From the mid-1970s onwards, Data Corp. has developed solutions for several publishing houses' archives. These activities, mainly supervised by Giering, began in 1975 with *The Boston Globe* and, in 1977, with *The Philadelphia Inquirer*. The adoption of the pre-existing full-text system caused some problems, though. Richard Giering tells of an author who wanted to research an article he knew about, concerning "The Who". He found nothing, since both "the" and "who" were designated as stop words. The solution lay in using different stop word lists for different fields.

We already had the capability for setting up different sets of stop words by field, by group. ... The solution was for the librarian to put 'The Who' in the index terms group. So now, a search for 'the Who' found that document (Giering in Bjørner & Ardito, 2004b).

The *Boston Globe*'s archive stopped its method of clipping via scissors and glue in 1977. The *Boston Globe* also gave rise to the business idea of selling the papers' articles, already digitally available, from an archive database. Mead Technology Labora-

tories started the NewsLib project. The successful endeavor was later outsourced and allocated to the sister company Mead Data Central. MDC started this new service as NEXIS, in 1980.

The World Wide Web and its Search Tools

The theory of information retrieval developed further; commercial retrieval systems conquered the (very small) market of onliners. This changed completely with the advent of Personal Computers. Initially used on a stand-alone basis, PCs and other, smaller computers were soon interconnected via telephone or data lines. The “IT boom” affected both commercial enterprises and private households.

The last milestone of our story so far is the advent of the internet, particularly its World Wide Web service, conceived by Timothy (‘Tim’) Berners-Lee (born 1955). The WWW made for a boom period in information retrieval theory and practice. Visible signs of this renewed upturn of information science are the search tools. These are nothing but information retrieval systems, but no longer focused on scientific information (or, more generally, specialist information) as in the previous decades, but on all sorts of information, which can be produced and retrieved by anyone. The WWW search engines made information retrieval into a mass phenomenon.

Companies link its employees’ workspaces and construct intranets (on the same technological basis as the internet). Here, too, large quantities of information are aggregated; the consequence is a need for information retrieval systems.

In an article on the current challenges for information retrieval, the authors emphasize the enormous importance of their discipline (Allan et al., 2003, 47):

During the last decade, IR has been changed from a field for information specialists to a field for everyone. The transition to electronic publishing and dissemination (the Web) when combined with large-scale IR has truly put the world at our fingertips.

Successful systems in the WWW include Yahoo! (founded in 1994 by Jerry Yang and David Filo), AltaVista (created in the laboratories of Digital Equipment Corp. in 1995), Google (launched in 1998 by Lawrence ‘Larry’ Page and Sergey Brin) as well as Teoma (co-founded by Apostolos Gerasoulis in 2000, on the basis of an idea by Jon Kleinberg). Higher-performance systems were developed for use by institutions. There, they were integrated into company-wide intranets and organizationally supervised in the context of knowledge management. Examples for such information retrieval systems are Autonomy, Convera, FAST or Verity. Research into search engines is heavily tech-driven; software development is the core focus. For the ASIS, this represented an incentive to change its name for a second time in 2000, into American Society for Information Science and Technology (ASIS&T).

We will now end our history of information retrieval—after the rather detailed reports on its early years—in order to deal with its individual subjects over the following systematic chapters.

One last thing to consider: research into and development of information retrieval are not to be regarded as complete by a long shot; rather, basic research and product development are in perpetual motion. On this subject, we will let Allan et al. have their say once more (2003, 47):

(I)nformation specialists and ordinary citizens alike are beginning to drown in information. The next generation of IR tools must provide dramatically better capabilities to learn, anticipate, and assist with the daily information gathering, analysis, and management needs of individuals and groups over time-spans measured in decades. These requirements require that the IR field rethink its basic assumptions and evaluation methodologies, because the approaches and experimental resources that brought the field to its current level of success will not be sufficient to reach the next level.

Conclusion

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- In 1945, Vannevar Bush introduced his vision of a comprehensive information research system (Memex), which saves full-texts and offers its users links between and within the documents in associative ways.
 - The Sputnik Shock (1957) exemplifies the information explosion while simultaneously demonstrating the inadequacies of the system of scientific communication.
 - In the Weinberg Report (1963), information retrieval is made public from basically the “highest level” (by U.S. President John F. Kennedy). A new scientific discipline—information science—is demanded.
 - Research into information retrieval can be observed since the 1950s. Early pioneers include Luhn, Salton and Garfield.
 - In Germany, systems for text analysis (PASSAT) and information retrieval (GOLEM) were developed by and in the sphere of Siemens in the 1960s.
 - In 1968, the American Documentation Institute renamed itself the American Society for Information Science (ASIS).
 - In the 1960s, preparatory work was made for the first online information systems. The pioneering companies were DIALOG (with Roger Summit), SDC Orbit (with Carlos A. Cuadra) and Lexis-Nexis (with Richard H. Giering). Commercial activities in the online business commenced in the years 1972-73.
 - In the 1980s and 90s, PCs and other, smaller computers conquered both companies and private households.
 - Information retrieval then entered a veritable boom period with the triumph of the internet, particularly the World Wide Web. Search tools in the WWW made information retrieval known to and usable by anyone. In companies’ intranets, too, retrieval systems came to be used more and more.
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B.2 Basic Ideas of Information Retrieval

Concrete and Problem-Oriented Information Need

We start with the user and assume that he has a need for action-relevant knowledge due to uncertainty and an anomalous state of knowledge. To begin with, let us take a look at some forms of information need that can be found in the following exemplary questions:

- Question Type A
 - Which city is the capital of Montana?
 - What is the URL of the iSchool caucus?
 - How much does a Walther PPK cost?
- Question Type B
 - What are the different interpretations of the Homunculus in Goethe's *Faust* Part II?
 - What is the relation between service marketing and quality management in business administration?
 - How have analysts been rating the company Kauai Coffee Company LLC in Kalaheo, HI?

Question Type A aims for the communication of a piece of factual information. The underlying information need is specific. In their analysis of information needs, Frants, Shapiro and Voiskunskii designate this type as a “Concrete Information Need” (CIN). Special cases of CIN are problems when navigating the WWW (as in the second question above).

Type B cannot be satisfied by stating a fact. Here, the information problem is resolved via the transmission of a more or less comprehensive collection of documents. Frants et al. call this Type B a “Problem Oriented Information Need” (POIN). CIN and POIN can be compared very easily, via a few characteristics (Frants, Shapiro, & Voiskunskii, 1997, 38):

CIN

1. Thematic borders are clearly defined.
2. The search query can be expressed precisely.
3. Generally, one factual information is enough to satisfy the need.
4. Once the information has been transmitted, the information problem is resolved.

POIN

1. Thematic borders are *not* clearly definable.
2. The search query formulation allows for *several* terminological variants.
3. Generally, *various* kinds of documents must be acquired. Whether they will satisfy the information need is an open question.
4. The transmission of the information may modify the information problem or uncover a new need.

The action-relevance of information for satisfying a CIN, that is to say, to answer a Type A question, can be exactly defined. Either the retrieved knowledge answers the question or it does not.

The situation is different for POINs, or Type B questions. In the hit lists, we generally find documents (or references to them) from commercial literature-based databases (bibliographical and full-text databases) as well as from the WWW and social media in Web 2.0. Here, we have several “hits”, which—more or less—answer aspects of what we asked. For Problem-Oriented Information Needs (POIN) and answers consisting of bibliographical information, the assessment of relevance will be different—also depending on who are asking and their respective context.

We would like to introduce a terminological nuance: We speak of an *objective* information need when discussing an objective matter of fact, i.e. when we abstract from a specific individual. Thus, an information need is satisfied when the transmitted information closes the knowledge gap—in the opinion of experts, or “as such”. However, any transmitted information that is objectively relevant may turn out to be irrelevant for a specific user. Either he already knows it, or he knows the author (and dislikes his work), or has no time to read such a long article, etc. We speak of a *subjective* information need when taking into consideration the specific circumstances of the searching subject.

In Web searches, it is useful to divide information needs into three. Broder (2002, 5-6) distinguishes between navigational, informational and transactional queries:

The purpose of [navigational queries] is to reach a particular site that the user has in mind, either because they visited it in the past or because they assume that such a site exists. ... The purpose of [informational queries] is to find information assumed to be available on the web in a *static form*. No further interaction is predicted, except reading. By *static form* we mean that the target document is not created in response to the user query. In any case, informational queries are closest to classic IR ... The purpose of [transactional queries] is to reach a site where further interaction will happen. This interaction constitutes the transaction defining these queries. The main categories for such queries are shopping, finding various web-mediated services, downloading various type of file (images, songs, etc.), accessing certain data-bases (e.g. Yellow Pages type data), finding servers (e.g. for gaming) etc.

These three different question types require different results for one and the same query. Navigational questions, for instance, cannot result in more than one link being put forward since the information need will be satisfied once the correct result has been transmitted.

Interplay of Information Indexing and Information Retrieval

Let the need for action-relevant knowledge be a given; now the user turns to a knowledge store and tries to resolve this deficit. The underlying problem here is to clearly

think and formulate something which one does not know at all, or at the very least not precisely. In other words: to formulate clearly and precisely, one would have to know what one does not know. Basic knowledge of the area in which one is to search is thus a fundamental requirement. The next step is to translate the information need (as clearly as possible) into a natural language. Let us take a simple example: A Julia Roberts fan would like to obtain some reviews of her performance in the movie *Notting Hill*. We already know this fan fulfils the basic condition of possessing relevant knowledge, otherwise she wouldn't even have heard of the film in the first place. The analysis of the question type is obvious as well: We are looking at a Problem-Oriented Information Need. The clear formulation of the query in everyday language is as follows: "I am looking for information about *Julia Roberts* in *Notting Hill*." The last step is to transform this formulation into a variant that can be used by retrieval systems. Here, there are two principal options: Either the machine assumes this task (disposing of the corresponding capacity of processing natural languages) or the user does it himself. The individual providers' retrieval systems each offer a comparable amount of commands, but they mostly use different syntaxes. On LexisNexis, for instance, we would formulate

HEADLINE:("Julia Roberts" w/5 "Notting Hill"),

whereas a sensible variant on DIALOG would be

(Julia (n) Roberts AND Notting (w) Hill)/TI,

and on Google we would be well advised to write

"Notting Hill" "Julia Roberts" (in this exact order).

An information service is a combination of a database (the entirety of all documentary units) and a retrieval system (a specialized software program). It is very helpful for the user to have background knowledge on both, the database and the retrieval system. Since the specialized information services' retrieval systems (as in the example of DIALOG and LexisNexis) generally use the Exact Match procedure, one must both have a thorough command of the search syntax of the retrieval system and enter the search arguments correctly (so that it can be found in the database). The machine can reduce any formulation problems via autocompletion, autocorrection, notifications (query enhancement or specification, if necessary) or help texts, but in the end it is always the user who determines the course of the retrieval process via his query formulation.

The retrieval system will recognize the terms or words, depending on its maturity level. Word-processing systems only register character sequences, using them as search arguments. If we search for "Java" (and our information need concerns the

Indonesian island), the search will be for “-j-a-v-a-” (and is guaranteed to retrieve the programming language of the same name in almost all results). A concept-oriented system, on the other hand, would recognize the ambivalence and ask back, perhaps like this:

- ☐ Java (Indonesian island)
 - ☐ Java (coffee)
 - ☐ Java (programming language)
- please select!

On the basis of the concepts or words, the documentary units (DUs) are addressed. These DUs are where the informational added value is stored, where the objects addressed in the documents are represented and formal aspects (such as year or author) of the documents are listed. Depending on the information service and document type, the complete documents may also be available digitally. The retrieval system filters the relevant documents from the total amount of all documentary units and displays them to the user in a certain order as retrieval results.

Documentary Reference Unit and Documentary Unit

How did the documentary units (DUs) get into the database? The original materials are the documents, which are segmented into documentary reference units. The selection as to which specific documentary reference units should be admitted to a database and which should not is determined via their so-called worthiness of documentation. This worthiness is based on a catalog of criteria, which might contain the following data: formal criteria (e.g. only documents in English, only HTML documents), content criteria (e.g. only documents about chemistry, only scientific documents), number of documentary reference units to be processed in a time unit (criteria: financial framework, personal resources, computer capacity), user needs (e.g. for the employees of one's own company or for the public at large) and any further criteria (such as the novelty of the knowledge contained in the document or simply avoiding spam). The unit formation relates to the targeted extent of the data set. For instance, one can analyze an omnibus volume by article or as a whole and a film by sequence or also as a whole. In documents on the WWW, one can understand the text of an HTML document as a unit, but one might also allocate the anchor texts, i.e. those (short) texts above a link, to the documentary reference unit that the links refer to. This is how the search engine Google operates, for example.

Relative to the decision one makes, the individual articles, film sequences, books or films would then make up the documentary reference units (DRUs). Then follows the information-practical work process, in which the DRUs are formally described, the content condensed (e.g. via an abstract), the treated objects expressed via con-

cepts (e.g. from a thesaurus, a nomenclature, a classification system). This process of Information Indexing can be accomplished intellectually, by human indexers, or automatically, via an indexing system. The result of this indexing is a documentary unit (also called a “surrogate”), which represents the documentary reference unit in the information service.

Pathology in the Era of Web 2.0

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Key Words: Web site; Web 2.0; Internet; Social software; Continuing education; Communities of practice

DOI: 10.1309/AJCPC9F25B4DEDH

Abstract

In the past few years, the term Web 2.0 has become a descriptor for the increased functionality of Web sites, including those with medical content. Most physicians do not know what Web 2.0 means or how it can impact their work lives. This review provides some background on the evolution of Web 2.0 and describes how its features are being incorporated into medical Web sites. Some potential applications of Web 2.0 in pathology and laboratory medicine are discussed, as are the issues that must be considered when adopting this new technology.

None of us is as smart as all of us.
—Japanese proverb

You may have heard about the Web sites Facebook and Twitter. Perhaps you have read entries in Wikipedia or watched videos on YouTube. Each of these Web sites has a different purpose, but all of them enhance communication and information sharing among a network of users through the exchange of text, images, sound, and video. These features, and the environment of collaboration that they create, are referred to as *Web 2.0*.

The term Web 2.0 was coined at the beginning of the

Figure B.2.1: Documentary Reference Unit (Full-Text Research Article as an Example; Excerpt). Source: American Journal of Clinical Pathology.

Our example of a documentary unit (Figure B.2.2) shows an article entry in a medical journal (Figure B.2.1). The bibliographical data are stated first: journal (“American Journal of Clinical Pathology”), year and month of publication (December, 2009), volume and issue number (Volume 132, Issue 6) as well as page numbers (pp. 824-828). In the abstract, the article’s topics are condensed to their essentials so that the reader is provided with a first impression of its content. Under “Publication Types”, we are told that the article is a review. The information service PubMed employs intellectual indexing via a thesaurus (called “MeSH”, short for “Medical Subject Headings”). The named descriptors (“controlled terms”) serve as information filters which are used to retrieve the paper via its content. Particularly important terms (such as Internet*) are marked by a star. We are further told in the DU that there has been a

comment on the article, and that it has been cited in two journals indexed by PubMed. The system alerts the user to thematically linked contributions (“related citations”). In the top right-hand corner, we find the link leading to the full text of the document.

The screenshot displays a PubMed article page. At the top left, there are links for 'Display Settings' and 'Abstract'. The article title is 'Pathology in the era of Web 2.0' by Schreiber WE and Giustini DM, published in 'Am J Clin Pathol' 2009 Dec;132(6):824-8. The abstract text is visible, discussing the increased functionality of Web 2.0 sites and its impact on medical content. Below the abstract, there are links for 'Comment in' and 'PMID: 19926572 [PubMed - indexed for MEDLINE] Free Article'. On the right side, there is a 'Send to' button and a 'Full Text FREE' button. Below these, there are sections for 'Related citations', 'Cited by 2 PubMed Central articles', 'All links from this record', and 'Recent activity'. The 'Related citations' section lists several articles, including 'Reflections on pathology and "Web 2.0"', 'Pilot study of linking Web-based supplemental interpretive information', 'HIV-associated resources on the internet', 'The medical school Web site: medical education's newest tool', and 'Informatics for practicing anatomical pathologists: marking a new'. The 'Cited by' section lists 'Pathologists in a net-savvy world' and 'Biomedical informatics and translational medicine'. The 'All links from this record' section lists 'Related Citations' and 'Cited in PMC'. The 'Recent activity' section has a 'Turn Off' button and a 'Clear' button.

Figure B.2.2: Documentary Unit (Surrogate) of the Example of Figure B.2.1 in PubMed.

Information filters and information summarization both serve to only provide the user those documents (where possible) that correspond to his information need. If we want to emphasize adequacy with regard to the objective information need, we speak of the transmitted knowledge’s “relevance”; referring to the subjective information need, of “pertinence”.

The goal of the entire process of information indexing (Input) and information retrieval (output) is to satisfy (objective or subjective) information needs by transmitting as much all relevant or pertinent information as possible, and only those kinds. It is clear that information retrieval and information indexing are attuned to each other, both representing parts of one system. Additionally, information scientists observe and analyze the users and usage of information services, and they evaluate both indexing quality and the quality of retrieval systems. Figure B.2.3 shows a schematic overview of the interplay of information indexing and information retrieval and of further tasks of information science.

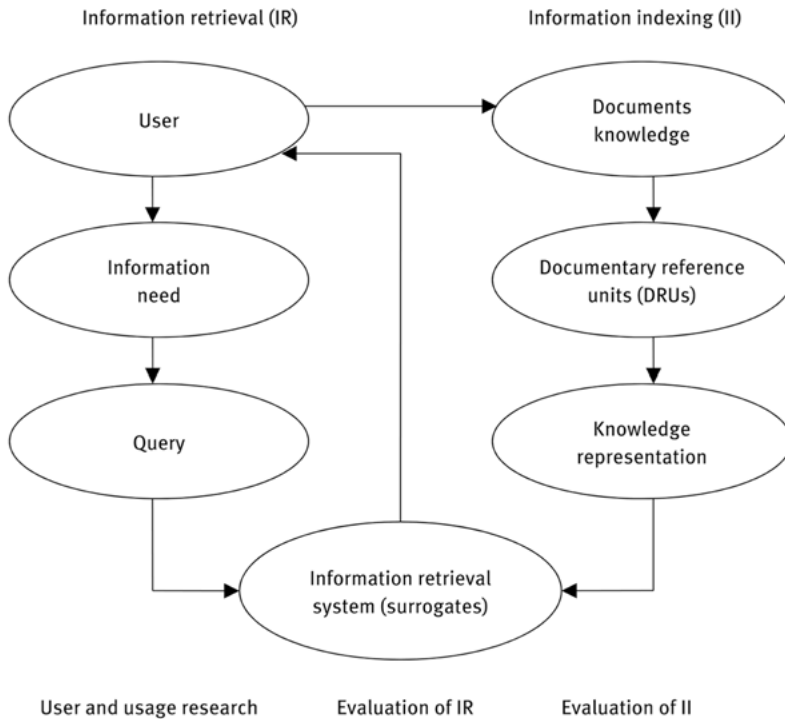


Figure B.2.3: Information Indexing and Information Retrieval.

Pull and Push Services

How does the relevant and pertinent information get to its receiver? There are two ways: either the user actively gets the information from out of the system, or he passively waits until the system provides him with it. Active user behavior is supported from the system side by the provision of pull services, passive behavior by push services. Sometimes the pull approach is referred to as “information retrieval” in a very broad sense and the use of push services as “information filtering”. In the final analysis, though, both aspects prove to be two sides of the same coin (Belkin & Croft, 1992).

Let there be an information need, say: the imminent launch of a research and development project in a company, which is meant to result in a new product as well as a patent. The first step, at time t_0 , is for the user to become active and request information. After this request, at t_1 , he has a basic amount of suitable information. Other people can work on the subject over the course of subject processing—that is to say, over the course of our exemplary research project: a competing enterprise submits a new invention for a patent, a crucially relevant journal article by a heretofore unknown scientist is published, press reports hint at the availability of govern-

ment grants for the project. All this we learn through the initiation of push services by the previously mentioned information services. Those query formulations that have been successfully worked out through the pull service are saved as “profile services”. Such “alerts” or SDIs (Selective Dissemination of Information; Luhn, 1958, 316) automatically search the databases when new documentary units have been admitted and notify the receiver periodically or in real time (via e-mail or the user’s individual homepage).

A special form of push service is represented by WWW services that observe the national or international stream of news on the Internet, discovering current topics, compiling articles about these topics from different sources and automatically creating a short overview on the subject. Such a “Topic Detection and Tracking” is being used by the search engines’ news services (such as Google News). It only ever searches through the new documentary units; in contrast to a “normal” profile service, there is no specific search request. This “search without search arguments” is oriented purely on the novelty and importance of a subject.

Information Barriers

Information filters and information condensation have a positive effect: they provide for a minimum of ballast being yielded. Information barriers, on the other hand, obstruct the flow of relevant information on its way from the complete current world knowledge to the user. Keeping the information stream as broad as possible—i.e. optimizing the recall—means knowing the information barriers and circumventing them. Engelbert (1976, 59) notes:

Information barriers represent an obstacle for the free flow of information to the user respective to the (social) objective information need and obstruct the development of the users’ subjective information needs ...

We will briefly sketch the individual barriers (Engelbert, 1976, 60-72). The political-ideological barrier slows down the information stream between countries with different forms of government, say: between liberal democracies and totalitarian regimes. The ownership barrier relates to the private ownership of knowledge, as when companies do not disclose their research results for competitive reasons or when other factors (e.g. in military research) speak in favor of secrecy. The legal barrier prohibits the free circulation of particular kinds of information (e.g. about people’s income or state of health, or, in official statistics, about individual companies) on the basis of legal norms. The time barrier mainly regards those pieces of information that need to reach the user quickly, as they will become obsolete and thus uninteresting otherwise. In the scientific field, one example might be slowly updated specialist information services in quickly developing disciplines. The effectiveness barrier is a necessary evil. It

states that knowledge management, e.g. in a company, must work effectively and thus cannot subscribe to the maximum of available external sources, particularly the commercial ones. Closely linked to this is the financing barrier. A corporate information department must make do with a given budget just like any other organizational unit.

The terminological barrier results from the continuing specialization of systems, which leads to an expert of one field being unable to understand or having trouble understanding the knowledge of other disciplines, since he is not accustomed to their jargon. This barrier exists within specialist languages (e.g. of chemistry or medicine), whereas the foreign language barrier is aimed at one's knowledge of natural languages. What use is the most relevant article if it is in Chinese, but the user only speaks English and has no access to a translator or translation service? The access barrier states that a user may know that an important document exists, but that he will not be able to acquire it within the means of reasonable effort. The barriers resulting from flaws in the information transmission process form the class of mistakes made by information systems and their employees' information activities, e.g. false indexing or inadequate ranking algorithms of search engines. The consciousness barrier means a lack of information literacy of the user; he does not even use information services or is unable to admit (to himself or others) that he has an information deficit. The resonance barrier assumes that the user has been forwarded documents that are actually relevant, only the user does not translate this knowledge into action. Engelbert (1976, 59) warns of the consequences of underestimating the effects of these information barriers:

Knowing about the existence of information barriers and understanding their modes of action is of great help for any information institution to operate successfully. Often, all endeavours prove fruitless or do not provide a sufficient improvement to information gathering in practice since the user, faced by one of the many barriers that stand between himself and the document's content, gives up trying to use the information.

Recall and Precision

The goal of information retrieval is to find all relevant or pertinent documents (or as many as possible) to satisfy an objective or subjective information need, and only these. Documents that do not fit the subject matter are ballast and thus an annoyance. On the one hand, we must circumvent information barriers so as not to restrict the amount of potentially important information from the outset, and on the other hand deal with information filters and information condensation in such a way that only those documents are yielded which promise the user action-relevant knowledge.

The quality of search results (Ch. H.4) is measured via the two parameters of “recall” and “precision”. We start with three amounts, where

- a = the number of relevant results retrieved,
- b = the number of non-relevant documentary units contained in the search results (ballast),
- c = the number of relevant documentary units that were not found (loss).

Recall is calculated as the quotient of the number of relevant documentary units retrieved and the total number of relevant documents:

$$\text{Recall} = a / (a+c).$$

Precision is the quotient of the number of relevant documentary units retrieved and the total number of retrieved data sets:

$$\text{Precision} = a / (a+b).$$

Precision can be exactly measured when numbers have been experimentally gathered for a and b. Recall, on the other hand, is a purely theoretical construct, since the value for c cannot be directly obtained. How can I know what I have *not* found? The construct of recall, connected to precision, is of use in a thought model. In everyday search situations, it is shown that heightening recall can lead to a simultaneous decrease in precision, i.e. we pay for a higher degree of completeness with a lower degree of exactness, that is to say: with ballast. If we get all data sets of a database, we will get the relevant, but also the non-relevant documents. The recall value relative to the given information system is an ideal “1”. The ballast is gigantic and the research results are useless. Conversely, if we try to heighten the precision, i.e. to avoid ballast at all cost, we will be left with lower completeness. A retrieval result entirely free from ballast (where precision equals 1) could be just as useless, since too much information is left out. As a rule of thumb, then, we can assume that there is an inverse relation between precision and recall (Cleverdon, 1972; Buckland & Gey, 1994). The searcher must balance out his ideal result in the interplay between completeness and ballast.

Here we must take a breath and remember the two sorts of information need there are. In the case of a factual question for an objective information need, the inversely proportional relation between recall and precision does not apply. Here, given a methodically sound search, both recall and precision equal 1. If we ask for the capital of North Rhine-Westphalia and are provided with the answer “Düsseldorf”, this will be both comprehensive and free of ballast. For all problem-oriented information needs, however, any “average” process of information retrieval will show the relation between precision and recall described above. Particularly inexperienced searchers will only succeed in achieving a “1” when combining their values for precision and recall. The searcher’s objective is to approximate the “Holy Grail” of information retrieval—values of “1” for both precision and recall—via elaborate searches, by using

the “art of searching” as well as the system’s elaborate search functionality. What we need are all relevant documents—nothing else.

Similarity

In information retrieval, we are interested in documents that are as similar to the query as possible. If the user is faced with an ideally suitable document, he may want to find further documents, similar to this first one. A retrieval system can help a user find further similar terms on the basis of search arguments already entered. Let a user be faced with an image (or a piece of music or a video); he now wishes to find further images (music files, videos) similar to this first one. If a user’s search behavior (or his shopping behavior in e-commerce) is similar to another user’s, a recommender system may make suggestions for documents or products. Ultimately, the logical consequence is for a retrieval system to organize its hit lists in such a way that the search results are ranked in descending order of similarity to the search request. We can already see, on these few examples: information science is very interested in similarity and the calculation thereof.

Similarity coefficients were developed by Paul Jaccard (1901), Peter H.A. Sneath (1957) and Lee R. Dice (1945), among others, in the context of numerical taxonomy (in biology; Sneath & Sokal, 1973, 131). Salton introduced the cosine formula to information science as a measurement for distance and similarity, respectively (Salton, Wong, & Yang, 1975). The calculation formulas in Jaccard and Sneath are identical, which means that we are looking at a single indicator.

In the calculation of similarity, we distinguish between two cases, which we will demonstrate on the example of the similarity between two documents. The similarity of documents is calculated via the share of co-occurring words. If absolute frequency values are available for the numbers of terms in the documents, and we thus have a (number of words in document D_1), b (number of words in document D_2) as well as g (number of words co-occurring in documents D_1 and D_2), we will calculate the similarity (SIM) of D_1 and D_2 via:

- $SIM(D_1 - D_2) = g / (a + b - g)$ (Jaccard-Sneath),
- $SIM(D_1 - D_2) = 2g / (a + b)$ (Dice),
- $SIM(D_1 - D_2) = g / \sqrt{a * b}$ (Cosine).

In the alternative case, we have weighting values for all words in documents D_1 and D_2 , which we have obtained, for instance, via the calculation of text-statistical values. Here, we can demonstrate more elaborate calculation options (Rasmussen, 1992, 422). In the numerator, we multiply the weight value of all individual word pairs from both documents and add up the values thus derived. If one word only occurs in one document (and not the other) the value at this place will be zero. Then we normalize the value by the denominator in analogy to the original formulas (applying Dice, we add up the squares of the weights of all words of D_1 and D_2).

Which coefficient is being used in each respective case should depend upon the characteristics of the documents, users, words etc. (as well as the system designer's preferences) to be calculated. Rasmussen (1992, 422) writes, for all three variants:

The Dice, Jaccard and cosine coefficients have the attractions of simplicity and normalization and have often been used for document clustering.

Let us play through an example for all three similarity measures (in the simple variants, with absolute numbers). Let document D_1 contain 100 words ($a = 100$), document D_2 200 ($b = 200$); 15 words co-occur in documents D_1 and D_2 ($g = 15$). Following Jaccard-Sneath, the similarity between the two documents is $15 / (100 + 200 - 15) = 0.053$; Dice calculates $2 * 15 / (100 + 200) = 0.1$, and the Cosine leads to $15 / \sqrt{100 * 200} = 0.106$.

Normalized similarity metrics such as Jaccard-Sneath, Dice and Cosine have values between 0 (maximal dissimilarity) and 1 (maximal similarity). If SIM is the similarity between two items, then $1 - \text{SIM}$ is their degree of dissimilarity (Egghe & Michel, 2002). Similarity metrics have strong connections to distance metrics (Chen, Ma, & Zhang, 2009).

Conclusion

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- A concrete information need, generally a factual question, is satisfied via the transmission of exactly one piece of information. Problem-oriented information needs generally regard a certain number of bibliographical references and can be satisfied (however rudimentarily) by retrieving several documents.
 - When translating an information need into a query, exact knowledge of the information retrieval systems' respective languages is a must. Aligning the search request with the documentary units is accomplished either via concepts or words.
 - In Information Indexing, documents are analytically divided into documentary reference units; they form the smallest units of representation (which always stay the same). The selection of what is admitted to an information system and what is not is controlled via the criteria of worthiness of documentation. Informational added value during input is created via a precise bibliographical description of the documentary reference units, via concepts that serve as information filters, as well as via a summary as a form of information condensation.
 - Pull services satisfy information needs by providing the user with a database of document units and the appropriate search functionality. Here, the user must become active himself. Push services, on the other hand, satisfy longer-term information needs by having a system automatically provide the user with ever-new information as it arises.
 - Information barriers are obstructions to the flow of information and they diminish recall. Such barriers should be recognized and removed where possible.
 - The basic measurements of the quality of information retrieval systems are recall and precision—recall describing the information's completeness and precision its appropriateness (in the sense of freedom from ballast). For objective information needs based on facts, both recall and precision can achieve the ideal value of 1, whereas in problem-oriented information needs we can

observe an inverse-proportional relation: increased recall is paid for with decreased precision (and vice versa).

- In information retrieval tasks, we have to find documents which are similar to the query. In information science, we measure similarity between documents, terms, users, etc. Frequently applied are indicators of similarity, which are normalized (to values between 0 and 1), e.g. Jaccard-Sneath, Dice and Salton's Cosine.
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B.3 Relevance and Pertinence

Relevance—Pertinence—Utility

The concept of relevance is one of the basic concepts of information science (Borlund, 2003; Mizzaro, 1997; Saracevic 1975; 1996; 2007a; 2007b; Schamber, Eisenberg, & Nilan, 1990), and it is simultaneously one of its most problematic: users expect an information system to contain relevant knowledge, and many information retrieval systems, including all major Internet search engines, arrange their search results via Relevance Ranking algorithms. Saracevic (1999, 1058) stresses that “relevance became a key notion (and key headache) in information science.”

Relevance is established between the poles of user and retrieval system. According to Saracevic (1999, 1059), relevance is

the attribute or criterion reflecting the effectiveness of exchange of information between people (i.e., users) and IR systems in communication contacts.

We previously distinguished between objective and subjective information needs. Correspondingly to these concepts, we speak of relevance (for the former) and pertinence (for the latter), respectively. Since relevance always aims at user-independent, objective observations, we can establish an initial approximation: A document (or the knowledge contained therein) is relevant for the satisfaction of an objective information need

- if it objectively serves to prepare for a decision or
- if it objectively closes a knowledge gap or
- if it objectively fulfils an early-warning function.

Soergel (1994, 589) defines relevance via the topic:

Topical relevance is a relationship between an entity and a topic, question, function, or task. A document is topically relevant for a question if it can, in principle, shed light on the question.

Pertinence incorporates the user into the equation, with the user’s cognitive model taking center stage. Accordingly, a document (or its knowledge) is pertinent for the satisfaction of a user’s subjective information need

- if it subjectively (i.e. in the context of the user’s cognitive model) serves to prepare for a decision or
- if it subjectively closes a knowledge gap or
- if it subjectively fulfils an early-warning function.

The search result can only be pertinent if the user has the ability to register and comprehend the knowledge in question according to his cognitive model. Soergel (1994, 590) provides the following definition:

Pertinence is a relationship between an entity and a topic, question, function, or task with respect to a person (or system) with a given purpose. An entity is pertinent if it is topically relevant and if it is *appropriate* for the person, that is, if the person can understand the document and apply the information gained.

Pertinence can change in the course of time (Cosijn & Ingwersen, 2000, 544):

Pertinence is characterised by the novelty, informativeness, preferences, information quality, and so forth of objects, that depend on the user's need at a particular point in *time*. In turn, the user's need changes as his understanding and state of knowledge (cognition) on the subject change during a session as well as over several sessions (...). Form, feature, and presentation of objects have a crucial impact on the assessments.

The preconditions for successful information retrieval are:

- the right knowledge,
- at the right time,
- in the right place,
- to the right extent,
- in the right form,
- with the right quality,

where “right” means that the knowledge, time etc. possess the characteristics of relevance or pertinence. Additionally, the expectation of the “right” information contains a strong emotional component (Xu, 2007). Here, a third term enters the fray: utility. Useful knowledge must animate the user to create new, action-relevant knowledge from the newly found information, on the basis of his own foreknowledge, and—if needed—to apply it practically. To quote Soergel (1994, 590) once more:

An entity has utility if it is pertinent and makes a useful contribution beyond what the user knew already. Utility might be measured in monetary terms (“How much is having found this document worth to the user?”) (...) A pertinent document may lack utility for a variety of reasons; for example, the user may already know it or may already know its content.

For certain research questions, it may make sense to differentiate between pertinence (in Soergel's sense) and utility; in all other cases, the two aspects may be combined and simply called pertinence. If in certain contexts it makes no difference whether to speak of pertinence or relevance, we will, broadly, speak of relevance.

Aspects of Relevance

Our definition of relevance is still rather unspecific. Saracevic (1975, 328) suggests taking into consideration all aspects of relevance:

Relevance is the A of a B existing between a C and a D as determined by an E.

For A, we might insert “measure”, “degree”, “estimate”, for B “relation”, “satisfaction”, for C “document”, “article”, for D “query”, “information need” and for E “person”, “user” or “algorithms of a ranking procedure”. Depending on what one specifically uses for A through E, different relevance aspects will emerge, as Saracevic (1996, 208) emphasizes:

Relevance can be and has been interpreted as a relation between any of these different elements. I called these different relations as different ‘views of relevance’.

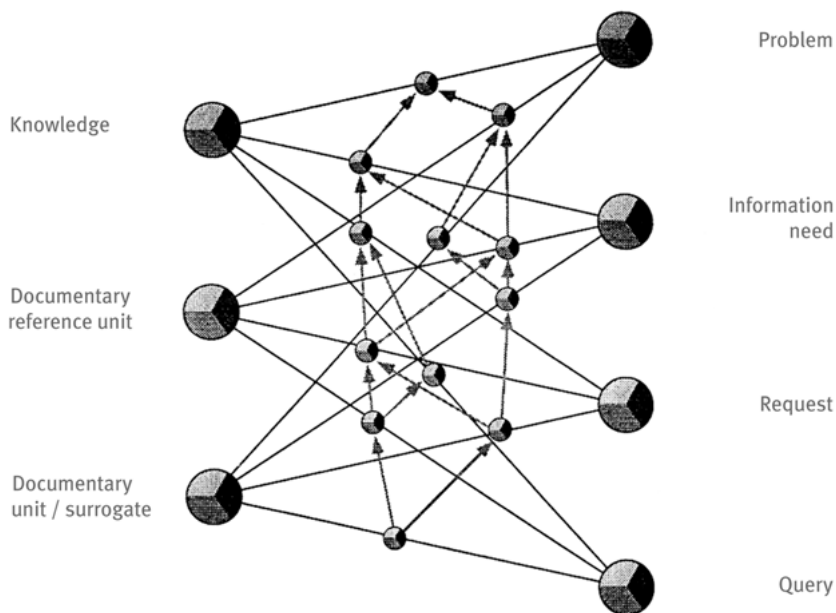


Figure B.3.1: Aspects of Relevance. Source: Modified from Mizzaro, 1997, 812. The Temporal Dimension is Not Represented. Black: Topic; Dark Grey: Task; Light Grey: Content.

Mizzaro (1997, 811 et seq.) works out a general framework for the aspects of relevance. System-side (i.e. Saracevic’s C), we must distinguish between the documentary reference unit, the surrogate or documentary unit and the knowledge being put in motion by the information. User-side (Saracevic’s D), we have a problematic situation, the information need, the natural-language query (request) as well as the formulation of the request in the system’s syntax (query). Mizzaro also sees a third, newly added dimension: the subject, expressed via the three manifestations “topic” (e.g. “the topic of relevance in information science”), “task” (e.g. “to write a chapter about relevance”) and “context” (a sort of residue category that admits everything which is neither topic nor task; e.g. “the time spent preparing a search about relevance”). As

the fourth parameter, Mizzaro introduces time. All four components are interwoven and generate the aspects of relevance.

Figure B.3.1 provides a synopsis of the aspects of relevance in Mizzaro's (1997, 812) model.

On the left hand side, there are the elements of the first dimension, and on the right side there are the elements of the second one. Each line linking two of these objects is a relevance (graphically emphasized by a circle on the line). The three components (third dimension) are represented by the grey levels used. ... Finally, the grey arrows among the relevances represent how much a relevance is near to the relevance of the information received to the problem for all three components, the one in which the user is interested, and how difficult it is to measure it.

Mizzaro's analysis shows that it is a little one-sided to only speak of “relevance” or “pertinence”. If we disregard the temporal dimension, there are thirty-six possible relations between the information retrieval system and its three categories, the user (with four categories) and the subject (again with three categories).

Relevance (in the strict user-independent sense) is the relation between the query with regard to the topic and the system-side aspects (information, documentary reference unit, surrogate), represented in Figure B.3.1 via the three lines starting from the query. In the literature (e.g. Cosijn & Ingwersen, 2000, 539), this is sometimes called “topical relevance”. A crucially important special aspect of topical relevance restricts itself to the documentary units in the IR system, and is thus an expression of “algorithmic relevance”. The objects here are the relevance values allocated by a system to a given query on the basis of its algorithms, i.e. Relevance Ranking with its weighting and sorting functions.

The topmost line in Figure B.3.1 runs between the problematic situation and the knowledge with regard to the thematic task in the specific user context. Here we are looking at the aspect of utility. The entire area between topmost and the three bottom-most lines represents the diverse aspects of pertinence.

Relevant or Not Relevant: The Binary Approach

If we want to analyze relevance we are in need of judgements made by assessing individuals. In this judgement, the relevance is being rated; in the case of topical relevance, the judgement is made by independent experts while neglecting the subjective user aspects—otherwise, it is made by the user himself, generally rather intuitively and unsystematically. Saracevic (1996, 215) here remarks that

(n)obody has to explain to users of IR systems what relevance is, even if they struggle (sometimes in vain) to find relevant stuff. People understand relevance intuitively.

An initially plausible supposition is that the experts' judgements, or the intuitive user behavior, either grant a document (or its content) relevance or not. A user clicks on a link in a list of search results from an Internet search engine, looks at the retrieved document and decides: "useful" or "useless". The binary approach of relevance judgements employs such a zero/one perspective, and our theoretical parameters Recall and Precision build on it.

Scientific endeavours to determine the effectiveness of retrieval systems have employed the binary perspective on relevance after the time of the classic Cranfield Studies (Cleverdon, 1978) as well as during the large-scale TREC (Text Retrieval Conferences). Experts assess the relevance of a document with regard to an information need, independently of other—better or worse—documents. If the document contains at least one passage that satisfies the information need, it is relevant; otherwise, it is not relevant (Harman, 1993). If the experts do not agree in their relevance assessment (which can very well be the case), they must discuss their opinions and arrive at a solution (or otherwise delete the document from their test data set).

Relevance Regions

In an empirical study Janes (1993) provides both end users (students of education and psychology) and budding information professionals (students of information science) with hit lists for certain queries that they must assess. On graph paper (100mm), the test subjects must each mark one value, where 0 represents no relevance and 100 represents maximum relevance. In roughly 50 cases, the number assigned by the end users is 100, while the second-highest value of the distribution is 0 (with 25 cases). The middle area (greater than 0 and smaller than 100) is also taken into consideration, but only lesser values are accrued in this section. If the same documents are shown to the information science students (with the same queries, of course), the result is slightly different. Here, too, a U-shaped distribution is created, as with the end users, only the values for the end points 100 and 0 are a lot more pronounced and there are far fewer decisions in favor of the middle section. Thus information professionals are far more likely to have a binary perspective on relevance than end users. Why? The experts know about the parameters Recall and Precision, and use them in their every-day work; the bivalence of relevance is thus prescribed by the information science paradigm pretty bindingly. Laymen do not know about these things and thus decide "independently of theory". The binary approach (but all the others, too) can thus be a statistical artefact. This is what Janes (1993, 113) emphasizes:

All of this raises an important question: Why should this be the case? There are two possible answers: Either it is a "real" phenomenon, or a statistical artefact. ... People have made these decisions and produced these judgments *because they have been asked to do so as part of a research*

study. We have no guarantee whatsoever that this is reflective of what people really do when evaluating information in response to their information needs.

It is possible that the relevance assessments represent a multi-stage process, which initially takes full advantage of the interval between 0 and 100, but then ends up in a 0/1 decision. If the user has checked and compared the documents, he will be in a better position to decide (binarily): “I need this” or “I don’t need this.” A new research question now arises: at which original relevance assessment (between 0 and 100) do users tend to ascribe (absolute) relevance to a result? An initial supposition would be around 50. However, two empirical studies (Eisenberg & Hu, 1987; Janes, 1991) show that this is not the case. The arithmetic mean of the “breakpoint” is at 41 in the Eisenberg-Hu study and at 46 in the Janes study; the minimum value is at 7 and 8, respectively, and the maximum value at 92 and 75. The standard deviation is very high, with 17 and 18, respectively. Apparently, there is no clear threshold value separating relevant from non-relevant documents. This separation rather depends upon the specific user context, i.e. the user’s processing capacity or the document’s environment in a hit list. Let us consider the example of two users searching for scientific literature; one does so in order to prepare for a short presentation in a basic seminar, the other to research for a PhD dissertation. The former will content himself with three to five articles. Once he has found these, his readiness to assess further documents from the hit list as relevant will decrease. (After all, he will then have to acquire, read, understand, excerpt and integrate them into his lecture.) The doctoral candidate, on the other hand, will invest far more time and, in all probability, employ a much lower threshold value between (more or less) relevant and not relevant. In this instance, documents that only peripherally touch upon “his” subject may also be relevant if they provide important details for his research question. That is why Borlund (2003, 922) points out that documents can lead to deviating relevance assessments with regard to the same information needs, when the context and/or the task is a different one.

In light of the above, it appears obvious that a user’s perception of relevance does not have to be binary in nature. It is far more appropriate to take the middle section (Greisdorf, 2000, 69) between the extremes of 0 and 100 into serious consideration and to speak of “relevance regions” (Spink & Greisdorf, 2001; Spink, Greisdorf, & Bateman, 1998) or to observe the entire spectrum in the interval [0,100] (Borlund, 2003, 918; Della Mea & Mizzaro, 2004).

It must be noted that our definition of relevance is independent of the specific user. This, of course, also holds for the classification of a document into relevance regions, relative to a certain information need. It may be difficult to work out, from time to time, which degree of relevance will objectively apply, since in determining relevance we are always dependent upon the relevance assessments of test subjects, and hence their subjective estimates. Empirical studies that do not work with the binary concept of relevance are thus elaborate in their test design, since large quantities of test subjects are required in order to achieve accurate results.

Relevance Distributions

How is information distributed when arranged according to relevance? Such distributions are the result of document sets arranged by relevance, but also, for instance, of lists of document-specific tags used in a broad folksonomy. We know about two ideal-typical curve distributions: the so-called “Power Law” and the inverse-logistic distribution.

Following the tradition of the laws of Zipf, Bradford and Lotka, there is a power law distribution adhering to the formula

$$f(x) = C / x^a$$

(Egghe & Rousseau, 1990, 293). C is a constant, x is the rank and a a value generally ranging from around 1 to 2 (in Figure B.3.2, the value is 2). For large sets, this power law proves to be correct in many instances (see, for example, Saracevic, 1975, 329-331). The maximum value in relevance distributions is 1, so C equals 1 in this instance, too. The law tells us that if the document ranked 1 has a relevance of 1, then the second-ranked document must have a relevance of 0.5 (where $a=1$) or of 0.25 (where $a=2$), the third 0.33 ($a=1$) or 0.11 ($a=2$) etc.

In a research project concerning the relevance of scientific documents with regard to some given topics (Stock, 1981), we found out that about 14.5% of all documents in the hit list had a weight of 100 (i.e. a relevance of 1), that very few items had values of between 99 and 31, and that about 66% of the documents had values between 30 and >0. The relevance judgment was made by the indexer. This example may stand for another kind of relevance distribution. Data by Spink and Greisdorf (2001, 165) lead us into the same direction. Here, we see several documents having a value of 1 or close to 1 in the upper ranks, followed by a relatively small amount of items in the middle and a large set of documents with low relevance weighings. In distributions of tags to a document, we can also find curve distributions that have a “long trunk” in addition to the long tail (Peters, 2009, 176-177). Kipp and Campbell (2006) note:

However, a classic power law shows a much steeper drop off than is apparent in all the tag graphs. In fact, some graphs show a much gentler slope at the beginning of the graph before the rapid drop off.

Looking at the function, one can see a graph that is similar to the logistic function, except it is reversed—we thus speak of an inverse-logistic function. The mathematical expression of this function is

$$f(x) = e^{[-C \cdot (x - 1)]^b},$$

where e is the Euler number, x is the rank, C' is a constant and the exponent b is approximately equal to 3.

For both views (Power Law and inverse-logistic distribution), there is evidence that they work in certain sets containing many documents. In smaller sets, these laws do not apply universally. The smaller the set, the smaller the probability that there will be a clear Power Law or a clear inverse-logistic distribution. In small hit lists, it is very probable that there will be no regular relevance distribution. If the output of a retrieval system consists of three results, where one discusses the research topic on half a page of a ten-page article and two only mention the topic in short footnotes, it is impossible to decide whether the distribution follows a Power Law or an inverse-logistic function.

Common to both distributions is the “long tail”, with its small Y values. The difference between both curve distributions is in their respective beginning: there is a clear drop in the Power Law’s Y values, whereas the top ranks in the inverse-logistic distribution are (nearly) equally relevant.

Figure B.3.2 shows the two relevance distributions. The Y axis represents the degree of relevance (normalized, i.e. mapped onto the interval $[0,1]$), and the X axis represents the documents ranked 1, 2, 3 etc. up to n , where the rankings are determined via relevance. In the example, we see a hypothetical distribution of 20 documents. When frequenting online search engines, one is often confronted with far greater quantities of search results. “In real life”, then, our rank no. 10 might correspond to a no. 100 or 1,000.

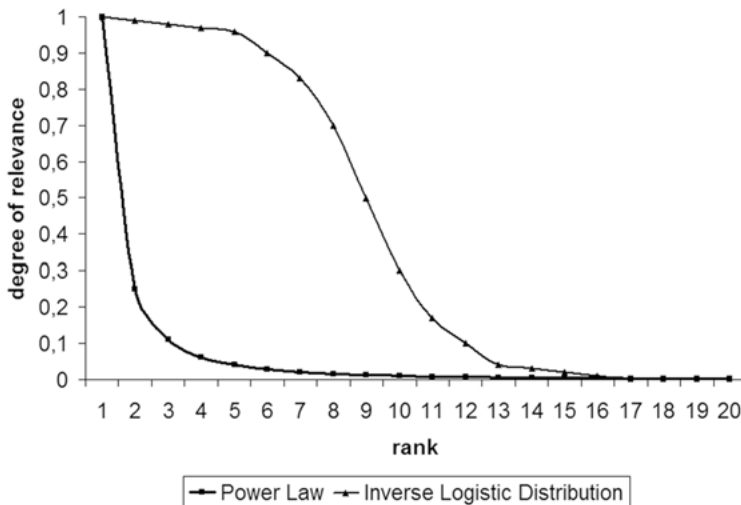


Figure B.3.2: Relevance Distributions: Power Law and Inverse-Logistic Distribution. Source: Following Stock, 2006, 1127.

Knowledge of the “correct” relevance distribution is fundamental for all development of search engines with Relevance Ranking. It is also important for the users of information retrieval systems. Where the Power Law applies, a user can concentrate on the first couple of results; they should pertain to the crucial aspects of the subject. If, on the other hand, the inverse-logistic approach applies, and the hit list is, in addition, a large one, the user must sift through pages of search results so as not to lose any information. Behavior that is to be considered rational for one approach becomes faulty when applied to the other (Stock, 2006).

Conclusion

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- A document (respectively: the knowledge contained therein) is *relevant* when it objectively (i.e. independently of the concrete specifications of the searching subject) serves to prepare for a decision, to close a knowledge gap or to fulfil an early-warning function. A document is *pertinent* if the user is capable of comprehending the knowledge it contains in accordance with his cognitive model, and if it is judged to be important. A document is *useful* if it is pertinent and the user creates new, action-relevant knowledge from it.
 - Relevance has several aspects; system-side, we distinguish between the documentary reference unit, the documentary unit (surrogate) and the contained knowledge; user-side, there is a problematic situation, an information need, a natural-language request and a query formulated in the system’s syntax; subject-side, there is the research topic, the task and the context. The fourth aspect is time.
 - When considering only the relations between subject, query and system aspects, excluding all other aspects, we speak of “topical relevance”. If, in addition, we concentrate on the documentary unit, we speak of “algorithmic relevance”.
 - The binary approach toward relevance knows only two values: relevant or not relevant. The parameters Recall and Precision use this approach. However, it can be empirically proven that users also assume the existence of values between the poles of “100% relevant” and “not relevant at all”, whereas information professionals tend to follow the binary approach.
 - It appears prudent to assume the existence of a middle section of partial relevance, and in consequence to speak of relevance regions or to take into consideration the entire spectrum of the interval [0,1].
 - Looking at empirically concrete distributions of documents via relevance (without consulting user assessments), there are at least two general forms of distributions: the Power Law and the inverse-logistic relevance distribution. These forms of distributions arise in large hit lists and do not (always) apply to small sets of retrieved documents. Knowledge of the “correct” form of relevance distribution affects, for instance, the construction of algorithms for Relevance Ranking as well as ideal user behavior when confronted with large hit lists.
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B.4 Crawlers

Entering New Documents into the Database

When a database is updated intellectually, the indexer's selections are steered by criteria of worthiness of documentation. When new documents are admitted into the database automatically, a software package, called a “robot” or a “crawler”, makes sure the documents are retrieved and copied into the system. A crawling process in the WWW always follows the same simple schema: the links contained within a quantity of known Web documents (“seed list”) are processed and the pages are searched and copied. The links on the pages will then be followed in turn. Arasu et al. (2001, 9) describe the characteristics of a crawler:

Crawlers are small programs that ‘browse’ the Web on the search engine's behalf, similarly to how a human user would follow links to reach different pages. The programs are given a starting set of URLs, whose pages they retrieve from the Web. The crawlers extract URLs appearing in the retrieved pages, and give this information to the *crawler control* module. This module determines what links to visit next, and feeds the links to visit back to the crawler. (...) The crawlers also pass the retrieved pages into a *page repository*. Crawlers continue visiting the Web, until local resources, such as storage, are exhausted.

Determining the worthiness of documentation of Web pages is problematic, though. Nobody knows, to any degree of accuracy, how many documents there are on the Internet. However, it is assumed that even the most powerful search engines can index and process only a fraction of the Web's content for retrieval. Crawlers—as “information suppliers” of the search engines—thus only take into account a part of the Web's contents. Due to this fact, crawlers and search engines have developed several policies for indexing as many documentation-worthy Web pages (and only those) as possible.

In the vast ocean of documents, there are those of good (useful) and those of bad (poor) quality, those that are current (fresh) or out-of-date (stale), those that are duplicates or spam and those that are hidden in the so-called “Deep Web”.

Croft, Metzler and Strohman (2010, 31) point out that apart from their technology, it is the collection of information that makes the work of search engines effective in the first place:

... if the right documents are not stored in the search engine, no search technique will be able to find relevant information.

What is meant to be searched and retrieved, in what way, and where? For their collection, search engines require copies of the documents, which are then indexed for the subsequent search process. Here, the Web crawler helps by asking Web servers for documents. Its goal is to retrieve URLs and to automatically copy pages; to do so, several request and working steps are needed, respectively.

In order to be able to access the Web page of a server, its specific address must be available. This is regulated via the standardized addressing of documents on the Internet, the so-called URL (uniform resource locator). A URL provides information about the page's type of transmission and source citation as well as the server name where the document is stored. Web servers and crawlers normally use a Hypertext Transfer Protocol (HTTP), which serves the straight-forward transmission of documents. These documents are represented via a formatting language, the Hypertext Markup Language (HTML).

On the Internet, every server is marked by a specific numerical Internet Protocol (IP) address. The host and server names, respectively, are translated via Domain Name Systems (DNS). From a DNS server we learn, for example, that the URL *www.uspto.gov* uses the IP address 151.207.247.130 and that the server's operating system is Apache.

The crawler initiates contact with a DNS server in order to learn the translated IP address. In practice, this translation is not performed by a single DNS server. Rather, several requests are often necessary to search the hierarchy of DNS servers.

Throughout the entire crawling process, the crawler must interact with several different systems. The crawling architecture requires several basic modules (Manning, Raghavan, & Schütze, 2008, 407 et seq.; Olston & Najork, 2010, 184 et seq.). The URL Frontier contains the seed set and consists mainly of URLs whose pages are retrieved by the crawler. The DNS Resolution Module specifies the Web server. The host name is translated into the IP address and a connection is established to the Web server. The Fetch Module retrieves the Web page belonging to the URL via HTTP. The Web page may then be saved in a repository of all collected pages. The Parsing Module extracts the HTML text and the links of this page.

Every extracted link is subject to a series of tests, which check whether it may be added to the URL Frontier. It is tested whether a page with the same content appears under a different URL. Undesired domains may be filtered out using the URL Filter. In addition, it is checked whether any Robots Exclusion Rules apply. URLs whose sites belong to a "Black List" are excluded.

The Duplicate Elimination Module finds out whether a URL is already present within the URL Frontier or whether it has been retrieved beforehand. All previously discovered URLs are retained and only the new ones are admitted. Subsequently, the extracted text is sent forward for indexing and the extracted links are led to the URL Frontier. When a page has been fetched, its corresponding URL is either deleted from the Frontier or it is led back to the Frontier for a continuous crawling process. The entire architecture of a crawler is exemplified in Figure B.4.1.

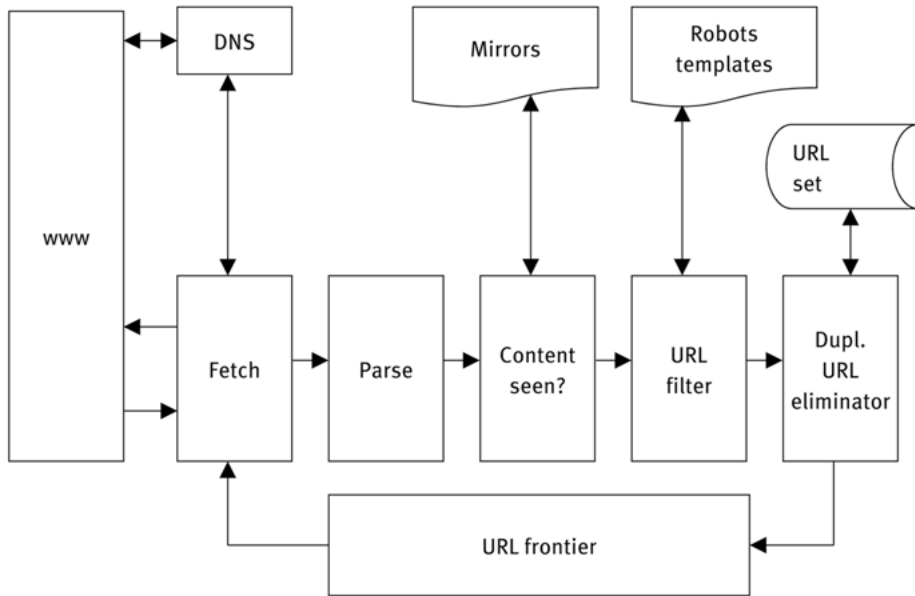


Figure B.4.1: Basic Crawler Architecture. Source: Modified from Manning, Raghavan, & Schütze, 2008, 407.

To prevent an aimless searching and copying of the pages, the crawler employs certain selection policies. The unknown overall quantity of Web pages alone means that the crawler must choose which pages it will visit, and in what way it will do so. Selection criteria are defined both before and during the crawling process. Restricted bandwidth costs and storage limitations are an issue at this point.

According to Baeza-Yates, Ribeiro-Neto and Castillo (2011, 528-529), Web crawlers often set themselves the following off-line limits:

- maximum amount of hosts/domains to be crawled,
- maximum depth resp. amount of links with regard to a starting set of pages,
- maximum total amount of pages,
- maximum amount of pages (or bytes) per host/domain,
- maximum page size and
- maximum out-links per page and selection of document types to be copied (e.g. only HTML or PDF).

FIFO Crawler and Best-First Crawler

How does the URL Frontier work? There are several possibilities of determining the order of the pages that are to be searched (Heydon & Najork, 1999):

- FIFO Crawler (“first in—first out”)

- Breadth-First Crawler: in a first step, all linked pages are located. Then the first among them is visited, then again the first one in the URL thus retrieved etc.,
- Depth-First Crawler: the first step is analogous to the Breadth-First Crawler, but in the second step, all links of the first page are processed before going on to further pages (which are then again processed “depth-wise”),
- Selection of links by chance;
- Best-First Crawler (the pages are navigated by—suspected—relevance).

The PageRank Crawler by Google (Googlebot) is one such Best-First Crawler. The URLs that are to be searched are arranged by the number and popularity of the pages that link to them (Cho, Garcia-Molina, & Page, 1998).

Recognizing Duplicates and “Mirrors”

It is taken to be a sign of quality for databases to state documentary reference units only once; in other words, duplicates are a no-no. In intellectually maintained databases, the indexers take care (perhaps via small programs that recognize duplicates) that every document is only processed once. How do crawlers on the Web deal with duplicates? Such “mirrors” are very frequent on the WWW: they are created for technical reasons in order to optimize access time; they serve commercial purposes when several online shops offer the same products, or they provide translations of Web pages. Mirrors always refer to entire Websites, not only to single pages within the sites.

In the literature, mirrors are defined via common path structures (Bharat et al., 2000, 1115):

Two hosts *A* and *B* are mirrors if for every document on *A* there is a *highly similar* document on *B* with the same path, and vice versa.

To recognize mirrors, all paths within the two observed hosts are compared by pairs.

According to Bharat and Broder (1999), there are five levels of mirroring:

- Level 1: Websites have the same path structure and display *identical* content,
- Level 2: Websites have the same path structure and display *the same* content (i.e. with deviations in formatting or other elements),
- Level 3: Websites have the same path structure and display *similar* content (i.e. with different advertising texts, for example),
- Level 4: Websites have a *similar* path structure and display similar content,
- Level 5: Websites have the same path structure and display *analogous* content (deviations mainly in the language).

Level 5 mirrors are translations of Websites; both versions are recognized by the crawler. Thus, the user can access the (semantically equivalent) material in different languages. Level 4 mirrors are a problematic case and, when in doubt, both should

be admitted. The mirrors of the remaining three levels are clearly duplicates, of which only one version should enter the database, respectively.

An alternative (or complementary) method for recognizing path structure works with the structure of outgoing links, which in the case of mirrors should be identical, or at least similar. Here, it must be noted that internal links name the same paths, but not the same host. Conversions must be performed. Furthermore, the relative frequency of matches between the out-links of both hosts is calculated. Since local changes are to be expected, a match of around 90% is a good indicator for mirrors (Bharat et al., 2000, 1118). An interesting fact is that the recognition of mirrors does not require an analysis of the documents' linguistic contents, as Bharat et al. (2000, 1122) emphasize:

(O)wing to the scale of the Web comparing the full contents of hosts is neither feasible, nor (as we believe) necessarily more useful than comparing URL strings and connectivity.

When searching for translated mirrors, language recognition is performed. However, this process does not search for aspects of content either, looking instead for statistical characteristics of the individual languages. When the mirror identification has been accomplished, the mirrors from Level 1 through Level 3 are deposited as aliases in the "Content Seen" block (in Figure B.4.1).

Freshness: Updating the Documentary Units

On the WWW, the question of the database's up-to-dateness comes up with force (Cho & Garcia-Molina, 2003). Webmasters constantly change the content of their pages, or even delete them. Allow us to provide an example for erroneous information due to out-of-dateness: on my Website, at a certain time t , I admit to supporting Liverpool Football Club. This page is recognized by the crawler and indexed in a search engine. In the meantime, my team has lost repeatedly, frustrating me in the process. At another time t' , I thus change my Website, switching my allegiance to Manchester United. Then, someone searches for information on Liverpool FC. He will find the URL of my Website in his hit list, click on it and find its current variant cheering on Manchester. His confusion would be great. Out-of-date information can never be entirely precluded, but a crawler can use strategies of refreshing in order to update the pages retrieved so far.

Let us play through the variants of a user being presented an out-of-date page from the perspective of search engines. The obsolete page may not even occur in any hit list; it may be noted but not clicked by the user; or it may be accessed without the user realizing (e.g. due to only minute changes) that it has grown out of date. Only when the user notices the problem is action called for. This scenario must be precluded as far as possible, using optimized refreshing strategies.

What error possibilities are there? We distinguish three variants:

- a change in content of a still operative Web page,
- a deleted page,
- a “forgotten” page.

The most frequent case is when the content of an existing Web page is changed. Should a crawler fail to notice that a page has been deleted; the link in the hit list will lead nowhere (HTTP 404—File not found). “Forgotten” Web pages represent a rather rare case. Here, a webmaster deletes the links to a page on the server (but lets the page itself continue to exist—now without any in-links). Such a page can still be available in the index of a search engine and thus be displayed for appropriate queries.

It is the crawler’s task to always index the current “living” content of a Web page. To do so, it must revisit previously crawled pages at intervals in order to be able to supply the search engine with up-to-date copies. But which interval is the right one in order for a fresh copy to avoid becoming a stale one (that no longer reflects the original’s current content)? Not all documents are updated with the same frequency. News sites or blogs are refreshed several times a day, other popular sites maybe once a day, once a week or once a month, whereas a personal homepage will likely be redesigned far more seldomly.

Croft, Metzler and Strohmman (2010, 38) mention that the HTTP Protocol disposes of a HEAD Request, where information about pages (e.g. Last-Modified) can be gathered. This method is cheaper than having to check the pages oneself, but it is still unprofitable because not all pages can be observed and processed continuously and at the same intervals. Rather, any excessive loading effort on the part of the crawlers and unnecessary strain placed on Web servers should be avoided.

At the moment when a page is being crawled, it is considered “fresh”, which of course it will continue to be only until it is modified. A characteristic of crawlers is that they must formulate queries, and are not automatically notified by the Web server once a certain page has just been changed. A date of modification alone is insufficient for the crawler, since it does not specify when it should next crawl the pages. The crawler requires a measurement that takes into account the various different modification frequencies of the pages in order to estimate the probable freshness of a page. According to Baeza-Yates, Ribeiro-Neto and Castillo (2011, 534), the following time-related metrics for Web pages p are used preferentially:

- Age: access time-stamp of a page (visit p) and the last-modified time-stamp of a page provided by the web server (modified p).
- Lifespan: point in time from which the page could no longer be retrieved (deleted p) and the time when the page had first appeared (created p).
- Number of changes during lifespan: changes p .
- Average change interval: $\text{lifespan } p / \text{changes } p$.

Another option for solving the update problem is provided by a sitemap. Sitemaps list certain information pertaining to URLs, e.g. a source’s probable frequency of modification, its last date of modification as well as the priority value (importance) of a

site. Using the sitemap information saves the crawler from having to take other, more elaborate query steps. Concerning the question why Web server administrators create sitemaps, Croft, Metzler and Strohman (2010, 44) argue:

Suppose there are two product pages. There may not be any links on the website to these pages; instead, the user may have to use a search form to get to them. A simple web crawler will not attempt to enter anything into a form (although some advanced crawlers do), and so these pages would be invisible to search engines. A sitemap allows crawlers to find this hidden content.

The sitemap also exposes modification times. ... The `changefreq` tag gives the crawler a hint about when to check a page again for changes, and the `lastmod` tag tells the crawler when a page has changed. This helps reduce the number of requests that the crawler sends to a website without sacrificing page freshness.

Politeness Policies

While many Web server administrators place great value on their pages being searched and retrieved by a broad public user base, other providers are opposed to this. The latter holds particularly true for providers of for-profit specialist information services, who are concerned with protecting their copyrights. Web server administrators can use the `robots.txt` file to control crawlers, who are thus told about which sources they can retrieve and process (and those they cannot). The great search engines generally adhere to these so-called “politeness policies” and follow the specified Allow and Disallow rules.

It is up to those who operate Websites on the WWW to state in the documents whether their sites may be processed by search engines. Here, a (not legally binding) standard applies (Koster, 1994; Baeza-Yates, Ribeiro-Neto & Castillo, 2011, 537). Pages are made inaccessible to search engines (via `<META NAME=“ROBOTS” CONTENT=“NOINDEX”>`), either in the meta-tags or in a text file (`robots.txt`), and any further crawling on the Website or the adoption of images is stated to be unwelcome.

For the Web server, it must be apparent that it is a crawler and not a “normal” user who is visiting the site. One of the reasons this is important is that any count of user visits should not be tainted by crawler activities. The crawler can identify itself as such in the User-Agent field of a query. There, it is also told which activities it may or may not perform, and where. According to Baeza-Yates, Ribeiro-Neto and Castillo (2011, 537), the Robot Exclusion Protocol comprises the following three types:

Server-wide exclusion instructs the crawler about directions that should not be crawled. ... Page-wise exclusion is done by the inclusion of meta-tags in the pages themselves ... Cache exclusion is used by publishers that sell access to their information. While they allow Web crawlers to index the entire contents of their pages, which ensures that links to their pages are displayed in search results, they instruct search engines not to show the user a local copy of the page.

A further politeness rule involves not making new and extensive crawler requests to the same host within a short period of time.

Focused Crawlers

In contrast to the crawlers of general Web search engines, topical crawlers (also called “focused crawlers”) restrict themselves to a certain section of the WWW (Chakrabarti, van den Berg, & Dom, 1999). Menczer, Pant and Srinivasan (2004, 380) observe:

Topical crawlers (...) respond to the particular information needs expressed by topical queries or interest profiles. These could be the needs of an individual user (...) or those of a community with shared interests (topical or vertical search engines and portals). ... An additional benefit is that such crawlers can be driven by rich context (topics, queries, user profiles) within which to interpret pages and select the links to be visited.

General crawlers work similarly to public libraries: their goal is to serve everyone. Topical crawlers, on the other hand, work like specialist libraries, concentrating on one area of expertise and attempting to adequately provide specialists in that area with knowledge.

Starting from the “seed list”, a topically relevant target must be located. In other words, the crawler must select the topically relevant pages and skip the irrelevant ones on the way. Here, it is possible that pages will indeed be needed as intermediary navigation pages on the way from the original to the target. However, these will likely be topically irrelevant and will thus be “tunnelled” and not saved.

In topical crawling, a knowledge organization system must cover the knowledge domain of the respective subject area. A page is deemed topically relevant when it contains terms from the KOS, either frequently or in key positions (e.g. in the Title Tag).

Deep Web Crawlers

The crawlers described so far serve exclusively to navigate to pages on the Surface Web, i.e. those that are firmly interlinked with other pages. Dynamically created pages, e.g. pages that are created on the basis of a query within a retrieval system, are not reached. The systems that can be reached via the Web, but whose pages lie outside the Web, make up the Deep Web. The databases located in the Deep Web are searched via their own search masks.

In the case of freely searchable Deep Web databases, crawling is possible in so far as their contents can be integrated into a surface search engine. (Whether such a procedure will adhere to copyright rules is another question.) Crawling the Deep Web is far more complicated than crawling the Surface Web. In addition to “normal” crawl-

ing, a Deep Web crawler must be able to perform three tasks (Raghavan & Garcia-Molina, 2001):

- “understand” the search mask of each individual Deep Web database,
- formulate optimal search arguments in order to access the entire database (if possible),
- “understand” hit lists and document displays.

An intelligently selected amount of search arguments should safeguard that (in the ideal case) all documentary units which are indeed available in the database will be retrievable. If the database provides a field for searching by year, the Deep Web crawler will have to enter, one by one, every year into the search field in order to be yielded the database’s entire content.

When only a keyword field is provided, adaptive strategies prove to be effective (Ntoulas, Zerkos, & Cho, 2005). In a first step, frequently occurring keywords that have already been observed elsewhere are arranged according to their frequency of occurrence and are then processed in the resulting order. Keywords are extracted from the documents thus yielded and are again arranged by frequency. In the second step, the n most frequent keywords for the research are processed. The greater n is, the greater the probability of “hitting” the total stock will be.

In the hit list, it must be noted whether the page has to be turned after a certain amount of hits and whether the list is restricted (e.g. never more than 3,000 documents displayed). Such restrictions, of course, will in turn affect the formulation of search arguments.

Once the stock of a Deep Web database has been crawled, the main task is accomplished. Since in many of these databases hardly any changes to the data sets can be observed, the only thing left to do is to have the crawler periodically process any updates that may arise.

Avoiding Spam

Like many e-mails, the WWW also contains spam. Spam pages do not have any actual relevance for certain queries, but pretend that they do. Gyöngyi and Garcia-Molina (2005, 2) define:

We use the term *spamming* (also, *spamindexing*) to refer to any deliberate human action that is meant to trigger an unjustifiably favourable relevance or importance for some web pages, considering the page’s true value.

Web crawlers should recognize spam and skip such pages. In order to accomplish this, one must know in what forms spam appears and how spam pages can be securely identified. Following Gyöngyi and Garcia-Molina (2005), we distinguish the following forms of spam:

- Techniques of Enhancing Relevance
 - Term Spamming,
 - Link Spamming;
- Techniques of Concealing Spam Information
 - “Hiding” Spam Terms and Links,
 - Cloaking,
 - Redirection.

When a term occurs on a Web page, search engines will—at least in principle—retrieve the document via this word. In Term Spamming, false or misleading words are introduced at various points: in the text body, in the title (this is seen as very useful since several search tools grant a high weighting to title terms), in other meta-tags such as the keywords (since these are ignored by most search engines, this will be modestly successful at best), in the anchor texts (since such texts are ascribed to both pages, spamming influences both the source page and the target page) or in the address (URL). Depending on the hoped-for effect, different methods are used in Term Spamming. When a select few words are frequently being repeated, the goal is to achieve a high weight value for these terms. However, the spammer can also load long lists of words in order to get his page to be retrieved via various aspects. The “interweaving” of spam terms into copied third-party pages misleads users into navigating to the spam page via the terms of the original page. A similar approach is pursued by Phrase Stitching, in which sentences or phrases of third-party texts are adopted by the spam page, which will now be retrieved via these “stolen” phrases.

Link Spamming exploits the fact that search engines regard the structure of links in the WWW as a criterion of relevance in the context of the link-topological models. The spammer will endeavour to increase the number of incoming links in particular. A legitimately useful resource (“honey pot”) may be developed, which is then used and linked to by others, while surreptitiously establishing links to one’s own spam sites that profit from the honey pot’s high value. An unsubtle variant of this involves infiltrating Web directories and inserting spam pages into misleading topical areas. It is equally simple to enter a comment in a blog, a message board, a wiki etc. and to assign a hyperlink to the text, leading to the spam page. The link exchange (“link my page and I’ll link yours”) leads to an increased number of in-links. Purchasing a(n expired) Web page may—at least for a certain time—keep the old links that lead to this page valid, thus increasing the score. An “elegant” form of Link Spamming involves operating a link farm, where the spammer can run a large number of pages, (including) on different domains. In this way, a spammer becomes at least partially able to control the number of incoming and outgoing links by himself.

How do spammers render their actions invisible? Both Term and Link Spamming cannot be detected by the naked eye: in the browser window, they use the same color for the symbols as they do for the background. In the case of spam links, a graphic is superimposed onto the anchor. This graphic is either small (1*1 pixel) and transparent, or it adopts the color of the background.

In Cloaking, a spammer works with two versions of his page; one version is for the crawler, and one for the Web browser. The precondition is that the spammer be able to identify the crawlers (e.g. via their IP addresses) and thus “slip” them the wrong page. Redirection is understood as the interposing of a page (“doorway”) in front of the target page, with the doorway page carrying the spam information.

Conclusion

- The linking between the retrieval system and the documents is created by indexers, who pointedly selected the right documentary reference units (for specialist information services in the Deep Web), or (for search engines in the WWW) by automatic crawlers.
 - A crawler starts with an initial quantity (the seed list) and processes the links contained within the Web pages, then the links contained in the newly retrieved pages etc.
 - Crawlers either use the “first in—first out” strategy, orienting themselves on breadth (Breadth-First Crawlers) and depth (Depth-First Crawlers), respectively, or they search for assumed relevance (Best-First Crawlers).
 - Webmasters use the “Robot Exclusion Standard” to define whether crawlers are welcome on their sites or not.
 - In the Web, it is common practice to host entire Websites in different locations, as so-called “mirrors”. Since it makes little sense to index pages twofold, such mirrors must be recognized. The strategies of comparing paths and matching outgoing links are particularly promising.
 - Data sets in databases must be updated. Particularly in the Web, individual pages are often changed and even deleted. When updating pages according to priority, crawlers observe the pages’ age, lifespan and the changes during their lifespan.
 - Topical crawlers, or focused crawlers, do not take into account every retrieved Web page but only those that concern a certain subject area. A knowledge organization system must be available for this subject area.
 - Crawling the Deep Web provides search tools in the Surface Web with Deep Web contents. Since Deep Web databases use search forms and specific output options, Deep Web crawlers must “understand” such conditions.
 - Web pages that are not relevant for a topic, but pretend that they are, are labelled as spam. Spammers use techniques such as the dishonest heightening of relevance (Term and Link Spamming) as well as techniques of concealing spam information. Crawlers should skip identified spam pages.
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B.5 Typology of Information Retrieval Systems

Weakly Structured Texts

The purpose of retrieval systems is to help users search and find documents via the documentary units stored within them. Depending on the document form, we initially distinguish between systems for text retrieval and those for retrieval of non-textual digital documents (multimedia documents). The latter process audio, graphic and film documents; they are, as of yet, far less advanced than their text retrieval counterparts. In the following, we assume that documents which are not digitally available have been either digitized or are represented by a “proxy”, the documentary unit. If the documentary units are available in the form of texts, we speak of “text retrieval”, if they are available non-textually, of “multimedia retrieval”. If a document is indexed purely on the basis of its own content, this is called “content-based information retrieval”. In cases where we use content-depicting terms (i.e. from a knowledge organization system) for the creation of metadata, we are practicing “concept-based information retrieval”. It is quite possible that retrieval systems—as hybrid systems—may dispose of both components, i.e. of content-based retrieval and of concept-based retrieval.

In the case of text documents, we can differentiate between structured, weakly structured and unstructured texts. For structured texts, the logical organization of the individual files occurs either in the form of charts (as in the relational database model) or objects (e.g. in the object-oriented database model). Weakly structured texts include all manner of published and unpublished textual documents, from microblog postings, private e-mails, internal reports, websites and scientific publications up to patent documents, all of which display a certain structure. Thus for instance, this chapter contains a number, a title, subheadings, graphics, captions, a conclusion and a bibliography. Structured data may be added to the texts via informational added values, i.e. controlled terms for representing content and author statements including catalog entries. Unstructured texts have no structure at all; in reality, they hardly occur at all. There are, however, retrieval systems that are unable to recognize (weak) structures and hence treat the documents as unstructured. The scope of information science is, in principle, wide enough to include all document forms, but there is a clear preference for weakly structured texts (Lewandowski, 2005, 59 et seq.).

Information science literature deals with texts’ formal structures (such as headings, footnotes etc.), but little attention is paid (as of now) to the syntactical structures of individual sentences in the context of retrieval systems. Let us take a look at the following sentence! Vickery and Vickery (2004, 201) provide the example.

The man saw the pyramid on the hill with the telescope.

There are four possible interpretations depending on how the single objects are taken to refer to one another. Alina and Brian Vickery demonstrate this in a picture (Figure B.5.1).

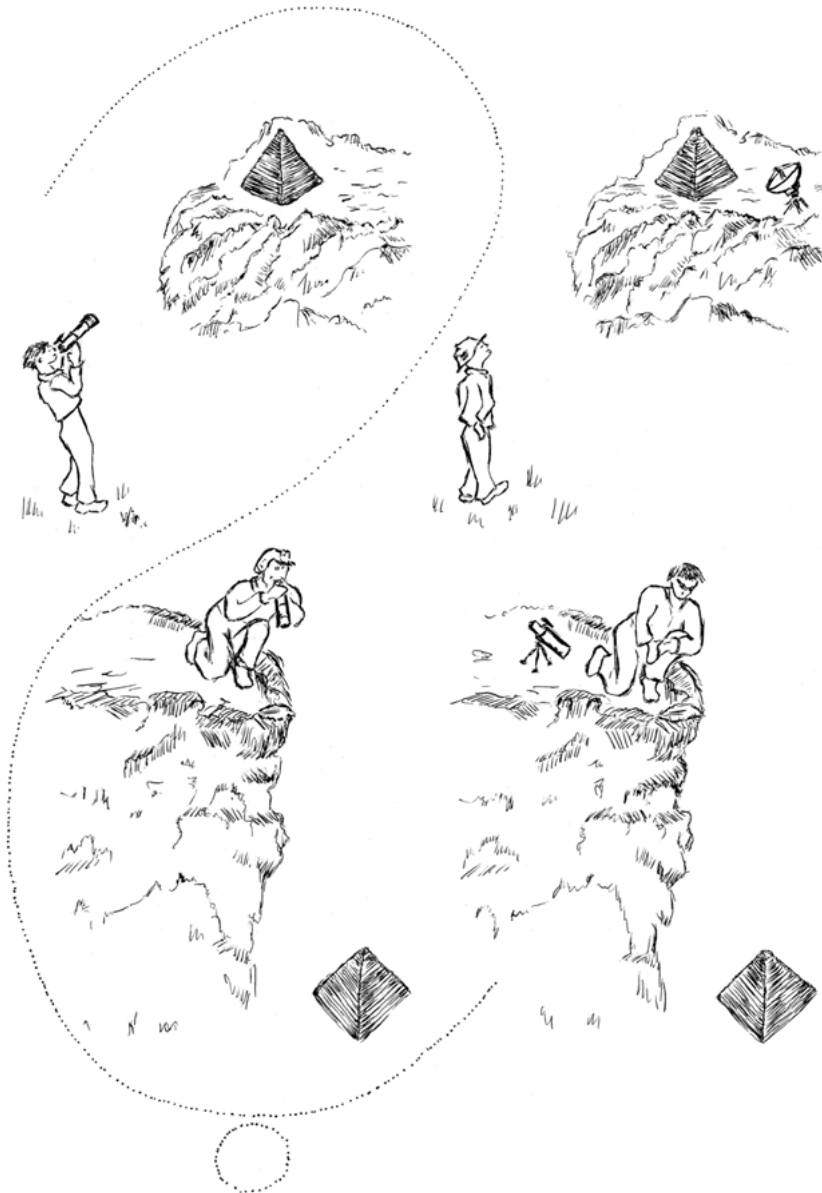


Figure B.5.1: Four Interpretations of "The man saw the pyramid on the hill with the telescope."
Source: Modified from Vickery & Vickery, 2004, 201.

The reader should realize which of the four cases is meant on the basis of the context. To exhaustively represent this via machines will probably remain out of information retrieval systems' reach for quite some time. If we process the sentence automatically, many systems will show the following results:

man—see—pyramid—hill—telescope.

For this reason, some providers of electronic information systems prefer the use of human indexers, since they will recognize the correct interpretation based on their knowledge of language and the real world.

Concept-Based IR and Content-Based IR

A fundamental decision when operating a retrieval system is whether to use a controlled vocabulary or not. Retrieval systems with terminological control are always concept-based. Terminological control is provided by KOSs such as nomenclature, classification, thesaurus or ontology. The controlled vocabulary has the advantage that both indexer and searcher dispose of the same amount of terms. Problems with homonyms (“Java” [island], “Java” [programming language]), synonyms (“PC”, “computer”) or the unbundling of compound terms (e.g., erroneously, “information science” into “information” and “science”) do not occur with terminological control. A disadvantage may be that the language of the standardized vocabulary, which after all is an artificial language with its own dictionary and grammar, might not be correctly used by everyone. Elements of uncertainty in using human indexers include their (perhaps misguided) interpretation of the text as well as possible lapses in concentration when performing their job. If one foregoes the use of human manpower for these reasons, working instead purely on the basis of content, the machine must be able to master the text, including all of its vagueness. For instance, in the theoretical case of a newspaper article headed “Super Size Mac”, reporting on new record profits posted by McDonald’s, it must recognize the correct content.

A great disadvantage of controlled vocabularies is the lack of ongoing updates in the face of continuous language development. KOSs tend to adhere to a terminological status quo, i.e. new subjects are recognized and adopted as preferred terms very late in the day. Working without terminological control, on the other hand, one will always be terminologically up to date with the current texts. Since there are dozens of classifications and far more than 1,000 different thesauri, the question arises as to the compatibility of the different vocabularies. If one wants to create transitions from one vocabulary to another, either concordances must be created or one must try to unify the different systems. In the end, a thought should be spared for the cost issue: Intellectual indexing is an expensive matter due to the labor costs for the highly qualified indexers (who 1. have expert knowledge, 2. speak foreign languages and 3. have

information science know how); automatic indexing, on the other hand, only involves programming (or buying the appropriate software) and its operation and is thus far more affordable.

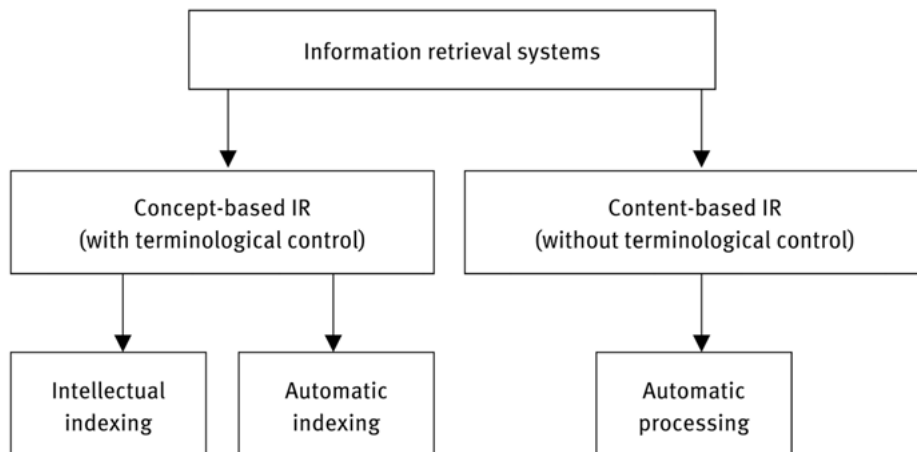


Figure B.5.2: Retrieval Systems and Terminological Control.

Automatization is an option for both the use of terminological control (as automatic indexing) and its renouncement (as the automatic processing of natural-language texts). According to Salton (1986, 656), there are no indications that human indexing is actually better than mechanical indexing.

No support is found in the literature for the claim that text-based retrieval systems are inferior to conventional systems based on intellectual human input.

Depending on where in the indexing and retrieval process terminological control is being used, we can make out four different cases (Chu, 2010, 62). The first case involves a controlled vocabulary for both indexing and retrieval; indexers and searchers must necessarily use the terms they are provided. In the second case, there is natural-language indexing and searching, with a controlled vocabulary working in the background—think, for example, of dialog steps in the case of recognized homonyms, of synonyms or during query expansion via hyponyms or hyperonyms. Thirdly, we could decide to only use a controlled vocabulary input-side; the user searches via natural language, and the system translates. In the last case, we only use the controlled vocabulary during retrieval, with the entire database content being natural-language. Such a search thesaurus—ideally used as a complement to natural-language search—allows the user to purposefully select search arguments; the system makes sure (e.g. by suggesting synonyms) that suitable documents are retrieved. The first case is used

mainly in heavily specialized professional information services; there, experts search their field or delegate to information intermediaries. Both are professions that require a good knowledge of the respective terminology. The other three cases are options for broad user groups like the ones found on the WWW. Chu (2010, 62-63) summarizes:

In consideration of the IRR (information representation and retrieval, A/N) features in the digital environment, the second approach appears more feasible than others. Both the third and fourth approaches store controlled vocabulary online for look-up or searching, which seems to be a viable option for employing controlled vocabulary.

Concept-based information retrieval requires concepts. We will get back to this subject starting in Chapter I.3.

Content-Based Text Retrieval

If one decides to use the variant of content-based search, there are two alternatives. The first consists of only providing the full text without any prior processing (this case is not mentioned in Figure B.5.2.). Such systems (e.g., “search” functions in word processing software) require the user to know the terminology of the documents intimately, as nothing precise can be retrieved otherwise. Due to the absence of relevance ranking, the hit lists must be kept small. No user would know what to do with a list of several hundred, or even thousand, search results. Except for very small data collections with only a few thousand documents, the “text only” method is practically unusable. Such a search function is found in office software; it is not a retrieval system. In the following, we will disregard such variants.

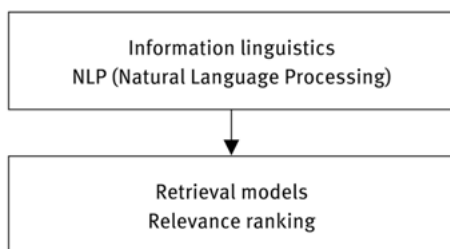


Figure B.5.3: Interplay of Information Linguistics and Retrieval Models for Relevance Ranking in Content-Based Text Retrieval.

The second variant leads to retrieval systems that process the documents. Here we distinguish between two steps (Figure B.5.3). The initial objective is to process the documents linguistically. Since in this case we are confronted with natural language, the procedure has come to be commonly called “Natural Language Processing”

(NLP); its object is the information-linguistic processing of natural-language texts. In the second step, there follows a ranking of the documents (retrieved via linguistic processing) according to relevance. For this step, several retrieval models are in competition with each other.

Information-Linguistic Text Processing

Natural Language Processing, or information linguistics, is a complex problem that will accompany us throughout many chapters (C.1 through C.5). At this point, we will make do with a brief sketch of the fundamental information-linguistic working steps (Figure B.5.4).

After recognizing the script (Arabic, Latin, Cyrillic etc.), the text's language will be identified. If we are dealing with WWW pages, the text will be partitioned into layout elements and navigation links, with all available structural information having to be identified. We are now at a crossroads, a fundamental decision influencing our further course of action. We can "parse" the text into n -grams (character sequences with n characters each) on the one hand, or into words on the other. A variant of the n -gram method applies to languages with highly irregular inflections. It initially follows the path to the basic forms, via the words, in order to then separate the n -grams.

Words are recognized by the separating characters (e.g. blanks and punctuation marks). Since stop words rarely contribute to the subject matter of a document, they are marked via a predefined list and—unless the user explicitly wishes otherwise—excluded from the search. There follows the recognition and correction (to be processed in a dialog with the user) of any input errors.

If names are addressed in the document ("Julia Roberts" – "Heinrich-Heine University Düsseldorf" – "Daimler AG"), the name will be indexed as a whole and allocated to the respective person or institution. This happens before forming the basic form, in order to preclude mistakes (such as interpreting the "s" in Julia Roberts as a plural ending and cutting it off).

A central position in information linguistics is occupied by morphological analysis. Here we can—depending on the language—either cut off suffixes and form stems (e.g. "retrieval" to "retriev") or elicit the respective basic forms (lexemes) via dictionaries or rulebooks (and then represent "retrieve"). Homonyms ("Java") are disambiguated and synonyms ("PC", "Computer") summarized. Compounds, i.e. words with several parts, such as "ice cream", must be recognized and indexed as precisely one term. In the reverse case scenario, the partition of compounds ("strawberry milkshake" into "strawberry", "milk", "shake", "milkshake", "strawberry shake", but not "straw"), the meaningful components must be entered into the inverted file in addition to the term they add up to. Depending on the language, this working step may be followed by addressing further language-specific aspects (such as the recognition, in the phrase "film and television director", of the term "film director").

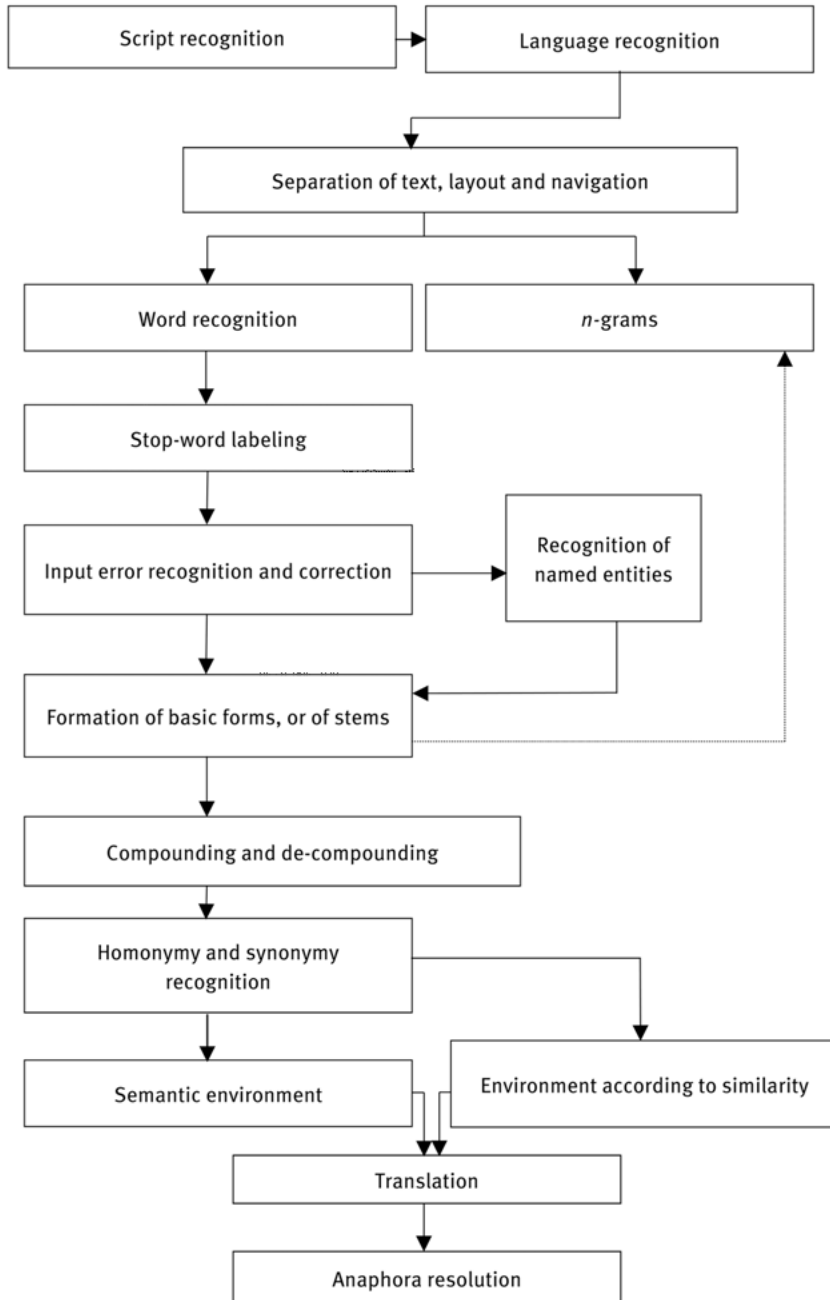


Figure B.5.4: Working Fields of Information-Linguistic Text Processing.

In the following step, we will enter the environment of already identified terms: the semantic environment on the one hand (e.g. hyponyms and hyperonyms), and the environment of statistical similarity (gleaned via analyses of the co-occurrence, in the documents, of two terms) on the other. The semantic environment can only be taken into consideration once we dispose of a knowledge organization system.

If a multilingual search is requested (e.g. search arguments in German, but results in German, English and Russian) algorithms for multilingual retrieval (e.g. automatic translation) must be available. The last working step, for now, deals with the solution of anaphora (e.g. pronouns) and ellipses (incomplete expressions). In the two sentences “Miranda Otto is an actress. She plays Eowyn.” the “she” must be correctly allocated to the name “Miranda Otto”.

In some places of our information-linguistic operating plan, it will be necessary to enter into a dialog with the user, e.g. in the cases of input errors, homonyms, perhaps in the semantic and statistical environment as well as during the translation. Without such a dialog component, an ideal retrieval would likely be impossible, unless a user always identifies himself to the system, which then records the characteristics and preferences of the specific individual. For instance, if a user has frequently asked for “Indonesia”, “Bali” or “Jakarta”, and is now asking for “Java”, he probably means the Indonesian island and not the programming language.

The information-linguistic working fields are applied to the documents in the database as well as to the queries (as quasi-documents). The goal is to find, as comprehensively and precisely as possible, those texts that correspond to the user’s intention. Following the maxim “Find what I mean, not what I say” (Feldman, 2000), we consciously chose the term “intention”: the object is not (only) which characters the user enters as his search argument—in the end, he must be presented with what he really needs.

Retrieval Models

Information science and IT know several procedures of modeling documents and their retrieval processes. Here, we will restrict ourselves to an enumeration of the most current approaches, before deepening our knowledge about them in Chapters E.1 through F.4. According to Salton (1988), information science disposed of four fundamental models prior to the days of the World Wide Web: the information-linguistic approach (NLP), which we just discussed, Boolean logic, the Vector-Space model as well as the probabilistic model. Apart from NLP and the Boolean model, all approaches fulfill the task of arranging the documents (retrieved via information-linguistic working steps) into a relevance-based ranking. The network model concentrates on the positions of agents in a (social) network and derives weighting factors from them.

With the advent of the WWW, the amount of retrieval models increases. Since the hypertext documents are interlinked, their position within the hyperspace can be determined in the context of link topology. In a final model, information about use cases as well as about characteristics of the user, e.g. his current position, are used for relevance ranking.

Boolean logic dates back to the English mathematician and logician George Boole (1815 – 1864), whose work “The Laws of Thought” (1854) forms the basis of a binary perspective on truth values (0 and 1) in connection with three functions AND, OR and NOT. The rigid corset of Boolean models always requires the use of functors; if the search argument is available in a document, it will be yielded as a search result; if it is not, the search will be unsuccessful. Since we fundamentally work with these zero-one decisions, any relevance ranking is impossible in principle. In extended Boolean models (Salton, Fox, & Wu, 1983), it is attempted to alleviate this problem via weight values. In order for this to work, the Boolean operators must be reinterpreted correspondingly.

Besides information-linguistic processing, an automatic, mechanical indexing requires an analysis of the terms that occur in the documents. According to Luhn (1957, 1958), this is performed via statistics, or in our terminology: via text statistics. Two fundamental weighting factors are a word’s frequency of occurrence in a document as well as the number of documents in a database in which the word appears. The first factor is called TF (term frequency) and relativizes the count of a word in a given text in relation to the total amount of words or to the number of the most frequent word (Croft, 1983) or it uses—as WDF (within document frequency weight)—logarithmic values (Harman, 1986). Generally, it can be said: the more often a word occurs in a text, the greater its TF or WDF will be, respectively. The second factor relativizes the weighting of a word in relation to its occurrence in the entire database. Since it is constructed to work in the reverse direction to the WDF, it is called IDF (inverse document frequency weight). The IDF is calculated as the quotient of the total number of documents in a database and the number of those documents containing the word (Spärck Jones, 2004 [1972]). The maxim here is: the more documents containing the word occur in the database, the smaller the word’s IDF will be. For every term in a document, the product $TF \cdot IDF$ will be calculated as a weight value.

There are two classic models that use text statistics: the Vector Space Model by Salton (Salton, Ed., 1971; Salton, Wong, & Yang, 1975; Salton, 1986; Salton & Buckley, 1988; Salton & McGill, 1983) and the probabilistic model, which was pioneered by Maron and Kuhns (1960) and crucially elaborated by Robertson (1977). In the Vector Space Model, both the documents and the queries are represented as vectors, where the space is generated by the respectively occurring words in the documents (including the query). If there are n different words, we are working in an n -dimensional space. The similarity between query and documents as well as that of documents to one another is determined via the respective angle of the document vectors. The smaller the angle, the higher the given text will appear in the ranking. The proba-

bilistic model asks for the probability of a given document matching a search query. In the absence of any additional information, the model is similar to the WDF-IDF approach. If additional information is available, e.g. when, following an initial search step, a user has separated those texts he deems important from those he does not, new weight values that influence relevance ranking can be created out of the words' probabilities of occurrence in the respective relevant and irrelevant documents.

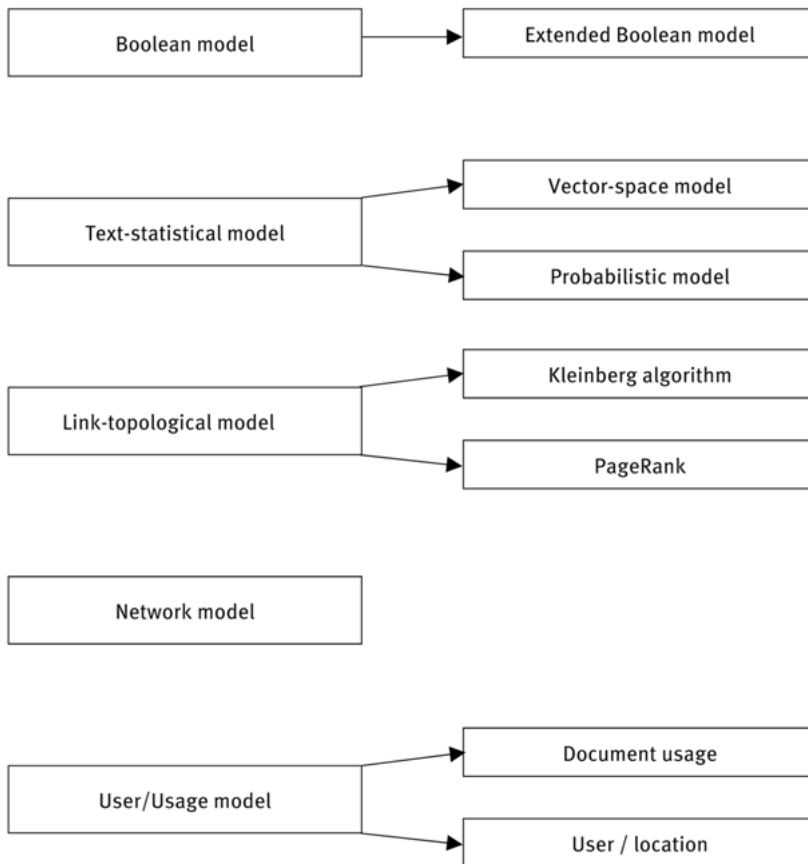


Figure B.5.5: Retrieval Models.

Documents in the World Wide Web are interlinked with one another. It is thus possible to regard the WWW as a space in which the individual documents are located. In this space, there exist documents that refer to many others, and there are documents to which many others refer. Such relations are made use of by the link-topological model. The algorithm of Kleinberg (1999) and PageRank, following Brin and Page (1998), both work in the context of this model. In the Kleinberg Algorithm, pages

are designated as “hubs” (based on their outgoing links) and “authorities” (based on their incoming links) according to their function. The weight is calculated on the basis of how frequently hubs refer to “good” authorities (e.g. those with many “good” incoming links) and vice versa.

PageRank only considers the authority value. A web document receives its PageRank via the number and the weight of its incoming links, which is calculated by adding the PageRanks of all these backlinks, each divided by the number of its outgoing links. In a theoretical model, the PageRank of a web page is the probability with which an individual surfing the Web purely by chance will locate this page.

In the Network Model, too, location is the key. Its basis lies in the theory of social networks (Wasserman & Faust, 1994) and in the “small worlds” theory (Watts & Strogatz, 1998). It can be shown that social systems—and subsystems, like scientific communities or the WWW—are not uniformly distributed, but are in fact heavily “clustered”. Within these clusters, we can detect central documents, or—in case of scientific authors, for instance—central names. The measure of centrality can be calculated and is suitable as a ranking criterion (Mutschke, 2003). The link-topological retrieval models by Kleinberg and Brin and Page can be regarded as a special case of the Network Model.

Insofar as search engines store information about the frequency of usage of given web documents (e.g. by reading protocols of users of the search engine’s tool bar), these may be suitable as a ranking criterion. Statements about the user (e.g. his current location) are useful for searches containing a geographic element (“where is the next pizza place?”). In the concrete example, a relevance ranking is the result of the calculation of the distance between the user’s location and the locations mentioned in the search results.

Let us emphasize that the named retrieval models are not mutually exclusive. It thus makes sense for practical applications to implement several models together.

So far, we have approached the functionality and the models of information retrieval only from the system side, ignoring the user. User-side, we distinguish between two fundamental retrieval approaches: searching or browsing. A user either systematically searches for documents, or he browses through document collections (Bawden, 2011). The latter is generally accomplished in the WWW by following links. Our retrieval models have nothing to do with this approach. They concentrate on the systematic search in databases with the help of specific retrieval systems.

Retrieval systems are not always intuitively easy to use, particularly by laymen. On top of this, a user may not exactly know what it is he is searching for at the beginning of his search. In this regard, it makes sense to model the retrieval process as an iterative sequence of several retrieval steps (see Figure B.5.6). In an initial feedback loop, problematic search terms—where present—are specified, e.g. by separating homonyms or summarizing synonyms. After viewing an initial list of search results, the user may be asked to inspect the first n (e.g. 20) hits and mark them, one by one, as either pertinent or not pertinent. Using this additional information, a second, rear-

ranged hit list is then created. The procedure of so-called “relevance feedback” can be repeated at will. In case the hit list is too small, the system will suggest further search entries (e.g. hyperonyms, hyponyms, related terms, all linked via OR) or the use of specific functions (e.g. truncation) to the user. In the reverse case, when a hit list is too rich, the system may suggest hyponyms for the purposes of specification or display a list of words that co-occur in documents with the original search argument.

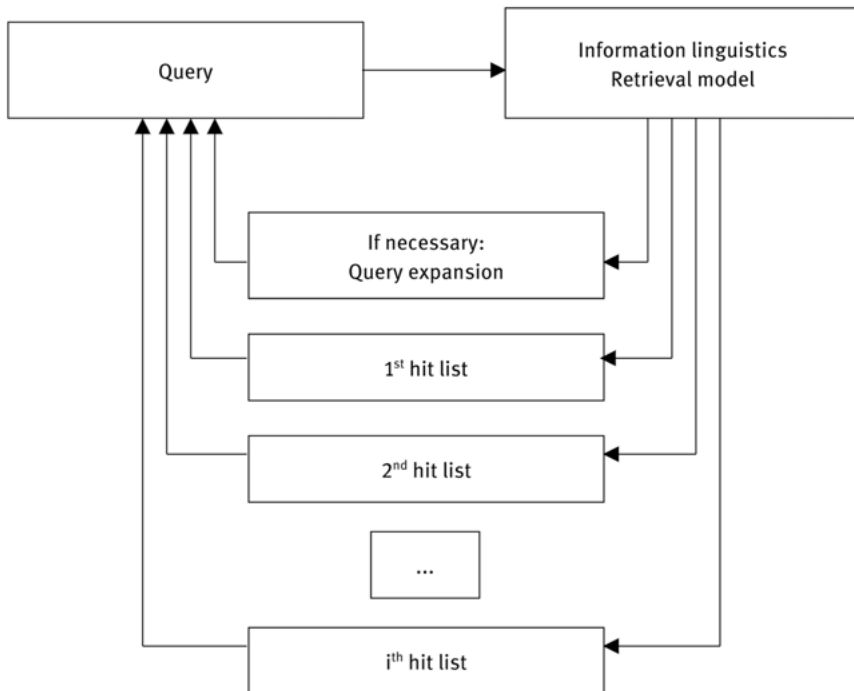


Figure B.5.6: Retrieval Dialog.

Since misgivings are often expressed to the effect that laymen in particular will not “accept” this (Lewandowski, 2005, 152), we must entertain the idea of automatizing the feedback loops as far as possible and only incorporating the user into a dialog when it is absolutely necessary.

Surface Web and Deep Web

Digital documents are either stored in a company-owned Intranet or “on the Internet”. This “on” is imprecise, however, since we must clearly distinguish two separate cases:

- the Surface Web: digital documents that are *on* the Web (i.e. that are firmly linked, resulting in no additional cost to the user),
- the Deep Web: digital documents stored in information collections (generally databases), the entry pages of which are reachable *via* the World Wide Web.

The majority of all information on the Web is evaluated by search tools—at least in principle. Information in specific data formats is not always recorded; documents without any incoming links cannot be directly indexed; problems may arise from the usage of frames. Undesirable information includes spam pages, which are Web documents with misleading descriptions or false statements in their metatags. Search tools take care not to include spam in their databases. We distinguish three types of search tools:

- search engines working on the basis of algorithms (with their own database and retrieval system) such as Google,
- intellectually compiled Web catalogs (hardly in use anymore) like the Open Directory Project or Yahoo!’s catalog,
- meta search engines, which have a retrieval system but no database of their own, collecting their content from third-party search tools instead.

Search engines use a single web page as the documentary reference unit, whereas Web catalogs generally only consider the home page of an entire website. Correspondingly, the extent of search engines’ databases is much greater than that of Web catalogs.

The second main class of digital online information is not stored on the Web itself but is only reachable via the Web. Of course, the home pages of such systems can also be found in search engines and Web catalogs, but the databases they store cannot. According to Sherman and Price (2001, 1-2), we are here confronted with the “Invisible Web”:

In a nutshell, the Invisible Web consists of material that general-purpose search engines either can not, or perhaps more importantly, *will not* include in their collections of Web pages. The Invisible Web contains vast amounts of authoritative and current information that’s accessible to you, using your Web browser or add-on utility software—but you have to know where to find it ahead of time, since you simply cannot locate it using a search engine.

Bergman (2001) describes this same aspect via the metaphor of the “Deep Web”. Since it is reachable via the WWW, it cannot be invisible, as Sherman and Price suggest. It simply requires different access paths than the Surface Web:

Searching on the Internet today can be compared to dragging a net across the surface of the ocean. While a great deal may be caught in the net, there is still a wealth of information that is deep, and therefore, missed. The reason is simple: Most of the Web’s information is buried far down on dynamically generated sites, and standard search engines never find it.

This “deep” information reachable via the Web is divided into two subclasses, separated by whether the offers are free of charge or not. There are singular free databases (such as the databases of national patent offices or library catalogs) and there are

commercial offers. The latter are the home of self-marketers, who are just as singular as the free databases mentioned above, except that they charge for their services. Another species at home here are content aggregators, which unify different information collections under one surface. Content aggregations may appear for publishing products (examples: Springer Link or ScienceDirect as full-text offers of journal articles by the two scientific publishers Springer and Reed Elsevier, respectively) or in the form of online hosts. The latter must be divided into three categories, according to technical criteria which require their own specific retrieval options. The categories are: legal information (e.g. LexisNexis and Westlaw), economic and press information, respectively (e.g. Factiva) and STM (science/technology/medicine) information (e.g. Web of Knowledge and Scopus).

Hybrid systems attempt to unify the two “worlds” of the WWW—its surface and its depths—within a single system, thus striving towards a kind of “cross-world retrieval” (Nde Matulová, 2009; Stock & Stock, 2004).

If we separate the WWW worlds according to the commercial aspect of the offers, we will have the search tools and free singular services on the one side and the commercial information providers on the other. Both worlds “coexist peacefully” and find different target groups: users who search in the free world are concerned with easily manageable queries in which personnel costs are negligible (i.e. in which no opportunity costs arise), whereas commercial services seek users with higher demands on retrieval functionality and content, and whose search times add up to a major revenue factor. Stewart (2004, 25) notes, from a business point of view:

Web searching is a simple method for conducting a simple search, and it is applied thousands of times each minute around the globe. But this is not a zero-sum game. Free web searching and fee-based searching can and will always co-exist.

But when it comes to complex searches on complex issues, with increased accuracy in search results, information aggregators have proven that their products provide a service that delivers real savings in time and money.

Conclusion

-
- Retrieval systems deal with weakly structured texts (apart from multimedia documents). Their structure consists mainly of formal characteristics (such as title, sections, and references) and, to a lesser degree, of syntactical structures discussed by linguistics.
 - Retrieval systems may be operated either with terminological control (“concept-based”) or without (“content-based”). Terminological control is achieved via the use of knowledge organization systems (such as nomenclature, Classification, thesaurus or ontology) and aligns the indexer’s language with that of the searcher.
 - Concept-based retrieval is developed either by human indexers or via an automated indexing process. Content-based retrieval systems always require automatic document processing, which in the case of text documents means, appropriately, text processing.

- Automatic text processing occurs in two subsequent steps: (1.) information linguistics (Natural Language Processing) and (2.) ranking the retrieved texts by relevance.
 - The basis of information-linguistic works is the recognition of script and language as well as the separation of text, layout and navigation (while simultaneously recording the weak structures in the text). Here the approaches divide into the variant of word recognition and the variant of *n*-grams (sequences of *n* characters).
 - Word recognition requires the labeling of stop words that do not carry meaning, input error correction, the recognition of personal names and a morphological analysis of flectional words that includes returning them to their stems, or to their lexemes.
 - On the level of concepts, homonyms and synonyms must be recognized, compounds formed and divided into their component parts, respectively, and the semantic environment of the term (such as hyponyms and hyperonyms) must be taken into consideration. If the objective is multilingual retrieval, foreign-language search arguments must be located. Finally, anaphora and ellipses must be solved.
 - Apart from the Boolean retrieval model, all models rank texts according to relevance (relevance ranking). The classical retrieval models take text statistics as their starting point, following an adage by Luhn. Important weight values are WDF (within document frequency weight) and IDF (inverse document frequency weight). Variants of the text-statistical model are the Vector Space Model and the probabilistic model.
 - The World Wide Web with its hypertexts and interlinking makes the link-topological model possible. Its variants (Kleinberg Algorithm and PageRank) provide the basis of ranking algorithms in Web search engines.
 - The link-topological model can be universalized into a general Network Model. Measures of centrality here serve as weight values. Information can be gleaned by using texts as well as the user as further ranking criteria.
 - The retrieval models are not mutually exclusive, but can be used together in an information retrieval system.
 - In order to optimize the alignment between a user's search argument and the stored documents, it can be advantageous to set up the retrieval process as a dialog. In this way, queries are specified and enhanced where needed, and further information (e.g. about texts already recognized as relevant by the user) can be transmitted to the system.
 - The "world" of digital documents on the Internet is divided: apart from the "Surface Web" recorded by search tools, there is the "Deep Web". Its databases can be reached via the WWW, but must be searched individually requiring further steps. Retrieval systems of the Surface Web are algorithmic search engines, Web catalogs and meta search engines. In the Deep Web, there are both subject-specific singular information collections and summarizing content aggregations. Many of the Deep Web offers are products offered by for-profit enterprises and thus charge a fee for their usage.
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B.6 Architecture of Retrieval Systems

Building Blocks of Retrieval Systems

Every system that is used to search and find information requires the storage of documents and of the knowledge contained within them. Storage media are either non-digital or digital. Non-digital stores have existed ever since information has been structured and made retrievable. They include index cards in alphabetical or systematic library catalogs, for example. In the following, we will be interested in digital stores exclusively.

Digital stores are within the purview of information technology. Looking at digital retrieval systems thus puts us in the intersection between computer and information science, called CIS. Since this is not a handbook on informatics, we will merely sketch a few important aspects of this area (a very good introduction into computer-science retrieval research is provided by Baeza-Yates & Ribeiro-Neto, 2011).

The building blocks at our disposal for the architecture of digital retrieval systems can be seen in a rough overview in Figure B.6.1. As the basic elements, we postulate an operating system and a database management system. In the retrieval system, we differentiate by file structure (document file and inverted file) and function (functionality of information linguistics and relevance ranking). Tools are used when structuring the databases via a well-defined field schema and—if provided—for the controlled vocabulary. Every retrieval system performs two tasks: in the indexing component (top left-hand corner of Figure B.6.1), the input is processed into the system; in the search system (top right-hand corner) the same goes for the output.

The documents to be recorded by the retrieval system (Ch. A.4), including the knowledge they contain, are retrieved either via the intellectual choice of a (human) indexer or by an automatic crawler (Ch. B.4). Crawlers locate new documents, avoid spam and check the documents for changes (update). The indexing component processes the documents. Here, the documentary reference units are transformed into documentary units. Surrogates are created depending on the tools being used (field schemas or knowledge organization systems). For documents without any documentary processing (such as web pages to be indexed by a search engine), certain further processing steps must be taken: character sets, scripts and the language being used are automatically recognized and noted. If the digital stores record bibliographic databases that include intellectual analysis of the documentary reference units (DRUs), these working steps are skipped since the previous indexers have already noted such information (e.g. in a field for the language). After this, the documentary units are saved, first in a (sequentially updated) document file and then in an inverted file that facilitates access to the document's contents by serving as an index.

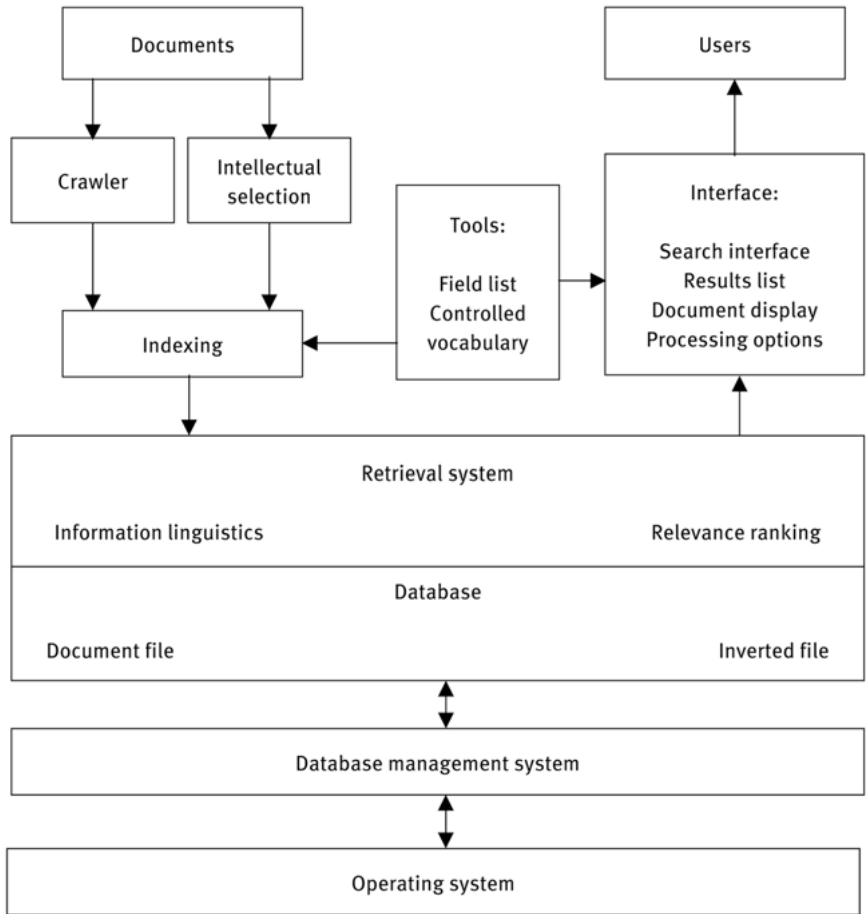


Figure B.6.1: Building Blocks of a Retrieval System.

The user interface is where the dialog with the user takes place. Here, four tasks must be performed: the user formulates his query in the search interface, the documentary units are presented to him in a hit list, the documentary units (as long as they do not coincide with the documentary reference unit, as is the case in search engines) are displayed and can then be locally processed further by the user. If field schemas or KOSs have been used during indexing, they will be offered to the user as search tools.

According to Croft, Metzler and Strohman (2010, 13), retrieval systems have two primary goals, quality and speed:

Effectiveness (quality): We want to be able to retrieve the most relevant set of documents possible for a query.

Efficiency (speed): We want to process queries from users as quickly as possible.

Character Sets

Since computer systems principally use only the digits 0 and 1 but natural languages have far richer sets of characters, standards are required to represent the characters of natural languages binarily. An early code that initially gained prevalence in the U.S.A. and later worldwide is called the “American Standard Code for Information Interchange” (ASCII). It was published in 1963 by the American Standards Organization and was subsequently adopted by the International Organization for Standardization (ISO) (ISO/IEC 646:1991). It is a 7-bit code, i.e. every natural-language character is represented via 7 binary digits. 7 bit can represent $2^7 = 128$ characters. Since many computer systems use 1 byte (8 bit) as the smallest unit, the 7-bit ASCII characters always lead with a zero. ASCII characters represent control commands (e.g. carriage return via 0001101), punctuation marks (e.g. ! via 0100001), digits (e.g. 1 via 0110001), capital letters (e.g. G via 1000111) and lower-case letters (e.g. t via 1110100). The ASCII 7-bit variant of a German greeting (“Guess Gott!”) is shown in Figure B.6.2.

```
10001111110010111101011100101111001111100110100000100011110111111010011101000100001
```

Figure B.6.2: Short German Sentence in the ASCII 7-Bit Code.

The addition of one bit in the 8-bit code enhances the 128 character restriction to 256. The standard ISO/IEC 8859 regulates the coding of the respective 128 new characters in various language-specific variants (ISO/IEC 8859:1987). There is, among others, a standard for Latin languages (ISO/IEC 8859-1) which by now also contains the German umlauts as well as the “ß” character (11011111). The tried-and-true ASCII 7-bit code survives (the “old” characters now lead with a zero). Figure B.6.3 presents the new coding of the sentence from Figure B.6.2 in the 8-bit code of ISO/IEC 8859-1 (“Grüß Gott!”). Language-specific variants exist for all European languages as well as for Arabic, Hebrew or Thai.

```
01000111011100101111110011011111001000000100011101101111011101000111010000100001
```

Figure B.6.3: The Sentence from Figure B.6.2 in the ISO 8859-1 Code.

The most comprehensive character set so far, which works with up to 4 Bytes (i.e. 32 bit), aims to incorporate all characters that are used around the world. Old build-up work is kept alive and incorporated into the new system. Unicode, or UCS (Universal Multiple-Octet Coded Character Set) (ISO/IEC 10646:2003), allows texts to be exchanged without any problems worldwide.

Storage and Indexing in Search Engines

Pages retrieved by the crawler (for the first time or repeatedly) are copied into the database of the search engine. Every documentary unit receives its own clear address in the database and is “parsed” into its smallest units. These units are the atomic pairs of word and location. In an AltaVista patent, we read (Burrows, 1996, 4):

The parsing module breaks down the portions of information of the pages into fundamental indexable elements or atomic pairs. ... (E)ach pair comprises a word and its location. The word is a literal representation of the parsed portion of information, the location is a numeric value.

The indexing module arranges the atomic pairs primarily by word, secondarily by location, with every word being admitted only once per documentary unit (but with varying numbers of locations). This index is accessible from the query. The stored addresses allow the results to be identified and yielded to the user.

Additional information (such as the size of the printing font) can be stored with the atomic pair. Likewise, field-specific information (e.g. from the TITLE tag in HTML documents) can be separated, allowing for access to words from precisely one field.

Storage and Indexing in Specialist Information Services

Specialist information databases in the Deep Web contain documentary units that are always structured in the same way. The structure is predetermined via a mandatory field schema, as Table B.6.1 shows on the example of a database for economic literature. These data describe document data; they are “metadata”.

The documentary reference units’ information is mapped onto the documentary unit field by field. This is accomplished purely intellectually in many (and particularly in high-quality) databases, but in favorable circumstances it can also be automated, e.g. for newspaper articles that are sent from a content management system directly to the retrieval system, where they are automatically indexed. Depending on subject and purpose of the database, a field is the smallest unit for admitting pieces of information of the same kind. In an initial rough breakdown, the fields are divided into (internal) management fields, formal bibliographic fields and content-depicting fields. Management fields include the number of the record as well as its identifier (date of creation, date of correction, name of reviser). In our example in Table B.6.1, taken from the ifo Literature Database, content is indexed via several methods that complement each other: a thesaurus (the Standard Thesaurus Wirtschaft for economic terms), further terms (worked out via the text-word method), a (pretty rough) subject area classification as well as a geographic classification. An abstract is written for every documentary reference unit. The content-depicting fields are DE, DN, IW, KL, KC, LC and AB. The task of the formal bibliographic fields is to describe a docu-

ment clearly. This includes statements on author, title, year etc. In the table, we see far more input fields than display fields in the formal bibliographic fields. All bibliographic data for a source (journal title, volume, page etc.) is entered analytically, but is summarized in a single field called “source” (SO) in the display. Depending on the output format (online database or printed bibliography), different variants of document displays can be created in this way.

Our example of a field schema is representative for many formally published scientific documents. Patents require more fields (i.e. on the legal status of the invention), fiction less.

Table B.6.1: Field Schema for Document Indexing and Display in the ifo Literature Database. Source: Stock, 1991, 313.

Input and Search Field	Field Name in the Display
Identifier	
Document Type	PU
Author	AU
Editor	
Author's Affiliation	
Article Title	TI
Journal	
Book Title	
Title of Book Series	
N° in Series	
Volume / Number	
Year	YR
Publishing Date	
Publishing Place	CI
Publisher	CO
ISSN	IS
ISBN	IB
Pages	
Language	LA
Controlled Terms	DE
Thesaurus ID N°s	DN
Further Subjects (Text-Word Method)	IW
Subject Classification	KL
Classification Code	KC
Country Code	LC
Abstract	AB
Call Number	SI
End Criterion	
Source	SO (is generated)

The field schema in Table B.6.1 shows rather vividly what we do *not* have in Web search engines, namely an analytic description of different types of information about the

document and from the document. Problems faced by Web search engines—e.g. not being able to tell whether the retrieval of a personal name means that the text is *by* that person or *about* them—cannot occur when using a field schema, since author, in the AU field, and person or persons thematized, in IW, are clearly separated from each other.

Document File and Inverted File

A practicable set-up of databases takes two files into consideration:

- the document file,
- the inverted file.

The document file is the central repository and records the documentary units, which are either intellectually entered or gleaned via automatic processing, in their entirety. (In this instance, we disregard the fact that search engines do not always copy the documents completely. In the case of very long documents, they only consider the first *n* Megabytes; where *n* might be around 100.) In order to minimize the sorting effort during output, Deep-Web databases deposit their documents according to the principle of FILO (“first in—last out”). This guarantees that the documentary units admitted last are ranked at the top of hit lists, thus creating an up-to-date ranking “by itself”.

The documentary units are stored one after the other in a retrieval system, via the allocation of an address. Where large amounts of data are available, it would be extremely time-intensive to search all databases sequentially, search argument by search argument. This is why in such cases a second additional file is created, called the inverted file (Harman, Baeza-Yates, Fox, & Lee, 1992).

An inverted file admits all words or phrases for a second time, only this time they are sorted alphabetically (or according to another sorting criterion). Zobel and Moffat (2006, 8) emphasize:

Studies of retrieval effectiveness show that all terms should be indexed, even numbers. In particular, experience with Web collections show that any visible component of a page might reasonable be used as a query term, including elements such as the tokens in the URL. Even stop-words ... have an important role in phrase queries.

Baeza-Yates and Ribeiro-Neto (2011, 340) emphasize the practical usefulness of inverted files:

An inverted file (or inverted index) is a word-oriented mechanism for indexing a text collection to speed up the searching task. The inverted file structure is composed of two elements: the *vocabulary* (also called lexicon or dictionary) and the *occurrences*. The vocabulary is the set of all different words in the text.

There are inverted files for all words, no matter where in the text they occur, and in addition, there are inverted files for certain fields (where available). Thus, the bibliographic database described above includes inverted files for author, title, year etc. Some providers of electronic information services offer a “basic index”, which generally consists of the entries for all content-related fields (such as title, keywords, abstract, continuous text).

Company

Document	2, 23, 45, 56
# Doc.	4
# Occurrence	7
Location (Character)	(2: 12) (2: 123) (23: 435) (45: 65) (45: 51) (45: 250) (56: 1032)
Word	(2: 4) (2: 28) (23: 99) (45: 13) (45: 17) (45: 55) (56: 432)
Sentence	(2: 1) (2: 3) (23: 15) (45: 1) (45: 1) (45: 2) (56: 58)
Paragraph	(2: 1) (2: 1) (23: 1) (45: 1) (45: 1) (45: 2) (56: 4)
Font	(2.4: 28) (2.28: 10) (23.99: 12) (45.13: 72) (45.17: 12) (45.55: 12) (56:432: 20)
Structure	(2.4: body/bold) (2.28: anchor) (23.99: body) (45.13: H1) (45.17: body) (45.55: body/italics) (56:432: H2)

ynapmoC

Document	2, 23, 45, 56
# Doc.	4
# Occurrence	7
Location (Character)	(2: 12) (2: 123) (23: 435) (45: 65) (45: 51) (45: 250) (56: 1032)
Word	(2: 4) (2: 28) (23: 99) (45: 13) (45: 17) (45: 55) (56: 432)
Sentence	(2: 1) (2: 3) (23: 15) (45: 1) (45: 1) (45: 2) (56: 58)
Paragraph	(2: 1) (2: 1) (23: 1) (45: 1) (45: 1) (45: 2) (56: 4)
Font	(2.4: 28) (2.28: 10) (23.99: 12) (45.13: 72) (45.17: 12) (45.55: 12) (56:432: 20)
Structure	(2.4: body/bold) (2.28: anchor) (23.99: body) (45.13: H1) (45.17: body) (45.55: body/italics) (56:432: H2)

Figure B.6.4: Inverted File for Texts in the Body of Websites.

For index entries, we distinguish between a word index and a phrase index. The word index records each word individually, whereas the phrase index stores entire related word sequences as a single entry. The decision as to which of the two forms are selected (or if both should be selected together) depends on the respective field contents. An author field can be meaningfully assigned a phrase index. Thus, the user can search for “Marx, Karl” instead of having to search—as in the word index—for “Marx AND Karl”. A title field, on the other hand, must be assigned a word index, since otherwise one would always have to know the exact and full title in order to search for it. When keywords are used that also occur in the form of multi-word terms (such as “Kidney Calculi”), it appears practicable to use both index forms in parallel. Someone who is familiar with the keywords can search directly for “Kidney Calculi”

via the phrase index, and a layperson may approach the controlled keywords incrementally, via “Kidney” or “Calculi” separately.

A comprehensive inverted file (as in Figure B.6.4) contains the following statements on every vocabulary entry: documents (or their addresses) containing the word or the phrase, position in the document (location of the entry’s first character, word number, sentence number, paragraph number, chain number in the keyword index) as well as structural information such as the printer font or further structural information in the text (such as H1 or H2 for headings respectively subheadings) or the keywords’ weight value. The positional information (such as 2:4 for the fourth word in the document with the address 2) form the basis of a retrieval that takes into consideration word proximity, and the structural information can be used in relevance ranking (i.e. to give an H1 word a higher weight than an H2 word). Additionally, the file can record frequency data for the entries (In how many documents does the word occur at least once? How often does the word occur in the entire database?). A very useful idea is the twofold admission of an entry (Morrison, 1968, 531): once with its normal string of letters (“company”) and once backwards (“ynapmoc”), since this will give you the option of left truncation (e.g. allowing you to search for “*company” in order to retrieve the word “BigCompany”, for example). Left truncation is now nothing other than (easily applicable) right truncation for the inverted file which is now arranged in the opposite direction.

Conclusion

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- Retrieval systems build on operating systems and database management systems. They contain the building blocks for input (intellectual document selection respectively crawling as well as indexing) and output (user interface for searching and finding information). Retrieval systems work with two files (document file and inverted file); they dispose of an information-linguistic functionality and a functionality for relevance ranking. If a retrieval system employs tools (such as field schemas or a KOS), they will be used both in indexing and in retrieval.
 - Since computer systems exclusively use the digits 0 and 1, all other characters must be coded via 0-1 sequences. The standards used are ASCII (7-bit code), ISO 8859 (8-bit code) and Unicode respectively UCS (code with up to 4 Bytes).
 - In search engines on the WWW, the pages retrieved by the crawler are copied into the retrieval system, where they are parsed. The smallest units are the atomic pairs consisting of word and location. The entire records are entered into a sequentially arranged document file, the atomic pairs (including additional information) into the inverted file.
 - Specialist information services in the Deep Web work with elaborate field schemas. These admit administrative information, formal-bibliographic statements and content-describing metadata.
 - Inverted files are—depending on the field—either word or phrase indices. In order to easily implement left truncation, index entries are entered both in their normal string of letters and in the reverse.
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Part C
Natural Language Processing

C.1 *n*-Grams

Words or Character Sequences?

For the following chapters (C.1 through C.5), we will assume that a retrieval system is meant to linguistically process natural-language text documents. In order to automatically index documents, it is necessary to parse them into the atomic pairs of word and location. A “word” can be a natural-language word recognized via blanks and punctuation marks. However, it can also be a formal word that brings character sequences in texts to the length n . The latter are n -grams, in which n stands for a number between 1 and—in general—6.

The parsing of texts into n -grams goes back to Shannon’s work on communication theory from the year 1948 (Shannon, 2001[1948]). Shannon introduced n -grams as “formal words” with a length of n .

1-grams (or monograms) are simply those characters that can be analyzed via their frequency of occurrence. 2-grams (bigrams) divide the original term into two characters, 3-grams (trigrams) into three, 4-grams (tetragrams) into four, 5-grams (pentagrams) into five characters.

n -grams refer, in a simple variant (1), to words that have been pre-identified (via blanks or punctuation marks) and parse them into character sequences of n . Of more use is a variant (2), which takes into consideration the blank spaces themselves. Here, the first and last letters of the words have the same probability of occurring in the n -gram as the letters in the middle have. Correspondingly, blanks are used to pad the beginning and end of the words so that each n -gram is always n characters in size. In an elaborate version (3), the n -grams run across the entire text, i.e. beyond word borders. It is useful to make a cut at sentence borders.

In the example

INFORMATION RETRIEVAL

there are eleven consecutive characters for INFORMATION and nine for RETRIEVAL. In variant (2), the following twelve respectively ten bigrams arise (let the star * stand for a blank space):

I, IN, NF, FO, OR, RM, MA, AT, TI, IO, ON, N,
R, RE, ET, TR, RI, IE, EV, VA, AL, L.

The example leads to 13 resp. 11 trigrams:

I, *IN, INF, NFO, FOR, ORM, RMA, MAT, ATI, TIO, ION, ON*, N,
R, *RE, RET, ETR, TRI, RIE, IEV, EVA, VAL, AL*, L,

to 14 resp. 12 tetragrams:

I, **IN, *INF, INFO, NFOR, FORM, ORMA, RMAT, MATI, ATIO, TION, ION*, ON**, N
 R, **RE, *RET, RETR, ETRI, TRIE, RIEV, IEVA, EVAL, VAL*, AL**, L

and to 15 resp. 13 pentagrams:

****I, ***IN, **INF, *INFO, INFOR, NFORM, FORMA, ORMAT, RMATI, MATIO, ATION, TION*,
 ION**, ON***, N****
 ****R, ***RE, **RET, *RETR, RETRI, ETRIE, TRIEV, RIEVA, IEVAL, EVAL*, VAL**, AL***, L****.

A term with k characters leads, in variant (2), to $k+1$ bigrams, $k+2$ trigrams, $k+3$ tetragrams as well as $k+4$ pentagrams. INFORMATION has eleven characters ($k=11$); the n -gram parsing thus leads to twelve ($11+1$) bigrams, thirteen ($11+2$) trigrams etc.

In variant (3), the n -grams glide through the whole text, as it were. In the form of gliding tetragrams, our example will be parsed into the following character strings:

***I, **IN, *INF, INFO, NFOR, FORM, ORMA, RMAT, MATI, ATIO, TION, ION*,
 ON*R, N*RE,
 RET, RETR, ETRI, TRIE, RIEV, IEVA, EVAL, VAL, AL**, L***.

The advantage of this variant (3) is obvious. If someone is looking for the phrase “Information Retrieval”, the additionally gleaned tetragrams ON*R and N*RE will provide important indicators for the desired phrase.

In a variant of the n -gram method, single neighboring letters are skipped. The resulting fragments, according to Anni Järvelin, Antti Järvelin and Kalervo Järvelin (2007), form so-called “ s -grams”. In practice, either one or two characters are skipped. Using bigrams and skipping one character, our example (via variant (1)) will look like this:

IF, NO, FR, OM, RA, MT, AI, TO, IN,
 RT, ER, TI, RE, IV, EA, VL.

Taking s -grams into consideration alongside n -grams may lead to increased retrieval effectiveness.

In contrast to retrieval systems that store natural-language words, when using n -grams we know the exact maximum number of possible terms. On the basis of 27 characters (26 letters and the blank space), the upper limit of index entries is 27^n terms, i.e.

729 terms for 2-grams,
 19,683 terms for 3-grams,
 531,441 terms for 4-grams and
 14,348,907 terms for 5-grams.

However, not all possible *n*-grams are actually active. For the English language, it is shown that around 64% of bigrams and only 16% of trigrams are in fact being used (Robertson & Willett, 1998, 49). This state of affairs is exploited in fault-tolerant retrieval, since the use of *n*-grams allows us to recognize and perhaps fix input errors (see Chapter C.5).

When choosing the road of natural-language words in information retrieval, the next milestones (Phase 1) will be word processing (incl. stop words, reduction to basic forms/lexemes, phrase formation and decompounding). After that, in Phase 2, there follow further information-linguistic procedures such as the processing of homonyms and synonyms, recognition of the semantic environment or solution of anaphora. Depending on the retrieval model, this is followed by a method for ranking the documents by relevance. If we take *n*-grams as our starting point in information retrieval, all word-processing milestones (at least in the majority of all cases) disappear (Phase 1) and relevance ranking directly follows the recognition of *n*-grams in query and documents. This sounds enticingly simple, since it represents an “abbreviation” of the working steps. It should be noted, however, that *n*-grams, for all that they may replace the first information-linguistic phase, will encounter great difficulties in the second. Semantic aspects such as hyponyms and hyperonyms elude the *n*-gram treatment.

n-grams do not presuppose any prior language recognition—to the contrary, it is via *n*-grams that we can identify languages. This means that *n*-grams work independently of languages. It would be a false conclusion, however, to suppose that they are equally applicable to all natural languages. The use of *n*-grams is very useful for languages that work without word separators (such as blank spaces)—these include Chinese (Nie, Gao, Zhang, & Zhou, 2000), Korean (Lee, Cho, & Park, 1999) and Japanese (Ando & Lee, 2003).

In languages that demarcate words via separators, *n*-grams also yield positive results (Robertson & Willett, 1998, 61). Even in “languages” that only have four letters (A, T, G, C for adenine, thymine, guanine and cytosine in DNA), *n*-grams can be successfully used to decrypt the genetic information (Volkovich et al., 2005). Languages in which the words tend to dramatically change form when inflected are thus relatively unsuited for *n*-gram-based retrieval. Problems arise, for instance, via umlauts (in German: Fuchs—Füchsin), vowel gradation (to sing—song) or circumfixation (to moan—bemoaned). The number of identical *n*-grams becomes too small in such cases to allow for recognition of the same word. Due to the highly pronounced infix structure during inflection, *n*-grams can only be used in Arabic, for example, after the words have been morphologically processed (via their basic forms/lexemes) (Mustafa, 2004).

Henrichs's Pentagram Index

An early use of the n -gram method has been documented in the 1970s. During a time when right truncation was not yet a matter of course and left truncation was entirely unsupported, n -grams provided the option of identifying meaningful terms even within longer words, and to make them retrievable. The procedure was introduced by Henrichs (1975) and uses pentagrams. We will illustrate Henrichs's parsing into 5-grams on the example of the German word WIDERSPRUCHSFREIHEITSBeweis (proof of consistency):

WIDER, IDERS, DERSP, ERSPR, RSPRU, **SPRUC**, PRUCH, RUCHS, UCHSF, CHSFR, HSFRE, SFREI, **FREIH**, **REIHE**, EIHEI, IHEIT, HEITS, EITSB, ITSBE, TSBEW, SBewe, **BEWEI**, EWEIS.

All meaningful pentagrams are entered into the alphabetical search list via the relation to the whole term. A pentagram is “meaningful” when another word that occurs in the database begins with the same sequence of characters. In this way, most of the candidates for additional search entries, such as TSBEW or CHSFR are purged, while syntactically useful new index entries such as WIDER, SPRUC, FREIH, REIHE and BEWEI survive. The user may now find, under B, next to *Beweis* (proof), the potentially interesting search term *Widerspruchsfreiheitsbeweis*. The procedure knows no semantic control, so that our exemplary term will also be listed under *Reihe* (chain) (which—even though *Reihe* is only found in the longer word by coincidence, and not by logic—is not that big of a problem, since the user will correctly identify the semantic trap and not use the term for his search).

If one uses n -grams not only as tools for users to formulate good search arguments (like Henrichs), but as a method of automatic information retrieval, models must be found that express similarities between queries and documents and facilitate search and relevance ranking. We will paradigmatically introduce two models, one of which works on the theoretical basis of the Vector Space Model (ACQUAINTANCE), the other on the basis of the probabilistic model (HAIRCUT). Note that neither model allows the use of Boolean operators.

ACQUAINTANCE: n -Grams in the Vector Space

The retrieval system ACQUAINTANCE was developed in the 1990s by Damashek in the United States Defense Department. Damashek (1995, 843) reports, in an article for *Science*:

I report here on a simple, effective means of gauging similarity of language and content among text-based documents. The technique, known as Acquaintance, is straightforward; a workable software system can be implemented in a few days' time. It yields a similarity measure that makes sorting, clustering, and retrieving feasible in a large multilingual collection of documents

that span an unrestricted range of topics. It makes no use of words per se to achieve its goals, nor does it require prior information about document content or language. It has been put to practical use in a demanding government environment over a period of several years, where it has demonstrated the ability to deal with error-laden multilingual texts.

ACQUAINTANCE performs three tasks: (1.) automatic language identification, (2.) the arrangement of similar documents into classes and (3.) information retrieval. In the third case, which is of principal interest to us here, we must distinguish between two variants: search via (a rather short) query or search via (a rather long) model document. The “classical” Vector Space Model (Ch. E.2) is enhanced by the idea of a mean document, the centroid.

ACQUAINTANCE uses n -grams (frequently: 5-grams) that glide throughout the entire text. The enumerated terms are the different n -grams as well as the total number of n -grams in a document. Every n -gram is allocated a weight value which is calculated as the relative frequency of the n -gram in the text (number of the given n -gram in the text divided by number of all n -grams in the text). In the Vector Space, the different n -grams correspond to the dimensions, the documents being represented by vectors. There are as many different dimensions in a database as there are different n -grams. Since ACQUAINTANCE does not employ any stop word lists, one goal must be not to use any general, less meaningful terms for indexing. Of central importance is the introduction of the centroid vector (Damashek, 1995, 845; Huffman & Damashek, 1995, 306; Huffman, 1996, 2). The centroid of a database is the “mean vector”. It refers to all dimensions; the respective weight value in the dimensions is the arithmetic mean of the weight values of the document vectors. Huffman and Damashek (1995, 306) write:

The creation of the centroid vector for a set of documents is straightforward and language independent. After each separate document vector is created, the normalized frequency for each n -gram in that document is added to the corresponding address in a centroid vector. When all documents have been processed, the centroid vector is normalized by dividing the contents of each vector address by the number of documents that the centroid characterizes. A centroid thus represents the “center of mass” of all the document vectors in the set.

The centroid characterizes features that are more or less shared by the documents in a document collection. Terms (i.e. n -grams) that occur less frequently have an extremely low weight in the centroid due to the mean value, whereas frequently occurring terms have a high arithmetic mean and thus a higher weight. When calculating the retrieval status value of documents, the centroid vector is always subtracted from the document vector, thus mostly freeing the document from non-selective general terms (Huffman & Damashek, 1995, 306):

(An) advantage of using a centroid vector is that it characterizes those features of a set of documents that are more or less common to all the documents, and are therefore of little use in dis-

tinguishing among the documents. The Acquaintance centroid thus automatically captures, and mitigates the effect of, those features traditionally contained in stop lists.

After the centroid has been subtracted from the document vector, the similarity between two documents M and N (of which one can be a query) is calculated via the cosine of the angle between the document vector M and the document vector N .

In retrieval tests in the context of the Text Retrieval Conferences (TREC), ACQUAINTANCE achieved varying results. For short queries, it fares worse than other systems on average, while achieving very good results when using entire documents as search queries. The longer the query (e.g. an entire model example), the simpler it is to compile a good statistical profile for the search and hence, the better the retrieval process works. Furthermore, ACQUAINTANCE deals well with garbled information. Huffman (1996, 10) reminds us of the sphere of ACQUAINTANCE's application:

In the defense and intelligence worlds, data is often received in garbled form. Sometimes the garbling can be quite severe, and a system that cannot deal gracefully with degraded data is very limited in its usefulness.

When a document is garbled to the extent of 10% (i.e. 10% of the characters in a text are replaced randomly by other characters for testing purposes), recall and precision are altered only minimally in comparison with non-garbled documents. Even with 20% garbling, ACQUAINTANCE still delivers useful results thanks to the statistical n -gram approach (Huffman, 1996, 13):

The system did perform quite well in the confusion track, which measures performance in an area where Acquaintance has a high degree of potential, namely working with garbled data. Even at a relatively high degree of garbling, the system's performance degraded quite gracefully. This type of behavior is quite important to users of document retrieval and filtering systems in the defense and intelligence fields.

HAIRCUT: n -Grams Processed Probabilistically

The experimental retrieval system HAIRCUT (Hopkins Automated Information Retriever for Combing Unstructured Text) is just as well-documented as ACQUAINTANCE. HAIRCUT was constructed by a research group at Johns Hopkins University, led by Mayfield and McNamee. The system has participated, in different variants, in various retrieval tests in the context of TREC. The authors are convinced of the capability of their system (Mayfield & McNamee, 2005, 2):

Through extensive empirical evaluation on multiple internationally developed test sets we have demonstrated that the knowledge-light, language-neutral approach used in HAIRCUT can achieve state-of-the-art retrieval performance.

HAIRCUT uses a probabilistic language model to find and arrange documents. A basic assumption behind this model is that a document is more likely to result from a query the more frequently the search argument occurs within it. In an initial test, the relative frequency of each search n -gram can be calculated for every document, and the resulting values be multiplied with each other. The attempt will fail, however, because if only a single n -gram derived from the search terms does not occur in the given document, it will enter the equation under the value of zero, thus bringing about a retrieval status value of zero for the document. We need a second determining value besides the relative frequency of the n -gram in the specific document. It proves useful, here, to draw upon the relative frequency of all data in the entirety of the database that contain the n -gram. Lastly, we need a smoothing parameter that steers the meaning of the two aspects.

Now we can introduce the language model of HAIRCUT. The premise is a search query Q , which is then parsed into the n -grams $q_1, q_2, \dots, q_i, \dots, q_n$. In all documents D , the relative frequency of q_i is calculated (via the quotient from the number of q_i in D and the total number of n -grams in D) and stored as $P(q_i | D)$. The procedure is analogous for the second determining value. We count the documents that contain q_i and divide the resulting number by the number of all documents in the database C . This will then be regarded as the relative frequency of q_i in C , thus $P(q_i | C)$. If q_i occurs at all in the database, $P(q_i | C)$ is principally greater than zero; the component $P(q_i | C)$ thus fulfils the desired condition. Let the (freely adjustable) smoothing parameter be α . In this model, the retrieval status value of a document is calculated as follows (McNamee & Mayfield, 2004, 76):

$$P(D | Q) = [\alpha * P(q_1 | D) + (1 - \alpha) * P(q_1 | C)] * \dots * [\alpha * P(q_i | D) + (1 - \alpha) * P(q_i | C)] * \dots * [\alpha * P(q_n | D) + (1 - \alpha) * P(q_n | C)].$$

Afterwards, the documents are arranged in descending order of their retrieval status value and yield an output ranked by relevance.

How does HAIRCUT work in practice? The texts (in both documents and queries) are input in Unicode. The punctuation marks serve as separators between sentences and are purged after the sentences have been recognized. Non-meaningful stop words must be deleted. This means that natural-language stop words are eliminated, but not stop n -grams. For instance, in English the word THE is ignored, but not the trigram THE, which occurs in MATHEMATICS, for instance (Teufel, 1988, 7). n -grams glide over the sentences, experimenting with the setting of both n and α .

HAIRCUT experiments with various languages. In European languages, it shows that the ideal length of n -grams lies at $n = 4$ and $n = 5$, respectively. The best precision value in English (at about 0.5) is reached by tetragrams, at $\alpha = 0.9$. They thus fare (slightly) better than the words. The influence of α turns out to be relatively low.

The ideal length of n correlates with the average length of a language's words. 4-grams achieve the best results in English and 5-grams in Finnish (a language with

long words). In languages with long words and many compounds, the precision values for words and n -grams are no longer as close to each other as they are in English. In Finnish (with an average word length of 7.2 characters), pentagrams achieve an increase in precision of 66% vis-à-vis words, in Swedish (word length: 5.3 characters) tetragrams outperform words by 34% and in German (5.9 characters) it is once more tetragrams that heighten precision by 28% (McNamee & Mayfield, 2004, 83).

Dependencies within Character Sequences

The statistical language model discussed so far has a theoretical problem. This is because it assumes that terms are independent of each other. This precondition does not hold for natural-language terms. If, for example, the character sequence BOSTON RED appears in a text on baseball, it is very probable that the next word will be SOX and not any other random word. The precondition of independence is equally void within n -grams. The letter N is far more likely to complete the word that begins with CONDITIO than any other letter. The probabilities of occurrence of individual n -grams vary considerably (Egghe, 2000).

Such information can be used to refine the statistical language model. We must now calculate the probability of a character within an n -gram (or of a word within a phrase or a certain environment), under the condition that the specific placement of the character (word) in the position k is dependent upon the characters in the preceding positions 1 through $k-1$ (Brown et al., 1992). In this place, we restrict our representation to n -grams for characters, but it can analogously be applied to words.

The probability of occurrence of a character z in the position k is contingent upon the preceding characters $z_1, \dots, z_{k-1}$: $P(z_k | z_{k-1})$. In this representation of conditional probability $P(z_k | z_{k-1})$, we call z_{k-1} the “history” and z_k the “prediction” (Brown et al., 1992, 468). The analogous contingent probability of course also applies to the character sequences within the history. We can now calculate the position-dependent probability of an n -gram up until the character z_k as the product of the contingent probability of the preceding character sequences:

$$P(z_k) = P(z_1) * P(z_2 | z_1) * \dots * P(z_k | z_{k-1}).$$

$P(z_1)$ is the probability of occurrence of the first character, $P(z_2 | z_1)$ the probability of z_2 following z_1 , $P(z_3 | z_2)$ the probability of z_3 following $z_1 z_2$ etc.

The respective probabilities are not given a priori, but must be gleaned linguistically via empirical studies on sample texts. To do so, it is calculated how often the entire character sequence $z_1 z_2 \dots z_{k-1} z_k$ occurs in the training corpus, as well as how frequently the character sequence $z_1 z_2 \dots z_{k-1}$ occurs (independently of which character follows z_{k-1}). Afterwards, the relative frequency is calculated (let C be the respective number of character sequences):

$$P(z_k | z_{k-1}) = C(z_1 z_2 \dots z_{k-1} z_k) / [C(z_1 z_2 \dots z_{k-1} z_{(1)}) + \dots + (z_1 z_2 \dots z_{k-1} z_{(n)})].$$

Brown et al. (1992, 469) call this method “sequential maximum likelihood estimation”.

Advantages and Disadvantages of *n*-Gram-Based Retrieval Systems

n-gram-based information retrieval systems have two great advantages vis-à-vis systems that build on natural-language words. Firstly, they work independently of languages and secondly, they can still be used for heavily garbled texts. Information-linguistic text processing can, to a large degree, be skipped.

The greater the amounts of search arguments, the better *n*-grams work in the Vector Space Model. Conversely, however, this also means that for short search arguments—which should be very frequent in end user systems—they can only be used with reservations. In the probabilistic model, shorter queries still yield adequate results.

We will meet *n*-grams again, under the special aspects of retrieval systems. They can be used, for instance, to identify languages as well as in fault-tolerant retrieval. *n*-grams are not only used in information retrieval, but also in spell-check programs and, related to these latter, in the processing of scanned and OCR-indexed documents (Harding, Croft, & Weir, 1997).

Conclusion

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- *n*-grams are formal words of the length *n*, where *n* is a number between 1 and 6. *n*-grams refer either to single words or glide through entire texts (or parts of a text, such as sentences).
 - When using *n*-grams, the maximum number of possible terms is a known quantity. For 26 letters and a blank space, the maximum limit of index entries lies at 27^n terms. However, by far not all possible *n*-grams are actually recorded. In the English language, only 16% of trigrams are in fact used.
 - For languages that work without word separators, such as Japanese, Korean and Chinese, *n*-grams are particularly useful. Also, a “language” such as DNA can be parsed via *n*-grams. Languages whose words tend to change form dramatically when inflected, however, cause problems.
 - As early as in the 1970s, pentagrams have been used in information retrieval practice as search aids.
 - An *n*-gram system with a theoretical anchoring in the Vector Space Model is ACQUAINTANCE by Damashek. Since this system works without a stop word list, the documents must be cleansed via the terms of a mean vector (centroid). ACQUAINTANCE yields satisfactory results in searches with model documents (even with garbled information), but fails when confronted with short queries.

- HAIRCUT by Mayfield and McNamee is anchored in probability calculation. It searches for documents whose n -grams are most likely to fit the n -grams of the query. In many European languages, an n value of five proves ideal.
 - The probabilistic model can be refined by not regarding the characters as independent but by calculating the dependencies within the character sequences via empirical linguistic analyses, and to then use them during retrieval.
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C.2 Words

When using natural-language words instead of formal words—i.e., *n*-grams—we must accomplish several tasks automatically:

- recognizing the writing system (in the case of alphabetic writing: recognizing the alphabet being used),
- recognizing the language,
- marking words with little meaning (stop words) via negative lists,
- forming the stems, or basic forms, of inflected words.

In further tasks (Ch. C.3), the objective will be to recognize phrases (such as “soft ice”) and personal names (“Miranda Otto”, “St. Louis, MO”, “Anheuser Busch”, “Budweiser” etc.) and to meaningfully separate compound terms (such as “maidservant”). During a concrete work sequence, it will be of advantage to identify phrases and personal names before marking stop words and forming word stems. Otherwise, serious problems may arise: company names may contain stop words, for instance (e.g. the “and” in “Fox and Wolf Ltd.”), and personal names can appear to have flecnional endings (e.g. a plural “s” in “Julia Roberts”).

Writing System Recognition

Writing systems can be encoded via Unicode. As of 2012, Unicode comprises around 100 scripts, including alphabetic writing and graphically oriented scripts, like Chinese. Unicode also features historical writing systems that are not in use anymore (e.g. ancient Egyptian hieroglyphs).

The availability of Unicode, however, does not mean that this code is always being used. Hence, when confronted with codes other than the standard examples (such as Unicode, ASCII or ISO 8859), it makes sense to use algorithms of automatic script recognition. The goal is to inform the retrieval system as well as the browser—for the display—about the writing system in which a document is rooted. Geffet, Wiseman and Feitelson (2005, 27) emphasize:

(The) goal ... is to develop a methodology that can be used by a browser to automatically determine which encoding was used in a document. This will allow the browser to choose the correct font for displaying the document, without requiring a trial-and-error search by the user.

An important piece of information for recognizing writing systems is the direction; i.e., from left to right in Latin and Cyrillic writing, and from right to left in Arabic and Hebrew, respectively. The latter direction is not always consequently applied in the languages that use it. For instance, words are sometimes written from right to left, but numbers from left to right. Figure C.2.1 demonstrates this via a magazine number by

Al Jazeera in Arabic. In a logically coded text, alignments are stated via directional information (*ltr* for “left-to-right” and *rtl* for “right-to-left”).

الاثنين 10، ذو القعدة 1426 هـ 12 ديسمبر 2005 م العدد 12128

Figure C.2.1: Reading Directions in an Arabic Text. Source: Al Jazeera.

In alphabet recognition, Geffet, Wiseman and Feitelson (2005) use (known) letter distributions for certain alphabets. The distribution of letters in a concrete document is compared to the known overall distributions; on the basis of this comparison, the document’s script can be identified correctly.

Language Identification

After having recognized the writing system used in a document, our next task is to identify its language. Unicode encodes scripts, not languages (Anderson, 2005, 27). Latin writing, for instance, is used equally for English, German or Albanian, whereas Arabic script is used for Arabic and Farsi.

In this chapter’s discussion of the recognition of language, we will concentrate on texts which are digitally available. In doing so, we exclude a huge research area: the recognition of spoken language. This area is immensely important for entering an oral dialog with retrieval systems and for searching audio files (or the texts within video files). We will come back to this particular problem in Chapter E.4.

Generally, texts are written in one language only. This allows us to draw on the entire text at hand for our analysis. However, many texts contain various different languages—e.g. on the level of sentences and paragraphs—such as verbatim quotes from sources written in another language than the main text. If we want to take into account such multilingualism as well, our analysis must begin on the level of sentences.

We will discuss three approaches that each provide (a more or less satisfactory) language recognition. An initial, easily applicable but very uncertain attempt is to use model types, e.g. “typical” language-specific letter combinations or other “typical” characteristics. Let us look at the following list (Dunning, 1994, 3):

Dutch	vnd
English	ery_
French	eux_
Gaelic	mh
German	_der_
Italian	cchi
Portuguese	_seu_
Serbo-Croatian	lj
Spanish	_ir_.

The sequence “cchi” does indeed occur very frequently in the Italian language; however, nobody would claim that texts discussing *zucchini* or *Pinocchio* must necessarily be written in Italian. Other “typical” language characteristics include certain diacritical signs (such as Å in Swedish) or rare punctuation marks (¿ in Spanish). This, too, is rather unreliable, since Å can occur in any language as part of the *Ångström unit*.

A second method uses word distributions as a linguistic criterion. Working with language-typical training documents, McNamee (2005) compiles a ranking of the 1,000 most frequent words in each language. Each word is additionally allocated its relative frequency. Figure C.2.2 shows Top 10 lists for four selected languages. The most frequent German word is “und”, with a relative frequency of 3.14%. McNamee’s language recognition works on the level of sentences. After a word’s frequency of occurring in a sentence has been counted, this number is immediately multiplied with the relative frequency of the word in the language-specific training documents. The values yielded for these words are then summarized sentence by sentence. The language with the highest value is the most probable language for the sentence.

McNamee (2005, 98) names an example: it involves the meaningless sample phrase “in die”. Instead of the usual 1,000, we will consider only the ten most frequent words of the individual languages according to Figure C.2.2. For Danish, the result is zero, since neither “in” nor “die” feature in their Top 10 lists. The remaining languages receive the following values:

German: $1 * 2.31 + 1 * 1.24 = 3.55$
 Dutch: $1 * 1.62 + 1 * 1.56 = 3.18$
 English: $1 * 1.50 + 1 * 0 = 1.50$.

The sentence fragment “in die” is thus classified as German.

Experiments using this method can occasionally yield good results. For the German language, the method achieves an effectiveness rating of 99.5%. Furthermore, very few foreign-language sentences are erroneously interpreted as being German (the exception being sentences in Danish, of which 1.6% have been misinterpreted as German). The result for the Portuguese version, however, yielding an effectiveness of only 80%, is hardly satisfactory.

The third method of language recognition we will discuss here uses *n*-grams. Let us take a look at the previously discussed system ACQUAINTANCE by Damashek (1995): here, a document is segmented into *n*-grams, and the frequency of the individual *n*-grams is then counted. The text *n*-grams are now compared to typical *n*-gram distributions in the individual languages. Hence, there must be an empirically definable language centroid. The calculation method, as in *n*-gram retrieval, is the cosine—however, in this instance the centroid vector is not subtracted. To incorporate the latter would be counterproductive, since it contains the frequent, general words—and they are the ones which are useful for language identification.

%	Danish	%	Dutch	%	English	%	German
4.35	og	4.19	en	6.16	the	3.14	und
3.71	hun	3.11	t	4.31	and	2.31	die
1.82	i	2.34	de	2.87	a	2.62	sie
1.70	de	2.25	van	2.06	of	1.92	zu
1.59	det	1.84	een	2.02	to	1.91	der
1.45	var	1.62	in	1.61	was	1.59	sich
1.38	han	1.56	die	1.57	he	1.30	er
1.38	at	1.50	is	1.50	in	1.24	in
1.25	saa	1.23	niet	1.04	it	1.19	nicht
1.24	en	1.22	ick	0.86	his	1.12	das

Figure C.2.2: The Ten Most Frequent Words in Training Documents for Four Languages. Source: McNamee, 2005, 98.

A relatively safe variant of language identification is derived from a combination of the three methods described above (typical language characteristics, typical word distributions, language centroid).

Stop Word Lists

A stop word is a word in a document or in a query that does not influence the retrieval process. It has the same probability of occurring in a document that is relevant for the query as it does in an irrelevant text. Stop words bear no knowledge, they are content-free. Wilbur and Sirotkin (1992, 45) claim:

These terms, such as “the”, “if”, “but”, “and”, etc., have a grammatical function but reveal nothing about the content of documents. By removing such terms one can save space in indices and also, in some cases, may hope to improve retrieval.

For Wilbur and Sirotkin (1992, 45), stop words are “noncontent words”. Frequently occurring words are hardly discriminatory and thus have a bad reputation in information retrieval. Hence, it must be discussed whether such words are to be applied in practice at all. Defining stop words and excluding them from the retrieval process as a matter of principle is a very dangerous step to take. The terms “be”, “not”, “or” and “to” can be interpreted as stop words; eliminating them, however, would leave us unable to retrieve the “Hamlet” quote “to be or not to be”. Stop words can thus be marked and ignored in the “normal” search process, but one must have the option of using a certain operator, or a phrase search, to incorporate them into one’s specific search needs.

We distinguish between three different approaches to gleaning stop words and subsequently using them in negative lists:

- determining general stop words in a language,
- determining stop words in a given document set,
- determining stop words within a single document.

Let us begin with the general stop words. Stop words are always language-specific, so we will need separate lists for German, English, French etc. If an information system works with bibliographical references or with factual information (but not with full texts), the stop word list will remain small. The information provider DIALOG, for instance, uses only nine entries for the English language (Harter, 1986, 88):

an, and, by, for, from, of, the, to, with.

Far longer negative lists must be compiled for the retrieval of natural-language full texts. Fox (1989) suggests a procedure that starts with the frequency values of words and processes the resulting ranking list intellectually: some frequent words do not enter into the list and some non-frequent words are added. The article by Fox (1989, 19)

reports an exercise in generating a stop list for general text based on the Brown corpus of 1,014,000 words drawn from a broad range of literature in English. We start with a list of tokens occurring more than 300 times in the Brown corpus. From this list of 278 words, 32 are culled on the grounds that they are too important as potential index terms. Twenty-six words are then added to the list in the belief that they may occur very frequently in certain kinds of literature. Finally, 149 words are added to the list because the finite state machine based filter in which this list is intended to be used is able to filter them at almost no cost. The final product is a list of 421 stop words that should be maximally efficient and effective in filtering the most frequently occurring and semantically neutral words in general literature in English.

The Brown Corpus counts grammatical basic forms, e.g. it summarizes “go”, “went” and “gone” into the entry “to go”. Fox, on the other hand, works with the individual word forms. His frequency list begins with:

the, of, and, to, a, in, that, is, was, he, for, it, with, as, not, his, on, be, at, by.

The list contains terms that are not semantically neutral, and which are thus unsuitable as stop words. These include

business, family, national, power, social, system, time, world.

Since the cross-section of the frequency list has been arbitrarily set at the value of 300, Fox has adopted further terms from this ranking (as “extra fluff words”). Their values are (slightly) below the threshold value of 300, e.g.

above (296 entries), *show* (289), *whether* (286), *either* (285), *gave* (285), *yet* (283), *across* (282), *taken* (281), *quite* (281).

The additional list (“nearly free words”) contains, for instance, individual letters and flectional forms of already represented words. Since “asked” occurs in the frequency list, the forms “ask”, “asking” and “asks” will be added. The same method has been used to compile a general stop list for the French language (Savoy, 1999).

A radical method of compiling general stop words is oriented on parts of speech. Consider the following sample sentence (Lo, He, & Ounis, 2005, 23):

I can open a can of tuna with a can opener.

The word “can” occurs three times in it, twice as a noun and once as a verb. Whereas it makes perfect sense for someone who is looking for cans and can openers to retrieve this sentence, it can safely be ruled out that anyone will be looking for “can” in the sense of “to be able to”. It is equally unlikely that someone will search the sentence via “I”, “a” or “with”. In a rough generalization, we can say that certain word classes (Hausser, 1998, 42) are more suitable for information retrieval than others. Articles, conjunctions, prepositions and particles are all function words bearing little to no content; they are thus stop words. Nouns and adjectives, having at least the potential to carry meaning, are not suitable as general stop words. What about verbs? According to Hausser (1998, 43), they are content words, i.e. not stop words. Auxiliary verbs and the most frequently used verbs, however, are hardly document-specific and thus become stop word candidates. Nominalized verbs must be regarded as potential bearers of content. The radical stop list thus contains all function words of a language, all auxiliary verbs as well as (at least the most frequently used) verbs. In the sample sentence, the first “can” is recognized as a verb and marked as a stop word, whereas the other two “can”s are kept as nouns. Compiling such a stop list is relatively simple; applying it to texts, however, is not as self-evident, since computational linguistics procedures (Fox, 1992, 105 et seq.) must first be used to determine the class of each word (at least for such words as our “can”, or nominalized verbs that allow for several interpretations).

One problem must not be overlooked in the case of all these stop lists: they contain pronouns. When solving anaphora in information retrieval (Ch. C.4), one must be provided the option of assigning the pronouns “their” related nouns. If these have been removed from the documents via a negative list, however, the desired anaphora solution becomes impossible.

In addition to the general stop words, we can identify domain-specific stop words that are applicable in certain knowledge domains. These are represented via area-specific document sets, e.g. via specialist databases (such as Medline for medicine). An interesting approach is pursued by Wilbur and Sirotkin (1992) and—building on this approach—by Yang and Wilbur (1996). They assume the existence, within the

domains, of document pairs that are closely thematically linked. Since for practical reasons it is hardly possible to consult human experts, the authors use the cosine similarity measurement. All document pairs whose similarity value crosses a threshold are regarded as similar. Of central importance is the introduction of term strength (Wilbur & Sirotkin, 1992, 47):

We define the *strength* of a term to be the probability that when it occurs in one member of a highly rated pair, then it also occurs in the other member.

Let the number of document pairs in which the term t occurs jointly be $g(t)$, and let the number of document pairs in which the same term t occurs only in the first document be $e(t)$. The term strength $s(t)$ of t is calculated via the following quotient (Yang & Wilbur, 1996, 359):

$$s(t) = g(t) / e(t).$$

The more frequently t occurs jointly in similar documents, the greater its term strength will be. If a term occurs in only one document, its term strength will be zero. A threshold value must be defined for the term strength $s(t)$. All terms whose term strength lies beneath this stop word threshold value are considered to be stop words and are added to the general stop list. Depending on the setting of the stop word threshold value, the amount of stop words can be enhanced massively. Yang and Wilbur (1996, 364) report:

The more aggressive stoplists were obtained by setting the threshold at word strength values of 0, 0.1, ..., 0.9, selecting the words whose strength value is less than or equal to the threshold, and adding these selected words to the standard stoplist.

In cases of very large (“aggressive”) stop word lists, reports indicate good retrieval results (Yang & Wilbur, 1996, 357 et seq.).

A useful method for long documents is to also search within the texts in order to find sections that fit the query (Ch. G.6). This is where document-dependent stop words come into play. Such stop words can indeed be suitable for retrieving the respective document as a whole, but they will not serve to retrieve the most appropriate passages within the document. Ji and Zha (2003, 324) emphasize:

(document-dependent stop words) are useful in discriminating among several different documents but are rather harmful in detecting subtopics in a document.

Let us assume—for the purposes of illustration—that an article discusses heart diseases under various aspects, e.g. relating to surgery. The word “heart” is surely useful for finding this specific article; within the article, however, the word will not serve to characterize the different sections, since it will appear equally often in all of them.

“Heart”, in our example, is a document-specific stop word. In this article, “surgery” appears almost as frequently as “heart”, but only in a single paragraph. “Surgery” is thus discriminatory on the level of passages, and is thus not a stop word in “passage retrieval”. According to Ji and Zha (2003, 324), a word can only be labeled a document-dependent stop word if two criteria are given: high frequency of occurrence in the document on the one hand, and equal distribution in the text on the other.

One thing we need to be careful about identifying words as document-dependent stop words is that relative high frequency alone is *not* sufficient for labelling as document-dependent stop word. The words are also required to be uniformly distributed in the whole document.

Word Form Conflation

Documents and queries contain words in morphological variants. The word forms of one and the same word are brought into a single form of expression via methods of “conflation”. Frakes (1992, 131) emphasizes the meaning of conflation for information retrieval:

One technique for improving IR performance is to provide searchers with ways of finding morphological variants of search terms. If, for example, a searcher enters the term *stemming* as part of a query, it is likely that he or she will also be interested in such variants as *stemmed* and *stem*. We use the term *conflation*, meaning the act of fusing or combining, as the general term for the process of matching morphological term variants.

We can distinguish two forms of word form conflation. Linguistic techniques work with morphological word form analysis. They determine a word form and then derive its basic form, its lexeme. Non-linguistic techniques forego elaborate linguistic analyses and develop rules that reduce the word forms to a common stem; this stem does not have to be a linguistically valid form. The linguistic methods use lemmatization to conflate the word forms into a basic form, whereas the non-linguistic methods use stemming to arrive at the stem.

When processing the word forms RETRIEVED and RETRIEVAL, for instance, lemmatization will lead us to the lexeme RETRIEVE, whereas a stemming algorithm will conflate the two words into the stem RETRIEV by cutting off the -AL and -ED endings. Galvez, de Moya-Anegón and Solana (2005, 524 et seq.) provide a short overview of the two basic techniques of conflating single word forms:

(T)his leads us to a distinction of two basic approaches for conflating uniterm variants:

- (1) *Non-linguistic techniques*. Stemming methods consisting mainly of suffix stripping, stem-suffix segmentation rules, similarity measures and clustering techniques.
- (2) *Linguistic techniques*. Lemmatisation methods consisting of morphological analysis. That is, term conflation based on the regular relations, or equivalence relations, between inflectional forms and canonical forms, represented in finite-state transducers (FST).

Both methods are error-prone. In lemmatization, several basic forms can be gleaned from one and the same word form. If we conflate the word form FLIER with both FLY (in the sense of a nominalized verb) and FLIER (as the common advertising method), one of the two solutions will always be wrong. In such cases, we speak of overconflation. In stemming, there is also—in addition to overconflation—underconflation, in so far as certain variants go unrecognized (Paice, 1996). Overconflation is a result of the over-enthusiastic cutting of character sequences, such as the -IER and -Y in our example, which leads different stems to fall together (like FLIER and FLY above to the stem FL). Underconflation occurs when the rules are unable to identify all forms of the same stem. Supposing that DIVIDING and DIVISION belong to the same stem, and that the algorithm only uses -ION and -ING separation, we will arrive—erroneously—at the two different stems DIVID and DIVIS. The methods of conflation thus have to be constructed so as to avoid over- and underconflation wherever possible.

Deriving Basic Forms (Lemmatization)

Lemmatization leads to basic forms. In order to walk this path of word form conflation, we must consult the linguistic subdiscipline of morphology (Pirkola, 2001, 331):

Morphology is the field of linguistics which studies word structure and formation. It is composed of inflectional morphology and derivational morphology. Inflection is defined as the use of morphological methods to create inflectional word forms from a lexeme. Inflectional word forms indicate grammatical relations between words. Derivational morphology is concerned with the derivation of new words from other words using derivational affixes. Compounding is another method for forming new words. A compound word (or a compound) is defined as a word formed from two or more words written together.

Morphology thus knows three principles of combination (Hausser, 1998, 39). Inflection (learn/s, learn/ed etc.) adjusts a word to its syntactic environment, derivation describes the combination of a word with an affix (a prefix such as un-, or a suffix such as -ment) to form a new word, and composition is the combination of several different words into a new one.

Closely related to composition is phrase building, i.e. the summary of several individual word forms into a unit (e.g. “high school”). We will deal with compounds and phrases in detail over the course of the next chapter.

Lemmatization is performed either by applying specific rules to word forms or by consulting dictionaries containing the lexemes in question as well as their possible inflectional forms and derivations. The goal is always to elicit the basic form. For nouns, this is the nominative singular, for verbs the present infinitive and for adjectives and adverbs it is the unmarked adverbial form.

According to Kuhlen (1977, 63), rule-bound procedures have the following aspiration:

Lexicographical basic forms must be created mechanically and without using a dictionary.

A very simple and exclusively rule-based lemmatization method for the English language is the S-Lemmatizer, which conflates singular and plural forms with each other (Harman, 1991, 8). It uses four rules that must be adhered to in the following order.

- Stop processing when a word form consists of 3 letters or fewer.
- When a word form ends with IES, but not EIES or AIES, replace IES with Y.
- When a word form ends with ES, but not AES, EES or OES, replace ES with E.
- When a word form ends with S, but not with US or SS, delete the S.

The S-Lemmatizer can be enhanced to include verbs via the -ING and -ED endings (see, for instance, Kuhlen, 1977, 72).

Dictionary-based procedures require two components; apart from the dictionary, there must be a recognition algorithm that categorizes and lemmatizes the word forms (Hausser, 1998, 49):

Online Dictionary. For every element (...) of the natural language, a lexical analysis must be defined and electronically stored.

Recognition Algorithm. Every unknown word form (e.g. 'books') must be automatically characterized by the system with regard to categorization and lemmatization:

Categorization. The interface is assigned the word form (here: noun) and its specific morphosyntactic characteristics (here: plural). The categorization is required for syntactic analysis.

Lemmatization. The interface is assigned the basic form (here: 'book'). The lemmatization is required for semantic analysis: the basic form facilitates access to the corresponding entries in a semantic dictionary.

If the dictionary contains the corresponding relations (e.g. hyperonyms and hyponyms, related terms), the semantic environment will be accessible; hence, it will be possible to make the transition from word to concept.

Stemming

Stemming is used to free word forms from their suffixes and, where needed, change them so that all meaningful stems of a word are summarized. Cutting off prefixes, on the other hand, hardly leads to an improved retrieval performance since various prefixes (such as anti- or un-) change the meaning of the stem. Hull (1996, 82) rejects the cutting of prefixes.

The detailed query analysis has revealed that prefix removal is generally a bad idea for a stemming algorithm. The derivational stemmer suffers in performance on a number of occasions for this reason. For example, prefix removal for the terms *overfishing* and *superconductor* is certainly undesirable. A quick scan over ... queries reveals several more instances where prefix removal causes problems: *antitrust*, *illegal* and debt *rescheduling*.

Stemming is thus reduced to suffixing. We can distinguish two different methods of processing suffixes: the Longest-Match approach and the iterative approach. As an example, let us consider the word “willingness”: in the iterative approach, the stemming algorithm runs through several steps. First the “-ness” is cut off and then the “-ing” is cut off, leaving us with the stem “will”. In the Longest-Match approach, the longest ending (in this case “-ingness”) is cut off in a single working step. As a paradigm for an iterative approach, we discuss the Porter algorithm, whereas the Lovins Stemmer should stand in as a Longest-Match example.

The Longest-Match Stemmer by Lovins (1968) contains a stemming procedure as well as working steps for re-coding stems. Lovins describes the principle of cutting off the longest ending (1968, 24):

The *longest-match* principle states that within any given class of endings, if more than one ending provides a match, the one which is longest should be removed.

The Lovins Stemmer uses a list of endings (Figure C.2.3 shows a cross-section of endings with anywhere between nine and eleven letters) as well as context-specific rules. Rule B, for instance, states that the received stem must contain at least three characters, C presupposes that the stem must contain at least four letters, E means that the ending cannot be cut off if it is preceded by an ‘e’, and A states that no restrictions apply. Lovins (1968, 30) uses a total of 29 rules.

Suffix	Rule
.11.	
alistically	B
arizability	A
izationally	B
.10.	
antialness	A
arisations	A
arizations	A
entialness	A
.09.	
allically	C
antaneous	A
antiality	A
arisation	A
arization	A
ationally	B
ativeness	A
eableness	E
entations	A
(etc.)	

Figure C.2.3: List of Endings to Be Removed in the Lovins Stemmer (Excerpt). Source: Lovins, 1968, 29.

Since there are pronunciation variants in the English language (such as “absorbing” and “absorption”), certain stems must be “re-coded”. Here, another set of rules (34 in total) applies; first, any duplicate consonants must be reduced to one. The pertinence of the procedure can be demonstrated on the examples of “metal” and “metallic”:

<i>Input</i>	<i>Longest-Match Stem</i>	<i>re-coded Stem</i>
metal	metal	metal
metallic	metall	metal.

In a further re-coding step, rules are defined for replacing letters, e.g. that “rpt → rb”:

<i>Input</i>	<i>Longest-Match Stem</i>	<i>re-coded Stem</i>
absorbing	absorb	absorb
absorption	absorpt	absorb.

Porter’s (1980) approach of forming word stems for the English language, widely used in practice, is an iterative stemmer (for current variants of the Porter Stemmer—also for other languages—see the homepage of “Snowball”; Porter & Boulton, undated). Several working steps follow after one another, each further processing the result of the preceding step. The rules that apply to each working step are based on preconditions themselves, the most important one being the m-condition.

<i>Step 1a</i>			<i>Example</i>		
SS	→	SS	caresses	→	caress
IES	→	I	ponies	→	poni
SS	→	SS	caress	→	caress
S	→		cats	→	cat
<i>Step 1b</i>					
(m>0)	EED	→	EE	feed	→ feed
				agreed	→ agree
(*v*)	ED	→		plastered	→ plaster
				bled	→ bled
(*v*)	ING	→		motoring	→ motor
				sing	→ sing
<i>Step 1c</i>					
(*v*)	Y	→	I	happy	→ happi
				sky	→ sky

Figure C.2.4: Iterative Approach in the Porter Stemmer. Working Step 1 (Excerpt). Source: Porter, 1980, 134.

This m-condition counts vowel-consonant sequences in a word stem. Vowels v are the letters A, E, I, O, U and Y—the latter under the condition that the Y follows a consonant. All other letters are consonants c. A sequence of letters ccc (where the number

of c is greater than zero) is described as C , a sequence vvv (also greater than zero) is called V . Hence, every word form can be represented as follows:

$$[C] VC VC \dots [V].$$

The consonants resp. vowels in the parentheses do not necessarily have to occur at this point in the word; the only important elements are the VC groups, which are enumerated word by word:

$$[C] VC^m [V].$$

The number m , meaning the number of VC groups, is the m -measure of a word or word stem (Porter, 1980, 133):

m will be called the *measure* of any word or word part when represented in this form.

Examples for $m = 0$ are “tree” or “by” (they do not contain a single VC group), for $m = 1$: “trouble” or “trees”, for $m = 2$: “troubles” or “private”, etc. The rules for the processing of suffixes always have the form

$$(\text{Condition}) S1 \rightarrow S2.$$

When a word form ends with the suffix $S1$, and the stem in front of $S1$ meets the condition, $S1$ is replaced by $S2$. The conditions either stand for certain letters (e.g. $*v*$ requires the stem to contain a vowel) or, in the multitude of cases, for a minimum amount of m . For instance, if the rule

$$(m > 1) \text{ EMENT} \rightarrow [\text{zero}]$$

applies, “ement” will be $S1$ and the empty space $S2$. The word form “replacement” would thus be reduced to “replac”, since this part of the word contains two VC sequences and thus meets the condition.

The Porter Stemmer has five iteration rounds. Step 1 (Figure C.2.4) mainly involves plural forms of nouns as well as the typical verb endings “-ed” and “-ing”. The subsequent rounds are more detailed (Porter, 1980, 137):

Complex suffixes are removed bit by bit in the different steps. Thus GENERALIZATIONS is stripped to GENERALIZATION (Step 1), then to GENERALIZE (Step 2), then to GENERAL (Step 3), and then to GENER (Step 4).

The Porter algorithm has no theoretical basis in linguistics, but it has proven to be extremely effective in practice (Porter, 1980, 136).

The algorithm is careful not to remove a suffix when the stem is too short, the length of the stem being given by its measure, *m*. There is no linguistic basis for this approach. It was merely observed that *m* could be used quite effectively to help decide whether or not it was wise to take off a suffix.

The Lovins and the Porter Stemmer are both general stemmers, i.e. they hold true for all word forms in the English vernacular. Stemmers exist for various languages other than English (Porter & Boulton, undated); for instance, positive results for information retrieval are reportedly achieved in the German language (Braschler & Rippinger, 2004).

General stemmers can deal only rudimentarily with the domain-specific terminology; for instance, chemistry or economics. Specialized word stem analyses are developed—according to a suggestion by Xu and Croft (1998)—via corpora, i.e. collections of relevant domain-specific documents. This leads to a corpus-based stemming. Xu and Croft (1998, 64) encourage this corpus-based approach, suggesting an appropriately varied usage of words for different subjects.

With stemming, an algorithm that adapts to the language characteristics of a domain as reflected in a corpus should ... be better than a nonadaptive one. Many words have more than one meaning, but their meanings are not uniformly distributed across all corpora.

The Porter Stemmer, for example, erroneously summarizes the words “policy” and “police”. In a politological environment, the probability of “policy” being adequate is higher than in a context that involves questions about building security services. One meaningful approach might be to apply corpus-based stemmers as instruments for adjusting general stemmers to the characteristics of a specific knowledge domain. We can distinguish between two ways of achieving the desired adjustment:

- co-occurrence of word variants in the text window (e.g. within 100 words) of a specialized corpus (Xu & Croft 1998),
- co-occurrence of word variants with other words in the corpus (Kostoff, 2003).

Xu and Croft (1998, 63) justify their procedure as follows:

Corpus-based stemming refers to automatic modification of equivalence classes to suit the characteristics of a given text corpus. ... The basic hypothesis is that the word forms that should be conflated for a given corpus will cooccur in documents from that corpus.

They are able to demonstrate, on the example of a document sample from the “Wall Street Journal”, that the words “policy” and “politics” almost never co-occur in the same text window and thus must not be conflated into a single word (in this corpus).

Kostoff points out that word forms do not have to co-occur in documents. Short texts and abstracts in particular are unlikely to contain different word forms of the same stem in the same text window. This leads to a point of constructive criticism of the Xu/Croft approach (Kostoff, 2003, 985):

Xu and Croft (1998) would have had a much more credible condition had the metric been co-occurrence similarity of each word variant with other (non-variant) words in the text, rather than high co-occurrence with other forms of the variant.

Instead of cancelling each other out, both approaches of corpus-based stemming ought to complement each other.

***n*-Gram Pseudo-Stemming**

Stemming does not have to categorically proceed via natural-language words; another option involves “pseudo-stemming” via *n*-grams. In a first variant, the *n*-gram pseudo-stemmer divides a word form into its *n*-grams. The similarity SIM of the two word forms is calculated via the number of different *n*-grams of Word₁ (a) and Word₂ (b) as well as the number (g) of shared different *n*-grams. Galvez, de Moya-Anegón and Solana (2005, 531) use the Dice coefficient to calculate the similarity:

$$\text{SIM}(\text{Word}_1 - \text{Word}_2) = 2g / (a + b).$$

Calculating the SIM values of all word pairs in a database leads to a matrix of word form similarities. In a further step, the word pairs are condensed into clusters. A threshold value must always be defined for the similarity. Now we can summarize those word forms whose similarity values lie above the threshold value into a cluster (“complete linkage”) or start with precisely one word pair and add word forms that are linked to (at least) one word above the threshold value. The latter procedure is then repeated until no further word can be added (“single linkage”). Since we are only concerned with suffix stemming, all word forms that do not begin with the same *n* letters are removed from the clusters recognized so far. The word forms that remain in a cluster have the same pseudo-stem and are considered to be *one* word.

There is a second variant of *n*-gram stemming that works with only a single stem, as is usually the case with stemming (Mayfield & McNamee, 2003, 416):

We would like to select an *n*-gram from the morphologically invariant portion of the word (if such exists). To that end, we select the word-internal *n*-gram with the highest inverse document frequency (IDF) as a word’s pseudo-stem.

The IDF weight value calibrates a word weight via the number of documents in a database that contain the word at least once (see Chapter E.1). Note: the smaller the number of documents containing the term in a database, the higher the IDF weight. An *n*-gram is linked to a word form in the sense of a pseudo-stem, if this *n*-gram will be the one that occurs in the fewest documents throughout the entire database. Since frequently occurring affixes have a low IDF weight, this procedure is very likely to

identify *the* characteristic *n*-gram of a word form. The example “juggling” (Figure C.2.5) is divided into seven tetragrams, of which “jugg” occurs in the fewest documents. Correspondingly, “jugg” is the pseudo-stem of “juggling”.

n-gram	Document frequency
*jug	681
jugg	495
uggl	6,775
ggli	3,003
glin	4,567
ling	55,210
ing*	106,463

Figure C.2.5: Document Frequency of the Tetragrams of the Word Form “juggling”. Source: Mayfield & McNamee, 2003, 415.

The advantages of *n*-gram stemmers vis-à-vis word-oriented stemmers include the pseudo-stems’ easy (automatic) construction as well as their independence of natural languages. A disadvantage is that they cannot take into account the specifics of individual languages.

The individual stemmers differ only minimally with regard to their respective performances in retrieval systems (Hull, 1996; Lennon et al., 1981); even the difference between stemming and lemmatization is minimal at best, as Kettunen, Kunttu and Järvelin (2005, 493) find out for the Finnish language.

Differences between stem generation and lemmatization were small and not statistically significant ...; their practical differences were not noticeable on any relevance level.

According to Harman (1991, 14), the use of stemmers in online retrieval systems is generally to be encouraged. However, one should always allow the users to deactivate stemming in order to prevent individual cases of overconflation.

Given today’s retrieval speed and the ease of browsing a ranked output, a realistic approach for online retrieval would be the automatic use of a stemmer, using an algorithm like Porter (1980) or Lovins (1968), but providing the ability to keep a term from being stemmed (the inverse of truncation). If a user found that a term in the stemmed query produced too many nonrelevant documents, the query could be resubmitted with that term marked for no stemming. In this manner, users would have full advantage of stemming, but would be able to improve the results of those queries hurt by stemming.

Word Processing for Cell Phone Keyboards

Retrieval systems are not only operated via QWERTY keyboards or country-specific variations thereof, but also via end devices with far fewer keys, such as cell phones. A roundabout way of entering words is the multiple typing of a key in order to identify the desired letter, i.e. hitting the number 2 three times in order to get a C. A simplification of this process consists of only hitting the respective key once and using dictionaries to determine which of the possible words is the one meant by the user. This is the underlying idea of the T9 system by Tegic Communications (King et al., 1999). For every natural language, a dictionary is stored that includes the stems of every word plus their respective frequency of usage. James and Reischel (2001, 366) describe T9:

In order to provide a one key press to one letter input, T9 in essence compares the sequences of ambiguous keystrokes to words in a large database to determine the intended word. This database also contains information about word usage frequency which allows the system to first present the most frequently used word that matches a given key sequence. Additional words that share the same key sequence are obtained by pressing a Next Word key.

If a word is not available in the database, the user can add it (now via the multiple-type procedure) and it will be available the next time he uses T9. T9 updates its list of suggestions with every new digit that is entered by continually creating additional information about the probable word stems. In the worst-case scenario, the user will only see the intended word after the last key has been struck (as in the example of the word “GOOD”, which only comes after the successive recommendations of “I”, “IN” and “INN”). Besides queries in retrieval systems, T9 also allows users to enter text into cell phones, e.g. in the Short Messaging Service (SMS).

Conclusion

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- The recognition of writing systems (or, more specifically, of alphabets) occurs either in the code itself (e.g. in Unicode) or via comparisons of letter distributions in the respective text with typical letter distributions in the alphabets.
 - Language recognition either uses (unspecific) models, language-specific word distributions or language-specific *n*-gram distributions. If a document contains several languages, the recognition must be performed on the level of individual sentences. Without adequate language recognition, the ensuing procedures of stop word lists as well as word form conflation become impossible (with the exception of *n*-gram-based methods).
 - A stop word has the same possibility of occurring in a document that is relevant to a query as it does in a non-relevant one. Stop words bear no content, which is why they must be marked and excluded from “normal” retrieval. However, stop words can be used to search via a specific operator as well as within phrases. Stop words are collected in negative lists.
 - General stop words of a language are frequently occurring words, auxiliary verbs and function words. Such lists are complemented by domain-specific stop words, which must be created for individual knowledge domains. Document-specific stop words do not contribute to a targeted

search of sections; they occur frequently in the given document and are equally distributed throughout all sections.

- Since documents and queries always contain word forms, i.e. morphological variants, it is necessary to conflate all forms of one and the same word. This is done either via lemmatization or via stemming.
 - Lemmatization leads the word forms back to their basic forms. There are purely rule-based approaches (such as the S-Lemmatizer for the English language) as well as approaches that in addition to certain rules always draw on a dictionary.
 - Stemming processes the suffixes of the word forms and unifies them in a stem. A Longest-Match stemmer (e.g. by Lovins) recognizes the longest ending and removes it. Iterative stemming (as in the Porter Stemmer) processes the suffixes over several rounds. The stems do not necessarily have to be linguistically valid word forms.
 - In addition to the general stemmers, corpus-based stemming takes into account characteristics of knowledge domains.
 - *n*-gram pseudo-stemmers are language-independent. They either work with all the different *n*-grams of a word form, uniting the forms of a stem via cluster-analytical procedures, or they use the *n*-gram that occurs in the smallest number of documents in a database.
 - Some cell phones do not have keys that clearly designate a single letter. For these systems, T9 stores a dictionary (which includes flectional forms) as well as values for the words' probabilities of occurrence.
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C.3 Phrases—Named Entities—Compounds—Semantic Environments

Compound Expressions

Some words only form a semantic unit when taken in conjunction with other words. While an exception in German (“juristische Person” / “legal entity”), it is common practice in English (“college junior” and “junior college”, “soft ice”, “high school”, “information retrieval”). Likewise, many technical languages use sequences of words to form a unit (“infectious mononucleosis”, “Frankfurt School”, “Bragg’s Law”). For the purposes of retrieval, it is expedient to search for such compound terms as units. Units that consist of several individual words are called “phrases”.

A subset of phrases, as well as several single-word expressions, relate to personal names (“Miranda Otto”), organizations (“Apple, Inc.”), place names (“New York City”) and products (“HP Laserjet 1300”). The component parts of the names must not only be recognizable as units, but we should also be able to retrieve variants, name changes and pseudonyms and understand them as pseudonyms—the phrases “Samuel Clemens” and “Mark Twain” being name and pseudonym of one and the same individual, for instance. Likewise, personal names must be excluded from the translation mechanisms of multilingual retrieval: a German-English translation of “Werner Herzog” into “Werner Duke” would miss the point of the search. Conflation takes into account the name as a whole (thus letting “Julia Roberts” keep the S at the end).

Certain languages, including German, allow the formation of compound expressions into a new word. Such compounding (e.g. *hairslide* out of *hair* and *slide*) is, at least potentially, never-ending since no reasons other than practical ones stand in the way of compounding n ($n = 1, 2, 3$ etc.) words into a new term. If compounds facilitate more precise searches—analogously to recognized phrases—it makes sense to create meaningful compounds out of individual words in the user’s input data. For instance, if the user enters *hair* NEAR *slide*, the system will change the query to *hairslide* OR (*hair* NEAR *slide*) (in so far as the system adheres to the Boolean Model).

The inverse of compound formation is compound decomposition. Here, retrieval systems are required to divide compound terms into their meaningful components (i.e. to recognize *hair* and *slide* in *hairslide*). Retrieval systems must thus take into consideration both the compound and its meaningful units (in a certain order if applicable) for a search. A user entry *hairslide* is thus reformulated (in a Boolean System) to *hairslide* OR (*hair* NEAR *slide*) as above. Non-Boolean systems following the Vector-Space Model or the probabilistic model, for example, enhance user input via the words’ respective meaningful compounds or meaningful components.

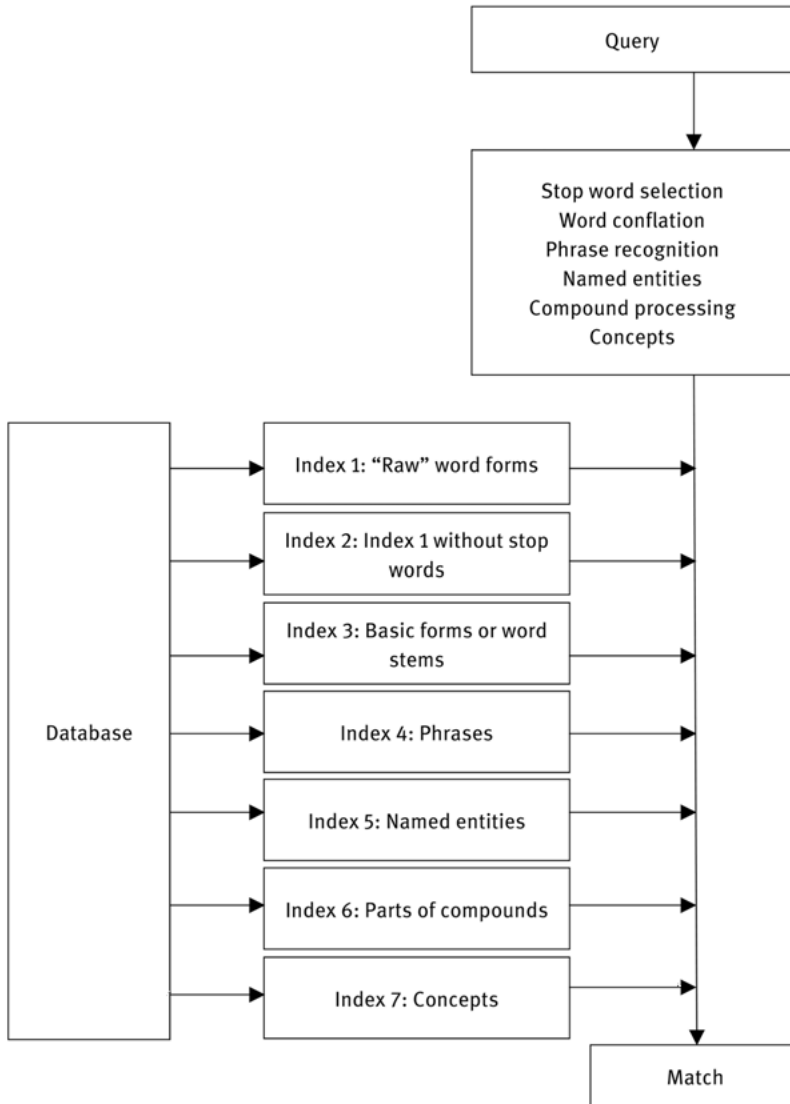


Figure C.3.1: Additional Inverted Files via Word Processing. Source: Heavily Modified after Perez-Carballo & Strzalkowski, 2000, 162.

Additional information gleaned from word processing may enter into its own respective inverted file (Figure C.3.1). Index 1, containing the “raw” word forms, will receive no attention unless the user insists upon it. The other indices are used to direct the search, with the individual indices subject to being switched on or off in accordance with the user’s wishes. For instance, if a query shows that compound decomposition leads to an abundance of ballast, this procedure can then be suppressed “at the push

of a button”. The result of word processing in a query should be presented to the user in such a way that the query may be optimized via dialog steps, even with regard to phrases, named entities, compounds and concepts.

In the preceding chapter our discussion of word processing did not go beyond the sphere of morphology. Now we cross the threshold of semantics, in so far as we must always take into account the meaning of the words as well as that of the phrases, names and compounds. Morphologically, nothing stands in the way of decomposing the word “raspberry” not only into “berry” but also into “rasp”; only a semantic analysis precludes such a problematic overdecomposition.

Phrase Building

There are concepts that consist of several individual words. Methods of Phrase Building attempt to recognize such relations and to make the phrases thus recognized searchable as a whole. We distinguish between two approaches to phrase building:

- statistical methods: the counting of joint occurrences of the phrase’s components,
- “text chunking”: the elimination of text components that cannot be phrases, so that all that is left in the end are the phrases—in the form of “large chunks”.

The goal is to build and maintain a phrase dictionary. If an entry is already stored in the dictionary, it can be directly identified in the document or in a query, respectively.

In the statistical method, a phrase is defined via its frequency of individual occurrence, via the co-occurrence of its components and via the proximity of the components in a document (Croft, Turtle, & Lewis, 1991, 32). Fagan (1987, 91) works with five parameters:

- Phrase Length: maximum number of components in a phrase (Fagan exclusively considers two elements),
- Environment: Text area in which the parts occur jointly, e.g. within a sentence or a query,
- Proximity: maximum number of words that may occur in the selected environment between the components of a phrase (Fagan chooses 1 while excluding stop words, but higher values up to around 4 may also be viable),
- Document Frequency of the Components (DFC): number of documents in which the component occurs at least once. Definition of a threshold value from which onward a component may become the first link of a phrase (in Fagan’s example of a database, that value is 55); if both components may potentially be a first link, their original order is kept within the text,
- Document Frequency of the Phrase (DFP): number of documents containing the phrase. Definition of a lower and an upper threshold value (Fagan experiments with a lower value of 1 and an upper value of 90).

A phrase may never consist of two identical elements.

Title: Word-Word Associations in Document Retrieval Systems

Word	Sentence N°	Word N°	DFC	First Link
word	1	1	99	yes
word	1	2	99	yes
associ	1	3	23	no
docu	1	5	247	yes
retriev	1	6	296	yes
system	1	7	535	yes

recognized phrases:

retriev system—(system retriev)

docu retriev—(retriev docu)

word associ

docu associ

Figure C.3.2: Statistical Phrase Building. Source: Fagan, 1987, 92 (slightly modified).

Figure C.3.2 demonstrates the procedure on the example of a title. The word forms run through a stemming routine, stop words being excluded. The column DFC shows the word frequency in the sample database. All frequencies except that of “associ” surpass the threshold value of 55, and are thus qualified as first phrase links. In the example, four phrases are recognized: three of which are useful and one of which (“docu associ”) is useless.

This initial phase of phrase identification may be followed by a second phase of normalization (Fagan, 1987, 97):

Normalization is beneficial, since it makes it possible to represent a pair of phrases like *information retrieval* and *retrieval of information* by the single phrase descriptor *inform retrief*. Similarly, the phrases *book review* and *review of books* can both be represented by the phrase descriptor *book review*.

In Figure C.3.2, normalization is indicated via parentheses.

This second phase in particular is very prone to errors. Statistically generated phrases tend to overconflate. “college junior” and “junior college” are statistically summarized despite not being equivalent in meaning (Strzalkowski, 1995, 399). Croft, Turtle and Lewis (1991, 34) point out:

Of course, two words may tend to co-occur for other reasons than being part of the same phrasal concept.

A procedure for chunking text components into phrases is introduced in a patent by LexisNexis (Lu, Miller, & Wassum, 1996). A strongly enhanced stop list, which simultaneously works as a decomposition list, is used to separate the text into larger chunks. After all one-word terms have been excluded, we are then left with the potential phrases. Apart from general stop words, the stop list comprises all auxiliary verbs,

adverbs, punctuation marks, apostrophes, various verbs and some nouns. The stop words mark incisions into the continuous text. Let us consider a short sample (Lu, Miller, & Wassum, 1996, 4-5):

Citing what is **called** newly **conciliatory comments** by the **leader** of the **Irish Republican Army's political wing**, the **Clinton Administration** announced **today** that it would issue him a **visa** to attend a **conference** on **Northern Ireland** in **Manhattan** on **Tuesday**. The **Administration** had been **leaning** against **issuing** the **visa** to the **official**, **Gerry Adams**, the **head** of **Sinn Fein**, **leaving** the **White House** caught between the **British Government** and a **powerful bloc** of **Irish-American legislators** who favored the **visa**.

The stop words are kept in normal type, while the resulting text chunks are set in bold. Since one-word terms (such as “citing”, “called”, “leader” etc.) cannot be phrases, they do not enter into the chunk list. An exception is formed by those words (such as *IRA*) that consist only of capital letters and thus, in all likelihood, represent acronyms. Recognized acronyms are regarded as synonymous with their longhand form (in this case: *Irish Republican Army*).

Multi-word concepts whose components always begin with a capital letter (e.g. *Clinton Administration*) are noted as phrases. Personal names with more than one component are compared with pre-existing lists. If the name occurs in the list, it will be allocated to the document. Name candidates that do not occur in lists are processed further in subsequent steps (see below). Phrase candidates that contain lower-case words and that only occur once in the text (such as “conciliatory comments”, “political wing”, “powerful bloc” and “Irish-American legislators”) should be skipped. If there are known synonyms for phrases, the different words will be summarized into one concept.

In our text, the following phrases emerge:

Irish Republican Army,
Clinton Administration,
Northern Ireland,
Gerry Adams,
Sinn Fein,
White House,
British Government.

Recognized phrases enter a phrase dictionary when they occur in more than five documents.

Name Recognition

Proper names are concepts that refer to a class containing precisely one element. Such a class is called a “named entity”. This includes all personal names, place names,

names of regions, companies etc., but also, for instance, names of scientific laws (e.g. the “Second Law of Thermodynamics” or “Gödel’s Theorem”). Such named entities should have already been recognized as units during the phrase building phase. In this section, we are dealing with “named entities” that unequivocally need to be recognized as such. However, name candidates of similar syntactic structure are often ambivalent: on the one hand, they refer to precisely one name, and on the other hand, they don’t. Wacholder, Ravin and Choi (1997, 202-203) provide some good examples. The *Midwest Center for Computer Research* is the name of precisely one organization, whereas the analog form, *Carnegie Hall for Irving Berlin*, contains two names: *Carnegie Hall* and *Irving Berlin*. When institutions use place names, these can form a part of their actual name (*City University of New York*) or merely serve as an informative addendum (*Museum of Modern Art in New York City*). Ambiguities in linked phrases are particularly complex: The components of *Victoria and Albert Museum* as well as of *IBM and Bell Laboratories* appear to be identical. However, the “and” in the first phrase is a part of the name, whereas the “and” in the second phrase is a conjunction that links two distinct names. The possessive pronoun can point to a single name (*Donoghue’s Money Fund Report*), or it can point to two of them (*Israel’s Shimon Peres*). For Wacholder, Ravin and Choi (1997, 203), it is clear that

(a)ll of these ambiguities must be dealt with if proper names are to be identified correctly.

Ambiguities also arise from the fact that people—particularly in earlier centuries—often did not know how to spell their names. Bowman (1931, 26-27) provides an example from the fifteenth century, in which four variants of one and the same name are noted in immediate succession:

On April 23rd, 1470, Elizabeth Blynkkynesoppe of Blynkkynsoppe, widow of Thomas Blynkyensoppe of Blynkkensope received a general pardon.

Homonyms and synonyms give rise to additional problems for the recognition of names. Homonyms arise when two different named entities share the same name. Synonyms are the result of one named entity being known under several different names: this can be due to name changes after marriage, linguistic variants in transliteration, different appellations in texts, typing errors or pseudonyms. Borgman and Siegfried (1992, 473) note, concerning the problem of synonymy:

Personal name matching would be a trivial computational problem if everyone were known by one and only one name throughout his or her life and beyond, and if that name were always reported identically. Neither assumption is valid, and the result is a complex computational problem.

Names have both inner characteristics (such as “Mr.” for personal names in English texts) and outer characteristics (specific context words depending on the type of

name). Knowledge of the correct internal and external specifics is a necessary aspect of any automatic name identification. McDonald (1996, 21-22) sketches his approach as follows:

The context-sensitivity requirement derives from the fact that the classification of a proper name involves two complementary kinds of evidence, which we will term ‘internal’ and ‘external’. *Internal evidence* is derived from within the sequence of words that comprise the name. This can be definitive criteria, such as the presence of known ‘incorporation terms’ (“Ltd.”, “G.m.b.H.”) that indicate companies; or heuristic criteria such as abbreviations or known first names often indicating people. ...

By contrast, *external evidence* is the classificatory criteria provided by the context in which a name appears. The basis for this evidence is the obvious observation that names are just ways to refer to individuals of specific types (people, churches, rock groups, etc.), and that these types have characteristic properties and participate in characteristic events. The presence of these properties or events in the immediate context of a proper name can be used to provide confirmation or critical evidence for a name’s category.

An important internal aspect for names is the capital spelling, in Latin script, of the first letter of some components (but not of all components, as *Wernher von Braun* exemplifies). The inner characteristics of names are often represented via indicator words. If such an indicator appears before or after a word, or within a previously identified phrase, name recognition will be initiated. Lu, Miller and Wassum (1996) provide lists of indicator words for companies (“Bros.”, “Co.”, “Inc.” etc.), public institutions (e.g. “Agency”, “Dept.”, “Foundation”), products (“7Up”, “Excel”, “F22”) and personal names. Indicators for personal names are first names. If a name-type-specific indicator occurs within a phrase, all phrase components after the indicator (for company names) or, for personal names, all phrase components before the indicator, will be deleted (Lu, Miller, & Wassum, 1996, 11-12). If a text contains the word sequence *charming Miranda Otto* multiple times, this sequence will become a phrase candidate. The indicator “Miranda” leads to the recognition that “Otto”, in this case, is a surname. The program deletes “charming” and saves *Miranda Otto*.

Homonymous names cannot be differentiated on the basis of internal characteristics alone. A personal name like “Paul Simon” is correctly formed, but it may refer to at least three individuals: a singer, a politician and a scientist. At this point, name identification is supplemented by methods that process external name characteristics. Fleischmann and Hovy’s (2002) approach uses subtypes for the individual name types, e.g. the classes of athlete, politician, businessman, artist and scientist for personal names. It analyzes whether class-specific words co-occur with the personal name in a shared text environment. Class-specific training documents allow for the extraction of relevant words, e.g. “star” and “actor” for the artist class or “campaign” for the politician class. The proximity between the personal name and the “typical” words is variable: it begins at one, then (if no results are found) successively increases in distance. In the sentence fragment “... of those donating to Bush’s campaign was actor Arnold Schwarzenegger ...”, with a minimum proximity of one, “Bush” will

be assigned to the politician class and “Arnold Schwarzenegger” to the artist class (which was true at the time of writing) (Fleischmann & Hovy, 2002, 3).

Where present, authority files are helpful. Large libraries in particular offer lists with norm data on certain named entities, e.g. the Library of Congress (LC Authorities) or the German National Library (PND: Personennamennormdatei / Personal Name Authority File).

Compound Decomposition

Compounds occur in varying frequencies across different languages. In languages with high compound frequencies, decompounding leads to better retrieval results (Airio, 2006). Compounds are essential in the German language, but the English language also features them and must be taken into consideration (“blackboard”, “gooseberry”, “psychotherapy”). If we regard phrases as compounds, we will have to decide which phrase components can be meaningfully searched on their own as well (e.g. only the second part of “high school” makes a good search argument, whereas both parts of “gold mining” do). Even artificial languages, such as specialist thesauri, feature multi-word concepts. The question arises as to which parts of descriptors that are compounds can be meaningfully searched on their own (Jones, 1971). Naturally, any recognized named entities must be excluded from decompounding. In the following sections, we will mainly discuss the German language, since it features a lot of compounds.

The decomposition of compound terms into their components can both raise the recall of a query and—undesirably—lower its precision. Let us regard three documents (Müller, 1976, 85):

- A: Ein *Brand* in einem *Hotel* hat acht Menschenleben gefordert.
(A *fire* in a *hotel* resulted in eight deaths.)
- B: Acht Menschen bei *Hotelbrand* getötet.
(Eight killed in *hotel fire*.)
- C: Das *Hotel*, in dem acht Touristen ihren Tod fanden, verfügte über keinen ausreichenden *Brandschutz*.
(The *hotel* in which eight tourists lost their lives had insufficient *fire* protection.)

When searching for “Hotelbrand” (“hotel fire”), retrieval systems without compound processing will only retrieve document B, whereas those featuring decompounding will extend recall to incorporate all texts—A, B and C.

Now consider these two documents (Müller, 1976, 85):

- D: Das *Golfspiel* ist in den Vereinigten *Staaten* ein verbreiteter Sport.
(*Golf* is a popular sport in the United *States*.)
- E: Die *Golfstaaten* senkten gestern den Ölpreis.
(The *Gulf States* lowered oil prices yesterday.)

Here compound processing for a query “gulf state” (“Golfstaat” in German) leads to increased ballast and hence, lower precision. An elaborated form of decompounding is thus required to foster recall-heightening aspects while suppressing precision-lowering variants.

Epenthesis are irregular in the German language. There are compounds with and without a linking “s” (e.g. “Schweinsblase” / “pig’s bladder” and “Schweinkram” / “filth”), using singular and plural forms (“Schweinebauch” / “pork belly”), and sometimes segments are abbreviated (“Schwimmverein” / “swimming team” as a compound of “Schwimmen” and “Verein”). If a dictionary containing the basic forms as well as previously recognized compounds with their various flectional endings is available, the longest respective matches can be marked from the right and the left via character-oriented partition. Here we also need a list of linking morphemes, i.e. those characters that occur between the individual components (such as -s- or -e- in German; Alfonseca, Bilac, & Pharies, 2008, 254). To demonstrate, we will now decompose the compound “Staubecken” (“Stau/ecken” / “reservoir” or “Staub/ecken” / “dusty corners”), going from right to left:

<i>Partition</i>	<i>Analysis</i>	
Staubecke-n	Staubecke	no entry
Staubeck-en	Staubeck	no entry
Staubec-ken	Staubec	no entry
Staub-e-cken	Staub	entry: Imperative Singular of “Stauben” (“to make dust”)
	cken	no entry rule out: “Stauben”
Staub-ecken	Staub	entry: Nominative Singular of “Staub” (“dust”)
	ecken	entry: Nominative Plural of “Ecke” (“corner”) note: Staub note: Ecke
Stau-b	Stau	entry: Nominative Singular of “Stau” (“traffic jam”)
	b	no entry rule out: “Stau”
Sta-ub	Sta	no entry
St-au	St	no entry
S-taub	S	no entry.

The partition leads to “Staub” (“dust”) and “Ecke” (“corner”). Repeating the decomposition from left to right, we will get a different result. This time, we will find the words “Stau” (“traffic jam” or “congestion”) and “Becken” (“basin”). To be sure of retrieving all decomposition variants, one would always have to partition from both directions (and would often be yielded false words due to overpartitioning). Such a procedure can be useful for compounds with more than two components, however,

in order to retrieve the shorter compounds contained within the larger ones. This is shown by the example of “Wellensittichfutter” (“budgie food”):

Wellensittich (retrieved as the longest match via right-hand partitioning)
 Futter
 Sittichfutter (retrieved as the longest match via left-hand partitioning)
 Welle
 Sittich.

“Welle” (“wave”) is an example of overpartitioning. This can be avoided by establishing a partial block on partitions for the lexeme “Wellensittich” (valid: “Sittich”; invalid: “Welle”). Any absolute blocks on partitions (e.g. for “Transport”, which in German has the character sequence of “Tran” (fish oil) and “Sport” (sports) must also be added to the dictionary entry. Hence, there are four cases that must be distinguished depending on which components are meaningful on their own in a given context (S) and which are not (N) (Müller, 1976, 124):

S//S	Both parts are meaningful words after partitioning (“Sittichfutter”),
N//S	Only the second link is a meaningful word after partitioning (“Wellensittich”),
S//N	Only the first link is a meaningful word after partitioning (“Auftraggeber” / “contracting authority”),
N//N	None of the links is a meaningful word after partitioning (“Transport”).

Of course dictionaries cannot predict all possible combinations of words in compounds. Faced with the threat of overpartitioning due to an excessive reliance on dictionaries and rules, it appears sensible to draw on further methods. In a patent for the company Microsoft, Jessee, Eckert and Powell (2004) describe a combined dictionary-statistics approach. This approach counts how often the basic forms, or stems, of the components of compound terms occur in that term’s environment. Additionally, the program notes how frequently these components are taken from the beginning and ending of the compound respectively. It is assumed that some components are more suited to be head segments, while others make better modifying segments. Now there are several possibilities for distinguishing “Staub/ecken” (“dusty corners”) from “Stau/becken” (“reservoir”): (1) if the word “Staub” (“dust”) occurs in the context of “Staubecken”, for instance, the case for a partitioning into “Staub” and “Ecken” is stronger than for the other variant; (2) if a word component—let us say “Stau” (which also means “damming”)—frequently appears at the beginning of a compound (e.g. in “Staumauer” / “dam wall”) within the context, the case for “Stau” and “Becken” will be stronger than for the other variant; (3) if a component (“Becken” / “basin”) frequently occurs at the end of a compound (e.g. in “Wasserbecken” / “water basin”)

within the context, the case for “Stau” and “Becken” will be stronger than for the other variant.

Finally, let us discuss a purely statistical approach to compounding. Driessen and Iijin (2005) require no intellectually compiled and maintained dictionaries, merely the data in an inverted file. Their algorithm works with the number of documents that contain the compound t_i ($DT_{(i)}$), as well as the number of texts containing all components of this compound individually ($DP_{(i,j)}$). A word component can mean any sequence of letters that corresponds to an entry in the index. Only when the inequality

$$DT_{(i)} < 3 * DP_{(i,j)}$$

is fulfilled can the compound t_i be partitioned. Driessen and Iijin exemplify their procedure via the Dutch compound “Basketbalkampioenschappen” (“basketball championship”). The only possible partition in the test database is into “basketbal” and “kampioenschappen”, since both words co-occur in documents more than three times as often as “Basketbalkampioenschappen”. Other partitioning variants (e.g. into “basketbal”, “kampioenschap” and “pen”) fail because their components co-occur too seldomly (in this example: never), thus rendering the value of the right-hand side of the inequality very low (in the example: zero).

Semantic Fields

Words are the linguistic expressions of concepts. A concept (Ch. I.3) is—as an abstraction—always a set (in the set-theoretical meaning of the word) whose elements are objects. These objects display certain characteristics. “Object” is very broadly defined here: in addition to real objects (e.g. a chair), it also refers to non-real objects (e.g. of mathematics). Concepts are not homonymous or synonymous, like words; they are additionally interlinked with other concepts via relations (Ch. I.4). Such relations occur both in natural languages (such as English, German etc.) and in technical languages (e.g. of chemistry, medicine or economics).

Concepts as well as their relations are noted in knowledge organization systems (KOSs). Natural-language KOSs are compiled in the context of linguistics, whereas the creation and maintenance of specialist KOSs is the task of information science. Our example for a natural-language KOS is WordNet, which is a lexical database for the English language. The selection criterion is the fact that WordNet has already been broadly addressed in different research discussions in the context of information retrieval. For a specialist-language KOS, we can think of the medical KOS MeSH (Ch. L.3), for example. Working with concept-oriented retrieval systems comprises two aspects:

- The clarification of homonymous and synonymous relations and
- The incorporation of the semantic environment into a query.

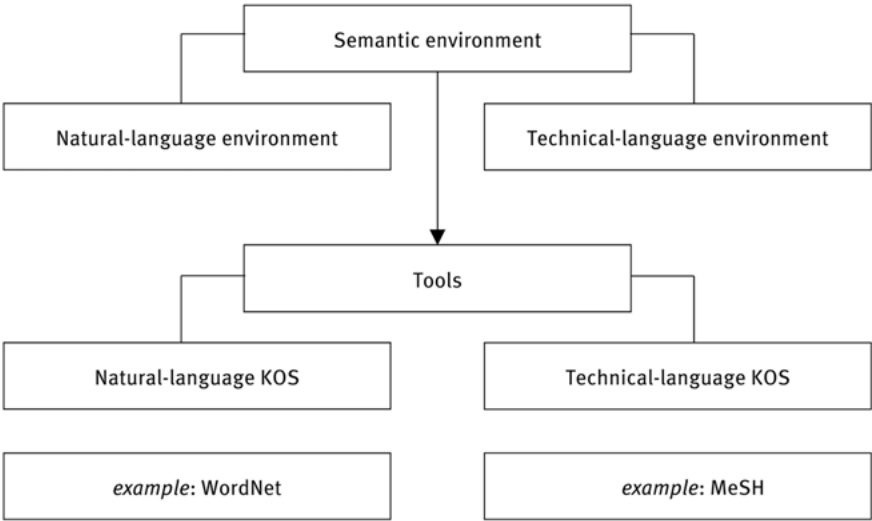


Figure C.3.3: Natural- and Technical-Language Knowledge Organization Systems as Tools for Indexing the Semantic Environment of a Concept.

The dictionary of a language—natural as well as specialist—can be regarded as a matrix, according to Miller (1995). The columns of this matrix are filled with all of a language’s words, whereas the rows will contain the different concepts (Table C.3.1). Our language in Table C.3.1 has n words and m semantic units (concepts). Most of the $n \times m$ cells are empty. The ideal case of a “logically clean” language would show every row and every column to only ever contain one pair $P(i,j)$. This is not the case for most natural and many specialist languages. Multiple entries in the same column, such as the pairs $P(3,1)$ and $P(3,2)$ in our example, express homonymy: the same word describes more than one concept. Several entries in the same row, such as the pairs $P(2,4)$ and $P(4,4)$ form synonyms: the same concept is expressed via different words.

Table C.3.1: Word-Concept Matrix.

		Words							
Concepts		W(1)	W(2)	W(3)	W(4)	...	W(i)	...	W(n)
	C(1)			P(3,1)					
	C(2)			P(3,2)					
	C(3)								
	C(4)		P(2,4)		P(4,4)				
	...								
	C(j)								
	...								
	C(m)								

Whenever several entries occur in the same row or column, retrieval systems enter into a “clarifying dialog” with their users or employ automatic procedures. For natural languages, the user will be asked whether he really wants to adopt all synonyms (rows) into his query. Since the synonyms, e.g. in WordNet, are not entirely identical, and the user might set great store by nuances of meaning, such an option is of essential value. For specialist languages with controlled vocabularies, further processing via preferred terms is a must. However, the user can decide at this point that he no longer wants to work within the controlled vocabulary but in the title, abstract or full text instead. These fields might then indeed contain the stated synonyms (e.g. the non-descriptors in a thesaurus). If there are homonyms for the search term (several entries in a column), the user (or, in the case of a purely automatic procedure: the system) must necessarily make a decision and choose precisely one concept. When using specialist KOSs during indexing, this situation is entirely unproblematic. An exact match is made possible by supplementing the descriptor with a homonym qualifier—a bracket term, as in *Java <programming language>*—that is recorded in the documentary unit. The case is different in environments where no intellectual indexing has been performed, e.g. in the entirety of the WWW. Here, the system has no exact information about which of the homonyms could be referred to in a specific document. Homonym clarification (however vague in this case) depends upon statements regarding the semantic environment of the concepts (see below).

The second aspect regards the exploitation of a concept’s relations to other concepts in order to enhance a query. Several information providers dispose of corresponding functionalities for incorporating concepts from the semantic environment of a starting concept into the search argument. Let us exemplify such a procedure via an example: for instance, we might be searching for literature on the subject *marketing for service companies*. We are not only interested in general literature on this subject but also in specific aspects, such as narrower concepts concerning marketing (e.g. communication policies) as well as service companies (e.g. business consulting). A search result informing us about the communication policy of business consultants will thus satisfy our information need. Graphically, we can imagine the research as presented in Figure C.3.4: we want an AND link between *Marketing*, or a hyponym (NT) thereof, and *Service Company* or a hyponym thereof. In a search that does not support semantic environment searches the user would have to strenuously locate and enter all hyponyms by himself. A system with semantic environment search functionality simplifies the process enormously.

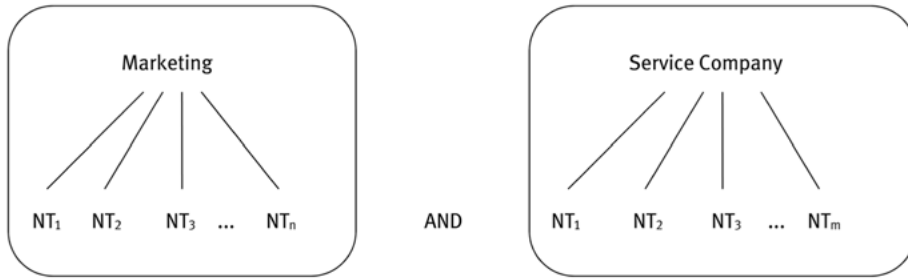


Figure C.3.4: Hierarchical Retrieval. Source: Modified Following Gödert & Lepsky, 1997, 16. NT: Narrower Term.

Natural-Language Semantic Environments: WordNet

We will discuss specialist-language KOS at length (Ch. L.1 through L.6)—at this point, however, we are going to concentrate on linguistic knowledge by sketching a portrait of WordNet (Miller, 1995; Fellbaum, 2005). In WordNet, a “synset” (set of synonyms) summarizes words that express exactly one concept. Synonyms in the dictionary of WordNet are, for instance, the words “car”, “automobile”, “auto” etc. In these synsets, the individual words that make up the sets of synonyms are regarded as equivalent. Words that are spelled the same but refer to different concepts (homonyms) are displayed and assigned to their respective synsets. After a search in WordNet Search (Version 3.1), it transpires that *car* is the name for five different concepts (or synsets):

- (1) {car, auto, automobile, machine, motorcar},
- (2) {car, railcar, railway car, railroad car},
- (3) {car, gondola},
- (4) {car, elevator car},
- (5) {cable car, car}.

WordNet’s synsets (around 117,000 at this time) are separated by word class. WordNet works with nouns, verbs, adjectives and adverbs. There are different relations for the various word forms. Nouns (Miller, 1998) are located in two hierarchical relations throughout various levels (hyponymy: “is-a relation”; and meronymy: “part-of relation”; see Ch. I.4). In contrast to nouns, verbs have much flatter hierarchical structures. Fellbaum (1998, 84) talks of “entailment”, which she graduates by time relation. In the case of simultaneity, we have the two hierarchical relations of +troponymy (“coextensiveness”, as in *talking* and *lisp*) and -troponymy (“proper inclusion”, as in *sleeping* and *snoring*). For activities that follow each other chronologically, Fellbaum differentiates between backward-facing preconditions (e.g. *knowing* and *forgetting*) and causes (such as *showing* and *seeing*). Adjectives, according to K.J. Miller (1998), are not located in any hierarchical-semantic structure but, preferentially, in similarity relations as well as antonymy. For instance, the relations “nominalizes” (a

noun nominalizes a verb) and “attribute” (a noun is an attribute for which an adjective has a value) are used beyond the borders of word classes. Concepts and their relations to each other span a “semantic network”. Using the example of *car* (synset 1), we show a small excerpt of this network (Navigli, 2009, 10:9) (Figure C.3.5).

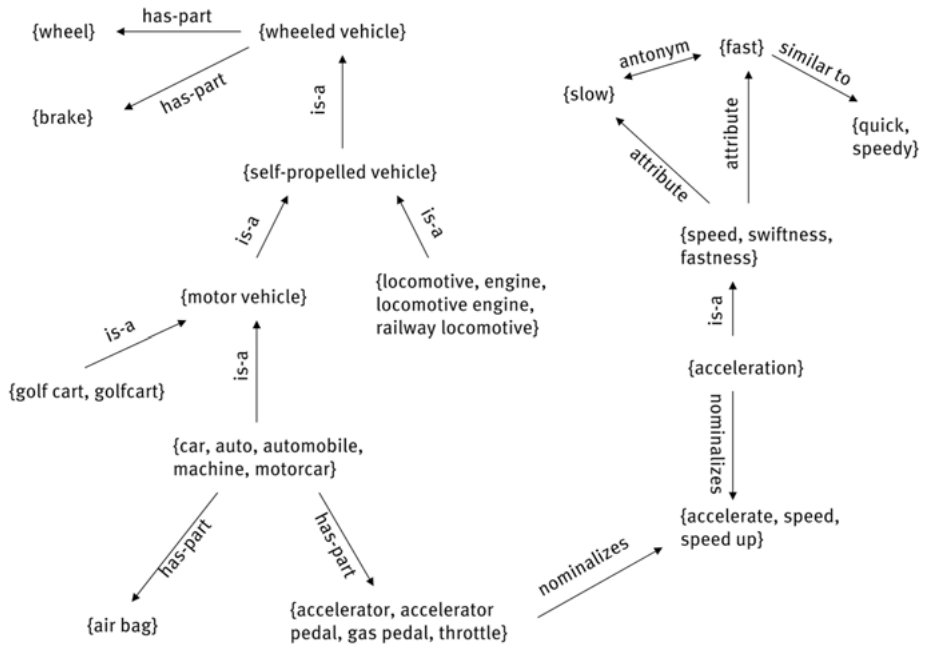


Figure C.3.5: Excerpt of the WordNet Semantic Network. Source: Modified from Navigli, 2009, 10:9.

The use of WordNet in retrieval systems will yield bad search results if WordNet is only used automatically (Voorhees, 1993; Voorhees, 1998). The system on its own is simply unable to select the correct synset for each homonym. The opposite is true, however, when the synsets to be used for search purposes are intellectually selected. Voorhees (1998, 301) summarizes her experiments with WordNet:

(T)he inability to automatically resolve word senses prevented any improvement from being realized. When word sense resolution was not an issue (e.g., when synsets were manually chosen), significant effectiveness improvements resulted.

This in turn leads us to enter into a man-machine dialog in order to select the correct concepts.

Semantic Similarity

How can the proximity—that is to say, the semantic similarity—between two concepts be quantitatively expressed? There are several methods (Budanitsky & Hirst, 2006), from among which we will introduce that of “edge-counting”. The basis is a semantic network, formed from a linguistic (such as WordNet, Figure C.3.5) or specialist perspective (such as MeSH). Concepts form the nodes whereas relations form the edges (Lee, Kim, & Lee, 1993; Rada et al., 1989). Resnik (1995, 448) expresses it as follows:

A natural way to evaluate semantic similarity in a taxonomy is to evaluate the distance between the nodes corresponding to the items being compared—the shorter the path from one node to another, the more similar they are. Given multiple paths, one takes the length of the shortest one.

In order to calculate the proximity between two concepts in a KOS we count the number of paths that must be run through in order to get from Concept C_1 to Concept C_2 in the semantic web as quickly as possible. Consider the following excerpt from a KOS (Fig. C.3.6).

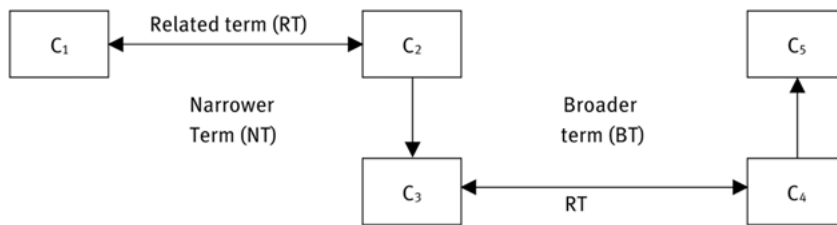


Figure C.3.6: Excerpt from a KOS.

In this semantic web, the shortest distance from C_1 to C_5 is four.

To express a semantic bond strength between the concepts via the respective relations, the relations must be weighted differently. The retrieval software Convera (Vogel, 2002), for instance, uses these standard settings:

- (Quasi-)Synonymy: 1.0,
- Narrower term: 0.8,
- Broader term: 0.5,
- Related term (“see also”): 0.4.

The system administrator can modify these settings and define as many further relations (with the same bond strength) as necessary. When applied to our example, the standard requirements yield the following picture (Fig. C.3.7).

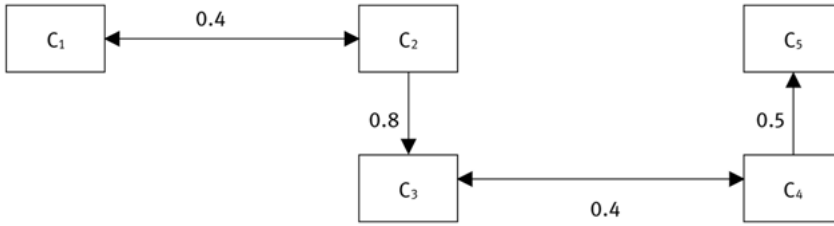


Figure C.3.7: Excerpt from a KOS with Weighted Relations.

There no longer is a 100% bond between C₁ and C₂, merely one of 40%, whereas C₂ and C₃ are linked at 80%, C₃ and C₄ at 40% and, finally, C₄ and C₅ at 50%. The bond strength between C₁ and C₅, then, is:

$$1/0.4 + 1/0.8 + 1/0.4 + 1/0.5 = 8.25.$$

Such proximity values are important for query expansion in so far as a threshold value can be defined: all concepts linked to an original concept are then allocated to the latter if their value is lower than or equal to the threshold value. Here it must always be noted that query expansion for non-transitive relations is only possible by one path. A further field of application for semantic proximity lies in the disambiguation of homonymous terms.

Disambiguation of Homonymous Terms

Recognizing the respective “right” semantic unit when dealing with homonymous words is important at two points in information retrieval:

- When a user enters a search atom for which there exist multiple concepts.
- When searching for appropriate documents for non-indexed text material following step 1 (while knowing the correct semantic unit).

The object is to solve ambiguities between words (Navigli, 2009). Such a task is divided into two steps (Ide & Véronis, 1998, 3). First we need a list of all the different concepts that can be expressed by a given word. The second step involves the correct allocation of (ambiguous) word and concept, both via the document context and via the relations of a KOS. Ambiguous user entries (*Java*) either lead to a man-machine dialog in the course of which the user marks the correct meaning of his search term, or to an automatic disambiguation. The more search atoms there are, the more elaborate the latter will be. If someone searches for *Java Programming Language*, the case is clear: they are not interested in coffee or Indonesian islands. Here, the machine exploits its knowledge of semantic similarity. The proximity between *Java* and *Programming Language* is one (*Java*, we will assume, is a hyponym of *Programming Language*),

whereas *Programming Language*'s proximity to *Java <Coffee>* and *Java <Indonesia>* is far greater. The system follows the shortest path between two concepts.

The case is far more complex for one-word queries. If statistical information is available concerning the probability of homonymous words occurring in a corpus, and if one of the concepts dominates the others, this frequent meaning could be automatically selected or at least placed high up in a list. If a user has already formulated several queries over the course of a longer session (or if we keep protocols about all of a user's queries), it will be possible to extract the required context statements from the previous queries. If, for instance, a user searches for "Vacation Indonesia", then for "Vacation Bali" and, finally, for "Java", we will be able to exclude the programming language and the coffee as unsuitable. Here, too, we exploit data about semantic similarity (in this case using the terms of the previously performed queries).

In the following, we claim that we have found the correct search term—using automatic procedures or a dialog. If a retrieval system uses a KOS and works with controlled terms, the next step will seamlessly lead to the correct documents. Homonym qualifiers of the sort *Java <Island>*, *Java <Coffee>* and *Java <Programming Language>*, which can be used in indexing as well as retrieval, solve all ambiguities.

The case is different for automatic indexing, and hence for all Web documents. Here, further terms are gleaned from the context of a word in an original document—e.g. a text window of two to six words (Leacock & Chodorow, 1998) from either side of the original term. At this point the measurement of semantic similarity comes back into play. Let us assume that we are searching for *Washington*, the U.S. State (and not the capital). Let the proximity between *Washington <State>* to *Olympia* (the state capital) in our KOS be two paths, and its proximity to *District of Columbia* ten paths. The competing term *Washington, DC* thus has a semantic proximity of one path to *District of Columbia*. If the text window contains both, *Washington* and *Olympia*, the document will be marked as a hit. If another document contains *Washington* and *District of Columbia*, on the other hand, that document is irrelevant. The "winners" are documents which contain words that are connected to the search term via short paths.

Conclusion

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- Concepts can consist of several words. The object is to identify multi-word concepts as coherent phrases as well as to meaningfully partition multi-word concepts (compounds) into their component parts. A special problem is the correct identification of named entities.
 - Recognized phrases, named entities (perhaps separated by type: personal, place, institutional and product names), compound parts and concepts all form their own inverted files, i.e. they can be pointedly addressed by the user if necessary.
 - Statistical methods of phrase building count the co-occurrence of the phrase components in text environments and use certain rules to identify the phrases. Procedures for "chunking" texts enhance general stop lists into partition word lists. Text components between the partition words are phrase candidates. If phrase dictionaries are available (intellectually compiled or as the

result of automatic procedures), phrase identification in a text occurs via alignment with the dictionary entries.

- Named entities are concepts that refer to a class containing exactly one element. The automatic identification of named entities occurs via name-internal characteristics as well as via external specifics (of the respective text environment). The internal name characteristics are pursued via indicator terms and rules. For personal names, first names are accepted as suitable indicators, whereas for companies it is designations concerning legal form. The identification of homonymous names depends upon an analysis of the external name specifics. From the occurrence of specific words in the text environment, a link is made to their affiliation with name types (for personal names: the profession of the individual in question, for example).
 - When decomposing compounds (or recognized phrases), we are faced with the problem of how to only extract the meaningful partial compounds or individual words from the respective multiword terms. When partitioning individual characters of a compound (from left to right and vice versa), the longest acceptable term will be marked in each case. “Acceptable” means that the word (excluding interfixes) can be found in a dictionary.
 - Intellectually compiled dictionaries determine, for every compound, into which parts a multiword term can be partitioned. However, this cannot be done for every compound, since their quantity is infinitely large (at least in principle). In addition, there are ambiguities that must be noted (the “Staubecken” problem).
 - The use of concepts (instead of words) in information retrieval is of advantage for two reasons: firstly, homonymous and synonymous relations are cleared up, and secondly, it becomes possible to incorporate the semantic environment of a search term into the query.
 - The word-concept matrix identifies both homonyms (words that refer to several concepts) and synonyms (concepts that are described via several words).
 - Concepts in KOSs (e.g. in WordNet or MeSH) form a semantic network via their relations.
 - The semantic similarity between two concepts within a KOS can be quantitatively described by counting the paths along the shortest route between the two concepts.
 - Homonymous words must be disambiguated in retrieval. Ambiguous user queries are cleared up either via further inquiries or—where sufficient information is available—automatically via semantic similarity. If a retrieval system uses a specialist KOS in indexing, the homonym qualifiers for the concepts render the search process unambiguous. In all other cases, words in the text environment will be checked for semantic similarity.
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C.4 Anaphora

Anaphora in Retrieval and Extracting

Natural language often uses expressions that refer to other words (or other denotations). In such cases the term referred to is called the “antecedent”, while the referring term is the “anaphor”. When anaphor and antecedent refer to the same object, we speak of “coreference” (Mitkov, 2003, 267). In the sentences

Miranda Otto is from Australia. She played Eowyn.

“she” is the anaphor and the name phrase “Miranda Otto” is the antecedent. Since both “she” and “Miranda Otto” refer to the same object, this is an example of coreference.

Anaphoric phrases use the following word classes:

- Pronouns
 - Personal Pronouns (he, she, it, ...),
 - Possessive Pronouns (my, your, ...),
 - Reflexive Pronouns (myself, yourself, ...),
 - Demonstrative Pronouns (this, that, ...),
 - Relative Pronouns (the students *who*, ...),
- Nouns
 - Replacement (this woman, the aforementioned),
 - Short Form (the Chancellor),
 - Metaphor (the Queen of the Christian Democrats),
 - Metonymy (the Brynhildr of Mecklenburg),
 - Paraphrasing (the former German minister of the environment),
- Numerals (the first-named, the two countries).

There are also ellipses: instances in which terms are left out entirely since the context makes it clear which antecedent is meant:

- Ellipsis (Egon chopped wood, just like Ralph) [the phrase left out is: “chopped wood”],
- Asyndeton, where conjunctions are omitted (I came, saw, conquered).

Anaphora and antecedents may occur within the same sentence, but anaphora can also cross sentence borders. Under certain circumstances there may be competing resolutions for an anaphor. Let us consider two examples:

(Ex. 1): The horse stands in the meadow. It is a stallion.

(Correct resolution of Ex. 1): The horse stands in the meadow. The horse is a stallion.

(Ex. 2): The horse stands in the meadow. It was fertilized yesterday.

(Correct resolution of Ex. 2): The horse stands in the meadow. The meadow was fertilized yesterday.

In Ex. 1, “it” refers to “the horse”, whereas in Ex. 2 it refers to “the meadow”. Sometimes a personal pronoun (“it”) may resemble an anaphor without functioning like one in the given context. Consider the sentences:

(Ex. 3): The horse stands in the meadow. It is raining.

(Correct resolution of Ex. 3): The horse stands in the meadow. It is raining.

The “it” in Ex. 3 refers neither to “horse” nor to “meadow”. Lappin and Leass (1994, 538) call such “it” forms “pleonastic pronouns”, which must never be confused with anaphora.

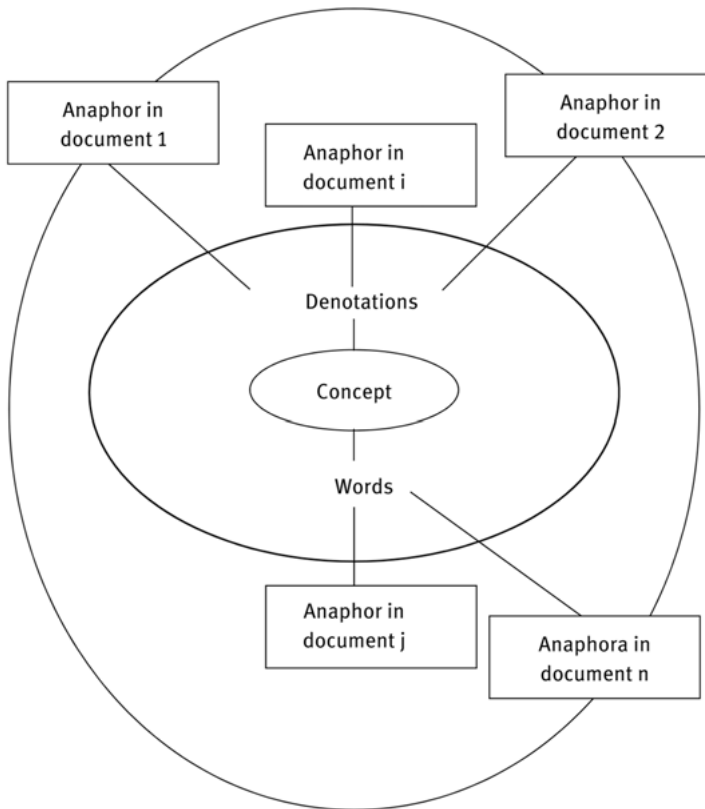


Figure C.4.1: Concepts—Words—Anaphora.

The goal of anaphor resolution is to conflate anaphora and their correct antecedents (Mitkov, 2002; 2003). In the same way that concepts summarize synonymous words, anaphor resolution summarizes words and their anaphora on a text-by-text basis.

Anaphor resolution is important for information retrieval for two reasons. On the one hand, anaphora cause problems for proximity operators (Ch. D.1), and on the

other hand they impact the counting basis of Text Statistics (Ch. E.1). The objective when dealing with proximity operators is to specify the Boolean AND for searches within precisely one sentence or paragraph, or within certain text windows (i.e. within a distance of five words at most). For instance, if we search for *horse w.s. Stallion* (w.s. meaning “within the same sentence”), we will not retrieve the document containing the above Example 1 unless we have resolved the anaphor correctly.

One of the core values of text statistics is the frequency of occurrence of a word (in word-oriented systems) or of a concept (in concept-oriented systems) in a document. In order to be counted correctly, the words (i.e. all words that express a concept) would have to be assigned their anaphora.

In knowledge representation, anaphora play a role during automatic Extracting (Ch. O.2). There, it is essential to select important sentences in a text for use in a Summary. It is possible that the selected sentences contain anaphora but not their antecedents, rendering the sentence on its own meaningless to the reader. In an (intellectually created) “ideal” system of anaphora resolution, the processing of anaphora produces better Extracts (Orasan, 2007). Anaphora resolution is regarded as a difficult enterprise whose success is far from being guaranteed at the moment.

The interplay between concepts, their corresponding denotations and words (more specifically: their lexemes or stems) and anaphora is exemplified in Figure C.4.1. A concept is described via different words; this relation is taken from the word-concept matrix of a knowledge organization system (Ch. I.3). In some cases, the words in the documents are replaced by anaphora; this relation is—at least theoretically—adopted from anaphora resolution.

Anaphora and Ellipses When Using Proximity Operators

Pirkola and Järvelin (1996) compile two parallel corpora: the original texts (of a newspaper article database) on the one hand and those same documents, now including their (intellectually) resolved referent terms, on the other. Identical queries are then posed to both corpora via the use of proximity operators, in order to determine their respective Recall. For most search arguments the study shows at least a slight rise in the Recall of the processed corpus, but the increases in relevant documents are most notable for named entities.

The authors distinguish between anaphora and ellipses (for the reference expressions) as well as between proximity on the level of sentences and of paragraphs (for the operators), respectively. With regard to searches for named entities, they report the following Recall increases:

	<i>Gains via Resolution</i>
Ellipses—Sentence Level	38.2%
Ellipses—Paragraph Level	28.8%
Anaphora—Sentence Level	17.6%
Anaphora—Paragraph Level	10.8%.

The gains are greater for personal names than for names of organizations—e.g. 23.3% for people as opposed to 14.8% for institutions via anaphora resolution and the use of a proximity operator on the sentence level. The most frequent anaphora for personal names are personal pronouns, and the most frequent ellipses are omissions of a person's entire name. Pirkola and Järvelin (1996, 208-209) report:

In newspaper text, a person is usually referred to through a personal pronoun (most often “*he*” and “*she*”), some other pronouns (e.g. “*who*”), substantive anaphora, an apposition attribute (e.g. *president* George Bush), the first name, the family name, or through a nick name (e.g. “*Jackie*”) or epithet (e.g. “*der Eiserne Kanzler*”). In our data, family name ellipsis and the personal pronoun *he/she* (...) were by far the most frequent. There were 308 person ellipses of which 305 were family name ellipses and 3 first name ellipses. Moreover, there were over 100 person name anaphora of which 81 were the pronouns *he/she*.

If a retrieval system offers proximity operators applied to full texts it appears useful to resolve anaphora and ellipses for the named entities, with preference given to personal names. The precondition is that the retrieval system must feature a building block for recognizing named entities.

Anaphora and Ellipses in Text Statistics

While counting the individual words (including their anaphoric and elliptic variants) after a successful anaphora resolution, we will glean—given a sufficiently long text—a higher value than by counting basic forms or stems without anaphora. Even the WDF weight value (within-document frequency weight; Ch. E.1), which relativizes the frequency of occurrence of each individual basic form (or of each individual stem) to the total number of the words, changes. Liddy et al. (1987, 256) describe the point of departure as follows:

In free-text document retrieval systems, the problem of correctly recognizing and resolving subsequent references is important because many of the statistical methods of determining which documents are to be retrieved in response to a query make use of frequency counts of terms. For this count to be a true measure of semantic frequencies, it would appear that the semantically reduced subsequent references should be resolved by their earlier, more fully specified referents in text.

The all-important question is: does anaphora resolution change the retrieval status value of the documents, and thus their output sequence in Relevance Ranking? Only if the answer is “yes” does the use of anaphora resolution make any sense for text statistics at all.

Extensive experiments on anaphora resolution in abstracts of scientific articles were performed at Syracuse University (Liddy et al., 1987; Bonzi & Liddy, 1989; DuRoss Liddy, 1990; Bonzi, 1991). Abstracts from two scientific databases (PsycInfo on psychology and INSPEC on physics) are twice confronted with queries: first with no processing, and then in a version where all anaphora have been intellectually resolved. The experiments do not attest to a general improvement of Relevance Ranking via the use of WDF; the results vary depending upon the specific query, the type of anaphor as well as the database. DuRoss Liddy (1990, 46) summarizes:

Results were mixed and highly inconclusive. ... (R)esolving anaphors had a positive effect for some queries and for some classes of anaphors, but a negative effect for other, and no effect at all for still others. Generally, it appears that resolution of the nominal substitutes and adverb classes of anaphors most consistently and positively improved retrieval performance. Also, PsycINFO documents were more positively effected by anaphor resolution than were INSPEC documents.

It should additionally be noted that personal, possessive and reflexive pronouns always lead to a retrieval optimization—even if only for very few queries (DuRoss Liddy, 1990, 47). For information retrieval, the Syracuse studies show that secure retrieval improvements can only be expected to result from the replacement of nouns (“nominal substitutes” via words such as “above”, “former” or “one”) and from personal, possessive and reflexive pronouns.

We must ask ourselves whether the central themes of a text in particular are often described anaphorically. Imagine, for instance, an author writing about Immanuel Kant but only using this exact name once in his text—he will speak with great rhetorical elegance of “the mentor of critical philosophy” and use many other epithets besides, or speak, simply, of “him”. In this case the WDF of “Immanuel Kant” would be very small; the article would yield no adequate retrieval status value for a search for “Immanuel Kant”. However, if in such a case the anaphora are correctly resolved the term weight will be increased significantly and the article’s place in the ranking will jump up. Indeed, this trend can be empirically verified. Important themes are referenced anaphorically far more often than peripheral ones (DuRoss Liddy, 1990, 49):

Anaphors were used by authors of abstracts to refer to concepts judged integral to the topic being reported on in 60% of their occurrences and to concepts peripheral to the topic in only 23% of their occurrences.

Summarizing the studies by Pirkola and Järvelin, as well as those by the Syracuse team, we will see that improvements, via anaphora resolution, to the performance of retrieval systems are possible in the following cases:

<i>Type of Antecedent</i>	<i>Anaphor</i>
Nouns	nominal substitutes, personal, possessive and reflexive pronouns;
Named entities	personal pronouns, elliptic elision of parts of a name.

Anaphora Resolution in MARS

Following what has been discussed so far we can see clearly that it would pay huge dividends in information retrieval to resolve at least certain anaphora and ellipses. Since elaborate anaphora analyses and resolutions could hardly be processed in an ongoing man-machine dialog due to the processing time necessary, the databases must be duplicated. The document file enhanced via the resolved anaphora and ellipses (including the related inverted files) serve to determine the weighting values of all terms and to search, the original file serves to display and output. Since there must always be the possibility of switching off functions (this also goes for anaphora processing, of course), the original inverted files must also be preserved.

At this point we will only introduce one of the many approaches to resolving anaphora (Mitkov, 2002; 2003), this approach being particularly relevant for the purposes of information retrieval and extracting: we are talking about the MARS (“Mitkov’s Anaphora Resolution System”) algorithm developed by Mitkov (Mitkov, 1998; Mitkov, Evans, & Orasan, 2002; Mitkov et al., 2007). MARS works without any deep world knowledge (it is “knowledge-poor”) and is suited for automatic use in information practice. However, its scope of performance is limited, only taking into account “pronominal anaphors whose antecedents are noun phrases” (Mitkov et al., 2007, 180). The system goes through five steps one after the other (Mitkov et al., 2007, 182). In the first step, the words are separated and their morphological forms determined (particularly gender and number). Step 2 identifies anaphora, pleonastic pronouns are excluded. Here, usage is made of lists of words that are typical for the pleonastic use of the pronouns (such as “to rain”). Step 3 determines the amount of possible antecedents for every identified anaphor. This is done via an alignment of gender and number in the corresponding sentence and in the two immediately preceding it. These can be several competing nouns, but of course only one of these nouns will be appropriate. Determining which one it is will be the task of steps 4 and 5. In step 4, a score is determined for each antecedent candidate expressing its proximity to the anaphor. The (rather extensive) list of antecedent indicators used here has been gleaned empirically. An indicator refers to certain verbs, for example (Mitkov, 1998, 870):

If a verb is a member of the Verb_set = {discuss, present, illustrate, identify, summarise, examine, describe, define, show, check, ...}, we consider the first NP (nominal phrase, A/N) following it as the preferred antecedent.

In the fifth step, finally, the candidate with the highest score is allocated to the anaphor as its antecedent.

Conclusion

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- Natural language uses expressions (anaphora) that refer to other words (antecedents) located elsewhere in the text. Anaphora are referring expressions such as pronouns, while ellipses are omissions of words. Anaphora can occur in the same sentence, but also beyond the borders of sentences.
 - There are two aspects in information retrieval where anaphora and ellipses play a role: the use of proximity operators and the counting basis of information statistics. In knowledge representation, anaphora resolution is important for automatic extracting.
 - When using proximity operators, increased recall via resolved anaphora and ellipses is particularly important for searching named entities. Here the most frequent anaphora are personal pronouns, and the most frequent ellipses are omissions of entire names.
 - It can be observed that authors make heavy use of anaphora, and particularly so when discussing the central themes of a document. If these are to be weighted to any degree of quantitative correctness, the anaphora must be resolved. Text statistics works better when both the replacements for nouns and their personal, possessive and reflexive pronouns are resolved.
 - Anaphora resolution is performed e.g. via the MARS algorithm, which works automatically throughout. This algorithm recognizes pleonastic pronouns and disposes of rules for detecting the (probable) right antecedent when faced with several competing ones.
 - It cannot be claimed with certitude that the resolution of anaphora and ellipses in information retrieval practice is always of advantage. However, anaphora resolution is a necessary step in the automatic development of extracts in text summarization.
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C.5 Fault-Tolerant Retrieval

Input Errors

Texts are not immune to typing errors; this goes for the users' search input as well as for the documents in the databasis. Fault-tolerant retrieval systems perform two tasks: firstly, they identify errors and secondly, they correct them. Input errors on the part of the users must be clarified via a dialog step (such as Google's "Did you mean: ..."; Shazeer, 2002); the correction is thus intellectual. Faulty documents, on the other hand, must always be corrected automatically.

The borders between an input error and a neologism are sometimes hardly to define. There are always new words created in natural languages (such as "cat-womanhood" or "balkanization"), which the system may at first sight misinterpret as errors (Kukich, 1992, 378-379). An analog case involves the creative use of spelling conventions, e.g. for artificial words ("info|telligence"), upper-case letters within a word ("ProFeed") or abbreviations, such as the "electronical" in "eMail".

We will distinguish three forms of input errors:

- Errors involving misplaced blanks,
- Errors in words that are recognized in isolation,
- Errors in words that are only recognized in context.

Blank errors contract neighboring words into one word ("... ofthe ...") or split up a word by attaching one (or several) of its characters to the word following or preceding it ("th ebook"). Observation tells us that blank errors frequently occur in function words, as Kukich stresses (1992, 385):

There is some indication that a large portion of run-on and split-word errors involves a relatively small set of high-frequency function words (i.e., prepositions, articles, quantifiers, pronouns, etc.), thus giving rise to the possibility of limiting the search time required to check for this type of error.

Errors within isolated words have three variants: they are typographical, orthographical or phonetic mistakes (Kukich, 1992, 387). Typographical mistakes ("teh" instead of "the") are merely slips (i.e. scrambled letters), and can be assumed that the writer does in fact know the correct spelling. In the case of orthographical errors ("recieve" instead of "receive"), this is probably not the case. Phonetic errors are a variant of orthographical errors, where the wrong form matches the correct form phonetically, but not orthographically ("nacherly" instead of "naturally" or—e.g. in the language of chatrooms—"4u" instead of "for you"). Damerau (1964, 172) distinguishes the following error types:

	<i>ALPHABET</i>
wrong letter (same word length)	<i>ALPHIBET</i>
omitted letter	<i>ALPABET</i>
scrambled letters (same word length, neighboring letters interchanged)	<i>ALHPABET</i>
additional letter	<i>ALLPHABET</i>
all others (“multiple error”).	

By far the most frequent case is that of a wrong letter in a word of the same length, followed by omitted letters, “multiple errors” and added letters:

- wrong letter: 59%,
- omitted letter: 16%,
- multiple error: 13%,
- added letter: 10%,
- scrambled letters: 2% (Damerau, 1964, 176).

Certain input errors can only be solved by observing the context, since, when taken in isolation, the word appears to be correct. If, for instance, the user wished to write “from” but accidentally typed “form”, the error will only be revealed by the word’s environment in the sentence (“... we went *form* Manhattan to Queens ...”). Here we can distinguish between syntactic and semantic errors (Kukich, 1992, 415). Syntactic errors cause a grammatical error in the sentence (“The study was conducted mainly *be* John Black”), and it should be possible, at least in theory, to label them via a syntax analysis. Semantic errors form a grammatically correct sentence in spite of the input error (“They will be leaving in about fifteen *minuets* to go to her house”). Great effort is required in order to correct such mistakes, and success is not guaranteed (Kukich, 1992, 429). Studies in the context of information retrieval mainly concentrate on identifying and correcting isolated words and phrases. In the following, we will introduce three approaches to providing fault-tolerant retrieval:

- Phonetic approaches (classical: Soundex),
- Approximate string matching (following Levenshtein and Damerau),
- *n*-Gram-based approaches.

Phonetic Approaches: Soundex and Phonix

The phonetically oriented approaches to error correction summarize words that sound the same. I.e., if a misspelled word is pronounced the same way as a correct one, the latter will be suggested to the user for correction. In the year 1917, Russell filed a patent application for his invention of a phonetically structured name register. It describes a procedure which would later, under the name of Soundex, decisively influence both phonetic retrieval of (identically-sounding) names and fault-tolerant retrieval. Russell no longer wished to arrange the names alphabetically into a file card or a book, but via their pronunciation (Russell, 1917, 1):

This invention relates to improvements in indexes which shall be applicable either to “card” or “book” type—one object of the invention being to provide an index wherein names are entered and grouped phonetically rather than according to the alphabetical construction of the names. A further object is to provide an index in which names which do not have the same sound do not appear in the same group, thus shortening the search. A further object is to provide an index in which each name or group of names having the same sound but differently spelled shall receive the same phonetic description and definite location.

In Russell’s procedure, the first letter of a word always survives; it serves as a sort of card box (in Figure C.5.1: the box marked “H”). All other letters are represented via numbers, where the following rules apply:

1. vocals: a, e, i, o, u, y,
2. labials and labio-dentals: b, f, p, v,
3. gutturals and sibilants: c, g (gh is ignored), k, q, x, s (excluding the s-ending), z (excluding the z-ending),
4. dentals: d, t,
5. palatal-fricatives: l,
6. labio-nasals: m,
7. dento- and lingua-nasals: n,
8. dental-fricatives: r.

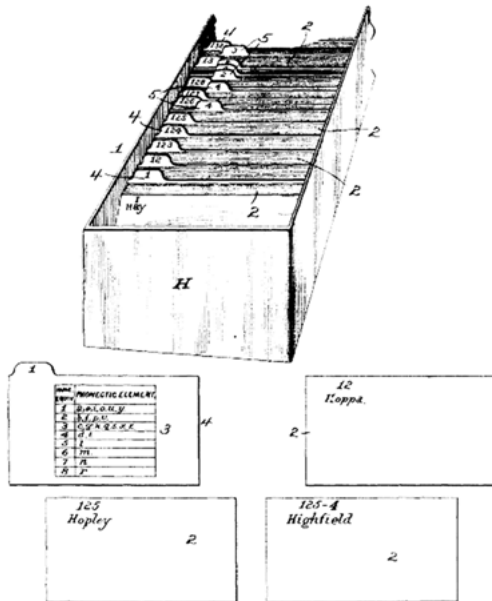


Figure C.5.1: Personal Names Arranged by Sound. Source: Russell, 1917, Fig.

In the case of two adjacent letters that belong to the same phonetic class, the second one will not be recognized (“ball” becomes “bal”). If several vowels occur in a word,

only the first one will be noted (“Carter” becomes “Cartr”). The stated rules allow the creation of a Soundex Code for every word. Consider these two examples from Russell (taken from Figure C.5.1)!

Hoppa	
1. H	oppa
2. H	opa (double allocation of a class)
3. H	op (one vowel only)
4. H	12 (code)
Highfield	
1. H	ighfield
2. H	ifield (gh deleted)
3. H	ifld (one vowel only)
4. H	1254 (code).

Apart from some minor changes, the Soundex Code has remained the same until today. H and W (neither of which are featured in Russell) have been added to the group of vowels, J (likewise ignored by Russell) is located in Group 2 and M and N are now united in the same class (Zobel & Dart, 1996, 167).

Soundex has two problems: it does not apply its own principles to the beginnings of words (and thus fails to link “Kraft” and “Craft” or “night” and “knight”), and it ignores the fact that certain letter combinations may represent different sounds depending on the word (“ough” sounds different in “plough” than it does in “cough”). Phonix (phonetic indexing), created by Gadd (1988; 1990), enhances Soundex by phonetic substitution (Gadd, 1988, 226):

Phonix (...) is a technique based on Soundex, but to which ‘phonetic substitution’ has been added as an integral part of both the encoding and the retrieval processes.

In a first working step, Phonix replaces certain sequences of letters with their phonetically analogous counterparts. Here a strict differentiation is made as to whether the sequence is located at the beginning, at the end or in the middle of the word. “KN” at the beginning is changed to “N” according to the rules, whereas “KN” at the end or in the middle stays the same. Phonix almost always eliminates vowels, W, H and Y as well as non-alphabetical characters (such as hyphens). In the case of several identical consonants following each other back-to-back, only the first one will be noted. The first letter after the phonetic substitution is stored as both letter and code, while the other letters are only represented via numbers:

1: B P; 2: C G J K Q; 3: D T; 4: L; 5: M N; 6: R; 7: F V; 8: S X Z.

We will present the different methods of Soundex and Phonix via an example (Gadd, 1988, 229) (“GH” is skipped).

	<i>Soundex (Russell)</i>	<i>Phonix</i>
KNIGHT	K714	N53
NIGHT	N14	N53
NITE	N14	N53.

Phonix thus works more precisely than Soundex with regard to the words' sound.

The phonetically oriented systems Soundex and Phonix pursue a rather coarse-meshed approach, since both summarize a lot of (often different) words into one code. Additionally, they do not provide for a ranking of several candidates (Kukich, 1992, 395). Phonetic systems are aligned toward a specific language, i.e. Soundex and Phonix toward English. It must further be noted that Soundex was originally conceived for phonetic searches of personal names. The use of input error control and the enhancement of the field of application to include words of all types necessarily lower the algorithms' precision.

On the basis of the phonetic algorithms, fault-tolerant information retrieval works with a second inverted file that records the respective codes. If a user input is ambivalent in terms of its Soundex or Phonix code (i.e. if the system retrieves several different entries in the codes' inverted file), the corresponding words can be presented to the user for selection. This procedure uncovers input errors, both of the user and of the documents' authors. For the purposes of the phonetic methods' precision, this procedure guarantees a complete overview of recognized word variants. However, users may be daunted by its long lists from which to select words.

If only input errors are meant to be considered, the first step will be to search for the search atoms entered; only in the case of very small hit lists (or even zero hits) will the phonetic code be used. If a code in the index matches the code of the (probably misspelled) input word letter by letter, the former word will either be directly used as a search argument or submitted to the user for confirmation.

The Damerau Method

The object of Damerau's and Levenshtein's methods is "*string matching that allows errors*, also called *approximate string matching*" (Navarro, 2001). Damerau (1964) uses letter-by-letter comparison to recognize and correct individual errors (i.e. all except "multiple errors"). The basis of comparison for potential errors is a dictionary. Each word of the dictionary, as well as each word of a text, is represented via numbers on a letter-by-letter basis, with multiply occurring letters only being recorded once. If the codes of word and dictionary entry match, the word is written correctly. If they do not, the error type identification will begin.

First, the number of all (i.e. including multiply occurring) letters of the text word and the dictionary entry will be counted. If the difference is greater than one, the algorithm will stop—the possibility of an individual error is ruled out. In the next step, the

codes are then examined analogously and those pairs that differ from one another in more than two code positions will not be processed further.

The remaining error candidates are compared letter by letter, with the program noting the differences by position. At this point the work branches off into three separate paths depending on whether (1.) the pairs have an identical number of letters (error types: wrong letter or scramble), or (2.) the input word is longer (error type: additional letter) or (3.) the dictionary entry is longer (error type: omitted letter).

If in case (1.) the words have an different letter at only one position, the input word will be corrected in accordance with the dictionary entry.

WRONG LETTER:

	12345678
Input:	ALPHIBET
Dictionary:	ALPHABET
	Only difference at position 5
Result	Correct Alphibet to Alphabet!

In the case of two different neighboring positions, the algorithm explores whether the same letters occur in the dictionary entry—in the reverse order—and, if so, corrects them.

SCRAMBLED LETTERS:

	12345678
Input:	ALHPABET
Dictionary:	ALPHABET
	Differences at positions 3 and 4. HP in the input corresponds to the reverse sequence PH in the dictionary
Result	Correct Alhpabet to Alphabet!

In case (2.), the first different letter in the input word is deleted and the others are moved one position to the left. If this results in a match, we can correct the input word.

ADDITIONAL LETTER:

	123456789
Input:	ALLPHABET
Dictionary:	ALPHABET
	First difference at position 3—Delete the L in input
Result	ALPHABET Match with dictionary: Correct Allphabet to Alphabet!

In case (3.) we proceed analogously, only here it is the first different letter in the dictionary word that is deleted.

OMITTED LETTER:

12345678
Input: ALPABET
Dictionary: ALPHABET
First Difference at position 4—Delete the H in dictionary entry
Result ALPABET Match with input: Correct Alpabet to Alphabet!

Damerau (1964, 176) claims that his method achieves a success rate of more than 80% in correcting individual mistakes.

Levenshtein Distance

Levenshtein introduces a proximity measurement between two sequences of letters—the Levenshtein Distance—into the literature. This measurement counts the editing steps between the words to be compared (Левенштейн, 1965; Levenshtein, 1966). Hall and Dowling (1980) employ the dynamic programming method in order to count the edit distance. Editing steps include ‘delete’, ‘insert’ and ‘substitute simple characters’ in sequences of letters. The Levenshtein distance is “the minimal number of insertions, deletions and substitutions to make two strings equal” (Navarro, 2001, 37).

We will demonstrate dynamic programming on an example by Navarro (2001, 46). It involves calculating the edit distance between *survey* and *surgery* (Table C.5.1). We create a matrix listing the two character strings, one in the rows and one in the columns, character by character. Now characters are compared to each other to see whether they are identical. A 0 will be entered if they are, a +1 if they are not. In *surgery* and *survey* the first three characters are identical, leading to a 0 in the diagonal. *g* and *v* are different—consequently, this will be the first editing step. The next letter (*e*) is the same again. In the following step, the *r* will be inserted; this is the second editing step. Since the *y* at the end is identical in both words, the edit distance remains 2.

In input errors, the correct form is identified as that word in the dictionary which has the shortest Levenshtein Distance to the false version.

Table C.5.1: Dynamic Programming Algorithm for Computing the Edit Distance between *surgery* and *survey* (Bold: Path to the Final Result). Source: Navarro, 2001, 46.

		Word						
		s	u	r	g	e	r	y
s u r v e y	0	1	2	3	4	5	6	7
	1	0	1	2	3	4	5	6
	2	1	0	1	2	3	4	5
	3	2	1	0	1	2	3	4
	4	3	2	1	1	2	3	4
	5	4	3	2	2	1	2	3
	6	5	4	3	3	2	2	2

Input Error Recognition and Correction via n -Grams

n -Grams partition texts or individual words into character strings of n elements (Ch. C.1). n -Grams are suitable for identifying and correcting input errors. Like the other methods, the n -gram method too requires a dictionary with the correct spellings of the words. If none is available, recourse may be taken to the entries of the database's inverted file. All entries in the dictionary, in the documents as well as in the user queries are transformed into n -grams, word by word. Zamora, Pollock and Zamora (1981) as well as Angell, Freund and Willett (1983) work with trigrams, while Pfeifer, Poersch and Fuhr (1996) experiment with bigrams.

The first step calculates, for every input word, its probability of being an input error. In this approach, all words with at least one n -gram that does not occur in the n -gram dictionary are definite errors. Zamora, Pollock and Zamora (1981, 308) state:

An important point is that unknown trigrams (those not encountered earlier and thus not in the trigram dictionary) are given an error probability of 1.00; unknown trigrams are considered to be highly diagnostic of error.

To calculate each specific error probability, we require information about the frequency of occurrence (O) for all n -grams in the dictionary. The error probability $P_{(n\text{-Gram})}$ of each n -gram of a word is thus

$$P_{(n\text{-Gram})} = 1 / (O_{(n\text{-Gram})} + 1),$$

so that rare n -grams provoke a higher error probability than frequent ones. The error probability of an entire word follows the highest value of the error probabilities of its n -grams. A threshold value is used to determine which words will enter the next phase—that of correcting the input errors.

The n -gram method uses a very simple procedure for correcting errors. Angell, Freund and Willett (1983, 255) describe

a simple but, seemingly, highly effective procedure for automatic spelling correction which involves the inspection of the words in a dictionary to identify that word which is most similar to an input misspelling, the degree of similarity being calculated using a similarity coefficient involving the number of trigrams common to a word in the dictionary and the misspelling, and the number of trigrams in the two words.

Each dictionary entry is partitioned into n -grams (here: trigrams), with blanks being used at beginning and end to bring the character strings to n -gram length if necessary. Let the number of a word's letters be m . Each entry T is thus represented via $m+n-1$ n -grams ($z(\text{lexicon}_i)$):

$$T = z(\text{lexicon})_1, z(\text{lexicon})_2, \dots, z(\text{lexicon})_{m+n-1}.$$

An analog procedure is applied for every word with a high error probability E :

$$E = z(\text{error})_1, z(\text{error})_2, \dots, z(\text{error})_{m'+n-1}.$$

We still require a statement (g) about the number of identical n -grams of T and E . Angell, Freund and Willet calculate the similarity between an erroneous word E and the dictionary entries T via the Dice Coefficient:

$$\text{Sim}(T, E) = 2g / (m+n-1 + m'+n-1).$$

The pair T, E with the highest similarity value should contain the correct allocation of E to T . In order to safeguard against misguided corrections, it is possible to introduce an additional threshold value for similarity (Sim), which only leads to error correction or display when exceeded.

To exemplify, we will consider an example (Angell, Freund, & Willett, 1983, 257). Let the word CONSUMMING, found in a document, have a high error probability; the entry CONSUMING is featured in the dictionary. How great is the similarity between CONSUMMING and CONSUMING? CONSUMMING has ten letters (thus $m = 10$) and, when using trigrams ($n = 3$), it is thus expressed via twelve n -grams ($m'+n-1 = 12$):

C, *CO, CON, ONS, NSU, SUM, UMM, MMI, MIN, ING, NG*, G.

The partitioning of CONSUMING results in the following eleven trigrams ($m+n-1 = 11$):

C, *CO, CON, ONS, NSU, SUM, UMI, MIN, ING, NG*, G.

Both strings have ten trigrams in common ($g = 10$):

C, *CO, CON, ONS, NSU, SUM, MIN, ING, NG*, G.

The similarity between CONSUMMING and CONSUMING is thus:

$$\begin{aligned} & 2 * 10 / (12 + 11) \\ & = 20 / 23 \\ & = 0.87. \end{aligned}$$

In the case of user input errors, the n -gram method can be used to offer the user a list of suggestions for correction, ranked by similarity.

In a comparative study on the example of personal name search, Pfeifer, Poersch and Fuhr (1996, 675) conclude that Soundex achieves extremely bad results. A variant of Phonix (with ending sound code) works far more effectively (second place in the test) and achieves similar results to the bigrams (ranked first; of similar quality as the

trigrams) as well as to the Damerau method (ranked third). These results suggest that combinations of different approaches may lead to a further improvement in input error identification and correction. Additionally, it appears useful to provide the user with suggestions for error correction ranked by relevance (Zobel & Dart, 1996) instead of only displaying one option for correction (which might even be wrong).

Conclusion

-
- Input errors (typographical, orthographical and phonetic mistakes) occur in documents as well as user queries. Whereas query mistakes can be intellectually corrected in a dialog, document mistakes can only be processed automatically.
 - Error types are blank errors (particularly involving function words), errors in words that are recognized in isolation, and errors in words that are only recognized via their context.
 - Phonetic approaches to input error correction use algorithms that stem from phonetic retrieval and which search for words based on their sound. The “classical” approach is Soundex, originally developed by Russell. An enriched version, Phonix, enhances Soundex by phonetic replacement.
 - Approximate string matching is performed either according to Damerau’s method or according to Levenshtein’s. Damerau’s approach recognizes and corrects individual mistakes (wrong letter, scrambled letters, additional letter, and omitted letter). The Levenshtein method calculates edit distances, i.e. the number of processing steps necessary to get from one word to the next. Both methods require a dictionary as the basis for comparison.
 - A dictionary (or access to the entries of an inverted file) is also required for the n -gram method. Error recognition and correction runs via comparisons of the n -grams of input word and dictionary entry.
-

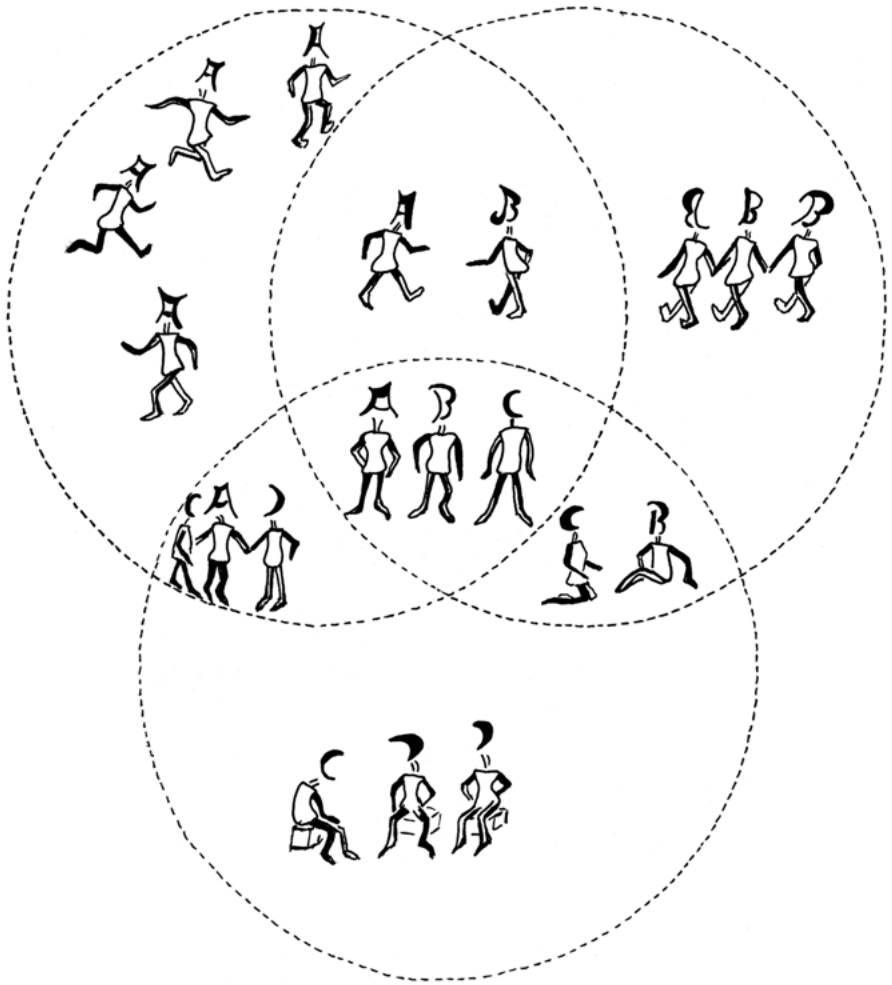
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Part D

Boolean Retrieval Systems



D.1 Boolean Retrieval

George Boole's 'Laws of Thought'

We now come to an approach that has become a classic in information retrieval: Boolean Retrieval Systems. Boolean systems distinguish themselves via their use of exactly described search atoms (e.g. *apples*, *pears*) as well as their use of operators (of particular importance are AND and OR). Let it be noted that the use of operators is not in keeping with an idiomatic use of language. Whereas one might say "I am looking for all the information about apples *and* pears," Boolean logic requires the formulation "apples OR pears".

The terminological and contentual origins of the models of 'Boolean Retrieval' lie in George Boole's work 'The Laws of Thought' (1854). Boole ascribes the assessment of logical statements to the binary decision "true" (1)—"false" (0), while stressing that this is exclusively a property of thought, not of nature (Boole, 1854, 318):

(T)he distinction between *true* and *false*, between *correct* and *incorrect* exists in the processes of the intellect, but not in the region of a physical necessity.

These are laws of *thought*, not laws of nature. Thought is expressed via certain signs ("signs of mental operations"; *ibid.*, 23), which are interconnected with one another where necessary. An initial combination option ("class I") introduces objects that are co-subsumed under several classes. The AND connection represents the algebraic product as a logical operator.

Let it ... be agreed, that by the combination xy shall be represented that class of things to which the names or descriptions represented by x and y are simultaneously applicable. Thus, if x alone stands for "white things", and y for "sheep", let xy stand for "white sheep" ... (*ibid.*, 20).

(T)he formal product of the expression of two classes represents that class of individuals which is common to them both ... (*ibid.*, 35).

The second operation option ("class II") connects parts into a whole:

We are not only capable of entertaining the conceptions of objects, as characterized by names, qualities, or circumstances, applicable to each individual of the group under consideration, but also of forming the aggregate conception of a group of objects consisting of partial groups, each of which is separately named or described. For this purpose we use the conjunctions "and", "or", &c. "Trees and minerals", "barren mountains, or fertile vales", are examples of this kind. In strictness, the words "and", "or", interposed between the terms descriptive of two and more classes of objects, imply that those classes are quite distinct, so that no member of one is found in another. In this and in all other respects the words "and", "or" are analogous with the sign + in algebra, and their laws are identical (*ibid.*, 23).

Here, Boole introduces the OR operator (as an algebraic sum). Since he additionally supposes that the classes joined via OR are disjunct, we are faced with an “exclusive or” (common abbreviation: “XOR”), as in: “either—or”.

A third operation option proceeds in the opposite direction to the algebraic sum and separates totalities into parts:

The above are the laws which govern the use of the sign +, here used to denote the positive operation of aggregating parts into a whole. But the very idea of an operation effecting some positive change seems to suggest to us the idea of an opposite or negative operation, having the effect of undoing what the former one has done. ... This operation we express in common language by the sign except, as, “All men *except* Asiatics”, “All states *except* those which are monarchical.” Here it is implied that the things excepted form a part of the things from which they are excepted. As we have expressed the operation by aggregation by the sign +, so we may express the negative operation above described by—minus. Thus if x be taken to represent men, and y , Asiatics, i.e. Asiatic men, then the conception of “All men except Asiatics” will be expressed by $x-y$ (ibid., 23 f.).

This introduces the NOT operator as the inverse function of the XOR operator, i.e. as an algebraic difference. It must be noted that we are faced with a dyadic operator, i.e. one with two argument slots, not with the monadic operator of negation.

The combinations via AND, XOR and NOT, respectively, are—just like the original arguments—either true or false. The truth or falsehood of the operation is always tied to the truth value of the single arguments, as well as to the corresponding truth value matrix of the connection. Boole (ibid., 55) names an example:

When $x = 1$ and $y = 1$ the given function $= 0$.

The line must be read as follows: if x is true (1) and y is also true (1), then the combination is false (0). All the possible truth value combinations of the original arguments are always checked. When there are two arguments, it follows that we must consider four combinations, three arguments means nine combinations etc.

The Boolean oeuvre affects both computer science—conveyed by Claude Shannon—as well as information science, with regard to search operators—conveyed mainly by Mortimer Taube (Smith, 1993).

“Atomic” Search Arguments

Before we turn to Boolean operators in retrieval systems, we must first clarify what exactly it is that such operators connect in the first place. The smallest units of a complex search argument are called the “search atoms”. Depending on the form of indexing (word or phrase index), the search atoms are individual words (*Boston*) or word groups merged in a phrase (*New York*). When determining a specific search atom, it is possible to fragment it via truncation (using wildcards). Depending on the

position in the atomic search argument that the truncation character is put in, we distinguish between truncation on the right (*hous**), truncation on the left (**house*) and internal truncation (*Sm\$th* when searching for *Smith* or *Smyth*). We can determine how many characters are to be replaced, as well as the amount of characters (alphabetical or numerical signs as well as the exact amount of characters required) to be replaced. The following list shows (using arbitrarily chosen characters for the respective truncation option) options for fragmentation. (The extent of factually realized options heavily varies from one retrieval system to another.)

- Open truncation on the right * (replaces nil to infinitely many characters)
*Example: hous** retrieves *house, houses, housing* etc.
- Open truncation on the left * (replaces nil to infinitely many characters)
*Example: *house* retrieves *farmhouse, lighthouse* etc.
- Open truncation on the right and the left *
*Example: *hous** retrieves the above examples as well as *farmhouses, lighthouses* etc.
- Limited truncation on the right ? (truncation on the left analogous) (replaces one or no character per occurrence)
Example: house? Retrieves *house* and *houses*, but not *housing*
- Precisely limited truncation on the right \$ (truncation on the left analogous) (replaces exactly one character)
Example: house\$ retrieves *houses*, but not *housing* or *house*
- Precisely limited internal truncation \$ (replaces exactly one character)
Example: Mil\$er retrieves *Miller* and *Milner*, but also *milder* or *milker*.

An open as well as limited internal truncation is possible in theory, but will only be put into practice in a rare number of cases.

- Specific truncation options:
 - Replacement of exactly one alphabetic character: @,
 - Replacement of exactly one numerical character: #,
 - Replacement of one of these characters only: [a,b,c],
 - Replacement of exactly one character but not this one: [^a], e.g. *19[^8]0* retrieves the decades 1900 through 1970 as well as 1990, but not 1980,
 - Replacement of exactly one character but not those that lie in the interval named: [^a-z], e.g. *198[^6-9]* retrieves the years 1980 through 1985 (however, this also includes the number 198a), but not 1986 through 1989.

Generous truncating can lead to undesirable ballast. Suppose someone wants to obtain a complete overview of apes, and formulates his search argument as **ape**—his search results will include anything concerning *apes*, but also anything about *capes, escape, apex* and *aperitif*.

Upper and lower-case spelling may be of importance in a search atom. Any retrieval system will thus differentiate the five cases of (1) only upper-case letters (*AIDS*), (2) only lower-case letters (*aid*), (3) first letter in upper case (*Aid*), (4) upper-case letter(s) within a word (*YouTube*) and (5) upper and lower-case letters without any differentiations (case insensitive).

The occurrence of special characters in search atoms must be taken into consideration in retrieval systems. This includes the hyphen (e.g. in family names, such as Hill-Wood), the ampersand (e.g. in company names like Barnes & Noble), apostrophes and the at-sign (particularly in e-mail addresses: info@hill.com). Problems arise when search atoms contain otherwise predefined characters, e.g. the at or Dollar sign (which we introduced as a truncating symbol above), or operator designations, such as the ‘and’ in the book title *Pride and Prejudice*. The solution in such cases is to unambiguously mark the special character as part of a search atom (e.g. via apostrophes). An e-mail address would thus be searched via *info“@”hill.com*, Jane Austen’s novel via *Pride “and” Prejudice*.

Atomic search arguments can be selected either via all parts of a document in the Basic Index (which is generally what happens in online search engines) or field-specifically. The indices supply dictionaries, which are offered to the user. In this way, he can find out before starting his search if and under what spelling a search argument is available in the database. Normally, once a sequence of search characters has been entered in the index file, its alphabetical environment will be displayed in numerical order. After that, the user can continue his search either via the listing number (such as *SEARCH T3*) or the retrieved entry (such as *SEARCH AU= Marx, Karl*).

The extent of search fields as well as browsing in the dictionary depends on the retrieval system’s stage of development. Professional electronic information services work with dozens or even hundreds of search fields, whereas search tools in the World Wide Web only know a select few corresponding options, and that solely under their ‘advanced’ setting.

Boolean Operators

Atomic search arguments are joined together into complex search arguments via operators and, as needed, parentheses. Operators are either the classical Boolean Operators or proximity operators, all of which represent an intensification of the Boolean AND.

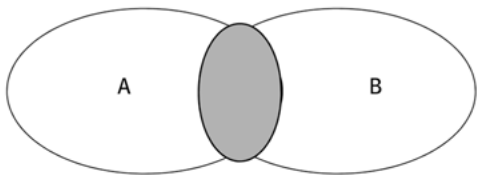
There are four Boolean Operators in information retrieval:

- AND (set theory: intersection, logic: conjunction),
- (inclusive) OR (set theory: union of sets, logic: disjunction),
- NOT: (set theory: exclusion set, logic: postsection),
- XOR in the sense of an exclusive OR (set theory: symmetrical difference, logic: contravalance).

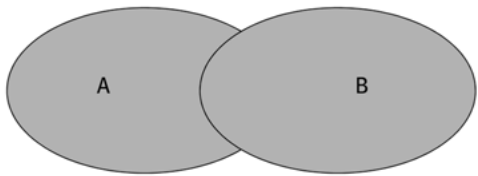
In the sense of formal propositional logic, we model the Boolean Operators via truth value charts, as George Boole did (Table D.1.1). For the purposes of easy demonstration, the Boolean Operators can be represented via Venn Diagrams (Figure D.1.1).

Table D.1.1: Boolean Operators.

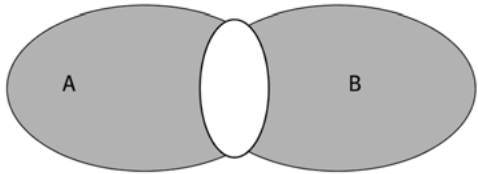
A	B	A AND B	A OR B	A NOT B	A XOR B
1	1	1	1	0	0
1	0	0	1	1	1
0	1	0	1	0	1
0	0	0	0	0	0
		Conjunction “both”	Disjunction “at least one”	Postsection “one without the other”	Contravalence “either one or the other”



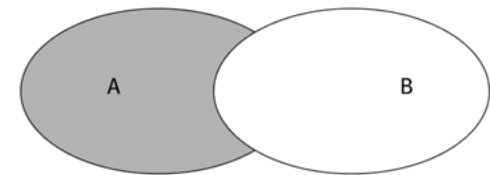
a) A AND B



b) A OR B



c) A XOR B



d) A NOT B

Figure D.1.1: Boolean Operators from the Perspective of Set Theory. a) AND, b) OR, c) XOR, d) NOT.

The value 1 is interpreted as “the search atom exists within the document” and the value 0 as “the search atom does not exist within the document”. A and B are the atomic search arguments, A AND B, A OR B, A NOT B as well as A XOR B the complex search arguments. Apart from these four operators, there are no further Boolean connections. Since contravalence can be derived from the other functors ($[A \text{ OR } B] \text{ NOT } [A \text{ AND } B]$), retrieval systems sometimes go without this operator.

When looking for A AND B, one will only find such documents that contain both A and B (described with “1” in the truth value chart); for all other possible combinations, the operator will give out the value 0. In set theory, AND represents the intersection. A search for A OR B will be successful if either A or B or both exist in a document; only when neither can be found will the resulting value be 0. Set-theoretically, the object of the OR is the formation of a union of sets. If a user searches for A NOT B, a 1 will result only in the second line of the truth value matrix, in case A occurs in the document while B does not. In set theory, the NOT gives way to what is called an exclusion set. Finally, A XOR B means that documents will be retrieved which contain A but not B, or B but not A. In everyday parlance, this corresponds to “either or”. XOR is a union of sets minus the intersection.

Let it be noted that certain systems use the monadic negation $\sim A$ instead of the dyadic NOT operator. The following truth value table holds for the negation (Table D.1.2).

Table D.1.2: Monadic not-Operator.

A	$\sim A$
1	0
0	1
Negation “not”	

The Boolean Operators can be combined at leisure. Since the operators’ binding strength is defined differently from retrieval system to retrieval system, parentheses must be set correspondingly. If, for instance, one searches for the descriptor (DE) *communism* and the authors (AU) *Karl Marx* or *Friedrich Engels* (including both of them as a team), one must formulate: $DE=Communism \text{ AND } (AU=Marx, Karl \text{ OR } AU=Engels, Friedrich)$.

The distributive, commutative and associative laws as well as DeMorgan’s laws (Korfhage, 1997, 53-63) hold for the Boolean Operators. The associative law concerns the parentheses while using identical operators. It reads as follows:

$$\begin{aligned}(A \text{ AND } B) \text{ AND } C &= A \text{ AND } (B \text{ AND } C) = A \text{ AND } B \text{ AND } C \\(A \text{ OR } B) \text{ OR } C &= A \text{ OR } (B \text{ OR } C) = A \text{ OR } B \text{ OR } C \\(A \text{ XOR } B) \text{ XOR } C &= A \text{ XOR } (B \text{ XOR } C) = A \text{ XOR } B \text{ XOR } C\end{aligned}$$

and holds for all operators except NOT. The commutative law regulates dealings with the sequence of operators:

$$A \text{ AND } B = B \text{ AND } A$$

$$A \text{ OR } B = B \text{ OR } A$$

$$A \text{ XOR } B = B \text{ XOR } A.$$

The commutative law does not hold for the postsection (NOT), since it depends upon the order of the search atoms. The distributive law exists in a variant for the AND operator and another for the OR, corresponding to the ‘bracketing out’ in algebra.

$$(A \text{ AND } B) \text{ OR } (A \text{ AND } C) = A \text{ AND } (B \text{ OR } C)$$

conjunctive distributive law

$$(A \text{ OR } B) \text{ AND } (A \text{ OR } C) = A \text{ OR } (B \text{ AND } C)$$

disjunctive distributive law.

In the postsection, we must use DeMorgan’s Laws due to the implicit negation, thus receiving analogs to the distributive laws. The operators AND and OR are switched, however:

$$(A \text{ NOT } B) \text{ AND } (A \text{ NOT } C)$$

$$= A \text{ NOT } (B \text{ OR } C)$$

$$(A \text{ NOT } B) \text{ OR } (A \text{ NOT } C)$$

$$= A \text{ NOT } (B \text{ AND } C).$$

Proximity Operators

Particularly in long texts, the use of the Boolean AND operator is critical. Documents might be retrieved in which the search atoms combined via AND occur in completely different contexts. Occasionally, databases work not only with a controlled vocabulary but with additional syntactic indexing in order to mark thematic contexts. An article might discuss the subject of anthropology in Kant in the first section, and anthropology in Hegel in the second, without ever discussing Kant and Hegel at the same time. Syntactic indexing can be used to represent this template, via two thematic chains 1 and 2, in a way that preserves the context: *Anthropology (1-2), Kant, Immanuel (1), Hegel, Georg Wilhelm Friedrich (2)*. To accentuate the AND to research thematic chains, we use proximity operators (Jacsó, 2004; Keen, 1992).

We distinguish between proximity operators based on counting words (occasionally even counting individual characters), which may occur between two search atoms at most, and those that exploit the structure of a text (sentences, paragraphs etc.). Both species further distinguish between whether the order of the atomic search

arguments is adhered to. The following list shows variants of proximity operators (the abbreviations vary from system to system).

Adjacency operators regard the word distance between two search atoms in the document. Here it is important to determine whether stop words should be included or not. Punctuation marks are ignored as a matter of course.

- Phrase: “A B” or A ADJ B (retrieves search atoms that are directly adjacent to each other, and in this exact order)

Example: “Miranda Otto” or “Miranda ADJ Otto” retrieves Miranda Otto

- Phrase without order: A (n) B (retrieves search atoms that are directly adjacent to each other)

Example: Miranda (n) Otto retrieves Miranda Otto and Otto, Miranda

- Interval of length n: A w/n B (retrieves search atoms with at most n words between them, where A and B occur in the same field)

Example: “Miranda Otto” w/10 Eowyn retrieves documents in which at most 10 other words occur between the phrase “Miranda Otto” and Eowyn, e.g. Eowyn is played by Miranda Otto or Miranda Otto plays Eowyn

- Interval of set length: A SAME B (retrieves search atoms that have at most n words between them, where A and B occur in the same field; the n is here kept constant by the system, e.g. n=10)

- Interval of length n with set order: A before/n B (retrieves search atoms that occur in the desired order and that have at most n words between them, where A and B occur in the same field)

Example: “Miranda Otto before/10 Eowyn” retrieves documents in which there are at most ten other words between the phrase “Miranda Otto” and Eowyn, which must occur in the specified sequence. For instance: Miranda Otto plays Eowyn, but not Eowyn is played by Miranda Otto.

The above operators can also be formulated negatively. The search argument A NOT w/n B correspondingly only retrieves such documents in which the search argument A occurs and B does not occur within ten words of A.

Syntactic operators regard the borders of sentences or paragraphs, or the thematic chains in syntactic indexing.

- Sentence: A w/s B (A and B occur in the same grammatical sentence or—in syntactic indexing—in the same indexing chain)

Example: Kant w/s Anthropology retrieves our above example (since there is a common chain with 1); it does not retrieve Kant w/s Hegel

- Paragraph: A w/p B (A and B occur in the same paragraph)

- Field: A w/seg B (A and B occur in the same database field or segment)

Like adjacency operators, syntactic proximity operators also allow for negations. A NOT w/s B, A NOT w/p B, A NOT w/seg B retrieve texts that contain the first search atom A, where the search argument B does not occur in the same sentence, paragraph and field, respectively.

Principally, proximity operators can be connected with each other and with Boolean operators.

Searches with proximity operators are prone to errors, since texts include rhetorical variants such as paraphrases, ellipses and anaphora. This phenomenon has been known for a long time. Mitchell (1974, 178) observes:

An article may say “National Science Foundation” in one paragraph and use the word “it” in succeeding sentences or paragraphs. Also, abbreviations and acronyms can be missed after they are defined.

This leads to the problem of anaphora and ellipses and their resolution (Pirkola & Järvelin, 1996) (Ch. C.4).

Hierarchical Search

Insofar as a database indexes content via a hierarchically structured knowledge organization system, it may be desirable to incorporate hyponyms, hyperonyms and related terms into the search. The host STN International offers several options for hierarchical and associative search:

- +NT: all hyponyms are connected via OR,
- +NT1: only the hyponyms of the next lowest level are taken into consideration,
- +BT: all hyperonyms are connected via OR,
- +RT: all related terms are connected via OR.

The different relations can also be incorporated into the search argument together.

When a database uses a classification system with hierarchical notation (e.g. the Dewey Decimal Classification DDC), hierarchical search is performed by truncating the notations. The DDC is a universal classification whose notations consist of digits. Each class (e.g. 382) has ten hyponyms at most (382.0 through 382.9). Open truncation on the right (DDC=382*) retrieves everything on class 382 including all hyponyms, limited truncation on the right (e.g. DDC=382? or DDC=382??) retrieves everything on 382 and the hyponyms on the next level (with one wildcard character), the next two levels (with two wildcards) etc.

Algebraic Operators—Frequency Operator

Algebraic operators are used in numerical fields (e.g. year, revenues, number of employees). The standard operators are A EQUALS [number], A LARGER THAN [number], A SMALLER THAN [number] as well as the interval A BETWEEN (number₁, number₂).

In full-text databases, it may be prudent to search for certain search atoms via their frequency of occurrence. Such a frequency operator can be formulated as ATLEAST n A, where n stands for the minimum number of A's occurrences. This search option is based on the premise—not always accurate—that the oftener a term occurs in a text, the more important it is. This operator is additionally challenged by synonyms and anaphora, insofar as these are not summarized and thus included in the search.

Advantages and Disadvantages of Boolean Systems

A disadvantage of Boolean retrieval is its difficult usability for laymen. Not only do the operators, which do not correspond to the vernacular, take getting used to—the correct translation of a search request via truncation, Boolean Operators, proximity operators, parentheses etc. is extremely complex. If the end user is allowed to place the search atoms next to each other haphazardly, with a Boolean AND to be included automatically, a lot of the variety of expressions will be lost.

Chu (2010, 113) notes the disadvantage that no relations outside of the four Boolean Operators can be expressed:

(I)t is difficult to express relationships other than the Boolean among terms (e.g., causal relationship) because such mechanism is simply not provided in the model. Suppose a user would like to find some information about “the application of computers in education”. A search query is then formed using the Boolean operator as *computers AND education*. The term *application* is not included in the query because the relationship is supposed to be expressed via Boolean operators, but none of the Boolean operators has this function. Consequently, by conducting the search *computers AND education* the user would get information on not only the use of computers in education, but also education about computers.

The problem Chu describes might be reduced via the skillful application of proximity operators, but the fundamental objection remains.

The Boolean model employs a strictly dichotomous principle: a document is retrieved or it is not retrieved, with no middle ground. Also, all terms are ranked equally, which leaves no space for weighting and Relevance Ranking. However, this only holds true for the “traditional” variant of the model described here. “Extended” Boolean systems (Chapter D.3) allow for the inclusion of weight values.

It must be emphasized that Boolean systems have shown their practicability for all commercial electronic information services, from the first online systems in the 1970s onwards.

Their great advantage is the multitude of expressions that can be modeled via the search requests. The flipside of this coin is the difficulty of usage by laymen mentioned above.

The Boolean systems are very popular with information professionals. When several large commercial providers of electronic information offered natural-lan-

guage systems with Relevance Ranking, these were unable to assert themselves in daily research practice. At best, they ran alongside Boolean retrieval, as an additional option. Taking stock of the advantages and disadvantages of Boolean systems as well as of other approaches, Frants et al. (1999, 94) arrive at the conclusion that the Boolean model does not fare any worse than its alternatives.

In this article, we have demonstrated that most of the criticism of Boolean systems is actually directed at methodology employed in operational systems, rather than the Boolean principle itself.

We should emphasize, that based on the existing evidence, we would not maintain ... that the Boolean search principle has greater potential or is superior to the other alternatives. We merely wish to emphasize that Boolean search is no worse than any other known approach.

Conclusion

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- The origins of Boolean retrieval are found in the “Laws of Thought” (1854) by George Boole (1815 – 1864). Boole introduced the operators AND, OR (in the sense of either-or) as well as NOT, in addition to the truth value matrix with the values 0 and 1.
 - We distinguish between search atoms and compound search arguments consisting of search atoms. Search atoms can be fragmented via truncation. They are searched either in the entire document or field-specifically.
 - AND, OR, NOT, XOR are Boolean Operators used in information retrieval, where the XOR can be omitted since it can be expressed via the other operators.
 - Proximity operators intensify the Boolean AND, incorporating the intervals between search atoms as adjacency operators, or as syntactic operators, incorporating the limits of sentences or paragraphs as well as syntactic chains. Synonyms, anaphora and ellipses lead to problems with proximity operators. Numerical fields use algebraic operators.
 - Advantages of Boolean systems are their decades-long successful application in the information industry as well as the ability of their functionality to construct complex atomic as well as composite search arguments. Their disadvantages are the difficulty of usage, particularly for user groups consisting of laymen, the restriction to four operators as well as (in the traditional variant) the dichotomous perspective on documents.
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D.2 Search Strategies

Menu and Commands Navigation

Users of information retrieval systems can be roughly placed into three groups (Wolfram & Hong, 2002): Information Professionals (trained searchers), informed laymen or “professional end users” (users who perform subject-specific search in their daily working life) as well as laymen or “end users” (the overwhelming majority of internet users).

Search History			
Database Details			
Set	Term Searched	Items	
S1	DT=B	428444	Display
S2	CS=DUSSELDORF	10184	Display
S3	AY=1995/PR	74060	Display
S4	S1 AND S2 AND S3	291	Display

Format
Short
Number of Records
10

Show Database Details for:
348: European Patents Fulltext

Bluesheet Rates

Figure D.2.1: Command-Based Boolean Search on the Example of Dialog Web. Source: Stock & Stock, 2003a, 26.

Depending on the type of user support, we distinguish between menu-based and command-based Boolean retrieval systems. The latter requires the user to fill every dialog field with a search command, a field symbol and a search atom. In Figure D.2.1, we see such a search via commands. File 348 (a patent database) is opened in the information provider DIALOG, the search steps are numbered from S1 through S3. The search atoms used,

- DT=B (document type: B, i.e., granted patent),
- CS=Dusseldorf (corporate source: Dusseldorf),
- AY=1995/PR (access year: 1995 is the year of priority),

are found via a command to open the dictionary. Search step 4 connects the three search atoms S1, S2 and S3 via the Boolean AND and retrieves all patents granted by the European Patent Office (DT=B) to Düsseldorf-based companies (CS=Dusseldorf) for the priority year 1995 (AY=1995/PR). Command-based Boolean retrieval offers the user group of experienced searchers ideal conditions for search and retrieval.

Professional end users and laymen, in particular, would be out of their depth when using search commands. A study of user requests for the search engine EXCITE

shows that less than 5% of all “advanced search” queries even contain Boolean operators, with some of these being used falsely to boot (Spink et al., 2001).

Figure D.2.2: Menu-Based Boolean Search on the Example of Profound. Source: Stock & Stock, 2003b, 43.

End users can resort to menu-based interfaces provided by information services. Here there are, again, two variants. The first predominantly addresses professional end users. This is shown in Figure D.2.2, on the example of a provider for market research information. In market research, the market sector, region and research subject (market volume, revenue, number of players etc.) are nearly always of importance. Correspondingly, these three fields are provided with directly accessible dictionaries. In addition, more search atoms can be added via further windows. The search system automatically connects the single arguments via AND during searches. Variants of this kind of menu navigation provide (e.g. in a drop-down menu) operator options (mostly the Boolean Operators AND, OR, NOT as well as, sometimes, a proximity operator). Compared to command-based systems, the amount of search options is significantly smaller, but after a short break-in period, professional end users will come to master these menu options without a problem.

The second variant of menu guidance addresses the information layman, the person surfing the World Wide Web. “Menu” is even an exaggeration, since the user is generally offered exactly one window in which to enter the search atoms. Such menu minimalism has been introduced successfully by the search engine Google. Where WWW search engines work with Boolean systems at all, the search atoms are always connected automatically. There has been a shift in the history of search engines. In the 1990s, several search arguments used to be connected via OR in the Boolean sense, but from around the year 2000 onward, the AND operator dominates (Proctor, 2002), since fewer documents are retrieved with this operator. This results in hit lists that do not contain vast and innavigable amounts of information.

The interpretation of unconnected search arguments via OR and AND, respectively, falls short, however, since users generally do not aim for exactly one operator option. Rather, the objective must be to interpret individual search queries in

such a way that all Boolean Operators are used as appropriately as possible. Frants and Shapiro (1991) propose analyzing the descriptors of documents that are already marked as positive. In the first step, the descriptors to be connected are selected via their frequency of occurrence and their proximity to each other, in the second step the operators AND and OR are added. The descriptors with the highest occurrence values are initially connected via OR, the search is performed (resulting in hit list 1) and the descriptors are disregarded in the following. The next most frequent descriptors are connected via AND according to proximity (hit lists 2 through n), and these hit lists are combined via OR (hit list m). The intersection of hit lists 1 and m provides the result to be given out. The algorithm (portrayed in a very simplified way here) hinges on two conditions: first, the documents must be indexed via controlled vocabularies, and second, the user must be willing and able to designate relevant documents after the first retrieval step. Neither condition is necessarily given, particularly in WWW search engines.

Another path is followed by Das-Gupta (1987). He tries to find meaningful interpretations of the vernacular AND. To simplify once more, Das-Gupta asks for the respective hyperonym(s) of the atomic search terms. If two search atoms have a common hyperonym, the operator in question will be, in all probability, OR. If they have different hyperonyms, the argument will be in favor of an AND (Das-Gupta, 1987, 245).

For example, in the phrase “cats and dogs”, the conjunction introduces two instances of the more global concept “animals”. In contrast, the conjunction in “antibiotics and sleeping sickness” introduces a single concept, that is the cure of sleeping sickness using antibiotics. The correct Boolean interpretation of the conjunction depends upon the function intended. In the first example, “cats OR dogs” preserves the intention, while in the second “antibiotics AND sleeping sickness” is necessary to maintain the coordination of concepts being performed.

Semantic similarity between adjacent search atoms speaks in favor of the Boolean OR, dissimilarity for AND. To implement this suggestion, a Boolean retrieval system needs information about semantic relations between concepts, e.g. a knowledge organization system.

Many end users face a problem when it comes to the logical NOT (mostly expressed in search engines by the minus (-) sign). Nakkouzi and Eastman (1990, 172) name an example:

Consider, for example, a query requesting information about “Lung Diseases AND NOT Cancer”. A document containing information about several lung diseases (including cancer) will not be retrieved. But this is probably not what the user intended.

A possible solution is offered by the hierarchy relation of knowledge organization systems. For the concept connected via NOT, one searches the hyperonym and all its hyponyms. Afterward, the undesired term is deleted and the rest of the terms are connected via OR. In our example, let disease be the hyperonym for cancer, with the

further hyponyms of inflammation and injury. The translation of the above query will be *lung disease AND (inflammation OR injury)*.

Information Profiles and Selective Dissemination of Information

A retrospective search will satisfy the ad-hoc information needs of a user in the manner of a pull service. If the information need stays the same over a longer period of time, the user will deposit his search request ('calibrated' via retrospective searches) as his 'information profile'. If new content is entered into the retrieval system's database following an update, a filtering mechanism will be activated that compares the existing user information profiles with the new documents. The system will act as a push service and inform the user about the current retrieval result.

This "Selective Dissemination of Information" (SDI) was conceived as early as 1958 by Luhn as part of a "Business Intelligence System". Luhn imagined an output of "action points", which might be single individuals, working groups or even larger units. Luhn (1958) notes:

Based on the document-input operation and the creation of profiles, the system is ready to perform the service function of selective dissemination of new information (*ibid.*, 316).

New information which is pertinent or useful to certain action points is selectively disseminated to such points without delay. A function of the system is to present this information to the action point in such a manner that its existence will be readily recognized (*ibid.*, 315).

An SDI service is open for the retrieval model that is deposited; hence it can proceed both according to the Boolean (Yan & Garcia-Molina, 1994) and to other models (Foltz & Dumais, 1992). Models that permit Relevance Ranking are of particular advantage for SDI systems whose documentary reference units represent internet documents. Thus, in case of large hit lists, the result can be restricted to the *n* first-ranked documents. Yan and Garcia-Molina (1999) describe the "Stanford Information Filtering Tool" (SIFT), which filters posts in Usenet while working with a combination of the Boolean and the Vector Space Models. The information profiles are managed in their own specific file. The search arguments deposited therein are used to search for new documents, and the hits are transmitted to the user. Yan and Garcia-Molina (1999, 530) describe SDI:

It is difficult to stay informed without sifting through huge amounts of incoming information. A mechanism, called *information dissemination*, helps users cope with this problem. In an information dissemination system, a user submits a long-term *profile* consisting of a number of standard queries to represent his information needs. The system then continuously collects new documents from underlying information sources, filters them against the user profile, and delivers relevant information to the user.

An SDI service should meet the following criteria (Yan & Garcia-Molina, 1994, 333):

It should allow a rich class of queries as profiles ...

It should be able to evaluate profiles continuously and notify the user as soon as a relevant document arrives, not periodically.

It should scale to a very large number of profiles and a large number of new documents.

It should efficiently and reliably distribute the documents to the subscribers.

For the second aspect, it would appear to make more sense to offer the user a choice between two options: output at set intervals or speedy, continual output. The output paths are specified as well. The system either transmits the results to the user's personal website or his mailbox.

Search Strategies

Search strategies in Boolean systems are closely related to the proverbial needle in the haystack. The objective is to retrieve, from many thousands of databases and the several billion documentary units contained therein, precisely the ones that satisfy one's information needs. The "Needle-in-the-Haystack-Syndrome" can be divided into three phases:

1. Searching and retrieving the relevant databases,
2. Searching and retrieving the relevant documentary units,
3. Modifying the search arguments.

In many technical areas, it is of fundamental importance to retrieve all centrally relevant documents. Let us consider, for instance, a search in preparation of a research project that is meant to lead to a patent. An erroneous search that fails to retrieve anything, even though relevant knowledge has been documented, will not only lead to the patent application's rejection for lack of innovation, but will have provoked an entirely superfluous research endeavor. Accordingly, the information filters must be calibrated to prevent relevant information slipping through the nets. These protective measures are called "hedges", as explained by Sanders and Del Mar (2005, 1162):

(Information professionals) have provided excellent search strategies, using filters they call "hedges" (as in hedging one's best) that help to separate the wheat (scientifically strong studies ...) from the chaff (less rigorous ones).

In Phase 1 of the search, the relevant databases must be found. We can distinguish between different types of databases according to the kinds of information they call up. Bibliographical databases exclusively account for the presence of documents. Generally, they apply elaborate methods of formal bibliographical description and content indexing to the individual documentary reference units. A disadvantage of these databases is the occasional media discontinuity after the search, if the full text

must be acquired separately. However, information practice is developing in such a way that the full text is either appended to the bibliographic data set as a facsimile (if available), or that a copy of the document can be ordered via a document delivery service. Full-text databases present the entire continuous text of the documentary reference unit's text in the documentary unit. Daily newspapers, legal documents, court decisions, patents and articles of scientific journals can thus be called up word for word. Full-text databases are complementors to bibliographical databases. Factual databases can be compared to reference books or handbooks. Here it is not the literature concerning a subject that is called up, but the subject itself. The breadth of factual databases is immense and spans all kinds of non-textual documents; it encompasses everything from company dossiers, balance sheets, biographies and data about materials up to the structural formulae of organic chemistry. Statistical databases are a special kind of factual databases, containing demographic or economic subjects (mostly arranged in time series). Thus one might survey the development of Germany's gross domestic product since reunification or the export volume of Riesling grapes from the state of Rhineland-Palatinate over the last twenty years. A characteristic of statistical databases is the option of econometric manipulation (e.g. calculating annual changes in grape exports, aggregating all species of grape or converting prices from Euros to Dollars). Further special forms of factual databases concern chemical substances, reactions and so-called "prophetic" substances (Markush structures) searched via structural formulae or chemical equations, as well as biosequences (gene sequences, which can be very long) with exact or fuzzy match.

There are aids for searching the relevant databases: an overview is provided by Gale's Directory of Databases. In addition, there are indexing databases available from the individual online information providers that point the user to appropriate databases. All individual databases have their special characteristics, their specific field schemata, their own thesauri, classification systems, output options and prices. All of these specifics are listed in the database descriptions, the so-called "bluesheets". (The name derives from the first online host, DIALOG, printing its database descriptions on blue paper.)

We will now sketch the search for a database on a concrete example (Figure D.2.3). In order to search within a host-specific index of databases, one opens the corresponding database (in DIALOG's system, this is done via b 411; "b" is for BEGIN, 411 the number of the database). The command SET FILES is used to intelligently restrict oneself to an area, let us say: all important daily newspapers, to an additional source (for instance, database 47) while neglecting an undesirable source (database 703), which would make the command *sf papersmj, 47 not 703*. Then the user will formulate the desired search argument. In our example, the subject is alcohol consumption in schools, to the exclusion of all other drug-related problems. The SELECT command employs truncation on the right (via ?), a proximity operator (4n: in the perimeter of 4 words), the Boolean NOT as well as a restriction to three fields (ti: title, lp: lead paragraph, de: descriptors). If one so wishes, one can save the search argument via

SAVE TEMP [name] for further usage. The DIALOG system lists all databases and adds the number of search results. (This is the extent of Figure D.2.3's representation of the dialog.)

Ranking the databases by number of search results (RANK FILES) is a useful option. The user then selects one or more databases from this list.

We now come to Phase 2, the search for relevant documents in the selected databases. In DIALOG's command terminology, we must work our way through the basic structure B-E-S-T:

- B BEGIN Call up database,
- E EXPAND Browse dictionary,
- S SELECT Search,
- T TYPE Output.

```
?BEGIN 411
File 411:DIALINDEX(R)
DIALINDEX(R)
(c) 1998 The Dialog Corporation plc

*** DIALINDEX search results display in an abbreviated ***
*** format unless you enter the SET DETAIL ON command. ***
?sf papersmj,47 not 703
  You have 16 files in your file list.
  (To see banners, use SHOW FILES command)
?s (college? or universit?) (4n) (liquor or drinking or alcohol)/ti,lp,de not
drug?/ti,lp,de
Your SELECT statement is:
  s (college? or universit?) (4n) (liquor or drinking or alcohol)/ti,lp,de not
drug?/ti,lp,de

  Items    File
  -----
  73    146: Washington Post Online_1983-1998/Oct 12
  13    387: The Denver Post_1994-1998/Oct 09
   3    471: New York Times Fulltext-90 Day_1998/Oct 12
   6    478: Houston Chronicle_1990-1998/Oct 12
  82    498: Detroit Free Press_1987-1998/Oct 08
```

Figure D.2.3: Host-Specific Database Search on the Example of DIALOG File 411. Source: DIALOG.

A user can definitely call up several databases at the same time (e.g. *BEGIN 3, 12, 134, 445*), as long as their field structure and content indexing are identical or at the very least similar. He must then, however, take care to delete the duplicates later. During the deletion process (*REMOVE DUPLICATES*), it must be determined which documents will be deleted and which should remain. Here the systems offer different variants. In DIALOG, the order specified in the BEGIN command decides which duplicates will remain. In other words, if a documentary unit is identical in databases 3 and 12, the one from file 12 will be deleted. An alternative would be to rank documentary units according to quality, perhaps asking for the presence of the full text or abstract in order to then put out the ones that display the most quality characteristics. Since duplicate control is not always reliable, it may make sense to inspect the (presum-

ably) detected duplicates prior to deletion. An interesting strategy might also be to pay particular attention to the documents that exist in several databases. To wit, it has been shown (Hood & Wilson, 2005) that those academic articles which populate a diverse amount of databases are cited more often than those that have only been indexed by one database. (Documents in Hood and Wilson's model collection which occur in two databases are cited 2.84 times on average, while those that occur in six databases are cited 25.53 times)

The EXPAND command is used to look at the field-specific indices, or the Basic Index, to see whether the desired search argument occurs at all, and if so, in which spelling variants. In our example from Figure D.2.1, we searched for Düsseldorf in the field Corporate Source (CS). The command *EXPAND CS=Duesseldorf* shows zero hits for the specific search atom. However, directly below we are presented with (the misspelled variant) Dusseldorf, which occurs in 10,184 documents. The user will correspondingly adopt this spelling variant.

If necessary, the search atoms are truncated on the right and combined into a search argument via Boolean and proximity operators as well as parentheses; this argument is then transmitted to the retrieval system via the search command (in DIALOG: SELECT).

A target-aimed search strategy is to initially search for the search atoms (with truncation where needed) individually (as in Figure D.2.1), and to then combine them via the search IDs. Thus the searcher always has the option of playing through different combination scenarios and choosing the most appropriate one. Here it is important to always know "one's" previous search history. Depending on the retrieval system, this history will be shown as a chart or only after an appropriate command (DISPLAY SET).

After any duplicates have been identified and deleted at this point, we now come to the display of search results retrieved so far. The documentary units are displayed wholly or partially. The display command generally requires three arguments: search step, desired data sets in the search step and output format. If TYPE is the display command, and we wish to view the first five documents in the free format (format 3) in the fourth search step, the command will be *TYPE s4/3/1-5*. If we want to see the title and descriptors of the first documentary unit, we will enter *TYPE s4/ti,de/1*.

We now come to the concluding Phase 3 of the Needle-in-the-Haystack-Syndrome, modifying the search results. We must distinguish between methods that enhance recall (in a hit list of insufficient size) from those that enhance precision (where the list is too large). If we raise recall, this will lead to greater sensitivity, and if, on the other hand, we raise precision, the list of search results will be more specific, as Haynes and Wilczynski (2004, 1043) point out:

Searchers who want retrieval with little non-relevant material can choose strategies with high specificity. For those interested in comprehensive retrievals or in searching for ... topics with few

citations, strategies with higher sensitivity may be more appropriate. The strategies that optimised the balance of sensitivity and specificity provided the best separation of eligible studies from others but did so without regards for whether sensitivity or specificity was affected.

Choosing the right search arguments—in the sense of “hedges”—is of central importance for a successful retrieval. One must consider (in this order) the descriptors of the controlled vocabulary (including hierarchical relations), title terms, document type (i.e. research article or review), the manner in which the subject is approached (e.g. theoretical work or empirical study), terms from the abstract as well as, possibly, words from the text. Among the secondary criteria are the publication year of the documentary reference unit as well as its language.

If the most appropriate descriptors for one’s information needs are unknown, there is a heuristic method that can be used to find them. Here the informetric functionality of the retrieval system is being used. The searcher first uses relevant words to search in the title field and requests a chart with the descriptors, ranked according to frequency (e.g. *RANK DE TOP 10*). The top-ranked descriptors are the ones that should then be used.

Recall-enhancing measures include the use of truncation, hierarchical search (incorporation of hyponyms and hyperonyms as well as related terms) as well as the consideration of further search atoms connected via OR. If precision-enhancing measures have been taken in a previous step, their removal will heighten recall.

Precision is attained via further search atoms connected via AND and NOT, the use of proximity operators as well as the removal of any previously used recall-enhancing methods.

Query Expansion: Building Blocks and Growing Citation Pearls

The development of an ideal search argument requires the use of strategies (Belkin & Croft, 1987). This is called “query expansion” (Efthimiadis, 2000). Occasionally, a distinction is made between query expansion (for short queries) and “query relaxation” (for longer ones) (Kumaran & Allan, 2008, 1840). Efthimiadis (1996, 126 et seq.) here distinguishes between the two forms of “building blocks strategy” and the “citation pearl growing strategy”, where the pearl that is meant to be grown is represented by a particularly relevant document.

In the building blocks strategy (Figure D.2.4), the user divides his information problem into different facets A, B, C etc., which may contain one or more terms. He then searches for appropriate words for the single facets, e.g. synonyms or quasi-synonyms, which are combined via the Boolean OR. In the course of the search dialog, further words are added, others may be removed. The facets are interconnected either via the Boolean AND (or NOT) or a proximity operator (represented in the graphic by the + sign). In search practice, it has proven effective to perform each step separately

and to save it in one's search history, in order to allow for modifications to the individual facets at all times.

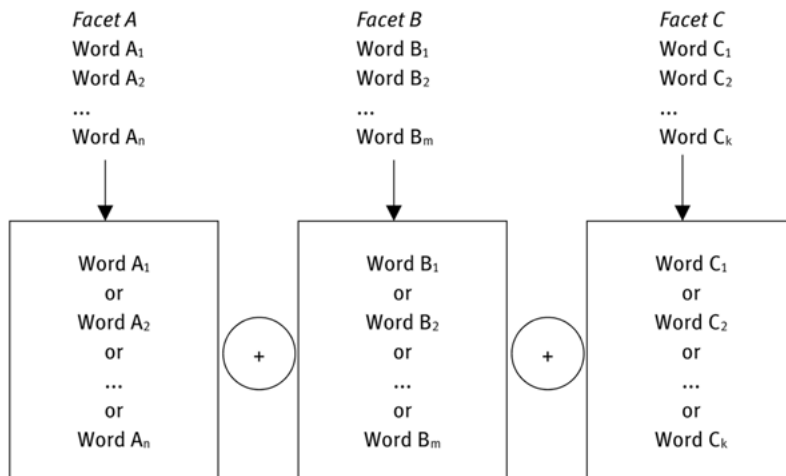


Figure D.2.4: The Building Blocks Strategy during Query Modification. Source: Following Efthimiadis, 1996, 127.

In the building blocks strategy, the searcher's expert knowledge informs the query formulation (vom Kolke, 1994, 125-128). For instance, a query for *emission control of diesel engines in cogeneration units* entices him to construct three blocks, *emission control*, *diesel engine* and *cogeneration unit*, in addition to their respective quasi-synonyms. However, if he knows that the emission control of diesel engines in cogeneration units proceeds in the same way as everywhere else, he will only use the blocks *emission control* and *diesel engine*.

In the strategy of growing citation pearls (Figure D.2.5), an intermediate goal is to retrieve an ideally appropriate document via the initial search, which is then terminologically "exploited". The most specific term $T_{\max}$ is gleaned from each of the facets A, B, C etc.; these terms must then be interconnected via AND as well as via proximity operators, as the latter lead to more precise results.

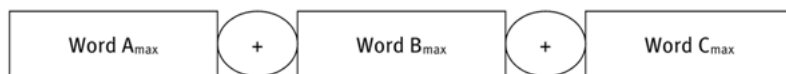


Figure D.2.5: Growing "Citation Pearls" during Query Modification. Source: Following Efthimiadis, 1996, 127.

Efthimiadis (1996, 128) describes the process of retrieving ever more "pearls" and the search arguments that can be gleaned from them:

The searcher then calls up one or more of these citations for online review, noting index terms (descriptors), and free text terms found in some of the relevant citations. These new terms are incorporated into subsequent query reformulations to retrieve additional citations. After adding these terms to the query and searching, one can again review more retrieved citations and continue this process in successive iterations until no additional terms that seem appropriate for inclusion are found.

Berrypicking

In view of the multitude and variety of electronic information resources, it would be erroneous to assume that one specialist database and one WWW search engine would suffice to retrieve everything that satisfies one's information need. Rather, there are diverse services, which must each be addressed with their own different query formulations. In the model by Bates (1989), the world of digital information is searched in the same way that a forest is searched for berries. One begins with the system that appears to fit the subject matter. A first hit list will provide the searcher with suggestions for modified searches in the same database, i.e. suitable descriptors and notations or the name of a certain author. In addition, it will be shown that this database alone cannot quench one's thirst for information. Further databases are accessed. When searching a database for scientific articles, one will be notified, for example, that there are patents concerning that same subject. Consequently, one will search a patent database. Here one is told that a certain company holds particularly many patents. A search engine yields the web presence of that company. Additionally, a news database is called up in order to browse press reports on the company. The "berrypicking" of information is pursued "until the basket is full", that is until the need for information that is not satisfied is left at a minimum.

Conclusion

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- Command-based retrieval systems permit an ideal usage of their system functionality, but they need the searcher to be able to master them. Menu-based systems guide the professional end user or the layman through certain (generally small) parts of the functionality.
 - Retrieval systems that allow one to enter search atoms without operators demand a Boolean interpretation of the search argument. Merely using AND or OR is not enough. When hierarchical KOSs are used, it becomes an option to distinguish between AND and OR for each search as well as to translate the vernacular NOT.
 - Selective Dissemination of Information (SDI) requires the creation of user-specific information profiles, which are then worked through one by one as new documents enter the database. The user is notified about the new information either via e-mail or via his website. (Apart from the Boolean model, SDI also allows for other retrieval models)
 - Search strategies resemble the proverbial needle in the haystack. They encompass the three phases of (1st) searching for and retrieving the relevant information services, (2nd) searching and

retrieving the relevant documents in the respective database, then (3rd) modifying the selected strategies via measures to enhance recall or precision.

- The ideal search argument is developed via search strategies. We distinguish between the Building Blocks Strategy and the Citation Pearl Strategy.
 - In view of the multitude and variety of information services, any search process requires the user to work his way through diverse databases. “Berry picking” represents a plastic model for this enterprise.
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D.3 Weighted Boolean Retrieval

Boolean Queries and Weighting

Regarding the question whether a document matches a search argument, the Boolean model only knows two manifestations: 0 and 1. This precondition clearly limits Boolean information retrieval. Boolean retrieval is particularly prevalent in bibliographical databases, whose “practical inadequacies”, according to Homann and Binder (2004, 98),

result from the implicit premise that search terms are either completely irrelevant or 100% relevant. Accordingly, any documents thus yielded are either designated as 100% relevant (exact hits) or as not relevant at all.

For some queries, the model works counter-intuitively (Fox et al., 1992). In the case of a query $A \text{ AND } B \text{ AND } C \text{ AND } D \text{ AND } E$, we will only retrieve documents that contain all five search atoms. However, a user might be satisfied with mere four (or fewer) hits. In a query $A \text{ OR } B \text{ OR } C \text{ OR } D \text{ OR } E$ a Boolean system yields documents if they contain at least one of the search atoms. Here, the user might surmise that documents rise in importance the more search terms they contain. However, Boolean models do not allow for a Relevance Ranking. Indexers can assign certain keywords to the documentary reference units during input, or not; they cannot gradate according to importance. In this same way the user is unable to allocate weightings to the search atoms when formulating his query.

It is the ambition of extended Boolean systems (Salton, Fox, & Wu, 1983) to break through these boundaries. These systems employ weight values that are allocated to the terms in the documents. These respective weighting values mainly stem from text-statistical methods (Ch. E.1) that are mapped onto the interval $[0,1]$. Partly, the systems also allow users to weight their query terms themselves—this, too, is converted to the interval $[0,1]$.

The goal of all weighted Boolean models is to combine the advantages of weighting and those of “classical” Boolean retrieval, as Bookstein (1978, 156) explains:

Weighted and Boolean retrieval schemes are among the most popular in both functioning and experimental information retrieval systems. Weighted schemes have the advantage of producing ranked lists of documents; Boolean systems allow the user to state his need with great precision. It is natural to try to combine these modes into a system providing the precision of a Boolean search and the advantages of a ranked output.

Weighting values for terms in documents (d) or queries (q) stem from the interval $[0,1]$; the two end values remain, allowing “classical” Boolean retrieval to be regarded as a special instance of the weighted Boolean model. When a user weights a search atom t with the weight value w (from $[0,1]$), we note this via $\langle t, q; w(t, q) \rangle$. Supposing a

user wants to search for *information retrieval* with a weight of 0.75, and for *Salton* with a weight of 0.25, the result will be the formulation

<“Information Retrieval”; 0.75> AND <Salton; 0.25>.

Analogously, the weighting value $w(t,d)$ is determined for every term t of a text document d . Since text statistics is able to calculate values greater than 1, the respective values must be mapped onto the interval $[0,1]$; i.e., normalized. This can be achieved by dividing all values by the greatest value, for instance.

The “Wish List” by Cater, Kraft and Waller

How do we interpret the standard operators AND, OR and NOT in weighted Boolean systems? Waller and Kraft (1979; similarly, Kraft & Waller, 1979), and Cater in cooperation with Kraft (1989) introduce a mathematical model that determines “legitimate Boolean expressions” in the context of weighted terms. This model has entered the information science literature as the “Waller-Kraft Wish List” (see also Cater & Kraft, 1987 for a system that fulfils all criteria of the wish list).

Let the value e be the retrieval status value (RSV) of a document with regard to specific search atoms and operators used; t is a term that occurs in a document d with the weight $w(t,d)$, respectively in a query q with the weight $w(t,q)$. The query term t in q is either identical to the term t in the document d or has otherwise been recognized as the equivalent to t in q (e.g. as a synonym of the query term or as a search result for a truncated search atom).

When considering weighting values from the query in addition, the query term weight $w(t,q)$ will be multiplied with the term weight in the document $w(t,d)$. If the search is not weighted, $w(t,q) = 1$ for all search atoms.

In its final variant (Cater & Kraft, 1989), the Wish List comprises eight aspects.

(1) In the case of a search with only one search atom, the product of the weighting of the search term and the weight of the term in the document is identical to the retrieval status value of the entire document d :

$$e(d) = w(t,q) * w(t,d).$$

When using several atoms t_1,q , t_2,q , t_3,q etc. to formulate one’s query, the document’s retrieval status value will depend upon the individual weight values of the corresponding document terms $w(t_1,d)$, $w(t_2,d)$, $w(t_3,d)$ etc. The individual weight values of the respective appropriate document terms are aggregated into the retrieval status value of the document for the respective query. In other words: only the weight values of the respective appropriate search arguments in a document influence the entire document’s RSV. This is what Waller and Kraft (1979, 240) call “separability”.

(2) The mechanisms for calculating the retrieval status value are constructed in such a way that they present the usual characteristics of Boolean algebra (provided we exclusively use the extreme values of 0 and 1). This criterion allows us to view weighted Boolean systems as generalizations of classical Boolean systems.

(3) Logically equivalent query formulations always lead to identical results (i.e. to identical e-values). To illustrate, let us assume the following search atoms: t_1 is given the weight of 0.75 by the user, t_2 0.4 and t_3 0.2. In that case, the two variant formulations

$$q_1 = (\langle t_1; 0.75 \rangle \text{ AND } \langle t_2; 0.4 \rangle) \text{ OR } \langle t_3; 0.2 \rangle$$

and

$$q_2 = (\langle t_1; 0.75 \rangle \text{ OR } \langle t_3; 0.2 \rangle) \text{ AND } (\langle t_2; 0.4 \rangle \text{ OR } \langle t_3; 0.2 \rangle)$$

are equivalent in the sense of the disjunctive distributive law. This criterion of consistency is “perhaps the most important one of all” for Waller and Kraft (1979, 240).

(4) The retrieval status value (e) of a document rises monotonously (for all t_j greater than zero), in step with the weighting values of the terms (Cater & Kraft, 1989, 23-24).

(5) Here, we are dealing with the special case of a user explicitly defining a weight of zero. Cater and Kraft insist on considering this weighting value during the calculation of e (1989, 24):

Weights of zero in queries should be treated as any other query weights. This requirement means that there is an essential difference between not including a term in a query and including it with a weight of zero. In the first case, one is not interested in the term, while in the second case, one is interested in the term, and is further interested in finding documents having weights near zero in that term.

(6) When a user lowers the weighting value for a search atom, its retrieval status value will also decrease. Further documents will be retrieved—where available—and listed at the end of the previous ranking.

(7) When new search atoms, connected via AND, are added to a query, one will not find any new documents (but the hit list can become smaller). When new atoms are created and connected via OR, the original documents in the hit list remain the same (but the hit list can increase in size).

(8) When a negation is performed in the search argument, DeMorgan’s laws also apply in weighted Boolean retrieval.

Lastly, let us point out that the Wish List by Cater, Kraft and Waller can only be used in the context of purely text-oriented Relevance Ranking models. When aspects from outside the texts (e.g. the linking of documents in the link-topological model)

are used for Relevance Ranking, certain aspects of the Wish List (such as separability) do not apply.

Since the definition of weighting values tends to be problematic, particularly for laymen users, the interfaces must guide the users intuitively. One possible option would be the provision of a search window for each term, with a slider depicted behind it. After the user pushes the slider into a certain position, the system will read the numerical value that corresponds to it and use it as a query weight value.

Minimum-Maximum Aggregation Model

In the case of several search atoms, how do we aggregate the weighting values of the respective appropriate terms into the document's retrieval status value? In the following, we will introduce three approaches:

- Aggregation via Minimum or Maximum,
- Arithmetical Aggregation,
- Aggregation via Weighted Minimum and Maximum.

In many-valued logic (e.g. that of Łukasiewicz from the year 1920), the AND and OR operators follow the minimum and the maximum of the values, respectively, and the negation follows the difference of 1:

$$\begin{array}{ll} \text{NOT(many-valued)} A = 1 - A & \text{(negation),} \\ A \text{ AND(many-valued)} B = \text{MIN}(A, B) & \text{(conjunction),} \\ A \text{ OR(many-valued)} B = \text{MAX}(A, B) & \text{(disjunction).} \end{array}$$

These thoughts are taken up by Zadeh (1965) in his “fuzzy logic”. Via Zadeh's work, they then enter the theory of information retrieval.

In a simple variant, fuzzy retrieval exclusively works with Minimum and Maximum (Min and Max Model, in short: MM Model). As already mentioned above, t is a term occurring in a document d with the weighting value $w(t,d)$. In a “naïve” version, the retrieval status value e of a document d is aggregated as follows:

$$\begin{aligned} e(d) [(t_1 \text{ AND } t_2 \text{ AND } t_3 \text{ AND } \dots)] &= \text{MIN} [w(t_1,d), w(t_2,d), w(t_3,d), \dots], \\ e(d) [(t_1 \text{ OR } t_2 \text{ OR } t_3 \text{ OR } \dots)] &= \text{MAX} [w(t_1,d), w(t_2,d), w(t_3,d), \dots], \\ e(d) [\text{NOT } t] &= 1 - w(t,d). \end{aligned}$$

An AND retrieves documents containing all search arguments connected via AND. The retrieval status value of the individual document follows the weight value of the term with the lowest weight (Minimum). In the OR connection, we select documents that satisfy at least one search argument. The document's retrieval status value now solely depends upon the maximum value of the weighting values of the search terms in the document. The negation NOT leads, in this model, to a retrieval status of 1

minus the weighting value of the term in the document. If the term does not occur in the document, the negation will result in a document value of 1. (For the naïve model, cf. Yager, 1987; Bookstein, 1980).

Arithmetic Aggregation

The fact that the MM model is, after all, too naïve is shown by the following example (Lee et al., 1993, 294). Let there be the following two documents:

$$d_1 = \{(\text{Thesaurus}; 0.40), (\text{Clustering}; 0.40)\} \text{ and} \\ d_2 = \{(\text{Thesaurus}; 0.99), (\text{Clustering}; 0.39)\}.$$

A user will now ask (without any weighting) for

Thesaurus AND Clustering.

Both d_1 and d_2 meet the search requirement and are yielded, due to the Minimum in the sequence (1st) d_1 (Minimum is 0.40), (2nd) d_2 (Minimum is 0.39). However, many people would intuitively argue that the second document is more relevant than the first one (due to the importance of the term *thesaurus*).

A possible arithmetical variant (which departs from fuzzy logic, however) is to no longer use the Minimum for the AND but the sum of the individual weighting values instead:

$$e(d) [(t_1 \text{ AND } t_2 \text{ AND } t_3 \text{ AND } \dots)] = [w(t_1, d) + w(t_2, d) + w(t_3, d) + \dots].$$

The example leads to a changed Relevance Ranking: (1st) d_2 ($0.99 + 0.39 = 1.38$), (2nd) d_1 ($0.40 + 0.40 = 0.80$).

As an alternative to the sum, one could also take into consideration the product

$$e(d) [(t_1 \text{ AND } t_2 \text{ AND } t_3 \text{ AND } \dots)] = [w(t_1, d) * w(t_2, d) * w(t_3, d) * \dots]$$

(Waller & Kraft, 1979, 242). This leads to the following retrieval status values (1st) d_2 ($0.99 * 0.39 = 0.3861$), (2nd) d_1 ($0.40 * 0.40 = 0.16$).

Mixed Minimum-Maximum Aggregation Model

A more elaborate variant of fuzzy retrieval is introduced by Fox and Sharat (1986) (see also Fox et al., 1992, 396, and Lee et al., 1993, 294) in the form of the MMM Model (“Mixed Min and Max Model”). Here, coefficients are introduced to “soften” the AND

and the OR (softness coefficients). The determinations of Maximum and Minimum remain, but they are combined with different weights, both for the AND and for the OR.

$$\begin{aligned} e(d) [(t_1 \text{ AND } t_2 \text{ AND } t_3 \text{ AND } \dots)] &= \gamma * \text{MAX} [w(t_1, d), w(t_2, d), w(t_3, d), \dots] \\ &+ (1 - \gamma) * \text{MIN} [w(t_1, d), w(t_2, d), w(t_3, d), \dots] \text{ for } 0 \leq \gamma < 0.5, \\ e(d) [(t_1 \text{ OR } t_2 \text{ OR } t_3 \text{ OR } \dots)] &= \gamma * \text{MAX} [w(t_1, d), w(t_2, d), w(t_3, d), \dots] \\ &+ (1 - \gamma) * \text{MIN} [w(t_1, d), w(t_2, d), w(t_3, d), \dots] \text{ for } 0.5 < \gamma \leq 1. \end{aligned}$$

Maximum and Minimum are each weighted with γ resp. $1 - \gamma$. The Minimum is weighted more highly for the AND, whereas the Maximum is given preference for the OR.

A Fuzzy Operator: ANDOR

The softener γ more or less dilutes the strict forms of the Boolean AND and OR, which depend upon the concrete value of γ . In the case of $\gamma = 0.5$ the AND and OR ratios are completely balanced. This leads Kraft and Waller to introduce the new functor ANDOR (Kraft & Waller, 1979, 179-180; Waller & Kraft, 1979, 243-244). Waller and Kraft (1979, 244) emphasize:

(h)ere the connector itself is fuzzy, as well as the weights. This has no parallel with the Boolean connectors except that it is a combination of the “AND” and “OR”.

The formula is constructed analogously to the elaborate MMM Model. The difference lies in the value for γ being left undefined:

$$e(d) [(t_1 \text{ ANDOR } t_2 \text{ ANDOR } t_3 \text{ ANDOR } \dots)] = \gamma * \text{MAX} [w(t_1, d), w(t_2, d), w(t_3, d), \dots] + (1 - \gamma) * \text{MIN} [w(t_1, d), w(t_2, d), w(t_3, d), \dots] \text{ for } 0 \leq \gamma \leq 1.$$

The user is called upon to define γ by himself. Values greater than 0.5 approach a fuzzy OR; values below incline toward a fuzzy AND.

When considering laymen users, as well as our slider for adjusting the term weights, we will have to introduce a second slider that regulates the operator’s respective AND and OR similarities.

Advantages and Disadvantages of Weighted Boolean Systems

We have already seen, in classical Boolean retrieval, that its usage is difficult for untrained users. This statement is doubly true for weighted Boolean retrieval. Here,

the user must not only be able to use the operators AND and OR, but also (at least in many cases) to assign weighting values to his search terms. As we learn from looking at the history of information retrieval, weighted Boolean systems have not managed to prevail in practice.

In the model of fuzzy logic, we are confronted with the arbitrariness of the Maximum and of the Minimum. This basic assumption is not entirely intuitive.

A clear advantage of weighted Boolean systems is their combination of the tried-and-true Boolean basic functionality with the weighting values in documents, as well as—where provided—in queries. This allows for the documents to be arranged in a Relevance Ranking.

Conclusion

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- In the classical Boolean model, documents and queries do not allow for any manifestations apart from 0 and 1. Weighted Boolean retrieval uses weighting values in the interval $[0,1]$ for search terms, for terms in the documents and for entire documents. This makes a relevance-ranked output possible.
 - Weighted Boolean systems unite the advantages of Boolean operators with those of the weightings.
 - The “Wish List” by Cater, Kraft and Waller describes a mathematical model for legitimate Boolean expressions that includes weighted terms and documents. Important aspects include the dependence of a document’s retrieval status value on the individual terms as well as the equivalence of the retrieval status values of documents in logically equivalent query formulations.
 - The fuzzy Boolean models have two variants: the naïve MM Model (Min and Max Model), which exclusively uses the Minimum of the values (for conjunction) respectively their Maximum (for disjunction), and the elaborate MMM model, which combines the Minimum and Maximum values.
 - Weighting values of terms in the query are taken into account by being multiplied with the term weighting in the document.
 - A new functor, itself fuzzy, is introduced in the form of ANDOR. Here, the user enters a numerical value of $[0,1]$ and then decides for himself whether the functor should work like an AND or like an OR.
 - When offering weighted Boolean systems to end users, one must make the (not always straightforward) theoretical relations intuitively understandable. One possibility would be the use of sliders to adjust weighting values for the search atoms as well as for the AND/OR combinations.
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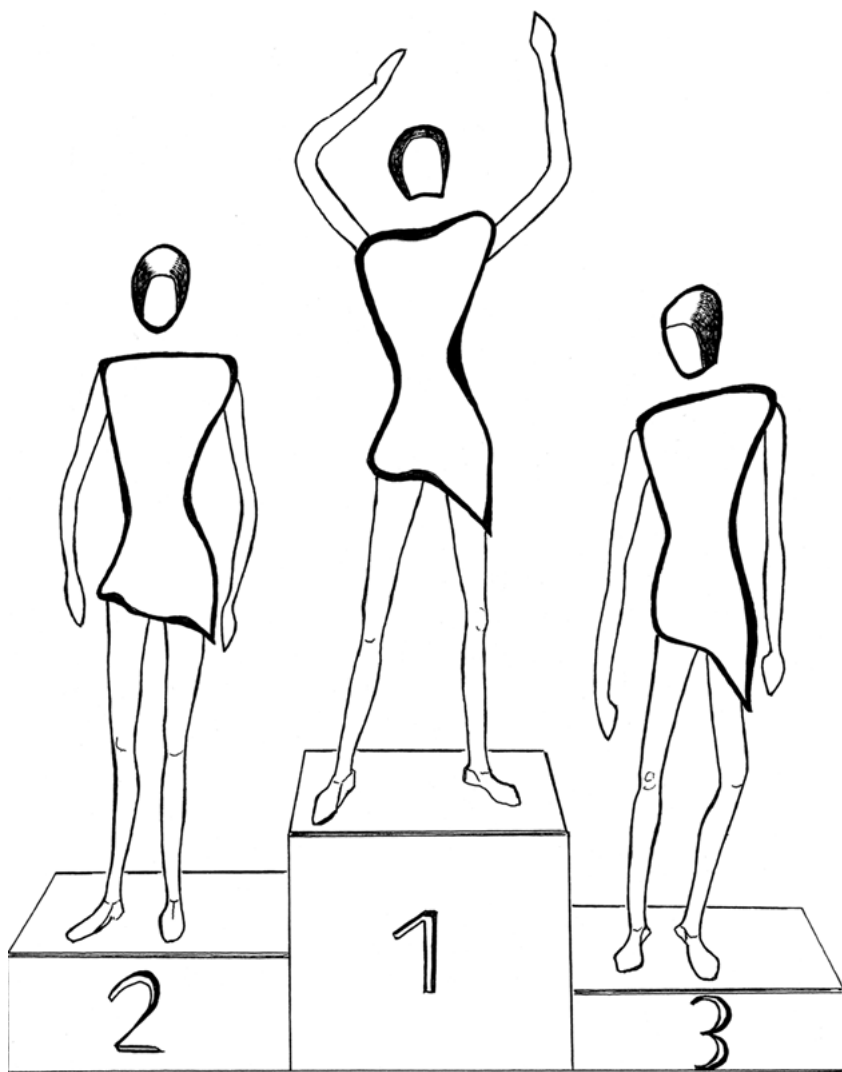
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Part E

Classical Retrieval Models



E.1 Text Statistics

Depending on the volume and maturity of the respective modules, the information-linguistic functionality makes it possible to locate natural-language documents (that more or less match a search query) in databases. Information linguistics alone is not able to bring the retrieved documents into a relevance-based order and yield them to the user. In the case of large amounts of search results (“large” begins at around ten documents, particularly for laymen), it is necessary to apply algorithms of Relevance Ranking.

Luhn’s Thesis: Term Frequency as Significance Factor

It is an obvious option to register the texts themselves in such a way that indicators for ranking by relevance can be derived from the structure of the terms in a text. The approach of using statistical term distributions in texts as the basis for Relevance Ranking goes back to Luhn. It is the words in the texts that represent the “ideas” of an author.

Since an author wants to be understood, he will put his ideas into words in such a way that his audience will be able to adequately comprehend them—on the basis of shared experiences. Luhn (1957, 311) describes the process of written communication:

The process assumes static qualities as soon as ideas are expressed in writing. Here the addressor has to make certain assumptions as to the make-up of the potential addressee and as to which level of common experience he should choose. Since the addressor has to rely on some kind of indirect feedback, he might therefore be guided by the degree to which the written expressions of ideas of others has raised the level of common experience relative to the concepts he wishes to communicate.

The author will choose that terminology which he shares with “his” circle of addressees. In this regard, it makes sense to use the terms of a natural-language text as the basis for automatic indexing (Luhn, 1957, 311):

It may be assumed that the means and procedures ... permit communication to be accomplished in a satisfactory manner. If it is possible to establish a level of common experience, it seems to follow that there is also a common denominator for ideas between two or more individuals. Thus the statistical probability of combinations of similar ideas being similarly interpreted must be very high.

Words in texts are not all equally well-suited for transporting ideas. The objective is to find the “significant words” (Luhn, 1958, 160):

It is here proposed that the frequency of word occurrence in an article furnishes a useful measurement of word significance.

The idea of a term's frequency in a text being a measure of its relevance has entered the history of information retrieval as "Luhn's Thesis", it forms the basis of text statistics.

Luhn's thesis does not state that term frequency and relevance correlate positively. Hence, it is not true that the more frequently a term appears, the more relevant it is. The distribution of words in a text follows a law; this law creates a connection between the ranking of a text's words, sorted by frequency (r), and the absolute frequency of the word in question (freq). The law, formulated by Zipf (1949), states that

$$r * \text{freq} = C$$

and it holds for all frequently occurring words in a text. The specific value of the constant C is text-specific. We are thus faced with a distribution of words in texts that is skewed to the left (indicated in Figure E.1.1 by the thick line). Words with a high frequency of occurrence tend to be function words that carry little or no meaning and are of no use for retrieval. Equally useless, according to Luhn, are those terms that occur extremely rarely. The significance of words in a text follows a bell curve that reaches its apex at the E-point (in Figure E.1.1) (Luhn, 1958, 160):

The presence in the region of highest frequency of many of the words ... described as too common to have the type of significance being sought would constitute "noise" in the system. This noise can be materially reduced by an elimination technique in which text words are compared with a stored common-word list. A simpler way might be to determine a high-frequency cutoff through statistical methods to establish "confidence limits". If the line C in the figure represents this cutoff, only words to its right would be considered suitable for indicating significance. Since degree of frequency has been proposed as a criterion, a lower boundary, line D, would also be established to bracket the portion of the spectrum that would contain the most useful range of words.

It is the task of text statistics to allocate relevance to the text words in such a way that their distribution will follow the dashed line in Figure E.1.1, with the most important terms being located near the E-point and the less important ones at an appropriate distance (to the right and left of E, respectively). If no stop word list is being used, it must further be made sure that the high-frequency words play no role in the determination of the significance of a text. Likewise, the low-frequency words must be paid special attention.

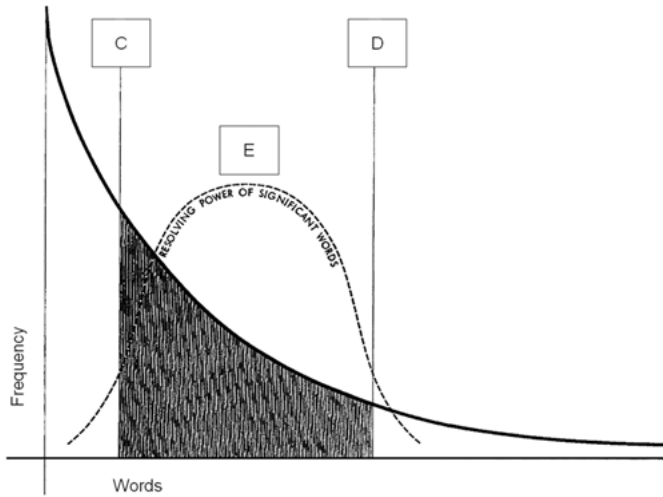


Figure E.1.1: Frequency and Significance of Words in a Document. Source: Luhn, 1958, 161.

What is a Term?

The weighting values of the terms form the basis for the document's place in the Relevance Ranking of the respective hit lists. "Term" (t), in natural-language texts, can have different meanings, depending on the degree of maturity of the information-linguistic functionality used in the retrieval system:

- t_1 unprocessed (inflected) word form,
- t_2 Basic Form (lexem) / Stem
- t_3 Phrase: for words that consist of several components: has the phrase been formed correctly?
- t_4 Decompounding: for words that represent a combination of several components: have the individual meaningful parts been recognized in addition to the compound?
- t_5 "named entities" are recognized as such,
- t_6 terms t_1 through t_5 with input error correction,
- t_7 concept (Basic Form / Stem with synonyms; in case of homonyms, separation into the different terms),
- t_8 concept including the anaphora of all of its words.

The ideal scenario t_8 assumes that each word form has run through a process of error recognition and correction, that all forms of one and the same word have been amalgamated into the basic form or the stem, that both phrasing and compound decomposition have been successfully completed, that the algorithm has recognized all per-

sonal names while performing the above, that all synonyms of the word have been summarized in the concept and, finally, that the system was able to correctly allocate all anaphora and ellipses.

If, in the following, we speak only of “term” (t), we leave open the question of which degree of maturity t_i the system has reached, since the weighting algorithms are analogously applicable to all kinds of t_i .

Document-Specific Term Weight (TF / WDF)

In the document-specific weighting of a term, we attempt to find a quantitative expression for this term’s importance in the context of a given document. Since all variants of the determination of document-specific term weight work with frequencies, the established shortcuts are either TF (“term frequency”) or WDF (“within-document frequency”). This weighting method is called a “bag of words model” (Manning, Raghavan, & Schütze, 2008, 107) because it ignores the ordering of terms in documents and applies only the terms’ number:

We only retain information on the number of occurrences of each term. Thus, the document *Mary is quicker than John* is, in this view, identical to the document *John is quicker than Mary*.

A very simple form of quantitatively defining a term in a document is to allocate 0 if the term does not occur and 1 if it occurs at least once. This does not involve a weighting in the actual meaning of the word, rendering such an approach unsuitable for the purposes of Relevance Ranking.

An equally easily applicable variant is to count the occurrence of terms in the document. This absolute frequency of the term in question is, however, dependent upon the total length of the document. In longer texts, the term’s frequency of occurrence is greater for the sole reason that the text is longer. This makes absolute term frequency an unsatisfactory weight indicator.

Relative term frequency, on the other hand, is a first serious candidate for document-specific word weight, as Salton (1968, 359) described it in the early years of retrieval systems:

Absolute word frequency statistics are normally rejected as a basis for the determination of word significance, since the length of the texts is then a disturbing factor. Instead, *relative frequency* measures are normally used, where the frequency of a given word in a given text is compared with the frequencies of all other words in the text.

If $\text{freq}(t,d)$ counts the frequency of occurrence of term t in document d , and L the total number of words in d , then the document-specific term weight in the variant “relative frequency” (TF-relH-Salton) is calculated via

$$\text{TF-relH-Salton}(t,d) = \text{freq}(t,d) / L.$$

Croft (1983) uses two modifications in his approach. First, he employs the frequency value of the word that occurs most often in d ($\text{maxfreq}(d)$) as the basis of comparison for relative frequency, and not the total number of words in the document. Secondly, he introduces a (freely modifiable) factor K ($0 \leq K < 1$), which allows for the relative meaning of the document-specific document weight to be specified. Croft's calculation formula (TF-relH-Croft) reads:

$$\text{TF-relH-Croft}(t,d) = K + (1 - K) * \text{freq}(t,d) / \text{maxfreq}(d).$$

Neither variant of relative frequency has managed to assert itself for wider usage. The winning variant is one which uses logarithmic values instead of percentages. We will call this variant WDF. Harman (1992b, 256) justifies the use of a logarithmic measurement:

It is important to normalize the within-document frequency in some manner, both to moderate the effect of high frequency terms in a document (i.e. a term appearing 20 times is not 20 times as important as one appearing only ones) and to compensate document length.

Since the logarithm base two (ld ; *logarithmus dualis*) has yielded satisfactory results in experiments (Harman, 1986, 190), Harman introduces the following formula as the calculation method:

$$\text{WDF}(t,d) = [\text{ld}(\text{freq}(t,d) + 1)] / \text{ld } L.$$

Why does the WDF formula calculate with a “+1” in the numerator? If a term does not occur in a document at all, i.e. if $\text{freq}(t,d) = 0$, then the numerator will read $\text{ld}(0 + 1)$, hence $\text{ld}(1) = 0$ and of course $\text{WDF}(t,d) = 0$; which, intuitively, is exactly what is to be expected at this point.

Which advantages does the logarithmic WDF weight have vis-à-vis the relative frequency? We will explain this via an example. Let a document d_1 have a length of 1,024 words, in which the term A occurs seven times. We will calculate WDF (after Harman) and TF-relH-Salton as follows:

$$\text{WDF}(A,d_1) = [\text{ld}(7 + 1)] / \text{ld } 1,024 = 3 / 10 = 0.3$$

$$\text{TF-relH-Salton}(A,d_1) = 7 / 1,024 = 0.0068.$$

Term B occurs 15 times in the same document:

$$\text{WDF}(B,d_1) = [\text{ld}(15 + 1)] / \text{ld } 1,024 = 4 / 10 = 0.4$$

$$\text{TF-relH-Salton}(B,d_1) = 15 / 1,024 = 0.0146.$$

The value area for the WDF is more “clinched” than it is for relative frequency, since it does not pull the weighting values apart quite as much.

Field- or Position-Specific Term Weight

Document-specific term weight is refined by having the term’s placement be considered at certain positions in the text. In the case of a structured document description, e.g. a bibliographic data set, the term may appear in the title, controlled terms (CT) and abstract fields, or in the full text. For weakly structured documents, on the other hand (e.g. in HTML), a term must be located in the different metatags (description, keywords), in the title tag or in the body. It is also possible to designate terms via a special label (occurrence in H1, H2, H3 etc., highlighted via italics or bold type or a larger font). The designation occurs via (freely adjustable) constant values, e.g. 2 for occurrence in the title, 1.7 for the CT field, 1.3 in the abstract and 1 in the text.

A simple procedure consists of determining a positional value P for a term (t) by adding the largest constant to the WDF as a factor:

$$P(t,d) * WDF(t,d).$$

If, for instance, a term occurs in the title and in the text, the value 2 will be used for P . If t occurs in the abstract (but not in either title or in the field of controlled terms), 1.3 will be used, etc. If, finally, t is only to be found in the continuous text, $P = 1$ and the WDF value will remain unchanged.

The simple procedure takes into consideration neither the term’s occurrence in several different fields, or positions, nor its frequency in them. A more complex procedure works with multiply weighted fields (Robertson, Zaragoza, & Taylor, 2004). We count the position-specific frequency of the term in the title [$\text{freq}(t\text{-Title},d)$], in the controlled terms field [$\text{freq}(t\text{-CT},d)$], in the abstract [$\text{freq}(t\text{-Abs},d)$] as well as in the continuous text [$\text{freq}(t\text{-Text},d)$] and then weight the respective frequency via the introduced constants. (Instead of title, abstract etc. we could of course also work with other text positions—provided the appropriate constant values have been introduced.) When calculating the position-squared WDF (PWDF), we must proceed analogously for the denominator of L , counting the number of title terms [$L(\text{Title})$], the controlled terms [$L(\text{CT})$], etc. one by one. It holds, of course, that $L(\text{Title}) + L(\text{CT}) + L(\text{Abs}) + L(\text{Text}) = L$. The PWDF of t in d , then, is calculated via:

$$\begin{aligned} \text{PWDF}(t,d) = \{ & \{ld (2 * [\text{freq}(t\text{-Title},d)] + 1,7 * [\text{freq}(t\text{-CT},d)] + \\ & 1,3 * [\text{freq}(t\text{-Abs},d)] + [\text{freq}(t\text{-Text},d)] + 1\} \\ & / \{ld (2 * L(\text{Title}) + 1,7 * L(\text{CT}) + 1,3 * L(\text{Abs}) + L(\text{Text}))\}. \end{aligned}$$

Term Weight via Inverse Document Frequency (IDF)

Terms in databases can be used to accurately search documents in a number of different degrees. A term that occurs in all documents of the database, for instance, is entirely non-discriminatory. A term's degree of discrimination rises together with the number of documents in which it does not occur—or, in the reverse: the less often it occurs in the documents of the database as a whole, the more selective it is. Robertson (2004, 503) emphasizes:

The intuition was that a query term which occurs in many documents is not a good discriminator, and should be given less weight than one which occurs in few documents, and the measure was an heuristic implementation of this intuition.

The measure is today called “inverse document frequency” (IDF); the term was introduced into information science in 1973 by Spärck Jones (a spelling variant is “Sparck Jones”). Luhn—as we have seen—wanted to ban both too-frequent and too-rare terms from the query process. This procedure is not always prudent, since frequent terms are also used for searches. In a legal database, the term “law” will occur very often, but it will still be required in a query for, say, “law AND internet”. For Spärck Jones, the solution lies not in the document and its term distribution but in the term distribution of the entire data basis (Spärck Jones 2004a [1972], 498-499; see also Spärck Jones, 2004b):

The natural solution is to correlate a term's matching value with its collection frequency. At this stage the division of terms into frequent and non-frequent is arbitrary and probably not optimal: the elegant and almost certainly better approach is to relate matching value closely to relative frequency. The appropriate way of doing this is suggested by the term distribution curve for the vocabulary, which has the familiar Zipf shape. Let $f(n) = m$ such that $2^{m-1} < n \leq 2^m$. Then where there are N documents, the weight of a term which occurs n times is $f(N) - f(n) + 1$.

In logarithmic notation, the Sparck Jones formula for the IDF weighting value for term t reads:

$$\text{IDF}(t) = [\text{ld } (N/n)] + 1,$$

where N is the total number of documents in the entire database and n counts those documents in which the term t occurs at least once. A variant of the IDF formula is a formulation without the addend 1:

$$\text{IDF}'(t) = [\text{ld } (N/n)]$$

(Robertson, 2004, 504). The designation “inverse” points to the structure of the formula: it is built in the fashion of a calculation of relative frequency (n/N), only the other way around—hence “inverse”.

Let us give another example: Our sample term A occurs in 7 documentary units, of which there are 3,584 in the database. According to Sparck Jones:

$$\text{IDF}(A) = [\text{ld } 3,584/7] + 1 = \text{ld } 512 + 1 = 9 + 1 = 10.$$

Since the term weight boils down to the product $\text{WDF} \cdot \text{IDF}$, we must rethink the “+1” in the IDF. We suppose that the term t is a very frequent word that occurs in all documents. In this case, $n=N$ and $n/N=1$. Since $2^0 = 1$, $\text{ld } 1 = 0$. If we add 1 at this point, we will be given out the neutral element of the multiplication, i.e. 1. The value of the product of $\text{WDF} \cdot \text{IDF}$ is exclusively dependent upon the WDF value in this case. If t also occurs frequently inside of a text, it will be assigned a large weight. Stop words in particular behave like the described t —and they are the ones who should not be granted a large weight. Hence, we can only use the original IDF formula if we dispose of an ideal stop word list. If we do not have this list, or if no stop word limitation is used in the weighting, we must absolutely use the modified formula IDF' . IDF' produces the value zero for t , which is what we would intuitively expect in this case.

The construction of the weighting factor IDF represents an important step for Relevance Ranking, since IDF occurs in various contemporary weighting schemas. Robertson (2004, 503) emphasizes:

The intuition, and the measure associated with it, proved to be a giant leap in the field of information retrieval. Coupled with TF (the frequency of the term in the document itself, in this case, the more the better), it found its way into almost every term weighting scheme. The class of weighting schemes known generally as $\text{TF} \cdot \text{IDF}$... have proved extraordinary robust and difficult to beat, even by much more carefully worked out models and theories.

The IDF weight, guided rather intuitively by Sparck Jones, can indeed be theoretically founded, as Robertson (2004) demonstrates in the context of probabilistic theory and Aizawa (2003) does in the context of information theory. The Vector Space Model, too, can build upon it without a hitch (Salton & Buckley, 1988).

TF*IDF

Each term in each document is allocated a weighting value $w(t,d)$, which is calculated as the product of

$$w(t,d) = \text{TF}(t,d) * \text{IDF}(t),$$

where “TF” means a variant from the area of document-specific term frequencies and “IDF” means one from the area of inverse document frequencies (for further variants see Baeza-Yates & Ribeiro-Neto, 2011, 73-74):

<i>Factor 1 (TF):</i>	<i>Factor 2 (IDF):</i>
TF(t,d) (absolute frequency)	IDF(t)
TF-relH-Salton(t,d)	IDF'(t)
TF-relH-Croft(t,d)	
WDF(t,d)	
P(t,d) * WDF(t,d)	
PWDF(t,d).	

Since IDF values are dependent upon the total amount of documents in the database, they cannot be deposited in the inverted file but must—at least in theory—be recalculated after every update. (In practice, however, one can hope that in case of a very large data basis, the IDF will only suffer minor changes. Correspondingly, the IDF values only have to be updated in longer intervals.) All variants of the TF values lead to a firm value and will be allocated to the term in the inverted file.

How does a hit list arranged via text-statistical criteria come about? First, we must distinguish whether the query is of the Boolean kind or not. In Boolean retrieval, there are two options: Path 1 allocates the weighting values to the terms and continues the (now weighted) Boolean search by observing the operators used (see Ch. D.3). Path 2 applies these Boolean operators to the search, which leads to an unranked hit list. Weighting values must be requested for all terms except those search arguments connected via NOT. The ranking is then compiled in the same way as for natural-language searches. A great disadvantage here is that the difference between the search arguments combined via AND and those connected via OR is lost. For this reason, Path 2 is less than ideal. The transition from a Boolean query to text-statistical Relevance Ranking is only possible in fields that represent the aboutness of a text (title, abstract, controlled terms, full text); for all other search arguments (e.g. the author and year fields) it is pointless and thus cannot be initiated.

In the case of natural-language queries (i.e. those that do not use Boolean operators), the search request will be registered and analyzed in an information-linguistic way. For every recognized term (t_1 through t_g —depending on system maturity), one must browse in the inverted file and calculate the weights. All documents that contain at least one of the search terms will be taken into consideration. The “accumulator” calculates the retrieval status value of the documents, relative to the query, and yields the documents in descending order of the calculated document weight (Harman, 1992a).

When talking about the accumulator, two variants must be distinguished. The basis for accumulation is the sum of the individual weights w of those terms that match the query, as Spärck Jones (2004a [1972], 499) proposed:

(T)he retrieval level of a document is determined by the sum of the values of its matching terms.

If t_1 through t_n are those terms that occur both in a document and in a query, then the retrieval status value of a document $e(d)$ is calculated according to the formula

$$e(d) = w(t_1, d) + w(t_2, d) + \dots + w(t_n, d).$$

Version 1 of the accumulation incorporates all retrieved documents into the calculation of the retrieval status value, which—if we were to express it in Boolean terminology—corresponds to a procedure following OR. The second version works analogously to the Boolean AND. Here, only those documentary units that contain all search arguments will be considered at first. Only these documents are ranked according to their retrieval status value. In the next step, the system chooses those documents that lack a query term, calculates the retrieval status value and enters these into the hit list—ranked beneath the hits from the first group. This procedure is repeated until, in the last group, the only documents that appear are those that have exactly one term in common with the query.

In Version 1, it can definitely come about that a document will be ranked at the very top even though not all search arguments appear within it. This can be a disadvantage of the procedure, in so far as it will be irritating to laymen in particular. Version 2, with its break after the first step, is extremely risky, since a (perhaps relatively unimportant) search term, or an unrecognized typo that does not occur in an otherwise highly relevant document, leads to information losses.

Conclusion

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- Text-statistical methods work with characteristics of term distribution, both in single texts and in the entire database, for the purposes of arranging the documents in a Relevance Ranking.
 - The historical starting point of text statistics is the thesis by Luhn, which states that a word's frequency of occurrence is a criterion for its significance. Frequency and significance do not correlate positively, however; rather, the significance follows a distribution in the form of a bell curve, in which both the frequent and the rare words are assigned only very little importance.
 - Weights can be applied to all manner of terms, independently of the degree of maturity of the information-linguistic functionality being used. (Of course the overall result will improve if information linguistics works ideally.)
 - Algorithms of document-specific term frequency (TF) work either with the relative frequency of a term's occurrence in a text or with a logarithmic measurement that sets the term frequency in relation to the total length of a text (WDF: within document frequency).
 - The field- and position-specific term weights additionally consider the locations in which the terms occur (i.e. in the title, the controlled terms' field, the abstract or only the body), in TF and WDF, respectively.

- Each term is allocated a database-specific degree of discrimination. Sparck Jones here introduces inverse document frequency (IDF), which is an expression of a term's occurrence in the documents of the entire database. Note: the rarer a term, the more discriminatory it is.
 - For every term in every text, a weighting value is calculated. This consists of the product of one of the variants of TF/WDF and one of the variants of IDF.
 - If text statistics works in the context of a Boolean retrieval system, it will render the weighting values back to the system, which now performs a Relevance Ranking in the manner of a weighted Boolean system.
 - If text statistics cooperates with a natural-language system, all documents in the database that coincide with at least one search term will be considered for the search. The sum of the weights of the query terms in the documents defines the retrieval status value, by which the Relevance Ranking is controlled.
 - An initial version of the Relevance Ranking considers all retrieved documents in one single step and then arranges them in descending order of their calculated retrieval status values (OR variant). A second variant searches (analogously to the Boolean AND) only for those texts that contain all search terms. Additionally, ranking within this group occurs solely by retrieval status value. Afterward, those documents that lack a search term will be processed, etc.
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E.2 Vector Space Model

Documents in n-Dimensional Space

The Vector Space Model is one of the fundamental theoretical approaches in information retrieval (Salton; 1991; Raghavan & Wong, 1986; Wong et al., 1987). There are no Boolean functors in this model. The user simply formulates his query (or points to a model document) and expects to be yielded documents that satisfy his information need. The Vector Space Model was developed in the 1960s and 1970s by Salton; to test his model, Salton experimented with the SMART system (System for the Mechanical Analysis and Retrieval of Text) (Salton, 1971; Salton & Lesk, 1965). The core idea of the model is to regard terms as dimensions of an n-dimensional space, and to locate both the documents and the user queries as vectors in said space. The respective value in a dimension is calculated via the methods of text statistics. Relevance Ranking is accomplished on the basis of the proximity between vectors; more precisely, via the angle between the query vector and the document vector.

	$Term_1$	$Term_2$	...	$Term_t$
Doc_1	$Term_{11}$	$Term_{12}$	...	$Term_{1t}$
Doc_2	$Term_{21}$	$Term_{22}$	...	$Term_{2t}$
...	...	...	...	...
Doc_n	$Term_{n1}$	$Term_{n2}$	...	$Term_{nt}$

Figure E.2.1: Document-Term Matrix. Source: Salton & McGill, 1983, 128.

According to this model, a retrieval system disposes of a large matrix. The documents are deposited in the rows of this matrix and the individual terms (with their respective weights) in the columns (Figure E.2.1). If the database contains n documents and t different terms, the matrix will comprise n*t cells. Since the documents generally only contain a limited amount of terms, many of the cells have a value of 0. Salton, Wong and Yang (1975, 613) “configure” the document space:

Consider a document space consisting of documents D_i , each identified by one or more index terms t_j ; the terms may be weighted according to their importance, or unweighted with weights restricted to 0 and 1. ... (E)ach document D_i is represented by a t-dimensional vector

$$D_i = (d_{i1}, d_{i2}, \dots, d_{it}),$$

d_{ij} representing the weight of the j th term.

In the weighted case (the normal scenario), the cells of the document-term matrix contain numbers stating the weighting value $w(t,d)$ of the corresponding term in the document. In Figure E.2.2, we see three documents Doc_1 , Doc_2 and Doc_3 in a three-dimensional space. The dimensions are formed out of the terms $Term_1$, $Term_2$ and $Term_3$.

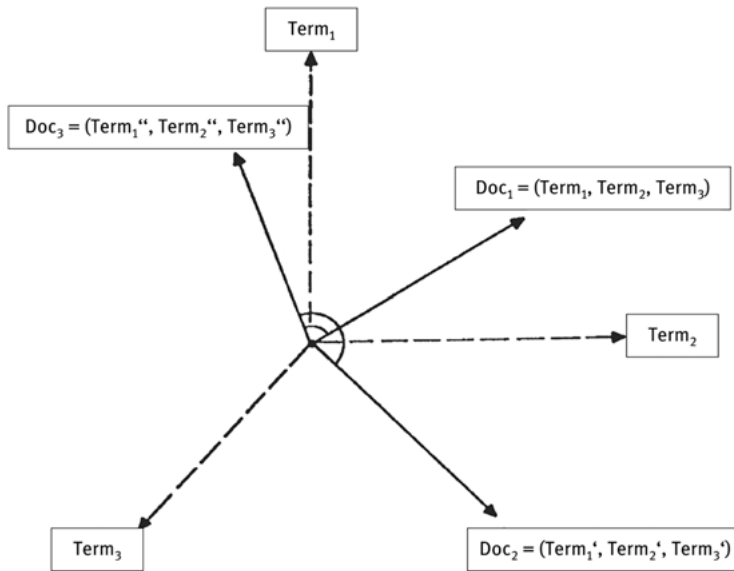


Figure E.2.2: Document Space (Three Documents and Three Terms). Source: Salton, Wong, & Yang, 1975, 614.

The procedure is analogous for queries; they are registered as a (pseudo) document. The vector of a query Q consists of the query terms and their weights (Salton & Buckley, 1988, 513):

$$Q = (QTerm_1, QTerm_2, \dots, QTerm_n).$$

Salton (1989, 351) points out that

- (i) in the vector-processing system, no conceptual distinction needs to be made between query and document terms, since both queries and documents are represented by identical constructs.

In user-formulated queries, the query vector Q will tend to be very small and only run through a very few dimensions. Since a user will hardly repeat his terms, the WDF value for all terms is the same. Thus the weight depends solely on the IDF value. Queries with a single search argument cannot be processed via this model at all, since only one dimension is available, and no angle can be created in this way. In practice, it has proved purposeful to enhance short search entries in order to provide the system with the option of creating a more expressive vector. Bollmann-Sdorra and Raghavan (1993) raise the concern that short queries have different structures than long text documents. Hence, an identical treatment of queries and documents is at the very least problematic. If a user works with long texts, or with complete documents as his queries, on the other hand, no problems are to be expected. The Vector

Space Model can bring its strengths to bear in cases of large amounts of query terms (as in the case of an entire model document and with the search option “More like this!”) in particular.

Let the query Q_j contain the (weighted) query terms $QTerm_{j1}$, $QTerm_{j2}$ etc. The similarity (Sim) between the query vector and the document vectors is calculated via their position in space, with the angles between the query vector and the document vectors being observed (Salton & McGill, 1983). The closer the vectors are to each other—in other words, the smaller the angle between them—the more similar the documents will be to each other. If the vectors lie on top of each other, the angle will be 0° and the similarity at its maximum. The minimum of similarity is reached in the case of a 90° angle.

Let the terms $Term_k$ be identical in document and query. To calculate the similarity according to the respective angle, Salton uses the cosine:

$$\text{Sim}(\text{Doc}_i, \text{Query}_j) = \frac{\sum_{k=1}^t \text{Term}_{ik} * QTerm_{jk}}{\sqrt{\sum_{k=1}^t (\text{Term}_{ik})^2 * \sum_{k=1}^t QTerm_{jk}^2}}$$

(Salton & McGill, 1983). In our case, the cosine only takes on the values between 0 and 1, since a weight may yield the minimum value of 0 but can never be negative. The cosine yields a result of 1 if there is an angle of 0° ; it becomes 0 if the maximum distance of 90° has been reached.

To exemplify, we will provide an example. We start with a query “A B” and assume that both terms have a weight of 1. In the database, there are three documents that contain either A or B (or—as in our case—both). The terms are weighted differently:

Document 1: $w(A) = 1$; $w(B) = 5$,
 Document 2: $w(A) = 5$; $w(B) = 1$,
 Document 3: $w(A) = 5$; $w(B) = 5$.

As the example has been so primitively chosen, we can directly recognize the similarities of the documents to the query from Figure E.2.3. The similarity is at its maximum in Document 3—the vectors lying on top of each other—whereas Documents 1 and 2 are less appropriate (at the same distance). If we calculate the cosine, we will receive the following ranking according to similarity:

1. $\text{Sim}(\text{Query 1, Document 3}) = 1$
2. $\text{Sim}(\text{Query 1, Document 1}) = 0.83$
3. $\text{Sim}(\text{Query 1, Document 2}) = 0.83$.

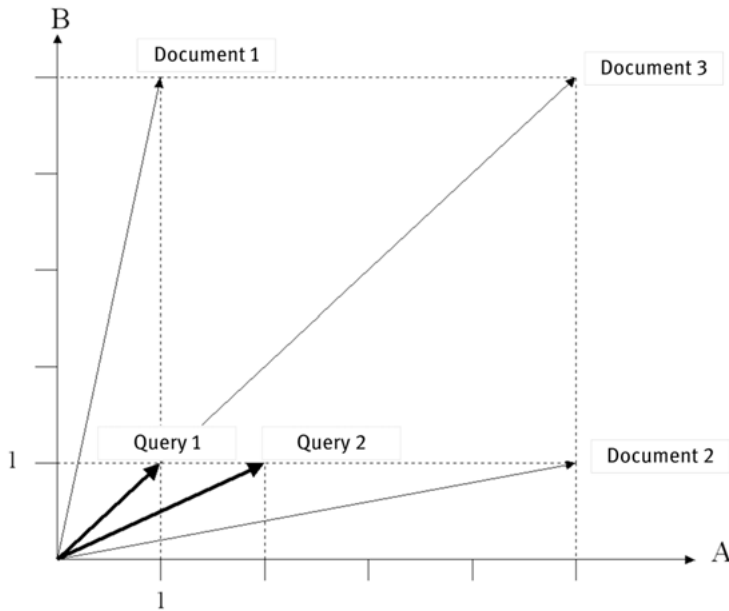


Figure E.2.3: Three Documents and Two Queries in Vector Space.

The numerator when calculating the similarity between Query 1 and Document 3 is $1 \cdot 5 + 1 \cdot 5$, i.e. 10. The denominator requires the calculation of the squares of the individual weighting values; $(1^2 + 1^2) = 2$ for the two query terms and $(5^2 + 5^2) = 50$ for the corresponding document terms. The multiplication leads to $2 \cdot 50 = 100$; the square root of 100 is ten. Dividing 10 by 10 gives you 1. Now we will weight Term A in Query 2 with a value of 2 (B remains at 1). The order of yielding will change to:

1. $\text{Sim}(\text{Query 2, Document 2}) = 0.96$,
2. $\text{Sim}(\text{Query 2, Document 3}) = 0.95$,
3. $\text{Sim}(\text{Query 2, Document 1}) = 0.61$.

Calculating the similarity between two documents proceeds along the same lines as the calculation of the similarity between a query and a document. In our three sample documents, we receive the following results (in which we impute that the documents contain no further terms):

- $\text{Sim}(\text{Document 1, Document 2}) = 0.385$,
- $\text{Sim}(\text{Document 1, Document 3}) = 0.830$,
- $\text{Sim}(\text{Document 2, Document 3}) = 0.830$.

Clustering Documents by Determining Centroids

Documents that are similar to each other can be summarized in certain classes (Salton, Wong, & Yang, 1975). A centroid is the mean vector of all its document vectors in the sense of a “center of gravity” (Salton, 1968, 141). Salton, Wong and Yang (1975, 615) describe the centroid:

For a given document class K comprising m documents, each element of the centroid C may then be defined as the average weight of the same elements in the corresponding document vectors.

For every dimension that occurs in a class in the first place, we must work out the respective term weights in the documents and calculate the arithmetic mean on this basis. Let the class K contain m documents. We then calculate the value of the centroid vector C of the dimension t according to

$$t(C) = (1/m) * [w(t)_1 + w(t)_2 + \dots + w(t)_m].$$

The documents are classed in such a way that their distance to “their own” centroid is minimal, but that to other centroids is at its maximum. In Salton’s model, it is envisaged that the clusters may intersect, i.e. that a document may belong to several classes (Salton, 1968, 139 et seq.).

It can be assumed that thematically similar documents are summarized within the clusters. If a query is close to a centroid vector, the user (at least in an initial step) will only be yielded the documents of the affected cluster, arranged between query and document vectors in accordance with the cosine.

Relevance Feedback

Salton assumes that an ideal search formulation may only succeed after several iterative attempts (Salton & Buckley, 1990). The user enters a first attempt at a query and is shown documents by the system. He marks certain documents in the hit list as relevant and others as irrelevant for the satisfaction of his information need. Salton (1968, 267) describes the procedure:

In essence, the process consists in performing an initial search and in presenting to the user a certain amount of retrieved information. The user then examines some of the retrieved documents and identifies each as being either relevant (R) or not relevant (N) to his purpose. These relevance judgments are then returned to the system and are used automatically to adjust the initial search request in such a way that query terms ... present in the relevant documents are promoted (by increasing their weight), whereas terms occurring in the documents designated as nonrelevant are similarly demoted. ... This process can be continued over several iterations, until such time as the user is satisfied with the results obtained.

The goal of Relevance Feedback is to remove the query vector from the areas of irrelevant documents and to move it closer to the relevant ones. Today's "classical" algorithm for Relevance Feedback in the context of the Vector Space Model has been developed by Rocchio (1971) (see also Chen & Zhu, 2002). The modified query vector q_m is calculated as the sum of the original query vector q , the centroid of those document vectors that the user has marked as relevant (D_r) and—with a minus sign—the centroid of the documents designated non-relevant (D_n). Baeza-Yates and Ribeiro-Neto (1999, 119) add three freely adjustable constants α , β and γ that can be used to weight the individual addends. The Rocchio algorithm looks as follows:

$$\bar{q}_m = (\alpha * \bar{q}) + \left(\frac{\beta}{|D_r|} * \sum_{\forall \bar{d}_i \in D_r} \bar{d}_i\right) - \left(\frac{\gamma}{|D_n|} * \sum_{\forall \bar{d}_i \in D_n} \bar{d}_i\right)$$

If no weight is desired, one will select the value of 1 for all three constants. If one wishes exclusively to focus on a positive feedback strategy, $\gamma = 0$. If one thinks that a user places more value on his positive judgments than on his negative ones, one must determine $\beta > \gamma$. Depending on how important the original query is deemed to be, one either works with $\alpha < 1$, $\alpha = 1$ (which is Rocchio's original path) or $\alpha > 1$.

The interplay between system and user is still trial-and-error in Rocchio's Relevance Feedback. It is not (at least not necessarily) a way that steadily approaches an optimum. In practice, however, a good user strategy—consisting of judging the "right" documents as relevant and the "wrong" ones as irrelevant—can yield optimal results. The good experiences made with Relevance Feedback are a strong argument for two things: firstly, to build the search process iteratively and to improve the search results afterward, and secondly, to actively incorporate the user into this process as far as possible. Search, in this sense, becomes a process of learning, as Chen and Zhu (2002, 84) point out:

Rocchio's similarity-based relevance feedback algorithm is one of the most popular query reformation methods in information retrieval and has been used in various applications. It is essentially an adaptive supervised learning algorithm from examples.

Term Independence

The Vector Space Model assumes that the individual dimensions and the terms thus depicted are independent of one another. Each dimension stands for itself—without creating a relation to other dimensions or even seeking to. The dimensions "Morning Star" and "Evening Star" form an angle of 90° (with a cosine of 0, meaning "no relation")—which is incorrect! In this model, a document is a linear combination of its

terms. This is also incorrect when put in this way. Rather, words and concepts are interrelated in various syntactical, semantic and pragmatic contexts. Terms depend on the occurrence or non-occurrence of other terms. We need only think of homonymy, synonymy and hierarchical term relations! Our above statement (Ch. E.1), that the Vector Space Model can work equally well with all manner of terms, should thus be rethought. This is because the more elaborate the information-linguistic functionality, the more strongly term interdependence will be taken into consideration. At t_1 , the level of word forms, ignorance of term interdependence is at its maximum. At t_2 , as well, little progress has been made; word form conflation may even have created further homonyms that used to be different from each other in their complete forms. Only once the concept level t_7 has been reached can we speak of a complete regard for the interdependences between terms. In the words of Billhardt, Borrajo and Maojo (2002, 237), this sounds as follows:

One assumption of the classical vector space model is that term vectors are pair-wise orthogonal. This means that when calculating the relevance of a document to a query, only those terms are considered that occur in both the document and the query. Using the cosine coefficient, for example, the retrieval status value for a document/query pair is only determined by the terms and the document have in common, but not by query terms that are semantically similar to document terms or vice versa. In this sense, the model assumes that terms are independent of each other; there exists no relationship among different terms.

The conclusion, at this point, is not that the Vector Space Model should be abandoned as a false approach, but rather to lift information linguistics to at least the semantic level t_7 . From a theoretical as well as a practical point of view, a collaboration between knowledge organization systems and the Vector Space Model is possible for the benefit of both (Kulyukin & Settle, 2001). Here, we are speaking of “context vectors” (Billhardt, Borrajo, & Maojo, 2002) or of the “semantic vector space model” (Liddy, Paik, & Yu, 1993; Liu, 1997). Synonyms and quasi-synonyms here form *one* dimension, while the respective homonyms, correspondingly, form several. When using thesauri with weighted relations, the queries can be enhanced by semantically similar terms (with careful, low weighting). Experiments with using WordNet in the Vector Space Model prove successful (Gonzalo et al., 1998; Rosso et al., 2004).

Latent Semantic Indexing

A variant of the connection between term interweaving and the Vector Space Model is LSI (latent semantic indexing) or LSA (latent semantic analysis), which has strongly influenced the debate on information retrieval since the late 1980s. LSI was conceived by a research team at Bell, consisting of Dumais, Furnas and Landauer in cooperation with Deerwester and Harshman (Deerwester et al., 1990; Berry, Dumais, & O'Brien, 1995; Landauer, Foltz, & Laham, 1998), and exclusively works with word forms that

occur in the text at hand. LSI thus does not use any syntactic analysis (e.g. morphology) and no semantics, neither linguistic knowledge (as in WordNet) nor world knowledge (as in specialized knowledge organization systems). However, LSI claims to correctly record deep-seated semantic relations (hence the name). Landauer, Foltz and Laham (1998, 261) emphasize:

LSA, as currently practiced, induces its representations of the meanings of words and passages from analysis of text alone.

The authors are aware of the limits of their approach—in particular, of their decision to forego world knowledge (Landauer, Foltz, & Laham, 1998, 261):

One might consider LSA's maximal knowledge of the world to be analogous to a well-read nun's knowledge of sex, a level of knowledge often deemed a sufficient basis for advising the young.

The basic idea of LSI is the summarizing of different words into a select few dimensions, where both terms and documents are located in the vector space. The dimensions are created on the basis of their words' co-occurrence in the documents. Dumais (2004, 194) describes the LSI Vector Space Model:

The axes are those derived from the SVD (singular value decomposition); they are linear combinations of terms. Both terms and documents are represented as vectors in this k -dimensional space. In this representation, the derived indexing dimensions are orthogonal, but terms are not. The location of term vectors reflects the correlations in their usage across documents. An important consequence is that terms are no longer independent; therefore, a query can match documents, even though the documents do not contain the query terms.

The calculations' starting point is a term-document matrix. All those terms that occur at least twice in the document collection are kept for further processing. In the first step, we note absolute document frequency, and in the next step we calculate the document weights w (via $WDF \cdot IDF$). The decisive step is Singular Value Decomposition (SVD). Here we determine, in the sense of factor analysis, those (few) factors onto which the individual documents prefer to load the terms. Only the k biggest factors are taken into consideration, the others are set to 0. The term-document matrix (with the weighted values) is split, so that we receive correlations between the terms and correlations between the documents. By restricting ourselves to k factors, the result is only an approximation of the original matrix. However, we do reach the goal of working with as few factors as possible. The factors now form the dimensions in the Vector Space; from here on in, "latent semantic indexing" works in the same way as the traditional Vector Space Model and determines the similarity of documents to queries via the cosine of the angle of their vectors in k -dimensional factor-vector space.

Dumais (2004, 192) describes the step of reducing the dimensions:

A reduced-rank singular value decomposition (SVD) is performed on the matrix, in which the k largest singular values are retained, and the remainder set to 0. The resulting reduced-dimension SVD representation is the best k -dimensional approximation to the original matrix, in the least-squares sense. Each document and term is now represented as a k -dimensional vector in the space derived by the SVD.

Determining the ideal value for k depends upon the make-up of the database; in every case, k is smaller than the number of original term vectors.

The resulting factors respectively factor dimensions are pseudo-concepts, based solely on statistical values, whose words and the documents that contain them have a high degree of correlation. Deerwester et al. (1990, 395) introduce them as follows:

Roughly speaking, these factors may be thought of as artificial concepts; they represent extracted common meaning components of many different words and documents. ... We make no attempt to interpret the underlying factors, nor to “rotate” them to some meaningful orientation. Our aim is not to be able to describe the factors verbally but merely to be able to represent terms, documents and queries in a way that escapes the unreliability, ambiguity and redundancy of individual terms as descriptors.

In practice, “latent semantic indexing” proves far less suitable for large, heterogeneous databases than for small, homogeneous document collections (Husbands, Simon, & Ding, 2005). To solve this problem, it is possible to divide heterogeneous data quantities into smaller, homogeneous ones and to only begin the factor analysis then (Gao & Zhang, 2005). Another problem remains. The ranking results of latent semantic indexing depend on the choice of the number of singular values k . Kettani and Newby (2010, 6) report:

The results of our analysis show that the LSI ranking is very sensitive to the choice of the number of singular values retained. This put the robustness of LSI into question.

Advantages and Disadvantages of the Vector Space Model

For laymen, the advantage of all variants of the Vector Space Model is that it does not work with Boolean Operators. For information professionals, on the other hand, that same fact tends to be a disadvantage, since no elaborate query formulation is possible: AND and OR are the same, and there is no NOT operator. The model does not allow for either Boolean, proximity or any other operators.

Since both the query and the documents are weighted, and a similarity measurement, the cosine, is calculated, the hit list (for at least two search arguments) is always ranked by relevance. A great advantage is the use of Relevance Feedback pro-

cedures that modify the original query iteratively and can lead to better results in search practice.

A further advantage can be made out in the fact that the documents are bundled into thematically similar clusters. The model thus achieves automatic classification (without any predefined knowledge organization system).

A great disadvantage of the original Vector Space Model, where word forms or stems are shown to be rooted in the text, is that in theory the model assumes the independence of the terms. This, however, is not the case in the reality of information retrieval.

Queries with few arguments form a very small vector in comparison with the document vectors, which refer to far more dimensions. Comparison thus becomes extremely difficult. This disadvantage, however, can be neutralized by enhancing the query and Relevance Feedback, respectively. Queries with only one search argument cannot be processed.

Further developments of the Vector Space Model have taken into consideration the model's intrinsic assumption of concept independence. Here, we can detect two different approaches in the forms of the semantic Vector Space and "latent semantic indexing". LSI is more suitable for small, homogeneous amounts of documents, particularly if no specialized KOSs are available. The semantic Vector Space Model can only bring its advantages to bear if an adequate knowledge organization system, such as WordNet or a specialized thesaurus, is available.

Conclusion

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- The Vector Space Model is a "classic" model of information retrieval. It was developed by Salton and experimentally tested in the context of the reference system SMART.
 - The Vector Space Model interprets terms from both texts and queries as dimensions in an n -dimensional space, where n counts the number of different terms. The documents and the queries (as pseudo-documents) are vectors in this space, whose position is determined by their dimensions (i.e. terms). The respective values in the dimensions are gleaned from text-statistical weightings (generally $WDF \cdot IDF$).
 - Relevance Ranking is determined via the angle between query and document vectors. One of the calculations performed is that of the cosine.
 - Documents that co-occur in delimitable areas of the Vector Space are summarized into classes. The "center of gravity" of such a class is formed by the centroid (a mean vector of the respective document vectors).
 - Using feedback loops between system and user, an original user query can be modified, and the retrieval result improved iteratively. Relevance Feedback takes into consideration (positively) terms from the relevant documents and (negatively) those from non-relevant texts.
 - Inherent to the Vector Space Model is the assumption that all terms are independent of one another (since they are at right angles to each others in the space). If one works with word forms in the texts, or with basic forms/stems, this assumption is not substantiated in the reality of the texts.

- If we work with concepts instead of words, the assumption of independence will have robbed of a lot of its sharpness. Context vectors in the semantic Vector Space Model now represent concepts and take into consideration synonymy, homonymy and, where applicable, hierarchical relations. The semantic Vector Space Model is thus combined with a linguistic or subject-specific knowledge organization system.
 - “Latent Semantic Indexing” (LSI) works on the basis of word forms that occur in the text, but it does not avoid the problems of the independence assumption. LSI summarizes terms into pseudo-concepts. These concepts are factors (in the sense of factor analysis) that correlate with word forms and with documents. The factors serve as dimensions in Vector Space, whereas both word forms and documents are vectors located therein. LSI achieves a massive reduction in the number of dimensions.
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E.3 Probabilistic Model

The Conditional Probability of a Document's Relevance under a Query

“Is this document relevant? ... probably”, is the title of a study by Crestani, Lalmas, van Rijsbergen and Campbell (1998). How probable is it that a document will be relevant for a search query? In the first article dedicated to the subject of probabilistic retrieval, the authors' motivation is to yield hit lists according to relevance. Maron and Kuhns (1960, 220) write:

One of our basic aims is to rank documents according to their relevance, given a library request for information.

This is the birth date of Relevance Ranking. Output is steered via the probability values that make a given document relevant for the search query at hand. The object is a conditional probability: the probability of a document's relevance *under the condition* of the respective query. If P is the probability of relevance, Q a query and D a document, we must always calculate

$$P(D | Q).$$

Following Bayes' theorem, the following connection applies:

$$P(D | Q) = [P(Q | D) * P(D)] / P(Q).$$

Both D and Q consist of terms t_i . $P(Q)$ is the probability of the respective query. This latter is a known quantity; hence we can set $P(Q)$ to 1 and thus neglect the denominator. In the numerator, $P(Q | D)$ is calculated, i.e. the conditional likelihood of a query being relevant under the condition of a document. We thus glean new query terms from documents of which we know—in the theoretical model—that they are relevant. $P(D)$ is the probability of the document, calculated via the terms that occur within it (generally via TF*IDF). In search practice, $P(Q | D)$ is, of course, not known. The system requires statements about it—made either by the user or gleaned from the system's own calculations. Thus it is clear that the probabilistic model must always run through a Relevance Feedback loop.

Figure E.3.1 shows the necessary program steps in an overview. Starting from (any) initial search, a hit list is created. Here, the paths divide into a Relevance Feedback performed by the user (by marking relevant and non-relevant documents) and into a pseudo-relevance feedback performed by the machine. In Relevance Feedback, relevance information is gleaned, i.e. notes on $(Q | D)$. We receive new search terms as well as their corresponding weights $P(Q | D)$. In the last step, the weighted old

search arguments, as well as—where applicable—new ones are used to modify the initial search.

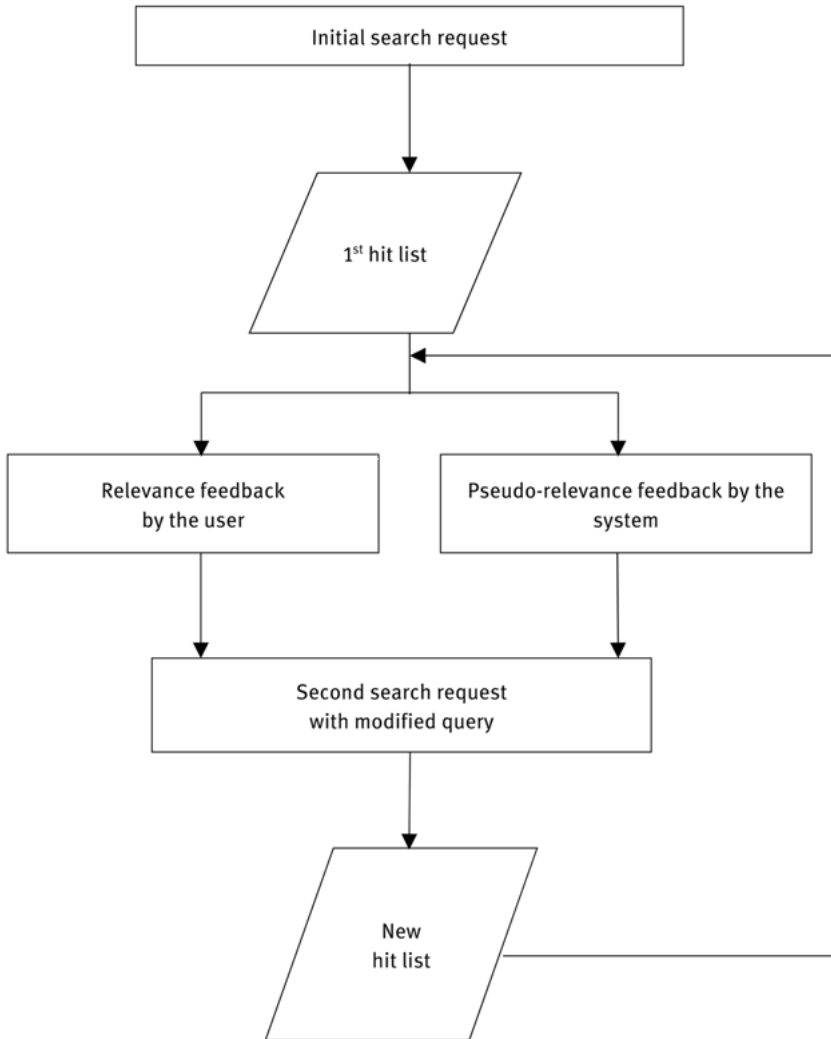


Figure E.3.1: Program Steps in Probabilistic Retrieval.

The model of probabilistic retrieval must take the documents' independence as a pre-requisite, since Bayes' theorem will not apply otherwise. Robertson (1977, 304) shows how this precondition is incommensurate with retrieval practice:

A retrieval system has to predict the relevance of documents to users on the basis of certain information. Whether the calculation of a probability of relevance, document-by-document, is an appropriate way to make the prediction depends on the nature of the relevance variable and of the information about it. In particular, the probability-ranking approach depends on the assumption that the relevance of one document to a request is independent of the other documents in the collection. ... There are various kinds of dependency between documents, at various levels between the relevance itself and the information about it.

Let us suppose that a user is presented with a rather long hit list, arranged via the documents' probability in relation to the search request. Let us further suppose that in a first scenario, the user has not found a single pertinent text among the top 20 ranked documents. In a second scenario, he has already noted 15 useful documents by the same stage. In Scenario 1, the user will "pounce" on the very first hit he encounters, and perceive it to be ultra-relevant. In Scenario 2, however, he is already saturated with information and will likely deem it to be of little relevance. The pertinence of a given text is thus clearly dependent upon the other documents; the documents are not independent. Hence, in the following we must discuss the probabilistic model without accounting for users' estimations. "Relevant" is what the model calculates, no matter what any user may think.

As in the simple Vector Space Model, so the simple probabilistic model (counterfactually) assumes that the individual terms are independent of one another.

Gleaning and Weighting Search Terms via Model Documents

In the following, we will use Bayes' theorem exclusively as the model's heuristic basis. Bayes' theorem works with probability values, i.e. values between 0 and 1. Values from $TF \cdot IDF$, as well as the terms' relevance values from the feedback loops, are not restricted to this interval. Of course, one could convert all values and map them onto the interval $[0,1]$. On the other hand, one can also work directly with the original values, since nothing changes in the ranking of the documents—and this is the only thing that counts. Robertson, Walker and Beaulieu (1999) write,

(t)he traditional argument of the probabilistic model relies on the idea of ranking; the function of the scoring method is not to provide an actual estimate of $P(R|D)$, but to rank in $P(R|D)$ order.

After the first iteration of the retrieval, there will be a hit list from which the user has selected both relevant and non-relevant documents. These (positive and negative) model documents must now be inspected closely. Overall, the user has judged N texts, of which R are relevant for his query and $N - R$ are irrelevant. The user does not necessarily have to use "relevant" and "non-relevant" for every document in the hit list, but can also opt for a neutral assessment. These "postponed" documents, however, do not enter into the further analysis. For all judged documents there follows term analy-

sis. Let the number of documents that contain a certain term t be n ; those which do not contain it are designated $N - n$. It is exclusively taken into account whether a term is at all present in the document, and not how often. The number of documents that contain the term t and that have been marked as relevant by the user is r . We can now set the respective relevance and non-relevance into relation with the occurrence ($t+$) and non-occurrence ($t-$) of the term t (Robertson & Sparck Jones, 1976; Sparck Jones, 1979; Sparck Jones, Walker, & Robertson, 2000):

	<i>Document relevant</i>	<i>Document not relevant</i>	<i>Sum</i>
$t+$	r	$n - r$	n
$t-$	$R - r$	$N - n - R + r$	$N - n$
<i>Sum</i>	R	$N - R$	N

The following four probabilities are united in this matrix:

- P1: The occurrence of term t is of fundamental importance for the relevance of the document: r / R ,
- P2: The non-occurrence of term t is of fundamental importance for the relevance of the document: $(R - r) / R$, in which it holds that: $P2 = 1 - P1$,
- P3: The occurrence of term t is of fundamental importance for the non-relevance of the document: $(n - r) / (N - R)$,
- P4: The non-occurrence of term t is of fundamental importance for the non-relevance of the document: $(N - n - R + r) / (N - R)$, where $P4 = 1 - P3$.

With these statements, it is possible to calculate the relative distribution of the terms in the documents. The following formula has asserted itself (Harman, 1992, 368; it was first introduced by Robertson and Sparck Jones in the year 1976):

$$w'(\text{old}) = \log \frac{\frac{r}{R-r}}{\frac{n-r}{N-n-R+r}}$$

A great problem of the Robertson-Sparck Jones formula must be taken into consideration. If a value of zero occurs in the denominator of the overall formula as well as in the denominator of the two components, the formula will be undefined (you cannot divide by zero). If, for instance, all relevant documents always contain the term t , then $R - r = 0$. t should really be very highly weighted, but due to the division by zero it does not get any w -value at all (Robertson & Sparck Jones, 1976, 136). An elegant solution to this problem is to add 0.5 to every value, which leads us to the following formula (Croft, Metzler, & Strohman, 2010, 249):

$$w' = \log \frac{\frac{r+0.5}{R-r+0.5}}{\frac{n-r+0.5}{N-n-R+r+0.5}}$$

For all terms t_i of the original search as well as the terms of the N assessed documents, the distribution value w' is calculated. A new search for t_i is then performed. The weighting value w of the terms in the retrieved documents (generally gleaned via a variant of TF*IDF) is multiplied with the distribution value w' . The retrieval status value e of the documents is—as known from text statistics—the sum of the weighting values (here: $w*w$) of those terms that also occur in the query. Since w can very well also include negative values, the corresponding documents are devalued when such undesired terms appear.

For the purposes of illustration, let us suppose that someone has searched for “Eowyn” (referring to the character from the “Lord of the Rings” movies as well as the actress portraying her). After viewing the first hit list, he marks 12 documents as relevant and 8 as not relevant. For each term in the 20 documents, a distribution matrix is formed; we demonstrate this on the example of the terms “Miranda” (as a reminder: Miranda Otto plays the character of Eowyn; the user does not necessarily know this, however) and “Círdan” (this character appears in the books, but not in the films). Let the following contingency matrix apply to “Miranda”:

	<i>Document relevant</i>	<i>Document not relevant</i>	<i>Sum</i>
Miranda+	10	1	11
Miranda–	2	7	9
<i>Sum</i>	12	8	20

(Miranda: $r = 10$; $R = 12$; $n = 11$; $N = 20$)

In the case of “Círdan”, these values arise:

	<i>Document relevant</i>	<i>Document not relevant</i>	<i>Sum</i>
Círdan+	1	7	8
Círdan–	11	1	12
<i>Sum</i>	12	8	20

(Círdan: $r = 1$; $R = 12$; $n = 8$; $N = 20$)

The following distribution values are the result:

$$w'_{(\text{Miranda})} = 1.32 \text{ and } w'_{(\text{Círdan})} = -1.59.$$

In all texts d_i that contain “Miranda”, their respective weight $w(\text{Miranda}, d_i)$ is multiplied by 1.32; in all texts d_i that contain “Círdan”, the weight $w(\text{Círdan}, d_i)$ is multiplied by -1.59 and results in a negative value. Documents with “Miranda” are upgraded

in the retrieval status; those with “Círdan” on the other hand are downgraded. Note that at his first attempt, the user can very well formulate his query vaguely (in our example, the user need not know anything about Miranda Otto or Círdan); by analyzing the model documents, the system locates suitable search terms for modifying the original query.

It is not required, and probably does not even make sense to take *all* terms of those documents retrieved in the first search into account for the next step. An alternative is to only consider those terms that surround the original search arguments within a certain window of text—that is to say, those that are at most n words away from the respective argument, or those that occur in the same sentence or, at least, the same paragraph.

For all (old and newly added) search terms, the weighting value ($w \cdot TF \cdot IDF$) (or a variant such as Okapi BM 25; Robertson, Walker, & Beaulieu, 1999) is calculated; the individual values are aggregated for the document (e.g. summarized) and make up the new retrieval status value.

Pseudo-Relevance Feedback

Users are not always ready or able to make a relevance assessment. Rather, they expect a “final” hit list to appear directly after entering their search arguments. The intermediate step required in the probabilistic model must be taken automatically in this case. After all, without relevance information the system cannot work, as Croft and Harper (1979, 285) emphasize:

A major assumption made in these models is that *relevance information* is available. That is, some or all of the relevant and non-relevant documents have been identified.

Croft’s and Harper’s idea, as simple as it is brilliant, is to designate the top-ranked documents from the initial search to be relevant and to use their term material for the second step. Croft and Harper (1979, 288) write:

The documents at the top of the ranking produced by the initial search have a high probability of being relevant. The assumption underlying the intermediate search is that these documents *are* relevant, whether in actual fact they are or not.

Since this procedure does not involve a user who actively participates in the Relevance Feedback, the automatically operating Croft-Harper procedure has come to be known as “Pseudo-Relevance Feedback”.

Since we are now dealing exclusively with documents marked as relevant, but no negative examples, we can only use the upper part of the compound fraction of the Robertson-Sparck Jones formula. Hence, the terms that are taken into consideration are those that occur particularly often in the relevant texts. R always equals N here;

r is the number of documents containing a certain term t , and $R - r$ is the number of texts that do not contain t . The weight w of t is calculated according to

$$w'(\text{old}) = \log [r / (R - r)] \text{ or (to avoid the problem of zero):}$$

$$w' = \log [(r + 0.5) / (R - r + 0.5)].$$

The procedure is risky, as Croft and Harper (1979, 288) stress while emphasizing that its success depends upon the type of query being used:

The effect of this search will vary widely from query to query. For a query which retrieved a high proportion of relevant documents at the top of the ranking, this search would probably retrieve additional relevant documents. For a query which retrieved a low proportion of relevant documents, this query may actually downgrade the retrieval effectiveness.

The selection of the right amount for N ($N = R$) remains a great risk factor. If the relevance distribution follows a Power Law, N must be kept relatively small—on the other hand, if an inverse-logistic distribution can be detected, N should be larger (much larger, even) (Stock, 2006). Udupa and Bhole (2011, 814) observe, in an empirical study,

that current PRF (pseudo-relevance feedback; A/N) techniques are highly suboptimal and also that wrong selection of expansion terms is at the root of instability of current PRF techniques.

We are in need of techniques that allow for a “careful selection” (Udupa & Bhole, 2011, 814) of the “right” terms from the pseudo-relevant documents. It appears to make sense not to select the terms on the level of the documents (Ye, Huang, & Lin, 2011) but to work with smaller units instead. For instance, those text segments that contain the original search arguments would qualify for this type of procedure.

Statistical Language Models

Language models compare the statistical distribution of terms in texts, entire databases or in the general everyday use of language. Ponte and Croft (1998), and later Liu and Croft (2005, 3) relate the statistical language models to information retrieval:

Generally speaking, statistical language modeling ... involves estimating a probability distribution that captures statistical regularities of natural language use. Applied to information retrieval, language modelling refers to the problem of estimating the likelihood that a query and a document could have been generated by the same language model, given the language model of the document either with or without a language model of the query.

Statistical language models in information retrieval can be theoretically linked with text statistics and probabilistic retrieval (Lafferty & Zhai, 2003). In appreciation of the text-statistical studies of Andrey A. Markov from the year 1913 (on the example of “Eugene Onegin”; Markov, 2006[1913]), the methods discussed here will be collectively referred to as “Hidden Markov Models” (HMM).

According to the approach by Miller, Leek and Schwartz (1999), a user’s search request depends, ideal-typically, on an ideally suitable model document that the user desires as the solution to his problem (which, in reality, almost certainly does not exist; moreover, the user will be able to picture more than one document). The system searches for texts that come close to the “ideal” and arranges them according to their proximity to it. Miller, Leek and Schwartz (1999, 215) describe their model as follows:

We propose to model the generation of a query by a user as a discrete hidden Markov process dependent on the document the user has in mind. A discrete hidden Markov model is defined by a set of output symbols, a set of states, a set of probabilities for transitions between the states and a probability distribution on output symbols for each state. An observed sampling of the process (*i.e.* the sequence of output symbols) is produced by starting from some initial state, transitioning from it to another state, sampling from the output distribution at that state, and then repeating these latter two steps. The transitioning and the sampling are non-deterministic, and are governed by the probabilities that define the process. The term “hidden” refers to the fact that an observer sees only the output symbols, but doesn’t know the underlying sequences of states that generated them.

Miller et al. work with two HMM states: documents and the use of the English language. The distribution of the document state is oriented on the relative frequency of the query terms in the documents. For every term t from the search request, we calculate for every document D that contains it:

$$P(t | D) = \text{freq}(t, D) / L.$$

L is the total length of the document. The second HMM state is the distribution of word forms in the English (or any other natural) language. The occurrence of the query term t is counted in all documents in which it appears. Also counted is the occurrence of the term in “English in general”. If the value for the latter is unknown, it can be estimated via the term’s overall occurrence in the database L_t . Here, too, the relative frequency is being calculated:

$$P(t | NT) = [\text{freq}(t, D_1) + \text{freq}(t, D_2) + \dots + \text{freq}(t, D_n)] / L_t,$$

where t occurs in documents D_1 through D_n and NT refers to the respective natural language (or the database). The probabilities $P(t | D)$ and $P(t | NT)$ are weighted and added via a factor α resp. $1 - \alpha$ (α lies between 0 and 1):

$$P(t | D, NT) = \alpha * P(t | D) + (1 - \alpha) * P(t | NT).$$

The probability of a document being located in close proximity to the ideal document is calculated as the product of the $P(t | D, NT)$ of all query terms:

$$P(Q | D \text{ is relevant}) = P(t_1 | D, NT) * P(t_2 | D, NT) * \dots * P(t_n | D, NT).$$

Liu and Croft (2005, 9-10) are convinced that the statistical language models can be successfully used in information retrieval:

Retrieval experiments on TREC test collections show that the simple two-state system can do dramatically better than the $tf*idf$ measure. This work demonstrates that commonly used IR techniques such as relevance feedback as well as prior knowledge of the usefulness of documents can be incorporated into language models.

Advantages and Disadvantages of the Probabilistic Model

An obvious plus of probabilistic models is their theoretical stringency. We must not forget that Relevance Ranking—historically speaking—starts here. Laymen users do not have to deal with Boolean Operators that they do not understand but can formulate their query in natural language. Another advantage is that Relevance-Feedback loops allow one to approach the ideal result step by step. However, all of these advantages—apart from the theoretical maturity—can also be found in the Vector Space Model.

Documents in hit lists are not independent—counter to the model assumptions. Whereas Relevance Feedback is of advantage when users are incorporated (at least as long as the users conscientiously determine relevance information), the automatically performed Pseudo-Relevance Ranking is risky because it cannot be safely taken for granted that the “right” documents are analyzed for feedback.

Conclusion

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- The probabilistic retrieval model starts with the question of how probable it is that a document is relevant with regard to the search request. The texts are then ranked in descending order of the probability value attained.
 - The probabilistic model is one of the classical approaches of retrieval research (especially retrieval theory). The first publication on the subject, published in 1960 by Maron and Kuhns, is regarded as the initial spark for systems with Relevance Ranking—in contrast to Boolean Systems, which do not allow any ranking.
 - Relevance Feedback is always employed, in order to glean relevance information about documents. Here, one can either ask the user or let the system automatically create a feedback loop.

- The model presupposes the documents' independence of one another, something which is not given in search practice: here, a user assesses the usefulness of a text on the basis of documents already sighted. Probabilistic retrieval thus always regards algorithmic relevance, and not pertinence.
 - In the case of user-side “real” Relevance Feedback, the user designates certain documents in a first hit list as relevant and certain others as not relevant. The terms from both positive and negative model documents are used—via the Robertson-Sparck Jones formula—to modify the original search request.
 - In system-side, automatically performed Pseudo-Relevance Feedback, the top-ranked results of the initial search are defined as relevant and regarded as positive model documents. This procedure is risky; the degree of risk depends upon the type of query and on the respective relevance distribution, especially for finding new search arguments.
 - Language models work with statistical distributions of terms in texts, databases and in general language use. Statistical language models, particularly the “Hidden Markov Models” (HMM), are a variant of probabilistic retrieval models. The HMM retrieval model by Miller et al. works with two states: that of the document in question and that of general language use. In both areas, the relative frequency of the respective search terms is calculated, weighted and added. HMM retrieval models are surprisingly simple while providing good retrieval results.
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E.4 Retrieval of Non-Textual Documents

Multimedia Retrieval

Other than written texts, the following digital media are of further interest to information retrieval:

- spoken texts,
- music and other audio documents,
- images,
- videos resp. movies, i.e. moving images with additional audio elements.

Written text, image and sound are the basic media; video documents represent a mixed form of (moving) image and sound. Where visual or audio documents are textually described in a documentary unit that serves as a surrogate, and this surrogate is being searched via written queries, we are dealing with a classic case of information retrieval. We will ignore this for now; in this chapter, our points of interest are image and sound themselves.

Both the queries and the documents can be stored in different media. We formulate a query (e.g. “John F. Kennedy in Berlin”) and aim for text documents, for spoken words (in our example: Kennedy’s famous speech from beginning to end), for images (contemporary photographs) as well as film sequences (Kennedy at different stations of his visit). Of course a user can also start with other documents, e.g. a piece of music (“Child in Time” by Deep Purple), and aim to retrieve texts (song lyrics and background information concerning the song’s origin) or a video (a performance of “Child in Time”). Whenever non-textual aspects are addressed during a search for documents, we speak of “multimedia retrieval” (Neal, ed., 2012; Lew, Sebe, Djeraba, & Jain, 2006).

The area of multimedia research and development is seeing a lot of work at the moment, particularly in terms of informatics and technology. On the subject of (moving) images, Sebe, Lew et al. (2003, 1) emphasize:

Image and video retrieval continues to be one of the most exciting and fastest-growing research areas in the field of multimedia technology.

Lew, Sebe and Eakins (2002, 1) point out, however:

Although significant advances have been made in text searching, only preliminary work has been done in finding images and videos in large digital collections.

The number of digital visual and audio documents rises rapidly due to the professional and private use of digital cameras and camcorders as well as the digital availability of various image, sound and video documents. Examples of broadly used information services on the World Wide Web include sharing services such as Flickr

(images), Last.fm (music) and YouTube (videos), as well as social networks like Facebook (text, images and other media) or MySpace (particularly for text and music). Provided the eventual availability of satisfactory and usable solutions, image and sound retrieval thus find their area of application in search engines on the World Wide Web, in company intranets (particularly of media companies) as well as in the private domain. In this chapter, we are interested in the specifics of searching and retrieving images, videos and audio files (particularly music) independently of (written) texts, i.e. based exclusively on content.

One thing that all image and sound retrieval has in common is the alignment of a search argument with the documents across *several* dimensions (similarity values having to be calculated for all of them). As an example, we will look at an image described via three (arbitrarily chosen) dimensions (Schmitt, 2006, 107):

There exist around three different groups of feature values for an image object. In the first group, there are the feature values for textures resulting from the use of Gabor filters. The second group, on the other hand, contains texture values that have been extracted via a Tamura procedure. Feature values extracted via color distribution make up the third group.

Every dimension is first aligned individually. A similarity value is calculated for each respective dimension (which is derived from the distance between the query and the media object) and cached as its retrieval status value (RSV). The steps that follow combine the retrieval status values of all dimensions into a single value that expresses the similarity between the query and the (complete) object. The challenges in image and sound retrieval are mainly found in the following aspects:

- Locating the most appropriate dimensions,
- Finding metrics for calculating similarities on each respective dimension,
- Calculating the distances between query and database object on the basis of the selected metric,
- Deriving a dimension-specific retrieval status value for each dimension,
- Combining the dimension-specific retrieval status values into a retrieval status value for the entire (image or sound) document.

Dimensions of Image Retrieval

“A picture says more than a thousand words”—if this saying is true, we will not find it very effective to exclusively use words to describe images. Clearly, we must attempt to work out some further dimensions of describing pictorial content (Smeulders et al., 2000; Datta et al., 2008). Jörgensen, in introducing “content-based image retrieval” (CBIR), roughly distinguishes between (1.) the basic dimensions (Jörgensen, 2003, 146 et seq.) that describe an image via measurable qualities (color, texture, form), (2.) dimensions of object detection (Jörgensen, 2003, 161 et seq.), such as the analy-

sis of salient figures, or 3D reconstruction, and (3.) “higher” dimensions (Jørgensen, 2003, 167 et seq.) like the allocation of descriptive general terms (e.g. soccer goal-keeper), “named entities” (Jens Lehmann) as well as terms that express emotions (Jens Lehmann’s joy after saving a penalty).

The basic dimension of color registers an image via the spectral values visible to the human eye, i.e. electromagnetic waves between 380 and 780 nm. It is a common way of operating with color spaces. Such spaces include the RGB Space (red, green, blue) used in computer displays and television sets, or the CMY Space (cyan, magenta, yellow) used in color printing. Both RGB and CMY colors can be mapped onto the HSI Space (hue, saturation, intensity). The hue describes the wave length via the name of the color (“red”), saturation describes the purity of the color (“red” as being 100% saturated, as opposed to “pink”, which contains parts of white) and intensity describes brightness. Color indexing consists of two steps: first, the color information is allocated to each pixel, then histograms of the individual colors as well as of the HSI dimensions are created. Depending on the lighting, images may represent otherwise identical objects as having different colors. Using suitable normalization, we can try to glean lighting-invariant color values.

The texture of an image is used to index its structure. “Texture” is rather hard to define (Sebe & Lew, 2001, 51).

Despite the lack of a universally accepted definition of texture, all researchers agree on two points: (1) within a texture there is significant variation in intensity levels between nearby pixels; that is, at the limit of resolution, there is non-homogeneity and (2) texture is a homogeneous property at some spatial scale larger than the resolution of the image.

Examples of textures include patterns on textiles or parquet flooring or on a furrowed field as well as cloud formations. Tools used to analyze texture include Gabor filters, which filter an image according to frequency ranges in order to determine its changes in grey tones or colors, and the procedure introduced by Tamura, Mori and Yamawaki (1978). The latter uses four subdimensions (granularity, contrast, direction and line similarity) and two derived subdimensions (regularity: dividing an image into split images and comparison via the four named dimensions; roughness: sum of granularity and contrast).

A particularly important dimension is the recognition of patterns in the images (“shape matching”). There are various approaches to accomplishing this. The objective—besides identifying the shape—is to normalize the position (if it is located at the edge of an image, for instance), to normalize the size (one and the same shape may figure in the background of one image, very small, and in the foreground of another, filling the entire frame) as well as to normalize the perspective (creating a rotation-independent view of the figure).

On the basis of the basic dimensions, it is possible to create certain further dimensions for information retrieval. Images are two-dimensional, but the objects they rep-

resent are mainly three-dimensional. Information concerning the third dimension must thus be retrieved in order to identify three-dimensional objects on the basis of two-dimensional images. This is called 3-D reconstruction. The procedure attempts to elicit a camera position in order to determine the “real” spatial proximity between objects in an image. The current state of research in image retrieval only allows for experimental simulations of higher dimensions; hardly any applicable results are available. The only area of research to have yielded some promising approaches is the recognition of emotions and other mental activities via facial expressions (Fasel & Luettn, 2003).

Application Scenarios for Image Retrieval

There are several starting points for searching images. The simplest case is the presence of a digital model image, which leads the user to search for “more like this!” In the second case, there is no image: the user himself drafts a sketch and expresses his wish to search for similar images (“sketch-based retrieval”; Jörgensen, 2003, 188). In the third case, the user searches verbally, which presupposes that the retrieval system is able to successfully use higher dimensions of image description.

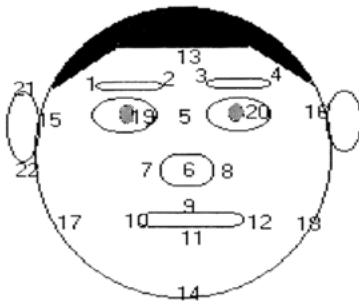


Figure E.4.1: Dimensions of Facial Recognition. Source: Singh et al., 2004, 76.

When is it of advantage to search for further images on the basis of model images or expressly drawn sketches? A first interesting application area is the search for a certain person—perhaps a criminal—on the basis of unmistakable biometric characteristics. Images of fingerprints as well as of the retina and iris are particularly useful (Singh et al., 2004, 63-67). The recognition of faces is a similar case (Singh et al., 2004, 68 et seq.). A figure, identified as a face, is divided into various dimensions (see Figure E.4.1): the position of the tip of the nose (N° 6), the midpoint between both eyes (N° 5), the middle of the hairline (N° 13) or of the eyebrows (between 1/2 and 3/4). Field trials

have shown that faces can only be satisfactorily identified when they are perfectly lit (BKA [German Federal Criminal Police Office], 2007).

Design marks are another exemplary application. Let us suppose a company has registered an image as a trademark. In order to ward off abuse it must keep a close eye on third parties' trademark applications. When a new application bears a similarity to their own image, the company must file a protest with the corresponding trademark office. On the other hand, it is also extremely useful for an individual or a company to know whether the design they wish to create and register is, as a matter of fact, new and unique. In both cases, image retrieval is of fundamental importance; in the former case, a model document (one's own trademark) is used to conduct the search, and in the second case the query is a sketch of one's proposed design. All systems listed in Eakins (2001, 333-343) have an experimental character; so far, no trademark office and no commercial trademark database offer image retrieval in practice (Stock & Stock, 2006, 1800).

Video Retrieval

Film and video retrieval are closely related to image retrieval; the former may be confronted with much larger quantities of data, but its retrieval tasks are easier to accomplish. After all, a lot of additional information is available: for instance, athletes are identifiable via the number on their jerseys as well as via their position in the starting lineup. Sebe, Lew and Smeulders (2003, 141) emphasize:

Creating access to still images had appeared to be a hard problem. It requires hard work, precise modeling, the inclusion of considerable amounts of a priori knowledge, and solid experimentation to analyze the contents of a photograph. Even though video tends to be much larger than images, it can be argued that the access to video is a simpler problem than access to still images. First of all, video comes in color and color provides easy clues to object geometry, position of the light, and identification of objects by pixel patterns, only at the expense of having to handle three times more data than black and white. And, video comes as a sequence, so what moves together most likely forms an entity in real life, so segmentation of video is intrinsically simpler than of a still image, again at the expense of only more data to handle.

In image retrieval, the documentary reference unit is easily identified. In video retrieval, there are several candidates (Schweins, 1997, 7-9): the individual images (frames) are probably not suitable, and so we must consider film sequences as DRUs. Here, there are two perspectives that must be distinguished: a technological perspective, with reference to individual shots, and a semantic perspective concentrating on scenes. A scene is defined via a unity of time, place, characters, storyline and sound. It generally consists of several shots, the continuity of which may be breached by the insertion of shots that belong to another scene: for instance, an interview (Figure E.4.2), filmed in shots 1 and 3, may be rendered more visually entertaining by briefly

showing the interviewee's surroundings (shot 2). An additional problem arises when the contents of image and sound diverge, as Schweins (1997, 9) describes:

This is the case, for instance, when the continuity is upheld by the soundtrack, whereas the camera position changes—or vice versa.

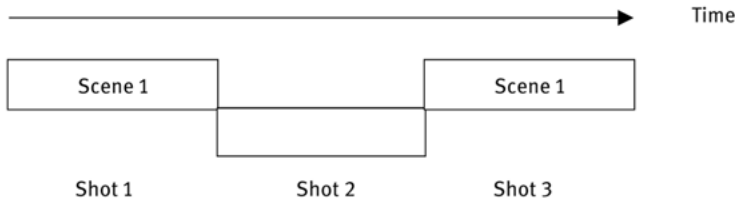


Figure E.4.2: Shot and Scene. Source: Modified Following Schweins, 1997, 9.

When two shots are separated from each other via a hard cut, segmentation can be simple. In many cases, however, film production uses soft transitions that make it difficult to draw clear lines. Smeaton (2004, 384) reports:

Fade in and fade out, dissolving, morphing, wipes, and many other chromatic effects are surprisingly commonplaces in TV and movies ... If a gradual transition takes place over, say, a four-second period, then the incremental difference between frames during the transition will be quite minor as the overall transition will span 100 frames at 25 fps (frames per second).

Video retrieval must be able to segment a movie into shots (“shot boundary detection”), and furthermore group these shots into scenes. It can be useful to select important individual images (“key frames”) and then search for them using the methods of image retrieval discussed above.

In addition to color, texture and shape, movies have another important characteristic: motion. Here, we must differentiate between the motion of the camera (panning or zooming) and “real” motion within the image. The latter can be used to determine objects. Let us consider, as an example, the scene in “Notting Hill” in which William (Hugh Grant) walks through the four seasons on Portobello Road. The scene is deftly cut and looks as though it were only a single shot. The background constantly changes, while the figure of William remains (more or less) the same. This relative constancy of a figure throughout many frames can allow us to positively identify it, and to regard the scene’s other characteristics of color, texture and shape as secondary.

How does one search for videos? Besides verbal search, search arguments include model images and model shots (Smeaton, 2004, 387-388):

Video shot retrieval systems have ... been developed using images as queries or even full shot-to-shot matching. Image-to-shot retrieval can be based on matching the query image against a shot keyframe using conventional image retrieval approaches. ... Searching through video can be

undertaken by matching a query directly against the video content, or by allowing searches on attributes automatically derived from the video.

Applications of video retrieval include both the private domain (searching and browsing through films one has shot or recorded oneself) and professional environments.

Spoken Queries

Spoken queries are particularly important when no keyboard is available, i.e. for people with disabilities, or—for all users—in situations where the voice-recognition capabilities of a (mobile) telephone are being used, i.e. in mobile information retrieval. The fundamental technology of “spoken query systems” is a unit of automatic natural language recognition and information retrieval. A particular problem of information linguistics in such application scenarios is homophony: words that sound the same but represent different concepts (“see”—“sea”).

In spite of great advances in recognizing spoken language, it is not impossible for words to be misinterpreted. Hence, it is extremely important to have a query dialog which uses Relevance Feedback in order to specify the query (or what the system thinks is the query). Crestani (2002, 107-108) describes an interactive retrieval system for spoken queries:

A user connects to the system using a telephone. After the system has recognized the user ..., the user submits a spoken query to the system. The Vocal Dialog Manager (VDM) interacts with the user to identify the exact part of the spoken dialogue that constitutes the query. The query is then translated into text and fed to the probabilistic IR system. ... The (system) searches the textual archive and produces a ranked list of documents. ... Documents in the ranked list are passed to the Document Summarization System that produces a short representation of each document that is then read to the user over the telephone using the Text-to-Speech module of the VDM.

In conclusion, the entire text document is transmitted to the user in a conventional manner (e.g. via SMS or e-mail). For short documents, it is also possible for the system to yield the entire text in spoken language.

Spoken Documents

Retrieval systems whose documents contain speeches also use spoken word recognition (SDR; “spoken document retrieval”). Such systems are important for archiving historical speeches, but also for broadcast news or spoken-word weblogs (podcasts). Documentary reference units do not have to be complete documents, but also small parts of documents. Individual topics, too, can be regarded as such (in the most basic

case: an excerpt from a speech of a certain length). Garofolo et al. (1997, 83) describe the main features of SDR:

In performing SDR, a speech recognition engine is applied to an audio input stream and generates a time-marked textual representation (transcription) of the speech. The transcription is then indexed and may be searched using an Information Retrieval engine. In traditional Information Retrieval, a topic (or query) results in a rank-ordered list of documents. In SDR, a topic results in a rank-ordered list of temporal pointers to relevant excerpts.

Since feedback with the speaker is not provided here (in contrast to spoken queries), any ambiguities will remain undetected. This applies not only to “true” homophones but also to similar sounds extending over several syllables or words. Sparck Jones et al. (1996, 402) name the relevant example of “Hello Kate” and “locate”, phrases whose last two syllables are homophones.

Two approaches toward transcribing spoken language can be identified: on the one hand, there are systems that try to recognize words, process them further (e.g. by working out basic forms) and then align them with the query; on the other hand, it is possible to work with phonemes (sounds). Here, too, the query must be transcribed and aligned via spoken language.

Dimensions of Music Information Retrieval

Searching and finding pieces of music have popular application areas as well as scientific and commercial ones. Anyone who is interested in music will occasionally search for specific titles, e.g. to find more information on a certain song, to buy it, to make it his ringtone etc. For information needs like these, it is important to formulate one’s query either via a recorded excerpt of the song or via its melody—sung, whistled or hummed by the user himself. Musicologists, on the other hand, search for themes or the influence of earlier composers on a current piece. For reasons of copyright protection, too, it makes sense to search music: a composer will check whether there are any plagiarisms of his work, or—in case of a new composition—make sure that none of his ideas have previously been recorded. Lastly, it is possible for developers to create recommender systems for pieces of music. Downie (2003, 295) sketches a possible popular application scenario for music information retrieval:

Imagine a world where you walk up to a computer and sing the song fragment that has been plaguing you since breakfast. The computer accepts your off-key singing, corrects your request, and promptly suggest to you that “Camptown Races” is the cause of your irritation. You conform the computer’s suggestion by listening to one of the many MP3 files it has found. Satisfied, you kindly decline the offer to retrieve all extant versions of the song, including a recently released Italian rap rendition and an orchestral score featuring a bagpipe duet.

Uitdenbogerd and Zobel (2004, 1053) see further fields of application for music information retrieval (MIR) systems that go beyond mere song searches:

People trying to find the name of a piece of music are not the only potential users of MIR technology. For example, composers and songwriters may question the source of their inspiration, forensic musicologists analyze songs for copyright infringement lawsuits, and musicians are often interested in finding alternative arrangements or performances of a particular piece.

Music information retrieval that is oriented on the content of the music itself (“content-based music retrieval”; CBMR) is performed in one of several ways: via systems that build on the pieces’ audio signals, via systems that work with symbolic representations (musical notation) or via hybrid systems that use both approaches together. Nearly all endeavors currently concentrate on Western music with its use of twelve semitones per octave. Byrd and Crawford (2002, 250) emphasize:

(N)early all music-IR research we know of is concerned with mainstream Western music: music that is not necessarily tonal and not derived from any particular tradition (“art music” or other), but that is primarily based on notes of definite pitch, chosen from the conventional gamut of 12 semitones per octave. ... Thus, we exclude music for ensembles of percussion instruments (not definite pitch), microtonal music (not 12 semitones per octave), and electronic music, i.e., music realized via digital or analog sound synthesis (if based on notes at all, often no definite pitch, and almost never limited to 12 semitones per octave).

An MIR system that works with audio signals must register these and process them further (if the objective is to arrive at specific notes) (Figure E.4.3). Audio signals within the audible range move between around 16 and 22,000 Hz; common audio formats include the non-compressed WAV (“wave”) as well as the compressed MP3 (“MPEG1 Audio Layer 3”; MPEG = “Moving Picture Experts Group”). The standard in digital music representation via the depiction of notes and time stamps is MIDI (“Musical Instrument Digital Interface”). In this scenario, the analogy to spoken language recognition is the translation of audio signals into the MIDI format—a process that has not been sufficiently mastered so far (Byrd & Crawford, 2002, 264).

Music information retrieval works with four basic dimensions: pitch, duration, polyphony and timbre. They are complemented by further dimensions: the registration of sung texts, e.g. as vocal scores or via the lyrics (in a case of classical retrieval), the identification of single instruments, or the recognition of individual singers. Another option is to use the music’s symbolical information, i.e. the notes, for the purposes of MIR.

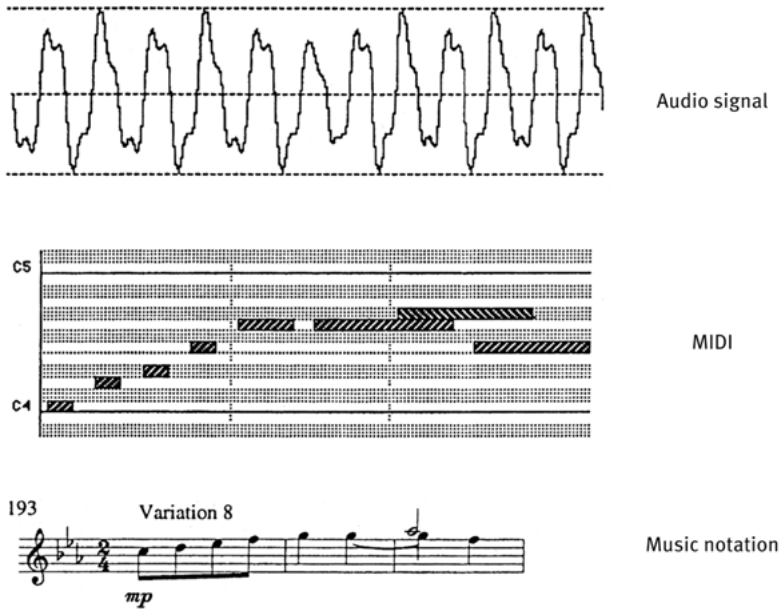


Figure E.4.3: Music Representation via Audio Signals, Time-Stamped Events and Musical Notation.
Source: Byrd & Crawford, 2002, 251.

The respective pitch of the individual sounds is a crucial dimension of content; if it is not specified by the notation or by the MIDI file, it must be extracted from the audio signals via specific wave forms. Such a “pitch tracking” process is prone to errors. For instance, it can be difficult to decide whether a specific sound is a harmonic tone or a single note, because many instruments produce overtones that always resonate. Uitdenbogerd and Zobel (2004, 1056) report:

For example, if the note A at 220 Hz is played on a piano, the wave-form of that note includes harmonics at double (440 Hz), triple (660 Hz), quadruple (880 Hz) frequency, and so on.

The content dimension of duration and rhythm is made up of several aspects, such as tempo, single note length and accent (Downie, 2003, 298):

Because the rhythmic aspects of a work are determined by the complex interaction of tempo, meter, pitch, and harmonic durations, and accent (whether denoted or not), it is possible to represent a given rhythmic pattern many different ways, all of which yield aurally identical results.

Considering duration to be the sole dimension of MIR appears insufficient. When combined with pitch, however, it might be possible to register melodies, as Suyoto and Uitdenbogerd (2005, 274) emphasize:

Duration information on its own is not useful for music retrieval. ... Rhythm seems to be insufficiently varied for it to be useful for melody retrieval. However, the combination of pitch and rhythm is sometimes needed in order for humans to distinguish or identify melodies.

The necessary interplay of pitch and rhythm can be demonstrated via an example. Ludwig van Beethoven's "Ode to Joy" and Antonín Dvořák's "The Wild Dove" begin with completely identical pitch. If "pitch" was the sole basis of retrieval, both fragments would thus be summarized as one—even though the pieces create distinct melodies due to their differences in note duration and measure (Byrd & Crawford, 2002, 257).

A melody is independent of the key in which it is played; the decisive factor is the respective musical "shape" (*Gestalt*). It was already clear to von Ehrenfels (1890, 259)

that the melody or tonal shape is something other than the sum of the individual sounds it is built on.

According to the theory of the production of ideas (Stock, 1995), the musical gestalt, or melody, cannot simply be fixed by being transposed into a key of one's choosing. It always depends upon the perceptivity of the recipient. Witasek (1908, 224) points out:

It happens all too often, particularly with people of little musical talent or training, that in listening to a somewhat difficult piece of music they will perceive the successive sounds, but no more than that; the shapes and melodies remain inaccessible to them because they do not possess the faculty of processing their sound perceptions in the corresponding way.

The shape is thus not made up of the totality of one's perceptions of the individual sounds; it is additionally dependent on the perceiving subject, and of this subject's ability to even conceive of such a shape. Once a subject has managed to conceive of a shape, however, this shape will be what is retained in the subject's memory, and not the individual sounds (independently of the key).

Two options of automatically registering melodies are being debated. Besides analyzing rhythm, one can describe a progression from "pitch" to "pitch" (higher, lower, equal) or, alternatively, describe the intervals (unison, second, flat third etc.). According to Uitdenbogerd and Zobel (2004, 1055), the interval technique appears to be much more promising:

Contour standardization reduces the melody to a string of characters representing "up", "down", and "same" pitch directions. This was shown to be insufficient with our collection for queries of even 20 notes. Our experiments showed that interval-based techniques are vastly more effective.

Harmonies are created by the simultaneous sounding of two or more "pitches". This is called "polyphony", in contrast to "monophony" (which has exactly one pitch). Notes

that are positioned on top of each other in a score are played at the same time and thus result in *one* chord, i.e. a single sound. Searches for monophonic pieces can be performed via the n -gram method: here, notes are expressed via their MIDI numbers and a window of the length n will glide over the piece of music. The search follows the normal patterns of n -gram retrieval. It is possible to enhance this approach via the polyphonic movement by working through the individual voices separately (Doraisamy & Rüger, 2003, 55):

For each window, we extract all possible monophonic pitch sequences and construct the corresponding musical words ... Note that the term “musical word” in this context does not necessarily subsume a melodic line in the musical sense; it is simply a monophonic sequence extracted from a sequence of polyphonic music data.

Timbre is contingent upon the characteristics of the instruments (Downie, 2003, 299):

The aural distinction between a note played upon a flute and upon a clarinet is caused by the differences in timbre. Thus, orchestration information, that is, the designation of specific instruments to perform all, or part, of a work, falls under this facet.

According to Downie (2003, 300), searches for timbre can be performed via a model sound search (e.g. by using a sequence of a flute playing as the query).

Music Information Retrieval via Model Documents

Shazam is an MIR system that is already being used in practice (Wang, 2006). In its database, it stores several million pieces of music. Using a (cell) phone, the user plays an excerpt of the title about which he wishes to receive further information, or which he wants to buy. After a maximum of 30 seconds, Shazam will present him with a search result (including the song title and name of the artist). The MIR system works exclusively with audio information, particularly with spectrograms. Both in terms of the database contents and in the query, each piece of music is exclusively analyzed for special characteristics, in the sense of “fingerprints” or “landmarks”. These are then deposited as a triple entry made up of “fingerprint”, location (expressed in seconds) and identification number. Lastly, it is noted at which point in time such a specific characteristic occurs in a piece.

In retrieval, alignment is achieved via a parallel analysis of the query histogram (whose chronology is noted along the Y axis) and the stored piece of music in question (X axis). If the query was identical to the music pattern and its related search result, the point cloud would have to express an entirely linear relation—in other words, it would have to form a diagonal in the graphic. The presence of static means that this ideal will never occur in practice; the search result will be that piece of music which

forms the highest value of linear correlation with the query. Wang and Smith (2001, 3) describe their idea:

The method is preferably implemented in a distributed computer system and contains the following steps: determining a set of fingerprints at particular locations of the sample; locating matching fingerprints in the database index; generating correspondence between locations in the sample and locations in the file having equivalent fingerprints; and identifying media files for which a significant number of the correspondences are substantially linearly related. The file having the largest number of linearly related correspondences is deemed the winning media file.

Query by Humming

When a user sings, whistles or hums an excerpt from a piece of music for the purposes of search, he performs a “query by humming” (Ghias et al., 1995). In contrast to searches via model documents, the MIR system must in such a case contend with further imperfections: a user may fail to sing correctly (false tempo, false note intervals), and background noises may create static. The first step consists in recognizing the borders between sung sounds (“note segmentation”). In analogy to the treatment of typos in written texts, the objective here is to identify and correct sounds that have been mistakenly inserted, excluded or replaced. Meek and Birmingham (2003) name some further error sources:

Transposition: the query may be sung in a different *key* or *register* than the target. Essentially, the query might sound “higher” or “lower” than the target.

Tempo: the query may be slower or faster than the target.

Modulation: over the course of a query, the transposition may change.

Tempo change: the singer may speed up or slow down during a query.

Non-cumulative local error: the singer might sing a note off-pitch or with poor rhythm.

Despite their proneness to errors, there already exist several “query by humming” systems, both in experimental phases and in practical usage.

Conclusion

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- The basic media in information retrieval are written text, images and sounds, as well as video (as a mixed form of moving images and sound). Written text is the object of “normal” information retrieval; multimedia retrieval means the search for non-textual documents.
 - “Content-based” image and sound retrieval processes the content of the image and sound documents themselves instead of working via the mediation of written texts. Generally, the alignment between queries and documents runs through several content dimensions.
 - In image retrieval, the basic dimensions are color, texture and form; application areas include the search for biometric characteristics or faces, as well as searches for design marks.

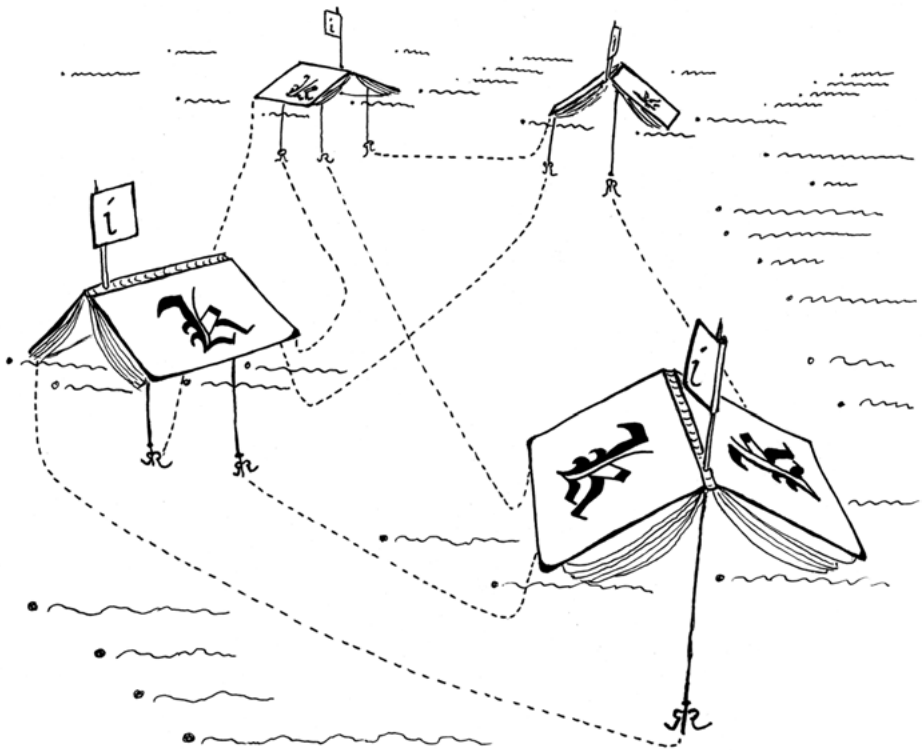
- In video retrieval, the documentary reference units are shots and scenes, both of which must be automatically segmented from the complete movie. In addition to the basic dimensions of image retrieval, video retrieval also has the dimension of motion. Searches for video sequences work by aligning either images or shots.
 - Spoken queries (e.g. by telephone) are directed toward retrieval systems, in which case the system will generate spoken replies. The recognition of spoken language is made more difficult by homophones. Errors in recognizing spoken language can be identified and corrected via a query dialog.
 - Historical speeches, radio broadcasts or podcasts are available as audio files and are the object of spoken-word document retrieval. Vagueness in recognizing spoken language and the problem of homophony are a problem here as well.
 - The content dimensions of music are pitch, rhythm, polyphony and timbre, while a further dimension, melody, is derived from a combination of pitch and rhythm. Music information retrieval occurs either via an excerpt from a piece (model document) or via a sung query (“query by humming”).
 - Query-by-humming systems must recognize and correct error sources on the part of the searcher as well as added and omitted sounds, changes in pitch and changes in tempo.
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Part F
Web Information Retrieval



F.1 Link Topology

Information Retrieval in the Surface Web

Text-statistical methods (including the Vector Space Model and the Probabilistic Model) are applicable to all sorts of texts. A distinction of documents in the World Wide Web is that they are interconnected via hyperlinks. The Web documents' link structure yields additional factors for Relevance Ranking in the WWW (Henzinger, Motwani, & Silverstein, 2002). Web Information Retrieval (Craswell & Hawking 2009; Croft, Metzler, & Strohman, 2010; Rasmussen, 2003; Yang, 2005) is a special case of general information retrieval in which additional subject matters—namely the location of the documents in hyperspace (hence link *topology*)—can be rendered serviceable for the construction of retrieval algorithms.

But it is not only link structure which distinguishes Web documents from documents in Deep Web databases. According to Lewandowski (2005a, 71-76; 2005b, 138-141), the following differences between retrieval in the Surface Web (“Web Retrieval”, paradigmatically represented by Google) and “traditional” information retrieval in information services of the Deep Web (PubMed, for instance) catch the eye:

	<i>Surface Web IR</i>	<i>Deep Web IR</i>
I. Documents		
Languages	many languages	generally unified indexing language
Formats	many formats	generally one format
Length	varies	in bibliographical databases: roughly identical
Parts	different parts (images, jump labels, ...)	exactly one data set (surrogate)
Linking	Hyperlinks	references and citations, links where useful
Spam	yes	no
Structure	weak	field structure
Content	heterogeneous	homogeneous
II. Universe		
Size	Web size unknown	(roughly) known
Coverage	not measurable	(roughly) measurable
Duplicates	yes: mirrors	no
III. Users		
Target Group	all Web users	generally specialists
Need	varies	specialized
Knowledge	low	professional end users: high; information professionals: very high
IV. IR System		
Interface	simple	often (very) complex
Functionality	low	high
Relevance	yes	no (additionally offered where necessary).
Ranking		

In databases of the Deep Web, Relevance Ranking is a “nice” additional offer that professional end users like to take advantage of. After being introduced, initial systems with a corresponding functionality (such as WIN, DIALOG Target or Freestyle) were adopted very rarely, or not at all, by information professionals as part of their daily work routine. In Web retrieval, on the other hand, Relevance Ranking is essential; we can hardly imagine an Internet search engine without a ranking algorithm.

First-generation search engines (Lewandowski, 2005b, 142), such as AltaVista, employ tried-and-true procedures and work with text statistics, the vector space model and probabilistic approaches. These do not adequately address the specifics of the Web, however—mainly, they fail to deal with the “negative creativity” of spammers. Spamming is not taken into account by the traditional retrieval theories, and it does not need to be, since each documentary reference unit (or at least every source) had been individually checked for quality.

The second generation of search engines takes into consideration the particularities of the World Wide Web. Here new standards are set by Google in particular (Barroso, Dean, & Hölzle, 2003). The ranking of Web documents is accomplished via two bundles of factors: query-dependent factors and query-independent factors. The following aspects are dependent upon a specific search request (Lewandowski, 2005b, 143):

- WDF and IDF (perhaps processed further via the Vector Space Model or the Probabilistic Model), possibly taking into account the position (in the title tag, in the meta-tags, its position in the body (top, middle, bottom)),
- Word distance (for more than one search argument: importance increases with proximity) (Vechtomova & Karamuftuoglu, 2008),
- Sequence of search arguments (arguments named first are more important),
- Structural information in the documents (text markup, font size, etc.),
- Consideration of anchor texts (of the linking documents),
- Language (documents in the user’s presumed language are given preference),
- Spatial proximity (for location-based queries: documents stemming from a source in the author’s environment are given a higher weight).

Query-independent factors are used to assign a weighting value to a specific Web page. Here the following factors can be meaningfully employed (Lewandowski, 2005b, 144):

- Incoming and outgoing links (link popularity),
- Placement of the page within a graph of Web documents,
- Placement of the page within the index hierarchy of the page (the higher up, the more important it is),
- Click popularity (the more it is frequented, the more important it is),
- Up-to-dateness (the more recent, the more important it is).

The central query-independent weighting factors are, without a doubt, link popularity and—in connection with it—the position of a page within the WWW’s entire document space. Arasu et al. (2001, 30) emphasize the meaning of this bundle of factors:

The link structure of the Web contains important implied information, and can help in filtering and ranking Web pages. In particular, a link from page *A* to page *B* can be considered a recommendation of page *B* by the author of *A*. Some new algorithms have been proposed that exploit this link structure ... The qualitative performance of these algorithms is generally better than the IR algorithms since they make use of more information than just the content of the pages.

At the center of the debate is the hyperlink (Yang, 2005, 41):

Hyperlinks, being by far the most prominent source of evidence in Web documents, have been the subject of numerous studies exploring retrieval strategies based on link exploitation.

A special case of Surface Web Retrieval is represented by Web 2.0 services. The algorithms for Relevance Ranking used there take into account service-specific criteria in addition to the weighting factors introduced above. On Facebook, with its “Edge-Rank”, these are the “friends” or “fans” of profiles, on Twitter they might be the number of a user’s followers or the amount of retweets of a microblog entry (see also Ch. F.3).

User Interfaces in Web Search Engines

Search engines in the Surface Web are distinguished by the fact that even untrained laymen are able to successfully use them to search and retrieve documents. Each search engine has a search interface that is designed as simply as possible. In Google, a search window is placed in front of an otherwise almost entirely empty screen. Simple search is enhanced by an advanced search function that includes several search options, e.g. regarding language, region, certain Web pages or file types (.doc, .pdf, .ppt, etc.). Search engines process all sub-databases (e.g. for images, maps or scientific-technological literature) as a “universal search”, but they also allow users to call up the respective specialist databases (in Google these would be Google Images, Google Maps and Google Scholar). It is of benefit for the user to retain search arguments once entered, even when switching databases. Web search engines process user input (e.g. in the case of typing errors). It should be possible to disable this option in order to search using “one’s own” words (in Google this is done via the “verbatim” option). Search engines permit the use of Boolean Operators. If a user foregoes the use of Boolean Operators, an operator (in the case of sufficiently large hit lists: AND) will be added. When a sufficient amount of characters is entered, it is possible to complete the search argument that the user may have meant. Such an autocompletion function works by aligning the still-incomplete character sequence of the user input with the entries in the inverted file and then recommending that entry which leads to the most results (Bast & Weber, 2006). Each newly entered character may change the recommendation.

Everything
Images
Maps
Videos
News
Shopping
More

Any time
Past hour
Past 24 hours
Past week
Past month
Past year
Custom range...

All results
Sites with images
Related searches
Dictionary
Reading level
Nearby
Custom Location
Translated foreign pages
Verbatim

Weather for New York, NY, USA

10°C | °F
Partly Cloudy
Wind: N at 21 km/h
Humidity: 48%

Mon	Tue	Wed	Thu
15° 4°	19° 13°	19° 6°	15° 1°

Detailed forecast: [The Weather Channel](#) - [Weather Underground](#) - [AccuWeather](#)

[New York Weather Forecast and Conditions](#)
www.weather.com/weather/today/New+York+NY+USNY0996
New York **weather** forecast and **weather** conditions. Today's New York **weather** plus a 36 hour forecast and Doppler radar from **weather.com**.
↳ 10 Day - 5 Day - Hourly - Weekend

[10 Day Weather Forecast for New York - weather.com](#)
www.weather.com/weather/tenday/New+York+NY+USNY0996
The **Weather Channel** - **weather.com**. United States (English). United States (English); United States (Spanish) - UK & Ireland (English) - France (French) - Brazil ...
↳ Tomorrow - Monthly - Text Forecast

[Weather Forecast: New York, New York - The Weather Network](#)
www.theweathernetwork.com/weather/usny0996
Current **Weather** Mon. Apr 2, 2012, 3:00 EDT **New York City** Central Park, New York, NY
Overcast. 7. °C. °C; °F. Overcast. Feels Like. Refers to how the outdoor ...

[Weather Forecast New York, NY | New York Weather | Wunderground](#)
www.wunderground.com/US/NY/New_York.html
Current conditions and five-day forecast, including ski and marine conditions.

[New York Weather - AccuWeather Forecast for NY 10007](#)
www.accuweather.com/forecast.asp
New York City Weather. Apr 01, 2012; 11:00 PM ET **New York City**, NY - High winds tomorrow, with gusts past 40 mph. more **Weather Videos** > ...

Figure F.1.1: Display with Direct Answer, Hit List and Further Search Options. Source: Google.

The search is followed by a search engine result page (SERP) in a second user interface. Apart from the output of a hit list, it is possible, from query to query, to directly satisfy one's information need without having to call up a document. In Figure F.1.1, we see the SERP for a search for *weather* "New York City" in Google. The data at the top of the screen may—depending on the information need—have already answered the question satisfactorily. Such a direct answer is particularly possible and purposeful in the case of factual questions (Chilton & Teevan, 2011).

The SERP's hit list is ranked by relevance. Each document is represented via its title, a URL and a so-called "snippet" (Turpin et al., 2007). The snippet contains either textual excerpts from the document that contain the search arguments, or content from the meta-tags. For instance, Google sometimes exploits the description tags for its snippets. Textual excerpts are either (more or less senseless) compilations of text passages, which contain the search arguments, or – as "rich snippets" (van der Meer et al., 2011) semantically enriched short texts. It would be equally possible to use more expressive abstracts (where available in the document or surrogate; see Ch. 0.1) or automatically creatable extracts (see Ch. 0.2). Since we know from user research

(Ch. H.3) that most users only really look at very few hits in a SERP, the central criterion of each Web search engine is its algorithm for Relevance Ranking.

Links and Citations

Links between documents were not created alongside the WWW but have existed since long before. They are found wherever a text refers to another text (Garfield, 1979) (Ch. M.2). Link topology assumes that the hyperlinks in the World Wide Web are at least similar to these references or citations within texts. What they are not is identical; link topology does not have a time axis, and individual links do not always behave like (scientific, technological or legal) formal citations. Smith (2004) is able to demonstrate that around 30% of all hyperlinks are used in a fashion analogous to academic citations, but that the remaining links serve other purposes, e.g. only to navigate, as unspecific pointers to related subjects or as advertising.

Link topology assesses relationships between Web documents as a criterion of relevance. In Figure F.1.2, we see an excerpt from the WWW which we will use to discuss the fundamental relationships between individual Web pages (Björneborn & Ingwersen, 2004). The letters represent singular Web pages as well as whole websites. Between A and B there is a simple neighborly link: A has an outgoing link toward B and B, correspondingly, has an incoming link from A. B has an internal (self-referential) link. A is not linked by any page whatsoever, whereas C links to no other page; both applies to I at the same time, i.e. I is isolated. There are reciprocal links between E and F, since E links to F and F links to E. The shortest path between A and G is 1, while there are three steps between A and H. Following Kessler (1963) one can say that, for instance, B and E are link-bibliographically coupled because both link D. According to Small (1973) it is true that e.g. C and D are co-linked since both are linked by B (Calado et al., 2006, 210). When constructing ranking algorithms, one must always distinguish whether A, B, C, etc. are individual pages or entire websites.

Of the link-topological algorithms for use in Relevance Ranking, we will examine two approaches more closely: the Kleinberg Algorithm (HITS: Hyperlink-Induced Topic Search) as well as the PageRank. The latter has become extremely relevant in retrieval practice, since it is of fundamental importance for the ranking algorithm of the search engine Google.

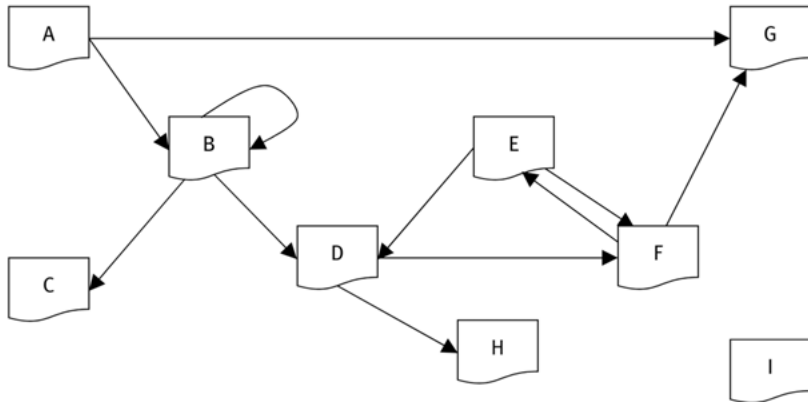


Figure F.1.2: Fundamental Link Relationships. Source: Following Björneborn & Ingwersen, 2004, 1218.

Kleinberg Algorithm

During his participation in the CLEVER (Clientside Eigenvector Enhanced Retrieval) project at IBM, Kleinberg developed his HITS Model, which—building on raw results of an initial text-statistical retrieval process—uses link-topological procedures to rank web pages (Kleinberg, 1997). Kleinberg (1999b, 1) emphasizes:

(W)e make use of well-studied models of discrete mathematics—the combinatorial and algebraic properties of graphs. The Web can be naturally modeled as a directed graph, consisting of a set of abstract nodes (the pages) joined by directional edges (the hyperlinks).

Building on results from citation analysis and the theory of social networks, Kleinberg introduces two types of Web pages: “Hubs” are focal points, distributors that link to many other pages, and “authorities” are Web pages that are linked to by many others; in short, hubs have many outlinks and authorities have many inlinks (Kleinberg, 1999b, 2-3):

In our ... work, we have identified a form of equilibrium among WWW sources on a common topic in which we explicitly build into the model this diversity of roles among different types of pages (...). Some pages, the most prominent sources of primary content, are the *authorities* on the topic; other pages, equally intrinsic to the structure, assemble high-quality guides and resources lists that act as focused *hubs*, directing users to recommended authorities. ... A formal type of equilibrium consistent with this model can be defined as follows: we seek to assign two numbers—a *hub weight* and an *authority weight*—to each page in such a way that a page’s authority weight is proportional to the sum of the hub weights of pages that link to it; and a page’s hub weight is proportional to the sum of the authority weights of pages that it links to.

Together, the most important hubs and authorities form a “community” with regard to a search topic. Gibson, Kleinberg and Raghavan (1998, 225) are able to show that (sufficiently large) thematic communities contain certain (smaller) communities that correspond to related topics (or homonyms):

The communities can be viewed as containing a core of central, “authoritative” pages linked together by “hub pages”; and they exhibit a natural type of hierarchical topic generalization that can be inferred directly from the pattern of linkage.

The precondition for using the Kleinberg Algorithm is the presence of an initial hit list, which results from an alignment of the query terms with the documents and is ranked via text-statistical procedures. The Kleinberg Algorithm is thus a variant of pseudo-relevance feedback.

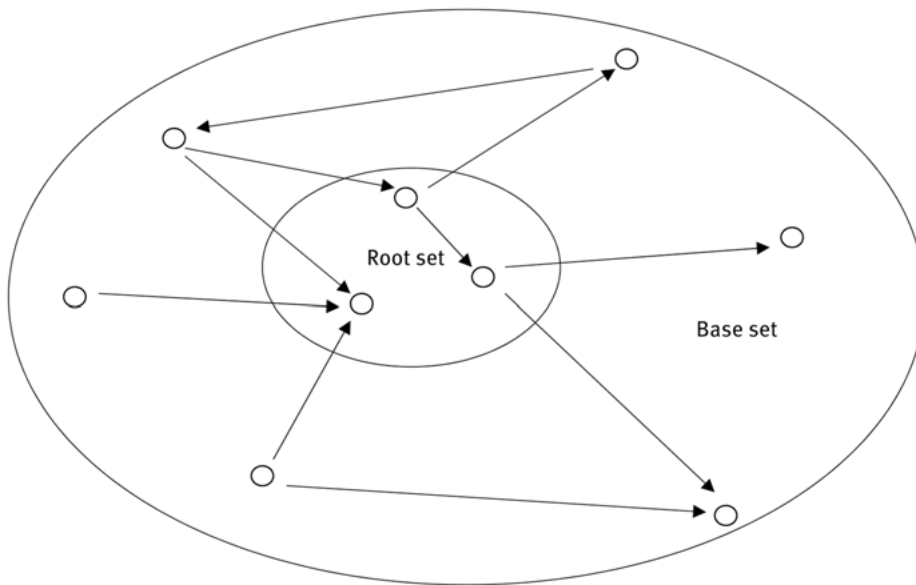


Figure F.1.3: Enhancement of the Initial Hit List (“Root Set”) into the “Base Set”. Source: Modified from Kleinberg, 1999a, 609.

Let the initial hit list be P , or the “root set” (Figure F.1.3). This initial hit list contains all search results retrieved in the first round that are found among the top 200 positions of the relevance-ranked set. (Setting the threshold at 200 is an arbitrary decision that can be varied depending on the specific environment.) The next step deals with the “neighborhood” of (at most) 200 pages from the “root set”. This neighborhood is determined via direct links. All pages that are linked to by a Web page from the “root set” are taken into consideration; i.e., all outlinks from P are being followed. Klein-

berg is much more rigorous when turning in the other direction: any individual page from P may only submit 50 pages that link to it. Kleinberg emphasizes (1999a, 609):

This latter point is crucial since a number of www pages are pointed to by several hundred thousand pages, and we can't include all of them in [the base set] if we wish to keep it reasonable small.

The initial quantity is enhanced into the “base set” by the amount of pages linked via outlinks and by the pages linked via inlinks—with the restriction mentioned above. This base set should display three fundamental characteristics (Kleinberg, 1999a, 608):

- the base set is relatively small,
- it contains many relevant documents (for the query),
- it contains the most (or at least very many) authorities.

The base set now created may contain several pages from one and the same Web domain; i.e., pages that have the same domain name in their URL. All links between such “intrinsic” pages are removed from the calculation of weights, since there is a danger of them being mainly navigation links. The pages (including its entire external links) remain, however. Kleinberg discusses the option of removing pages that are linked to again and again by different pages of a domain, since they might be advertising pages or unspecific references (“This site is designed by...”)—in the end, however, he keeps them in the base set.

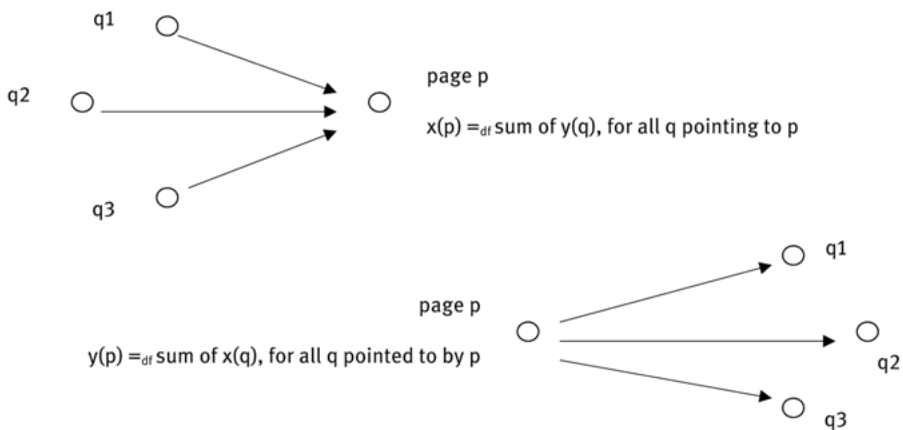


Figure F.1.4: Calculating Hubs and Authorities. Source: Kleinberg, 1999a, 612.

The authority weight of a web page $x(p)$ is the sum of the hub values of those pages that have placed an outlink on p . Let these pages be q_1 through q_n . It now holds that

$$x(p) = y(q_1) + y(q_2) + \dots + y(q_n).$$

Analogously, the hub weight $y(p)$ of a page p is determined via the sum of the authority values of those pages that p links to. Let these pages again be q_1 through q_n . The hub weight $y(p)$ follows the formula

$$y(p) = x(q_1) + x(q_2) + \dots + x(q_n).$$

Table F.1.1: Link Matrix with Hub and Authority Weights.

	A	B	C	D	E	F	G	H	I	Sum Hub
A	-	1	0	0	0	0	1	0	0	2
B	0	-	1	1	0	0	0	0	0	2
C	0	0	-	0	0	0	0	0	0	0
D	0	0	0	-	0	1	0	1	0	2
E	0	0	0	1	-	1	0	0	0	2
F	0	0	0	0	1	-	1	0	0	2
G	0	0	0	0	0	0	-	0	0	0
H	0	0	0	0	0	0	0	-	0	0
I	0	0	0	0	0	0	0	0	-	0
Sum Authority	0	1	1	2	1	2	2	1	0	10

The totality of documents in the base set can be represented as a matrix, where the rows display the outlinks and the columns display the inlinks. Table F.1.1 considers the example from Figure F.1.2 and regards pages A through I as the results of an initial search. (The self-reference by B has been left unconsidered, since the object here are individual Web pages and not sites.)

There are reciprocal relationships between hubs and authorities. “Good” hubs link to good authorities, and conversely, “good” authorities are linked by good hubs (Kleinberg, 1999a, 611):

Hubs and authorities exhibit what could be called a *mutually reinforcing relationship*: a good *hub* is a page that points to many good authorities; a good *authority* is a page that is pointed to by many good hubs.

Since good hubs refer to good authorities and vice versa, both hub and authority weights can only be calculated iteratively. In each round of iteration, the calculation proceeds via two steps. In the first step, the values are normalized by being mapped onto the interval $[0,1]$. Let the sum of all values squared be 1, which means that each value must be multiplied by

$$\frac{1}{\sqrt{q_1^2 + q_2^2 + \dots + q_n^2}}$$

(Kleinberg, 1997, 9). Numerical “outliers” are thus caught and no longer unduly influence the calculations. The normalized values are then added in step two.

In the first round of iteration, no weighting information is available yet. We thus work with the absolute amount of inlinks (when calculating the authority value) as well as the number of outlinks (when calculating the hub value). From the second round onward, the values gleaned in round 1 are applied and the procedure is repeated until both hub values and authority values approach a threshold value. Kleinberg (1999a, 614) deems 20 rounds of iteration to be sufficient.

The result of this procedure is two lists that arrange the pages into rankings according to their final hub and authority weights, respectively. The first ten pages of each represent the “core” of the thematic community (Gibson, Kleinberg, & Raghavan, 1998, 226-227):

We declare the 10 pages with the highest $a()$ values together with the 10 pages with the highest $h()$ values to be the core of a community.

The exact maximum number (here: 10) is eminently arbitrary, but it should be “processable” to users. It cannot be assumed that the top community retrieved in this manner—basically adhering to the most general term of the query—will be at the top of any hierarchy. A search for “linguistics” in 1998, for instance, results in a top community that is dominated almost exclusively by computational-linguistic Web pages. Correspondingly, Gibson, Kleinberg and Raghavan (1998, 231) warn of equating the top community with the most general discussion of the topic:

(T)he principal community can at first appear to be a *specialization*, rather than a generalization, of the initial topic; but in reality, HITS has focused on this community because it represents a more “Web-centric” version of the topic.

There may be delimitable subclusters within the base set that correspond to communities. If our initial search had been for a homonymous word (such as “Java”), we could hope for its individual meanings to be available in different communities. In the matrix, these are areas in which the density of the values 1 is high, e.g. in the cells D-F, E-D, E-F, F-E and F-G in Table F.1.1. Kleinberg (1999a, 622 et seq.) further pursues his approach of analyzing the direct link relationships. He segments the different subclusters—where available—via comparisons with the respective mean vectors that make up the cluster components. Other elegant forms of cluster formations would adhere to link-bibliographical coupling or follow co-links (Ding et al., 2001, 2-3).

At many different points the HITS algorithm introduces arbitrarily selected values (e.g. 200 following the initial research) in order to separate the “good” from the “bad”

rankings and to then neglect documents concerning the latter. Bharat and Henzinger (1998, 106) here speak of “pruning” with the goal of

eliminating non-relevant nodes from the graph. ... All nodes whose weights are below a threshold are pruned.

To determine threshold values, Bharat and Henzinger propose two alternatives, either the mean or a certain part of the maximum value (e.g. $1/10 \cdot \text{maximum value}$). Both pruning methods appear more plausible than arbitrarily selected threshold values.

PageRank

“Page”Rank is a pun on the ranking of the Web *page* and the name of one of its inventors (Larry *Page*). Google’s PageRank is distinct from the HITS algorithm in two fundamental aspects:

- PageRank is independent of user queries, and
- the algorithm only considers authorities and not hubs.

Link-topological Relevance Ranking is designed for large databases. It’s all in the name “Google”, which is clearly inspired by the term “googol” (10^{100}). An explicit ambition of the PageRank is to be resistant to all (then-)known spam methods. Brin and Page (1998, 108) emphasize:

We have built a large-scale search engine which addresses many of the problems of existing systems. It makes especially heavy use of the additional structure present in hypertext to provide much higher quality search results. We chose our system name, Google, because it is a common spelling of googol ... and fits well with our goal of building very large-scale search engines.

Like Kleinberg’s algorithm, PageRank too takes citation analysis as its starting point. However, it modifies it at a decisive point (Page, 1998, 4):

Although the ranking method of the present invention is superficially similar to the well known idea of citation counting, the present method is more subtle and complex than citation counting and gives far superior results.

If n counts the number of inlinks (in analogy to citations), a Web page A has the following weight according to the primitive variant of citation analysis:

$$w_{\text{CitationAnalysis}}(A) = n.$$

This method corresponds to Kleinberg’s authority value of a Web page in its first round of iteration.

The more elaborate version of counting inlinks draws upon the model of a so-called “random surfer”. This ideal-typically introduced character clicks his way through the WWW at random, simply following the links without any guidance. Sometimes, though, he aborts this procedure and restarts elsewhere (Brin & Page, 1998, 110):

PageRank can be thought of as a model of user behavior. We assume there is a “random surfer” who is given a Web page at random and keeps, clicking on links, never hitting “back” but eventually gets bored and starts on another random page.

The probability of a random surfer visiting a page A is its PageRank PR. The probability of the random surfer aborting his click sequence and restarting elsewhere is d. This attenuation factor d is set by Google at 0.85, but in principle it is open to all values in the interval [0,1]. A page is much more likely to be reached by the random surfer when many other pages link to it. And it becomes even more probable when the pages that link to it have many inlinks themselves. These two aspects are of central importance for the calculation of the PageRank:

- a Web page A has a high PageRank if many other pages link to it (which is the same method used to count the number of citations),
- a Web page A has a high PageRank if pages that have a high PageRank themselves link to it (this has no corresponding feature in citation analysis).

Let the Web page A have n inlinks that go out from the pages $T_1, \dots, T_n$. For every one of the linking pages t_i , the number of outlinks $C(t_i)$ is counted. If the database contains N Web pages in total, the PageRank of page A is

$$PR(A) = d/N + (1 - d) * [PR(T_1)/C(T_1) + \dots + PR(T_n)/C(T_n)].$$

(In this formula, d has not yet been set at 0.85—instead, it is a freely chosen value in the interval [0,1].) Page (1998, 4) notes:

The ranks form a probability distribution over web pages, so that the sum of ranks over all web pages is unity.

Up to this point, the PageRank is a probabilistic (link-topological) model. In later versions, Brin and Page forego the stringent probabilistic foundation of their model and ignore the value N (the number of documents in the database), which makes the PageRank variant that ends up being used in Google look like this (Brin & Page, 1998, 110):

$$PR(A) = (1 - d) + d * [PR(T_1)/C(T_1) + \dots + PR(T_n)/C(T_n)].$$

A problem is caused by Web pages (such as some PDF documents) that have no outlinks (but do have inlinks). A certain amount of PageRank will be assigned to them, which will seep away uselessly. Page, Brin, Motwani and Winograd (1998, 6) refer to

such one-way streets as “dangling links”, which are removed from the system before the PageRank is calculated and re-added later:

Because dangling links do not affect the ranking of any other page directly, we simply remove them from the system until all PageRanks are calculated. After all the PageRanks are calculated, they can be added back in, without affecting things significantly. Notice the normalization of the other links on the same page as a link which was removed will change slightly, but this should not have a large effect.

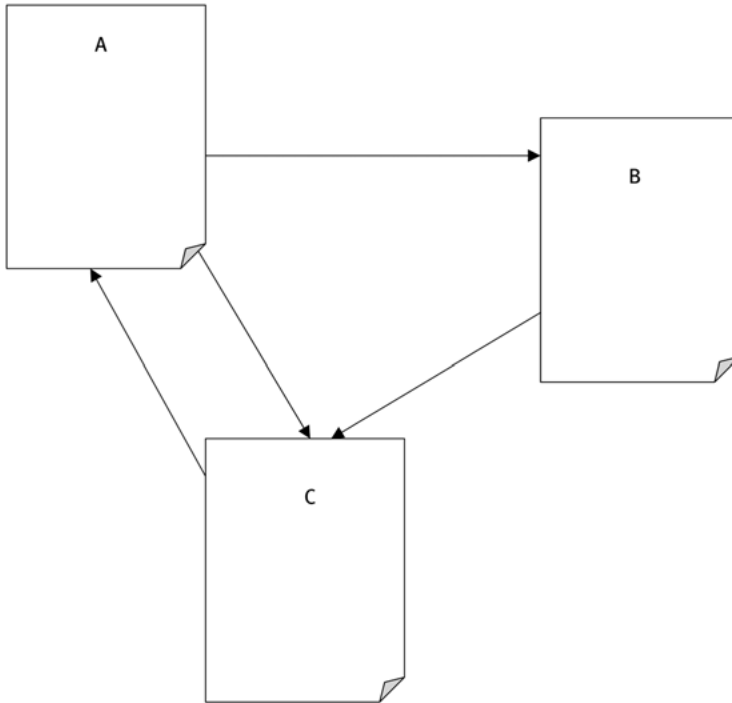


Figure F.1.5: Model Web to Demonstrate the PageRank Calculation. Source: Modified from Page, 1998, Fig. 2.

If the PageRank of the linking pages is unknown, there is a cold-start problem. Then the PageRank calculation proceeds iteratively. In the first round, the linking pages' PageRanks are set at 1; from round 2 onward, we will be working with the results of the round that went before. We now calculate the PageRank, including the damping factor 0.85, for the three documents from Figure F.1.5.

Round 1:

$$\text{PR}(A) = 0.15 + 0.85 * (1/1) = 1$$

$$\text{PR}(B) = 0.15 + 0.85 * (1/2) = 0.575$$

$$\text{PR}(C) = 0.15 + 0.85 * (1/1 + 1/2) = 1.425$$

Round 2:

$$\text{PR}(\text{A}) = 0.15 + 0.85 * (1.425/1) = 1.361$$

$$\text{PR}(\text{B}) = 0.15 + 0.85 * (1/2) = 0.575$$

$$\text{PR}(\text{C}) = 0.15 + 0.85 * (0.575/1 + 1/2) = 1.064$$

... etc. ...

Round 12:

$$\text{PR}(\text{A}) = 1.16$$

$$\text{PR}(\text{B}) = 0.64$$

$$\text{PR}(\text{C}) = 1.19.$$

After about 100 rounds of iteration, satisfactory results are achieved even for large document sets (Page, 1998, 5).

It would be a serious error to assume that the weighting values according to the PageRank reflect the quality of a Web page. The PageRank gives a quantitative expression to the placement of a page on the Web—nothing more. Thus for instance, the entry page for “Star Wars” has a much higher PageRank than a page featuring the English text of the cave allegory from Plato’s “Republic”.

Conclusion

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- Web Information Retrieval is a special form of general information retrieval, which, due to the hypertext structure of the World Wide Web, is able to use the links between Web documents as a ranking criterion.
 - User interfaces for searching Surface Web Search Engines are built simply enough for layman users to use instinctively. Advanced Search interfaces permit the use of more elaborate search arguments. Universal Search goes through all sub-databases in one lookup, whereas specialist databases (such as Google Scholar) only grant access to subareas. Web search engines use fault-tolerant retrieval and utilize autocompletion of search arguments.
 - The Search Engine Result Page (SERP) directly satisfies the user’s information need, where possible, and yields a hit list that is ranked by relevance. A title, a URL and a snippet are displayed for every document.
 - The first generation of Web search engines (mid- to late 1990s) principally considers “classical”, text-oriented algorithms. The ranking algorithms prove non-resistant to spam, however. With the second generation of search engines (starting in the late 1990s), search engines re-orientate toward the specifics of the WWW, and thus to link topology.
 - Weighting factors of Web documents can be classified as query-dependent aspects (e.g. WDF and IDF, word proximity, anchor texts) and query-independent aspects (particularly the position of a Web page in the Web graph).
 - In Web 2.0 services, Relevance Ranking mainly orients itself on user- and usage-specific factors.
 - Hyperlinks are regarded as analogs to references and citations. However, they distinguish themselves from the latter via their lack of a temporal relation and with regard to goals pursued (e.g. advertising or navigation) that do not occur in citations. Outlinks, in this analogy, correspond to the references and inlinks to citations.
 - The Kleinberg Algorithm (HITS) is a procedure that employs pseudo-relevance feedback. An initial text-oriented search is given link-topological further processing in a second step. The (pruned)

initial hit list is enhanced by those pages that are linked by documents within it and parts of those that link to it. The objective is to retrieve hubs (documents with many outlinks) and authorities (documents with many inlinks), which together form a community.

- The “pruning” of arranged lists is performed either via arbitrarily determined threshold values or via statistical calculations (e.g. the distribution’s mean serving as the threshold value).
 - The PageRank of a Web page (used by Google), developed by Brin and Page, is calculated via the number of its inlinks as well as via the PageRank of the linking pages. The underlying model relies on the figure of a random surfer who arbitrarily follows links, but who from time to time interrupts his sequence in order to begin anew elsewhere. The probability of him aborting and pursuing links leads to the introduction of a damping factor d . Under these conditions, the PageRank of a Web page is the probability of the random surfer retrieving it.
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F.2 Ranking Factors

Learning to Rank

Various weighting factors guide the ranking algorithms of Web retrieval systems. Apart from text-statistical factors such as TF*IDF and the procedures that build on them (such as the Vector Space Model or the Probabilistic Model), as well as link-topological parameters (such as PageRank or the Kleinberg Algorithm), there are a number of further ranking factors. These are outfitted in various ways, always relative to the retrieval system's area of application. A Web search engine such as Google, for instance, will take into consideration structural information in documents (such as font size), anchor texts, and the Web page's path length, whereas a microblogging service will tend to concentrate on criteria relating to each individual blogger (i.e. their number of followers or followees). The objective in constructing such composite weighting factors is, ideally, to arrive at a relevance-ranked hit list. Relevance assessments are gleaned via retrieval tests with their variants of Recall and Precision (Ch. H.4). Systems such as TREC have large quantities of documents, queries and relevance judgments at their disposal. They thus make it possible to simulate different combinations of ranking factors, with the goal of finding the best combination. A sub-task involves finding the “right” ranking factors (Geng, Liu, Qin, & Li, 2007). For example: when a ranking factor is CPU-intensive (thus requiring a lot of processing time) but hardly provides any notable ranking advantages compared to other factors, it will not be used in practice. Where learning mechanisms (which automatically search for an ideal solution) are used to combine ranking factors, we speak of “learning to rank” (Burges et al., 2005; Liu, 2009). In this chapter we will sketch a few weighting factors for general retrieval (i.e. non-personalized retrieval; Ch. F.3) that have not previously been discussed. This will be a brief selection; on its website, Google reports using over 200 weighting criteria (for a selection, see Acharya et al., 2003).

Structural Information in Documents

The initial goal of structural recognition in documents is to separate formal bibliographical elements from those pertaining to content. Only the content is suitable for text-statistical analysis. Lewandowski (2005, 59) emphasizes the importance of recognizing structural information:

When indexing Web documents, incorporating the document structure is of particular importance. Apart from field characteristics explicitly denoted in the document text, this mainly involves structural features implicitly contained in the document, whose primary purposes are not document indexing and which the authors are often unaware of using.

Here Lewandowski is thinking mainly of a document's layout and navigation elements.

An initial challenge involves separating the content from the rest in HTML tables (Lewandowski, 2005, 217 et seq.). Tables, i.e. entries within a `<table>` tag, can be both “real” content-bearing elements and “unreal” ones serving other purposes. Tables in upper and lower racks are particularly likely to contain navigation information, as are those on the left-hand side of the screen. Furthermore, elements placed at the right and at the bottom may be (content-free) advertisements. Navigational elements are recognizable by their presence, in unchanged appearance, on nearly every page within a website. Except on the page that is highest in the folder hierarchy, the terms from this table section can be removed from text-statistical analysis as content-free. Passages that are not recognized as content (such as advertising) are not allocated to the documentary reference unit.

When using HTML, there are two explicit content-descriptive tags (Lewandowski, 2005, 62): the title tag and the hierarchy of headings `<h1>` through `<h6>`. Equally content-related, but fallen into disrepute through abuse, are the keywords and the description in the meta-tags. An “expressive” URL may point to the content of its page.

Similarly related to content, if oriented on navigation, is the anchor text that accompanies the respective links (which we will come back to). Implicit content-related structural information is hidden in layout tags such as `<b>` (bold) or `<i>` (italics), in the font size, in statements regarding a larger or smaller font (relative to the standard size) as well as—crucially—in line breaks (indicating paragraphs) and jump labels (indicating individual chapters). If the content area contains a table, the latter must be incorporated into the analysis (Wang & Hu, 2002). Such tables may even be paid special attention as they express relations in a clear and precise manner.

Structural information in documents serve as weighting factors for ranking: they unify approaches of TF*IDF with their corresponding, structurally identified position. Thus the occurrence of terms in the hierarchy of headings can be weighted correspondingly—the further up, the higher the weighting value. It is also possible to weight italicized, bold or differently-sized words more highly than words in the continuous text. Terms in the title tag, in tables and in the URL are dealt with in analogous fashion. For documents with several paragraphs it is worth considering whether to grant a higher weight to the terms in the first and last paragraphs, as in these the author frequently uses terms that are of crucial importance for his work.

Anchors

Anchors are the texts that appear above a link in the browser, and which correspondingly reside as character sequences between the `<a>` and `</a>` tags in the source code. In the example

Mardi Gras in New Orleans

"Mardi Gras in New Orleans" is the anchor text which points to the page <http://www.mardigrasneworleans.com/>. If anchors are well written (not merely noting: "click here"), they can provide compressed information about the content of the linked pages. However, they also provide avenues for possible abuses, such as the anchor text "miserable failure" on some Web pages linking to the homepage of a former U.S. President.

The anchor terms are treated as though they were terms of the linked page. A primitive procedure involves weighting terms that occur in the anchors with randomly selected values. A more elaborate procedure works with a weighted TF*IDF calculation. The weighting results from values relative to whether the anchor is directed toward a page within the same site (low weight) or to an external page (higher weight) (Kraft & Zien, 2004, 667). To calculate term frequency (TF), we regard the entirety of all anchor texts that link to the same page as one single (pseudo-)document. Following a suggestion by Hawking, Upstill and Craswell (2004, 512), the absolute frequency of the terms is counted and used as TF. This unusual value (as opposed to relative frequency or the WDF value) is justified on the grounds that every anchor represents an individual "vote" for the linked page, i.e. that every word counts. Numerical outliers are counteracted by using a logarithmic measurement. For the IDF, the authors use a variant of the standard IDF value. Let the absolute frequency of a term t in the set of pseudo-documents be TF_d , n the amount of documents that contain the term t at least once, N the total amount of documents in the database and α the weighting factor following Kraft and Zien. Following Hawking, Upstill and Craswell (2004, 513), the weighting of term t in an anchor text is calculated as follows:

$$\text{Anchor Weight}(t) = \alpha * \log(TF_d + 1) * \log [(N - n + 0,5) / (n + 0,5)].$$

Path Length

In a Google patent, Dean et al. (2001) discuss the path length of URLs as a weighting factor for Web pages. Path Length PL determines the number of forward slashes (/) respectively dots (.) above the minimum. If a URL contains the designation of Internet service (www) the minimum amount of dots will be 2 (<http://www.xyz.com>), otherwise it will be 1 (<http://xyz.com>). The URL

<http://www.phil-fak.uni-duesseldorf.de/infowiss>

thus has a path length of 2. A maximum path length of 20 is postulated. The weight of a Web page d according to its position in the folder hierarchy is calculated via the formula

$$w(\text{Path Length})(d) = \log(20 - PL) / \log(20).$$

If the path length is 0, the result will be a weighting value of 1; for a path length of 1 the value of a Web page decreases to 0.98, for 2 to 0.96 etc. up to 19, where the result will be a factor of 0. From a path length of 20 onward the formula is undefined.

Such a weighting grants a methodical advantage to websites with flat hierarchies. Websites that are heavily structured and have various hierarchy levels are at a disadvantage due to their deep-lying pages with long path lengths. Academic websites in particular tend to be organized in deep folder hierarchies with pages (e.g. in the PDF format) featuring research reports and publications. These, however, are hardly any less important than the (possibly content-poor) entry page of the respective faculty. From this perspective, weighting via path length—and thus ranking via position in a folder hierarchy—appears highly questionable.

Freshness

Search engines have serious problems with dates and sometimes with the up-to-date-ness of their data (Lewandowski, Wahlig, & Meyer-Bautor, 2006) (see also Ch. B.4). However, with some types of information it may be very useful to rank the pages in question via their freshness. This applies to product information, or to news portals. In such instances, but in other contexts, too, it annoys the user to receive out-of-date information. In a Google patent application, Henzinger (2004, 1) emphasizes:

Frequently, web documents that are returned as “hits” to the users include out-of-date documents. If the freshness of web documents were reliably known, then the known freshness could be used in the ranking of the search results to avoid returning out-of-date web documents in the top results. Currently, however, a reliable freshness attribute for web documents does not exist.

Webmasters can state the attribute “Last Modified” in the Hypertext Transfer Protocol (HTTP), but they are not obligated to do so and there is also a danger of false entries being made. It would thus be very risky to rely on data from one single page. Henzinger does not rely on a Web page’s up-to-dateness statements at all but attempts to estimate the freshness of a page *p* via statements on those pages that link to it. Since this results in a multitude of last-modified statements, the danger of a miscalculation is reduced considerably. If one of the pages in the document set has no date, one can still gather up-to-dateness information on the condition that different versions of the pages have been stored over a longer period of time. If at the beginning of the observation period there is a link to the page *p*, and if this link is removed from a given version onward, the crawl date of this first linkless version will be noted down. Henzinger defines up-to-dateness, as a “freshness score”, via a threshold value—e.g. two years. Counted are those documents whose last-modified statements or whose last version

with the link to p are above the threshold value (number of “old” pages), as well as those documents whose freshness is beneath the threshold value (number of “fresh” pages). The “freshness score” of page p is the quotient from the number of new and old pages:

$$\text{Freshness Score}(p) = \text{Number of fresh pages} / \text{Number of old pages}.$$

If the number of “old” pages dominates, the “freshness score” of p will be smaller than one. In the reverse scenario, the degree of currentness will be a value greater than one. In practice, it can prove useful to employ logarithmic measurements.

Types of Web Queries

Broder (2002) distinguishes between three types of Web searches: navigation, information and transaction queries. Rose and Levinson (2004) refine the categorization of fundamental search goals. Their approach is set off from Broder’s mainly via its distinction between navigation, information goals and the need for resources (not for information purposes), e.g. games or music downloads. Depending on the target group of a specialist search engine, other query types are possible as well. A scientific search engine could, for instance, make meaningful distinctions between research reports, homepages of scientists, congresses and conferences, and entry pages of scientific faculties (Glover et al., 2001, 99). Lewandowski, Drechsler and von Mach (2012, 1773) identified the following general query intents:

Informational, where the user aims at finding some documents on a topic in which he or she is interested.

Navigational, where the user aims at navigating to a Web page already known or where the user at least assumes that a specific Web page exists.

Transactional, where the user wants to find a Web page where a further transaction (e.g., downloading software, playing a game) can be performed.

Commercial, where the query has “commercial potential” (i.e., the user might be interested in commercial offering). This also can be an indicator whether to show advertisements on the search engine results page (SERP).

Local, where the user is searching for information near his or her current geographic position.

The (perhaps hierarchically structured) list of fundamental search goals is offered to the user to check in addition to the query input line. For every category, attributes that are of significance to the system must be designated in order to serve as the basis for searches and rankings respectively. If, for example, a user has checked “homepage” in the scientific search engine, applicable attributes will include search terms in the <title> tag as well as path length. The various attributes can be weighted differently (Glover et al., 2001, 100). Another method involves prototyping user behavior

to derive regularities. Lee, Liu and Cho (2005) demonstrate that users' clicking habits can be used to distinguish navigation goals from information goals. When a majority of users searching for a certain Web page click on the first result of the hit list, and if this remains their only click on average, the page will very likely be the goal of a navigation search. This information is allocated to the page after a satisfactory amount of observed cases, from which point onward it can be explicitly targeted as a document feature during retrieval.

Usage Statistics

Some Web pages are visited often, others more seldom or even not at all. Such usage statistics can be used as ranking criteria for a page. The more frequently a page is visited, the higher its weighting factor will be. The precondition for introducing such a ranking criterion is the availability of usage statistics. Here, two paths can be pursued:

- Statistics of clicked links in a search engine's results lists,
- Statistics of all visited pages of selected servers (or users).

An advantage of using results lists is that all information is available as a sort of "by-product" of the search protocols and can thus be directly processed further. A disadvantage is that only those pages are being considered that occur at least once in a hit list and have been clicked on as well. Toolbars, installed by certain users, allow search engines to gather information about all page views outgoing from a server. Here an advantage lies in the fact that not only page views following a search are being protocolled, but all URL input as well as clicks are registered, too. A disadvantage is that these toolbar users are hardly a representative sample of all Web users.

The search engine Direct Hit used the former method. With regard to counting the clicks in hit lists, its developer Culliss speaks of a user-centric approach—as opposed to the author-centric approach (counting the links in and on pages, as in the PageRank and the Kleinberg Algorithm). A user is shown a hit list—as is common in search engines. The system notes all URLs from this list that are selected by the user, and allocates the clicked pages a score of +1 (relative to the query). Provided enough queries, Culliss (1997) envisages as accurate a representation of a user-adequate Web page as possible. If the query is resubmitted, the pages' usage score becomes a criterion for Relevance Ranking.

The procedure can be refined via the additional storage of (previous) users' profiles. This is a fundamental idea behind the search engine Ask Jeeves. Even if a current user does not submit any search profile, he can still profit from the search expertise of registered users (Culliss, 2003, 3):

A searcher who does not possess certain personal data characteristics, such as being a doctor, for example, could also choose to see articles ranked according to the searching activity of previous searchers who were doctors.

As an example of employing usage statistics regarding the evaluation of Web pages selected by toolbar users, we introduce Google's solution (Dean et al., 2001). Here, two parameters are used to calculate a document's usage score: the number of different users of a document as well as the number of all page views, both within a single unit of time (i.e. within a month) respectively. Page views from automatic agents are not counted in this approach.

Dwell Time

One idea would be to use dwell time as a ranking criterion. Dwell time is the amount of time that users take to view a document from a hit list. One could assume, intuitively, that short dwell times (with correspondingly swift returns to the search engine result page) are evidence in favor of the document's irrelevance (for notes on this, see Morita & Shinoda, 1994, 276). A universally valid correlation between display time and relevance could not be detected, however (Kelly & Belkin, 2004). Long dwell times in particular do not turn out to indicate document relevance, as Guo and Agichtein (2012, 569) discover:

A "long" page dwell time does not necessarily imply result relevance. In fact, a most frustrating scenario is when a searcher spends a long time searching for relevant information on a seemingly promising page that she clicked, but fails to find the needed information. Such a document is clearly non-relevant.

Language

Where a retrieval system identifies a user-preferred language, or several, it appears sensible to grant higher ratings to documents written in one of these languages—on condition that the texts' language has been recognized (Ch. C.2). User language identification as well as a corresponding ranking by language is introduced in a patent for Google by Lamping, Gomes, McGrath and Singhal (2003). The first step—recognizing preferred and, possibly, less preferred languages—is performed via the query terms, via the search interface as well as via characteristics of the search results (Lamping et al., 2003, 2):

Search query characteristics are determined from metadata describing the search query. User interface characteristics are determined also using the search query metadata, as well as client-side and server-side preferences and the Internet protocol (IP) address of the client. Search results characteristics are determined based on an evaluation of each search result.

Search term analysis yields a user language that is then noted down as the preferred one. For short queries in particular it is possible that no exhaustive language recogni-

tion can be performed. In such cases the other tools of language recognition will be used. If the client uses an interface that points to non-Anglophone provenance (e.g. an internet address with a German .de domain), the language in question will be designated as “preferred” and English as “less preferred”. In case of English-language interfaces and a multitude of English search results, English is taken to be the preferred language. If up to this point no user language has been found, weighting by language will not be implemented.

In the second step, the initial hit list is re-ranked by language. Each document has an initial score s_i that lies in the interval $[0,1]$. For each of the n results (1, ..., i , ..., n) of the original list, three distinctions are made (Lamping et al., 2003, 8):

- no recognized preferred language: unchanged adoption of the old score, $e_i = s_i$,
- document in a preferred language: change of the old score into the new retrieval status value $e_i = [(s_i + 1) / 2]$,
- document in a less preferred language: change of the old score into the new retrieval status value $e_i = [(s_i * 2 + 1) / 3]$.

When a query recognized by the IP address as German meets a German document, the former’s score will be raised—e.g. from 0.8 to 0.9. An English-language page with the same value of 0.8 as its initial score—deemed “less preferred” under these conditions—will change its retrieval status value to 0.87.

Ranking by Distance: Geographical Information Retrieval

The ranking of documents according to their distance from the user’s stated location requires the availability of the following aspects:

- location of a Web page,
- location of the query,
- geographical knowledge organization system,
- coordinates of the locations.

The location of documents is extracted from the texts. Here we must distinguish threefold: by the offer’s location (e.g. a pizza parlor in Kerpen-Sindorf), by the page’s content (“Pizza delivery in Sindorf”) and by the service area of the provider (e.g. pizza delivery to all districts of Kerpen, but not to any other towns). Wang et al. (2005, 18) define these three categories of location:

Provider location: The physical location of the provider who owns the web resource ...

Content location: The geographic location that the content of a web resource describes. ...

Serving location: The geographic scope that a web resource can reach.

Content location and serving location are of relevance for search. The objective is to recognize and extract address information featured on a Web page. Several kinds of aspects are candidates for extraction:

- town and street names,
- zip codes,
- dialing codes.

Location data are incorporated into various different hierarchies. Zip codes delineate areas in other ways than dialing codes and administrative units. It is important to align different knowledge organization systems, particularly when disambiguating homonymous names. For instance, there are several towns called *Kerpen* in Germany, but only one with the zip code 50170 and the dialing code 02273, respectively. When location data from different KOSs overlap, place recognition can increase in precision. For instance, when the place Kerpen is found on a Web page alongside the zip code 50170 (but no information is available about the street), it becomes clear that the address can only be in the district of Sindorf and not any of the other parts of Kerpen. The goal of location extraction is to get the most exact localization possible, preferably down to the street and address level.

For queries it must first be clarified whether there actually is a reference to location that would justify a ranking by distance. The occurrence of a place name in the query alone does not always point to an actual location. A query formulated “Ansel Adams Yosemite” may contain a place name (“Yosemite”), but the object of this search would have to be the photographer A. Adams and his book “Yosemite” (Gravano, Hatzivassiloglou, & Lichtenstein, 2003, 327). Ranking by distance would be completely unnecessary in this instance. To clear up this problem we can either offer decidedly location-oriented search engines (such as Google Maps) or outfit a general search tool with a proximity operator to be used in the query when searching for locations. Operators that point to a “local” instead of a distance-independent “global” search include “near”, “north of”, “only five minutes’ drive from” or “at most 25 km long” (Delboni, Borges, & Laendler, 2005, 63). All natural-language operators of this kind must be translated into search radii by the retrieval system. The next locus to be retrieved is the user’s current or hypothetical location. A hypothetical location is a point of reference from where the user wants to reach a destination, irrespective of his current location when searching. A typical example for such a query is “Hotel near Cologne Cathedral” by a user currently in New York City.

The user’s current location becomes important when he performs a search for services relative to this location. This might be a list of the nearest pizza parlors. To comply, the system must know the user’s location. This necessary data is either entered directly by the user (and stored for further use in any personalized location searches thenceforward), derived—in the case of queries via mobile end devices—by identifying the pertinent radio cell or (more precisely) calculated via a “global positioning system” (GPS) (Rao & Minakakis, 2003). In the latter case, geographical information retrieval (GIR) becomes Mobile Information Retrieval in a special variant.

Alignment between a location-oriented, “local” search and localized Web pages is performed via geographical knowledge organization systems coupled to geographi-

cal information systems (GIS). Riekert (2002, 587) emphasizes the interplay of these two aspects:

Spatial references can be specified in basically two ways: (1) textually, i.e., by indicating a geographic name, (2) geometrically, i.e., by specifying coordinates.

A KOS in which every concept comes with coordinates (latitude and longitude) is called a “gazetteer”. Every building, every administrative unit, every dialing code, every zip code, etc. are thus allocated their respective coordinates.

In order to perform a ranking by distance, both the coordinates of the user’s (hypothetical or actual) location and the coordinates of documents’ locations must be given. Alignment between the query and the documents is performed exclusively via the locations’ respective latitudes and longitudes. Sometimes the documents are ranked by the linear distance between locations (Watters & Amoudi, 2003, 144). This can be problematic, e.g. if there is a river between two locations and no (real) bridge on the (hypothetical) line that links them. Retrieval results will become more precise if at instances like these the system draws upon a route planner defining the exact distance between the two points.

Ubiquitous Retrieval

When a retrieval system uses other types of context-specific information besides geographical information (such as the time of day as well as certain real-time information), this is called ubiquitous retrieval. Where providers—mostly organized locally—offer region-specific information (e.g. local weather reports, opening hours of museums or restaurants, delay notices in public transportation), we often speak of a “ubiquitous city”. Examples are Oulu in Finland, as a small town (Gil-Castineira et al., 2011), and Seoul in South Korea, as a metropolis (Shin, 2009). Access to this information and to these services is provided either via mobile devices or at stationary access points such as the “media pillars” in Oulu and Seoul. Ubiquitous retrieval allows users to retrieve necessary information in a ubiquitous city, or it offers them the information in the form of a push service. Ubiquitous retrieval is always context-aware retrieval, with the context changing continuously. Kwon and Kim (2005, 167) write:

One of the most critical technologies in a variety of application services of ubiquitous computing is to supply adequate information or services depending on each context through context-awareness. The context is characterized by being continuously changed and defined as all information related to the entities such as users, space and objects [...]. Ubiquitous computing applications need to be context-aware, adaptive behavior based on information sensed from the physical and computational environment.

In a pull service, the user searches for certain context-specific information by himself: where is the next pizza parlor? What will the weather be like tonight? What's on at the theatre today? In Oulu the system uses maps and shelf plans to lead library users with mobile internet access to the exact location of the book they have found in the online catalog (Aittola, Parhi, Vieruaho, & Ojala, 2004). In the push variant, the retrieval system acts proactively and informs the user if the context should change (Brown & Jones, 2001). We must distinguish between two change variants:

- The context has changed for the user.
- The context in the database has changed.

The former case is given e.g. when the user has changed locations and another pizza parlor is now closer to him. In the second case, a new weather report may have been published or the theatre's ticketing system reports that only very few tickets for that night's performance are available.

Ranking in Social Media

Social media are (Linde & Stock, 2011, 260):

- Sharing services (e.g. YouTube for videos and Flickr for images),
- Social bookmarking services (e.g. Del.icio.us for all kinds of bookmarks and BibSonomy for scientific documents),
- Knowledge bases (wikis, weblogs and microblogs),
- Social networks (e.g. Facebook).

Each of these types of services requires different factors for ranking document lists. Here we will exemplarily name some bundles of criteria.

The sharing service Flickr introduced its “interestingness rating” for ranking search results. A Yahoo! patent application (Butterfield et al., 2006) provides a list of five general ranking criteria of “interestingness” for ranking hit lists: (a) the number of tags attached to a document, (b) the number of users who tagged the document, (c) the number of users who retrieved the document, (d) time (the older the document is, the less relevance it has) and (e) the relevance of metadata. Additionally, the patent application mentions two further ranking criteria for a “personalized interestingness rank”: (f) user preferences (e.g. designated favorites) and (g) the user's place of residence.

In the social bookmarking service BibSonomy, an algorithm called “FolkRank” is implemented in order to “focus the ranking around the topics defined in the preference vector” (Hotho, Jäschke, Schmitz, & Stumme, 2006, 419). The FolkRank tracks the idea of “super-posters” or “super-authors” who publish a huge amount of content and might thus be considered experts in a particular field. Accordingly, matching search results published by super-posters are ranked higher than others.

Since Twitter has neither a comprehensive retrieval system nor an elaborate ranking functionality at the moment, we will introduce an approach from the litera-

ture that discusses criteria for the ranking of hit lists in microblogs. Alhadi, Gottron, Kunegis and Naveed (2011) emphasize that otherwise customary weighting values such as TF or WDF are counterproductive given the brevity of the texts (at most 140 characters on Twitter). A tweet with only one word (which would be rather content-poor) would thus receive a high WDF value and rise to the top of the hit list in a search for this particular word. Alhadi et al. (2011) bank on other factors in their LiveTweet. Their model orients itself on the probability of a tweet's being retweeted. "We consider a tweet to be interesting—and therefore of good quality—if it is retweeted" (Alhabib et al., 2011). Accepted relevance criteria are:

- the availability of exclamation and question marks, URLs, usernames (presence of @) and hash-tags (#),
- the (historically observed) probability for all terms of a tweet that the tweet in question was retweeted,
- positive and negative terms or emoticons such as *great* or *bad*, but also: :-) or :-(,
- sentiments (where affective terms—derived from a list—occur in the tweet; Ch. G.7),
- retweeting probabilities of the author and his followers.

Additionally, they discuss the freshness of tweets and the author's position in the network of all other users as possible ranking criteria.

In social networks, the posts on the respective users' pages are given out in a certain order. A patent for Facebook (Zuckerberg et al., 2006) names both the preferences of the viewing user as well as the preferences of the subject user (i.e. the user who has entered the document) as ranking criteria. The EdgeRank used by Facebook currently takes three factors, or "edges" (all types of documents that are entered into Facebook, i.e. posts, images, videos etc.) into consideration:

- Affinity (the preferences of user and edge creator, as named in the patent),
- Weight (dependent on the edge type),
- Time Decay (freshness: the newer, the more relevant).

Ranking by Bid: Sponsored Links

A special kind of ranking takes its cue from the price offered by advertisers for clicks on an ad text. Serious search engines on the internet list such "sponsored links" separately from the editorial hit list, mostly at the top or on the right-hand side of the screen. In meta-search engines, which collate results from various different search tools, some originally "sponsored links" might be ranked among the editorial results (Nicholson et al., 2006). Otherwise, it can be assumed that the large Web search engines perform a strict separation of "organic" (i.e. algorithmically compiled) and "paid" hit lists.

The basic idea of "sponsored links" probably goes back to initiatives by the companies RealNames (Teare, Popp, & Ong, 1998) and GoTo.com (Davis et al. 1999). Over-

whelming commercial success is achieved by Google's variant ("AdWords") (Linde & Stock, 2011, 335-339). All systems share the common approach of allowing advertisers to buy, or bid on, search arguments. These are then yielded for appropriate queries, i.e. always in the appropriate context. The RealNames version works with a fixed price and search arguments are only allocated once, so that no ranking is possible. GoTo (later called Overture, now belonging to Yahoo!) allows several advertisers to buy one and the same search argument. The ranking is then built on the basis of the respective price per click each company has offered to pay (Davis et al., 1999):

The system and method of the present invention ... compares this bid amount with all other bid amounts for the same search term, and generates a rank value for all search listings having that search term. ... A higher bid by a network information provider will result in a higher rank value ...

Google refines the approach by using so-called "performance information" about the bid price when ranking the ad texts (Kamangar, Veach, & Koningstein, 2002). The "click-through rate" of the displayed ads is of importance here. This click rate is the relative frequency of clicks on the ad text, compared to all displays. The ranking position of the "sponsored links" in Google AdWords is calculated on the basis of the product of bid price per click, click rate and other factors. Whereas the ranking of ad texts in GoTo is a pure price ranking, Google has at least installed rudimentary aspects of a "Relevance" Ranking next to the price. Of course geographical information retrieval can also be used for ad texts, and location-dependent advertising can be programmed to appear only in a certain geographical window between searcher and advertised object (Yeh, Ramaswamy, & Qian, 2003).

Conclusion

- Retrieval systems in the WWW use various criteria to rank documents by relevance. Search engines use other ranking factors than social media. Automatized learning procedures (learning to rank) are used in hopes of finding suitable ideal ranking criteria or bundles of criteria.
- Web documents are mainly in the HTML format. Here the objective is to cleanly separate the document's content from the layout and the navigation elements. Only the content elements are required for text-statistical processing. Explicitly content-descriptive HTML tags (such as the title and the hierarchy of headings) as well as implicit tags in the text (e.g. font size) may provide clues to important text words that must, correspondingly, given a higher weight. Structural features of terms in documents (e.g. occurrence in a certain field, font size) determine the positional value of the term in the document and refine the TF*IDF calculation.
- The terms in anchor texts that link to a certain document are allocated to the linked document and weighted correspondingly to their position (internal or external link), their absolute frequency as well as the IDF value.
- Path length may designate a weighting factor. Here it holds that the higher up a page is in the folder hierarchy, the more important it is.

- Currentness, too, can be consulted as a weighting factor for Web documents. A “freshness” weighting makes particular sense for pages where time is an important dimension.
 - Apart from the formulated query, there are (implicitly “meant” but seldomly described) search goals, e.g. according to Broder navigation, information or transaction goals. Systems can preformulate all such general goals and offer them to the user to pick and choose.
 - In certain circumstances usage statistics of Web pages may be used as ranking criteria. The more often a page is clicked, the more important it may be. Instances of usage are gleaned either by observing the clicked links in the hit lists of a search engine (click history) or by evaluating all page views for users of a toolbar.
 - Sometimes short dwelling times on documents are taken as indicators for the respective document’s lack of relevance with regard to the query. Conversely, however, long dwelling times do not turn out to be reliable indicators of relevance.
 - User characteristics may reveal a user’s preferred languages. Web pages in such a preferred language may receive a higher retrieval status value than pages in other languages, thus rising to the top of the Relevance Ranking.
 - Geographical Information Retrieval (GIR) allows the ranking of documents with location-dependent content according to the content’s distance from the user’s location. GIR requires the recognition and extraction of addresses on Web pages, knowledge of the user’s (hypothetical or actual) location, as well as a “gazetteer”, which unifies geographical knowledge organizations systems (e.g. for administrative units, zip codes or dialing codes) with the respective coordinates (latitude and longitude).
 - A special form of GIR is Mobile Information Retrieval, which takes into consideration the user’s changing locations. Localization occurs either by identifying the radio cell in question or via a “global positioning system” (GPS).
 - Ubiquitous retrieval always leads from the context of a query (place and time), a context which may change continuously. Pull services allow for context-specific queries, whereas push services provide users with context-specific information.
 - Social media require their own respective bundles of ranking criteria: ranking via interestingness in sharing services, ranking of bookmarks via (among other methods) the bookmarking user’s status (in the FolkRank), ranking via the probability of being retweeted in microblogs (LiveTweet) as well as ranking by affinity, weight and time delay in the EdgeRank.
 - “Sponsored Links”, i.e. texts and links with advertising content, are ranked either exclusively via the offered price per click or via a combination of price and other factors (e.g. the ad link’s click-through rate).
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F.3 Personalized Retrieval

Simulation of Reference Interviews

Users have an information need when sampling Web search engines. This need is motivated by a “problematic situation” or an “anomalous state of knowledge”. These goals underlying the actual formulated queries are not generally made explicit. However, the “why?” behind a query is of fundamental importance for the success of a search. Rose and Levinson (2004, 13) discuss the problem via an example:

Searching is merely a means to an end—a way to satisfy an underlying goal that the user is trying to achieve. (By “underlying goal”, we mean how the user might answer the question “why are you performing that search?”) That goal may be choosing a suitable wedding present for a friend, learning which local colleges offer adult education courses in pottery, seeing if a favourite author’s new book has been released, or any number of other possibilities. In fact, in some cases the same query might be used to convey different goals—for example, the query “ceramics” might have been used in any of the three situations above.

A user wishing to retrieve information at the reference desk of his library will first be interviewed by the reference librarian, who needs to understand the user’s information need (Curry, 2005). There is no librarian in an internet search engine; the inquiry or research interview must thus be simulated. There are several options for performing user-oriented, “personalized” searches:

- Entering a fundamental query type,
- Creating a user profile,
- Saving and analyzing earlier queries made by a user,
- Saving and analyzing similar queries made by other users,
- Taking into account the point of presence or a preferred language of the user,
- Taking into account the location when dealing with distance-critical queries,
- as a special case of these: Localizing a mobile location.

The object of personalized information retrieval is not (objective) relevance but always (subjective) pertinence. Pertinence always depends upon the user, his tasks (Li & Belkin, 2008), goals, attitudes (Thatcher, 2008), interests etc. (Pitkow et al., 2002, 51). Personalized retrieval is a variant of “adaptive hypermedia” (Brusilovsky, 2001, 87):

Adaptive hypermedia is an alternative to the traditional “one-size-fits-all” approach in the development of hypermedia systems. Adaptive hypermedia systems build a model of the goals, preferences and knowledge of each individual user, and use this model throughout the interaction with the user, in order to adapt to the needs of that user.

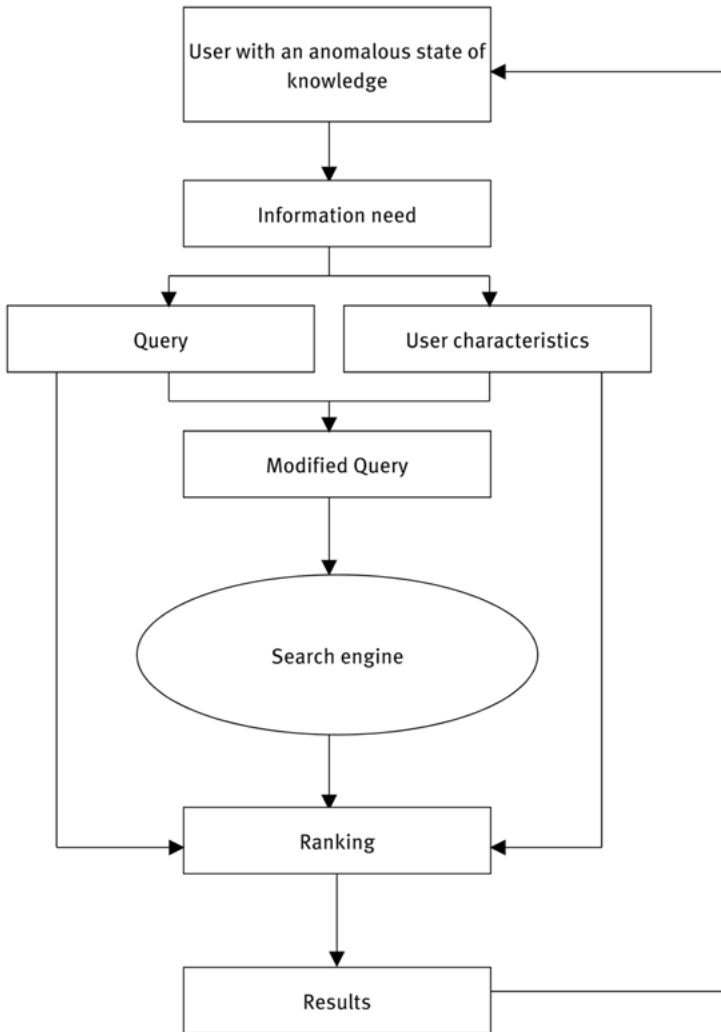


Figure F.3.1: User Characteristics in the Search Process.

We will summarize all user-specific aspects under the term “user characteristics”. User characteristics arise at two points in the search process: firstly, in the modification of the user-formulated query, and secondly, as an additional criterion of Relevance Ranking. It must further be safeguarded that the information (e.g. a location reference) necessary for accessing certain user characteristics in the first place is extracted from the Web pages (in the example: by sorting the pages with the corresponding regional reference by their distance to the user’s location).

User Characteristics

We will only speak of a “personalized search” when both the queries are modified in the image of a given profile and Relevance Ranking incorporates the profile as a weighting factor. The user profile can be gleaned just as easily via voluntary user disclosure as by observing his previous searches. Linden (2004, 1) writes:

Personalized search generates different search results to different users of the search engine based on their interests and past behavior.

In a patent for Ask Jeeves, Culliss (2003, 3) describes the manifold data that a comprehensive user profile displays:

Demographic data includes ... items such as age, gender, geographic location, country, city, state, zip code, income level, height, weight, race, creed, religion, sexual orientation, political orientation, country of origin, education level, criminal history, or health. Psychographic data is any data about attitudes, values, lifestyles, and opinions derived from demographic or other data about users.

Personal interest data includes items such as interests, hobbies, sports, profession or employment, areas of skill, areas of expert opinion, areas of deficiency, political orientation, or habits. Personal activity data includes data about past actions of the user, such as reading habits, viewing habits, searching habits, previous articles displayed or selected, previous search requests entered, previous or current site visits, previous key terms utilized within previous search requests, and time or date of any previous activity.

Gross, McGovern and Colwell (2005, 2) propose searching all files on the user’s computer in addition to register and store the central themes:

(T)he electronic agent ... performs an analysis of information contained in the user’s computer. ... Examples of the data analyzed include all system and non-system files such as ... machine configuration, e-mail, word processing documents, electronic spreadsheets, presentation and graphic package documents, instant messenger history and stored PDF documents. The agent analyzes the user’s data by scanning the words used in the documents ...

When searching from mobile end devices, information about the user’s current location is added via global positioning systems (GPS). In a patent application for Microsoft, Teevan, Dumais and Horwitz (2004, 3) emphasize the geographical aspect of personalization:

(The user model) can be sourced from a history or log of locations visited by a user over time, as monitored by devices such as the Global Positioning System (GPS). When monitoring with a GPS, raw spatial information can be converted into textual city names, and zip codes. ... Other factors include logging the time of day or day of week to determine locations and points of interest.

Culliss (2003), Gross et al. (2005) and Teevan et al. (2004) describe the “transparent user”, who may be provided with search results that match his profile perfectly, but who can also be ideally addressed via “suitable” advertising—not to mention the potential for abuse when person-related data are passed on to third parties.

The condition for personalized searching is that a specific user be identified, ideally via registration and a password. The user profile is made up of statements by the user himself, possibly a scan of his computer, his locations (determined via GPS) as well as his search history. During the automatic compilation of his search history, all queries are deposited in a personal historical database, alongside hit lists and clicked links (centrally). All events are time-stamped, which allows for the identification of searches that follow one another closely. Furthermore, it is worth considering whether the viewed documents should be analyzed text-statistically and the most important terms from them selected. The union of these terms represents an approximation of the thematic user profile. If a search engine provides a controlled vocabulary (i.e. from WordNet or a specialist KOS) with the corresponding relations, we will glean a user-specific semantic web from the thematic user profile (Pretschner & Gauch, 1999).

A specific personalized search is processed doubly: there is a “normal” search via the query terms on the one hand, and a comparative search on the basis of the user profile on the other. Gross, McGovern and Colwell (2005) compare the thematic user profile with the average word distributions of a language. Those user terms that are noticeably above average language use point to central interests and can be applied to the initial results in a second search. Homonyms occurring in the query are generally disambiguated in case the user has already performed frequent searches in the respective thematic environment. If the user has researched “Bali” at an earlier time and never displayed any interest in programming languages, his current query “Java” is probably also concerning an Indonesian island and not IT literature.

Personalized Ubiquitous Retrieval

Personalization can be linked with ubiquitous retrieval (Jones & Brown, 2004, 235 et seq.). Ubiquitous Retrieval (Ch. F.2) does not generally distinguish between different users. However, the information gleaned from a user profile can also be used in ubiquitous retrieval. The user profile thus becomes a fundamental point of reference in context-aware retrieval. E.g. if a user, having frequently researched Mozart’s “Magic Flute” in the recent past, arrives in a city where a theatre has scheduled a performance of this very opera that same night, a personalized ubiquitous retrieval system, as a push service, would be able to offer the user tickets for this event.

Conclusion

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- A query in a library begins with a reference interview in which the librarian seeks to know the user's information need and goals. This interview must be simulated in retrieval systems.
 - Besides the query, Personalized Information Retrieval always takes a user's specific characteristics into account—both when reformulating search arguments and during Relevance Ranking.
 - User characteristics are deposited in the retrieval system as a “user profile”. They can contain demographic and psychographic data, descriptions of personal interests and activities just as they may store past queries while allocating the user's clicked search results, or derive important themes by indexing the content of all files on the user's PC. The user's location in space is determined via user input (in case of a stationary end device) or via global positioning systems (for mobile end devices).
 - Personalized Ubiquitous Retrieval connects ubiquitous retrieval with further contexts, i.e. those aspects that have been gleaned from specific user profiles.
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F.4 Topic Detection and Tracking

Detecting and Tracking Current Events

In the World Wide Web, the Deep Web (e.g. information services by news agencies) as well as in other media (e.g. radio or television), there are documents regarding current events and whose content can frequently be found in various different sources (sometimes in slight variations). This is particularly true for news, but also for certain entries to weblogs (Zhou, Zhong, & Li, 2011), for texts in Bulletin Board Systems (Zhao & Xu, 2011) and for large amounts of e-mails organized by topic (Cselle, Albrecht, & Wattenhofer, 2007). In contrast to “normal” information retrieval, the search here does not start with a user’s recognized information need, but instead with a new event that must be detected and represented. The user is offered information about these events via specialized news systems, such as Google News, in the manner of a push service. It is as if a user had tasked an SDI service with keeping him up to date around the clock. Allan, who has fundamentally shaped this area of research, calls this domain of information retrieval “topic detection and tracking” (TDT). Allan (2002b, 139) defines this research area:

Topic Detection and Tracking (TDT) is a body of research and an evaluation paradigm that addresses event-based organization of broadcast news. The TDT evaluation tasks of tracking, cluster detection, and first story detection are each information filtering technology in the sense that they require that “yes or no” decisions be made on a stream of news stories before additional stories have arrived.

Google News restricts its offer to documents available in the WWW, which are offered for free by news agencies or the online versions of newspapers. It does not take into consideration the commercial offers of News Wires, most articles in the print versions of newspapers and magazines (which still make up the majority of all news) as well as all information transmitted non-digitally (via radio broadcasting). Google News places its focus on new articles and those that are of general interest. Bharat (2003, 9) reports:

Specifically, *freshness*—measurable from the age of articles, and *global editorial interest*—measurable from the number of original articles published worldwide on the subject, are used to infer the importance of the story at a given time. If a story is fresh and has caused considerable original reporting to be generated it is considered important. The final layout is determined based on additional factors such as (i) the fit between the story and the section being populated, (ii) the novelty of the story relative to other stories in the news, and (iii) the interest within the country, when a country specific edition is being generated.

Four fundamental concepts are of importance at this point:

- A “story” is a definable passage (or an entire document) in which an event is discussed.
- A “topic” is the description of an event in the respective stories, according to Fiscus and Doddington (2002, 18): “a seminal event or activity, along with directly related events and activities.”
- The arrangement of current stories into a *new* topic is “topic detection”.
- The adding of stories to a *known* topic is “topic tracking”.

Topic detection and tracking consists of several individual tasks (of which the first five follow Allan, 2002a):

- Story Segmentation: Isolating those stories that contain the respective topic (in documents discussing several events),
- New Event Detection: Identifying the first story that addresses a new event,
- Cluster Detection: Summarizing all stories that contain the same topic,
- Topic Tracking: Analysis of the ongoing news stream for known topics,
- Link Detection: Analysis tool for determining the topical similarity of two stories,
- Allocating a Title to a Cluster: Either the title of the first story or allocation of the first *n* terms, arranged by weight, from all stories belonging to the cluster,
- Extract: Writing a short summary of the topic (as a form of automatic extracting),
- Ranking the Stories: Where a cluster contains several stories, ranking them by importance.

An overview on the working steps is shown in Figure F.4.1.

Topic Detection

In the case of news from radio and television, the audio signals must first be translated into (written) text. This is accomplished either via intellectual transcription or by using speech recognition systems. In a news broadcast (e.g. a German “Tageschau” transmission running fifteen minutes), several singular stories are discussed, each counting as individual units via a segmentation of the overall text (Allan et al., 1998, 196 et seq.). Dealings with an agency’s news stream are analogous. To simplify, we can assume in this case that each news document contains exactly one story.

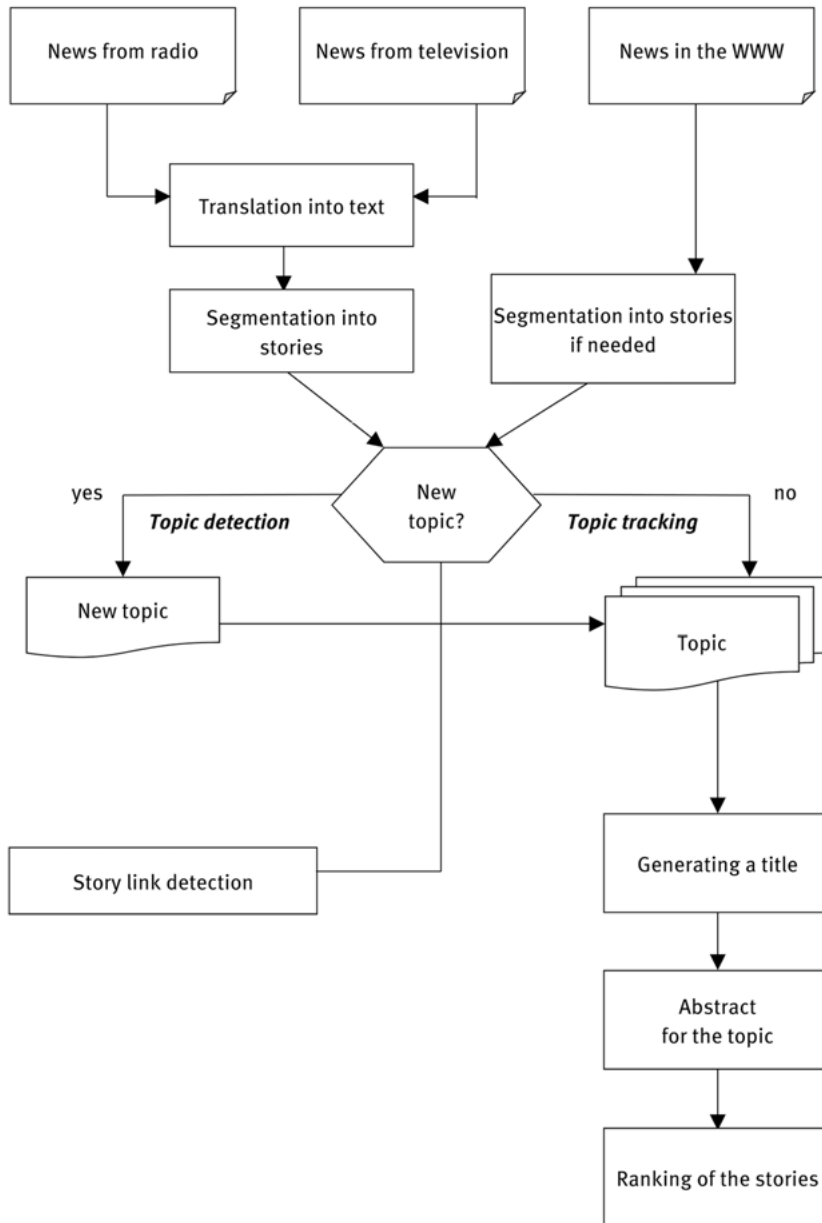


Figure F.4.1: Working Steps of Topic Detection and Tracking.

The decisive question in analyzing the news stream is: does the recently arrived story discuss a new topic, or does it address one that is already known? A (recognized) topic is expressed via the mean vector (centroid) of its stories. In *topic detection* we calcu-

late the similarity, or dissimilarity, between the current story and all others in the database. If no similarity can be detected (i.e. if a new topic is at hand), this first story is taken to be representative of the new topic. On the other hand, when similarities are observed between the current story and older ones, we are dealing with a case of *topic tracking*. What follows is a comparison between the current story and all previously known topics. A central role in topic detection and tracking is assumed by “story link detection”, whose algorithm finds out whether the current story is dealing with a new topic or a known one.

At this point, Allan et al. (2005) use the Vector Space Model and a variant of TF*IDF to determine the term weight. For every story from the news stream, a weighting value is calculated for every single term. $tf_{t,s}$ is the (absolute) frequency of occurrence of a term t in the story s , df_t counts all stories in the database that contain the term t and N is the number of stories in the database. The term weight w of t in s is calculated as follows:

$$w_{t,s} = [tf_{t,s} * \log((0,5 + N) / df_t)] / [\log(N + 1)].$$

Allan et al. suggest taking into consideration the first 1,000 terms, arranged by weight, for inclusion in a story vector. With the exception of a few long news texts, all words of a story should be acknowledged. The similarity between the story vector and all other story vectors in the database is analyzed by calculating the cosine. The authors have empirically determined a value of $\text{Sim}(s_1, s_2) = 0.21$, which separates new stories from old ones. If the highest similarity between the new story and a random old one is below 0.21, the current story will be registered as a new topic; if it is above that number, it will then be asked to what known topic the new story belongs.

In practice, it is shown that this general approach is not enough to separate new stories from old ones with any reliability. The performance of TDT systems improves via the addition of complementary factors.

When identifying a topic, the central roles are played by personal names (“named entities”) on the one hand and further words (“topic terms”) on the other. Two stories address the same topic if they frequently contain both “named entities” and the rest of the words in combination. Kumaran and Allan (2005, 123) justify this approach as follows:

The intuition behind using these features is that we believe every event is characterized by a set of people, places, organizations, etc. (named entities), and a set of terms that describe the event. While the former can be described as the *who*, *where*, and *when* aspects of an event, the latter relates to the *what* aspect. If two stories were on the same topic, they would share both named entities as well as topic terms. If they were on different, but similar, topics, then either named entities or topic terms will match but not both.

A representative example of a “mismatch” is shown in Figure F.4.2. In this case, the system has not made use of the distinction between “named entities” and “topic

terms”. Due to the high TF value of “Turkey” and “Turkish”, respectively, and the very high IDF value of “Ismet Sezgin”, the Vector Space Machine claims that the above story is similar to the below, and hence not new. In fact, the above text is new. Not a single “topic term” co-occurs in both reports, leaving Kumaran and Allan (2005, 124) to conclude:

Determining that the topic terms didn’t match would have helped the system to avoid this mistake.

It thus appears to make sense to calculate the similarity (cosine) between two stories separately for “named entities” and “topic terms”. Only when both similarity values exceed a threshold value will a story be classified as belonging to an “old” topic.

Topic not seen before

Turkey has sent 10,000 troops to its southern border with Syria amid growing tensions between the two neighbors, newspapers reported Thursday. Defense Minister **Ismet Sezgin** denied any troop movement along the border, but said **Turkey’s** patience was running out. **Turkey** accuses Syria of harboring Turkish Kurdish rebels fighting for autonomy in **Turkey’s** southeast; it says rebel leader Abdullah Ocalan lives in Damascus.

Closest Story due to Named Entities

A senior **Turkish** government official called Monday for closer military cooperation with neighboring Bulgaria. After talks with President Petar Stoyanov at the end of his four-day visit, Turkish Deputy Premier and National Defense Minister **Ismet Sezgin** expressed satisfaction with the progress of bilateral relations and the hope that Bulgarian-**Turkish** military cooperation will be promoted.

Figure F.4.2: The Role of “Named Entities” and “Topic Terms” in Identifying a New Topic. Source: Kumaran & Allan, 2005, 124.

A very important aspect of news is their relation to time and place. Makkonen, Ahonen-Myka and Salmenkivi (2004, 354 et seq.) work with “temporal” and “spatial similarity” in addition to “general similarity”. To determine the temporal relation, it is at first required to derive exact data from the statements in the text. Let us suppose a piece of news bears the date of May 27th, 2003. Phrasings in the text such as “last week”, “last Wednesday”, “next Thursday” etc. need to be translated into exact dates, i.e. “2003-05-19:2003-05-25”, “2003-05-21” and “2003-05-29”, respectively. Stories that are similar to each other, all things being equal, without overlapping in terms of their date, probably belong to different topics. Reports on Carnival processions in Cologne

for the years 2011 and 2012 hardly differ in regard to the terms that are used (“millions of visitors”, “float”, “candy” etc.), but they do bear different dates.

The spatial relation is pursued via a geographical concept system. The similarity between different spatial statements in two stories can be expressed via the path length between identified geographical concepts in a geographical KOS (Makkonen, Ahonen-Myka, & Salmenkivi, 2004, 357 et seq.). If one source talks about Puchheim, and another, thematically related source talks about the county Fürstentfeldbruck, both stories will be classified as similar to each other due to their path length of 1 (Puchheim is part of Fürstentfeldbruck county). Another pair of news reports likewise discusses similar topics, with one talking once more about Puchheim and the other about Venlo, in the Netherlands. Since the path length in this instance is, say, 8, nothing can lead us to suppose that they are discussing the same event.

Topic Tracking

When tracking topics, we assume the existence of a large number of known topics. The similarity calculation now proceeds by aligning the topic vectors, i.e. the topic’s respective name and topic centroids, with the new stories admitted into the database. In addition, comparisons can be made between temporal and spatial relations.

In the first story, the centroid is identical to the vector of this story. Only when there is at least a second story can we meaningfully speak of a “mean vector”. The centroid changes as long as further stories are identified for the topic. If the centroid is used to determine the title, e.g. by designating the first ten topics of the centroid, ranked by weight, as “title”, the title can indeed change as long as new stories are allocated to the topic.

Reports about events are written in different languages, given international interest. Tracking known topics beyond language borders becomes a task for multilingual topic tracking. Larkey et al. (2003) use automatic translation, which however does not lead to satisfactory results.

If a topic consists of several stories, these must be ranked. In a patent by Google, Curtiss, Bharat and Schmitt (2003, 3) pursue the path of developing quality criteria for the respective sources:

(T)he group of metrics may include the number of articles produced by the news source during a given time period, an average length of an article from the news source, the importance of coverage from the news source, a breaking news score, usage patterns, human opinions, circulation statistics, the size of the staff associated with the news source, the number of news bureaus associated with the news source, the number of original named entities the source news produces within a cluster of articles, the breadth of coverage, international diversity, writing style, and the like.

If a TDT system comprises all relevant sources, it appears obvious that one should grant the *first* story the “honor” of being prominently named in the top spot. The rest of the stories can be arranged according to the Google News criteria.

Conclusion

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- In topic detection and tracking, one analyzes the stream of news from WWW, Deep Web (particularly databases from news agencies and newspapers) as well as broadcasting (radio as well as television). The goals are (1) to identify new events and (2) to allocate stories to already known topics.
 - When discovering a new topic, one analyzes the similarity between a current story and all previously stored stories in the database. If there is no match, the story will be introduced as the first representative of a new topic.
 - If similarities with previously stored stories arise, a second step will calculate the similarity of the current story to known topics and allocate the story to a topic.
 - For the concrete calculation of similarities between stories, as well as between story and topic, TF*IDF as well as the Vector Space Model suggest themselves. A topic is represented via the centroid of all stories that discuss the respective event.
 - In news, “named entities” play a central role. It proves pertinent to select two vectors for every document, one for personal names and one for the “topic terms”. Only when both vectors display similarities to other stories can it be concluded that the current story belongs to a known topic.
 - In addition, it must be considered whether to draw on concrete temporal and spatial relations as discriminatory characteristics in stories.
 - If several stories are available for a topic, these will be ranked. The top spot should be occupied by that story which first reported on the event in question. Afterward, the ranking can be arranged via quality criteria of the sources.
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Part G

Special Problems of Information Retrieval



G.1 Social Networks and “Small Worlds”

Intertextuality and Networks

Documents do not stand in isolation, but always bear relation to other documents. With regard to textual documents, this is described as “intertextuality” (Rauter, 2006). In scientific, technical and legal texts intertextuality is expressed via quotations and citations, in Web documents via links. The texts’ authors are also situated in contexts, having possibly collaborated with other authors or seen their writings quoted by colleagues in their field. Likewise, the topics discussed in documents are of course also interconnected in networks. In this chapter we will attempt to derive criteria for Relevance Ranking from the position of a document, a subject or an author. The underlying idea is that a “prominent” text or author in the network, or a “prominent” subject, will be weighted more highly, all other things being equal, than a document, author or topic located at the network’s outskirts.

We have already dealt with aspects of networks in the context of link topology (Ch. F.1). In this chapter, we significantly expand our perspective by not restricting ourselves to only one type of relation (like the links in Ch. F.1), or to a select few indicators (such as hubs and authorities).

In the theory of social networks, we speak of “actors”; in the case of information retrieval, these actors are documents, authors (as well as any further actors derived from them, such as institutions) and topics. The actors are interconnected, in our case via formal links. The goal of network models is to work out the structure underlying the network. Wasserman and Faust (1994, 4) posit the following four basic principles of perspective in social networks:

Actors and their actions are viewed as interdependent rather than independent, autonomous units.

Relational ties (linkages) between actors are channels for transfer or “flow” of resources (either material or nonmaterial).

Network models focusing on individuals view the network structural environment as providing opportunities for or constraints on individual action.

Network models conceptualize structure (social, economic, political, and so forth) as lasting patterns of relations among actors.

An actor in a network is described as a “node” (NO), and the connection between two nodes as a “line” (LI). Each graph consists of nodes and lines. A graph is “directed” when all lines describe a direction. Directed graphs are information flows between citing and cited documents or between linking and linked Web pages (Thelwall, 2004, 214-215). A graph is “undirected” when the relation between two nodes expresses no direction. Undirected graphs of interest to information retrieval are “bibliographic coupling” and the “co-citation” of (formally citing) texts (Ch. M.2) as well as co-authorship. Building on information concerning the authors’ affiliations, we can

investigate the connections between institutes, enterprises, regions (i.e. cities) as well as countries. Where the topics (words or concepts) are regarded as actors, the co-words (e.g. those neighboring each other or co-occurring in a text window or in the title) or co-concepts (co-descriptors when using thesauri, co-notations when using a classification system) form the nodes of an undirected subject graph.

Table G.1.1: Examples for Graphs of Relevance for Information Retrieval.

Graph	Actor
Directed:	
References and Citations	Article
In-Links and Out-Links	Web Page
Undirected:	
Co-Authors	Author
Co-Words	Subject
Co-Concepts	Subject
Bibliographic coupling	Article
Co-citations	Article
Link-bibliographic coupling	Web Page
Co-links	Web Page

Nodes in a graph can be more or less firmly interlinked. The density (D) of a graph (G) is calculated by dividing the number of available lines in the graph (#LI) by the maximum amount of lines. If n nodes are in the graph, a maximum of $n * (n - 1) / 2$ lines can occur in an undirected graph and $n * (n - 1)$ lines in a directed graph. The graph's density is calculated as follows:

$$D(G\text{-undirected}) = \#LI / [n * (n - 1) / 2] \text{ and}$$

$$D(G\text{-directed}) = \#LI / [n * (n - 1)].$$

The density of a graph is between 0 (when there is no line) and 1 (when all nodes are interconnected). The diameter of a graph is the greatest path length within the graph. The average path length is the arithmetic mean of the path lengths between all nodes of the graph.

Three bundles of aspects and the parameters associated with them are of interest to information retrieval: the “prominence” of an actor in a network, an actor in a particularly visible position, and the emphasized role of an actor in a “small world” network.

Following the starting point of the observation, we distinguish between complete graphs and graphs that begin with a node (ego-centered graphs). In the complete graph the respective values are determined in independence of any search activity; in ego-centered graphs the network is only created on the basis of a user query, and

the parameters are only calculated for the hit list. Related to information retrieval, the parameters for a complete graph are calculated via the contents of the entire database (which can be extremely time-consuming). For the ego-centered or—more appropriately—“local” graph, only an initial search result (which may be reduced to a maximum amount of n documents) serves as the basis of parameter calculation. Yaltaghian and Chignell (2004) demonstrate on the basis of co-links that the (more easily calculable) procedure involving local graphs shows satisfactory results.

Actor Centrality

The prominence of an actor is described via his level of centrality (C). Wasserman and Faust (1994, 173) emphasize:

Prominent actors are those that are extensively involved in relationships with other actors. This involvement makes them more visible to the others.

Different measurements are used to calculate centrality. The simplest of these is the “degree” (CD), in which the numbers of lines in a node are counted and set in relation to all other nodes in the network. If m lines are outgoing from a node N and the entire (undirected) network has n nodes, the “degree” (CD) of N will be

$$CD(N) = m / (n - 1).$$

A heavily simplified variant foregoes the reference to network size and only works with the number m :

$$CD'(N) = m.$$

In Figure G.1.1, the actor C has four connections and the network comprises nine nodes in total. The “degree” CD of C is thus

$$CD(C) = 4 / (9 - 1) = 0.5 \text{ and} \\ CD'(C) = 4, \text{ respectively.}$$

In the case of a directed graph (all methods of calculation remaining equal), we must distinguish between the “in-degree” (number of connections that point to one node) and the “out-degree” (number of connections outgoing from one node). The Kleinberg Algorithm works with (weighted) “in-degrees” and “out-degrees” of Web pages, whereas the PageRank restricts itself to the (weighted) “in-degrees” of a Web page.

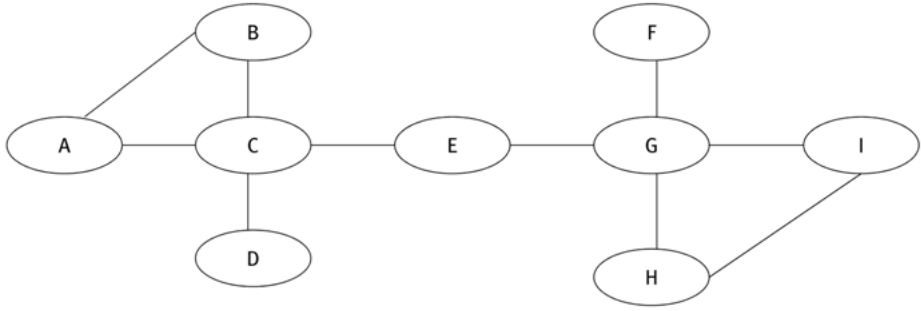


Figure G.1.1: Centrality Measurements in a Network. Source: Mutschke, 2004, 11.

A further centrality measurement is “closeness” (CC). This involves the proximity of an actor to all other actors in the network. To calculate the “closeness” of a node N we must count and add (in both the directed and the undirected scenario) the respectively shortest paths $PA_1, \dots, PA_i$ between N and all other nodes in the network. The “closeness” is the quotient of the number of nodes in the network minus 1 and the sum of the path lengths:

$$CC(N) = (n - 1) / (PA_1 + \dots + PA_i).$$

In our exemplary network in Figure G.1.1, the “closeness” for nodes A and E is:

$$\begin{aligned} CC(A) &= 8 / 21 = 0.38, \\ CC(E) &= 8 / 14 = 0.57. \end{aligned}$$

As a final centrality measurement, we will introduce “betweenness” in undirected graphs. Consider the paths between C and I in our example: there is no direct path, only the route via E and G. We can say, figuratively, that E and G “control” the path from C to I (Wasserman & Faust 1994, 188). The “betweenness” measures this “control”; the more “control” an actor exercises, the higher his “betweenness” will be. We now record all possible paths between any two actors (e.g. between A and all others) in the network as g_A . Then we record those connections between A and the rest that proceed via the node N as g_{nA} . The absolute “betweenness” values CB' for a node N are the sum of the quotients from g_{nA} and g_A , g_{nB} and g_B etc. for all nodes except N:

$$CB'(N) = (g_{nA} / g_A) + \dots + (g_{nB} / g_B) + \dots + (g_{nX} / g_X).$$

As the maximum number of paths in the (undirected) network (without N) is $(n - 1) * (n - 2) / 2$, we will normalize the “betweenness” via this value:

$$CB(N) = CB'(N) / [(n - 1) * (n - 2) / 2].$$

Node E in our example “controls” all paths from A, B, C, D on the one hand and from F, G, H, I on the other, but not the paths within the two sub-networks. Each node has seven goals in total, of which four can only be reached via E. We thus calculate $4/7$ for all paths; the sum for all nodes is $8 * 4/7$, i.e. 4.57. With due regard to the normalizing value $[(9 - 1) * (9 - 2) / 2] = 28$, we calculate a “betweenness” for E:

$$CB(E) = 4.57 / 28 = 0.163.$$

In large networks it may be of advantage to not take into consideration all nodes and paths but only those that cross certain threshold values. A method uses the minimum number of lines that must be outgoing from a node (“k-cores”; Wasserman & Faust, 1994, 266-267). If we put $k = 1$, the result will be the original network; at $k = 2$ all nodes with only a single line (i.e. with a “betweenness” of zero) will be excluded. As k rises, the “more important” actors in each case remain. When choosing a 2-core in Figure G.1.1, the nodes D and F drop out of the network since they only have one line each.

We can also use threshold values for the paths. Mutschke (2004, 15) suggests leaving only those lines in the graph that represent the m “best relationships”. The parameter m in the m -path Model determines a minimum rank for a node. Mutschke (2004, 15-16) describes this idea:

A 1-path network thus only features lines to that co-actor (or to those co-actors) of an observed actor that has the highest degree among all its co-actors. A 2-path network also “accepts” connections to co-actors with the second-highest degree etc. The m -value thus corresponds (in the case of differing degrees) to the maximum number of “best” co-actors that an actor in an m -path network is connected to. If two co-actors share the same degree value, both connections to these co-actors will be admitted to the network.

In the case of a 1-path threshold value, the lines between A and B as well as between H and I in Figure G.1.1 will fall away. Mutschke (2004, 16) hopes to use this method to separate a research area’s “centers of excellence” (in a co-author analysis).

Depending on the point of departure (complete graph or hit list) and any used threshold values, additional ranking criteria are gleaned via “degree”, “closeness” and “betweenness” (or a combination of all three measurements). In the case of co-authorship, this founds a ranking via author centrality (Mutschke, 2004, 34 and 36).

An experimental re-ranking of hit lists from the search engine Google via parameters of social networks (among them “degree”, “closeness” and “betweenness”), applied to the link structures of Web documents, shows a significant improvement to the retrieval quality in nearly all cases (Yaltaghian & Chignell, 2002).

Degree of Authors

If our goal is to determine the centrality of an author in a network of scientists, it appears sensible to start by searching his name in thematically relevant databases. According to White (2001, 620) we must distinguish between four different aspects:

- the author's citation identity: all scholars cited by the author in his writings,
- the citation image-makers: all scholars who cite the author and his writings, respectively,
- the citation image: all scholars who are co-cited with the author,
- the co-authors: all scholars who have co-published with the author at least once.

The result is thus four ego-centered graphs, two of them undirected (citation image and co-authors) and two directed (citation identity and citation image-makers). The numbers of scholars in the undirected graphs each form the "degree" of the ego. In the directed graphs, the object is either to transmit information (in the direction cited document $\rightarrow$ citing document) or, conversely, to award reputation (in the direction citing document $\rightarrow$ cited document). We will examine the variant involving reputation. Here we are interested in the number of "out-degrees" for citation identity, i.e. the number of different authors cited by the ego in his writings. For the "image-makers", on the other hand, we measure the number of "in-degrees", i.e. the number of authors who cite the ego at least once in their writings, thus awarding him reputation via the corresponding reference.

An easily performable method of measuring the "degree" of (formally citing) documents is to use the "times cited" ranking option in citation databases such as Web of Science or Scopus. "Times cited" yields the simple "degree" (CD') of a text on the basis of its citations.

One parameter derived from the number of an author's publications and citations is the h-index by Hirsch (2005). If the number of an ego's publications is N_p and if the number of their citations is known to us, h will be the number of articles that have been cited at least h times. Hirsch (2005, 16569) defines:

A scientist has index h if h of his or her N_p papers have at least h citations each and the other ($N_p - h$) papers have $\leq h$ citations each.

If an ego has written 100 articles, for instance, 20 of these having been cited at least 20 times and the other 80 less than that, then the ego's h-index will be 20. A parameter competitor to the h-index is the average number of citations per publication. This parameter, however, discriminates against highly productive authors, rewarding low output. The h-index has the methodical disadvantage, though, that it is hardly possible to compare scientists with different research ages (time period after their first publication). Apart from this flaw, the h-index is a measurement of a scientist's influence in the social network of his colleagues (Hirsch 2005, 16569):

Thus, I argue that two individuals with similar h s are comparable in terms of their overall scientific impact, even if their total number of papers or their total number of citations is very different. Conversely, comparing two individuals (of the same scientific age) with a similar number of total papers or of total citation count and very different h values, the one with the higher h is likely to be the more accomplished scientist.

Hirsch (2005, 16571) derives the m -index with research age in mind. Let the number of years after a scientist's first publication be t_p . The m -index is the quotient of h -index and research age:

$$m_p = h_p / t_p.$$

An m -value of 1 would mean, for instance, that a scientist has reached an h -value of 10 after 10 research years. Both the h -index and the m -index should be suitable criteria for Relevance Ranking in scientific databases. The retrieval status value of a document will thus be weighted with the corresponding h - or m -values of its author. For documents with multiple authors, it is worth considering whether to follow the h - or m -value of its most influential author. No experimental findings concerning the effects of the named values on the quality of retrieval results are available yet.

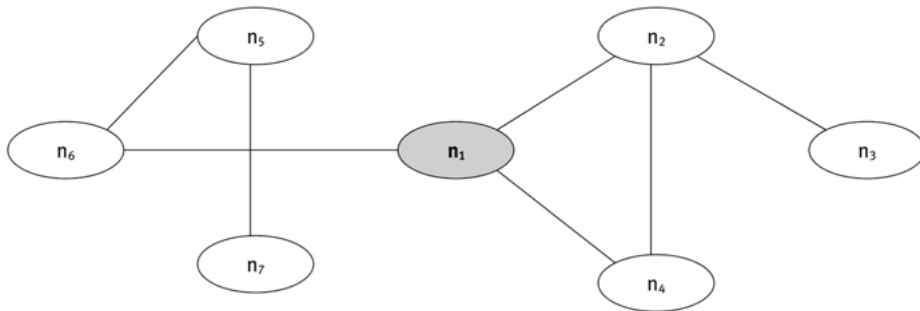


Figure G.1.2: Cutpoint in a Graph. Source: Wasserman & Faust, 1994, 113.

Cutpoints and Bridges

Web pages, (citing) documents, authors and topics (words as well as concepts) can take particularly prominent positions in the network. An actor is a “cutpoint” when it (and only it) connects two otherwise disjointed sub-graphs with each other. In Figure G.1.2, the node n_1 is in such a cutpoint position, since no other node connects the sub-graphs n_5, n_6, n_7 with n_2, n_3, n_4 . If the cutpoint n_1 is removed, the graph will break into two pieces.

Similar to the cutpoint is the “bridge”, only here it is not a node that is being observed but a line. In Figure G.1.3, the line between n_2 and n_3 is a bridge. If the bridge is removed, the sub-graphs will no longer be connected.

Cutpoints and bridges can serve as “brokers” between thematic environments. Searches for authors, topics or documents in their position as “cutpoints” or “bridges” will yield those actors which are equally at home in different areas and which are particularly useful for an interdisciplinary entry into a subject area.

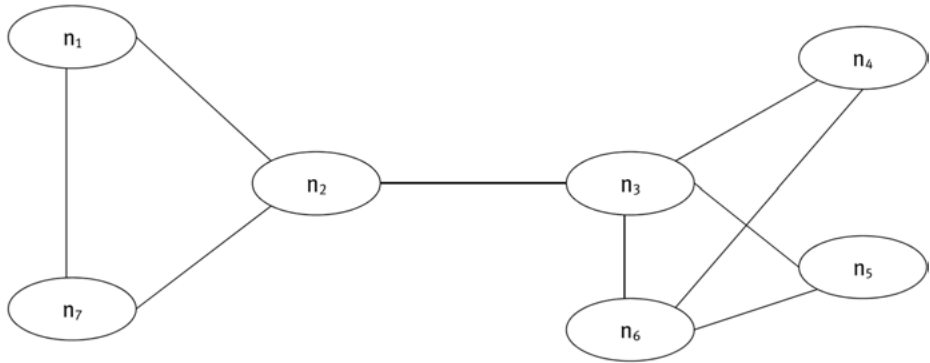


Figure G.1.3: Bridge in a Graph. Source: Wasserman & Faust, 1994, 114.

“Small World” Networks

In an experiment by Milgram (1967) that has since become a classic, the investigator asks some randomly selected persons in Nebraska and Kansas to redirect a letter so that it will reach a target individual in Boston, unknown to them, and whose address is also unknown. Some of these letters do indeed reach their destination. The number of persons featured in the transmission chain is around six in each case. Apparently, paths in social networks are short. In scientific as well as in popular contexts, the concept of “six degrees of separation” becomes a synonym for “small worlds”. Examples include the “Erdős Numbers” in mathematics, which express the proximity between any given mathematician and the prominent Hungarian mathematician Erdős. All of Erdős’ co-authors have an Erdős Number of 1, all of these co-authors’ co-authors, who themselves have not published together with Erdős, get an Erdős Number of 2, etc. An example in popular culture is the network of film actors. In the game “Six Degrees of Kevin Bacon”, the objective is to find the shortest path between the actor Kevin Bacon and any other actor. Here one searches for collaborations in movies: all actors who have performed together with Bacon are one line apart, those who only performed with these former are two lines apart, etc. As the network of

movie actors is a particularly small “small world”, there are hardly any path lengths exceeding four.

Watts and Strogatz (1998) explain the makeup of a “small world”. On a scale between a completely ordered network (Figure G.1.4, left-hand side) and a network whose paths are randomly determined (right), the “small worlds” lie in the middle. “Small worlds” have two prominent characteristics:

- their graph density is high,
- the graph’s diameter is small.

Watts and Strogatz (1998, 441) emphasize:

One of our main results is that ... the graph is a small-world network: highly clustered like a regular graph, yet with small characteristic path length, like a random graph.

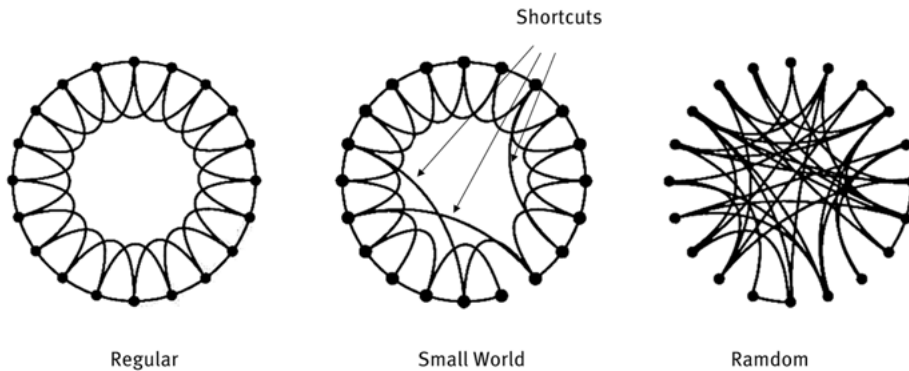


Figure G.1.4: “Small World” Network. Source: Watts & Strogatz, 1998, 441.

Kleinberg (2000, 845) points out that the diameter of a “small world” is exponentially smaller than its size:

A characteristic feature of small-world networks is that their diameter is exponentially smaller than their size, being bounded by a polynomial in $\log N$, where N is the number of nodes. In other words, there is always a very short path between any two nodes.

In a “small world” network there are necessarily “shortcuts” between actors that are far apart. Such shortcuts or “transversals” are necessary to establish these short paths.

We know that the WWW, or parts of it, represents “small worlds” (Adamic, 1999). Björneborn and Ingwersen (2001, 74) describe the “small Web world”:

Web clusters consist of closely interlinked web pages and web sites, reflecting cognate subject domains and interest communities. ... A human or digital agent exploring the Web by following

links from web page to web page has the possibility to move from one web cluster to another “distant” cluster using a single transversal link as a short cut.

“Small Web Worlds” thus contain (more or less) self-contained sub-graphs (“strongly connected components”, SCC) that are connected among one another via “shortcuts”. The “degree” measurement is useful for calculating the prominence of an actor in such an SCC subcluster. For the eminently important “shortcuts”, on the other hand, the “degree” is completely unsuitable. The shortcuts do not necessarily lead from the actors with high “degrees”, but (also) from actors on the periphery. In this context Granovetter (1973) speaks of “weak ties”. It is the strength of these weak ties that accounts for the “small worlds”.

How can we use these insights into “small worlds” in information retrieval? We already know that the Web consists of small worlds. That other documents (scientific articles, patents, court rulings, newspaper articles etc.) are likewise located in small worlds can only be assumed. Research into this subject is in its infancy.

An initial area of application for “small worlds” is in determining and displaying entire sub-graphs, i.e. the strongly interconnected network components SCC (Adamic, 1999, 447-448). The initial hit list of a query is analyzed for SCC via the respective values for graph density. The isolated sub-graphs can be ranked either via the number of actors (this is Adamic’s suggestion) or via the graphs’ density. In one scenario the largest graph is at the top of the output list, in the other it is the graph whose actors are most strongly interconnected. The user chooses a graph and the system will display the individual actors according to their prominence. If the actors are Web pages (Khazri, Tmar, & Abid, 2010) this procedure will be analogous to the determination of communities in the Kleinberg Algorithm. If subjects (words or concepts) are used as actors (Chee & Schatz, 2007), on the other hand, the user will first gain an overview of the different thematic complexes that touch upon his query. The case is similar for authors, except that here the user will see author networks. In the case of a citation database we calculate the “strongly connected components” via direct citations, co-citations or bibliographic coupling. The result will be clusters in which the documents are interconnected via strong formal citation relations. Depending on what has been collected in a database (weblinks, subjects, authors, citations), the user can be offered several options for determining the SCC. This procedure always requires some form of system-user interaction; if the system offers several structuring options, an experienced user will profit far more than a research layman.

Conclusion

- Documents do not stand in isolation but make reference to one another in the sense of intertextuality. In so far as the documents themselves as well as their authors and the subjects they discuss can be located in networks, it is possible to derive novel retrieval options from their positions in the network.
 - In directed graphs, actors and lines that are important for information retrieval are references and citations as well as in-links and out-links, respectively. In undirected graphs they are co-authorship, co-subjects (co-words as well as co-concepts), co-citations (co-links) and bibliographic coupling (link-bibliographic coupling).
 - Graphs as a whole are described via their density and their diameter.
 - The prominence of an actor in a network is its level of centrality. This can be expressed quantitatively via “degree”, “closeness” and “betweenness”. All three parameters (or a combination of them) are suitable as factors in Relevance Ranking.
 - The degree of scientific authors is determined via four aspects: citation identity, citation image-makers, citation image and co-authors.
 - The h-index sets the number of an author’s publications in relation to the amount of times his work has been cited. A scientist has an index h when h of his publications have been cited at least h times. The m-index additionally takes into consideration the author’s research age.
 - Actors and lines with a particularly prominent position in the network are cutpoints and bridges. They serve to interlink otherwise unconnected sub-graphs with one another. They are particularly suited as entry points to interdisciplinary problems.
 - “Small worlds” are networks with high graph density and small diameters. They contain “strongly connected components” (SCC) as well as “shortcuts” between these SCC. Information retrieval of “small worlds” allows one to identify the SCC and grants the user an initial overview of the search topic. In a second step, the central actors within the SCC are searched.
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G.2 Visual Retrieval Tools

Lists of search terms are generally ranked alphabetically—particularly in Deep Web information services. Displays of documentary units (DU) are normally arranged in descending chronological order (in Deep Web information services) or in descending order of their relevance to the search argument (in Web search engines). However, there are alternatives to the usual list ranking that attempt to process search tools and results visually. The most prominent example of visual retrieval tools are probably tag clouds, which are offered by many Web 2.0 services.

Visual Search Tools

Generally, tag clouds (Peters, 2009, 314-332) are alphabetically ranked representations of tags in services that work with folksonomies (Ch. K.1 – K.3). In principle, all sorts of terms (keywords from a nomenclature, descriptors from a thesaurus, notations from a classification system, concepts from an ontology, text words, citations, etc.) can be visualized in this way. In general usage, we speak of “term clouds”. Term clouds are used for different sets of documents:

- entire information services (representation of the most important terms in the whole database),
- results (representation of the most important or—in the case of very few hits—all terms in a current set of search results),
- documents (representation of the most important or—in the case of very few terms—all terms of a specific document).

The frequency of a term in each respective document set is visualized via font size: the more frequent the term, the larger its font. Empirical studies show that users remember terms with larger fonts more easily than those with smaller fonts (Rivadeneira, Gruen, Muller, & Millen, 2007, 997). Term size is calculated via the following formula (Sinclair & Cardew-Hall, 2008, 19):

$$\text{TermSize}(t_i) = 1 + C * [\log(f_i - f_{\min} + 1)] / [\log(f_{\max} - f_{\min} + 1)].$$

f_i is the frequency of term i 's occurrence in the data set, $f_{\min}$ is the minimum frequency in the top n terms (Sinclair and Cardew-Hall worked with $n = 70$), $f_{\max}$ is the maximum frequency in the top n terms, and C is a scaling constant to determine the maximum text size.

Since a term cloud is built, in most cases, on a selection of terms, targeted searches are not possible. On the example of tag clouds, Hearst and Rosner (2008) report:

(I)t seems that the main value of this visualization is as a signal or marker of individual and social interactions with the contents of an information collection, and functions more as a suggestive device than as a precise depiction of the underlying phenomenon.

Peters (2009, 314) emphasizes the function of tag clouds as a browsing tool:

Folksonomies' ability to support browsing through information platforms rather than specific searches via search masks and queries is often seen as one of their great advantages. ... In the visualization of folksonomies, it is important that the user gets an impression of the information platform in its entirety and find entry points for this browsing activities.

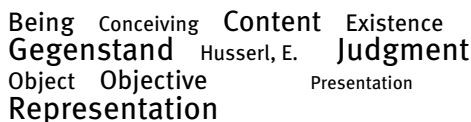
As a “visual summary” (Sinclair & Cardew-Hall, 2008, 26), term clouds are useful for browsing through a document set's term material. Sinclair and Cardew-Hall (2008, 27) name three salient characteristics of a term cloud:

It is particularly useful for browsing or non-specific information discovery. ...

The tag cloud provides a visual summary of the contents of the database. ...

It appears that scanning the tag cloud requires less cognitive load than formulating specific query terms. That is, scanning and clicking is ‘easier’ than thinking about what query terms will produce the best result and typing them into the search box.

Figure G.2.1 shows a term cloud that represents the term material on the subject of “Gegenstand” in the primary literature of Alexius Meinong (the method of knowledge representation used is the Text-Word Method, Ch. M.1).



Being Conceiving Content Existence
Gegenstand Husserl, E. Judgment
Object Objective Presentation
Representation

Figure G.2.1: Term Cloud. Example: Database Graz School (Stock & Stock, 1990). Meinong. Primary Literature on the Subject *Gegenstand*.

A term cloud shows no relations between terms. This is achieved via the construction of term clusters (Knautz, 2008). Semantic networks can be woven out of terms that are connected via syntagmatic relations (i.e. that occur in the same document). Here it is of no consequence which method of knowledge representation has been used to compile the terms. In the case of folksonomies, a tag cloud becomes a semantic tag cluster. Where descriptors from a thesaurus are used, we are looking at a “statistical thesaurus” (Stock, 2000; Knautz, 2008). A particularity of these networks is that the user can influence their resolution, or granularity, via their settings.

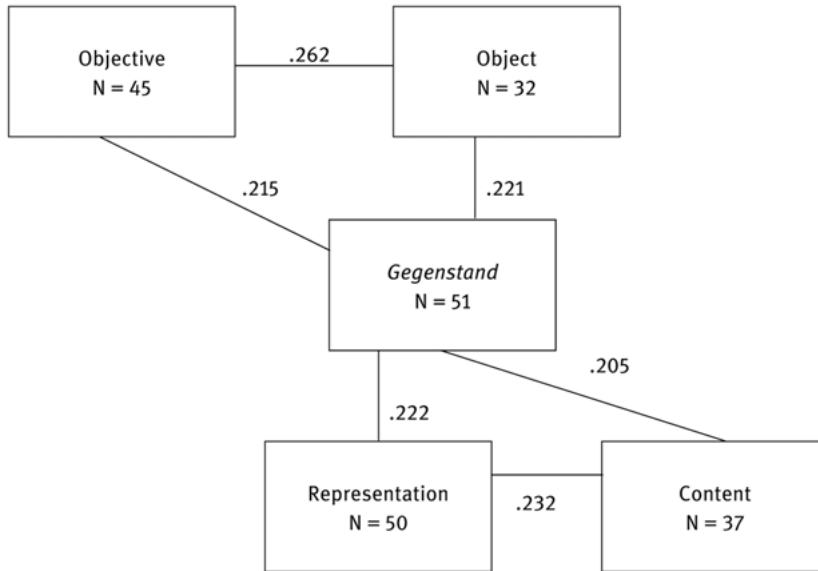


Figure G.2.2: Statistical KOS—Low Resolution. Example: Database Graz School; Search Argument: Author: Meinong, A. Cluster around *Gegenstand*; Pre-Defined Subject Similarity: $SIM > 0.200$.

We calculate the similarity SIM between two subjects via one of the traditional methods. In the following example, the Jaccard-Sneath variant has been used. The user has the option of changing the SIM value, e.g. via a scroll bar. In our example, we use the Graz School database, in which the semantic thematic networks are represented graphically (Stock & Stock, 1990). Suppose a user is interested in the writings of Alexius Meinong. Let the set of search results be $N = 217$, and yielded in the form of a network where the terms represent the nodes and the similarity between terms is expressed by the lines. Additionally, the system notes every term's frequency of occurrence in the search results. The number above the lines shows the degree of similarity according to Jaccard-Sneath. An alternative would be to use stroke width to visualize the degree of similarity (Knautz, Soubusta, & Stock, 2010). Figures G.2.2 and G.2.3 show clusters starting from the subject of *Gegenstand* ('object' in the sense of Meinong's object theory).

In Figure G.2.2, the user has set the scroll bar to $SIM > 0.200$. All subjects having a similarity with the search topic greater than 0.2 are thus displayed. Likewise, the connections between subjects are only displayed if they exceed the threshold value. Since a similarity value of greater than 0.2 is very large in the context of this set of results, the user only sees the basic structure of the network. Figure G.2.3 shows the network's aspect after the user lowers the threshold value to $SIM > 0.110$. The semantic network gets richer, but potentially also more cluttered. The user has three options for continuing his search:

- Clicking on a node in order to view the documents available for it. (A click on *Gegenstand* thus leads to 51 texts by Meinong).
- Clicking on a line in order to view those documents that discuss both terms on the nodes together.
- Clicking on an excerpt from the current network not in order to be displayed any documents, but to see a more specific semantic network on the basis of the selected terms.

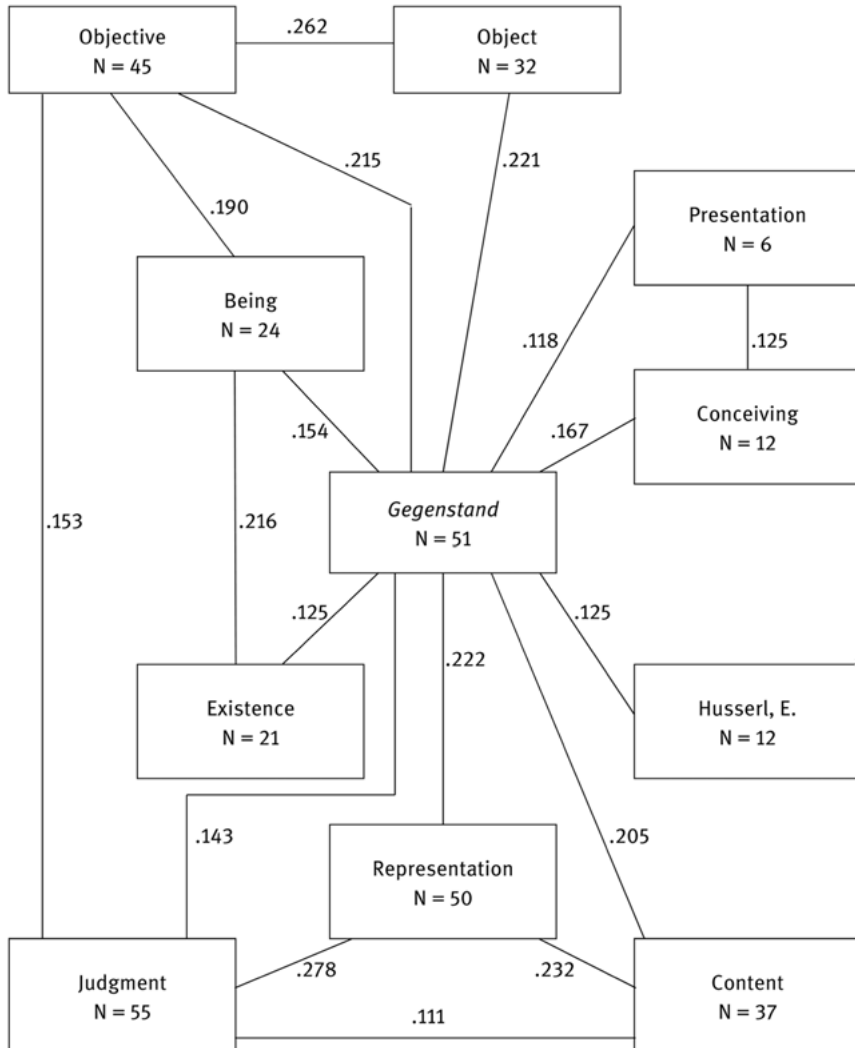


Figure G.2.3: Statistical KOS—High Resolution. Example: Database Graz School; Search Argument: Author: Meinong, A. Cluster around *Gegenstand*; Pre-Defined Subject Similarity: SIM > 0.110. Source: Stock, 2000, 33.

The arrangement of the terms in the cluster is not random. Specific software is required to place the subclusters as close to each other as possible while keeping semantically unrelated parts far apart (Kaser & Lemire, 2007).

In an empirical study that uses the tags of a folksonomy to form clusters, tag clusters yielded far better evaluation results than tag clouds did (Knautz, Soubusta, & Stock, 2010, 7). Advantages of tag clusters include (Knautz, Soubusta, & Stock, 2010, 8):

- clustering offers a more coherent visual distribution than alphabetical arrangements; ...
- tag clusters offer the possibility of visualizing even large result sets after an initial search. In this matter the user gains an additional thematic overview of the content;
- the steep structure of tag clouds is dissolved so that users and providers are able to actively interact with the visualization;
- by using tag clusters users are able to adapt the result set to their query. Thus users can independently generate small subsets of all documents relevant for their information need.

It should be possible to generalize results obtained via folksonomies for all manner of terms (controlled terms, classification notations, etc.).

Visualization of Search Results

The “classic” search engine results page (as in Google, for instance) is the list, which possesses a series of advantages (Treharne & Powers, 2009, 633):

- A rank-ordered list has several advantages. Its format is lean, ubiquitous and scalable; consistent, simple and intrinsic; and user and task inclusive.

The challenge in visualizing the representation of search results is to make the visualization easier to understand for the user than the well-established list form. Sebrechts et al. (1999, 9) emphasize:

- The utility of visualization techniques derives in large part from their ability to reduce mental workload.

Possible forms of visualization include graphs or sets. In set-theoretical representation (Figure G.2.4.a), the documents that make up the search results are clustered into hierarchically ranked classes via methods of quasi-classification (Ch. N.2). When search results are presented in graphs (Figure G.2.4.b), the connections between documents must first be analyzed. For Web documents, this can be achieved by analyzing the links between items. Furthermore, anchor texts can be evaluated in order to label the lines.

Up to date, search engines on the WWW that used search result visualization (e.g. Kart00, which was discontinued in 2010) have not succeeded on the market.

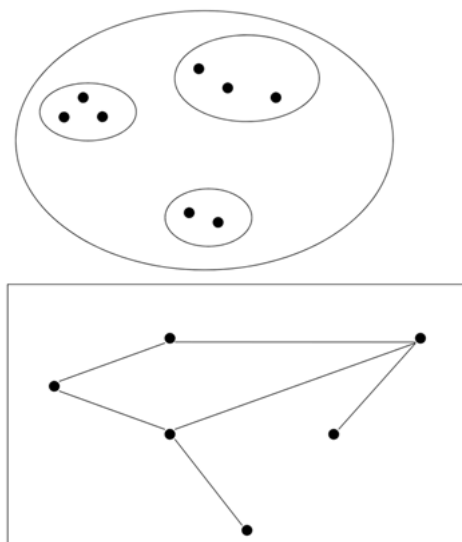


Figure G.2.4: Visualization of Search Results. a) Above: as Sets, b) Below: as Graphs (Undirected in this Case).

Visual Display of Informetric Results

Informetric searches (Ch. H.2) do not seek single documents. Instead, they condense retrieved document sets into new information. These sets might be all writings by a certain scientist, the patents of an enterprise, or all publications from a given geographical region. Selecting the respective document set is the prerogative of the user. A last step in informetric analyses involves visualizing the results (White & McCain, 1997). “Visualization refers to the design of the visual appearance of data objects and their relationships” (Börner, Chen, & Boyack, 2003, 209). Following an informetric analysis of scientific literature, the visualization represents science “maps” or an “atlas of science” (Börner, 2010). The visualization can be static (as in our graphics) or interactive. According to Börner, Chen and Boyack (2003, 210),

(i)interaction design refers to the implementation of techniques such as filtering, panning, zooming, and distortion to efficiently search and browse large information spaces.

Börner, Chen and Boyack (2003, 238) believe that knowledge visualization can help assess “scientific frontiers, forecast research vitality, identify disruptive events/technologies/changes, and find knowledge carriers.”

Our example of a science map comes from Haustein (2012). It concerns a scientific journal (vol. 2008 of the *European Journal of Physics B*), which is visualized as a map in connection with its most-cited journals and their *Web of Science* subject categories

(Haustein, 2012, 45). The underlying document set contains all articles of this journal that were published in the year 2008. All references in the document set were surveyed. In order to be displayed, a cited journal needed at least 145 references. Additionally, all journals named in the map were allocated the subject categories under which they are filed in *Web of Science* (Figure G.2.5).

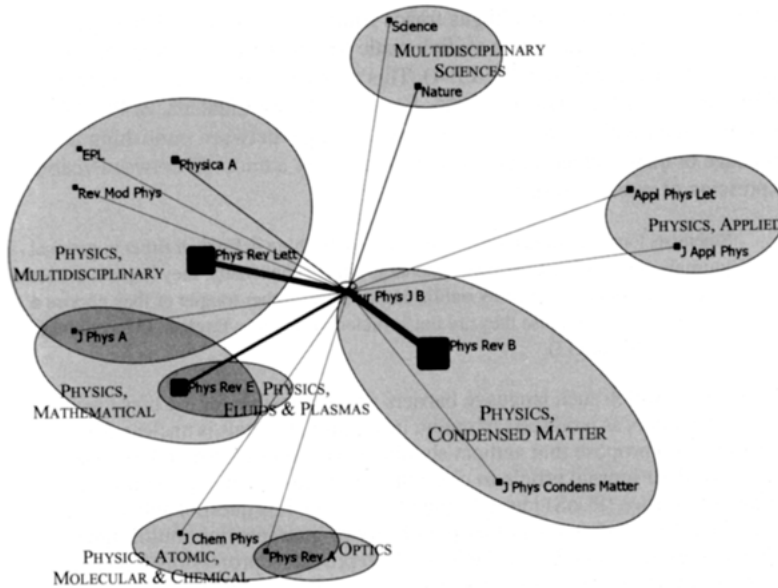


Figure G.2.5: Mapping Informetric Results: Ego Network of “The European Physical Journal B” with its Most Frequently Cited Journals and *Web of Science* Subject Categories in 2008. Source: Haustein, 2012, 45.

A further visualization variant does not use imaginary science maps, but ties information to geographical locations via actual maps. The results of informetric analyses are here combined with information services for maps, in the sense of mash-ups. Our example is taken from a work by Leydesdorff and Bornmann (2012). The informetric analyses were performed at the patent database of the U.S. Patent and Trademark Office (USPTO), and the map taken from Google Maps. In Figure G.2.6, the marks indicate all locations in the northern Netherlands from where inventors submitted their patents to the USPTO in the year 2007. Leydesdorff’s tool (at www.leydesdorff.net) is interactive and allows zooming in and out on the map.

In our third example, we leave the sphere of science and technology and turn to user-generated content in the form of photos. The biggest photosharing service on the Web is Flickr. It allows both geotagging (i.e. marking the location of the photograph on a map) and the adoption of GPS and chronological information stored on

the camera. If this information is available, one can reconstruct the time and place that a photograph was taken. Manifold options for informetric analyses arise for user-generated content. Crandall, Backstrom, Huttenlocher and Kleinberg (2009, 767) use images from Flickr to mark typical tourist paths:

Geotagged and timestamped photos on Flickr create something like the output of a rudimentary GPS tracking device: every time a photo is taken, we have an observation of where a particular person is at a particular moment of time. By aggregating this data together over many people, we can reconstruct the typical pathways that people take as they move around a geospatial region.

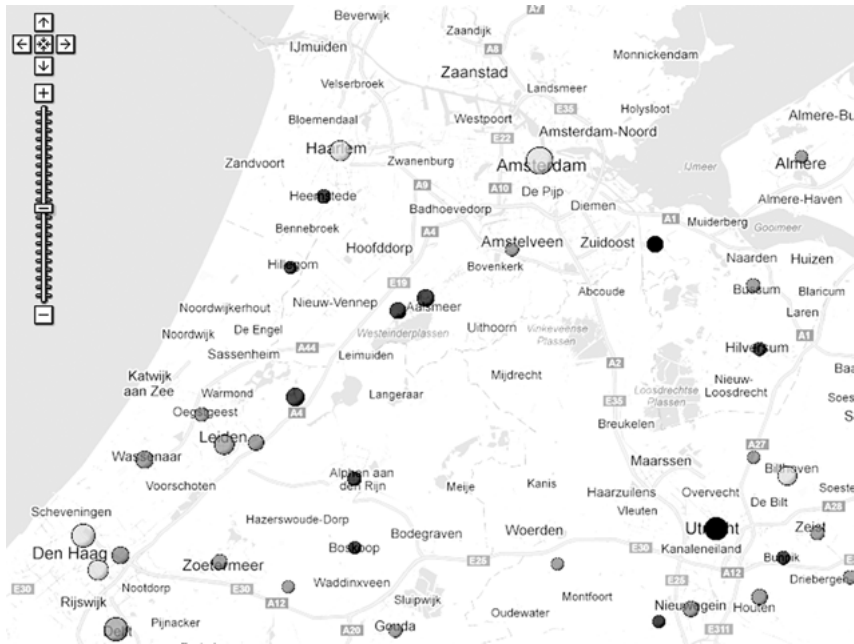


Figure G.2.6: Mash-Up of Informetric Results and Maps: Patents of Inventors with a Dutch Address in 2007. Source: Leydesdorff & Bornmann, 2012, and www.leydesdorff.net.

Figure G.2.7 shows the results for Manhattan. The image stems from the connection between locations where photographers have taken pictures within half an hour. Some obvious touristic focal points are established: the area south of Central Park as well as the financial center in the south of Manhattan. Brooklyn Bridge, too, is clearly recognizable.



Figure G.2.7: Visualization of Photographer Movement in New York City. Source: Crandall, Bachstrom, Huttenlocher, & Kleinberg, 2009, 768 (Source of the Photos: Flickr).

Conclusion

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- In current information services, the display of search terms and results is dominated by lists. Some alternatives attempt to drop the list form in favor of visualization.
 - Search terms can be visually processed (mostly in alphabetical order) in term clouds, with the terms' font size showing their frequency. The most prominent examples are the tag clouds in folksonomies.
 - Semantic term clusters can be derived from the syntagmatic relations of the terms. Here it becomes possible for users to vary the granularity of the semantic network by changing the degree of similarity.
 - The visualization of search results is achieved, for instance, via their representation as graphs or sets.
 - A broad field is the visualization of the results of informetric analyses. Here document sets are described in their entirety. Visualizations of scientific and technical information lead to science maps or atlases of science. A combination of informetric results and map services lends itself to the visualization of information anchored geographically.
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G.3 Cross-Language Information Retrieval

Translating Queries or Documents?

In Cross-Language Retrieval, a user formulates his search argument in an initial language, aiming to retrieve documents that are written in other languages besides the original (Nie, 2010; Oard & Diekema, 1998; Grefenstette, 1998). The user then has the option of either checking the desired target languages or of searching documents in all languages. Which texts must search tools translate for this purpose? Is it enough to translate the queries, or should the documents be translated, or both? If there existed a clear and unambiguous translation for every term in every language, we could restrict our efforts to the easiest task of all—translating the queries. McCarley (1999, 208) rejects this idea of unambiguity in mechanical translation, claiming that in the ideal scenario, both queries and documents should be translated:

Should we translate the documents or the queries in cross-language information retrieval? ... Query translation and document translation become equivalent only if each word in one language is translated into a unique word in the other language. In fact machine translation tends to be a *many-to-one* mapping in the sense that finer shades of meaning are distinguishable in the original text than in the translated text. ... These two approaches are not mutually exclusive, either. We find that a hybrid approach combining both directions of translation produces superior performance than either direction alone. Thus our answer to the question ... is *both*.

To test the unambiguity of mechanical translation, we let Google Translate render an excerpt from Goethe's *Faust* (from the *Prologue in Heaven*) into English, and then retranslated the result into German using the same method (Figure G.3.1).

German Original Text (Goethe):

Von Zeit zu Zeit seh ich den Alten gern,
Und hüte mich, mit ihm zu brechen.
Es ist gar hübsch von einem großen Herrn,
So menschlich mit dem Teufel selbst zu sprechen.

Translation into English by Google Translate:

From time to time I like to see the Old,
And take care not to break with him.
It is very pretty from a great lord,
So human to speak with the devil himself.

Retranslation into German, also Google Translate:

Von Zeit zu Zeit Ich mag das Alte zu sehen,
Und darauf achten, nicht mit ihm zu brechen.
Es ist sehr hübsch von einem großen Herrn,
So menschlich mit dem Teufel selbst zu sprechen.

Figure G.3.1: Original, Translation and Retranslation via a Translation Program. Source: Google Translate.

The software struggles, for instance, with the character “der Alte” (“the Old”), since it is incapable of drawing the connection between “der Alte” and “ihn” (“him”) (in line 2). The translation is linguistically unsatisfactory, but (with the exception of “the Old”) it more or less does justice to the content. In the retranslation into German, line 4 even remains completely unchanged.

We speak of “cross-language information retrieval” (CLIR) when only the queries, but not the documents, are translated. Kishida (2005, 433) defines:

Cross-language information retrieval (CLIR) is the circumstance in which a user tries to search a set of documents written in one language for a query in another language.

In contrast to this, there are multilingual systems (“multi-lingual information retrieval”; MLIR), in which the documents themselves are available in translation and in which query and documents are thus in the same language. Ripplinger (2002, 33) emphasizes:

This makes CLIR systems different to multilingual IR systems which offer the user query formulation and document searching in different languages where the query and the document language have to be the same.

The user is not necessarily required to understand a retrieved foreign-language document in every detail. A suitable area of application would be the search for images, or for charts with predominantly numerical content (Nie, 2010, 23). Here, a rudimentary understanding of the image’s (or chart’s) title is enough (alternatively, one can let a translation program like *Google Translate* render the meaning in a very general way); the desired content (the image, the cell) can still be used without a problem.

In felicitous individual cases where multilingual parallel corpora are available (e.g. documents from the European Union in various semantically equivalent versions in different languages), procedures for cross-language information retrieval can be derived without any direct translation.

Why are CLIR and MLIR so important? Here we should keep in mind that the knowledge stored in documents exists independently of language. Knowledge is potentially relevant, no matter what language an author has used to formulate it. The work “Ethnologue” states that there are currently around 7,000 living languages (Lewis, 2009).

Which working steps must cross-language information retrieval perform (Figure G.3.2)? A user formulates a query in “his” language and additionally defines further target languages. The better the natural language processing (as a reminder, there are: conflation, phrase identification, decompounding, personal name recognition, homonymy, synonymy), the better the query will be translated. Without a language-specific information-linguistic query processing, it would be extremely difficult for CLIR to work satisfactorily. The language of the documents is known, i.e. has been identified via language recognition or defined via meta-tags. For the documents,

too, elaborate information-linguistic procedures must be used. Matching is used to retrieve, separately for each language, documents that must then be entered into a single, relevance-ranked list. Specific problems in CLIR include the translation of queries as well as “mixed” Relevance Ranking (Lin & Chen, 2003) (a similar variant of which is used in meta-search engines, since these must also rank documents of differing provenance in a single hit list). The rest is normal information retrieval.

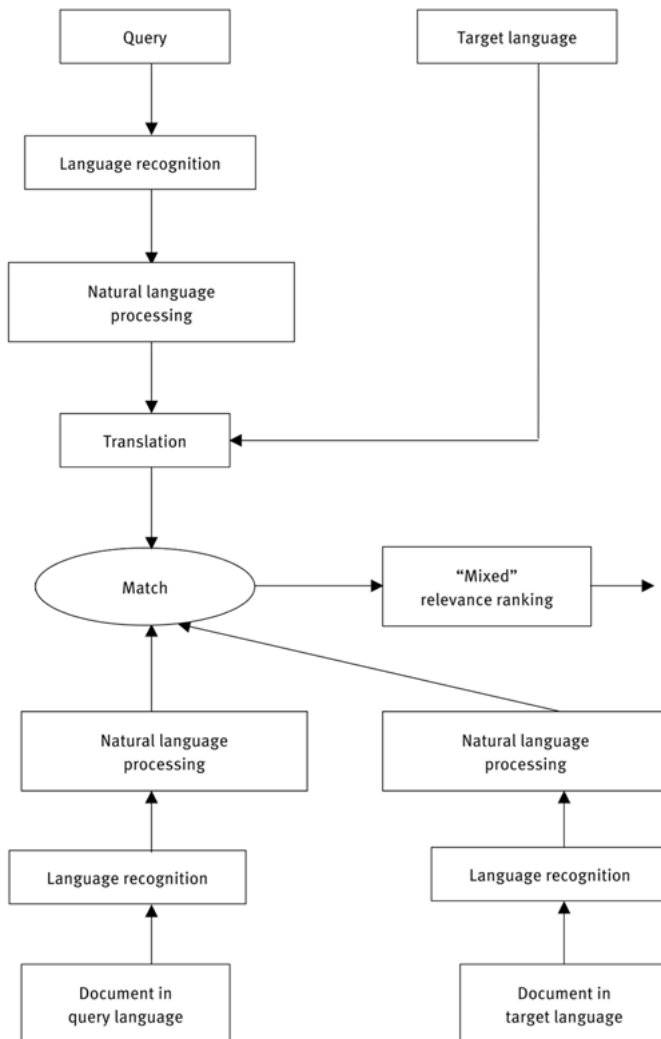


Figure G.3.2: Working Steps in Cross-Language Information Retrieval.

In dealing with cross-language retrieval, we will address three approaches:

- Translating the query via a dictionary,
- Translating the query via a thesaurus (i.e. via language and world knowledge),
- MLIR via usage of parallel corpora.

Machine-Readable Dictionaries

Many approaches to cross-language retrieval use the simple method of translating queries via machine-readable dictionaries (Hull & Grefenstette, 1996). Roughly speaking, this is a mixture of techniques of mechanical translation and information retrieval (Levow, Oard, & Resnik, 2005). The challenge is to find the right translating variant in each instance (Braschler, 2004, 189):

The main problem in using machine-readable dictionaries for CLIR is the ambiguity of many search terms.

Even presupposing that the language-specific information-linguistic processing is performed optimally, we are still confronted with the ambiguity of the translation. Pirkola et al. (2001, 217-218) demonstrate this on a simple example. The Finnish word “kuusi” is homonymous, referring to both “spruce” (the tree) and the number “six”. The English word “spruce” has two meanings (the aforementioned conifer as well as the adjective meaning “neat and tidy”), while “six” has four (“six in cricket”, “knocked for six”, “sixes and sevens” as well as the numeral). Let us suppose that the correct meaning of “kuusi” in a given text is “spruce”. In the monolingual case of Finnish (as in English), there is a further potential translation. In the multilingual Finnish-English scenario, however, there are five further candidates. The degree of potential ambiguity rises in cross-language systems. Pirkola et al. (2001, 217) emphasize:

The effectiveness of a CLIR query depends on the number of relevant search key senses in relation to the number of irrelevant senses in the CLIR query. The proportion is here regarded as an ambiguity measure of *degree of ambiguity* (DA). In the spruce-example above, the degree of ambiguity is increased from 1:1 in both Finnish and English retrieval to 1:5 in Finnish to English retrieval.

Where no further—incorrect—candidates for translation are extant, the degree of ambiguity is $DA = 0$. In all cases where it holds that $DA > 0$, the mono- and multilingual ambiguity must be resolved.

Ballesteros and Croft (1998a) suggest counting the co-occurrence of (translated) terms in documents in the target language. The procedure is only applicable when there are at least two search atoms in the original query. The pair of terms which co-

occurs most frequently in the corpus is deemed to be the most probable translation. Ballesteros and Croft (1998a, 65) write:

The correct translations of query terms should co-occur in target language documents and incorrect translations should tend not to co-occur.

Furthermore, it can be of advantage to use procedures of Relevance Feedback. The initial query is used to find model documents, whose most important terms must then be added to the original query. According to Ballesteros and Croft (1998a, 1998b), it is possible to implement such a “local”, i.e. query-dependent, Relevance Feedback both before and after the translation. The authors report positive results (Ballesteros & Croft, 1997, 85):

Pre-translation feedback expansion creates a stronger base for translation and improves precision. Local feedback after MRD (machine readable dictionary, a/n) translation introduces terms which de-emphasize irrelevant translations to reduce ambiguity and improve recall. Combining pre- and post-translation feedback is most effective ...

A particular problem arises when there is a dictionary linking languages A and B and another one for B and C, but none for A and C. Here the approach of transitive translation is of use (Figure G.3.3) (Ballesteros, 2000).

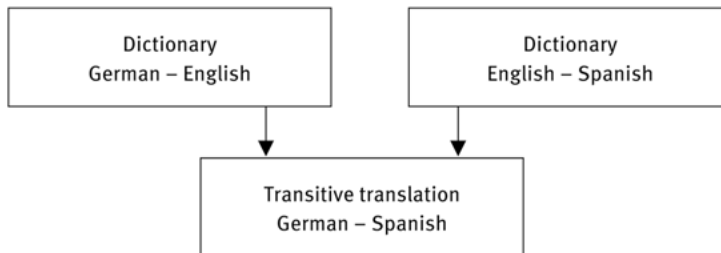


Figure G.3.3: Transitive Translation in Cross-Language Retrieval.

Useful results can be expected when at least the phrasing in the original language is taken into account, when term co-occurrence in the corpora as well as Relevance Feedback are used, and if the ambiguity in the transitive case also increases relatively to bilingual translation.

Ballesteros and Croft use automatic Pseudo-Relevance Feedback. However, it is also possible to let the user participate in choosing the correct translations. Petrelli, Levin, Beaulieu and Sanderson (2006, 719) report positive experiences, but also “side effects”:

By seeing the query translation, users were more engaged with the search task and felt more in control. We regarded this interaction proposal as fundamental although it uncovered potential weaknesses in the translation process that could undermine CLIR acceptability.

Specialist Thesauri in CLIR

A very elegant path of cross-language retrieval can be tread when a multilingual specialist thesaurus is available (Salton, 1969, 10 et seq.). Here the descriptor is defined independently of language; it is allocated words (alongside non-descriptors) in the respective supported languages. The thesaurus AGROVOC, for instance, uses 22 languages. The descriptor with the AGROVOC identification number 3032 is called *food* in English, *produit alimentaire* in French and *Lebensmittel* in German (see Ch. L.3).

The documents (from one of the supported thesaurus languages) are indexed—intellectually or automatically—via descriptors in the respective languages. Additionally, the system will note the language of the text. The (cross-language) descriptor set summarizes the various descriptors (in different languages) into the respective concept. The user searches via the descriptors in his language and states the desired target languages. The documents are easily retrievable via the cross-language descriptor set. Salton's (1969, 25) results are positive:

An experiment using a multi-lingual thesaurus in conjunction with two different document collections, in German and English respectively, has shown that cross-language processing (for example, German queries against English documents) is nearly as effective as processing within a single language.

For this procedure it is vital that a comprehensive multilingual specialist thesaurus be available. Constructing a multilingual specialist thesaurus takes both time and effort and is problematic due to the inherent ambiguities. The smaller and the more well-defined the subject area, and the greater the consensus—across language areas—about terminology, the more promising it becomes to use a multilingual thesaurus.

Corpus-Based Methods

We will now address methods that forego explicit translations. Instead, we will exploit the availability of parallel corpora, i.e. collections of documents that are stored in different languages. Institutions frequently establish parallel Web presences in different languages, e.g. for all official documents of the European Union. The basic idea behind using such parallel corpora for cross-language retrieval is the following (Figure G.3.4): An initial search is performed in the user's language (language 1) and leads to a relevance-ranked hit list in which the documents are written in language 1.

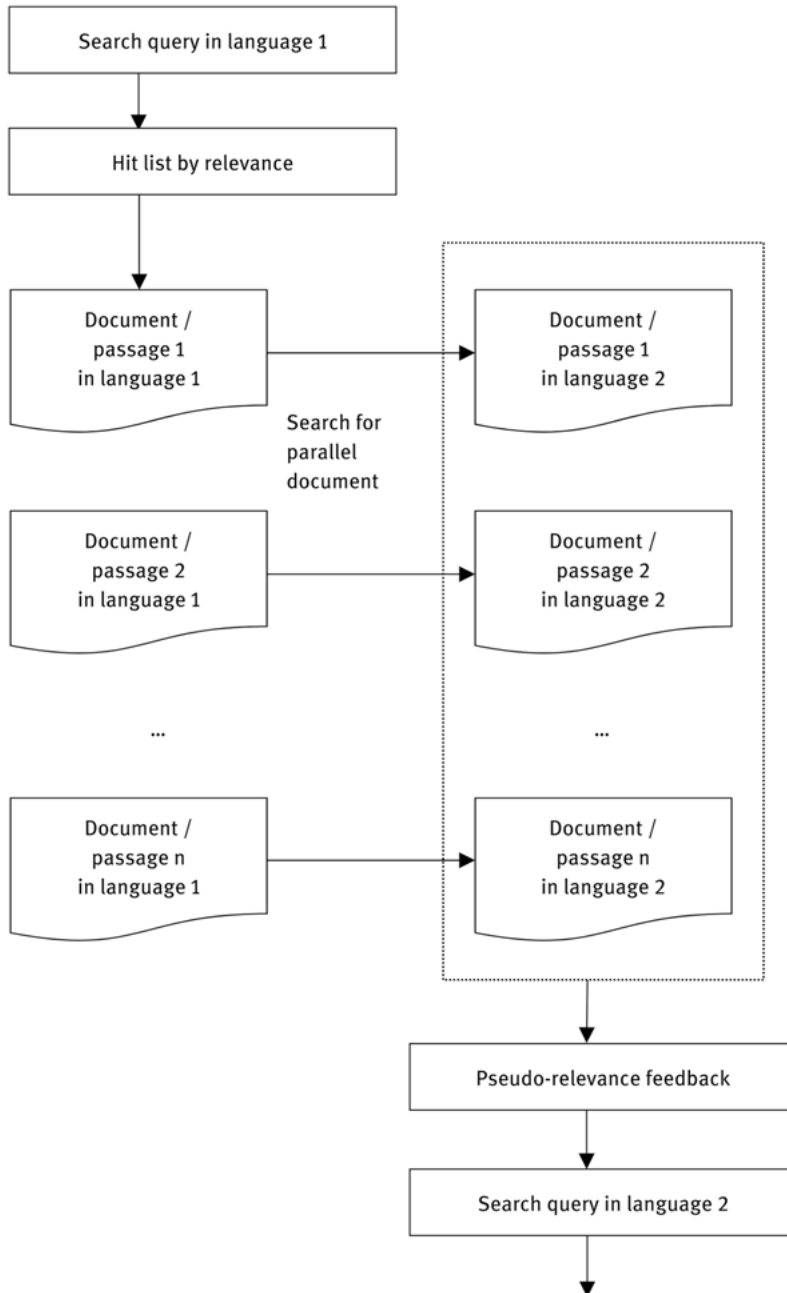


Figure G.3.4: Gleaning a Translated Query via Parallel Documents and Text Passages.

In the following, we are interested only in the top n texts of the hit list (n can be between 10 and 20, for instance). The decisive step is the accurate retrieval of parallel

documents in a different language (language 2). Within the retrieved pairs of parallel documents, the objective is to find that passage in the text which best suits the query (Li & Yang, 2006). (How this retrieval for text passages works will be discussed in Chapter G.6). The passages from the parallel documents, i.e. in language 2, are analyzed in the sense of Pseudo-Relevance Feedback, via the Croft-Harper Formula. We crop the resulting word list down to the top m terms and use these as search arguments for a parallel search in language 2. Subsequently, we perform a normal research for documents in language 2.

Conclusion

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- Cross-language information retrieval (CLIR) is performed by translating only the search queries and not the documents. In multi-lingual information retrieval (MLIR), on the other hand, documents are already available in translation.
 - Even if a user does not speak the language of a document, cross-language retrieval makes sense because a foreign-language document may contain language-independent content, such as images or charts.
 - The problem of ambiguity is exacerbated in cross-language retrieval, as the multiple translations in original and target language add up.
 - A frequently used procedure in CLIR is the use of machine-readable dictionaries. Certain procedures (e.g. analyzing the translated query terms for co-occurrence in documents or via local feedback) can be used to minimize translation errors.
 - When using multilingual specialist thesauri, CLIR achieves results comparable in quality to those of monolingual information retrieval. The construction of the respective multilingual thesauri is very elaborate, however; additionally, the documents must be indexed via the thesauri (intellectually or automatically).
 - If a retrieval system disposes of parallel corpora, i.e. documents of identical content in different languages, it can work without any explicit translation of search queries. On the basis of search results in the original language, parallel documents (and the most appropriate text passages within them) are searched in the target languages. Via Pseudo-Relevance Feedback, a search query is created in the target language.
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G.4 (Semi-)Automatic Query Expansion

Modifications of Queries

It appears rather improbable for a user to be able to directly search his desired documents with a single query formulation. Far more often the initial query will fail to produce the desired results. It then becomes necessary to modify the initial formulation, and the user or the system (or both) are asked to reconsider the search strategy. This process is called “query expansion” (Ch. D.2). If only the information system (and not the user) solves such problems, we speak of “automatic query expansion” (Carpineto & Romano, 2012). If there is a dialog between the system and the user to find the optimal search arguments, we label this as “semi-automatic query expansion”. For Efthimiadis (1996, 122), the search goes through two stages:

For the sake of simplicity the online search can be reduced to two stages: (1) initial query formulation, and (2) query reformulation. At the initial query formulation stage, the user first constructs the search strategy and submits it to the system. At the query reformulation stage, having had some results from the first stage, the user manually or the system automatically, or the user with the assistance of the system, or the system with the assistance of the user tries to adjust the initial query and improve the final outcome.

There are three options for performing query modifications: intellectually, automatically and semi-automatically, that is to say interactively. Several methods are used to determine the “good” search arguments in each case, which are shown in Figure G.4.1.

In many cases it proves useful to work via similarities, as van Rijsbergen (1979, 134) stresses:

If an index term is good at discriminating relevant from nonrelevant documents, then any closely associated index term is also likely to be good at this.

Similarities regard not only words and concepts but also documents in their entirety, which are used as model documents for further search.

Documents are similar to one another if they show overlap via internal, i.e. structural and content-related factors, or via external factors, such as corresponding user behavior. Similarity search takes document-internal factors so seriously, that users are given the option of starting with a model document and searching for any further similar documents (“More like this!”). External factors are, for instance, documents looked at (or, in e-commerce, bought) at the same time or explicit document ratings by users. Such external document factors are processed in recommender systems (Ch. G.5), which then suggest further documents for the user to look at, again starting with a document. Search arguments can also be similar, e.g. by neighboring one another in paradigmatic and syntagmatic relations or by being used together in search queries. If the user is offered such similar terms, he can use them to modify his query in order

to optimize the retrieval result. The system-side offer of similar documents or similar search arguments always leads into a dialog between user and retrieval system, i.e. to an iterative search strategy.

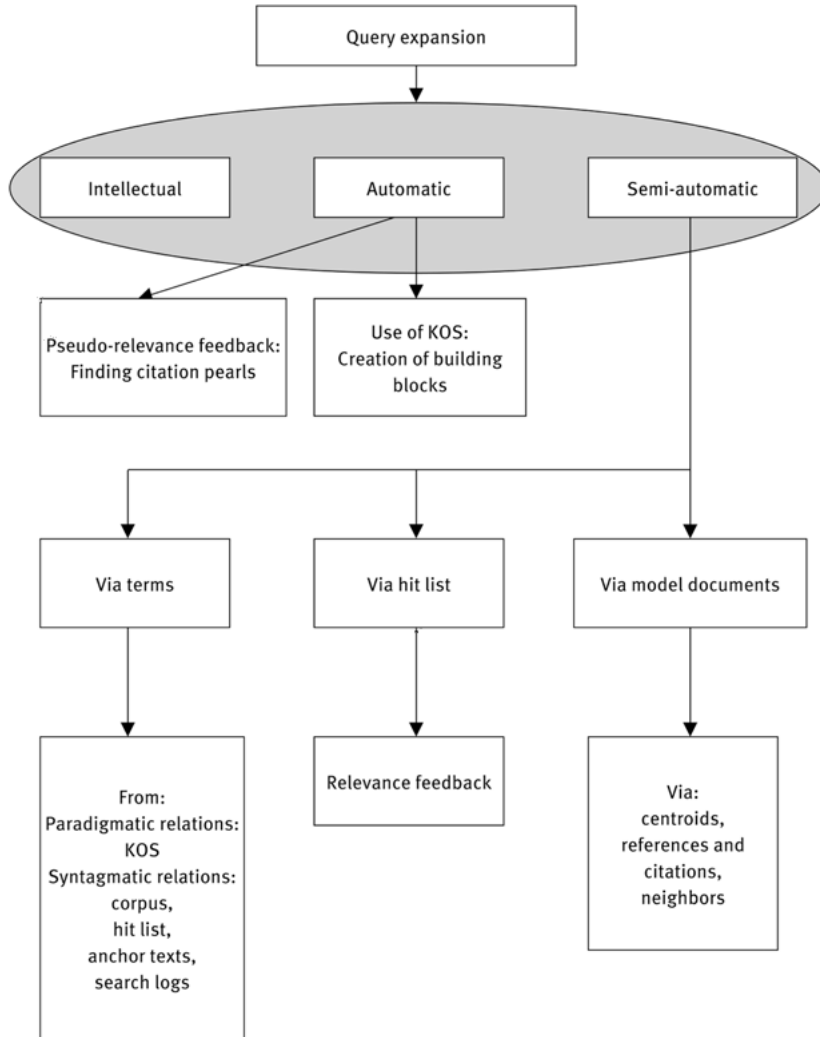


Figure G.4.1: Options for Query Expansion. Source: Modified from Efthimiadis, 1996, 124.

When a user contacts colleagues or other people's documents during a query (re)formulation, we speak of "collaborative IR" (CIR) (Hansen & Järvelin, 2005; Fidel et al., 2000).

Automatic Query Expansion

If the retrieval system automatically expands the query, the intellectual paths must be simulated. An analogy to the block strategy is to use paradigmatic relations of KOSs (as “ontology-based query expansion”; Bhogal, Macfarlane, & Smith, 2007), i.e. synonymy, hierarchy or other relations. The system can admit the concepts neighboring each other (i.e. the hyponyms or the hyperonyms on the next hierarchy level) or—if the relation is transitive (Weller & Stock, 2008)—concepts into the respective facet via path lengths exceeding one (Järvelin, Kekäläinen, & Niemi, 2001), e.g. sister concepts or hyponyms of all hierarchical levels. Here all linguistically oriented KOSs, such as WordNet, can be used as well as all specialist KOSs. Central roles are taken by threshold values for semantic similarity and by the exclusion of homonyms in query expansion (Ch. I.4).

An analogy to the strategy of finding “pearls” is pseudo-relevance feedback (Ch. E.3). In an initial search, the top-ranked documents are regarded as “pearls” and their most frequent terms are used in the expanded search.

Relevance Feedback as Query Modification

We now come to the semi-automatic variants of query modification, in which the retrieval system presents the user with suitable further terms or model documents. Here the user consciously selects and incorporates these words or documents into his search.

An initial option is to show the user a number of documents and to ask, for every result, whether it solves the information problem or not. This is the method used by Relevance Feedback (Salton & Buckley, 1990), which plays an important role both in the Vector Space Model (Ch. E.2) and in the Probabilistic Model (Ch. E.3). Multiple feedback loops enable the user to adjust the results lists to his information need as well as to optimize them by selecting suitable “pearls”.

Proposing New Search Arguments

Which sources can generate suggestions for suitable search terms? As in the fully automatic procedure, it makes sense to use a KOS—it may even be more useful here (since it is more selective and thus more precise) than in the automatic case (Voorhees, 1994), as the user can decide which of the proposed terms to choose. If the expanded query contains misleading homonyms (e.g., expanding a search on *Indonesia* with the term *Java*), these should be identified during alignment with the KOS.

If the retrieval system is not supported by KOSs, or if one wants to additionally use the full texts for query modification, one may proceed via an analysis of syntagmatic term relations and thus the word co-occurrences. Possible sources are:

- Documents in the corpus (in the entire database or—in folksonomy-based Web 2.0 services—in the tags field),
- Anchor texts of documents in the database,
- Documents in the hit list,
- Anchor texts pointing to documents in the hit list,
- Queries (from search logs).

The database-specific co-occurrence of words leads to a similarity thesaurus, or a statistical thesaurus. Such “clumps” of similar words (Bookstein & Raita, 2001) can be gleaned via a frequency distribution of the words in text windows that also contain the original word. The user thus receives individual words similar to his initial search atoms. Similarity is calculated via one of the relevant procedures: Jaccard-Sneath, Dice, or Cosine.

Zazo et al. (2005, 1166-1167) discuss the possibility of expanding an entire query, i.e. several search terms at once, via a similarity thesaurus. They start with the Vector Space Model, but define the roles of documents and terms in the reverse: documents represent the dimensions and terms the vectors. Zazo et al. (2005, 1166) write:

This turns the classic concept of information retrieval systems upside down (...). To construct the similarity thesaurus, the terms of the collection are considered documents, and the documents are used as index terms (...); in other words, the documents can be considered capable of representing the terms.

The initial query is formulated as a vector (as is generally customary in the model). The Cosine calculation provides us with a ranking of similar words. Zazo et al. (2005, 1171) report on satisfactory results:

The technique described obtains good results in query expansion. A thesaurus of similarity between terms was constructed, taking advantage of the possibilities the documents have to represent the terms. ... The main characteristic resides ... in the fact that the expanded terms were chosen and weighted taking into consideration the terms of the whole query, and not each individual term separately.

In Web Information Retrieval there is a further option for suggesting terms from documents for query expansion: anchor texts. The compilation of a similarity thesaurus only via anchor texts takes far less effort, since the amount of data in the anchor texts is far smaller than the data set of the texts of all documents. Additionally, anchor texts are generally formulated succinctly, thus bearing the fundamental terms (at least most of the time). The terms can be gleaned either from the co-occurrence of words and an initial search argument in all anchor texts of the database or by summarizing all anchor texts that link to one and the same document into a single document

beforehand, then determining term similarities (Kraft & Zien, 2004). Anchor texts can be gleaned either from the entire database (statically) or from the up-to-date hit list (dynamically) in each case.

Of course it is also possible to glean terms for query modification dynamically from all text parts of the respective retrieved documents in the results list. Rose (2006, 798) describes the (by now defunct) option “Prisma” from the search engine AltaVista:

After entering a query, the system displays a list of words and phrases that represent some of the key concepts found in the retrieved documents. Users may explore related topics by replacing their original query with one of the Prima suggestions. Alternatively, they can narrow their search by adding one of the Prisma suggestions (with implicit Boolean “AND”) to their existing query.

The user is offered words and phrases for the continuing restriction of the hit list. The search dialog thus yields ever smaller document sets, thus heightening Precision. However, the user can also add the suggestion terms to the original search argument via OR (or start an entirely new search via one of the terms). Now the Recall will rise; the extent of the search results increasing through iterative usage.

The last option of gleaning new search terms to be introduced here works with saved queries from the search engine’s log file (Whitman & Scofield, 2000). Huang, Chien and Oyang (2003, 639) describe this approach:

Using this method, the relevant terms suggested for original user queries are those that co-occur in similar query sessions from search engine logs.

A “query session” comprises all subsequent search steps of a user until the point of his quitting the search engine. All search terms occurring therein are regarded as a unit. Let us take a look at the following three search formulations (Huang, Chien, & Oyang, 2003, 639):

- 1st User: “search engine”, “Web search”, “Google”;
- 2nd User: “multimedia search”, “search engine”, “AltaVista”;
- 3rd User: “search engine”, “Google”, “AltaVista”.

If a fourth user then goes on to request “search engine”, the system will offer him the additional terms “AltaVista” and “Google” (which both co-occur twice with the initial query term).

Similar Documents: More Like This!

Let us assume that the user has found a document that is perfect for his information need after his initial query. Since this one document is not enough for him he wants

to find more, preferably similar ones to this model document (“More like this!”). The retrieval system’s task is easy to formulate: “finding nearest neighbours” (Croft, Wolf, & Thompson, 1983, 181). We see the following options for retrieving ideal neighbors on the basis of model documents:

- co-occurrence in a cluster,
- common terminology,
- connections via references or citations (Ch. M.2),
- neighborhood in a social network (Ch. G.1).

When using the Vector Space Model it is possible to cluster documents, i.e. to summarize similar documents into a specific document set. The texts of such a set cluster around a centroid, the mean vector of all vectors of its documents (Griffiths, Luckhurst, & Willett, 1986). If the model document belongs to such a document set, all documents of the same centroid will be yielded as results, ranked by their similarity (Cosine) to the model document.

If a system—such as LexisNexis’ “More like this”—uses terms to calculate the similarity between a model document and other texts, weighting values will be calculated for all terms of the model (via TF*IDF) and the top *n* terms will be used as search arguments. LexisNexis shows the user the retrieved “core terms” and allows him to add further words as well as to remove certain suggested terms.

When the retrieval system contains documents that are connected via certain lines, we can establish their neighborhood via an analysis of the corresponding social network (in the sense of Ch. G.1). Simple procedures build on direct citation and link relations, i.e. on directed graphs. Such methods are applicable in the World Wide Web as well as in contexts where formal citation is used (in scientific literature, in patents as well as in legal practice). Electronic legal information services (such as LexisNexis and Westlaw) offer navigation options along the references (i.e. to the cited literature) and citations (to the citing literature) of court rulings. Analogous options are available in patent databases (references and citations of patent documents) as well as many scientific databases (here on scientific articles and books, but also on patents).

Undirected citation graphs come about via analysis of bibliographic couplings (Kessler, 1963) and of co-citations (Small, 1973) (Ch. M.2). Bibliographic couplings are used to search for “related records” in the information service “Web of Science”. The advantage of this use of references is the language independence of query expansion, as Garfield (1988, 162), the creator of scientific citation indices, emphasizes:

The *related records* feature makes it easy to find papers that do not share title words or authors. You can locate related papers without the need to identify synonyms. If you choose, you can instantly modify your search to examine the set of records related to the first related record.

Starting from the model document’s references, “bibliographic coupling” searches for documents that have a certain amount of bibliographic data in common with the model. The documents are ranked via their degree of similarity. Starting with the co-

citation approach, Dean and Henzinger (1999) use co-links to determine the similarity between a model Web page and other pages. Two Web pages are thus similar when many documents send out links to both at the same time.

Conclusion

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- Whenever a user is unable to directly reach the goal of his search with a query formulation, the query will be modified. The objective is to find the “good” search arguments in each case. Query modifications can be performed intellectually, automatically or in a man-machine dialog.
 - Automatic query expansion tries to simulate an intellectual query modification. If knowledge organization systems are available in a retrieval system, the paradigmatic relations can be exploited in order to add further terms to the facets. In Pseudo-Relevance Feedback, the top-ranked texts are regarded as pearls and their terminology is used to search further.
 - Semi-automatic procedures are based on machine-made suggestions, which are viewed by the user and purposefully used in further search steps.
 - A Relevance Feedback performed by the user starts with ranked documents in a hit list, then lets the user evaluate some documents and reformulates the initial search on the basis of the words in both the positive and the negative documents.
 - Retrieval systems can suggest new search arguments to the user. These arguments are either gleaned from a KOS or from word co-occurrences. When an entire database is used to recognize similarities between words, one is working with a “statistical thesaurus”. As they are often formulated very precisely, anchor texts are also suitable as a source for word co-occurrences in Web Information Retrieval. The restriction to terms that occur in documents and appear as results of an initial search allows an iterative heightening of Precision (while using an AND link).
 - Once a user has found an ideally suitable document, i.e. a pearl, he will want to find further documents that are as similar to this model as possible. This similarity is determined via co-occurrence in a cluster (in the Vector Space Model), via co-occurring terms, via connections such as references and citations as well as via neighborhood in a social network.
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G.5 Recommender Systems

Certain information systems tender explicit recommendations to their users (Ricci, Rokach, Shapira, & Kantor, 2011, VII):

Recommender Systems are software tools and techniques providing suggestions for items to be of use to a user. The suggestions provided are aimed at supporting their users in various decision-making processes, such as what items to buy, what music to listen, or what news to read. Recommender systems have proven to be valuable means for online users to cope with the information overload and have become one of the most powerful and popular tools in electronic commerce.

Apart from suggested tags (when indexing documents via folksonomies) or search terms (during retrieval), the user is generally displayed documents that are not explicitly searched for. The range of document types is large, encompassing everything from products (for sale in e-commerce), scientific articles, films, music, all the way to individual people. The goal and task of recommender systems is defined thus (Heck, Peters, & Stock, 2011):

The aim is personalized recommendation, i.e. to get a list of items, which are unknown to the target user and which he might be interested in. One problem is to find the best resources for user *a* and to rank them according to their relevance.

Recommendable documents are identified by the systems when user behavior (clicking, buying, tagging etc.) is analyzed or explicit recommendations by other customers are communicated (i.e. ratings in the form of stars or “likes”). These methods are called “collaborative filtering” and “content-based recommendation”, respectively, and are used in recommender systems. Riedl and Dourish (2005, 371) discuss the motivations for working with recommender systems:

In the everyday world ... the activities of others give us cues that we can interpret as to the ways in which we want to organize our activities. A path through the woods shows us the routes that others have travelled before (and which might be useful to us); dog-eared books in the library may be more popular and more useful than pristine ones; and the relative levels of activity in public space help guide us towards interesting places. These ideas helped spark an interest in the ways in which computing systems might use information about peoples’ interests and activities to form recommendations that help other people navigate through complex information spaces or decisions.

We distinguish between collaboratively compiled recommender systems and user-centered content-based recommendations. The latter are recommendations only on the basis of documents viewed (or products bought in e-commerce) by the same user; collaborative recommender systems draw on experiences of or recommendations by other users as well. Perugini, Goncalves and Fox (2004, 108) draw a clear line between these two kinds of recommendation:

Content-based filtering involves recommending items similar to those the user has liked in the past; e.g. ‘Since you liked *The Little Lisper*, you also might be interested in *The Little Schemer*’. Collaborative filtering, on the other hand, involves recommending items that users, whose tastes are similar to the user seeking recommendation, have liked; e.g., ‘Linus and Lucy like *Sleepless in Seattle*. Linus likes *You’ve Got Mail*. Lucy also might like *You’ve Got Mail*’.

In systems that operate collaboratively, we must further differentiate between explicit user recommendations and implicit patterns derived from user behavior. We are thus confronted with the following spectrum of recommender systems, which are not mutually exclusive but complement each other (in the form of hybrid systems) (Adomavicius & Tuzilin, 2005):

- user-specific systems (“content-based”);
- collaborative systems,
 - explicit (ratings),
 - implicit (user behavior);
- hybrid systems.

Collaborative Filtering

The first collaborative filtering system was probably Tapestry by Goldberg, Nichols, Oki and Terry (1992). The authors define “collaborative filtering” as follows (Goldberg, Nichols, Oki, & Terry, 1992, 61):

Collaborative filtering simply means that people collaborate to help one another perform filtering by recording their reactions to documents they read. Such reactions may be that a document was particularly interesting (or particularly uninteresting). These reactions, more generally called *annotations*, can be accessed by others’ filters.

A purely collaborative recommender system abstracts completely from the documents’ content and exclusively works with the current user’s similarity to other users. Balabanovic and Shoham (1997, 67) describe such a system:

A pure collaborative recommendation system is one which does no analysis of the items at all—in fact, all that is known about an item is a unique identifier. Recommendations for a user are made solely on the basis of similarities to other users.

Such a system can be constructed e.g. in the framework of the Vector Space Model, with users being represented by the vectors and documents by the dimensions. The specific vector of the user *U* is calculated via his viewed (or bought, or positively rated) documents or products. The similarity to other users is derived by calculating the cosine of the vectors in question. Via a similarity threshold value (Cosine), the system recognizes those other users that are most similar to the original user. All that

these former view, buy or rate positively (and which the respective user has not yet viewed, bought or rated) is offered to the original user as a recommendation.

In the case of user-generated content in Web 2.0, the connections between users, documents and tags can be exploited for collaborative filtering. In social bookmarking services (such as CiteULike), users enter document surrogates into the service and add tags to them (at least in many cases). On this example, we will demonstrate how recommender systems work for finding experts. Two paths offer themselves:

- Calculating the similarities between bookmarking users and tagging users (Heck & Peters, 2010),
- Calculating the similarities between the authors of bookmarked and tagged articles (Heck, Peters, & Stock, 2011).

The basis of calculations are, in each case, the bookmarked documents or the allocated tags (Marinho et al., 2011). If two users have bookmarked many of the same documents, this is an indicator of similarity. If two users have used many of the same tags, this is a further indicator for similarity, although it leads to different results. In Figure G.5.1, we see a cluster with names of CiteULike users that are similar to the user “michaelbusmann”, and which can thus be recommended to him as “related experts”. For visualization, Heck and Peters (2010) chose the form of a social network displaying the connections between names. The disadvantage of this method of recommendation is that the users are only featured in the network under their aliases, and not their real names. In a further step, the recommender system can recommend to the user michaelbusmann those documents (for bookmarking or reading) that were bookmarked by the largest number of users featured in the network in Figure G.5.1 (but not by michaelbusmann). The problem of aliases now no longer arises.

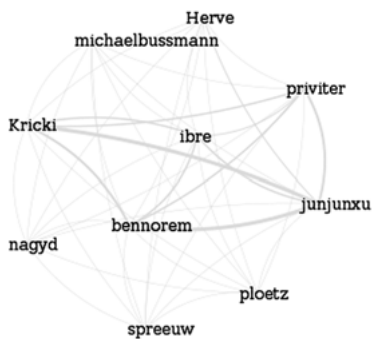


Figure G.5.1: Social Network of CiteULike-Users Based on Bookmarks for the User “michaelbusmann” (Similarity Measurement: Dice, $SIM > 0.1$, Data Source: CiteULike). Source: Heck & Peters, 2010, 462.

We will stay with CiteULike in order to address the similarity between authors. The problem of aliases is not an issue in the case of author names, as the anonymous

tagging users remain in the background and only their actions are of import. In Figure G.5.2 we see (again in the form of a social network) author names outgoing from R. Zorn, arranged according to common tags in their articles bookmarked on CiteULike. Such a network could be submitted to R. Zorn to recommend authors with similar research subjects as him. Indeed, Heck asked several authors whether such recommendations would be useful for their work when looking for cooperation partners for scientific projects. The results are positive (Heck, Peters, & Stock, 2011):

Looking at the graphs almost all target authors recollected important colleagues, who didn't come to their mind first, which they found very helpful. They stated that bigger graphs like [Figure G.5.2, A/N] show more unknown and possible relevant people.

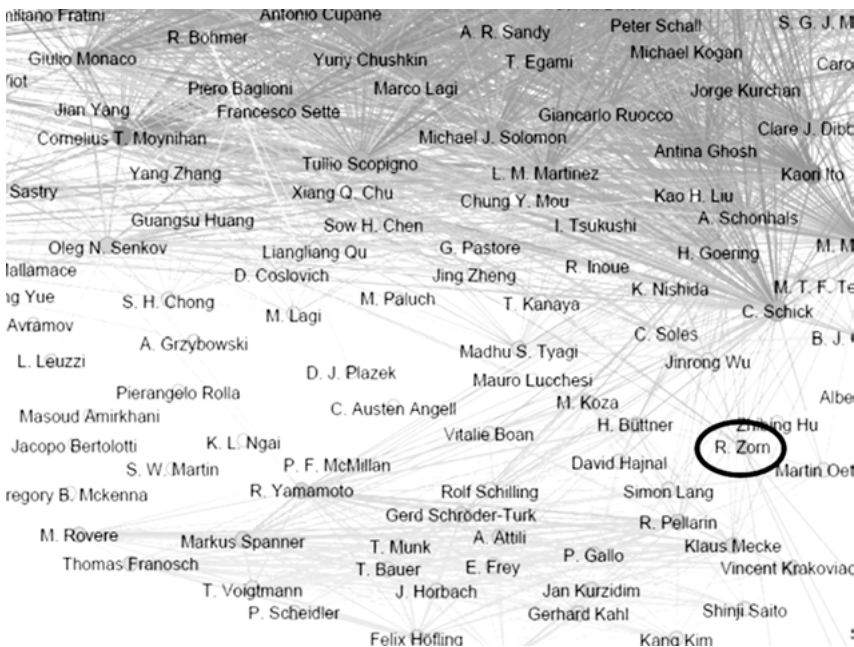


Figure G.5.2: Social Network of Authors Based on CiteULike Tags for the Author “R. Zorn” (Similarity Measurement: Cosine, $0.99 \geq \text{SIM} \geq 0.49$, Data Source: CiteULike). Source: Heck, Peters, & Stock, 2011.

Content-Based Recommendation

A purely user-specific recommender system concentrates on user profiles and the documents' content. According to Balabanovic and Shoham (1997, 66-67), the circumstances are as follows:

Text documents are recommended based on a comparison between their content and a user profile. ...

We consider a pure content-based recommendation system to be one in which recommendations are made for a user based solely on a profile built up by analyzing the content of items which that user has rated in the past.

User-specific recommender systems thus have a lot in common with the profiling services (alerts, SDIs) of suppliers of specialized information. The user submits explicit ratings or demonstrates certain preferences via his buying or clicking behavior. The objective is to retrieve documents that are as similar to these preferences as possible, either by aligning document titles or terms, or via neighborhood in directed or undirected graphs (Kautz, Selman, & Shah, 1997).

Hybrid Recommender Systems

Hybrid systems take into account both: the behavior of the actual user (via a user profile) and that of the other customers (Kleinberg & Sandler, 2003). As an example of a recommender system with a hybrid character, we will introduce Amazon's approach, substantially conceived by Linden (Linden, Jacobi, & Benson, 1998; Linden, Smith, & York, 2003).

The basic idea on Amazon is item-to-item filtering. In the user-specific subsystem, the list of already bought products is regarded as the "user profile". This list can be edited by the customer at any time; it is possible to remove individual documents from it. When performing new searches and calling up a data set, the user is displayed further documents. All recommendation processes on Amazon are founded on a chart that contains the similarity values of all documents among each other. There are several options of expressing similarities, of which Linden, Smith and York (2003, 79) have chosen the Cosine:

It's possible to compute the similarity between two items in various ways, but a common method is to use the cosine measure ..., in which each vector corresponds to an item ..., and the vector's M dimensions correspond to customers who have purchased that item.

In both forms of recommendation—user-centered and collaborative—Amazon works with similarity values and ranks the recommendations in descending order following the Cosine, always starting with model documents: those documents just viewed, in the collaborative scenario, and those that have been bought (and were not deleted in the list) in the case of user-centered recommendations. An example for collaborative item-to-item filtering is given in Figure G.5.3. The item searched was a book called "Modern Information Retrieval" by Baeza-Yates and Ribeiro-Neto. Recommendations were those other books that were bought by the same users who bought "Modern Information Retrieval".

Customers Who Bought This Item Also Bought



Figure G.5.3: Collaborative Item-to-Item Filtering on Amazon. Recommendations Subsequent to a Search for “Modern Information Retrieval” by Baeza-Yates and Ribeiro-Neto. Source: Amazon.com.

Problems of Recommender Systems

Recommender systems are not error-proof. Resnick and Varian (1997) discuss error sources as well as the consequences of such technologies beyond algorithms and implementations. One latent error source that is always present is the user, who actively collaborates in the system (e.g. via his ratings) or does not. What is his incentive for leaving an explicit rating in the first place? Do those who actively vote represent the average user, or are they “a particular type”? If the latter is true, then their ratings cannot be generalized. Voting abuse can hardly be ruled out. Resnick and Varian (1997, 57) discuss possible fraud:

(I)f anyone can provide recommendations, content owners may generate mountains of positive recommendations for their own materials and negative recommendations for their competitors.

Recommender systems often touch upon the user’s privacy. The more information is available concerning a user, the better his recommendations will be—and the more the providers of the recommender systems will know about him. Resnick and Varian (1997, 57) comment:

Recommender systems ... raise concerns about personal privacy. ... (P)eople may not *want* their habits or views widely known.

Conclusion

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- Recommender systems offer users personalized documents that are new to them and might be of interest.
 - Recommender systems work either user-specifically (by aligning a user profile with the content of documents) or collaboratively (by aligning the documents or products of similar users). Hybrid systems unite the advantages of both variants in a single application.
 - In the case of user-generated content, similarities between users and between documents can be calculated via tag co-occurrences or co-bookmarked documents, and then be used for collaborative filtering.
 - Problems of recommender systems lie in possible false explicit ratings as well as in significant intrusions into the user's privacy.
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G.6 Passage Retrieval and Question Answering

Searching Paragraphs and Text Excerpts

Relevance Ranking—either for Web documents or for other digital texts—normally orients itself on the document, which is regarded as a whole. Salton, Allan and Buckley (1993, 51) call this the “global” approach. They complement it with an additional “local” perspective, which views text passages in a document as units, and aligns Relevance Ranking with them, too. Kaszkiel, Zobel and Sacks-Davis (1999, 436) mainly see the following areas of application for text passage retrieval:

Passage retrieval is useful for collections of documents that are of very varying length or are simply very long, and for text collections in which there are no clear divisions into individual documents.

Examples include patent documents, which form single documentary reference units but vary between a few and a few hundred pages in length. Scientific works in the World Wide Web include both 15-page articles and entire PhD-dissertations (with hundreds of pages of text) as single units. For legal texts and other juridical documents (jurisdiction, commentaries, articles), it has proven useful to introduce the sub-unit “paragraph” to a documentary reference unit “law”.

Passages are thus sub-units of documentary reference units. According to Callan (1994, 302), sub-units are either discourse locations, semantic text passages or excerpts, i.e. windows of a certain size:

The types of passages explored by researchers can be grouped into three classes: discourse, semantic, and window. *Discourse* passages are based upon textual discourse units (e.g. sentences, paragraphs and sections). *Semantic* passages are based upon the subject or content of the text (...). *Window* passages are based upon a number of words.

Hearst and Plaunt (1993) speak of “text tiling” when discussing semantic text units. The basic idea is to partition a text in such a way that each unit corresponds to a topic that is discussed in the document (Liu & Croft, 2002, 377).

“Passage retrieval” is concretely applied in three contexts:

- When ranking documents according to the most appropriate text passages,
- When ranking text passages within a (long) document,
- During retrieval of the most appropriate text passage (from all retrieved documents), as the answer to a factual question of the user.

Ranking Documents by Most Appropriate Passages

The first option of using passage retrieval still treats documents as wholes, but ranks them by the retrieval status value of the most appropriate passage. Kaszkiel, Zobel and Sacks-Davis (1999, 411) emphasize:

In this approach, documents are still returned in response to queries, providing context for the passages identified as answers.

The advantage of this procedure is that text length cannot gain any significance in Relevance Ranking. Consider, for instance, two documents that are search results for a query. Let the first text be 30 pages in length, while discussing the desired topic on 4 of them; the second text only has two pages, but the search topic is discussed throughout. Ranking the documents via the Vector Space Model, the second document will receive a higher retrieval status value than the first one, since it displays a (probably much) higher WDF value for the search terms. (As a reminder: WDF weighting relativizes term weight vis-à-vis text length.) However, the first document would probably be of more value to the user. Kaszkiel and Zobel (2001, 347) lament this distorting role played by text length in Relevance Ranking:

(L)ong documents that have only a small relevant fragment have less chance of being highly ranked than shorter documents containing a similar text fragment, although ... long documents have a higher probability of being relevant than do short documents.

Kaszkiel and Zobel (1997; 2001) report of good experiences in randomly chosen passages (“arbitrary passages”). This method involves a variant of a window, of variable or fixed size, which glides over a text. Kaszkiel and Zobel (2001, 355) define

an *arbitrary passage* as any sequence of words of any length starting at any word in the document. The locations and dimensions of passages are delayed until the query is evaluated, so that the similarity of the highest-ranked sequence of words, from anywhere in the document, defines the passage to be retrieved; or, in the case of document retrieval, determines the document’s similarity.

When working with fixed-size text windows, the dimensions of a passage are already known before each query. It can be shown that the discriminatory power of the window size depends upon the number of query terms. Long queries (ten or more terms) can be better processed via small windows (between 100 and 200 words), short queries via larger ones (between 250 and 350 words) (Kaszkiel & Zobel, 2001, 362):

For short queries, the likelihood of finding query terms is higher in long passages than in short passages. For long queries, query terms are more likely to occur in close proximity; therefore, it is more important to locate short text segments that contain dense occurrences of query terms.

When using variably-sized text windows, their dimensions only become known once the query is processed in the document. According to Kaszkiel and Zobel (2001, 359), it is true that:

A variable-length passage is of any length that is determined by the best passage in a document, when the query is evaluated.

This procedure requires a lot of computing. Starting with each word of a text, we create windows with respective lengths of two words, three words, etc. up until the end of the document. A text with 1,000 words leads to around 500,000 passages. The retrieval status value of all these passages is calculated individually (Kaszkiel & Zobel, 1997, 180).

Kaszkiel and Zobel's procedure works out the retrieval status value of precisely one passage, namely that of the best one, and uses this value as the score for the entire document.

Ranking Passages within a Document

We assume that a user has localized a document that he thinks will satisfy his information need. He must now find precisely those passages that prominently deal with his problem. Particularly in the case of long texts, the user is thus dependent upon "within-document retrieval". The system identifies the most appropriate text passages and ranks them according to their relevance to the query; it thus offers the user a content-oriented tool for browsing within a document.

In the context of a probabilistic approach, Harper et al. (2004) introduce a procedure that yields a ranking of (non-overlapping) text passages for long documents. The starting point is a fixed-size text window (set at 200 words). The objective is to glean the probability of relevance (P) of a text window for a query with the terms $t_1, \dots, t_i, \dots, t_m$. A "mixture" of the query term's relative frequency in the text window as well as in the entire document is used as the basis of the calculation. The retrieval status value of each text window is the product of the values of each individual query term. n_{iw} is the frequency of occurrence of term i in the text window, n_{id} is the corresponding frequency of i in the entire document, α is an arbitrarily adjustable value in the interval between greater than zero and one (e.g. 0.8). n_w counts all terms in the text window (here always 200), n_d all terms in the document. P is calculated as follows for a query with m terms:

$$P(\text{Query} | \text{Window}) = [\alpha * n_{1w} / n_w + (1 - \alpha) * n_{1d} / n_d] * \dots \\ * [\alpha * n_{mw} / n_w + (1 - \alpha) * n_{md} / n_d].$$

Discourse locations may be used to display the text window with the highest P-value. This involves expanding a text window so that a number of passages form a unit—the next text window is then only given out if it does not overlap with the one preceding it. Finally, the user is yielded a list of discourse locations relevance-ranked within the document following the query.

It may prove useful during within-document retrieval to cleanse the original query of document-specific stop words. Terms that are suitable for the targeted retrieval of a document are not always equally suitable for marking the best text passage. Words that are spread evenly throughout the entire document are fairly indiscriminatory.

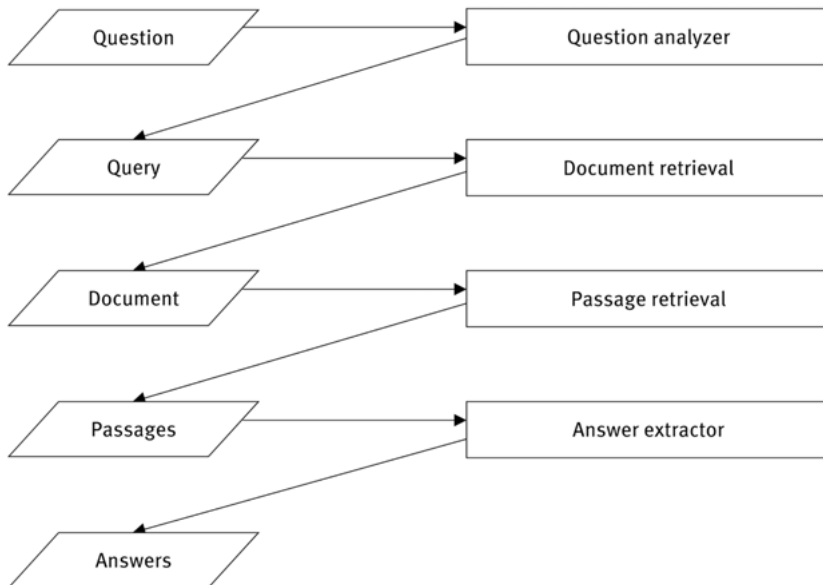


Figure G.6.1: Working Steps in Question-Answering Systems. Source: Tellex et al., 2003, 42.

Question-Answering Systems

In the case of a concrete information need aiming for factual data, the search process is finished once the respective subject matter has been transmitted. Here, the user does not desire a list display of several documents; he does not even need one (complete) document, but only that text passage which answers his question. We can use passage retrieval to answer factual questions in a Question-Answering System (Corrada-Emmanuel & Croft, 2004). Two further building blocks are added to document and passage retrieval. The question analyzer constructs a query out of the colloquial query (question) by eliminating stop words and performing term conflation. Finally, we need the answer extractor, which extracts the best answer from the retrieved pas-

sages (see Figure G.6.1). Cui et al. (2005, 400) describe the typical architecture of a question answering system (QA):

A typical QA system searches for answers at increasingly finer-grained units: (1) locating the relevant documents, (2) retrieving passages that may contain the answer, and (3) pinpointing the exact answer from candidate passages.

Since users are definitely interested in receiving not only the “pure” factual information but also its context of discussion, transmitting the most appropriate discourse passage (e.g. a paragraph) while emphasizing the query terms can satisfy their respective specific information needs (Cui et al., 2005, 400).

Passage retrieval searches (via Boolean, probabilistic or Vector Space-based models) those passages that contain the query terms as frequently and as closely beside one another as possible. The simple fact of the query terms’ occurrence, however, does not mean that the correct answer to the user’s question can be found there. Consider the following example (Cui et al., 2005, 400)! The query is:

What percent of the nation’s cheese does Wisconsin produce?

The used query terms—after eliminating stop words—are italicized. Two passages (sentences) match the query:

S1. In *Wisconsin*, where farmers *produce* roughly 28 percent of the *nation’s cheese*, the outrage is palpable.

S2. The number of our *nation’s* *producing* companies who mention California when asked about *cheese* has risen by 14 percent, while the number specifying *Wisconsin* has dropped by 16 percent.

Only sentence S1 answers the factual question. Sentence S2 contains the same search terms, but they are in another context and they do not provide the user with any relevant knowledge. A possible way of finding the best passage is in comparing the distance between words in queries and sentences. Disregarding stop words (such as “in”, “where” etc.), we see that there is a distance of 0 (direct neighborhood) between *Wisconsin* and *produce* in the query. In S1, that distance is 1 (only *farmers* lies in between), and in S2 it is 10. The shorter the distance between all search terms in the sentence, the higher the sentence’s score. An alternative consists of using the lengths of the longest similar subsequences between the question and the passage as the score (Ferrucci et al., 2010, 72). In a direct comparison between the query and S1, this would be *nation’s cheese* and thus 2, whereas S2 only reaches a value of 1. This alternative is used in the IBM system Watson DeepQA (named after the founder of IBM, Thomas J. Watson), which achieved fame through its successes on the TV quiz Jeopardy!.

However, Watson draws on several further features in calculating the score for the best answer (here called “evidence profile”) (Ferrucci et al., 2010, 73):

The evidence profile groups individual features into aggregate evidence dimensions that provide a more intuitive view of the feature group. Aggregate evidence dimensions might include, for example, Taxonomic, Geospatial (location), Temporal, Source Reliability, Gender, Name Consistency, Passage Support, Theory Consistency, and so on. Each aggregate dimension is a combination of related feature scores produced by the specific algorithms that fired on the gathered evidence.

Watson thus not only uses passage retrieval but also stores additional sources of knowledge.

Conclusion

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- From the “global” perspective, information retrieval orients itself on a document as a whole; from the “local” perspective, it takes into account the passages best suited for the respective query, as sub-units of documentary reference units.
 - Text passages are divided into discourse locations (sentences, paragraphs, chapters), semantic locations (the area of a topic) and text windows (text excerpts of fixed or variable length).
 - Passage retrieval, applied to documents, transmits the retrieval status value of the best passage to the entire document, thus aligning the texts’ Relevance Ranking with the best respective passage. This form of Relevance Ranking works independently of document length.
 - When working with text windows (of fixed size), the respective number of words depends upon the extent of the query. Short queries demand larger windows than extensive query formulations.
 - If a retrieval system uses variably-sized text windows, the specific window size is only determined once all text passages in the document have been parsed, which means that this procedure requires a lot of computation. However, the window finally identified will be the ideally appropriate one.
 - Particularly for longer documents, it makes sense to perform an in-document ranking of the individual passages’ relevance for the query. Here it may prove important to cleanse the query of document-specific stop words.
 - Question-Answering Systems serve to satisfy a specific information need by gathering factual information. This factual information may—in lucky cases—already be the most appropriate passage of a relevant document. Failing that, it can also be found via a further processing step in which possible answers are weighted. A popular example of a Question-Answering System is Watson, which managed to beat its human competitors on Jeopardy!.
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G.7 Emotional Retrieval and Sentiment Analysis

Documents do not contain only the sort of content that can be assessed rationally; they can also engage the reader/viewer emotionally. Likewise, it is possible for the documents' creators to let their feelings, judgments or moods influence their work. In this chapter, we will take a closer look at two of these aspects. Documents that express feelings, or that provoke them in the viewer, are retrieved via "emotional information retrieval" (EmIR); documents that contain (positive or negative) attitudes toward an object are processed by the means of sentiment analysis.

Emotional Laden Documents

Some documents are emotional laden (Newhagen, 1998; Neal, 2010), since they either represent emotions or provoke them in the recipient. Research results by Knautz (2012) show that information services representing the emotions of documents always need to work with two dimensions: those emotions represented in the document, and those that are induced in the viewer. These emotions are not necessarily the same. For instance: a video clip showing a person who is furious (expressed emotion: anger) might induce viewers to feel the opposite emotion, namely delight (because the person's anger is so exaggerated as to appear comical). Knautz (2012, 368-369) reports:

To explain emotional media effects, we made use of the appraisal model ... in which emotions are the result of a subjective situation assessment. The subjective importance of the event for the current motivation of the subject is crucial for triggering the emotion.

According to the *commotion model* by Scherer (...), emotions can however, also arise when looking at depicted emotions via induction (appraisal process), empathy or emotional contagion. Scherer's model was examined in relation to video, music, and pictures, and it was shown that the emotions felt not only emerge in different ways, but also that they can be completely different from the emotions depicted. This aspect is fundamental for any system that tries to make emotional content searchable.

The emotional component is particularly noticeable in images, videos or music, but it can also be detected in certain texts (e.g. poems, novels, and weblogs) and on Web pages. In many search situations it makes sense to investigate the documents' emotional content in addition to their aboutness, e.g. when searching for photographs for a marketing campaign (e.g. *scenery to put a smile on your face*), or for music to underscore a film sequence (meant to express grief). Emotional retrieval shares points of contact with "affective computing", i.e. information processing relating to emotions (Picard, 1997).

In information science, there are four ways in which it is at least principally possible to make emotions retrievable in documents (Knautz, Neal et al., 2011). Firstly, one can proceed in a content-based manner (Ch. E.4) and derive the emotions from

the document itself. There are reports of positive experiences with pattern recognition of emotional facial expressions in images, but other than that we are still far away from any operable systems. While there are a few experimental retrieval systems for emotions in music, images and video, their functionality is extremely limited. What's more, content-based methods are principally unable to grasp the second dimension of emotions induced in the viewer. The second path leads us to the indexing of documents with terms taken from a knowledge organization system (Ch. L.1 – L.5). Here we would need a KOS for emotional objects and a translation guideline for using the controlled vocabulary to express certain feelings relating to the document. We know from experiments involving the content representation of images by professional indexers that inter-indexer consistency (i.e. the congruence between different indexers in choosing appropriate terms) is very low (Markey, 1984), so this path seems bumpy at best. A possible third path involves not working with a single indexer and a predefined KOS, but letting the documents' users tag them freely. When large numbers of users create a broad folksonomy (Ch. K.1), the result is a very useful surrogate based on the taggers' "collective intelligence". The fourth path is also based on the "crowd-sourcing" approach, only here the taggers are given some predefined fundamental descriptions of feelings in the sense of a controlled vocabulary. We know from emotion research (Izard, 1991) that every emotion has both a quality (e.g. joy) and a quantity (i.e. slight joy or great joy). Following an idea by Lee and Neal (2007) for emotional music retrieval, the quantity of an emotion can be rather practically defined via a scroll bar. Scroll bars are also suitable for emotional image retrieval (Schmidt & Stock, 2009) and emotional video retrieval (Knautz & Stock, 2011). Indexers are called upon to define the intensity of both the emotions expressed in the document and those they experience themselves.

Basic Emotions

When predefining emotions, the result should be a list of commonly accepted basic emotions. For Power and Dalglish (1997, 150), these are anger, disgust, fear, joy and sadness, while Izard (1991, 49) adds some further options such as interest, surprise, contempt, shame, guilt and shyness. Describing emotions in images, Jørgensen (2003, 28) uses the five Power/Dalglish emotions as well as surprise. Both Lee and Neal (2007) and Schmidt and Stock (2009) restrict their lists to the five "classical" fundamental emotions.

Are taggers capable of arriving at a consensus (even if it is only statistically relevant) regarding the quality and quantity of emotions in documents in the first place—a consensus which could then be used in retrieval systems? Schmidt and Stock (2009) show their test subjects images, a list of fundamental emotions, and a scroll bar. Their experiment leads to evaluable data for more than 700 people. For some images there is no predominant feeling, while others achieve some clear results. Con-

cerning the photograph in Figure G.7.1 (top), the collective intelligence is of one mind: the image is about joy.

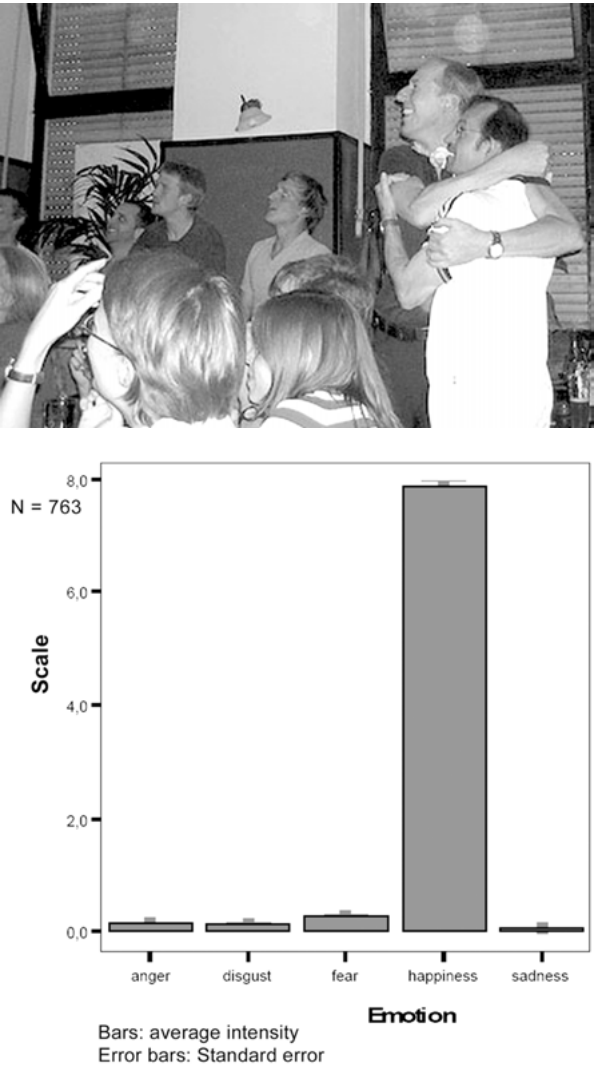


Figure G.7.1: Photo (from Flickr) and Emotion Tagging via Scroll Bar. Source: Schmidt & Stock, 2009, 868.

On a scale from 0 (no intensity) to 10 (highest intensity), the test subjects nearly unanimously vote for *joy*, at an intensity of approximately 8, while the other emotions hardly rise above 0. More than half of the images (which, it must be noted, have been deliberately selected with a view to their emotionality) achieve such clear results:

- one single fundamental emotion (two in exceptional cases) receives a high intensity value,
- the consistency of votes is fairly high (i.e. the standard deviation of the intensity values is small),
- the distance to the intensity value of the next highest emotion is great.

In Schmidt and Stock's (2009) study, no distinction is made yet between expressed and felt emotions. A further investigation into the indexing consistency of emotional laden documents, this time on the example of videos and involving the distinction between expressed and felt emotions, again shows a considerable consistency in different users' emotional estimates (Knautz & Stock, 2011, 990):

The consistency of users' votes, measured via the standard deviation from the mean value, is high enough (roughly between 1 and 2 on a scale from 0 to 10) for us to assume a satisfactory consensus between the indexers. Some feelings—particularly love—are highly consent-inducing.

Emotional Retrieval

Systems of emotional retrieval create an additional point of access to documents. In addition to their aboutness, users can now also inquire into documents' "emotiveness". Indexing is used to make sure that basic emotions represented in the work and induced in the user are allocated to the document. When enough users tag a document via scroll bar, and if an emotion is indeed present (which does not always have to be the case, after all), the users' collective intelligence should be capable of correctly and clearly naming the emotion. In the scroll-bar method, it is the indexing system's task to safeguard the identification of the basic emotions (via the average intensity values, their standard deviation and the distance to the next closest emotion).

As with all methods that rely on the collaboration of users (Web 2.0; "collaborative Web"), the latter are motivated by the predominant critical success factor. By transposing experiences from the world of digital gaming to indexing and retrieval systems ("gamification"), it becomes possible to tie users to the system via incentives. Such incentives might be points awarded for successful user actions ("experience points") or rewards for the completion of tasks ("achievements") (Siebenlist & Knautz, 2012, 391 et seq.). Incentives are used to motivate users to collaborate with the system (Siebenlist & Knautz, 2012, 402):

Extrinsic methods (e.g. collecting points and achievements) are linked to intrinsic desires (e.g. the wish to complete something). ... (G)ame mechanics offer us a means to motivate users and generate data.

As per usual, information services display the documents, but they also ask their users to participate in improving their indexing: "Has the image (the video, the Web page, the text) touched any emotions inside you? Are any emotions displayed on the

image? If so, adjust the scroll bars!” Once a critical mass of taggers is reached, emotions can be precisely identified. Here we have a typical cold-start problem, however. As long as this critical mass of different taggers has not been reached, the system at best only has some vague pointers to the searched-for emotions. In addition to these pointers, the system is dependent upon gleaning additional information—content-based this time—and, for instance, extrapolating the emotion represented in images or videos from their color distributions (Siebenlist & Knautz, 2012).

We will now demonstrate the indexing and retrieval component of an emotional retrieval system on the example of MEMOSE (Media Emotion Search). MEMOSE works with ten emotions: love, happiness, fun, surprise, desire, sadness, anger, disgust, fear, and shame. For the purpose of emotional tagging, the system works with a user interface consisting of one picture and 20 scroll bars on one Web page. The 20 scroll bars are split into two groups of 10 scroll bars each (for the 10 shown emotions and the same 10 felt emotions). The value range of every scroll bar goes from 0 to 10; the viewers are asked to use the scroll bar to differentiate the intensity of every emotion. A zero value means that the emotion is not shown on the image or not felt by the observer at all.

MEMOSE’s retrieval tool is designed to access the tagged media in the manner of a search engine. The user must choose the emotions to be searched for, and he also has the opportunity to search via topical keywords. After the query is submitted, the results are shown in two distinct columns distinguishing between shown and felt emotions (see Fig. G.7.2). Both columns are arranged in descending order regarding the selected emotions. For every search result, there is a display of a thumbnail of the corresponding picture alongside scroll bars that show the intensity of the selected emotions. The retrieval status value (RSV) of the documents (that satisfy both topic and emotion) is calculated via the arithmetic mean of the emotion’s intensity. By clicking on the thumbnail, the picture is opened in an overlay window filled with further information, such as: associated tags, information about the creator and so on. When the user clicks on “Memose me”, he is able to tag the document by emotion. The evaluation of MEMOSE is positive (Knautz, Siebenlist, & Stock, 2010, 792):

We concluded from the results of the evaluation that the search for emotions in multi-media documents is an exciting new task that people need to adapt. Especially the separated display of shown and felt emotions in a two-column raster was at first hard to cope with. And—not unimportant for Web 2.0 services—our test persons told about MEMOSE as an enjoyable system.

Sentiment Analysis and Retrieval

Texts may contain assessments—i.e. of persons, political parties, companies, products etc. Sentiment analysis (Pang & Lee, 2008) faces the task of determining the tenor of acceptance and—provided a value judgment could be identified in the first

place—to allocate it into one of the three classes “positive”, “negative”, or “neutral”. Wilson, Wiebe and Hoffmann (2005, 347) define:

Sentiment analysis is the task of identifying positive and negative opinions, emotions, and evaluations.

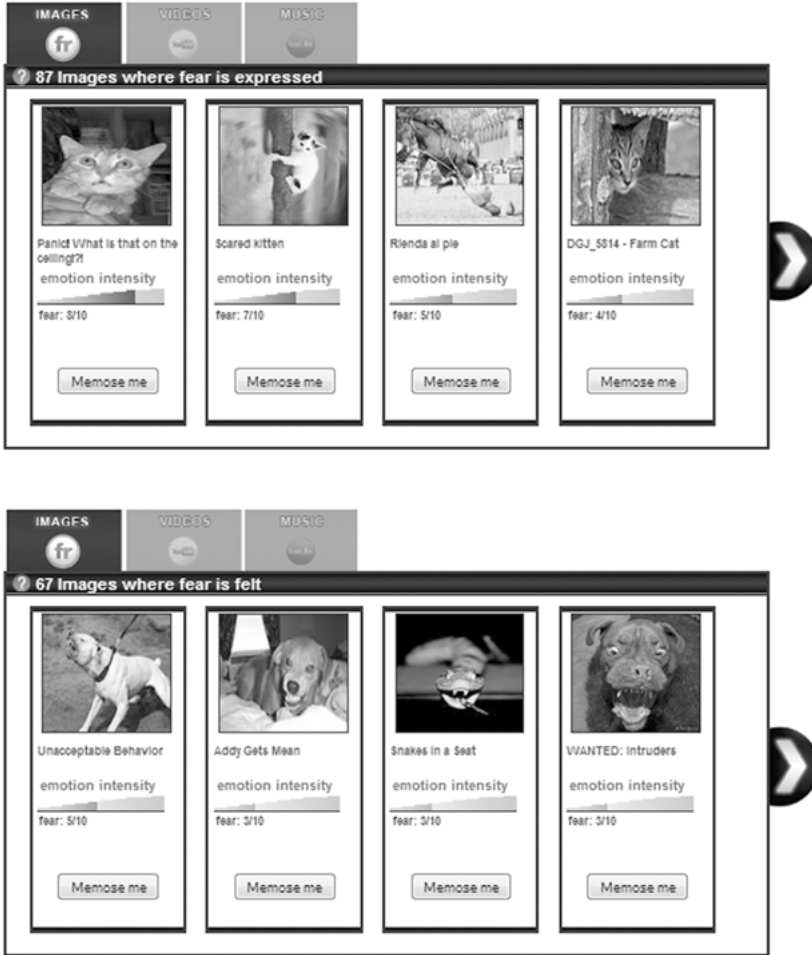


Figure G.7.2: Presentation of Search Results of an Emotional Retrieval System. Search Results for a Query on Depicted and Felt Fear. Source: MEMOSE.

Sentiment analysis is occasionally also called “semantic orientation”. Taboada et al. (2011, 267-268) exemplify the goals and advantages of this method:

Semantic orientation (SO) is a measure of subjectivity and opinion in text. It usually captures an evaluative factor (positive or negative) and potency or strength (degree to which the word, phrase, sentence, or document in question is positive or negative) towards a subject topic, person, or idea (...). When used in the analysis of public opinion, such as the automated interpretation of on-line product reviews, semantic orientation can be extremely helpful in marketing, measures of popularity and success, and compiling reviews.

In principle, all text documents can be used as sources for sentiment analyses. However, two document groups are predominantly used in practice:

- Press reports (newspapers, magazines, newswires) (Balahur et al., 2010),
- Social Media (weblogs, message boards, rating services, product pages of e-commerce services) (Zhang & Li, 2010).

Sentiment analyses of press reports are used in press reviews and media resonance analyses: which sentiment is used to thematize a company, a product, a politician, a political party etc. in the press—perhaps over the course of a certain time period? Sentiment analyses in social media lead to analyses of sentiments for companies, products etc. by users in Web 2.0 services, also sometimes represented over time. The texts of press reports are generally subject to formal control and written in a journalistic language. Texts by users in Web 2.0 services are unfiltered, unprocessed and written in the user's respective language (including any spelling or typing errors and slang expressions).

In media resonance and social media analyses, a complete document serves as the source of a sentiment analysis. If a document contains several ratings, the text will be segmented into its individual topics at the beginning of the analysis. This is done either via passage retrieval (Ch. G.6) or topic detection (Ch. F.4). Sentiment analysis starts at different levels. Abbasi, Chen and Salem (2008, 4) emphasize:

Sentiment polarity classification can be conducted at document-, sentence-, or phrase- (part of sentence) level. Document-level polarity categorization attempts to classify sentiments in movie reviews, news articles, or Web forum postings (...). Sentence-level polarity classification attempts to classify positive and negative sentiments for each sentence (...). There has also been work on phrase-level categorization in order to capture multiple sentiments that may be present within a single sentence.

A Web service that allows its users to rate a product with stars (say, from zero to five) provides sentiment analysis with additional information by, for instance, classifying a rating of four or five stars as “positive”, zero and one stars as “negative” and the rest as “neutral”. If these explicit votes conform to the result of automatic sentiment analysis, the probability of a correct allocation will rise.

The level of sentences and phrases in particular is very problematic for automatic analysis. In the model sentence *Product A is good but expensive* there are two sentiments: *Product A is good* (positive) and *Product A is expensive* (negative) (Nasukawa & Yi, 2003, 70). Constructions such as *Product A beats Product B in terms of quality* or

In terms of quality, *A clearly beats B* mean a positive rating for A and a negative one for B. The exact meaning of the negation in the context of a sentence occasionally causes problems. In the sentence *Camera A shoots bad pictures* the rating is clearly negative, whereas the sentence *It is hard to shoot bad pictures with this camera* is, equally clearly, positive—but both sentences contain the identical phrase *bad pictures* (Nasukawa & Yi, 2003, 74). Adverbs strengthen (*absolutely, certainly, totally*) or weaken (*possibly, seemingly*) the adjectives or verbs in their area of influence (*Product A is surely unsuitable—Product B is possibly unsuitable*) or even turn the adjective’s meaning into its opposite (*The concert was hardly great*) (Benamara et al., 2007).

Sentiment Indicators

Sentiment analysis goes through three steps:

- Compiling a list of general or domain-specific indicator terms for sentiments,
- Analyzing the documents via a list of indicator terms,
- Allocating the retrieved ratings to the document.

Sentiment indicators are often domain-specific and must be created anew for each area of application. Semi-automatic procedures prove useful here. In a first phase, one searches documents that are typical for the domain by starting with words whose value orientation is definitely positive or negative (Hatzivassiloglou & McKeown, 1997). The terms *good, nice, excellent, positive, fortunate, correct, and superior*, for instance, have a positive evaluative orientation, while *bad, nasty, poor, negative, unfortunate, wrong, and inferior* are negative (Turney & Littman, 2003, 319). The two lists with positive and negative indicator terms serve as “seed lists”. All terms that are linked to a seed list term via “and” or “but” are marked in the document. Those terms found via “and” frequently, but not always (Agarwal, Prabhakar, & Chakrabarty, 2008) bear the same evaluative orientation as the original term, while those connected via “but” bear the opposite (with the same amount of uncertainty). Hatzivassiloglou and McKeown (1997, 175) justify their suggestion:

For most connectives, the conjoined adjectives usually are of the same orientation: compare *fair and legitimate* and *corrupt and brutal* ... with **fair and brutal* and **corrupt and legitimate* ... which are semantically anomalous. The situation is reversed for *but*, which usually connects two adjectives of different orientations.

The resulting list of indicator terms must be intellectually reviewed and, if necessary, corrected in a second phase. Exceptions to the general usage of “and” and “but” must be taken into account. In the sentence *Not only unfunny, but also sad*, the algorithm would allocate “unfunny” and “sad” into different classes due to the “but”, even though both are meant negatively (Agarwal, Prabhakar, & Chakrabarty, 2008, 31).

A domain-independent list of terms and their respective sentiments is submitted by Baccianella, Esuli and Sebastiani (2010). SentiWordNet names three values for every synset from WordNet (Ch. C.3), which state the degree of “positivity”, “negativity” and “neutrality” (the sum of these values for each term is always 1). Such a list can serve as source material or as a corrective for individual indicator term lists.

In the second working step, the derived list of indicator terms is applied to the respective document sets. Systems either work fully automatically (and thus, due to the above-mentioned linguistic problems, with a degree of uncertainty regarding the level of sentences and phrases) or only provide heuristics for subsequent intellectual sentiment indexing. This path (or even an exclusively intellectual variant) is chosen by commercial service providers in the area of press reviews and Web monitoring. In case of very large quantities of data, there is a possibility—however “unclean”—of an automatic variant. Godbole, Srinivasaiah and Skiena (2007a, 2007b) connect indicator lists (for sports, criminality, the economy etc.) with certain “named entities” (for which synonym lists are available) on the sentence level. The document basis consists of news that are accessible online. If the “named entity” and an indicator term co-occur in a sentence, the result is a quantitatively expressed hit: +1 (for a positive indicator, e.g. *is good*), -1 (for a negative indicator, e.g. *is not good*). For a modifier (such as *very*) the score changes, e.g. *very good* will be weighted with +2. At this point, there are several options for creating parameters (Godbole, Srinivasaiah, & Skiena, 2007a, 0053 et seq.):

- average sentiment: the arithmetic mean of the raw values,
- “polarity”: the relative frequency of positive sentiments with regard to all evaluative text passages,
- “subjectivity”: the relative frequency of text passages with sentiments in relation to all text passages.

“Subjectivity” shows how emotional laden news relating to a topic are (independently of their orientation), “polarity” and average sentiment give pointers to the sentiments’ orientation.

When one wants to make recognized sentiments searchable in a retrieval system, the scores (or merely their orientation) must be stored in a special field in the documentary unit. This gives users the option of only searching for positively (or negatively) slanted documents, or for neutral ones.

In the sense of informetric analyses, it is possible to create rankings by sentiment, e.g. a ranking of the “top positive” movie actors (in descending order of average sentiment values) or of “top negative” politicians (in ascending order of sentiment values). Time series show how estimations change over time.

Rating Web Pages: Folksonomy Tags with Sentiment

Web pages leave an impression on their users, for good or ill. When tagging such pages in social bookmarking services (such as Delicious), users sometimes report their experiences. If content is indexed exclusively on the basis of the documents' aboutness, tags such as *awful* or *useful* are generally beside the point. However, such terms are ideal for positive-negative evaluations of sources in sentiment retrieval. Yanbe et al. (2007, 111) emphasize:

One of the interesting characteristics of social bookmarking is that often tags contain sentiments expressed by users towards bookmarked resources. This could allow for a sentiment-aware search that would exploit user feelings about Web pages.

In their analysis of sentiment tags in the Japanese service Hatena Bookmarks, Yanbe et al. (2007) find evaluative tags that have been attached to Web resources thousands of times. Positive ratings are particularly widespread, whereas negative ratings are rather rare (Yanbe et al., 2007, 112):

Only one negative sentiment tag was used more than 100 times (“it’s awful”; more than 4,000 times, A/N). This means that social bookmarkers usually do not bookmark resources to which they have negative feelings.

The number of positive and negative evaluations of each Web resource on a social bookmarking service can be determined via two lists with positive and negative words, respectively; multiple allocations of tags must be taken into account. Subtracting the number of negative votes from the positive votes results in a document-specific sentiment value. If this is greater than zero, the tendency is toward “thumbs up”, if it is below zero—“thumbs down”.

Conclusion

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- Documents are sometimes emotional laden, as they either represent feelings or induce them in the recipient. Searching for and finding of such documents are tasks of emotional information retrieval (EmIR).
 - Tagging via controlled vocabularies in particular is a useful method of knowledge representation for EmIR. The system predefines some basic emotions (such as anger, disgust, fear, joy, sadness), whose perceived intensity is adjusted by the user via a scroll bar.
 - Sentiment analysis uncovers positive or negative evaluations of people, companies, products etc. that may be contained in documents. Objects of analysis are generally press reports as well as social media in Web 2.0 (blogs, message boards, rating services, product pages in e-commerce).
 - In social bookmarking services, evaluative tags such as “awful” or “great” can be viewed as indicators toward the user’s sentiment in rating a Web page.

- Sentiment analysis and retrieval are used for press reviews and media resonance analyses and Web 2.0 monitoring services, as well as for rating Web pages.
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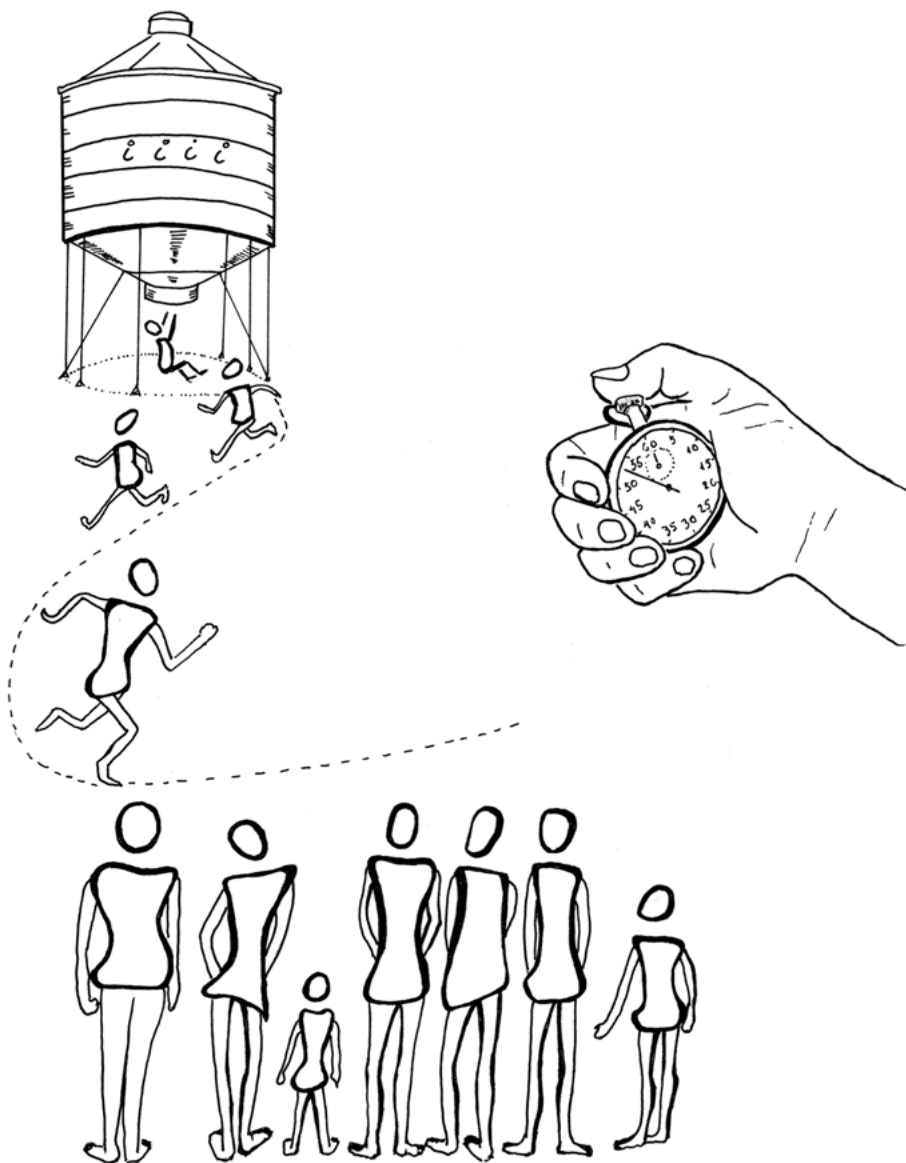
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Part H

Empirical Investigations on Information Retrieval



H.1 Informetric Analyses

Subjects and Research Areas of Informetrics

According to Tague-Sutcliffe, “informetrics” is “the study of the quantitative aspects of information in any form, not just records or bibliographies, and in any social group, not just scientists” (Tague-Sutcliffe, 1992, 1). Egghe (2005b, 1311) also gives a very broad definition:

(W)e will use the term ‘informetrics’ as the broad term comprising all-metrics studies related to information science, including bibliometrics (bibliographies, libraries, ...), scientometrics (science policy, citation analysis, research evaluation, ...), webometrics (metrics of the web, the Internet or other social networks such as citation or collaboration networks).

According to Wilson, “informetrics” is “the quantitative study of collections of moderate-sized units of potentially informative text, directed to the scientific understanding of information processes at the social level” (Wilson, 1999, 211). We should also add digital collections of images, videos, spoken documents and music to Wilson’s units of text. Wolfram (2003, 6) divides informetrics into two aspects,

system-based characteristics that arise from the documentary content of IR systems and how they are indexed, and usage-based characteristics that arise from the way users interact with system content and the system interfaces that provide access to the content.

We will follow Tague-Sutcliffe, Egghe, Wilson and Wolfram (and others, for example Björneborn & Ingwersen, 2004) in calling this broad research of empirical information science “informetrics”. Informetrics therefore includes *all* quantitative studies in information science. When a researcher performs scientific investigations *empirically*, concerning for instance the behavior of information users, the scientific impact of academic journals, the development of a company’s patent application activity, Web pages’ links, the temporal distribution of blog posts discussing a given topic, the availability, recall and precision of retrieval systems, the usability of Web sites, etc., he is contributing to informetrics. We can make out three subject areas of information science in which such quantitative research takes place:

- information users and information usage,
- evaluation of information systems and services,
- information itself.

Following Wolfram, we divide his system-based characteristics into the categories “information itself” and “information systems”. Figure H.1.1 is a simplistic graph of subjects and research areas of informetrics as an empirical information science.

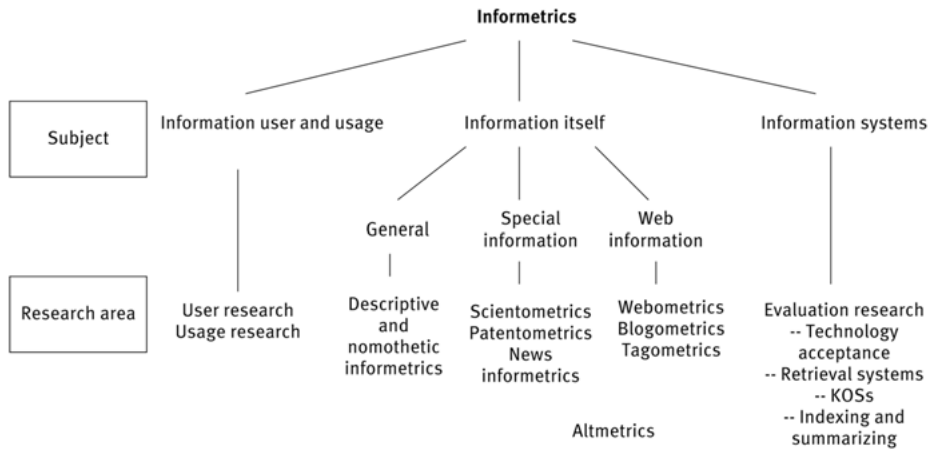


Figure H.1.1: Subjects and Research Areas of Informetrics. *Source:* Modified from Stock & Weber, 2006, 385.

The term “informetrics” (from the German “Informetrie”) was coined by Nacke (1979) in the Federal Republic of Germany and by Blackert and Siegel (1979) in the German Democratic Republic.

Nomothetic and Descriptive Informetrics

“Information itself” can be studied in various ways. Generally, we work descriptively with information, information flows or content topics and try to derive informetric regularities by generalizing the descriptive propositions or using mathematical models (Egghe & Rousseau, 1990). Examples include the laws of Lotka, which is a power law (Egghe, 2005a), or the “inverse logistic” distribution of documents by relevance (Stock, 2006). Following the Greek notion of “nomos” (law), we will call this kind of empirical information science “nomothetic informetrics” (Stock, 1992, 304). Some typical nomothetic research questions are “What kind of distribution is adequate to multi-national authorship?” or “Are there any laws for the temporal distribution of, say, blog posts?”

In contrast to nomothetic informetrics, there is descriptive informetrics, which analyses individual items such as individual documents, subjects, authors, readers, editors, journals, institutes, scientific fields, regions, countries, languages and so on. Typical descriptive research questions are “What are the core subjects of the publications of Albert Einstein?” or “How many articles did Einstein publish per year throughout his entire (scientific) life?” If there are any known informetric laws, researchers can compare the findings of their descriptive work to these laws. In this way, they can

make a distinction between “typical” individual distributions (if the individual’s data approximate one of these laws) and non-typical distributions.

Methods of data gathering in informetrics concerned with information itself comprise citation analysis (Garfield, 1972; Garfield, 1979; Cronin & Atkins, Ed., 2000; for problems of citation analysis, see MacRoberts & MacRoberts, 1996) and publication analysis (Stock, 2001), including subject analyses of publications. The challenge of descriptive and nomothetic informetrics is the creation of a meaningful set of search results for analysis (see Chapter H.2).

The study of information itself has also been called “bibliometrics” (Pritchard & Wittig, 1981; Sengupta, 1992). This term is sometimes used in the context of scientometrics. However, since “bibliometrics” refers to books (the Ancient Greek “bíblos” meaning “book”), it is more appropriate to use the term “informetrics” as the broadest term, since it contains all kinds of information.

Scientometrics

According to van Raan, “(s)cientometric research is devoted to quantitative studies of science and technology” (van Raan, 1997, 205). The main subjects of scientometrics are individual scientific documents, authors, scientific institutions, academic journals and regional aspects of science. Scientometrics exceeds the boundaries of information science. “We see a rapid addition of scientometric-but-not-bibliometric data, such as data on human resources, infrastructural facilities, and funding” (van Raan, 1997, 214). Information science oriented scientometrics examines aspects of information and communication, in contrast to economics, sociology and psychology of science. Those aspects may include productivity (documents per year), topics of the documents (words, co-words), reception (readers of the documents) and formal communication (references and citations, information flows, co-citations) (for journal informetrics, see Haustein, 2012).

Scientometrics concentrates solely on scientific information. There are other kinds of special information, most importantly patent information and news information. Quantitative studies of patent information may be called “patentometrics”, “patent bibliometrics” (Narin, 1994) or “patent informetrics” (Schmitz, 2010), empirical studies of news “news informetrics”. Patentometrics, scientometrics and news informetrics are capable of producing some interesting indicators for economics (for patentometrics, see Griliches, 1990).

Webometrics and Web Science

In short, webometrics is informetrics on the World Wide Web (Björneborn & Ingwersen, 2001; Cronin, 2001; Thelwall, Vaughan, & Björneborn, 2005). According to

Björneborn and Ingwersen (2004, 1217), webometrics consists of four main research areas:

- (1) Web page content analysis; (2) Web link structure analysis; (3) Web usage analysis (including log files of users' searching and browsing behavior); (4) Web technology analysis (including search engine performance).

There are clear connections to other informetric activities. Web page content analysis is a special case of subject analysis, Web link structure study (Thelwall, 2004) has its roots in citation analysis, Web usage analysis is part of a more general user and usage research, and Web technology analysis refers to information systems evaluation.

Web Science is a blend of science and engineering (Berners-Lee et al., 2006, 3):

(T)he Web needs to be *studied* and understood, and it needs to be *engineered*. At the micro scale, the Web is an infrastructure of artificial languages and protocols; it is a piece of engineering. But the linking philosophy that governs the Web, and its use in communication, result in emergent properties at the macro scale (some of which are desirable, and therefore to be engineered in, others undesirable, and if possible to be engineered out). And of course the Web's use in communication is part of a wider system of human interactions governed by conventions and laws.

As scientometrics is not only information science (but consists additionally of sociological, psychological, juridical and other methods), so does Web Science as well. Other branches besides information science (such as the social sciences and computer science) contribute to Web Science and form a multidisciplinary research area.

Webometrics and Web Science find their subjects on the World Wide Web. However, this is only one of the Internet's services. If we include all services such as e-mail, discussion groups and chatrooms, it is possible to speak about "cybermetrics". We can define special branches of webometrics as well. Ergo, analyzing the blogosphere informetrically leads to "blogometrics", which also represents a kind of special information, namely blog posts, podcasts and vodcasts (video podcasts). A lot of tools for analyzing microblogging posts and users have emerged with the advent of Twitter and (in China) Weibo, e.g. measuring information flows within tweets during scientific conferences (Weller, Dröge, & Puschmann, 2011). "Tagometrics" studies keywords (tags), their development and their structure in folksonomies.

The use of informetric tools in combination with "new" media (such as blog posts, tweets, bookmarks or tags in sharing services) is called "altmetrics" (for "alternative metrics") or, in relation to scientometrics and Web 2.0 services, "scientometrics 2.0" (Priem & Hemminger, 2010; Haustein & Peters, 2012).

There are close relations between general descriptive and nomothetic informetrics and special applications such as scientometrics and webometrics. In general informetrics, for example, co-citation analysis is a way of mapping the intellectual structure of a scientific field. In webometrics, co-link analysis also leads to the production of a map, but this map does not necessarily represent intellectual or cognitive

structures (Zuccala, 2006). Thus the application of informetric methods in specialized fields of empirical information science is not necessarily the same, sometimes merely proceeding by analogy. In the early days of webometrics, links between Web pages and citations were seen as two sides of the same coin. Web pages “are the entities of information on the Web, with hyperlinks from them acting as citations” (Almind & Ingwersen, 1997, 404). Today we have to recognize specific differences between links and citations, e.g. that links are time-independent and citations are not. They are “actually measuring something different and therefore could be used in complementary ways” (Vaughan & Thelwall, 2003, 36). Another example is the study of citations in microblogging services. While in classical informetrics a citation is a formal mention of another work in a scientific publication, in Twitter or Weibo a citation is—by analogy—a link pointing to external URLs or a retweet that “cites” other users’ tweets (Weller, Dröge, & Puschmann, 2011).

User and Usage Research

The topics of user research are humans and their information behavior (Wilson, 2000). Information-seeking behavior on the Web (especially the usage of search engines) is well documented (Spink, Wolfram, Jansen, & Saracevic, 2001; see Ch. H.3). Typical research questions include the users’ information needs, the search process, the formulation of queries, the use of Boolean operators, the kinds of questions (e.g. concrete versus problem-oriented), the topics searched for and the number of clicked hits in a search engine’s list of results. Similar studies have been conducted on both users and usage of libraries’ and commercial information providers’ services. User research distinguishes between user groups, such as information professionals, professional end users and end users (Stock & Lewandowski, 2006) or between author, reader and editor of scientific journals (Schlögl & Stock, 2004; Schlögl & Petschnig, 2005).

Methods of user research include observations of humans in information-gathering situations, questionnaires and surveys as well as analyses of log files. Methods of usage research include log files, statistics of downloads, numbers of interlibrary loan cases, lending numbers (in libraries) and data on social tagging in STM bookmarking systems (Haustein et al., 2010). The results of user and usage research can be applied to performance and quality studies of information services.

Evaluation Research

Retrieval systems are a special kind of information system and fulfill certain functions. Any overall inspection of retrieval systems has to consider the aspects of information systems’ and information services’ evaluation and include models of technology acceptance (see Ch. H.4).

Knowledge representation analyzes knowledge organization systems (like thesauri or classification systems) and the indexing and abstracting process. All KOSs and all working steps of knowledge representation are objects of evaluation (see Ch. P.1 and P.2).

Conclusion

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- We summarize all empirically-oriented information-scientific research activity under the term “informetrics”. The subjects of informetrics are information itself, information systems as well as the users and usage of information.
 - Descriptive informetrics describes concrete objects, while nomothetic informetrics aims at discovering laws.
 - Scientometrics analyzes information from the scientific arena. Further specific kinds of information that can be informetrically analyzed include patent writs (patent informetrics) and news items (news informetrics).
 - Webometrics applies informetric approaches to the World Wide Web. In the area of Social Media in particular, e.g. Twitter and blogs, there are tools for analyzing information and its users.
 - User or usage research addresses information research behavior as well as users’ dealings with information systems.
 - Retrieval and knowledge representation evaluation are subareas of the evaluation of information systems and services.
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H.2 Analytical Tools and Methods

Online Informetrics

The goal of “normal” search is the retrieval of specific documentary units that satisfy an information need. An appropriate analogy here would be the search for the proverbial “needle in the hay”. We will put the search for individual needles aside for now and turn to larger units (Wilson, 1999a). The object now is certain *quantities* of documentary units that are qualified as a whole. These can be, for instance, a company’s range of patents, a scholar’s or a working group’s texts, publications from a university, a city, region or country, or concerning a certain subject matter. In “normal” searches, the results we get are entries that have been entered into the system in the exact same manner in which it is retrieved for us now (excepting the layout). In informetric analyses, on the other hand, we create new information: information that has never been explicitly entered into the database and which only transpires via the informetric search and analysis procedure itself. Wormell (1998, 25) views informetric analysis as a tool for the analysis of database contents.

Informetric analysis offers many new possibilities for those who want to explore online databases as analytical tools. Informetric analysis assumes that databases can be used not only for finding facts and accessing documents, but also for tracing the trends and developments in society, science, and business.

The goals of informetric analyses are:

- to support “normal” retrieval of single documentary units (via the refinement of retrieval strategies),
- to support information retrieval systems’ evaluation (via quantitative statements on content and usage of retrieval systems; Wolfram, 2003),
- to support descriptive and evaluative procedures (by providing raw data) in the service of scientometrics, patentometrics, news informetrics, etc. (Persson, 1986), diffusion research (to broaden knowledge) as well as competition and industry analyses, respectively (Stock, 1992; 1994).

The basis are documentary units from publications (scientific works, patents, newspaper articles etc.), where the field entries contained within them (including for example author, affiliation, title terms, citations, volume, source, descriptors, notations) are evaluated quantitatively. Depending on what is at the center of the information need, we distinguish between publication analyses (e.g. “who has published the most about the subject X?”) (Stock, 2001), citation analyses (e.g. “which of a scientist’s works is cited the most?”) (Garfield, 1979) and subject analyses (e.g. “which research subjects are of central importance for the group of scientists X and how do they cohere?”) (Stock, 1990). Wilson (1999a, 117) observes:

Informetric research can be classified several ways, for example, by the type of data studied (e.g., citations, authors, indexing terms), by the methods of analysis used on the data obtained (e.g., frequency statistics, cluster analysis, multidimensional scaling), or by the types of goals sought and outcomes achieved (e.g., performance measures, structure and mapping, descriptive statistics).

Informetric analyses draw on the contents of specialist and general databases (such as Web of Science, Scopus or Derwent World Patents Index, for instance). In this book, we will pass by empirical methods and results while concentrating on the informetric analysis functionality.

This functionality is made available online (as “online informetrics”) by several commercial information services (such as DIALOG, Questel, STN International with command-oriented interfaces or Thomson Reuters’ “Web of Knowledge” with menu navigation). Since not all functions are available online at all times, many informetric topics require additional software for further offline processing, such as the program FUN for publication and citation analysis (Järvelin, Ingwersen, & Niemi, 2000) or HistCite for the analysis and representation of knowledge domains (Garfield, 2004).

Selecting the Document Set

The decisive aspect in informetric search is the selection of the quantity of documentary units that is meant to be analyzed. This depends on the retrieval strategy on the one hand, but also on the selected databases’ content. It would be illusory to suppose that specialized databases are ideally complete. Even if one summarizes several relevant databases of an information provider, the ideal result will still be a ways off. Wilson (1999b) intellectually produced an ideally complete data collection and compared it with the contents of an information provider (DIALOG). Wilson (1999b, 651) observes:

It is apparent that the databases are giving only samples, albeit quite large ones, of the Exhaustive Collection, but not approximations of it.

It can be shown, however, that the respective sample is a fairly good approximation of the parameters of the complete collection—the absolute values are of little use, then, but some derived values (such as rankings) may turn out to be serviceable. Wilson (1999b, 664) sums up the results of her case study:

I conclude that with respect to size-invariant features in both classes of distribution, the Dialog sample matches closely those of the (...) Exhaustive Collection, i.e. it ‘estimates these population parameters’ well.

If one works with several databases, duplicates may arise that have to be removed without fail (unless the intersection itself is interpreted as an informetric indicator). Additionally, one must decide which documents will remain and which are to be deleted. Ingwersen and Hjortgaard Christensen (1997, 208) suggest implementing the process of duplicate removal in differing database order (“reversed duplicate removal”) in order to be able to exploit the advantages of single databases.

When dealing with a data set from a cluster of files, the removal of duplicates is mandatory for any subsequent analysis. Operationally the removal is easy. However, the fundamental question is: From which databases do we want to remove the duplicates, and in which file to keep them? Awareness of this issue, and thus of the so-called Reversed Duplicate Removal (RDR) technique, is crucial for the outcomes of the subsequent analysis.

RDR is useful when two (or more) databases are suitable for informetric purposes, but each in a different way. If, for instance, a database A offers a field for the authors’ affiliation and the other (B) does not, we will call up the databases in the order A, B and evaluate the affiliation statements informetrically. If on the other hand database A has a rather bad indexing and B a more exhaustive one, we will take a second step and call up the databases in the order B, A so as to implement a subject analysis.

Hood and Wilson (2003) provide a list of all possible problems that searchers should always keep in mind when performing informetric analyses. The following difficulties may arise on the micro level, the single database level and on the level of its data sets (Hood & Wilson, 2003, 594-596):

- Naming Variants: abbreviation standards, differences between American and British English, transliteration standards used in non-romance languages,
- Typos,
- (lack of) Consistency in Subject Indexing,
- Name Assignment / Synonyms: How are different variants of a person’s name summed up?
- Name Assignment / Homonyms: How are similar-sounding names of different persons—such as ‘Miller, S’—disambiguated?
- Journal Titles: Due to individual journals’ occasional lack of clear identifiability, one might use the ISSN instead of the title,
- Time Designations: In time intervals, there are variants—e.g. ‘1983-1984’, ‘1983-84’, ‘83-84’—that must be summed up,
- Statements on the Authors’ Affiliation / Synonyms: Problems arise due to differing statements regarding institutions. This is because the authors or the patent applicants each state a different variation of their employer’s name, such as “Nat. Phys. Lab., Taddington, UK”
“Nat. Phys. Labs.; Taddington, UK”
“National Phys. Lab., Taddington, UK”
“NPL, Taddington, UK” etc. (Ingwersen & Hjortgaard Christensen, 1997, 214).
Breitzman (2005, 1016) reports of assignation problems with patent holders:

Those unfamiliar with patent analysis may not recognize the significance of the assignee name problem. Most large organizations patent under between 10 and 100 names, many of which are not obviously related to the parent name. For example, of the 300+ names that Aventis Pharmaceuticals patents under, only 2% of the patents are assigned to a name with 'Aventis' in the assignee string.

The variants must be recognized as synonyms and intellectually summed up,

- Affiliation / Homonyms: As with personal names, similar-sounding designations of different institutions, locations or countries must be disambiguated. If, for instance, one is to perform an informetric analysis of "Wales" (in the U.K.), any documents by authors from "New South Wales" (in Australia) must be deleted (in so far as the database constructs a word index),
- Field Structure: Databases sometimes sum up different statements in a single field, which can then no longer be cleanly disambiguated. If, for instance, year, volume, issue number, page numbers and other statements are summed up in the field "source"—and in no particular order at that—it can be difficult, impossible even, to cleanly extract single pieces of information (such as the year).

The macro level regards a database as a whole, while simultaneously putting it into a context with other databases (Hood & Wilson, 2003, 596-599):

- Coverage: In specialist databases, it would be important to know the ratio of the literature indexed within the field in question relative to the overall amount of literature available. It is difficult to measure a specific degree of coverage, however, since there are no reliable data on what documents are to be assigned to this particular field. Many databases prefer certain languages (mostly English) or journals from certain countries (mostly the U.S.). There are databases that restrict themselves to the indexing of certain document types (thus Scopus, for instance, only analyzes journal and proceedings articles, to the detriment of books),
- Duplicates: Typos can lead to documentary reference units being registered multiple times. The duplicates must be removed, as in cross-database retrieval,
- Time: The time period in which the database has registered documents,
- Time Delay in the Indexing Process: The time that passes between the release of a documentary reference unit and the appearance of said unit in the online database,
- Fields and Field Entries: Which analyzable fields are there in the first place? Are the available fields always filled with content? When using multiple databases, the field abbreviations (and the field contents, of course) must match,
- Database Policy: Database producers change their policy from time to time, i.e. registering further journals or document types or adding full texts to the bibliographical documentary units. When a certain sample increases significantly over exactly one year, this can thus mean two things: first, a "real" rise of published content in that year or, second, the indexing of new journals,

- **Inverted File:** For compound expressions, it is important that there is an index of phrases. If, for instance, one uses a word index in a ranking of names, the given names will be separated from the surnames and the finished list should begin with the most common given name.

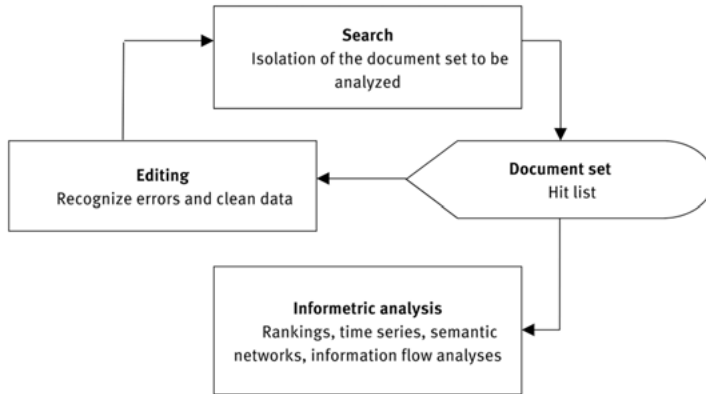


Figure H.2.1: Working Steps in Informetric Analyses.

Databases have been set up to allow information seekers to perform their “normal” searches. The goal is to establish relevant documentary units. When designing their databases, the producers often did not consider informetric analyses. Therefore, one must take care during informetric analyses to draw sustainable samples via elaborate retrieval strategies. Then one must process them in such a manner that (halfway) reliable statements can be made about the related basic population (see Figure H.2.1). Hood and Wilson (2003, 604) summarize:

The main problem is that most electronic databases are designed as Information Retrieval tools, and not as Informetric tools. Electronic databases can and are used as data sources for informetric studies, but the data usually requires significant manipulation or cleaning up. There is no such thing as clean data in electronic databases.

In the following, we will distinguish between four basic forms of informetric analyses: rankings, time series, semantic networks and information flow analyses (Stock, 1992).

```

                                Dialog Response
                                -----
Started processing RANK
...Ranking 100 of 232 records
...Ranking 200 of 232 records
Completed Ranking 232 records
DIALOG RANK Results
-----
RANK: S33/1-232   Field: IC=   File(s): 342
(Rank fields found in 226 records -- 117 unique terms) Page 1 of 15

RANK No.  Items  Term
-----
    1      51   C11D
    2      37   G08G
    3      25   B21B
    4      21   C07C
    5      19   B01J
    6      16   A61K
    7      14   G01C
    8      14   G06F

P = next page      Pn = Jump to page n
P- = previous page M = More Options      Exit = Leave RANK

```

Figure H.2.2: Command-Based Informetric Analysis on the Example of DIALOG. Source: Stock & Stock, 2003, 27.

Rankings

Rankings sort certain entries according to how frequently they appear in the retrieved document set. Such a ranking will typically take the form of a Power Law or an Inverse Logistic Distribution. If an empirically gathered distribution strays from the expected values, this may be an indicator of a “special” document set. There are two alternatives for explaining such a particularity: first, one might have searched a document set that doesn’t represent any unified sample (but a random set instead), or, second, the document set does indeed deviate from the expected scenario.

If the correct document set has been selected, the ranking will be searched via the command RANK and the argument field’s abbreviation, further parameters if needed, direction (ascending or descending order) as well as the number of field content to be retrieved.

We will exemplify the formation of a ranking via an example—performed with the information provider DIALOG (Stock & Stock, 2003). Our search concerns citations of European patents whose priority date is 1995 and which have been filed by a Düsseldorf-based enterprise. We are interested in which technological areas in Düsseldorf are most effective. In the start screen of DialogWeb, we select search via commands. File 348 (European Patents Fulltext) offers itself for our search, since it contains a field for the location of the patent applicant (CS). We search for granted patents (DT=B), for company seats in Düsseldorf (CS=Dusseldorf) and for the year of our priority date (AY=1995/PR), receiving 291 hits (Figure D.2.1). As our next step

is to search for citations of these patents, we proceed to search via File 342 (Derwent Patents Citation Index). The field abbreviation for cited patents is CT. We thus have to rename the numbers of our Düsseldorf patents (in the field PN of File 348) to CT. This is done via the MAP command and the field prefix option MAP PN/CT=. The provisional result is saved under a number (such as SC004). The resulting list will contain all patent numbers of the Düsseldorf patents as well as their family members in the European Patent Office. This is how we were able to generate 465 search terms for the 291 search results. Now we change databases (B 342) and perform our stored search (EXECUTE STEPS SC004). We receive 232 patents in total, all of which having cited (at least) one of the original patents. The question of the most-cited technological areas can be answered via the RANK command. We want to represent the areas as four-digit numbers of the International Patent Classification and thus formulate RANK (IC 1-4). The result is shown in Figure H.2.2. Our 1995 Düsseldorf patents turn out to have been most effective in the IPC classes C11D (detergents) and G08G (traffic control systems).

Results Analysis

177 records. TS=(informetr*)

Select field to rank by:	Analyze:	Set display options:	Sort by:
☒ Author ☐ Country/Territory ☐ Document Type ☐ Institution Name	☐ First 500 records ☒ All (up to 2000 records)	Show the top 10 results. Minimum record count (threshold): 2	☒ Record count ☐ Selected field

Rank results by the selected field.

Use the checkboxes below to view the records.

Note: The number of records displayed may be greater than the listed Record Count if the original set contained more records than the number of records analyzed.

VIEW RECORDS	Field: Author	Record Count	% of 177	Bar Chart	SAVE ANALYSIS DATA TO FILE
☐	Egghe, L	20	11.3 %	■	SAVE ANALYSIS DATA TO FILE
☐	Rousseau, R	14	7.9 %	■	
☐	Burrell, QL	10	5.6 %	■	
☐	Wilson, CS	9	5.1 %	■	
☐	Bookstein, A	7	4.0 %	■	
☐	Ingwersen, P	7	4.0 %	■	
☐	Wormell, I	7	4.0 %	■	
☐	Bar-Ilan, J	6	3.4 %	■	
☐	Wolfram, D	6	3.4 %	■	
☐	Hood, WW	5	2.8 %	■	

Figure H.2.3: Menu-Based Informetric Analysis on the Example of Web of Knowledge. Source: Web of Knowledge.

In Figure H.2.3 we see an example of a menu-based informetric analysis, as provided, for example, by Web of Knowledge. After isolating the hit list to be analyzed, one can literally generate rankings at the push of a button (without any way of editing the list, however). Thus it is possible, for example, to arrange the document set in a ranking according to the number of citations. The top spots are accordingly taken by the most-cited documents for the given subject. The “Analyze” button generates

rankings for authors, institutions etc. The example in Figure H.2.3 shows the most productive authors on the subject “Informetrics”.

Informetric Time Series

The question concerning informetric time series is: how does a subject change over time? Some sample questions include: how have an enterprise’s patent application activities developed over the last 20 years? How many publications per year are available for subject X since 1980? How many articles per year has an author Y written throughout his academic career? In the selected document set, the only thing of interest is the content of the year field with its related number of documentary units. If one wants to compare several document sets (i.e. the patents of the two most patent-intensive companies in a technological area) with one another, there are two options. The first one initially creates a ranking and then generates a time series for the two top companies. The second variant connects both steps. The host STN International offers the TABULATE command to directly process rankings further (STN, 1998). If, for instance, we want to know the top ten researchers (via their number of publications) in the area of nanoelectronics, and when they have published their results, we will open the relevant physics databases in STN, formulate a search for “nano? AND single electron”, delete the duplicates, create rankings (in STN: ANALYZE) for the author and year fields and enter the command TABULATE (with the output format, the field contents to be entered into the chart, the number of values as well as the sorting direction for both fields). The primary sorting key is the author field, the sorting is in descending order of frequency; correspondingly, the secondary sorting key is the year field, which is ranked in descending order of the field contents. The result is a chart as displayed in Figure H.2.4. An additional useful editing option is the graphical representation of the time series.

AU	1997	1996	1995	1994	1992	1998	1993	1991	1982	1985	1986
MATSUMOTO, K.	10	12	4	0	0	1	0	0	0	0	0
CHOU, S.Y.	4	1	2	0	4	0	0	0	0	0	0
AHMED, H.	2	2	2	0	0	1	0	0	0	0	0
BANDYOPADHYAY, S.	2	3	0	2	0	0	0	0	0	0	0
DOROGI, M.	0	3	3	0	0	0	0	0	0	0	0
ISHII, M.	1	3	2	0	0	0	0	0	0	0	0
LEOBANDUNG, E.	3	1	2	0	0	0	0	0	0	0	0
LINGJIE GUO	4	0	2	0	0	0	0	0	0	0	0
YOKOYAMA, N.	4	0	1	0	0	1	0	0	0	0	0
ANDRES, R. P.	2	1	2	0	0	0	0	0	0	0	0

Figure H.2.4: Informetric Time Series on the Example of STN International via the TABULATE Command. Source: STN, 1998, 7.

Semantic Networks

Searchable aspects in documentary units, such as authors, descriptors, notations, affiliation statements (institutions, cities, and countries), citations and words in the continuous text often do not arise in isolation. They stand in certain relations to one another: authors co-publish an article (co-authors), for instance, and analogously one can analyze relations between research institutes, cities or countries via the given affiliation. Normally, a documentary unit is allocated several text words, keywords, descriptors of a thesaurus, or notations, which thus become co-subjects (or specific co-text words, co-keywords, co-descriptors and co-notations). When a documentary unit has multiple bibliographical references (e.g., in a bibliography), these are all citations. All words in the continuous text (or in a text window with an overseeable number of terms) are analyzed as co-words, a little more elaborately, but using the same basic principle.

After selecting the document set, the following values must be calculated for the pairs $A - B$, which are to be analyzed: (a) the number of documents containing A, (b) the number of documents containing B, as well as (g) the number of documents containing both A and B. We can use one of the typical formulae to calculate the similarity SIM between A and B (Jaccard-Sneath, Dice, and cosine).

The semantic network which results is an undirected graph, whose nodes are used to enter elements of documentary units (authors, subjects etc.), and whose lines are used to enter the similarities SIM. The size of a semantic network can be “set up” by determining threshold values for a, b, g and SIM. Changing this value for SIM changes the graph’s resolution behavior. With a high threshold value, we see a select few nodes, basically the net’s skeleton; with a lower threshold, the cluster becomes richer, but sometimes also less navigable. For examples of semantic networks of subjects see Fig. G.2.2 and G.2.3.

Information Flow Analyses

Analyses of information flows answer the questions: does information flow from A to B? Information flows can be retraced when they are documented. Academic research, technological development and legal practice have systematically established such documenting systems. Generally, references (footnotes, bibliographies etc.) document information flows. In a reference, scientific articles, patents or court rulings hint at where certain pieces of information come from (see Ch. M.2). As far as databases contain references, information flows can be reconstructed via information retrieval. For all three documenting systems, there are data collections with references: in the academic field, Web of Science, Scopus and Google Scholar must be named, but many specialist databases or digital libraries (such as the ACM Portal) also document references. In the technological arena, references are increasingly being stored, and in

law, citation databases such as Shepard's Citation Index or Westlaw's KeyCites have been a matter of course for decades (at least in the U.S.).

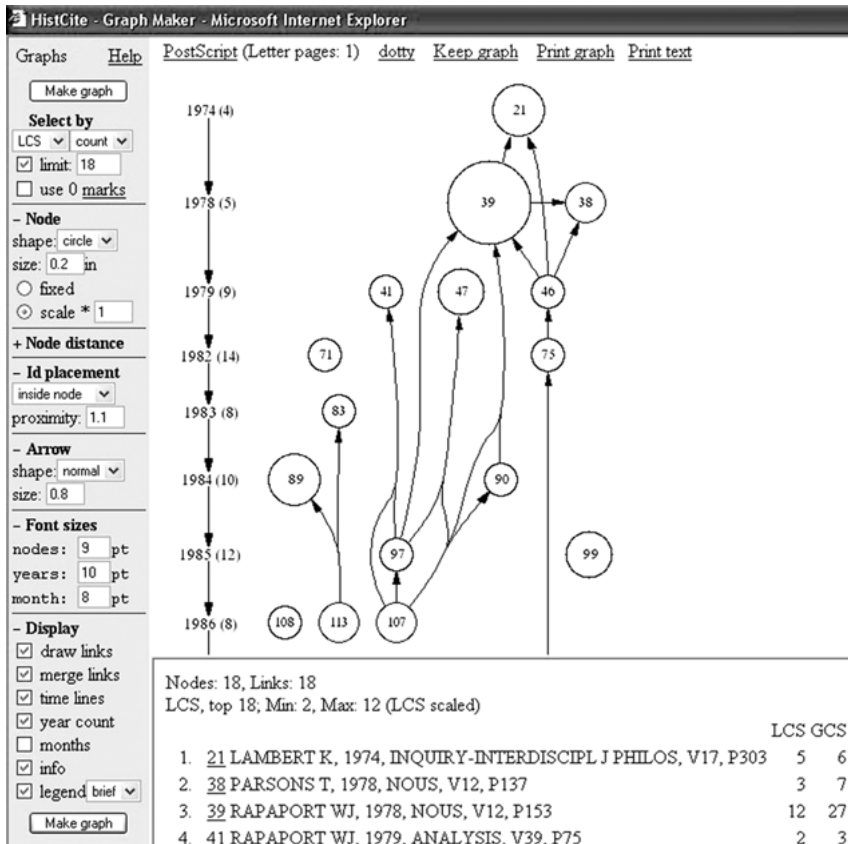


Figure H.2.5: Information Flow Analysis of Important Articles on Alexius Meinong Using Web of Science and HistCite. Source: Garfield, Paris, & Stock, 2006, 398.

Simple information flow analyses as part of a “normal” search for documentary units follow information flows both forwards and backwards along their timeline and additionally compare reference apparatuses. The precondition for a successful retrieval in citation databases is a initial document that is relevant for a certain information problem. Following the information flows “backwards” means searching for all cited literature. This is not a big problem, and could also be accomplished without electronic retrieval. The situation is different for “forwards” searches. Here, we search all articles, patents or rulings that cite our initial document. Such retrieval is only possible in citation databases. The next form is also exclusive to such information services: it concerns the comparison of references in different documents. We are looking for

documentary units that have as many references in common with the initial document as possible. All retrieval strategies of the citation databases complement other strategies and retrieve documents that could not be found via any other strategy.

We now come, finally, to informetric information flow analyses. In contrast to semantic networks, information flow representations are directed graphs in which the line displays either the direction of the reference (i.e. from citer to cited) or of the citation (from cited to citer). Our example in Figure H.2.5 is a graph of references of important articles on the philosopher Alexius Meinong (Garfield, Paris, & Stock, 2006). The starting point is node N° 21, an article by K. Lambert from the year 1974. The information flow graph exemplifies the position of the single articles in the network of information flows. Thus article N° 39 (by W.J. Rapaport) holds a central position in this scientific discussion. The graph in Figure H.2.5 has been created via a search in Web of Science, additionally using the software HistCite (Garfield, 2004). Similar software is provided by further information service providers such as STN International (with STN AnaVist) or Questel.

Conclusion

-
- Informetric analyses via databases lead to new information, which has been “slumbering” in the databases but never explicitly been entered into them. What is characterized are certain sets of documents, not individual documentary units.
 - Special attention must be paid to the selection of the document set to be analyzed. Since both general and specialist databases mainly do not guarantee completeness, one tends to have to search across databases. During the removal of duplicates which is now necessary, it can be useful to call up the databases in different order (reversed duplicate removal) in order to exploit the advantages of the respective single databases.
 - On the micro-level (database and its documentary units) as well as on the macro-level (database as a whole and in its context vis-à-vis other databases), there are problems that can (mainly) be solved via intellectual processing.
 - For all the insecurity concerning the literature, it can be shown that each respective ideally isolated sample is pretty effective in approaching the parameters of the basic population. Absolute values have hardly any meaning, but derived parameters such as rankings do.
 - Rankings arrange entries according to their frequency of occurrence in the isolated document set. The most productive authors or the most-cited documents on a given subject are particularly easy to determine, for example.
 - Informetric time series show the development of topics over time. The central sorting argument is the statement in the year field of the database records.
 - The co-occurrence of statements (such as co-authors, co-citations or co-subjects) is the basis for the representation of semantic networks. The strength of the overlap (or similarity/distance, respectively) can be calculated via different procedures (Jaccard-Sneath, Dice, and cosine).
 - Information flow analyses use references and citations to retrace how information has flowed between documents in academic research, technology (in patents) as well as court rulings.
 - Commercial information services such as DIALOG, Questel, STN International, Scopus or Web of Knowledge provide informetric basic functionality online. In addition, for more elaborate investigations we need special analysis software (such as HistCite).
-

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H.3 User and Usage Research

Information Behavior

User and usage research is a sub-discipline of empirical information science (informatics) that describes and explains the information behavior of users (Case, 2007; Cole, 2012; Ingwersen & Järvelin, 2005, Ch. 3). As Dervin and Nilan (1986, 6) emphasize, certain results of user research can be used to improve retrieval systems:

Information systems could serve users better—increase their utility to their clients and be more accountable to them. To serve clientele better, user needs and uses must become a central focus of system operations.

For Spink (2010) information behavior is a human instinct. The phylogeny of this instinct is the evolution of information behavior of the whole human tribe; the ontogeny is the development of the information behavior of an individual person. Spink (2010, 4) writes,

Information behavior is a behavior that evolved in humans through adaption over the long millennium into a human ability, while also developing over a human lifetime.

According to Wilson (2000), there are three distinct levels of human behavior with regard to information. “Information behavior” is the most general level, comprising all sorts of human information processing, communication, gathering etc. “Information-seeking behavior” relates to all variants of human information searching habits (i.e. questions to colleagues, browsing in books, visits to libraries, using commercial information services in the Deep Web or search engines). The narrowest concept is “information search behavior”, which exclusively refers to information seeking in digital systems (Wilson, 1999a, 263):

(I)nformation searching behaviour being defined as a sub-set of information-seeking, particularly concerned with the interactions between information user (with or without an intermediary) and computer-based information systems, of which information retrieval systems for textual data may be seen as one type.

In contrast to the system-centered approach of retrieval research, which exclusively concentrates on the way documents are processed by the search systems’ algorithms, the person-centered approach of user research places users themselves at the forefront of the investigation (Wilson, 2000, 51). It is of great importance for information science research to integrate the system-oriented and the user-oriented approaches into a single field of research (Ingwersen & Järvelin, 2005).

Wilson (1999a) developed a comprehensive model of information behavior, which we have modified for our purposes (Fig. H.3.1). All steps of the process mainly

serve one objective: uncertainty resolution, or uncertainty reduction (Wilson, 1999b). The starting point is a recognized information need of a user. Depending on the “circumstances”, information-seeking behavior can be triggered or not. These “circumstances”, according to Wilson (1999a, 256), are subject to intervening variables, e.g. psychological or demographical in nature. For instance, technophobia on the part of a user can have a great influence at this point (Nahl, 2001, 8-10). Hence information behavior also encompasses—following Case (2007, 5)—“purposive behaviors that do not involve seeking, such as actively *avoiding* information.”

If information-seeking behavior is triggered, it is either via information search behavior (query formulation in an information retrieval system, random serendipitous finds, scanning and processing of results lists and documents) or via other information-seeking behavior (e.g. questions to colleagues, library visits)—or via both forms of information-seeking behavior working in parallel. Successful search processes lead to information usage, which, in the ideal case, satisfies the user’s information need. The model is designed to work via feedback, i.e. several iterations are necessary before the uncertainty posited by the task and the circumstances can be optimally reduced. Wilson (1999a, 268), too, emphasizes the value of such feedback:

When feedback is explicitly introduced into the models of information behaviour the process can be seen to require a model of the process that views behaviour as iterative, rather than one-off and the idea of *successive* search activities introduces new research questions.

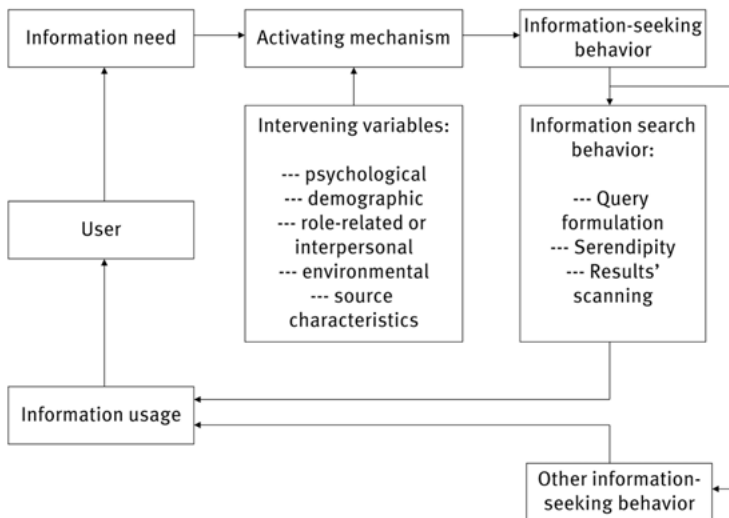


Figure H.3.1: Model of Information Behavior. Terminology and Intervening Variables are Adopted from Wilson, 1999a, 258.

Apart from human information behavior, we must take into account corporate information behavior. This involves corporate policy and its implementation with regard to the processing of information within the enterprise. Knowledge-intensive businesses in particular heavily rely on optimal dealings with (internal as well as external) information. Digital information systems become vital parts of the corporate memory. The core of the structure of the so-called “IT-supported organizational memory” (Stein & Zwass, 1995, 85) is information retrieval (Walsh & Ungson, 1991, 64) and knowledge representation. The fundamental question for institutions is: How should we organize our corporate knowledge management? We can observe three organization strategies for dealing with information (Linde & Stock, 2011, 184-185). (1.) Institutions rely on end user activities. Here, information professionals or knowledge managers make suitable information services available within the institution in short-term projects, train employees, and then withdraw from the daily running of the activities. The professional end users, i.e. the staff, look for information and produce and store documents on their own. (2.) Institutions bundle information know-how in an individual work unit. Generally, the task of this unit is to both manage internal knowledge and to consult just-in-time external knowledge. The objective is not only to search documents and make them available to specialists, but to process found information. Organizational variant (3.) is a compromise between (1.) and (2.). End users assume light information search as well as information production and storage tasks, the results of which flow directly into their work. The difficult or company-critical work of performing complex retrieval tasks (say, searches for patents), of operating intranets and retrieval systems, of summarizing information from internal documents, etc. is left to information professionals. The respective variant of corporate knowledge management is determined via methods of user and usage research. The necessary individual working steps are then determined. Knowledge management is always aligned with the company’s development.

Methods of User and Usage Research

The object of examination of user and usage research is man himself, in his capacity as a being that works with information. In the ideal scenario, one can distinguish between three large groups of users (Stock & Lewandowski, 2006, 1079):

- Information professionals,
- Professional end users,
- End users (information laymen).

An information professional is an expert in search and retrieval as well as in the representation, storage and supply of documents and knowledge. In general, he is an information scientist or comes from a similar academic background (e.g. library science). The professional end user is a specialist in an institution who processes, ad hoc, simple information needs that arise in the workplace. To do so, he uses search

engines in the Web as well as those information services in the Deep Web that correspond to his profession (a doctor would know and use Medline, an engineer Compendex). The layman end user, finally, is the typical “googler” who searches in the Surface Web for private and, occasionally, professional reasons. Depending on the level of his information literacy (Ch. A.5), he may from time to time venture into the Deep Web. To rank users according to their information literacy, one must first perform an information literacy test and then apply methods of user and usage research. There is a template available for aspects of retrieval competence (Cameron, Wise, & Lottridge, 2007).

Three bundles of methods are generally used to register the users scientifically (separated by user group) (Lewandowski & Höchstötter, 2008, 312-314):

- Observation of users in a laboratory situation,
- User survey,
- Analysis of the log files of search tools.

In laboratory studies, interactions between user and search system can be observed in detail. The methods used are keystroke tracking, eye-tracking, tracking of the areas on the screen viewed by the user, and filming of the user throughout the entire process. Occasionally, the users are asked to speak their thoughts (“thinking aloud”) (Ericsson & Simon, 1980). Fixations focused on a specific area of the screen (say, about 150 – 600 milliseconds) registered via eye-tracking are associated with cognitive load (as a consequence of interest, or of trouble understanding) (Duchowski, 2007). Filming the test subjects makes it possible to observe their facial expressions for indicators of emotion (joy, surprise, anger, etc.).

User surveys depend on users’ truthful answers. They employ questionnaires, which are filled out by the user in private, or interviews, in which the user is quizzed by an interviewer following a guideline. Depending on whether the questions are predetermined, based on a guideline, or improvised, we speak of standardized, semi-standardized, and non-standardized interviews, respectively (Fank, 2001, 249). It is also possible to develop so-called “personas”, i.e. ideal-typical persons that the test subjects are asked to identify themselves with (Cooper, 2004). A useful step is to make the instrument that is being used (questionnaire or interview guideline) undergo a pretest (with a few test users) in order to safeguard the instrument’s quality. Questionnaires have the advantage of reaching many test subjects at little cost; possible disadvantages include questions being misunderstood (or not understood at all) and the rate of response being low. Interviews have the advantage that interviewees’ reactions can be observed and misunderstandings resolved, while new aspects may be introduced by the interviewees. Their disadvantages are the time and effort they take as well as the danger of test subjects sometimes giving what they think is the expected answer instead of what they really mean.

Analyses of search log files (Jansen, 2009) provide detailed information about information search behavior. Their disadvantage is that they provide no information about the users themselves. Typical analyses of log data include traffic analysis (hits,

page views, visits, and visitors), analysis of the countries (states, cities) of the users, analysis of entry page and exit page analysis as well as following links to pages, analysis of errors (e.g., the 404 error: “page not found”), etc. (Rubin, 2001).

Information Needs

Fundamental human needs are of physiological nature (food, water, sex, etc.), and aim for safety, love, esteem and self-actualization (e.g., problem-solving or creativity) (Maslow, 1954). Information needs are not featured in Maslow’s theory of needs. However, many fundamental human needs require information in order to be satisfied. This begins with information seeking in order to locate sources of food, water, and sex, and ends with problem-solving, which can only be achieved via knowledge (to be sought and retrieved). For Wilson (1981, 8), the object is “information-seeking towards the satisfaction of needs.” Case (2007, 18) states that

(e)very day of our lives we engage in some activity that might be called information seeking, though we may not think of it that way at the time. From the moment of our birth we are prompted by our environment and our motivations to seek out information that will help us meet our needs.

The information-related motives that result from our fundamental human needs shall henceforward be abbreviated to “information needs”.

An individual’s information need is the starting point of any search for information. The objective is to satisfy the information need by transforming retrieved information into new knowledge by combining them with the searcher’s pre-knowledge (Cole, 2012). According to Taylor (1968, 182), there are four levels to the human need for information: the actual, visceral need, perhaps unconscious to the user (Q1), the conscious, within-brain description of the need (Q2), the formalized need; a rational statement of the need (Q3), and the compromised need in language / syntax the user believes is required by the information system (Q4). Q1 through Q4 may be interpreted to be present in every user at all times, but they can also be regarded as four consecutive phases (Cole, 2012, 19-20). Apart from these psychological factors, there are also cognitive ones (Davis & Shaw, eds., 2011, 30-31) that result from anomalous states of knowledge and from uncertainty (see Ch. A.2).

Dervin and Nilan (1986, 17) list the following descriptions of “information need” found in the literature:

- 1) a conceptual incongruity in which the person’s cognitive structure is not adequate to a task (...); 2) when a person recognizes something wrong in his or her state of knowledge and wishes to resolve the anomaly (...); 3) when the current state of possessed knowledge is less than needed (...), 4) when internal sense runs out (...), and 5) when there is insufficient knowledge to cope with voids, uncertainty, or conflict in a knowledge area (...).

Information seeking may also serve to entertain the seeker (Case, 2007, 115):

In a behavior possibly related to anxiety (as a relief mechanism), or possibly unrelated (as a result of a human need for play and creativity), people tend to seek out entertainment as well as information.

This entertainment may be derived from sources that are needed to satisfy the user's needs for play and creativity (i.e. a search for digital games on a search engine), or from the search itself. The user is entertained by searching on search engines, browsing through hit lists or following links in digital documents.

Corporate Information Needs

Information need analyses in institutions have the goal of describing strengths and weaknesses of corporate information systems. It is important for corporate knowledge management to know which information types and services have meaning for employees. Koreimann (1976, 65) defines:

Under the concept of information need analysis, we subsume all those procedures and methods that are suited for solving concrete business tasks in the framework of an enterprise.

Gust von Loh (2008) distinguishes objective information needs (being the information need that arises from a certain job position) from subjective information needs (the need articulated by the holder of a position). Objective information needs result “from the tasks that arise in the enterprise, and thus do not depend on the person of the employee” (Gust von Loh, 2008, 133). In information need analyses, objective information needs can be derived deductively from strategy requirements and job descriptions in the enterprise. Subjective information needs, on the other hand, can be deduced empirically via methods of user and usage research. The information supply comprises all information that is available in the institution. By evaluating log files, one can empirically deduce the information demand, i.e. the extent to which the offered information is, in fact, used. To build an information supply from the ground up, we consider all information deemed relevant for the company in question. Users will be asked whether they require certain information at all, and—if so—in which form. For instance, if an enterprise does research, it makes sense to offer its employees access to patent databases. Whether they require a free system with a very basic functionality, like Espacenet, or one with a far greater functionality, such as Questel's PlusPat database, will be determined via surveys of the parties concerned (or a representative sample). Some of the staff will not be aware of the existence of certain information services (e.g. Espacenet or PlusPat) in the first place. Here it is the task of knowledge management to create a demand for these services (Mujan, 2006, 33). Gust

von Loh (2008, 133) reports of an information need being created in employees via the simple implementation of an information need analysis:

In my case, it happened that e.g. certain staff did not know all functions of the specially developed database management program. Via questions concerning the delivery program, several found out for the first time that the option of creating such a program even existed. Particularly those employees who had not been in the company for long did not know about this function.

Fig. H.3.2 shows an overview of the individual objects of examination in a corporate information need analysis.

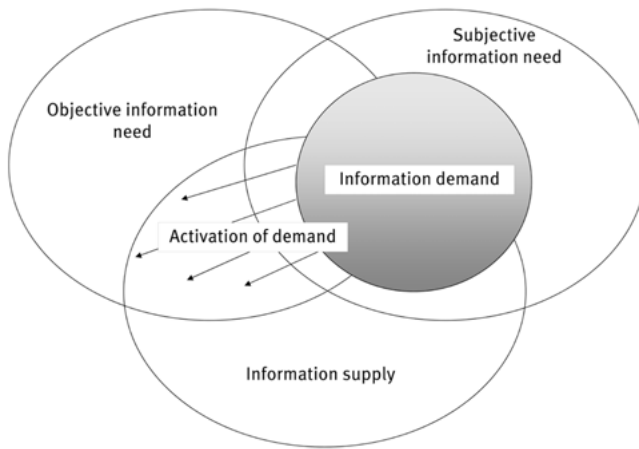


Figure H.3.2: Variables of the Analysis of Corporate Information Needs. Arrows: Information Marketing. Source: Modified following Mujan, 2006, 33, and Bahlmann, 1982.

The Search Process

Kuhlthau (2004) investigates the information search behavior of high school seniors doing homework. From this, she derives a model of the search process. Kuhlthau's study relates to libraries, but it is clear that the model can also be applied to information search processes on the internet (Luo, Nahl, & Chea, 2011). Kuhlthau's model describes six phases of the search process (task initiation, topic selection, prefocus exploration, focus formulation, information collection, search closure). All phases incorporate three realms (Kuhlthau, 2004, 44):

- the affective (feelings),
- the cognitive (thoughts),
- the physical, sensorimotor behavior (actions) (Fig. H.3.3).

	1	2	3	4	5	6
	Task initiation	Topic selection	Prefocus exploration	Focus formulation	Information collection	Search closure
Feelings	uncertainty	optimism	confusion, frustration, doubt	clarity	sense of direction, confidence	relief
Thoughts		ambiguity $\xrightarrow{\text{increase interest}}$ specificity				
Actions		seeking relevant information $\xrightarrow{\text{increase interest}}$ seeking pertinent information				

Figure H.3.3: Kuhlthau's Model of the Information Search Process. Source: Modified following Kuhlthau, 2004, 45.

Task initiation, which is dominated by “feelings of uncertainty and apprehension” (Kuhlthau, 2004, 44), comes at the beginning of the search process. In the second phase—topic selection—these uncertain feelings mutate toward optimism, while the thoughts revolve rather vaguely around “weighting topics against criteria of personal interest, project requirements, information available, and time allotted” (Kuhlthau, 2004, 46). After initial searches for relevant information, the feelings revert back to uncertainty and doubt. Frustration abounds. Inspecting the information, the test subjects realize that they have “to form a focus or a personal point of view” (Kuhlthau, 2004, 47). After this prefocus exploration comes the search process's turning point, focus formulation. Here, optimism regains the upper hand, in connection with “increased confidence and a sense of clarity” (Kuhlthau, 2004, 48). The individuals are now capable of finding the right search arguments on the basis of their own pre-knowledge. Their personal focus lets them search pertinent information, i.e. information that satisfies the (now recognized) subjective information need. Stage 5 (information collection) relates to all interactions between the user and the information system, such as query formulation and dealing with hit lists and documents. Stage 6 ends the search process (Kuhlthau, 2004, 50):

(F)eelings of relief are common. There is a sense of satisfaction if the search has gone well or disappointment if it has not. ... Thoughts concentrate on culminating the search with a personalized synthesis of the topic or problem. Actions involve a summary search to recheck information that may have been initially overlooked.

Under what circumstances do users end their search? Some stop when all they find anymore is redundant information, others due to the feeling that they have achieved

a satisfactory result. However, some users end the search process when the deadline for the report looms and they have no more time for search.

Query Formulation and Results Scanning and Processing

Log file analyses have been used to determine which search atoms and search strategies are used by users in performing targeted information searches with search tools. The two “classical” studies regard the search engines AltaVista (Silverstein et al., 1998) and Excite (Spink et al., 2001; Spink & Jansen, 2004a). Newer studies are rare, since search engine providers keep their log files under wraps. Studies on query formulation research, for instance:

- the number of search atoms (typically: between 1 and 3) (Silverstein et al., 1998, 7; Spink et al., 2001, 230; Taghavi et al., 2012),
- the number of Boolean and proximity operators (typically: none; rarely: phrases “...” and “AND”) (Silverstein et al., 1998, 7; Spink et al., 2001, 229),
- search entries (disregarding stop words, the most frequent search atom at the end of the last century was “sex”) (Silverstein et al., 1998, 8; Spink et al., 2001, 231),
- query modification (adding terms, deleting terms, totally changing the query, etc.) (Silverstein et al., 1998, 10; Spink et al., 2001, 228-229); typically, less than half of the users change their search arguments at all (Spink & Jansen, 2004b; Xie & Joo, 2010, 270),
- the number of queries in a session (typically: between 1 and 3) (Silverstein et al., 1998, 10; Spink et al., 2001, 227-230).

Spink and Jansen (2004b) summarize the empirical results regarding query formulation:

In summary, most Web queries are short, without query reformulation or modification, and have a simple structure. Few search sessions include advanced search techniques, and when they are used, many include mistakes.

Besides performing targeted searches, users browse in information collections, sometimes finding information that they hadn’t even been looking for. This regards browsing in libraries (Björneborn, 2008) as well as on the internet (McCay-Peet & Toms, 2011). From a Persian fairy tale about “The Three Princes of Serendip”, who never did reach their destination (Serendip, i.e. Ceylon) but managed (by happy accident) to return home as rich men, Horace Walpole called such occurrences of fortunate discovery “serendipity”. Information systems can increase the degree of serendipity by producing “a creative association between two disparate pieces of information” (McCay-Peet & Toms, 2011). This can be realized by offering term clouds or term clusters in search tools, for example.

Following a search, the user is shown his results on a search engine results page (SERP). Here, various empirical studies are possible:

- viewed pages with hits: the average number of screens (at 10 hits each) in the AltaVista study is 1.39 (Silverstein et al., 1998, 10) and 2.35 for Excite (Spink & Jansen, 2004b). Spink et al. (2001, 229) observe “that a large percent of users do not go beyond the first page,”
- which pages are shown on the first SERP in particular? How many organic hits are there, and how many ads (“sponsored links”) (Höchstötter & Lewandowski, 2009)? What are the SERP positions of clicked pages?,
- click-through data: clicked links to the documents. For certain authors, such data represent an indicator for “implicit user feedback” (Agichtein, Brill, & Dumais, 2006),
- viewed areas of the SERP (registered via eye-tracking methods) (Lorigo et al., 2008),
- dwell time: how long does a user view a document? (Morita & Shinoda, 1994),
- link depth: how many links do users follow going out from SERPs (Egusa et al., 2010)? Do they also use links in the resulting documents and, further, in the documents clicked from there?

Journal Usage

Certain document types allow us to make statements on their type of usage. This is particularly the case for formally published documents in science, technology, and law. The data is aggregated on the basis of usage measurements of specific documents (e.g. journal articles), and provides information about the way a scientific journal, a scientist’s oeuvre, or a company’s patent portfolio is being used. In the following, we will exemplarily describe some methods for registering journal usage. The necessary kind of data is obtained via several different user activities:

- clicking on links to the article,
- downloading articles,
- reading articles,
- bookmarking articles,
- discussing articles in blog posts or microblogs (e.g. Twitter),
- citing articles (presuming that the reader is an author who has published a work of his own).

Data sources for registering download activity include the usage statistics of individual libraries (local usage) or of publishers (global usage). COUNTER (Counting Online Usage of NeTworked Electronic Resources) is a standard for collecting data used locally in the form of scientific journals (Shepherd, 2003). COUNTER exclusively works on the journal level, not taking into account articles. For instance, the Journal Report 1 states the number of successful full-text article requests by month and journal. The

data sources are statements by the publishers, who report them to subscribing libraries (Haustein, 2012, 171-181). Global data on the worldwide usage of journals could be determined via the same methods, but only if publishers or editors are willing to offer such data. This is the case for the journal PloS (Public Library of Science), which publishes detailed download data—even on the level of articles (Haustein, 2012, 187-189). Most other scientific publishers, however, regard usage data as a business secret. The project MESUR (Bollen, Rodriguez, & van de Sompel, 2007), in which various publishers released their usage data for research purposes (Haustein, 2012, 189-191), was another notable exception. Meaningful parameters for the usage of scientific journals include the Download Immediacy Index (DII) and the Download Half Life (Schlögl & Gorraiz, 2011). The DII is the quotient of the number of all downloads of articles from a journal J in the year t and the total number of articles published in J in t. Download Half Life is the median age of all downloaded articles from J in t. Both parameters are analogy constructions to parameters of citation analysis.

Downloading an article does not necessarily mean that the user has read it. The only way of registering reading habits is to perform surveys. Potential readers are asked whether and how often they read a scientific journal. Additionally, they can be asked whether results published in such a journal are brought to bear on their own professional practice (Schlögl & Stock, 2004). Such surveys can uncover detailed information, but they also have disadvantages. Thus only very small samples are collected, and the interviewees merely offer their subjective assessments (Haustein, 2012, 166).

The reader perception of an article or an entire journal can be derived from the user activities of bookmarking and tagging documents (Haustein & Siebenlist, 2011). Here we are talking about Web 2.0. Informetric methods specific to Web 2.0 are called “alternative metrics” (altmetrics). Haustein (2012, 199) justifies the use of social bookmarking as usage data:

In analogy to download and click rates, usage can be indicated by the number of times an article is bookmarked. Compared to a full-text request as measured by conventional usage statistics, which does not necessarily imply that a user reads the paper, the barrier to setting a bookmark is rather high. Hence, bookmarks might indicate usage even better than downloads, especially if users took the effort to assign keywords. By tagging, academics generate new information about resources: tags assigned to a journal article give the users' perspective on journal content.

Possible data sources are Web 2.0 services that save bookmarks for scientific literature (e.g. BibSonomy, CiteULike, Connotea or Mendeley) (Haustein, 2012, 198-212). Further altmetric approaches regard the number of comments or mentions of an article in the blogosphere as a usage indicator. Peters, Beutelspacher, Maghferat and Terliesner (2012)

analyzed linking behavior in blog posts and tweets, number of comments assigned to blog posts and share of publications found in social bookmarking systems.

A classical usage indicator is the number of citations received by a document or a journal. A citation is the formal mention of a document in another document (such as our bibliographies at the end of each chapter). From the perspective of the citing party, it is a reference, from that of the cited party, a citation. A citation can only come about if the reader of the cited article is himself a (scientific) author who has managed to publish his work in a scientific journal or as a book. The people who only read a document do not contribute to this indicator at all. The Impact Factor, introduced by Garfield (1972), is an indicator of central importance for journal scientometrics. It takes into consideration both the number of publications in a scientific journal and the amount of these publications' citations. The Impact Factor IF of journal J is calculated as a fractional number. The numerator is the number of citations over exactly one year t that name articles from journal J that appeared over the two preceding years (i.e. $t - 1$ and $t - 2$). The denominator is the number of citable source articles in J for the years $t - 1$ and $t - 2$. Let the number of source articles from J for $t - 1$ be $S(1)$, the number for $t - 2$ be $S(2)$, and the number of citations of all articles from J from the years $t - 1$ and $t - 2$ in the year t be C . The Impact Factor for J in t will be:

$$IF(J,t) = C / [S(1) + S(2)].$$

Apart from the traditional IF, there are weighted indicators (that take into account the meaning of a citing source, in analogy to the PageRank for Web pages) as well as normalized indicators (that try to subtract out discipline-specific and document-type-induced biases) (Haustein, 2012, 263-289). The Impact Factor is exclusively defined for academic journals. It is not applicable to individual articles. Likewise, it is impossible to derive the influence of an individual article from the Impact Factor of a journal. (The fact that an article appears in a journal with a high Impact Factor does not automatically make it a highly influential article—rather, it could be in the typical “long tail” of informetric distributions.)

Like the Impact Factor, the Immediacy Index (II) is calculated by Thomson Reuters for their information service Web of Science. The II only incorporates citations and publications from a pertinent year under review (t) into the formula. Let $S(t)$ be the citable source articles of journal J in the year t ; $C(t)$ are all citations of articles from the year t in the same year, i.e.

$$II(J,t) = C(t) / S(t).$$

Literature half-life, i.e. the literature's median age, is the established time-based indicator (Burton & Kebler, 1960). Citing half-life relates to the age of the references, and thus to the up-to-dateness of a journal's citations. Cited half-life, on the other hand, relates to the age of the citations and is thus an indicator for how long the results of a journal stay “fresh”, i.e. cited.

Haustein (2012, 367) emphasizes that it is impossible to comprehensively capture the usage of an academic journal via a single indicator (not even if it is the Impact Factor). Their multi-dimensional approach to journal evaluation is based on a multitude of research methods and indicators that each registers a specific user behavior.

Conclusion

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- Information behavior comprises all kinds of human information processing (including actively avoiding information), information-seeking behavior is restricted to the retrieval of information and, finally, information search behavior—the narrowest concept—exclusively regards retrieval in digital systems.
 - We distinguish between human information behavior and corporate information behavior. Particularly in knowledge-intensive enterprises, it is important that internal and external information be dealt with optimally.
 - In human information behavior, there are three ideal-typical user groups: information professionals, professional end users, and end users (laymen). Alternatively, or complementarily, it is possible to work out the users' degree of information literacy empirically.
 - Observing users in a laboratory situation, interviewing users, and analyzing the log files of search engines have proven to be useful methods of user and usage research in information science practice.
 - Human information needs are derived from fundamental human needs. Without information seeking, many primary human needs cannot be satisfied at all. An individual information need is the starting point of a search for information. Apart from psychological factors (such as the actual, visceral need), cognitive factors are dominant in the context of anomalous states of knowledge and uncertainty. However, information seeking can also serve to entertain the user.
 - Studies of corporate information needs distinguish the objective information need that arises as a result of the job position from the subjective information need of the person holding the position. The goals are to optimize both the deployment of information in an institution and the way it is used by the staff.
 - In Kuhlthau's process model of information search, six phases (task initiation, topic selection, prefocus exploration, focus formulation, information collection, search closure) are identified, each giving rise to various different feelings, thoughts and actions.
 - User studies on query formulation record the number of search atoms, Boolean operators or the modification of queries, among other things. User studies on hit lists and documents count viewed pages with hit lists, click-through data, clicked pages' rankings, viewed page areas, dwell-times and link depth.
 - Apart from click rates, studies on the usage analysis of scientific journals use numerical values concerning downloads, reading behavior, bookmarkings, comments and blog posts, as well as references and citations. Classical usage parameters of scientific journals are the Impact Factor, the Immediacy Index as well as Half-Life.
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H.4 Evaluation of Retrieval Systems

How can the quality of a retrieval system be described quantitatively? Is search engine X better than search engine Y? What aspects determine the success of a retrieval system? Answering these questions is the task of evaluation research. Croft, Metzler and Strohman (2010, 297) define “evaluation” as follows:

Evaluation is the key to making progress in building better search engines. It is also essential to understanding whether a search engine is being used effectively in a specific application.

Evaluation regards both system criteria and user assessments. Tague-Sutcliffe (1996, 1) places the user at the center of observation:

Evaluation of retrieval systems is concerned with how well the system is satisfying users not just in individual cases, but collectively, for all actual and potential users in the community. The purpose of evaluation is to lead to improvements in the information retrieval process, both at a particular installation and more generally.

Retrieval systems are a particular kind of information system. Hence, we must take into account evaluation criteria for information systems in general as well as for retrieval systems in particular.

A Comprehensive Evaluation Model

We introduce a comprehensive model that allows us to span a theoretical framework for all aspects of the evaluation of retrieval systems (similarly: Lewandowski & Höchstötter, 2008, 318; Saracevic, 1995). The model takes into account different dimensions of evaluation. The methods of evaluation derive from various scientific disciplines, including information systems research, marketing, software engineering, and—of course—information science.

A historical point of origin for the evaluation of information systems in general is the registration of technology acceptance (Davis, 1989), which uses subdimensions (initially: “ease of use” and “usefulness”, later supplemented by “trust” and “fun”) in order to measure the quality of the information system’s technical make-up (dimension: IT system quality). In the model proposed by DeLone and McLean (1992), the technical dimension is joined by that of information quality. Information quality concentrates on the knowledge that is stored in the system. The dimension of knowledge quality consists of the two subdimensions of document quality (more precisely: documentary reference unit quality) and the surrogates (i.e. the documentary units) derived from these in the information system. DeLone and McLean (2003) as well as Jennex and Olfman (2006) expand the model via the dimension of service quality. When analyzing IT service quality, the objective is to inspect the services offered by

the information system and the way they are perceived by the users. The quality of a retrieval system thus depends upon the range of functions it offers, on their usability (Nielsen, 2003), and on the system's effectiveness (measured via the traditional values of Recall and Precision). In an overall view, we arrive at the following four dimensions of the evaluation of retrieval systems (Figure H.4.1):

- IT service quality,
- Knowledge quality,
- IT system quality,
- Retrieval system quality.

All dimensions and subdimensions make a contribution to the usage or non-usage—i.e., the success or failure—of retrieval systems.

Evaluation of IT Service Quality

The service quality of a retrieval system is described, on the one hand, via the processes that the user must implement in order to achieve results. On the other hand, it involves the attributes of the services offered and the way these are perceived by the user.

Suitable methods for registering the process component of an IT service include the sequential incident technique and the critical incident technique. In the sequential incident technique (Stauss & Weinlich, 1997), users are observed while working through the service in question. Every step of the process is documented, which produces a “line of visibility” of all service processes—i.e., displaying the service-creating steps that are visible to the user. If the visible process steps are known, users can be asked to describe them individually. This is the critical incident technique (Flanagan, 1954). Typical questions posed to users are “What would you say is the primary purpose of X?” and “In a few words, how would you summarize the general aim of X?”

A well-known method for evaluating the attributes of services is SERVQUAL (Parasuraman, Zeithaml, & Berry, 1988). This method works with two sets of statements: those that are used to measure expectations about a service category in general (EX) and those that measure perceptions (PE) about the category of a particular service. Each statement is accompanied by a seven-point scale ranging from “strongly disagree” (1) to “strongly agree” (7). For the expectation value, one might note that “In retrieval systems it is useful to use parentheses when formulating queries” and ask the test subject to express this numerically on the given scale. The corresponding statement when registering the perception value would then be “In the retrieval system X, the use of parentheses is useful when formulating queries.” Here, too, the subject specifies a numerical value. For each item, a difference score $Q = PE - EX$ is defined. If, for instance, a test subject specifies a value of 1 for perception after having noted a 4 for expectation, the Q value for system X with regard to the attribute in question will be $1 - 4 = -3$.

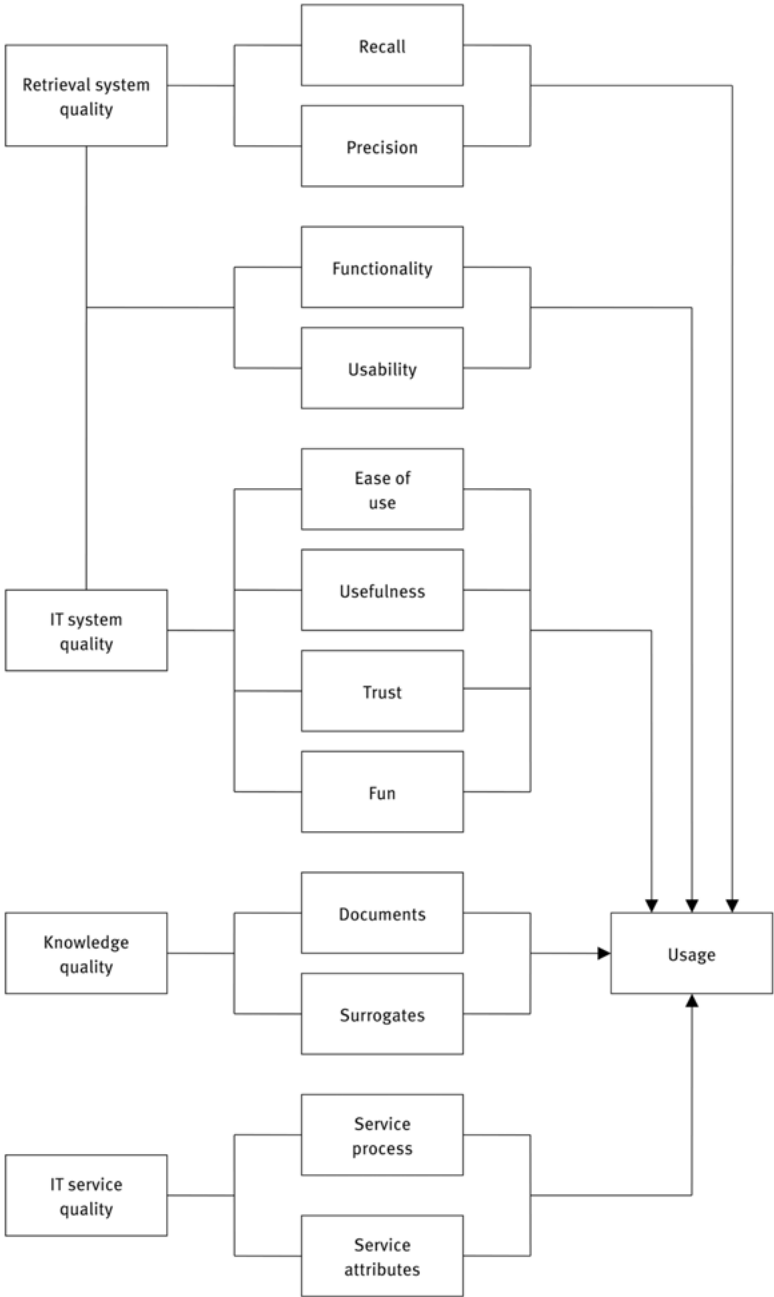


Figure H.4.1: A Comprehensive Evaluation Model for Retrieval Systems. Source: Modified Following Knautz, Soubusta, & Stock, 2010, 6.

Parasuraman, Zeithaml and Berry (1988) define five service quality dimensions (tangibles, reliability, responsiveness, assurance, and empathy). This assessment is conceptualized as a gap between expectation and perception. It is possible to adopt SERVQUAL for measuring the effectiveness of information systems (Pitt, Watson, & Kavan, 1995). In IT SERVQUAL, there are problems concerning the exclusive use of the difference score and the pre-defined five quality dimensions. It is thus possible to define separate quality dimensions that are more accurate in answering specific research questions than the pre-defined dimensions. The separate dimensions can be derived on the basis of the critical processes that were recognized via sequential and critical incident techniques. It was suggested to not only apply the difference score, but to add the score for perceived quality, called SERVPERF (Kettinger & Lee, 1997), or to work exclusively with the perceived performance scoring method. If a sufficient amount of users were used as test subjects, and if their votes were, on average, close to uniform, SERVQUAL would seem to be a valuable tool for measuring the quality of IT systems' attributes.

In SERVQUAL, both the expectations and the perceptions of users are measured. Customers value research (McKnight, 2006) modifies this approach. Here, too, the perception values are derived from the users' perspective, but the expectation values are estimates, on the part of the developers or operators of the systems, as to how their users will perceive the attribute in question. The difference between both values expresses "irritation", i.e. the misunderstandings between the developers of an IT service and their customers.

Evaluation of Knowledge Quality

Knowledge quality involves the evaluation of documentary reference units and of documentary units. The quality of the documents that are depicted in an information service can vary significantly, depending on the information service analyzed. Databases on scientific-technological literature (such as the ACM Digital Library) contain scientific articles whose quality has generally already been checked during the publication process. This is not the case in Web search engines. Web pages or documents in sharing services (e.g. videos on YouTube) are not subject to any process of evaluation. The information quality of such documents is extremely hard to quantify. Here we might consider subdimensions such as believability, objectivity, readability or understandability (Parker, Moleshe, De la Harpe, & Wills, 2006).

The evaluation of documentary units, i.e. of surrogates, proceeds on three subdimensions. If KOSs (nomenclatures, classification systems or thesauri) are used in the information service, these must be evaluated first (Ch. P.1). Secondly, the quality of indexing and of summarization is evaluated via parameters such as the indexing depth of a surrogate, the indexing effectiveness of a concept, the indexing consistency of surrogates, or the informativeness of summaries (Ch. P.2). Thirdly, the update

speed is evaluated, both in terms of the first retrieval of new documents and in terms of updating altered documents. Both aspects of speed express the database's freshness.

Evaluation of IT System Quality

The prevalent research question of IT system quality evaluation is "What causes people to accept or reject information technology?" (Davis, 1989, 320). Under what conditions is IT accepted and used (Dillon & Morris, 1996)? Davis' empirical surveys lead to two subdimensions, perceived usefulness and perceived ease of use (Davis, 1989, 320):

Perceived usefulness is defined ... as "the degree to which a person believes that using a particular system would enhance his or her job performance." ... *Perceived ease of use*, in contrast, refers to "the degree to which a person believes that using a particular system would be free of effort."

Following the theory of reasoned action, Davis, Bagozzi and Warshaw (1989, 997) are able to demonstrate that "people's computer use can be predicted reasonably well from their intentions" and that these intentions are fundamentally influenced by perceived usefulness and perceived ease of use. The significance of the two subdimensions is seen to be confirmed in other studies (Adams, Nelson, & Todd, 1992) that draw on their correlation with the respective information system's factual usage. In the further development of technology acceptance models, it is shown that additional subdimensions join in determining the usage of information systems. On the one hand, there is the "trust" that users have in a system (Gefen, Karahanna, & Straub, 2003), and on the other hand, the "fun" that users experience when using a system (Knautz, Soubusta, & Stock, 2010). The trust dimension is particularly significant in e-commerce systems, while the fun dimension is most important in Web 2.0 environments. Particularly critical aspects of retrieval systems' usefulness are their ability to satisfy information needs and the speed with which they process queries. Croft, Metzler and Strohman (2010, 297) describe this as the effectiveness and efficiency of retrieval systems:

Effectiveness ... measures the ability of the search engine to find the right information, and efficiency measures how quickly this is done.

We will return to the question of measuring effectiveness when discussing Recall and Precision in the below.

When evaluating IT system quality, questionnaires are used. The test subjects must be familiar with the system in order to make correct assessments. For each subdimension, a set of statements is formulated that the user must estimate on a 7-point

scale (from “extremely likely” to “extremely unlikely”). Davis (1989, 340), for instance, posited “using system X in my job would enable me to accomplish tasks more quickly” to measure perceived usefulness, or “my interaction with system X would be clear and understandable” for the aspect of perceived ease of use. In addition to the four subdimensions (usefulness, ease of use, trust, and fun), it must be asked if and how the test subjects make use of the information system. If one asks factual users (e.g. company employees on the subject of their intranet usage), estimates will be fairly realistic. A typical statement with regard to registering usage is “I generally use the system when the task requires it.” When test subjects are confronted with a new system, estimates are hypothetical. It is useful to calculate how the usage values correlate with the values of the subdimensions (and how the latter correlate with one another). A subdimension’s importance rises in proportion to its correlation with usage.

Evaluation of Retrieval System Quality I: Functionality and Usability

Retrieval systems provide functions for searching and retrieving information. Depending on the retrieval system’s purpose (e.g. a general search engine like Google vs. a specialized information service like STN International), the extent of the functions offered can vary significantly. When evaluating functionality, the object is the “quality of search features” (Lewandowski & Höchstötter, 2008, 320). We differentiate between the respective ranges of commands for search, for push services, and for informetric analyses (Stock, 2000). Table H.4.1 shows a list of functionalities typically to be expected in a professional information service.

Table H.4.1: Functionality of a Professional Information Service. Source: Modified Following Stock, 2000, 26.

Steps	Functionality
<i>Selection of Databases</i>	Database Index
	Database Selection
	– precisely one database
	– selecting database segments
	– selection across databases
<i>Looking for Search Arguments</i>	Browsing a Dictionary
	Presentation of KOSs
	– verbal
	– graphic
	– display of paradigmatic relations for a concept
	– display of syntagmatic relations for a concept
	Statistical Thesaurus
	Dictionary of Synonyms (for full-text searches)

Search Options

- thesaurus (in the linguistic sense)
- Dictionary of Homonyms
- dialog for clarifying homonymous designations
- Field-Specific Search
- search within fields
- cross-field search in the basic index
- Citation Search
- search for references (“backward”)
- search for citations (“forward”)
- search for co-citations
- search for bibliographic coupling
- Grammatical Variants
- upper/lower case
- singular/plural
- word stem
- Fragmentation
- to the left, to the right, in the center
- number of digits to be replaced (precisely n digits; any number of digits)
- Set-Theoretical Operators
- Boolean operators
- parentheses
- Proximity Operators
- directly neighboring (with and without a regard for term order)
- adjacency operator
- grammatical operators
- Frequency Operator
- Hierarchical Search
- search for a descriptor including its hyponyms on the next n levels
- search for a descriptor including its hyperonyms on the next n levels
- search for a descriptor including all related terms
- Weighted Retrieval
- Cross-Database Search
- duplicate detection
- duplicate elimination
- Reformulation of Search Results into Search Arguments
- mapping in the same field
- mapping into another field
- mapping with change of database
- hit list
- sorting of search results
- marking of surrogates for sorting and output
- output of reports in tabular form
- output of the surrogates in freely selectable format
- particular output formats (e.g. CSV or XML)
- Ordering of Full Texts
- provision in the original format

Display and Output

	– link to the digital version
	– link to document delivery services
<i>Push Services</i>	– creation of search profiles
	– management of search profiles
	– delivering of search results
<i>Informetric Analysis</i>	– rankings
	– time series
	– semantic networks
	– information flow graphs

How does a retrieval system present itself to its users? Is it intuitively easy to use? Such questions are addressed by usability research (Nielsen, 2003). “Usable” retrieval systems are those that do not frustrate the user. This view is shared by Rubin and Chisnell (2008, 4):

(W)hen a product or service is truly usable, the user can do what he or she wants to do the way he or she expects to be able to do it, without hindrance, hesitation, or questions.

A common procedure in usability tests is task-based testing (Rubin & Chisnell, 2008, 31). Here an examiner defines representative tasks that can be performed using the system and which are typical for such systems. Such a task for evaluating the usability of a search engine might be “Look for documents that contain your search arguments verbatim!” Test subjects should be “a representative sample of end users” (Rubin & Chisnell, 2008, 25). The test subjects are presented with the tasks and are observed by the examiner while they perform them. For instance, one can count the links that a user needs in order to fulfill a task (in the example: the number of links between the search engine’s homepage to the verbatim setting). An important aspect is the difference between the shortest possible path to the goal and the actual number of clicks needed to get there. The greater this difference is, the less usable the corresponding system function will be. An important role is played by the test users’ abandonment of search tasks (“can’t find it”) and by their exceeding the time limit. Click data and abandonment frequencies are indicators for the quality of the navigation system (Röttger & Stock, 2003).

It is useful to have test subjects speak their thoughts when performing the tasks (“thinking aloud”). The tests are documented via videotaping. Use of eye-tracking methods provides information on which areas of the screen the user concentrated on (thus possibly overlooking a link). In addition to the task-based tests, it is useful for the examiner to interview the subjects on the system (e.g. on their overall impression of the system, on screen design, navigation, or performance).

Benchmarks for usability tests are generally set at a minimum of ten test subjects and a corresponding number of at least ten representative tasks.

Evaluation of Retrieval System Quality II: Recall and Precision

The specifics of retrieval systems are located in their significance as IT systems that facilitate the search and retrieval of information. The evaluation of retrieval system quality is to be found in the measurements of Recall and Precision (Ch. B.2) as well as in metrics derived from these (Baeza-Yates & Ribeiro-Neto, 2011, 131-176; Croft, Metzler, & Strohman, 2010, 297-338; Harman, 2011; Manning, Raghavan, & Schütze, 2008, 139-161). Tague-Sutcliffe (1992) describes the methodology of retrieval tests. Here are some of the questions that must be taken into consideration before the test is performed:

- To test or not to test? Which innovations (building on the current state of research) are aimed for in the first place?
- What kind of test? Should a laboratory test be performed in a controlled environment? Or are users observed in normal search situations (Ch. H.3)?
- How to operationalize the variables? Each variable to be tested (users, query, etc.) must be described exactly.
- What database to use? Should an experimentation database be built from scratch? Can one draw on pre-existing databases (such as TReC)? Or should a “real-life” information service be analyzed?
- What kind of queries? Should informational, navigational, transactional, etc. queries (Ch. F.2) be consulted? Can such different query types be mixed among one another?
- Where to get queries? Where do the search arguments come from? How many search atoms and how many Boolean operators are used? Should further operators (proximity operators, numerical operators, etc.) be tested?
- How to process queries? It is necessary for the search processes to run in a standardized procedure (i.e. always under the same conditions). Likewise, the test subjects should have at least a similar degree of pre-knowledge.
- How to design the test? How many different information needs and queries are necessary in order to achieve reliable results? How many test subjects are necessary?
- Where do the relevance judgments come from? One must know whether or not a document is relevant for an information need. Who determines relevance? How many independent assessors are required in order to recognize “true” relevance? What do we do if assessors disagree over a relevance judgment?
- How many results? In a search engine that yield results according to relevance, it is not possible (or useful) to analyze all results. But where can we place a practicable cut-off value?
- How to analyze the data? Which effectiveness measurements are used (Recall, Precision, etc.)?

Effectiveness measurements were already introduced in the early period of retrieval research (Kent, Berry, Luehrs, & Perry, 1955). In the Cranfield retrieval tests, Clever-

don (1967) uses Recall and Precision throughout. Cranfield is the name of the town in England where the tests were performed.

Table H.4.2: Performance Parameters of Retrieval Systems in the Cranfield Tests. Source: Cleverdon, 1967, 175.

	Relevant	Non-relevant	
Retrieved	a	b	$a + b$
Not retrieved	c	d	$c + d$
	$a + c$	$b + d$	$a + b + c + d = N$

Table H.4.2 is a four-field schema that differentiates between relevance/non-relevance and retrieved/not retrieved, respectively. The values for a through d stand for numbers of documentary units. To wit, a counts the number of retrieved relevant documentary units, b the number of found irrelevant DUs, c the number of missed relevant DUs, and, finally, d is the number of not retrieved irrelevant DUs. N is the total number of documentary units in the information service. Recall (R) is the quotient of a and $a + c$; Precision (P) is the quotient of a and $a + b$; finally, Cleverdon (1967, 175) introduces the fallout ratio as the quotient of b and $b + d$.

Even though Cleverdon (1967, 174-175) works with five levels of relevance, a binary view of relevance will prevail eventually: a documentary unit is either relevant for the satisfaction of an information need or it is not.

It is possible to unite the two effectiveness values Recall and Precision into a single value. In the form of his E-Measurement, van Rijsbergen (1979, 174) introduces a variant of the harmonic mean:

$$E = 1 - \frac{1}{\alpha \left(\frac{1}{P} \right) + (1 - \alpha) \left(\frac{1}{R} \right)}$$

α can assume values between 0 and 1. If α is greater than 0.5, greater weight will be placed on Precision; if α is smaller than 0.5, Recall will be emphasized. In the value $\alpha = 0.5$, effectiveness is balanced on P and R in equal measure. The greatest effectiveness is reached by a system when the E-value is 0.

Since 1992, the TREC conferences have been held (in the sense of the Cranfield paradigm) (Harman, 1995). TREC provides experimental databases for the evaluation of retrieval systems. The test collection is made up of three parts:

- the documents,
- the queries,
- the relevance judgments.

The documents are taken from various sources. They contain, for instance, journalistic articles (from the “Wall Street Journal”) or patents (from the US Patent and Trademark Office). The queries represent information needs. “The topics were designed to mimic a real user’s need, and were written by people who are actual users of a retrieval system” (Harman, 1995, 15). Table H.4.3 shows a typical TReC query.

Table H.4.3: Typical Query in TReC. Source: Harman, 1995, 15.

Query	
<i>Number:</i>	066
<i>Domain:</i>	Science and Technology
<i>Topic:</i>	Natural Language Processing
<i>Description:</i>	Document will identify a type of natural language processing technology which is being developed or marketed in the U.S.
<i>Narrative:</i>	A relevant document will identify a company or institution developing or marketing a natural language processing technology, identify the technology, and identify one or more features of the company’s product.
<i>Concept(s):</i>	1. natural language processing; 2. translation, language, dictionary, font; 3. software applications
<i>Factors:</i>	Nationality: U.S.

The relevance judgments are made by assessors. For this, they are given the following instruction by TReC (Voorhees, 2002, 359):

To define relevance for the assessors, the assessors are told to assume that they are writing a report on the subject of the topic statement. If they would use any information contained in the document in the report, then the (entire) document should be marked relevant, otherwise it should be marked irrelevant. The assessors are instructed to judge a document as relevant regardless of the number of other documents that contain the same information.

Each document is evaluated by three assessors in TReC. Inter-indexer consistency is at an average of 30% between all three assessors, and at just under 50% between any two assessors (Voorhees, 2002). This represents a fundamental problem of any evaluation following the Cranfield paradigm: relevance assessments are highly subjective.

In order to calculate the Recall, we need values for c , i.e. the number of documents that were not retrieved. We can only calculate the absolute Recall for an information service after having analyzed all documents for relevance relative to the queries. This is practically impossible in the case of large databases. TReC works with “relative Recall” that is derived from the “pooling” of different information services. In the TReC experiments, there are always several systems to be tested. In each of them, the queries are processed and ranked according to relevance (Harman, 1995, 16-17):

The sample was constructed by taking the top 100 documents retrieved by each system for a given topic and merging them into a pool for relevance assessment. ... The sample is then given to human assessors for relevance judgments.

The relative Recall is thus dependent upon the pool, i.e. on those systems that just happen to be participating in a TREC experiment. Results for the relative Recall from different pools cannot be compared with one another.

An alternative method for calculating Recall is Availability. This method stems from the evaluation of library holdings and calculates the share of all successful loans relative to the totality of attempted loans (Kantor, 1976). The assessed quantity is known items, i.e. the library user knows the document he intends to borrow. Transposed to the evaluation of retrieval systems, known items, i.e. relevant documents (e.g. specific URLs in the evaluation of a search engine) are designated as the test basis. The documents were not retrieved via the search engine and they are available at the time of analysis. Relevant queries are then constructed from the documents that must make the latter searchable in a retrieval system. Now the queries are entered into the system that is to be tested and the hit list analyzed. The question is: are the respective documents ranked at the top level of the SERP (i.e. among the first 25 results)? The Availability of a retrieval system is the quotient of the number of retrieved known items and the totality of all searched known items (Stock & Stock, 2000).

When determining Precision in systems that yield their results in a relevance ranking, one must determine a threshold value up to which the search results are analyzed for relevance. This cut-off value should reflect the user behavior to be observed. From user research, we know that users seldomly check more than the first 15 results in Web search engines. Here, the natural decision would thus be to set the threshold value at 15. Does it make sense to use the Precision measurement? In the following example, we set the cut-off value at 10. In retrieval system A, let the first five hits be relevant, and the second five non-relevant. The Precision of A is $5 / 10 = 0.5$. In retrieval system B, however, the first five hits are non-relevant, whereas those ranked sixth through tenth are relevant. Here, too, the Precision is 0.5. Intuitively, however, we think that system A works better than system B, since it ranks the relevant documentary units at the top.

A solution is provided by the effectiveness measurement MAP (mean average precision) (Croft, Metzler, & Strohman, 2010, 313):

Given that the average precision provides a number for each ranking, the simplest way to summarize the effectiveness of rankings from multiple queries would be to average these numbers.

MAP calculates two average values: average Precision for a specific query on the one hand, and the average values for all queries on the other. We will exemplify this via an example (Figure H.4.2). Let the cut-off value be 10. In the first query, five documents (ranked 1, 3, 6, 9 and 10) are relevant. The average Precision for Query 1 is 0.62.

The second query leads to three results (ranked 2, 5 and 7) and an average Precision of 0.44. MAP is the arithmetic mean of both values, i.e. $(0.62 + 0.44) / 2 = 0.53$. The formula for calculating MAP is:

$$\text{MAP} = \frac{1}{n} \sum_{i=1}^n \frac{1}{m} \sum_{j=1}^m P(R_{kj})$$

$P(R_{kj})$ is the Precision for the ranking position k , m is the cut-off value (or—if the hit list is smaller than the cut-off value—the number of hits) and n is the number of analyzed queries.

Ranking for Query 1 (5 relevant documents)

Rank	1	2	3	4	5	6	7	8	9	10
relevant?	r	nr	r	nr	nr	r	nr	nr	r	r
P	1.0	0.5	0.67	0.5	0.4	0.5	0.43	0.38	0.44	0.5

Average Precision: $(1.0 + 0.67 + 0.5 + 0.44 + 0.5) / 5 = 0.62$

Ranking for Query 2 (3 relevant documents)

Rank	1	2	3	4	5	6	7	8	9	10
relevant?	nr	r	nr	nr	r	nr	r	nr	nr	nr
P	0	0.5	0.33	0.25	0.4	0.33	0.43	0.38	0.33	0.3

Average Precision: $(0.5 + 0.4 + 0.43) / 3 = 0.44$

Mean Average Precision: $(0.62 + 0.44) / 2 = 0.53$

Cut-off: 10; r : relevant; nr: not relevant; P: precision

Figure H.4.2: Calculation of MAP. Source: Modified Following Croft, Metzler, & Strohman, 2010, 313.

Su (1998) suggests gathering user estimates for entire search result lists. “Value of search results as a whole is a measure which asks for a user’s rating on the usefulness of a set of search results based on a Likert 7-point scale” (Su, 1998, 557). Alternatively, it is possible to have a SERP be rated by asking the test subjects to compare this list with another one. The comparison list, however, contains the search results in a random ranking. This way, it can be estimated whether a hit list is better than a random ranking of results.

Evaluation of Evaluation Metrics

Precision, MAP, etc.—there are a lot of metrics for measuring the effectiveness of retrieval systems. Della Mea, Demartini, Di Gaspero and Mizzaro (2006) list more than 40 effectiveness measurements known in the literature. Are there “good” meas-

urements? Can we determine whether measurement A is “better” than measurement B? What is needed is an evaluation of evaluation in information retrieval (Saracevic, 1995). Metrics are sometimes introduced “intuitively”, but without a lot of theoretical justification. Why, for instance, are the individual Precision values of the relevant ranking positions added in MAP? Saracevic (1995, 143) even goes so far as to term Recall a “metaphysical measure”.

A huge problem is posed by uncritical usage of the dichotomous 0/1 view of relevance. In relevance research (Ch. B.3), this view is by no means self-evident, as a gradual conception of relevance can also be useful. Saracevic (1995, 143) remarks:

(T)he findings from the studies of relevance on the one hand, and the use of relevance as a criterion in IR evaluations, on the other hand, have no connection. A lot is assumed in IR evaluations by use of relevance as the sole criterion. *How justifiable are these assumptions?*

Furthermore, the assessors’ relevance assessments are notoriously vague. To postulate results on this basis, e.g. that the Precision of a retrieval system A is 2% greater than that of system B, would be extremely bold.

What is required is a calibration instrument on which we can measure the evaluation measurements. Sirotkin (2011) suggests drawing on user estimates for entire hit lists and to check whether the individual evaluation measurements match these. Such matches are dependent upon the cut-off value used, however. For instance, the traditional Precision for a cut-off value of 4 is “better” than MAP, whereas the situation is completely reversed when the cut-off value is 10 (Sirotkin, 2011).

“The evaluation of retrieval systems is a noisy process” (Voorhees, 2002, 369). In light of the individual evaluation metrics’ insecurities, it is recommended to use as many different procedures as possible (Xie & Benoit III, 2013). In order to glean useful results, it is thus necessary to process all dimensions of the evaluation of retrieval systems (Figure H.4.1).

Conclusion

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- A comprehensive model of the evaluation of retrieval systems comprises the four main dimensions IT service quality, knowledge quality, IT system quality and retrieval system quality. All dimensions make a fundamental contribution to the usage (or non-usage) of retrieval systems.
 - The evaluation of IT service quality comprises the process of service provision as well as important service attributes. The process component is registered via the sequential incident technique and the critical incident technique. The quality of the attributes is measured via SERVQUAL. SERVQUAL uses a double scale of expectation and perception values.
 - Knowledge quality is registered by evaluating the documentary reference units and the documentary units (surrogates) of the database. The information quality of documentary reference units is hard to quantify. Surrogate evaluation involves the KOSs used, the indexing, and the freshness of the database.

- Evaluation of IT system quality builds on the technology acceptance model (TAM). TAM has four subdimensions: perceived usefulness, perceived ease of use, trust, and fun. The usefulness of retrieval systems is expressed in their effectiveness (in finding the right information) and in their efficiency (in doing so very quickly).
 - Elaborate retrieval systems have functionalities for calling up databases, for identifying search arguments, for formulating queries, for the display and output of surrogates and documents, for creating and managing push services, and for performing informetric analyses. A system's range of commands is a measurement for the quality of its search features. The quality of the system's presentation, as well as that of its functions, is measured via usability tests.
 - The traditional parameters for determining the quality of retrieval systems are Recall and Precision. Both measurements have been summarized into a single value: the E-Measurement. Experimental databases for the evaluation of retrieval systems (such as TREC) store and provide documents, queries, and relevance judgments. The pooling of different databases provides an option for calculating relative Recall (relative always to the respective pool). Calculating Precision requires the fixing of a cut-off value. In addition to traditional Precision, MAP (mean average precision) is often used.
 - As there are many evaluation metrics, we need a calibration instrument to provide information on "good" measuring procedures, "good" cut-off values, etc. However, as of yet no generally accepted calibration method has emerged.
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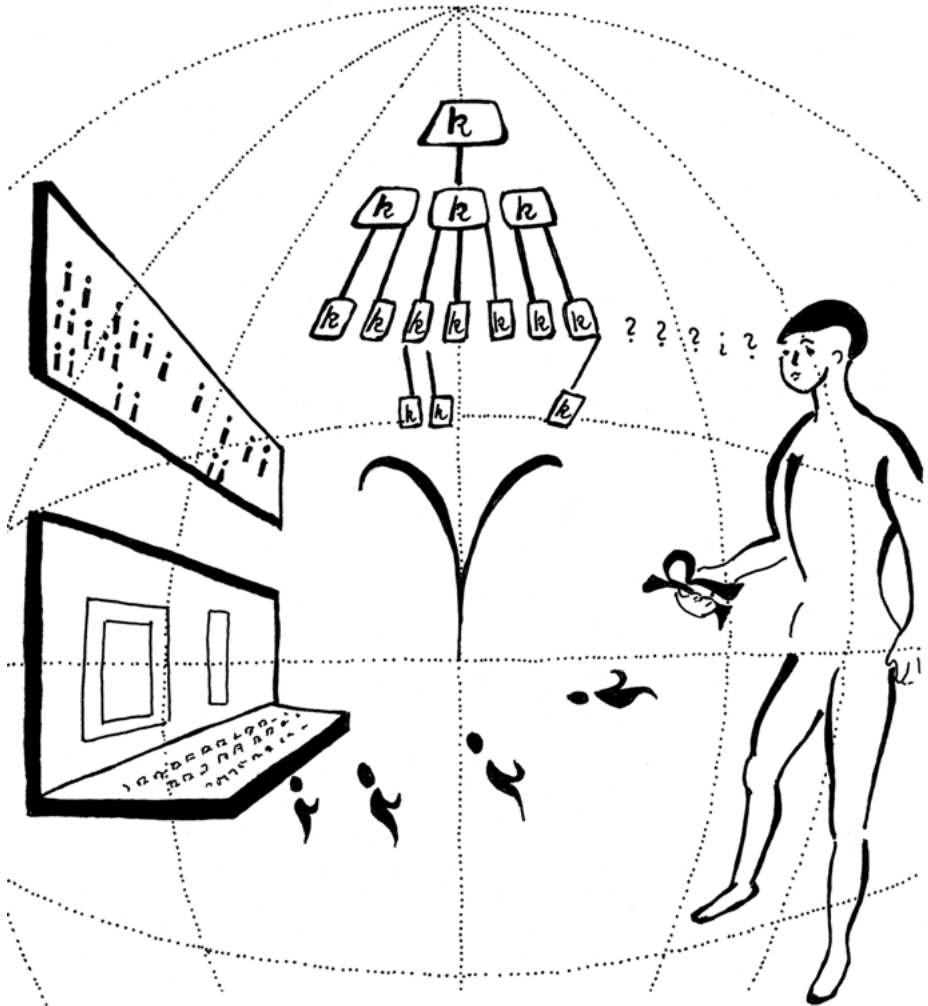
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Knowledge Representation



Knowledge Representation

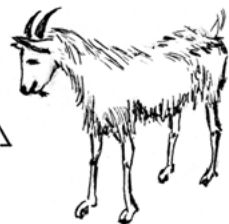
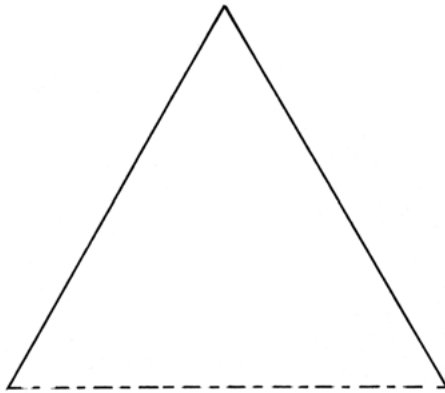
Part I

Propaedeutics of Knowledge Representation

$$\{X_1, X_2, \dots, X_i, \dots, X_n\}$$

$$f(x) \wedge f'(x) \wedge [g(x) \vee g'(x)]$$

Goat



I.1 History of Knowledge Representation

Antiquity: Library Catalogs and Hierarchical Concept Orders

The history of knowledge representation goes a long way back. Particularly philosophers and librarians face the task of putting knowledge into a systematic order—the former being more theoretically-minded, the latter concerned with the practical aspect. We thus have to pursue two branches in our short history of knowledge representation—one dealing with the theoretical endeavors of structuring knowledge, the other with the practical task of making knowledge accessible, be it via a systematic arrangement of documents in libraries or via corresponding catalogs.

The book generally regarded as *the* fundamental work for the history of classification, which over many centuries has been identical with the history of knowledge representation proper, is the “History of Library and Bibliographical Classification” by Šamurin (1977). According to Šamurin, our story begins with the catalogs found in the libraries of Mesopotamia and Egypt. Reports suggest that libraries have existed since around 2750 B.C., in Akkad (Mesopotamia) as well as in the 4th Dynasty (2900 – 2750 B.C.) in the ancient Egyptian El Giza (Giza). Of the Assyrian king Assurbanipal (668 – 626 B.C.)’s library, it is known that it had a catalog (Šamurin, 1977, Vol. 1, 6):

The library of Nineveh (Kuyunjik) had a stock of more than 20,000 clay tablets with texts in the Sumerian and Babylonian-Assyrian languages, and possessed a catalog of clay tablets with entries in cuneiform script.

Of the Egyptian library of the Temple of Horus in Edfu (Apollinopolis Magna), there has even survived the “catalog of the boxes containing books on large parchment scrolls”, i.e. a part of a classification system (Šamurin, 1977, Vol. 1, 8-9).

The high point of antique, practically-oriented knowledge representation (Casson, 2001) had probably been reached with the catalog of the Ptolemies’ library in Alexandria. Their librarian Callimachos (ca. 305 – ca. 240 B.C.) developed the “Pinakes” (*Pinax*: ancient Greek for *tablet* and *index*), which probably represented a systematic catalog and a bibliography at the same time (Šamurin, 1977, Vol. 1, 15; Schmidt, 1922). The “Pinakes”, according to Rudolf Blum (1977, 13), are a

catalog of persons who have distinguished themselves in a cultural field, as well as of their works.

According to reconstructions, Callimachos divided the authors into classes and sub-classes, ordered them alphabetically, according to the authors’ names, within the classes, added biographical data to the names, appended the titles of the respective author’s works and, finally, stated the first words of each text as well as the texts’ length (number of lines) (Blum, 1977, 231). The documentation totals 120 books

(Löffler, 1956, 16). The great achievement of Callimachos is not that he catalogued documents in the sense of an inventory—this had been common practice for a long time—but that he indexed the documents, albeit rudimentarily, in terms of content via a prescribed ordering system. For the first time, the content was at the center of attention; in our opinion, this is the hour of birth of content indexing and knowledge representation. Blum (1977, 325-326) emphasizes:

Callimachos did not structure the Alexandrine library—this had been achieved by Zenodotos, who was the first to alphabetically structure authors and, partly, works—but he tried to index it completely and reliably.

Blum (1977, 330), in consequence, does not only speak of a catalog, but also of “information brokering”:

What he (=Callimachos) thus did as a scholar was to disseminate information from literature and about literature.

The Alexandrine catalogs and the Pinakes by Callimachos were a “fundamental and scarcely emulated paragon” (Löffler, 1956, 20) for the cataloguing technology of Hellenism and of the Roman Empire.

The most important theoretical groundwork for knowledge representation has been developed by Aristotle (384 – 322 B.C.). Here we can find criteria according to which terms must be differentiated from one another, and according to which concepts are brought into a hierarchical structure. Of fundamental importance are the conceptions of genus and species (Granger, 1984). In the “*Metaphysics*” (1057b, 34 et seq.), we read:

Diversity, however, in species is a something that is diverse from an certain thing; and this must needs subsist in both; as, for instance, if animal were a thing in species, both would be animals: it is necessary, then, that in the same genus there be contained those things that are diverse in species. For by genus I mean a thing of such a sort as that by which both are styled one and the same thing, not involving a difference according to accident, whether subsisting as matter or after a mode that is different from matter; for not only is it necessary that a certain thing that is common be inherent in them, (for instance, that both should be animals,) but also that this very thing—namely, animal—should be diverse from both: for example, that the one should be horse but the other man.

A concept definition thus invariably involves stating the genus and distinguishing between the species (Aristotle, *Topics*, Book 1, Ch. 8):

The definition consists of genus and differentiae.

It is important to always find the nearest genus, and not to skip a hierarchy level (*Topics*, Book 6, Ch. 5):

Moreover, see if he (“a man”) uses language which transgresses the genera of the things he defines, defining, e.g. justice as a ‘state that produces equality’ or ‘distributes what is equal’: for by defining it so he passes outside the sphere of virtue, and so by leaving out the genus of justice he fails to express its essence: for the essence of a thing must in each case bring in its genus.

What powers the distinction between the species of a genus? Aristotle keeps apart two aspects—on the one hand, the coincidental characteristics of an object (e.g. that horses have a tail and humans don’t), and on the other hand, its essential ones, the specific traits that make the difference (in the example: that humans have reason and horses don’t). In the Middle Ages, this Aristotelean thesis was expressed in the following, easily remembered form (Menne, 1980, 28):

Definitio fit per genus proximum et differentiam specificam.

The definition thus proceeds by stating the respective superordinate generic term (genus proximus) and the fundamental difference to the other hyponyms within the same genus (differentia specifica). This task is not easy, as Aristotle emphasizes in his Topics (Book 7, Ch. 5):

That it is more difficult to establish than to overthrow a definition, is obvious from consideration presently to be urged. For to see for oneself, and to secure from those whom one is questioning, an admission of premises of this sort is no simple matter, e.g. that of the elements of the definition rendered the one is genus and the other differentia, and that only the genus and differentiae are predicated in the category of essence. Yet without these premises it is impossible to reason to a definition.

The rule that the hyperonym must always be stated in the definition inevitably results not only in the exact concept definition, but additionally brings with it a hierarchical concept order, such as a classification.

Lasting influence on later classification theories has been exerted by Porphyry (234 – 301) with his work “Introduction to Aristotle’s Categories” (Porphyrios, 1948). He himself did not create any classification schema, but his deliberations allow for a consistently dichotomous formation of the species definitions. This dichotomy is a special case, which can in no way be generalized and cannot be explained by any remarks of Aristotle’s, either.

Middle Ages and Renaissance: Combinational Concept Order and Memory Theater

The classification and cataloguing practice of medieval libraries was unable to build on the achievements of antiquity, since Alexandrian expert knowledge on libraries, for instance, had been lost. The catalogs were now mere shelf catalogs, with no atten-

tion paid to the works' contents at all. We will thus only consider two theoretical conceptions from this time, both of which attempt to organize all concepts, or even the entirety of knowledge.

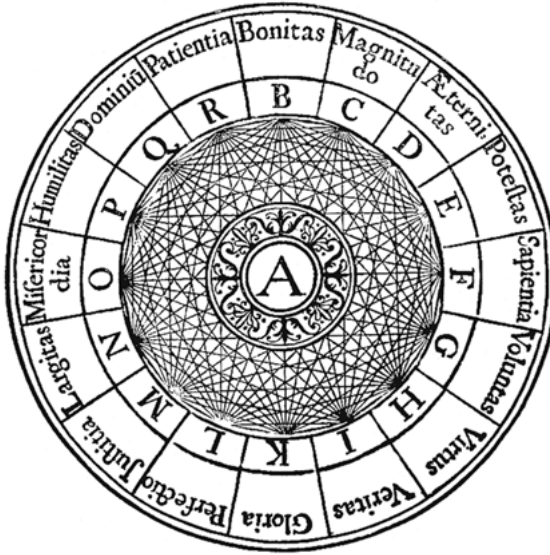


Figure I.1.1: Example of a Figura of the Ars Magna by Llull. Source: Lullus, 1721, 432A.

Llull (1232 – ca. 1316) built a system which—consolidated by combinational principles—helps to recognize and systematize all concepts (Henrichs, 1990; Yates, 1982; Yates, 1999). His “Ars Magna” was created, in several variants, between 1273 and 1308. For the different levels of being (from inanimate nature via plants etc. up to the angels and, finally, God; Yates, 1999, 167), Llull constructs discs (*figurae*), each of which represent categorical concepts, where the concepts are always coded via letters (e.g. C for ‘Magnitudo’ on Disc A). Figure I.1.1 shows Disc A (*Absoluta*) at the top. Added to this are further discs, structured concentrically. Turning the discs results in combinations between concepts, whose meaningful variants are recorded as *Tabulas*. Two aspects in particular are important in Llull’s work: the coding of the concepts via a sort of artificial notation (Henrichs, 1990, 569) and his claim of being able to represent not only the entirety of knowledge via these combinations, but also to offer heuristics for locating regions of knowledge previously disregarded. Yates (1982, 11) emphasizes:

There is no doubt that the Art is, in one of its aspects, a kind of logic, that it promised to solve problems and give answers to questions (...) through the manipulation of the letters of the figures. ... Llull ... claimed that his Art was more than a logic; it was a way of finding out and “demonstrating” truth in all departments of knowledge.

Camillo's primary aim was not to cleverly arrange knowledge (which has, by all means, been achieved in the audience and the segments—according to magical and alchemical principles, respectively), but to make the knowledge “catch the visitor's eye”, allowing him to easily attain and retain it. The user is the actor, since it is he who stands on the stage, and the images in the circles look to him. Camillo's memory theater is, in the end, a “staging of knowledge” in opposition to “dead attic knowledge” (Matussek, 2001b, 208). Camillo makes it clear that the knowledge organization system is an expression of a (collective) memory, and that one of the tasks at hand is also the interactive presentation of the knowledge.

From the Modern Era to the 19th Century: Science Classification, Abstract, Thesaurus, Citation Index

In the following centuries, the dominant endeavors are of creating classifications, particularly in the sciences, both from a theoretical as well as from a practical, library-oriented perspective. It is worth mentioning, in this context, the general science classifications by Leibniz (1646 – 1716) (the so-called “faculty system”, Šamurin, 1977, Vol. 1, 139 et seq.), Bacon (1561 – 1626) (Šamurin, 1977, Vol. 1, 159 et seq.) as well as those by the encyclopedists Diderot (1713 – 1784) and d'Alembert (1717 – 1783) (Šamurin, 1977, Vol. 1, 187 et seq.).

Classifications of lasting influence were created in the natural sciences in the 18th and early 19th centuries. Here, von Linné (1707 – 1778), with his work “Systema naturae” (1758) and de Lamarck (1744 – 1829) with his “Philosophie zoologique” (1809) must be noted. The (typically Aristotelian) variant of stating genus and species, still prevalent to this very day, is introduced into biological nomenclature (e.g.: *Capra hircus* for domestic goat; Linnaeus, 1758, Vol. 1, 69). Lamarck (1809, Vol. 1, 130) introduces the idea that the organizing principle be arranged from the more complicated towards the simpler organisms. His organizing principle leads to a classification according to vertebrates and invertebrates and, in the first class, to mammals, birds, reptiles and fish as well as, for the invertebrates, from mollusks to infusoria (Lamarck, 1809, Vol. 1, 216).

Methodically, classification research hardly develops at all in the modern era; progress is unambiguously concentrated on the side of content.

Classifications and other forms of concept orders serve as information filters; they support the process of finding precisely what the user requires in a (huge) quantity of knowledge. The modern era is when a second mode of dealing with knowledge is being developed: information aggregation. Here the goal is to represent long documents via a short, easily understandable text. Such a task was accomplished by the first scientific journals. The book market had grown so large in the 17th century that individual scholars found it hard to stay on top of ongoing developments. Early journals, such as the “Journal des Scavans” (founded in 1665), printed short articles that

either briefly summarized books or reported on current research projects and discoveries. Over the following two decades, the number of journals rose sharply, and comprehensive research reports soon established themselves as the predominant type of article. The consequence was that journal literature also became unparsable. This, in turn, led to the foundation of abstract journals (such as the “Pharmaceutisches Central-Blatt”, 1830) and the “birth” of Abstracts (Bonitz, 1977).

The year 1852 saw the publication of a thesaurus of the English language—marking the first time that concepts and their relations had been arranged in the form of a vocabulary. This work by Roget (1779 – 1869) generally leans towards lexicography, but also—if only secondarily—towards knowledge representation. It unites two aspects: that of a dictionary of synonyms and that of a topical reference collection (Hüllen, 2004, 323):

Roget’s *Thesaurus* (is) a topical dictionary of synonyms.

The topical order is kept together via hierarchical relations between classes, sections, groups etc., e.g. (Hüllen, 2004, 323):

Class III. *Matter*,
 Section III: *Organic matter*,
 Group 2°: *Sensation (1) General*,
 Subgroup 5: *Sound*,
 Numbered entry articles: 402-19.

The articles each contain the main entry (e.g. ODOUR), followed by its synonyms (smell, odorament, scent etc.), where, to some extent, explanations are made and homonyms disambiguated. The order of the main entries within each respective superordinate class follows various different relations, such as antonyms, arrangement according to *absolute* (for *Simply Quantity* into *Quantity*, *Equality*, *Mean* and *Compensation*) and *relative* (here: into *Degree* and *Inequality*) or according to *finite* (for *Absolute Time* into *Period* and *Youth*, among others) and *infinite* (here: into *course* or *age*, respectively). Roget artificially limited his thesaurus to 1,000 main entries; the concept order ends with *Religious Institutions*: 1,000 *Temple* (Roget, 1852, XXXIX).

Poole (1878) attempted (without any lasting success) to use thesauri as vocabularies for catalogs.

A completely different path from that of classifications and thesauri is taken by citation indexing as a form of knowledge representation. Here, bibliographic statements within the text, in the footnotes or the bibliography are analyzed as bearers of knowledge. Following predecessors in the form of citation indices of Biblical passages in Hebrew texts from the 16th century (Weinberg, 1997), one of the “great” citation indices was created in 1873. “Shepard’s Citations”—constructed by Shepard (1848 – 1900)—comprise references to earlier rulings in current court cases, where the verdict is qualified by a specialist (e.g. as “cited positively” or “cited negatively”) (Shapiro, 1992).

Decimal Classification, FID and Mundaneum

Classification research and practice experienced a methodical revolution in the form of the “Decimal Classification” (1876) of the American librarian Dewey (1851 – 1931), working in Amherst (Gordon & Kramer-Greene, 1983; Rider, 1972; Vahn, 1978; Wiegand, 1998). The basic idea is simple (Binswegen, 1994). Knowledge is always divided into at most ten hyponyms, represented via decimal digits. The books’ arrangement in the library follows this classification, so that thematically related works are located in close proximity to each other. The decimal principle may not lead to easily memorable, “expressive” notations, but it does permit easy dealings with the system, as it is freely expandable downwards. Dewey (1876) emphasizes, in the introduction to his classification:

Thus all the books on any given subject are found standing together, and no additions or changes ever separate them. Not only are all the books on the subject sought, found together, but the most nearly allied subjects precede and follow, they in turn being preceded and followed by other allied subjects as far as practicable. ... The Arabic numerals can be written and found more quickly, and with less danger of confusion or mistake, than any other symbols whatever.

Dewey’s Decimal Classification derived the structure of its content from the state of science in the mid-19th century. The first hierarchy level shows the following ten classes:

- 000 General
- 100 Philosophy
- 200 Theology
- 300 Sociology
- 400 Philology
- 500 Natural science
- 600 Useful arts
- 700 Fine arts
- 800 Literature
- 900 History.

We will try to exemplify the structure of Dewey’s classification via three examples (all from Dewey, 1876):

- 000 General
 - 010 Bibliography
 - 017 Subject Catalogues
- 500 Natural science
 - 540 Chemistry
 - 547 Organic
- 600 Useful arts
 - 610 Medicine
 - 617 Surgery and dentistry.

Šamurin (1977, Vol. 2, 234) acknowledges the position of Dewey in the history of knowledge representation:

Notwithstanding the ... criticism that met and continues to meet the Decimal Classification, its influence on the subsequent developments in library-bibliographic systematics has been enormous. Not one halfway significant classification from the late 19th and throughout the 20th century could ignore it. ... Dewey's work concludes a long path, on which his numerous predecessors proceeded, gropingly at first, then more confidently.

Dewey's work was received positively in Europe. On the instigation of the Belgian lawyer Otlet (1868 – 1944) (Boyd Rayward, 1975; Levie, 2006; Lorphèvre, 1954) and the Belgian senator and Nobel laureate La Fontaine (1854 – 1943) (Hasquin, 2002; Lorphèvre, 1954), the first “International Bibliographical Conference” was convened in Brussels in 1895. Here, it was decided that Dewey's classification should be revised and introduced in Europe. Since that time, there exist two variants of the Decimal Classification: the “Dewey Decimal Classification” DDC (new editions and revisions of Dewey's work) as well as the European “Classification Décimale Universelle” CDU (first completed in 1905; translated into English as the “Universal Decimal Classification” UDC and appearing, in 1932/33, in an abridged German edition as the “Dezimalklassifikation” DK; McIlwaine, 1997). The CDU contains, compared to the original, an important addition: the “auxiliary tables”, which function as facets (e.g. for statements pertaining to time and place), where the auxiliary tables' notations can be appended to each and every notation of the main tables. This enhances, on the one hand, the expressiveness of the classification, but only marginally increases the total number of notations. The auxiliary table for places uses the notation (43) for Germany. We are now able to always create a relation to Germany by appending (43), e.g.

Chemistry in Germany: 54(43)
or Surgery / Dentistry in Germany: 617(43)

“On the side”, Otlet and La Fontaine founded, in 1895, the “Institut International de Bibliographie”, which would be rechristened, in 1931, into “Institut International de Documentation” IID and, in 1937, into “Fédération Internationale de Documentation” FID (FID, 1995; Boyd Rayward, 1997). In 1934, Otlet published one of the fundamental works of documentation (Day, 1997). From 1919 onward, Otlet and La Fontaine tried to unite the entirety of world knowledge—structured classificatorily—in one location. However, their plan of this “Mundaneum” failed (Rieusset-Lemarié, 1997).

Faceted Classification

What began as an auxiliary table in the CDU—located more on the fringes—was declared the principle by the Indian Ranganathan (1892 – 1972). His “Colon Classifica-

tion” (Ranganathan, 1987[1933]) is faceted throughout, i.e. we no longer have a system of a main table but instead as many subsystems as dictated by necessary facets. This system surmounts the rigidity of the Decimal Classifications in favor of a synthetic approach. The notation does not come about via the consultation of a system location within the classification, but is built from notations from the respective appropriate facets when editing the document. Apart from one basic facet for the scientific discipline, Ranganathan works with five further facets:

Who?	Personality	(Separator: ,)
What?	Material	(;)
How?	Energy	(:; hence, “Colon Classification”)
Where?	Place	(.)
When?	Time	(').

The important aspect of Ranganathan’s system is not the specific design of the facets, as these differ by use case, but the idea of representing knowledge from several different perspectives simultaneously.

Present

We understand the “present” as the endeavors in knowledge representation from the second half of the 20th century onwards. Methods that break off from the classification and also point to alternative paths for structuring knowledge are developed in parallel to information retrieval. Here, we can be relatively brief, as we will address all these methods and tools over the following chapters.

In a decimal classification, the concepts are generally very finely differentiated and made up of several (conceptual) components (e.g. DK 773.7: Photographical procedures that use organic substances; procedures with dye-forming organic compounds and photosensitive dyes, e.g. diazo process). In this case, one speaks of “precombination”. However, one can also deposit the components individually (e.g. “photography”, “organic substance”, “dye” etc.), thus facilitating a “postcoordinated” indexing and search. Only the user puts the concepts together when searching, whereas in the precombined scenario, he will find them already put together. Postcoordinated approaches in knowledge representation can be found from the mid-1930s of the 20th century onward (Kilgour, 1997); they lead, via Taube (1953)’s “Uniterm System”, to the predominant method of knowledge representation from ca. 1960 onward, the thesaurus. In the late 1940s and 50s, there are early endeavors to construct terminological control and thesauri (or early forms thereof) for the purposes of retrieval (Bernier & Crane, 1948; Mooers, 1951; Luhn, 1953; for the history, cf. Roberts, 1984); from 1960 onward, this method is regarded as established (Vickery, 1960). Mooers (1952, 573) introduces “descriptors” (or “descriptive terms”) as means of indexing the messages of a document:

To avoid scanning all messages in entirety, each message is characterized by N independently operating digital descriptive terms (representing ideas) from a vocabulary V , and a selection is prescribed by a set of S terms.

The paragon for many documentary thesauri should be the “Medical Subject Headings”—“MeSH” in short—of the American National Library of Medicine, whose first edition appeared in 1960 (NLM, 1960). We already know of a thesaurus in Roget’s work, but that had been linguistically oriented, whereas MeSH (and other thesauri) introduced the thesaurus principle into the information practice of specialist disciplines. Lipscomb (2000, 265-266) points out the significance of MeSH:

In 1960, medical librarianship was on the cusp of a revolution. The first issue of the new *Index Medicus* series was published. ... A new list of subject headings introduced in 1960 was the underpinning of the analysis and retrieval operation. ... MeSH was a pioneering effort as a controlled vocabulary that was applied to early library computerization.

MeSH contains a dynamically developing quantity of preferred terms (descriptors), which are interconnected via relations and which form the basis of postcoordinated indexing and search, respectively.

Concepts—including those in specialist languages—are often expressed by diverse synonymous terms. Nomenclatures are needed, which add up the common variants into exactly *one* concept and define it. Chemical Abstracts Services (CAS) set standards with their CAS Registry Number (Weisgerber, 1997). This system, which allocates an unambiguous number to every substance and biosequence, has existed since 1965. Thus, the Registry Number 58-08-2 stands for caffeine and its synonyms (146 in total) as well as the structural formula (which is also graphically retrievable). Weisgerber (1997, 358) points out:

Begun originally in 1965 to support indexing for *Chemical Abstracts*, the Chemical Registry System now serves not only as a support system for identifying substances within CAS operations, but also as an international resource for chemical substance identification for scientists, industry, and regulatory bodies.

With regard to Shepard’s Citations, Garfield introduced his idea of a scientific citation index in 1955. As opposed to the legal citation index, Garfield does not qualify the references, instead only noting the occurrence of the bibliographic statement, since such an assessment can hardly be performed (automatically) in an academic environment (Garfield & Stock, 2002, 23). In 1960, Garfield founded the Institute for Scientific Information, which would go on to produce the “Science Citation Index” and, later, the “Social Sciences Citation Index” as well as the “Arts & Humanities Citation Index” (Cawkell & Garfield, 2001).

Not every discipline has a vocabulary that is shared by all its experts. Particularly in the humanities, the terminology is so diverse and usage of terms so inconsistent,

that the construction of classification systems or thesauri is not an option. This gap in information practice was filled by Henrichs with his text-word method (Hauk & Stock, 2012). In philosophical documentation, Henrichs (1967) works exclusively with the term material of the text in front of him, registering selected terms (understood as “search entry points” into the text) and their thematic relations in the surrogates.

In the environment of research into artificial intelligence, the idea of using ontologies developed around 1990. This is a generalization of the approach of classification and thesaurus, where (certain) logical derivations are included in addition to the concept orders. Since ontologies generally represent very complex structures, their application is restricted to reasonably assessable areas of knowledge. One of the best-known definitions of this rather IT-oriented approach of knowledge representation is from Gruber (1993, 199):

A body of formally represented knowledge is based on a *conceptualization*: the objects, concepts, and other entities that are presumed to exist in some area of interest and the relationship that hold them (...). A conceptualization is an abstract, simplified view of the world that we wish to represent for some purpose. Every knowledge base, knowledge-based system, or knowledge-level agent is committed to some conceptualization, explicitly or implicitly.

An *ontology* is an explicit specification of a conceptualization. The term is borrowed from philosophy, where an ontology is a systematic account of Existence. For knowledge-based systems, what “exists” is exactly that which can be represented.

Ontologies are the basic methods of the so-called “semantic Web” (Berners-Lee, Hendler, & Lassila, 2001).

With the advent of “collaborative” Web services, the indexing of content is also being approached collectively. The users become indexers as well. They use folksonomies (Peters, 2009), methods of knowledge representation that know no rules and whose value comes about purely from the mass of “tags” (being freely choosable keywords) and their specific distribution (i.e., a few very frequent tags and a “long tail” of further terms or relatively many frequent terms as the “long trunk” as well as the “long tail”).

With the folksonomies, the methods of knowledge representation now span all groups of actors dealing with knowledge and its structure:

- Experts in their field and in the application of knowledge representation (nomenclature, classification, thesaurus, ontology),
- Authors and their texts (text-word method, citation indexing),
- Users (folksonomy).

Nowadays, we can find approaches combining the social elements of folksonomies with elaborately structured ontologies, called the “social semantic Web” (Weller, 2010).

Conclusion

- Knowledge representation has a long history, which reaches back into antiquity. Endeavors to structure knowledge can be observed both in terms of practical (mainly in libraries) and theoretical deliberations (philosophy).
 - A fundamental development is the definition of concepts by stating the nearest hyperonym as well as the essential differences to its co-hyponyms in Aristotle. This is also the basis of the hierarchical concept order.
 - In the Middle Ages, Llull's combinational concept order with coded terms is noteworthy.
 - The Renaissance offers the schema of a memory theater by Camillo, which includes the idea of not merely structuring knowledge but—in order to better acquire it—to present it interactively.
 - In the 18th and early 19th century, progress was made in terms of the content of concept orders. The nomenclatures and classifications by Linné and Lamarck, influenced by Aristotle, are most important here.
 - Early forms of information aggregation via Abstracts can be found in the abstract journals of the 19th century. A thesaurus (which was mainly linguistically oriented) was introduced by Roget, a citation index for legal literature by Shepard.
 - Important influence on classification theory and practice was wielded by Dewey's Decimal Classification as well as its European variant (as CDU, UDC or DK, respectively).
 - Since around the second half of the 20th century, postcoordinated indexing and retrieval by using a (documentary) thesaurus has achieved widespread prevalence. Standards were set by the "Medical Subject Headings" (MeSH).
 - Text-oriented documentation methods—with no regard to prescribed concept orders—include Henrichs' text-word method and Garfield's citation indexing of academic literature.
 - Ontologies enhance concept orders, such as classifications and thesauri, via the option of allowing for certain logical conclusions.
 - Folksonomies allow users to describe documents according to their preferences via freely choosable keywords (tags).
 - The social semantic Web is a combination of the social Web (folksonomies) and the semantic Web (ontologies).
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1.2 Basic Ideas of Knowledge Representation

Aboutness and Ofness

The object of knowledge representation is—without any restriction—explicit objective knowledge. In the case of subjective knowledge, this knowledge must be externalized in order to be made accessible to scientific observation (and processing). This externalization is clearly limited by implicit knowledge. Explicit objective knowledge is always contained in documents. For Henrichs (1978, 161), the object is

knowledge in the objectivized sense, i.e. a (1.) represented (and thus fixed) knowledge and (2.) systematized (i.e. in some way contextualized) knowledge, which is thus formally (via the semi-otic system being used and its material manifestation) and logically (structurally and contentually) accessible and hence publicly available.

In knowledge representation, the documents from which the respective knowledge is to be determined are available. The object at hand is the documents' content. What is “in” a book, an article, a website, a patent, an image, a blog entry, an art exhibit or a company dossier?

In daily life, we are often confronted with the task of having to describe something to a conversational partner: the film we've just seen or the book we're reading at the moment, etc. We answer the question as to what the film or the book is about via one or several term(s), or we provide a short summary in complete sentences. Intuitively, we talk about the content of the medium. To describe “what it's about”, we will use the term “aboutness” (Lancaster, 2003, 13).

Why is it such a crucial endeavor of knowledge representation to identify “aboutness”? The core indexing activity is the indexer's decision as to which topic the document discusses and which terms are to be selected as key terms. This cannot be an arbitrary decision, but must be made with a view to the potential user, whose goal is to successfully satisfy his information need. Similar difficulties are encountered by people writing an abstract. One may be intuitively able to understand the meaning of “aboutness”; however, it is far more difficult—if not downright impossible—to describe exactly and in detail what happens during this decision process. Documents have a topic. Does the indexer make the right decision? Are we solely dependent upon his intuitive, implicit knowledge?

A documentary reference unit contains knowledge (or suppositions, assumptions etc.—depending) about specific objects. These objects must be identified so that they can be searched. The aboutness, or the topics discussed, must always be separated into the objective (“What?”) and the knowledge of the person processing the knowledge (“Who?”). Ingwersen (1992, 227) defines “aboutness”:

Fundamentally, the concept refers to “what” a document, text, image, etc., is about and the “who” deciding the “what”. ... (A)boutness is dependent on the individual who determines the “what” during the act of representation. Aboutness is divided into author aboutness, indexer aboutness, user aboutness and request aboutness.

Ingwersen is concerned with exposing the cognitive diversities in interpreting documentary reference units. Since every actor is anchored in his own social context, every representation has its own cognitive origin. Likewise, the various styles of representation depend upon the conventions of the specialized knowledge domain or of the media. The typology of aboutness is specified by Ingwersen (2002, 289):

- Author aboutness, i.e., the *contents as is*;
- Indexer aboutness, i.e., the *interpretation of contents* (subject matter) with a purpose;
- Request aboutness, i.e., the user or intermediary interpretation or *understanding of the information need* as represented by the request;
- User aboutness, i.e., *user interpretation* of objects, e.g., as *relevance feedback* during interactive IR or the use of information and giving credit in the form of references.

The author processes his subjects and formulates the results in a natural language (or in an image, a film, a piece of music etc.). The indexer’s interpretation is based upon the documentary reference unit and is different from the author’s. On the one hand, the indexing is oriented on the entire document, but on the other hand new perspectives are added to the content. Since indexing terms, or descriptors, are limited, the process is one of reduction (Ingwersen, 2002, 291).

A human indexer will concentrate on the text (or image etc.) and describe it via certain keywords. A human abstractor will attempt to condense the fundamental topics of the original into a short summary. Indexer and abstractor do not always work in the same way; their performance is subject to fluctuations. There are definitely some unfortunate results of “bad abstract days”. Both the indexer and the abstractor can be replaced by machines. The performance of these does not waver; whether or not they can reach the level of quality of human processing shall remain an open question for now. The topics worked out by indexer and abstractor are offered to the users; here, lists of indexing terms as well as the retrieval systems’ user interfaces are of great usefulness. “User aboutness” refers to the topics of the user’s subjective information need, i.e. the knowledge he thinks he needs in order to solve his problem. In the end, the user will formulate his information need with regard to a topic by formulating his query in a specific manner.

Starting from the precondition that when we read a document, our inner experience tells us what the text is about, Maron correlates human behavior with the document: the objective is asking questions and searching for information. The indexer’s task is to predict and answer the question about the expected search, respectively the searching behavior of the users who would be satisfied by the document in question. Maron (1977, 40) goes even further, reducing aboutness to the selection of terms:

I assert that for indexing purposes the term (or terms) you select is your behavioural correlate of what you think that document is *about*—because the term is the one you would use to ask for that document. So we arrive at the following behavioural interpretation of *about*: The behavioural correlate of what a document is *about* is just the index term (or terms) that would be used to ask for that item. How you would ask is the behavioural correlate (for you) of what the document in question is *about*.

Statements about documents, terms and document topics are arranged in a document-term matrix and their relations are observed. Maron distinguishes between three kinds of aboutness: “S-about” concerns subjective experience, the relation between the document and the result of the reader’s inner experience. This rather psychological concept is objectified in “O-about” by incorporating an external observer; it relates to the current or possible individual behavior of questions or searches for documents. In “R-about”, the focus is on a user community’s information search behavior during retrieval. The goal of this operational description of aboutness is to uncover probability distributions, e.g. the search behavior of those who are satisfied by certain documents. R-aboutness is explained by those terms that lead to the information need being satisfied and which would be used by other users. It is the relation between the number of users whose information needs have been satisfied by a document if they used certain terms when formulating their query and the number of users whose information need has been (initially) obliterated by the document.

Bruza, Song and Wong (2000) doubt that the qualitative features of aboutness can be linked to probability-theoretical quantities. Some of the qualities of aboutness, according to Bruza et al., can be found in the context of a given retrieval system, but not in the user’s perspective. As a solution, the enumeration of common-sense characteristics and their logical relations is developed. Even the perspective of “non-aboutness” is taken into consideration and formalized. Lancaster (2003, 14) remarks:

In the information retrieval context, nonaboutness is actually a simpler situation because the great majority of items in any database clearly bear no possible relationship to any particular query or information need (i.e., they are clearly “nonabout” items).

What and for the benefit of whom does the author want to communicate; what does the reader expect from the document? These two central questions are taken on by Hutchins, who observes aboutness from the linguistic side. The linguistic analysis of the structure of texts shows that the sentences, paragraphs and the entirety of the text represent something given, something already known and something completely new, respectively. Hutchins (1978, 173-174) writes:

In any sentence or utterance, whatever the context in which it may occur, there are some elements which the speaker or writer assumes his hearer or reader knows of already and which he takes as “given”, and there are other elements which he introduces as “new” elements conveying information not previously known.

Given elements stand in relation to the preceding discourse or to previously described objects or events, and are called “themes” by Hutchins. Unpredictable, new elements are called “rhemes”. A characteristic of all texts is their ongoing semantic development. This is also the route taken by the author when he writes a document. Writing about a given subject, he presupposes a certain level of knowledge, e.g. linguistic competence, general knowledge, a social background or simply interest. He uses these presuppositions to establish contact with the reader. The reader, in turn, shows his interest by wanting to learn something new, on the one hand, and by not being overtaxed by too much foreknowledge that he does not possess. Aboutness, according to Hutchins, refers to the author’s text. Accordingly, the indexer should not try to communicate his own interpretation of the text to the users, but instead minimize any and all obstacles between the text and its readers. Indexers should only use such topics and terms that have been presupposed by the text’s author for his readers (Hutchins, 1978, 180):

My general conclusion is that in most contexts indexers might do better to work with a concept of “aboutness” which associates the subject of a document not with some “summary” of its total content but with the ‘presupposed knowledge’ of its text.

Hutchins also admits some exceptions for abstracting services and specialized information systems, though. The task of abstracting is to summarize. Specialized information systems are tailored to a certain known user group; the user’s wishes ought to be known.

Salem (1982) concentrates on in-depth indexing, and thus analyzes a document according to two viewpoints. The core, which is normally the central topic spanning the document’s entire content, is represented by a term. The “graduated” aboutness contains many themes represented by different key terms (strong relation to the core) and other terms having little or nothing to do with the core. According to Salem, the indexer’s difficulty lies in identifying the core terms and those terms graduated toward the core subject. Via an empirical study, he finds out that the amount of terms with strong relations to the core should exceed the number of terms without any relation to the core.

If a document has several semantic levels—as is generally the case for images, videos and pieces of music, as well as in some works of fiction—we must distinguish between these on the level of the topics. In information science literature, the distinction between ofness and aboutness has been introduced by Shatford (-Layne) (1986). Ofness relates to the pre-iconographical level discussed by Panofsky, whereas aboutness is concerned with the iconographical level (Lancaster, 2003; Layne, 2002; Shatford-Layne, 1994; Turner, 1995). The third level of iconology does not play any role for information science, since it involves specialist (e.g. art-historical) questions. Our example of “The Last Supper” in Chapter A.2 would thus have the ofness value “Thirteen people, long table, hall-like room”, and—from a certain perspective—the

aboutness “Last supper of Jesus Christ with his disciples, shortly after the announcement that one of his stalwarts will betray him.” While the ofness can be evaluated relatively clearly, it is highly complicated to correctly index the aboutness (Svenonius, 1994, 603). In databases that analyze images or videos (e.g. of film or broadcasting companies), both aspects must be kept strictly separate.

Objects

The aboutness and ofness of a document show what it is about. This “what” are the topics—more specifically, the objects thematized. The concept of the object (in German *Gegenstand*) must be viewed very generally; it involves real objects (such as the book you are holding in your hand), theoretical objects (e.g. the objects of mathematics), fictional objects (such as Aesop’s fox) and even physically and logically impossible objects (e.g. a golden mountain and a round rectangle). Following the Theory of Objects by Meinong (1904), we are thus confronted by three aspects: (1st) the object (which exists independently of any subject), (2nd) the experience (the object’s counterpart in the experiencing subject) as well as (3rd) a psychological act that mediates between object and experience.

According to Meinong, there are two important kinds of objects (*Gegenstände*): single objects (Meinong calls them *Objekte*) and propositions (*Objektive*). Propositions link single objects to complex state of affairs. On the side of experience, imagination corresponds with the single object and thinking with the proposition. The psychological act runs in two directions: from single object to imagination, as experience; and from imagination to single object, as fantasy. The analogous case applies to judgement and supposition on the level of propositions and thinking, respectively.

Objects (*Gegenstände*) in documents must thus always be analyzed from two directions: as specific single objects, and jointly with other objects in the context of propositions. In surrogates, single objects are described via concepts, propositions via sentences. The first aspect leads to information filters, the second to information condensation.

Knowledge Representation—Knowledge Organization—Knowledge Organization System (KOS)

When representing the thematized objects in documents, we are confronted by three aspects: knowledge representation, knowledge organization and knowledge organization system. Kobsa (1982, 51) introduces a rather general definition of knowledge representation:

Representing the knowledge *K* via *X* in a system *S*.

Knowledge (K) is fixed in documents, which we divide into units of the same size, the so-called documentary reference units. S stands for a (these days, predominantly digital) system, which represents the knowledge K via surrogates. These surrogates X are of manifold nature, depending on how one wishes to represent the K in S. X in a popular Web database for videos will look entirely different from X' in a database for academic literature. The object is knowledge representation via language, or more precisely: via concepts and statements, regardless of whether the knowledge is retrieved in a textual document, a non-textual document (e.g. an image, a film or a piece of music) or in a factual document. Here, we in general work with concepts, not with words or non-textual forms of representation. This generally distinguishes the approach of knowledge representation, as “concept-based information retrieval”, from “content-based information retrieval”, in which a document is indexed not conceptually but via its own content (via words in the context of text statistics or via certain characteristics such as color distributions, tones etc. in non-textual documents).

When we speak of “representation”, we are not referring to a clear depiction in the mathematical sense (which is extremely difficult to achieve—if at all—in the practice of content indexing), but, far more simply, of replacement. This reading goes back to Gadamer (2004, 168):

The history of this word (representation; A/N) is very informative. The Romans used it, but in the light of the Christian idea of the incarnation and the mystical body is acquired a completely new meaning. Representation now no longer means “copy” or “representation in a picture”, or “rendering” in the business sense of paying the price of something, but “replacement”, as when someone “represents” another person.

X replaces the original knowledge of the document; the surrogate is the proxy—the representative—of the documentary reference unit in the system S.

In the context of artificial intelligence research, knowledge representation is also understood as automatic reasoning. This means derivations of the following kind: *When this is the case, then that will also be the case.* For instance, if in a system we know that *All creatures that have feathers are birds*, and, additionally, that *Charlie has feathers*, the system will be able to automatically deduce *Charlie is a bird* (Vickery, 1986, 153). We will pay particular attention to this subject in the context of ontologies.

When we forego the aspects of automatic reasoning and information condensation (i.e., information summarization), we speak of “knowledge organization”. Knowledge is “organized” via concepts. The etymological root of the word “organize” goes back to the Greek *organon*, which describes a tool. Kiel and Rost (2002, 36) describe the “organization of knowledge”:

The fundamental unit, which ... is organized, are concepts and relations between them. ... Here, organization is far more than simply an order of the concepts found in or ascribed to the documents.

The fact that knowledge is present in documents does not automatically mean that it is straight-forwardly available to every user at all times. Safeguarding this accessibility, or availability, requires the organization of knowledge, as Henrichs (1978, 161-162) points out:

Organization here means: measures toward controlling all processes of providing access/availability, i.e. also the dissemination of such knowledge, and at the same time the institutionalization of this dissemination.

In certain knowledge domains, it is possible to not only organize knowledge, but also to order it via a predetermined system. In areas that do not conform to normal science, this option is not available. Here, one can only use such organization methods that do not approach the knowledge via preset organization systems, but either derive the knowledge directly from the documents (such as e.g. the text-word method or citation indexing) or allow the user to freely tag the documents (folksonomy). The knowledge organization system (KOS), sometimes called “documentary language”, is always an order of concepts which is used to represent documents, including their oneness and aboutness (Hodge, 2000). Common types of knowledge organization systems include the nomenclature, the classification and the thesaurus. The ontology has a special status, since it has both a concept order and a component of automatic reasoning. Furthermore, ontologies attempt to represent the knowledge itself, not the documents containing the knowledge.

The narrowest concept is that of the knowledge organization system (KOS); knowledge organization comprises all such systems as well as further user- and text-oriented procedures. The most general concept we use is knowledge representation, which unites knowledge organization, ontologies as well as information condensation in itself.

Methods of Knowledge Representation

The various methods offered by knowledge representation can be arranged, via the main actors (author, indexer and user), into three large groups (Fig. I.2.1). Knowledge organization systems, such as nomenclature, classification, thesaurus and ontology, require professional specialists—for two completely different tasks:

- Constructing specific tools on the basis of one of the stated methods (e.g. the development and maintenance of the “Medical Subject Headings” MeSH),
- The practical application of these tools, i.e. indexing (e.g. analyzing the topics of a medical specialist article via the concepts in MeSH).

A note on terminology on the subject of “ontology”: This concept is not used in a unified manner in the literature (Gilchrist, 2003). There are authors who use “ontology” synonymously to “knowledge organization system”, i.e. subsuming thesauri

and classifications. We do not follow this terminology. An “ontology” (in the narrow sense), to us, is a specific knowledge organization system, which is available in a machine-readable, formal language and also disposes of mechanisms of automatic reasoning. When a nomenclature, a classification system or a thesaurus is realized in a machine-readable language, we call this KOS a “simple KOS” (or SKOS)—in contrast to a “rich” KOS, i.e. an ontology. When knowledge organization systems (SKOS or ontologies) are linked they form “networked KOS” (or “NKOS”).

Knowledge organization systems represent their own (artificial) languages; like any other language, they have a lexicon and rules. They can be used everywhere where normal science conditions are given.

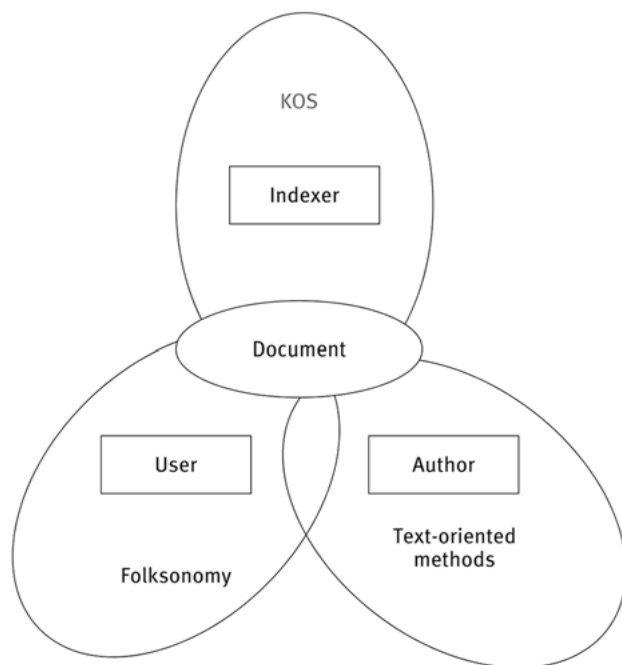


Figure I.2.1: Methods of Knowledge Representation and their Actors.

The author has created a document, which now speaks for itself. If the document is a text, then specific search entries to the text can be created by marking certain keywords (in the simplest case: the title terms; but also far more elaborately, in the context of the text-word method). If the document contains references (e.g. in a bibliography), these can be analyzed as a method of knowledge organization in the context of citation indexing. Text-linguistic methods of knowledge organization are used as a supplement to the knowledge organization systems. In non normal science areas of knowledge (such as philosophy), they play a fundamental role, since no knowledge

organization systems can be created there. These methods cannot be applied to non-textual documents, of course.

Folksonomies are a sort of free tagging systems, where the users of the documents assign keywords to documents, unbound by any rules. Where large amounts of users work on this method of knowledge organization certain regularities of distribution of the individual tags (e.g. the Power Law and the inverse-logistic distribution) provide for interesting options of search and retrieval. Folksonomies complement every other method of knowledge representation. In areas of the World Wide Web that contain large masses of document types (such as images or videos) that cannot easily be searched via algorithmic procedures, folksonomies often represent the only way (mainly for economic reasons) of indexing the documents' topics in the first place.

It is by no means impossible to use different methods as well as different specific tools in order to represent a singular document. In such a case, we speak of “polyrepresentation” (Ingwersen & Järvelin, 2005, 346). Thus it is possible, for instance, to employ different knowledge organization systems from the same specialist discipline (e.g. an economic thesaurus and an economic classification system). Analogously, in many cases it will be recommended to use the same method but employing tools from different disciplines. Such a procedure is particularly suited for documents whose knowledge is situated in fringe areas of several different disciplines (e.g. in a text about economic aspects of a certain technology, using both an economic and a technological thesaurus). Larsen, Ingwersen and Kekäläinen (2006, 88) plead for the use of polyrepresentation:

There are many possibilities of representing information space in cognitively different ways. They range from combining two or more complementing databases, over applying a variety of different indexing methods to document contents and structure in the same database, including the use of an ontology, to applying different external features that are contextual to document contents, such as academic citations and inlinks.

Indexing and Summarizing

The word “indexing” goes back to the “index” of a book (Wellisch, 1983; on how to create a book index, cf. Mulvany, 2005). In the context of knowledge representation, we use “indexing” with a much broader meaning: indexing means the practical work of representing the thematized single objects in a documentary reference unit on the surrogate, i.e. the documentary unit, via concepts. Here, the indexer makes use of the methods of knowledge representation and the respective specific tools. From the document, he extracts that knowledge which it is about (aboutness and ofness), and thus determines those concepts that will let the surrogate “get caught” in an information filter (like a nugget of gold in a sieve). Indexing is performed via knowledge organization systems (nomenclature, classification, thesaurus, ontology), via rule-governed

further methods of knowledge organization (text-word method, citation indexing) or in complete freedom of any rules in the case of folksonomies. Indexing is performed either intellectually, by human indexers, or automatically, via systems. If humans perform the work, the process always incorporates—besides the explicit knowledge from the documents—the implicit (fore)knowledge of these people. In case of automatic indexing, the system creator's implicit knowledge has been embedded within the system. It is thus unsurprising, even to be expected, that in intellectual indexing the surrogates of one and the same document indexed by two people (or by the same person at different times) should be different. If the document is automatically indexed, by the same system, the surrogates will of course be identical—however, they may all be equally “off” or even false.

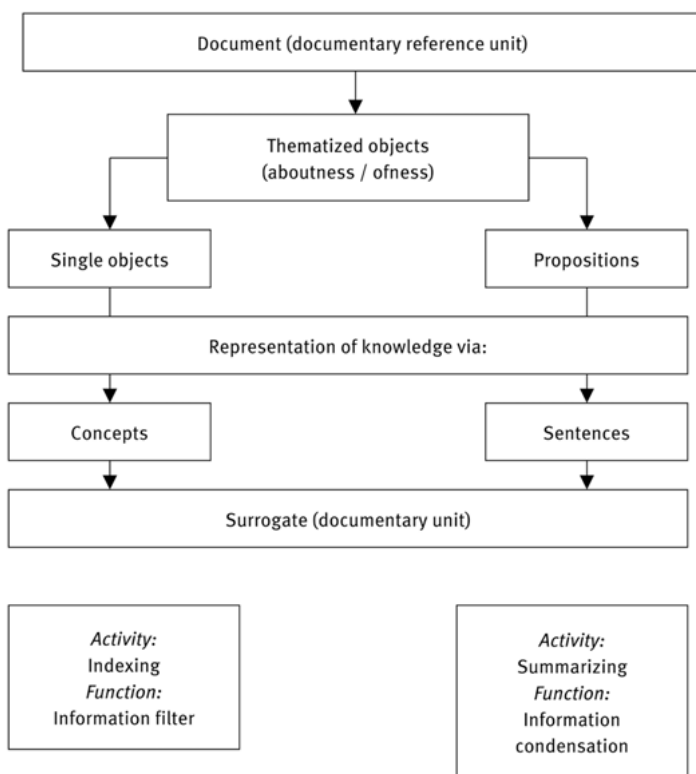


Figure I.2.2: Indexing and Summarizing.

Content condensation is accomplished via summarizing. The thematized facts of the document are represented briefly but as comprehensively as possible, in the form of sentences—e.g. as a “short presentation”, i.e. an abstract or an extract. Depending on the size of the documentary reference unit, the reduction of the overall content can be considerable. Abstracting pursues the primary goal of helping the user consolidate

a yes/no decision—i.e. “do I really need this document or not?”—with regard to the surrogates found via the information filters. Summaries thus do not help the search for knowledge—like concepts do—but only serve to represent the most basic facts in a cropped, “condensed” way.

Like indexing, summarization is performed either intellectually (as abstracts) or automatically (as extracts). Also like indexing, we are always confronted by the implicit knowledge of the abstractors and the system designers, respectively.

Conclusion

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- The knowledge discussed in a document and which is meant to be represented is its aboutness (its topics). Centrally important knowledge is formed by the core subjects. In the case of non-textual documents, as well as some fictional texts, the difference between ofness (subjects on the pre-iconographical level) and aboutness (subjects on the iconographical level) must be noted.
 - Objects (*Gegenstände*) are either single objects (*Objekte* according to Meinong) or several single objects in a context (proposition or state of affair). Single objects are described via concepts, propositions via sentences.
 - In knowledge representation, the object is to represent the knowledge found in documents in information systems, via surrogates. Knowledge representation comprises the subareas of information condensation, information filters as well as automatic reasoning.
 - Information filters are created in the context of knowledge organization. Apart from knowledge organization systems (nomenclature, classification, thesaurus, ontology), one works with text-oriented methods (citation indexing, text-word method) as well as user-oriented approaches (folksonomy).
 - It is possible, even in some cases recommended, to apply several methods of knowledge representation (e.g., a classification system and a thesaurus), as well as different tools (e.g., two different thesauri), to documents in equal measure (polyrepresentation).
 - Indexing means the representation of objects’ aboutness via concepts, summarizing means representing the thematized propositions in the form of sentences. Indexing consolidates information filtering, summarizing (abstracting or extracting) consolidates information condensation.
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I.3 Concepts

The Semiotic Triangle

A knowledge organization system (KOS) is made up of concepts and semantic relations which represent a knowledge domain terminologically. In knowledge representation, we distinguish between five approaches to knowledge organization systems: nomenclatures, classification systems, thesauri, ontologies and, as a borderline case of knowledge organization systems, folksonomies. Knowledge domains are thematic areas that can be delimited, such as a scientific discipline, an economic sector or a company's language. A knowledge organization system's goal in information practice is to support the retrieval process. We aim, for instance, to offer the user concepts for searching and browsing, to index automatically and to expand queries automatically: We aim to solve the vocabulary problem (Furnas, Landauer, Gomez, & Dumais, 1987). Without KOSs, a user will select a word A for his search, while the author of a document D uses A' to describe the same object; hence, D is not retrieved. Concept-based information retrieval goes beyond the word level and works with concepts instead. In the example, A and A' are linked to the concept C, leading to a successful search. Feldman (2000) has expressed the significance of this approach very vividly from the users' perspective: "Find what I mean, not what I say."

We are concerned with a view of concepts which will touch on known and established theories and models but also be suitable for exploiting all advantages of knowledge organization systems for information science and practice. If we are to create something like the 'Semantic Web', we must perforce think about the concept of the concept, as therein lies the key to any semantics (Hjørland, 2007). Excursions into general linguistics, philosophy, cognitive science and computer science are also useful for this purpose.

In language, we use symbols, e.g. words, in order to express a thought about an object of reference. We are confronted with the tripartite relation consisting, following Ogden und Richards (1985[1923]), of the thought (also called reference), the referent, or object of reference, and the symbol. Ogden and Richards regard the thought (reference) as a psychological activity ("psychologism"; according to Schmidt, 1969, 30). In information science (Figure I.3.1), *concept* takes the place of (psychological) thought. As in Ogden and Richards' classical approach, a concept is represented in language by designations. Such designations can be natural-language words, but also terms from artificial languages (e.g. notations of a classificatory KOS). The concept *concept* is defined as a class containing certain objects as elements, where the objects have certain properties.

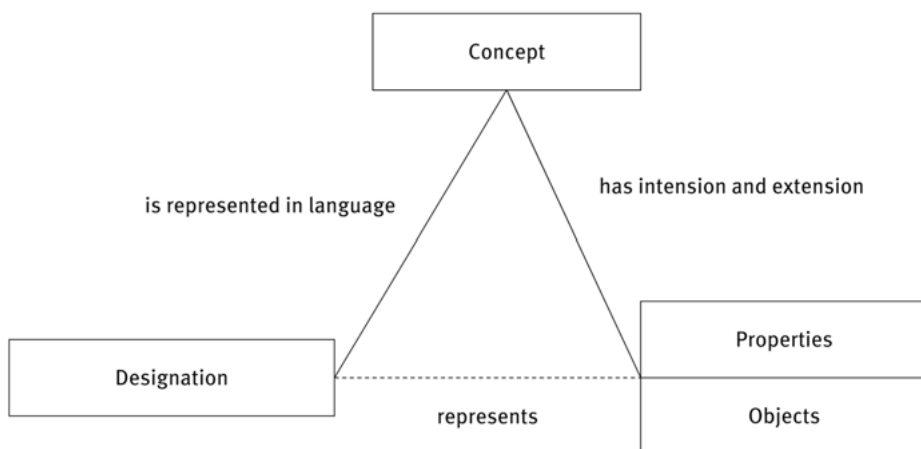


Figure I.3.1: The Semiotic Triangle in Information Science.

We will discuss the German guidelines for the construction of KOS, which are very similar to their counterparts in the United States, namely ANSI/NISO Z39.19-2005. The norms DIN 2330 (1993, 2) and DIN 2342/1 (1992, 1) understand a concept as “a unit of thought which is abstracted from a multitude of objects via analysis of the properties common to these objects.” This DIN definition is not unproblematic. Initially, it would be wise to speak, instead of (the somewhat psychological-sounding) “unit of thought”, of “classes” or “sets” (in the sense of set theory). Furthermore, it does not hold for each concept that all of its elements always and necessarily have “common” properties. This is not the case, for example, with concepts formed through family resemblance. Lacking common properties, we might define *vegetable* as “is cabbage vegetable or root vegetable or fruit vegetable etc.” But what does ‘family resemblance’ mean? Instead of *vegetable*, let us look at a concept used by Wittgenstein (2008[1953]) as an example for this problem, *game*. Some games may have in common that there are winners and losers, other games—not all, mind you—are entertaining, others again require skill and luck of their players etc. Thus the concept of the game cannot be defined via exactly one set of properties. We must admit that a concept can be determined not only through a conjunction of properties, but also, from time to time, through a disjunction of properties.

There are two approaches to forming a concept. The first goes via the objects and determines the concept’s extension, the second notes the class-forming properties and thus determines its intension (Reimer, 1991, 17). Frege uses the term ‘*Bedeutung*’ (meaning) for the extension, and ‘*Sinn*’ (sense) for the intension. Independently of what you call it, the central point is Frege’s discovery that extension and intension need not necessarily concur. His example is the term *Venus*, which may alternatively be called *Morning Star* or *Evening Star*. Frege (1892, 27) states that *Evening Star* and *Morning Star* are extensionally identical, as the set of elements contained within them

(both refer to Venus) are identical, but are intensionally non-identical, as the Evening Star has the property *first star visible in the evening sky* and the Morning Star has the completely different property *last star visible in the morning sky*.

The extension of a concept M is the set of objects O_1, O_2 etc. that fall under it:

$$M =_{df} \{O_1, O_2, \dots, O_i, \dots\},$$

where “ $=_{df}$ ” is to mean “equals by definition”. It is logically possible to group like objects together (via classification) or to link unlike objects together (via colligation, e.g. the concept *Renaissance* as it is used in history) (Shaw, 2009; Hjørland, 2010a; Shaw, 2010).

The intension determines the concept M via its properties f_1, f_2 etc., where most of these properties are linked via “and” ($\wedge$) and a subset of properties is linked via “or” ($\vee$) (where $\forall$ is the universal quantifier in the sense of “holds for all”):

$$M =_{df} \forall x \{f_1(x) \wedge f_2(x) \wedge \dots \wedge [f_g(x) \vee f_{g'}(x) \vee \dots \vee f_{g''}(x)]\}.$$

This definition is broad enough to include all kinds of forming concepts such as concept explanations (founding on ANDing properties) and family resemblance (founding on ORing properties). It is possible (for the “vegetable-like” properties) that the subset of properties f_1, f_2 etc., but not f_g , is a null set, and it is possible that the subset $f_g, f_{g'}$ etc., but not f_1, f_2 etc., is a null set (in the case of concept explanation).

Concept Theory and Epistemology

How do we arrive at concepts, anyway? This question calls for an excursion into epistemology. Hjørland (2009; see also Szostak, 2010; Hjørland, 2010b) distinguishes between four different approaches to this problem: empiricism, rationalism, hermeneutics and pragmatism. We add a fifth approach, critical theory, which can, however—as Hjørland suggests—be understood as an aspect of pragmatism (see Figure I.3.2).

Empiricism starts from observations; thus one looks for concepts in concrete, available texts which are to be analyzed. Some typical methods of information science in this context are similarity calculations between text words, but also between tags when using folksonomies and the cluster-analytical consolidation of the similarity relations.

Rationalism is skeptical towards the reliability of observations and constructs a priori concepts and their properties and relations, generally by drawing from analytical and formal logic methods. In information science, we can observe such an approach in formal concept analysis (Ganter & Wille, 1999; Priss, 2006).

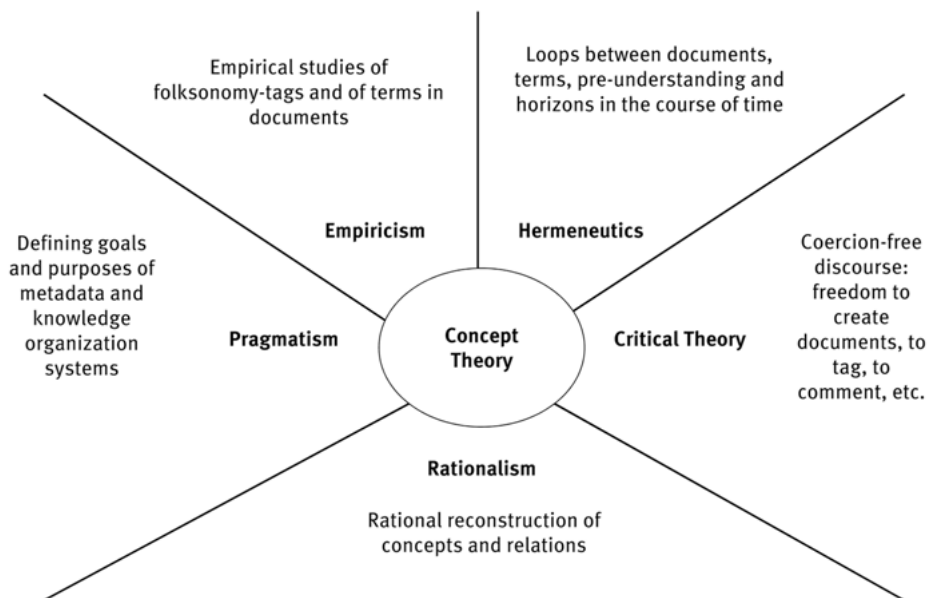


Figure I.3.2: Epistemological Foundations of Concept Theory.

Hermeneutics (called ‘historicism’ in Hjørland, 2009, 1525) captures concepts in their historical development as well as in their use in a given “world horizon”. In understanding this, human’s “being thrown” into the world plays an important role (Heidegger, 1962[1927]). A text is never read without any understanding or prejudice. This is where the hermeneutic circle begins: The text as a whole provides the key for understanding its parts, while the person interpreting it needs the parts to understand the whole (Gadamer, 1975[1960]). Prejudices play a positive role here. We move dynamically in and with the horizon, until finally the horizons blend in the understanding. In information science, the hermeneutical approach leads to the realization that concept systems and even bibliographical records (in their content-descriptive index fields) are always dynamic and subjects to change (Gust von Loh, Stock, & Stock, 2009).

Pragmatism is closely associated with hermeneutics, but stresses the meaning of means and goals. Thus for concepts, one must always note what they are being used for: “The ideal of pragmatism is to define concepts by deciding which class of things best serves a given purpose and then to fixate this class in a sign” (Hjørland, 2009, 1527).

Critical Theory (Habermas, 1987[1968]) stresses coercion-free discourse, the subject of which is the individual’s freedom to use both words and concepts (as for example during tagging) at his or her discretion, and not under any coercion. Each of the five epistemological theories is relevant for the construction of concepts and relations in information science research as well as in information practice, and should

always be accorded due attention in compiling and maintaining knowledge organization systems.

Concept Types

Concepts are the smallest semantic units in knowledge organization systems; they are “building blocks” or “units of knowledge” (Dahlberg, 1986, 10). A KOS is a concept system in a given knowledge domain. In knowledge representation, a concept is determined via words that carry the same, or at least a similar meaning (this being the reason for the designation “Synset”, which stands for “set of synonyms”, periodically found for concepts) (Fellbaum, Ed., 1998). In a first approach, and in unison with DIN 2342/1 (1992, 3), synonymy is “the relation between designations that stand for the same concept.” There is a further variant of synonymy, which expresses the relation between two concepts, and which we will address below.

Some examples for synonyms are *autumn* and *fall* or *dead* and *deceased*. A special case of synonymy is found in paraphrases, where an object is being described in a roundabout way. Sometimes it is necessary to work with paraphrases, if there is no name for the concept in question. In German, for example, there is a word for “no longer being hungry” (*satt*), but none for “no longer being thirsty”. This is an example for a concept without a concrete designation.

Homonymy starts from designations; it is “the relation between matching designations for different concepts” (DIN 2342/1, 1992, 3). Our paradigmatic example for a homonym is *Java*. This word stands, among others, for the concepts *Java (island)*, *Java (coffee)* and *Java (programming language)*. In word-oriented retrieval systems, homonyms lead to big problems, as each homonymous—and thus polysemous—word form must be disambiguated, either automatically or in a dialog between man and machine (Ch. C.3). Varieties of homonymy are homophony, where the ambiguity lies in the way the words sound (e.g. *see* and *sea*), and homography, where the spelling is the same but the meanings are different (e.g. *lead* the verb and *lead* the metal). Homophones play an important role in information systems that work with spoken language, homographs must be noted in systems for the processing of written texts.

Many concepts have a meaning which can be understood completely without reference to other concepts, e.g. *chair*. Menne (1980, 48) calls such complete concepts “categorematical”. In knowledge organization systems that are structured hierarchically, it is very possible that such a concept may occur on a certain hierarchical level:

... with filter.

This concept is syncategorematical; it is incomplete and requires other concepts in order to carry meaning (Menne, 1980, 46). In hierarchical KOSs, the syncategoremata are explained via their broader concepts. Only now does the meaning become clear:

Cigarette
... with filter

or

Chimney
... with filter.

One of the examples concerns a filter cigarette, the other a chimney with a (soot) filter. Such an explication may take the incorporation of several hierarchy levels. As such, it is highly impractical to enter syncategoremata on their own and without any addendums in a register, for example.

Concepts are not given, like physical objects, but are actively derived from the world of objects via abstraction (Klaus, 1973, 214). The aspects of concept formation (in the sense of information science, not of psychology) are first and foremost clarified via definitions. In general, it can be noted that concept formation in the context of knowledge organization systems takes place in the area of tension between two contrary principles. An economical principle instructs us not to admit too many concepts into a KOS. If two concepts are more or less similar in terms of extension and intension, these will be regarded as one single “quasi-synonymous” concept. The principle of information content leads in the opposite direction. The more precise we are in distinguishing between intension and extension, the larger each individual concept’s information content will be. The concepts’ homogeneity and exactitude will draw the greatest profit from this. Komatsu (1992, 501) illustrates this problematic situation (he uses “category” for “concept”):

Thus, economy and informativeness trade off against each other. If categories are very general, there will be relatively few categories (increasing economy), but there will be few characteristics that one can assume different members of a category share (decreasing informativeness) and few occasions on which members of the category can be treated as identical. If categories are very specific, there will be relatively many categories (decreasing economy), but there will be many characteristics that one can assume different members of a category share (increasing informativeness) and many occasions on which members can be treated as identical.

The solution for concept formation (Komatsu, 1992, 502, uses “categorization”) in KOSs is a compromise:

The basic level of categorization is the level of abstraction that represents the best compromise between number and informativeness of categories.

According to the theory by Rosch (Mervis & Rosch, 1981; Rosch, 1975a; Rosch 1975b; Rosch & Mervis, 1975; Rosch, Mervis, Gray, Johnson, & Boyes-Braem, 1976; Rosch, 1983), we must distinguish between three concept levels: the superordinate level, the basic level and the subordinate level:

Suppose that basic objects (e.g., *chair*, *car*) are the most inclusive level at which there are attributes common to all or most members of the category. Then total cue validities are maximized at that level of abstraction at which basic objects are categorized. That is, categories one level more abstract will be superordinate categories (e.g., *furniture*, *vehicle*) whose members share only a few attributes among each other. Categories below the basic level will be subordinate categories (e.g., *kitchen chair*, *sports car*) which are also bundles of predictable attributes and functions, but contain many attributes which overlap with other categories (for example, *kitchen chair* shares most of its attributes with other kinds of chairs) (Rosch, Mervis, Gray, Johnson, & Boyes-Braem, 1976, 385).

Thus many people agree that on the basic level, the concept *chair* is a good compromise between *furniture*, which is too general, and *armchair*, *Chippendale chair* etc., which are too specific. In a knowledge organization system for furniture, the compromise looks different, as we must differentiate much more precisely: Here we will add the concepts from the subordinate level. If, on the other hand, we construct a KOS for economic sciences, the compromise might just favor *furniture*; thus we would restrict ourselves to a superordinate-level concept in this case.

Concepts whose extension is exactly one element are individual concepts, their designations are proper names, e.g. of people, organizations, countries, products, but also of singular historical events (e.g. *German Reunification*) or individual scientific laws (*Second Law of Thermodynamics*). All other concepts are general concepts (Dahlberg, 1974, 16). We want to emphasize categories as a special form of general concepts. Moving upwards through the levels of abstraction, we will at some point reach the top. At this point—please note: always in the context of a knowledge domain—no further step of abstraction can be taken. These top concepts represent the domain-specific categories. Fugmann (1999, 23) introduces categories via the concepts' intension. Here a concept with even fewer properties no longer makes any sense. In faceted KOSs (Broughton, 2006; Spiteri, 1999), the categories form the framework for the facets.

According to Fugmann (1999), the concept types can be distinguished intensionally. Categories are concepts with a minimum of properties (to form even more general concepts would mean, for the knowledge domain, the creation of empty, useless concepts). Individual concepts are concepts with a maximum of properties (the extension will stay the same even with the introduction of more properties). General concepts are all the concepts that lie between these two extremes. Their exposed position means that both individual concepts and categories can be used quite easily via methods of knowledge representation, while general concepts can lead to problems. Although individual concepts, referring to named entities, are quite easy to define, some kinds of KOSs, e.g. classification systems, do consider them for controlled vocabulary (in classification systems, named entities have no dedicated notations).

Vagueness and Prototype

Individual concepts and categories can generally be exactly determined. But how about the exactitude of general concepts? We will continue with our example *chair* and follow Black (1937, 433) into his imaginary chair exhibition:

One can imagine an exhibition in some unlikely museum of applied logic of a series of “chairs” differing in quality by at least noticeable amounts. At one end of a long line, containing perhaps thousands of exhibits, might be a Chippendale chair; at the other, a small nondescript lump of wood. Any “normal” observer inspecting the series finds extreme difficulty in “drawing the line” between chair and not-chair.

The minimal distinctions between neighboring objects should make it nearly impossible to draw a line between chair and not-chair. Outside the “neutral area”, where we are not sure whether a concept fits or not, we have objects that clearly fall under the concept on the one side, and on the opposite side, objects that clearly do not fall under the concept. However, neither are the borders between the neutral area and its neighbors exactly definable. Such blurred borders can be experimentally demonstrated for a lot of general concepts (Löbner, 2002, 45).

As a solution, we might try not searching for the concept’s borders at all and instead work with a “prototype” (Rosch, 1983). Such a prototype can be regarded as “the best example” for a Basic-Level concept. This model example possesses “good” properties in the sense of high recognition value.

If we determine the concept intensionally, via a prototype and its properties, the fuzzy borders are still in existence (and may cause the odd mistake in indexing these border regions), but on the plus side, we are able to work satisfactorily with general concepts in the first place. If we imagine a concept hierarchy stretching over several levels, prototypes should play a vital role, particularly on the intermediate levels, i.e. in the Basic Level after Rosch. At the upper end of the hierarchy are the (superordinate) concepts with few properties, so that with all probability, one will not be able to imagine a prototype. And the at the bottom level, the (subordinate) concepts are so specific that the concept and the prototype will coincide.

Stability of Concepts

No concept (and no KOS) remains stable over time. “Our understandings of concepts change with context, environment, and even personal experience” (Spiteri, 2008, 9). In science and technology, those changes are due to new observations and theories or—in the sense of Kuhn (1962)—to scientific revolutions. A good example is the concept *planet of the solar system*. From 1930 to 2006 the extension of this concept consisted of nine elements, now there are only eight (Pluto is no longer accepted as *planet* in astronomy). There are new ideas leading to new concepts. *Horseless car-*

riages were invented, then renamed *Automobiles* (Buckland, 2012). For some concepts their meaning changes over time. Some years ago a *printer* meant a human, now a *printer* is a machine as well. And, finally, there are concepts which were used in some times, but more or less forgotten today. Good examples are concepts of old occupations such as *wainwright* or *cooper*.

Conceptual History is a branch of humanities and studies the development of the meaning of ideas. It is related to Etymology which is the study of the history of words. Both disciplines are useful auxiliary sciences of information science.

Definition

In knowledge representation practice, concepts are often only implicitly defined—e.g. by stating their synonyms and their location in the semantic environment. It is our opinion that in knowledge organization systems, the used concepts are to be exactly defined, since this is the only way to achieve clarity for both indexers and users.

Definitions must match several criteria in order to be used correctly (Dubislav, 1981, 130; Pawłowski, 1980, 31-43). Circularity, i.e. the definition of a concept with the help of the same concept, which—as a mediate circle—can now be found across several definition steps, is to be avoided. To define an unknown concept via another, equally unknown concept (*ignotum per ignotum*) is of little help. The inadequacy of definitions shows itself in their being either too narrow (when objects that should fall under the concept are excluded) or too wide (when they include objects that belong elsewhere). In many cases, negative definitions (*a point is that which has no extension*) are unusable, as they are often too wide (Menne, 1980, 32). A definition should not display any of the concept's superfluous properties (Menne, 1980, 33). Of course the definition must be precise (and thus not use any meaningless phrases, for example) and cannot contain any contradictions (such as *blind viewer*). Persuasive definitions, i.e. concept demarcations aiming for (or with the side-effect of) emotional reactions (e.g. to paraphrase Buddha, *Pariah is a man who lets himself be seduced by anger and hate, a hypocrite, full of deceit and flaws ...*; Pawłowski, 1980, 250), are unusable in knowledge representation. The most important goal is the definition's usefulness in the respective knowledge domain (Pawłowski, 1980, 88 et seq.). In keeping with our knowledge of vagueness, we strive not to force every single object under one and the same concept, but sometimes define the prototype instead.

From the multitude of different sorts of definition (such as definition as abbreviation, explication, nominal and real definition), concept explanation and definition via family resemblance are particularly important for knowledge representation.

Concept explanation starts from the idea that concepts are made up of partial concepts:

$$\text{Concept} =_{\text{df}} \text{Partial Concept}_1, \text{Partial Concept}_2, \dots$$

Here one can work in two directions. Concept synthesis starts from the partial concepts, while concept analysis starts from the concept. The classical variant dates from Aristotle and explains a concept by stating genus and differentia. Aristotle works out criteria for differentiating concepts from one another and structuring them in a hierarchy. The recognition of objects' being different is worked out in two steps; initially, via their commonalities—what Aristotle, in his “Metaphysics”, calls “genus”—and then the differences defining objects as specific “types” within the genus (Aristotle, 1057b 34). Thus a concept explanation necessarily involves stating the genus and differentiating the types. It is important to always find the nearest genus, without skipping a hierarchy level.

Concept explanation works with the following partial concepts:

Partial Concept₁: Genus (concept from the directly superordinate genus),

Partial Concept₂: Differentia specifica (fundamental difference to the sister concepts).

The properties that differentiate a concept from its sister terms (the concepts that belong to the same genus, also called hyponyms) must always display a specific, and not an arbitrary property (accidens). A classical definition according to this definition type is:

Homo est animal rationale.

Homo is the concept to be defined, *animal* the genus concept and *rational* the specific property separating man from other creatures. It would be a mistake to define mankind via *living creature* and *hair not blond*, since (notwithstanding jokes about blondes) the color of one's hair is an arbitrary, not a fundamental property.

The fact that over the course of concept explanations, over several levels from the top down, new properties are always being added means that the concepts are becoming ever more specific; in the opposite direction, they are getting more general (as properties are shed on the way up). This also means that on a concept ladder, properties are “inherited” by those concepts further down. Concept explanation is of particular importance for KOSs, as their specifications necessarily embed the concepts in a hierarchical structure.

In concept explanation, it is assumed that an object wholly contains its specific properties if it belongs to the respective class; the properties are joined together via a logical AND. This does not hold for the vegetable-like concepts, where we can only distinguish a family resemblance between the objects. Here the properties are joined via an OR (Pawłowski, 1980, 199). If we connect concept explanation with the definition according to family resemblance, we must work with a disjunction of properties on certain hierarchical levels. Here, too, we are looking for a genus concept, e.g. for Wittgenstein's *game*. The family members of *game*, such as board game, card game, game of chance etc. may very well have a few properties in common, but not all. Con-

cepts are always getting more specific from the top down and more general from the bottom up; however, there are no hereditary properties from the top down. On those hierarchy levels that define via family resemblance, the concepts pass on some of their properties, but not all.

Let us assume, for instance, that the genus of *game* is *leisure activity*. We must now state some properties of games in order to differentiate them from other leisure activities (such as *meditating*). We define:

Partial Concept₁ / Genus: Leisure Activity

Partial Concept₂ / Differentia specifica: Game of Chance v Card Game v Board Game v ...

If we now move down on the concept ladder, it will become clear that *game* does not pass on all of its properties, but only ever subsets (as a game of chance needs not be a card game). On the lower levels, in turn, there does not need to be family resemblance, but only “normal” (conjunctive) concept explanation. For each level, it must be checked whether family resemblance has been used to define disjunctively or “normally”. We have to note that not all concept hierarchies allow for the heredity of properties; there is no automatism. This is a very important result for the construction of ontologies.

Modeling Concepts Using Frames or Description Logics

How can a concept be formally represented? In the literature, we find two approaches, namely frames and description logics (Gómez-Pérez, Fernández-López, & Corcho, 2004, 11 and 17).

One successful approach works with frames (Minsky, 1975). Frames have proven themselves in cognitive science, in computer science (Reimer, 1991, 159 et seq.) and in linguistics. In Barsalou’s (1992, 29) conception, frames contain three fundamental components:

- sets of attributes and values (Petersen, 2007),
- structural invariants,
- rule-bound connections.

Among the different frame conceptions, we prefer Barsalou’s version, as it takes under consideration rule-bound connections. We can use this option in order to automatically perform calculations on the application side of a concept system (Stock, 2009, 418-419).

The core of each frame allocates properties (e. g., *Transportation*, *Location*, *Activity*) to a concept (e. g., *Vacation*), and values (say, for location *Kauai* or *Las Vegas*) to the properties, where both properties and values are expressed via concepts. After Minsky (1975), the concept is allocated such attributes that describe a stereotypical

situation. There are structural invariants between the concepts within a frame, to be expressed via relations (Barsalou, 1992, 35-36):

Structural invariants capture a wide variety of relational concepts, including spatial relations (e.g., between *seat* and *back* in the frame for *chair*), temporal relations (e.g., between *eating* and *paying* in the frame for *dining out*), causal relations (e.g., between *fertilization* and *birth* in the frame for *reproduction*), and intentional relations (e.g., between *motive* and *attack* in the frame for *murder*).

The concepts within the frame are not independent but form manifold connections bound by certain rules. In Barsalou's *Vacation*-frame, there are, for example, positive (the faster one drives, the higher the travel cost) and negative connections (the faster one drives, the sooner one will arrive) between the transport attributes. We regard the value for the location *Kauai* on the attribute level, and the value *surfing* on the activity level. It is clear that the first value makes the second one possible (one can surf around Kauai, and not, for example, in Las Vegas).

A formulation in description logic (also called terminological logic; Nardi & Brachman, 2003) and the separation of general concepts (in a TBox) and individual concepts (in the ABox) allow us to introduce the option of automatic reasoning to a concept system, in the sense of ontologies. So-called "roles" describe the relations between concepts, including the description of the concepts' properties. General concepts represent classes of objects, and individual concepts concrete instances of those classes. If some of the values of the properties are numbers, these can be used as the basis of automatic calculations. "Reasoning in DL (description logics, A/N) is mostly based on the subsumption test among concepts" (Gómez-Pérez, Fernández-López, & Corcho, 2004, 19). Therefore, the amount of automatic reasoning is very limited. Let us look to a simple example. We introduce the concept *budgerigar* into the TBox. A typical role of *budgerigar* is *has wings* with a numerical value of 2 (a budgie has two wings). Additionally, we introduce the individual concept *Pete* into the ABox and allocate *Pete* to the class *budgerigar*. Now, the formal concept system is able to perform a step of automatic reasoning: *Pete has two wings*.

Barsalou (1992, 43) sees (at least in theory) no limits for the use of frames in knowledge representation. (The lesson is the same for the application of description logics.) Some groundwork, however, must be performed for the automatized system:

Before a computational system can build the frames described here, it needs a powerful processing environment capable of performing many difficult tasks. This processing environment must notice new aspects of a category to form new attributes. It must detect values of these attributes to form attribute-value sets. It must integrate cooccurring attributes into frames. It must update attribute-value sets with experience. It must detect structural invariants between attributes. It must detect and update constraints. It must build frames recursively for the components of existing frames.

Where the definition as concept explanation at least leads to *one* relation (the hierarchy), the frame and the descriptions logics approaches lead to *a multitude* of relations between concepts and, furthermore, to rule-bound connections. As concept systems absolutely require relations, frames—as concept representatives—ideally consolidate such methods of knowledge representation. This last quote of Barsalou's should inspire some thought, though, on how not to allow the mass of relations and rules to become too large. After all, the groundwork and updates mentioned above must be put into practice, which represents a huge effort. Additionally, it is feared that as the number of different relations increases, the extent of the knowledge domain in whose context one can work will grow ever smaller. To wit, there is according to Nardi and Brachman (2003, 10) a reverse connection between the language's expressiveness and automatic reasoning:

(T)here is a tradeoff between the expressiveness of a representation language and the difficulty of reasoning over the representation built using that language. In other words, the more expressive the language, the harder the reasoning.

KOS designers should thus keep the number of specific relations as small as possible, without for all that losing sight of the respective knowledge domain's specifics.

Conclusion

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- It is a truism, but we want to mention it in the first place: Concept-based information retrieval is only possible if we are able to construct and to maintain adequate Knowledge Organization Systems (KOSs).
 - Concepts are defined by their extension (objects) and by their intension (properties). It is possible to group similar objects (via classification) or unlike objects (via colligation) together, depending on the purpose of the KOS.
 - There are five epistemological theories on concepts in the background of information science: empiricism, rationalism, hermeneutics, critical theory and pragmatism. None of them should be forgotten in activities concerning KOSs.
 - Concepts are (a) categories, (b) general concepts and (c) individual concepts.
 - Categories and individual concepts are more or less easily to define, but general concepts tend to be problematic. Such concepts (such as *chair*) have fuzzy borders and should be defined by prototypes. Every concept in a KOS has to be defined exactly (by extension, intension, or both). In a concept entry, all properties (if applicable, objects as well) must be listed completely and in a formal way. It is not possible to work with the inheritance of properties if we do not define those properties.
 - Concepts are objects of change: there are new concepts, concepts which are no longer used and concepts which changed their meaning over time.
 - In information science, we mainly work with two kinds of definition, namely concept explanation and family resemblance. Concepts, which are defined via family resemblance (*vegetable* or *game*), do not pass all properties down to their narrower terms. This result is important for the design of ontologies.

- Concepts can be formally presented as frames with sets of attributes and values, structural invariants (relations) and rule-bound connections. Alternatively, we can work with general concepts (TBox), individual concepts (ABox) and roles (describing relations) in the context of description logics.
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I.4 Semantic Relations

Syntagmatic and Paradigmatic Relations

Concepts do not exist in independence of each other, but are interlinked. We can make out such relations in the definitions (e.g. via concept explanation) and in the frames. We will call relations between concepts “semantic relations” (Khoo & Na, 2006; Storey, 1993). This is only a part of the relations of interest for knowledge representation. Bibliographical relations (Green, 2001, 7 et seq.) register relations that describe documents formally (e.g. “has author”, “appeared in source”, “has publishing date”). Relations also exist *between* documents, e.g. insofar as scientific documents cite and are cited, or web documents have links. We will concentrate exclusively on semantic relations here.

In information science, we distinguish between paradigmatic and syntagmatic relations where semantic relations are concerned (Peters & Weller, 2008a). This differentiation goes back to de Saussure (2005[1916]) (de Saussure uses “associative” instead of “paradigmatic”). In the context of knowledge representation, the paradigmatic relations form “tight” relations, which have been established (or laid down) in a certain KOS. They are valid independently of documents (i.e. “in absentia” of any concrete occurrence in documents). Syntagmatic relations exist between concepts in specific documents; they are thus always “in praesentia”. The issue here is that of co-occurrence, be it in the continuous text of the document (or of a text window), in the selected keywords (Wersig, 1974, 253) or in tags when a service applies a folksonomy (Peters, 2009). We will demonstrate this with a little example: In a knowledge organization system, we meet the two hierarchical relations

Austria – Styria – Graz and Environs – Lassnitzhöhe;
Cooking Oil – Vegetable Oil – Pumpkin Seed Oil.

These concept relations each form paradigmatic relations. A scientific article on local agricultural specialties in the Styria region would be indexed thusly:

Lassnitzhöhe – Pumpkin Seed Oil.

These two concepts thus form a syntagmatic relation. Except for in special cases (in varieties of syntactic indexing), the syntagmatic relation is not described any further. It expresses: The following document deals with Pumpkin Seed Oil and Lassnitzhöhe. It does not state, however, in which specific relations the terms stand towards one another.

The syntagmatic relation is the only semantic relation that occurs in folksonomies (Peters, 2009; Peters & Stock, 2007). In the sense of a bottom-up approach of building KOSs, folksonomies provide empirical material for potential controlled vocabularies,

as well as material for paradigmatic relations, even though the latter is only “hidden” in the folksonomies (Peters & Weller, 2008a, 104) and must be intellectually revealed via analysis of tag co-occurrences. Peters and Weller regard the (automatic and intellectual) processing of tags and their relations in folksonomies as the task of “tag gardening” with the goal of emergent semantics (Peters & Weller, 2008b) (Ch. K.2).

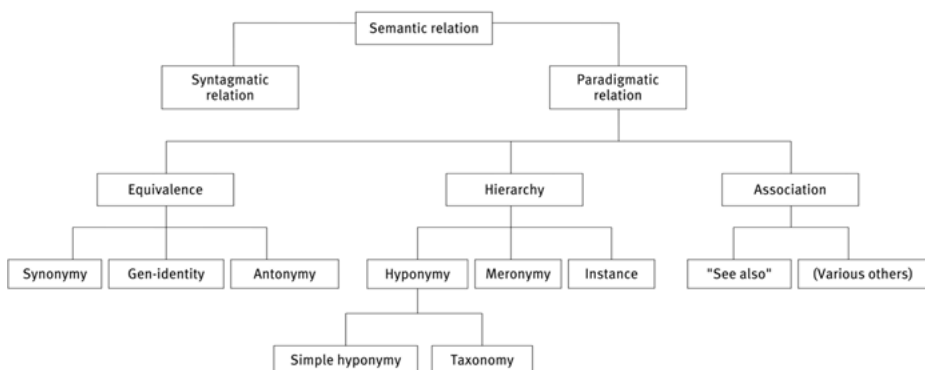


Figure I.4.1: Semantic Relations.

Paradigmatic relations, however, always express the type of connection. From the multitude of possible paradigmatic relations, knowledge representation tries to work out those that are generalizable, i.e. those that can be used meaningfully in all or many use cases. Figure I.4.1 provides us with an overview of semantic relations.

Order and R–S–T

Relations can be differentiated via the amount of their argument fields. Two-sided relations connect two concepts, three-sided relations three etc. It is always possible to simplify the multi-sided relations via a series of two-sided relations. *To heal*, for instance, is a three-sided relation between a person, a disease and a medication. This would result in three two-sided relations: person–disease, disease–medication and medication–person. We will assume, in this article, that the relations in question are two-sided.

The goal is to create a concept system in a certain knowledge domain which will then serve as a knowledge organization system. KOSs can be characterized via three fundamental properties (where x , y , z are concepts and ρ a relation in each case). Reflexivity in concept systems asks how a concept adheres to itself with regard to a relation. Symmetry occurs when a relation between A and B also exists in the opposite direction, between B and A . If a relation exists between two concepts A and B , and

also between B and C, and then again between A and C, we speak of transitivity. We will demonstrate this on several examples:

R Reflexivity	$x \rho x$ “... is identical to...”
Irreflexivity	$\neg(x \rho x)$ “... is the cause of...”
S Symmetry	$(x \rho y) \rightarrow (y \rho x)$ “... is equal to...”
Asymmetry	$(x \rho y) \rightarrow \neg(y \rho x)$ “... is unhappily in love with...”
T Transitivity	$[(x \rho y) \wedge (y \rho z)] \rightarrow (x \rho z)$ “... is greater than...”
Intransitivity	$[(x \rho y) \wedge (y \rho z)] \rightarrow \neg(x \rho z)$ “... is similar to...”

An order in a strictly mathematical sense is irreflexive (-R), asymmetrical (-S) and transitive (T) (Menne, 1980, 92). An order that has as its only relation *is more expensive than*, for example, has these properties: A certain product, say a lemon, is not more expensive than a lemon (i.e. -R); if a product (our lemon) is more expensive than another product (an apple), then the apple is not more expensive than the lemon but cheaper (-S); if, finally, a lemon is more expensive than an apple and an apple more expensive than a cherry, then a lemon, too, is more expensive than a cherry (T).

For asymmetrical relations, we speak of an inverse relation if it addresses the reversal of the initial relation. If in $(x \rho y)$ ρ is the relation *is hyponym of*, then the inverse relation ρ' in $(y \rho' x)$ is *is hyperonym of*.

In so far as a KOS has synonymy which of course is always symmetrical (if x is synonymous to y , then y is synonymous to x) it will never be an order in the mathematical sense. An open question is whether all relations in knowledge organization systems are transitive as a matter of principle. We can easily find counterexamples in a first, naïve approach to the problem. Let us assume, for instance, that the liver of Professor X is a part of X and Professor X is a part of University Y, then transitivity dictates that the liver of Professor X is a part of University Y, which is obviously nonsense. But attention! Was that even the same relation? The liver is an organ; a professor is part of an organization. Only because we simplified and started from a general part-whole relation does not mean that transitivity applies. Intransitivity may thus mean, on the one hand, that the concept order (wrongly) summarizes different relations as one single relation, or on the other hand, that the relation is indeed intransitive.

Why is transitivity in particular so important for information retrieval? Central applications are query expansion (automatically or manually processed in a dialog between a user and a system; Ch. G.4) or (in ontologies) automatic reasoning. If someone, for example, were to search for stud farms in the Rhein-Erft district of North-Rhine Westphalia, they would formulate:

Stud Farm AND Rhein-Erft District.

The most important farms are in Quadrath-Ichendorf, which is a part of Bergheim, which in turn is in the Rhein-Erft district. If we expand the second argument of the search request downwards, proportionately to the geographical structure, we will arrive, in the second step, at the formulation that will finally provide the search results:

Stud Farm AND (Rhein-Erft District OR Bergheim OR ... OR Quadrath-Ichendorf).

Query expansion can also lead to results by moving upwards in a concept ladder. Let us say that a motorist is confronted with the problem of finding a repair shop for his car (a Ford, for instance) in an unfamiliar area. He formulates on his mobile device:

Repair Shop AND Ford AND ([Location], e.g. determined via GPS).

The retrieval system allocates the location to the smallest geographical unit and first takes one step upwards in the concept ladder, and at the same time back down, to the sister terms. If there are no results, it's one hierarchy level up and again to the sister terms, and so forth until the desired document has been located.

A query expansion by exactly one step can be performed at every time. If we imagine the KOS as a graph, we can thus always and without a problem incorporate those concepts into the search request that are linked to the initial concept via a path length of one. (Whether this is always successful in practice is moot. The incorporation of hyperonyms into a search argument in particular can expand the search results enormously and thus negatively affect precision.) If we want to expand via path lengths greater than one, we must make sure that there is transitivity, as otherwise there would be no conclusive semantic relation to the initial concept.

Equivalence

Two designations are synonymous if they denote the same concept. Absolute synonyms, which extend to all variants of meaning and all (descriptive, social and expressive) references, are rare; an example is *autumn* and *fall*. Abbreviations (*TV*–*television*), spelling variants (*grey*–*gray*), inverted word order (*sweet night air*–*night air, sweet*) and shortened versions (*The Met*–*The Metropolitan Opera*) are totally synonymous as well. Closely related to total synonymy are common terms from foreign languages (*rucksack*–*backpack*) and divergent language use (*media of mass communication*–*mass media*).

After Löbner (2002, 46), most synonymy relations are of a partial nature: They do not designate the exact same concept but stand for (more or less) closely related

concepts. Differences may be located in either extension or intension. Löbner's (2002, 117) example *geflügelte Jahresendpuppe* (literally *winged end-of-year doll*, in the German Democratic Republic's official lingo) may be extensionally identical to *Weihnachtsengel* (*Christmas angel*), but is not intensionally so. As opposed to true synonymy, which is a relation between designations and a concept, partial synonymy is a relation between concepts.

In information practice, most KOSs treat absolute and partial synonyms, and furthermore, depending on the purpose, similar terms (as quasi-synonyms) as one and the same concept. If two terms are linked as synonyms in a concept system, they are (right until the system is changed) always a unit and cannot be considered in isolation. If the concept system is applied to full-text retrieval systems, the search request will be expanded by all the fixed synonyms of the initial search term. Synonymy is reflexive, symmetrical and transitive.

Certain objects are "gen-identical" (Menne, 1980, 68-69). This is a weak form of identity, which disregards certain temporal aspects. A human being in his different ages (Person X as a child, adult and old man) is thus gen-identical. A possible option in concept systems is to summarize concepts for gen-identical objects as quasi-synonyms. There is, however, also the possibility of regarding the respective concepts individually and linking them subsequently.

If gen-identical objects are described by different concepts at different times, these concepts will be placed in the desired context via chronological relations. These relations are called "chronologically earlier" and—as the inversion—"chronologically later". As an example, let us consider the city located where the Neva flows into the Baltic Sea:

Between 1703 and 1914: Saint Petersburg
 1914–1924: Petrograd
 1924–1981: Leningrad
 Afterwards: Saint Petersburg again.

Any neighboring concepts are chronologically linked:

Saint Petersburg [Tsar Era] is chronologically earlier than *Petrograd*.
Petrograd is chronologically earlier than *Leningrad*.

The chronological relation is irreflexive, asymmetrical and transitive.

Two concepts are antonyms if they are mutually exclusive. Such opposite concepts are for example *love–hate*, *genius–insanity* and *dead–alive*. We must distinguish between two variants: Contradictory antonyms know exactly two shadings, with nothing in between. Someone is pregnant or isn't pregnant—*tertium non datur*. Contrary antonyms allow for other values between the extremes; between love and hate, for example, lies indifference. For contradictory antonyms it is possible, in retrieval, to incorporate the respective opposite concept—linked with a negating term such as

“not” or “un-”—into a query. Whether or not contrary antonyms can meaningfully be used in knowledge representation and information retrieval is an open question at this point. Antonymy is irreflexive, symmetrical and intransitive.

Hierarchy

The most important relation of concept systems, the supporting framework so to speak, is hierarchy. Durkheim (1995[1912]) assumes that hierarchy is a fundamental relation used by all men to put the world in order. As human societies are always structured hierarchically, hierarchy is—according to Durkheim—experienced in everyday life and, from there, projected onto our concepts of “the world”.

If we do not wish to further refine the hierarchy relation of a knowledge organization system, we will have a “mixed-hierarchical concept system” (DIN 2331:1980, 6). It is called “mixed” because it summarizes several sorts of hierarchical relation. This approach is a very simple and naïve world view. We distinguish between three variants of hierarchy: hyponymy, meronymy and instance.

Hyponym-Hyperonym Relation

The abstraction relationship is a hierarchical relation that is subdivided from a logical perspective. “Hyperonym” is the term in the chain located precisely one hierarchy level higher than an initial term; “hyponym” is a term located on the lower hierarchy level. “Sister terms” (first-degree parataxis) share the same hyperonym; they form a concept array. Concepts in hierarchical relations form hierarchical chains or concept ladders. In the context of the definition, each respective narrower term is created via concept explanation or—as appropriate—via family resemblance. If there is no definition via family resemblance, the hyponym will inherit all properties of the hyperonym. In case of family resemblance, it will only inherit a partial quantity of the hyperonym’s properties. Additionally, it will have at least one further fundamental property that sets it apart from its sister terms. For all elements of the hyponym’s extension, the rule applies that they are also always elements of the hyperonym. The logical subordination of the abstraction relation always leads to an implication of the following kind (Löbner, 2002, 85; Storey, 1993, 460):

If x is an A, then x is a B
iff A is a hyponym of B.

If it is true that *bluetit* is a hyponym of *tit*, then the following implication is also true:

If it is true that: x is a bluetit, then it is true that: x is a tit.

The abstraction relation can always be expressed as an “IS-A” relation (Khoo & Na, 2006, 174). In the example

Bird – Songbird – Tit – Bluetit

(defined in each case without resorting to family resemblance), it is true that:

The bluetit IS A tit.
The tit IS A songbird.
The songbird IS A bird.

Properties are added to the intension on the journey downwards: A *songbird* is a *bird* that sings. The *bluetit* is a *tit* with blue plumage. Mind you: The properties must each be noted in the term entry (keyword entry, descriptor entry, etc.) via specific relations; otherwise any (automatically implementable) heredity would be completely impossible.

If we define via family resemblance, the situation is slightly different. In the example

Leisure Activity – Game – Game of Chance

it is true, as above, that:

A game of chance IS A game.
A game IS A leisure activity.

As we have delimited *game* via family resemblance, *game of chance* does not inherit all properties of *game* (e.g. not necessarily *board game*, *card game*), but only a few. The hyponym’s additional property (*is a game of chance*) is in this case already present as a part of the hyperonym’s concepts, which are linked via OR. The clarification is performed by excluding the other family members linked via OR (for instance, like: *is precisely a game requiring luck*).

It is tempting to assume that there is reciprocity between the extension and intension of concepts in a hierarchical chain: To increase the intension (i.e. to add further properties on the way down) would go hand in hand with a decrease of the number of objects that fall under the concept. There are certainly more birds in the world than there are songbirds. Such a reciprocal relation can be found in many cases, but it has no general validity. It is never the case for individual concepts, as we could always add further properties to those without changing the extension. The intension of *Karl May*, for instance, is already clearly defined by *author*, *born in Saxonia*, *invented Winnetou*; adding *has business relations with the Münchmeyer publishing house* would not change the extension in the slightest. We can even find counterexamples for general concepts, i.e. concepts that display an increase in extension as their intension is aug-

mented. The classical example is by Bolzano (1973[1837]). Dubislav (1981, 121) gives a lecture on this case:

Let us use with *Bolzano* the concept of a “speaker of all European languages” and then augment the concept by adding the property “living” to the concept “speaker of all living European languages”. We can notice that the intension of the first concept has been increased, but that the extension of the new concept thus emerging contains the extension of the former as a partial quantity.

Of course we have to assume that there are more speakers of all living European languages than speakers of all European languages which include dead languages such as Gothic, Latin or Ancient Greek.

We can make out two variant forms of the abstraction relation: taxonomy, and non-taxonomical, “simple” hyponymy. In a taxonomy, the IS-A relation can be strengthened into IS-A-KIND-OF (Cruse, 2002, 12). A taxonomy does not just divide a larger class into smaller classes, as is the case for simple hyponymy. Let us consider two examples:

? A queen IS A KIND OF woman.
(better: A queen IS A woman).
? A stallion IS A KIND OF horse.
(better: A stallion IS A horse).

In both cases, the variant IS A KIND OF is unrewarding; here, we have simple hyponymy. If we instead regard the following examples:

A cold blood IS A KIND OF horse.
A stetson IS A KIND OF hat,

we can observe that here the formulation makes sense, as there is indeed a taxonomical relation in these cases. A taxonomy fulfills certain conditions, according to Cruse (2002, 13):

Taxonomy exists to articulate a domain in the most effective way. This requires “good” categories, which are (a) internally cohesive, (b) externally distinctive, and (c) maximally informative.

In taxonomies, the hyponym, or “taxonym”, and the hyperonym are fundamentally regarded from the same perspective. *Stallion* is not a taxonym of *horse*, because *stallion* is regarded from the perspective of gender and *horse* is not. In the cases of *cold blood* and *horse*, though, the perspectives are identical; both are regarded from a biological point of view. The hyponym-hyperonym relation is irreflexive, asymmetrical and transitive.

Meronym-Holonym Relation

If the abstraction relation represents a logical perspective on concepts, the part-whole relation starts from an objective perspective (Khoo & Na, 2006, 176). Concepts of wholeness, “holonyms”, are divided into concepts of their parts, “meronyms”. If in an abstraction relation it is not just any properties which are used for the definition but precisely the characteristics that make up its essence, then the part-whole relation likewise does not use any random parts but the “fundamental” parts of the wholeness in question. The meronym-holonym relation has several names. Apart from “part-whole relation” or “part-of relation”, we also speak of “partitive relation” (as in DIN 2331:1980, 3). A system based on this relation is called “mereology” (Simons, 1987).

In individual cases, it is possible that meronymy and hyponymy coincide. Let us consider the pair of concepts:

Industry – Chemical Industry.

Chemical industry is as much a part of industry in general as it is a special kind of industry.

Meronymy is expressed by “PART OF”. This relation does not exactly represent a concept relation but is made up of a bundle of different partitive relations. If one wants—in order to simplify, for example—to summarize the different part-whole relations into a single relation, transitivity will be damaged in many cases. Winston, Chaffin and Herrmann (1987, 442-444) compiled a list of (faulty) combinations. Some examples may prove intransitivity:

Simpson's finger is part of Simpson.
 Simpson is part of the Philosophy Department.
 ? Simpson's finger is part of the Philosophy Department.
 Water is part of the cooling system.
 Water is partly hydrogen.
 ? Hydrogen is part of the cooling system.

The sentences marked with question marks are false conclusions. We can (as a “lazy solution”) do without the transitivity of the respective specific meronymy relations in information retrieval. In so doing, we would deprive ourselves of the option of query expansion over more than one hierarchy level. But we do not even need to make the effort of differentiating between the single partitive relations. The elaborated solution distinguishes the specific meronymy relations and analyzes them for transitivity, thus providing the option of query expansion at any time and over as many levels as needed.

We will follow the approach, now classical, of Winston, Chaffin and Herrmann (1987) and specify the part-whole relation into meaningful kinds. Winston et al. dis-

tinguish six different meronymy relations, which we will extend to nine via further subdivision (Figure I.4.2) (Weller & Stock, 2008).

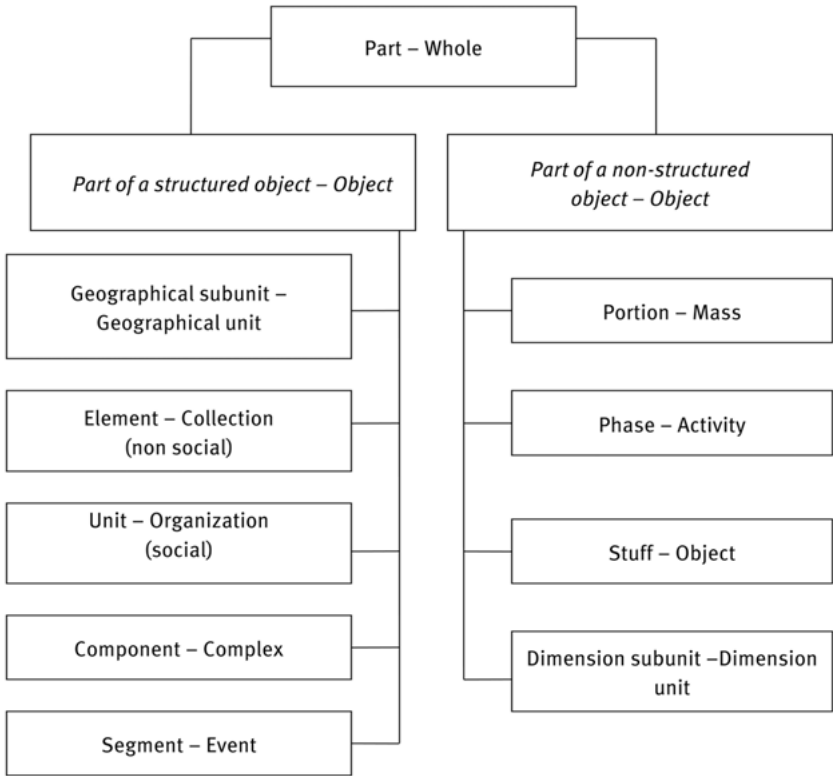


Figure I.4.2: Specific Meronym-Holonym Relations.

Insofar as wholenesses have a structure, this structure can be divided into certain parts (Gerstl & Pribbenow, 1996; Pribbenow, 2002). The five part-whole relations displayed on the left of Figure I.4.2 distinguish themselves by having had wholenesses structurally subdivided. Geographical data allow for a subdivision according to administrative divisions, provided we structure a given geographical unit into its subunits. *North-Rhine-Westphalia* is a part of *Germany*; the locality *Kerpen-Sindorf* is a part of *Kerpen*. (Non-social) uniform collections can be divided into their elements. A *forest* consists of *trees*; a *ship* is part of a *fleet*. A similar aspect of division is at hand if we divide (uniform) organizations into their units, such as a *university* into its *departments*. Johansson (2004) notes that in damaging uniformity there is not necessarily transitivity. Let us assume that there is an association Y , of which other associations X_i (and *only* associations) are members. Let person A be a member of X_i . In case of transitivity, this would mean that A , via his membership in X_i , is also a part of Y . But

according to Y's statutes, this is absolutely impossible. The one case is about membership of persons, the other about membership of associations, which means that the principle of uniformity has been damaged in the example. A contiguous complex, such as a *house*, can be subdivided into its components, e.g. the *roof* or the *cellar*. Meronymy, for an event (say a *circus performance*) and a specific segment (e.g. *trapeze act*), is similarly formed (temporally speaking in this case) (Storey, 1993, 464).

The second group of meronyms works independently of structures (on the right-hand side in Figure I.4.2). A wholeness can be divided into random portions, such as a *cup* (after we drop it to the floor) into *shards* or—less destructively—a *bread* into servable *slices*. A continuous activity (e.g. *shopping*) can be divided into single phases (e.g. *paying*). One of the central important meronymy relations is the relation of an object to its stuff, such as the *aluminum parts* of an *airplane* or the *wooden parts* of my *desk*. If we have a homogeneous unit, we can divide it into subunits. Examples are *wine (in a barrel)* and *1 liter of wine* or *meter–decimeter*.

All described meronym-holonym relations are irreflexive, asymmetrical and transitive, insofar as they have been defined and applied in a “homogeneous” way.

We have already discussed the fact that in the hierarchical chain of a hyponym-hyperonym relation the concepts pass on their properties (in most cases) from the top down. The same goes for their meronyms: We can speak of meronym heredity in the abstraction relation (Weller & Stock, 2008, 168). If concept A is a partial concept (e.g. a *motor*) of the wholeness B (a *car*), and C is a hyponym of B (let's say: an *ambulance*), then the hyponym C also has the part A (i.e. an *ambulance* therefore has a *motor*).

Instance

In extensional definition, the concept in question is defined by enumerating those elements for which it applies. In general, the question of whether the elements are general or individual concepts is left unanswered. In the context of the instance relation, it is demanded that the element always be an individual concept. The element is thus always a “named entity”.

Whether this element-class relation is regarded in the context of hyponymy or meronymy is irrelevant for the instance relation. An instance can be expressed both via “is a” and via “is part of”. In the sense of an abstraction relation, we can say that:

Persil IS A detergent.

Cologne IS A university city on the Rhine.

Likewise, we can formulate:

Silwa (our car) IS PART OF our motor pool.

Angela Merkel IS PART OF the CDU.

Instances can have hyponyms of their own. Thus in the last example, *CDU* is an instance of the concept *German political party*. And obviously our *Silwa* has parts, such as chassis or motor.

Further Specific Relations

There is a wealth of other semantic relations in concept orders, which we will, in an initial approach, summarize under the umbrella term “association”. The associative relation as such therefore does not exist; there are merely various different relations. Common to them all is that they—to put it negatively—do not form (quasi-)synonyms or hierarchies and are—positively speaking—of use for knowledge organization systems.

In a simple case, which leaves every specification open, the associative relation plays the role of a “see also” link. The terms are related to each other according to practical considerations, such as the link between products and their respective industries in a business administration KOS (e.g. *body care product* SEE ALSO *body care product industry* and vice versa). The unspecific “see also” relation is irreflexive, symmetrical and intransitive.

For other, now specific, associative relations, we will begin with a few examples. Schmitz-Esser (2000, 79-80) suggests the relations of usefulness and harmfulness for a specific KOS (of the world fair ‘Expo 2000’). Here it is shown that such concept relations have persuasive “secondary stresses”. In the example

Wind-up radio IS USEFUL FOR communication in remote areas.

there is no implicit valuation. This is different for

Overfishing IS USEFUL FOR the fish meal industry.
Poppy cultivation IS USEFUL FOR the drug trade.

A satisfactory solution might be found in building on the basic values of a given society (“useful for whom?”) in case of usefulness and harmfulness (Schmitz-Esser, 2000, 79) and thus reject the two latter examples as irreconcilable with the respective moral values or—from the point of view of a drug cartel—keep the last example as adequate.

Whether a specification of the associative relation will lead to a multitude of semantic relations that are generalizable (i.e. usable in all or at least most KOSs) is an as yet unsolved research problem. Certainly, it is clear that we always need a relation *has_property* for all concepts of a KOS. This solution is very general, it would be more appropriate to specify the kind of property (such as “has melting point” in a KOS on materials, or “has subsidiary company” in an enterprise KOS).

Relations between Relations

Relations can be in relation to each other (Horrocks & Sattler, 1999). In the above, we introduced meronymy and formed structure-disassembling meronymy as its specification, and within the latter, the component-complex relation, for example. There is a hierarchy relation between the three above relations. Such relations between relations can be used to derive conclusions. If, for example, we introduced to our concept system:

Roof is a component of *house*,

then it is equally true that

roof is a structural part of *house*

and

roof is a part of *house*;

generally formulated:

$(A \text{ is a component of } B) \rightarrow (A \text{ is a structural part of } B) \rightarrow (A \text{ is a part of } B).$

Relations and Knowledge Organization Systems

We define knowledge organization systems via their cardinality for expressing concepts and relations. The three “classical” methods in information science and practice—nomenclature, classification, thesaurus—are supplemented by folksonomies and ontologies. Folksonomies represent a borderline case of KOSs, as they do not have a single paradigmatic relation.

Nomenclatures (keyword systems) distinguish themselves mainly by using the equivalence relation and ignoring all forms of hierarchical relation. In classification systems, the (unspecifically designed) hierarchy relation is added. Thesauri also work with hierarchy; some use the unspecific hierarchy relation, others differentiate via hyponymy and (unspecific) meronymy (with the problem—see Table I.4.1—of not being able to guarantee transitivity). In thesauri, a generally unspecifically designed associative relation (“see also”) is necessarily added. Ontologies make use of all the paradigmatic relations mentioned above (Hovy, 2002). They are modeled in formal languages, where terminological logic or frames are also accorded their due consideration. Compared to other KOSs, ontologies categorically contain instances. Most ontologies work with (precisely defined) further relations.

Table I.4.1: Reflexivity, Symmetry and Transitivity of Paradigmatic Relations.

	Reflexivity	Symmetry	Transitivity
Equivalence			
– Synonymy	R	S	T
– Gen-identity	–R	–S	T
– Antonymy	–R	S	–T
Hierarchy			
– Hyponymy			
– Simple hyponymy	–R	–S	T
– Taxonomy	–R	–S	T
– Meronymy (unspecific)	–R	–S	?
– Specific meronymies	–R	–S	T
– Instance	R	–S	–T
Specific relations			
– “See also”	–R	S	–T
– Further relations	Depending on the relation		

Table I.4.2: Knowledge Organization Systems and the Relations They Use.

	Folksonomy	Nomenclature	Classification	Thesaurus	Ontology
Term	Tag	Keyword	Notation	Descriptor	Concept
Equivalence	–	yes	yes	yes	yes
– Synonymy	–	yes	yes	yes	yes
– Gen-identity	–	yes	–	–	yes
– Antonymy	–	–	–	–	yes
Hierarchy	–	–	yes	yes	yes
– Hyponymy	–	–	–	yes	yes
– Simple hyponymy	–	–	–	–	yes
– Taxonomy	–	–	–	–	yes
– Meronymy (unspecific)	–	–	–	yes	–
– Specific meronymies	–	–	–	–	yes
– Instance	–	–	–	as req.	yes
Specific Relations	–	–	–	yes	yes
– “See also”	–	as req.	as req.	yes	yes
– Further relations	–	–	–	–	yes
Syntagmatic relation	yes	yes	yes	yes	no

The fact that ontologies directly represent knowledge (and not merely the documents containing the knowledge) lets the syntagmatic relations disappear in this case. If we take a look at Table I.4.2 or Figure I.4.3, the KOSs are arranged from left to right, according to their expressiveness. Each KOS can be “enriched” to a certain degree and lifted to a higher level via relations of the system to its right: A nomenclature can become a classification, for example, if (apart from the step from keyword to nota-

tion) all concepts are brought into a hierarchical relation; a thesaurus can become an ontology if the hierarchy relations are precisely differentiated and if further specific relations are introduced. An ontology can become—and now we are taking a step to the left—a method of indexing if it introduces the syntagmatic relation, i.e. if it allows its concepts to be allocated to documents, while retaining all its relations. Thus the advantages of the ontology, with its cardinal relation framework, flow together with the advantages of document indexing and complement each other.

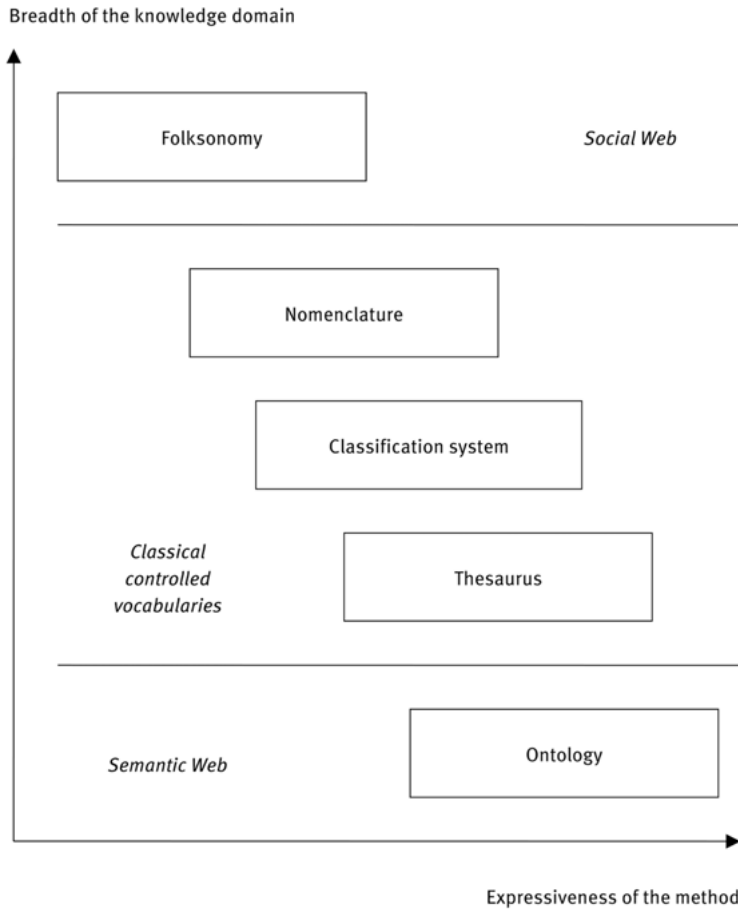


Figure I.4.3: Expressiveness of KOSs Methods and the Breadth of their Knowledge Domains.

Conclusion

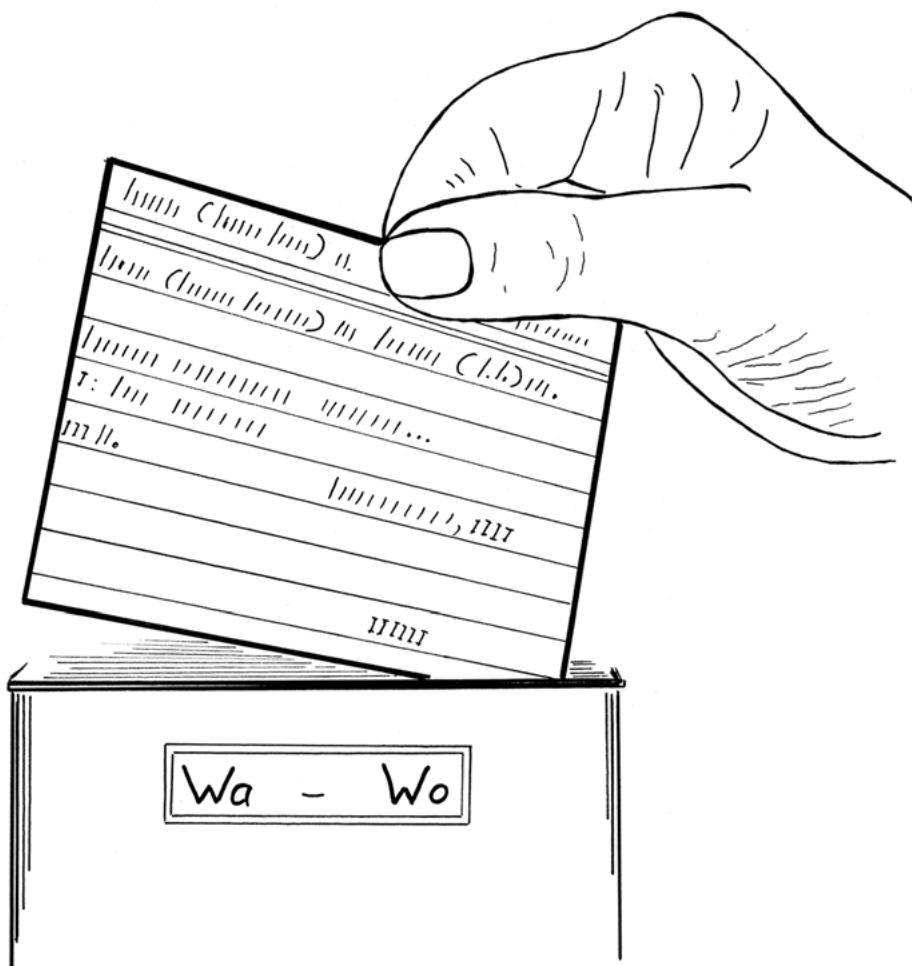
- Concept systems are made up of concepts and semantic relations between them. Semantic relations are either syntagmatic relations (co-occurrences of terms in documents) or paradigmatic relations (tight relations in KOSs).
 - There are three kinds of paradigmatic relations: equivalence, hierarchy, and further specific relations.
 - Especially for hierarchic relations, transitivity plays an important role for query expansion. Without proofed transitivity, it is not possible to expand a search argument with concepts from hierarchical levels with distances greater than one.
 - Equivalence has three manifestations: synonymy, gen-identity and antonymy. Absolute synonymy, which is very sparse, is a relation between a concept and different words. All other kinds of synonymy, often called “quasi-synonymy”, are relations between different concepts. Gen-identity describes an object in the course of time. Contradictory antonyms are useful in information retrieval, but only with constructions like *not* or *un-*.
 - Hierarchy is the most important relation in KOSs. It consists of the (logic-oriented) hyponym-hyperonym-relation (with two subspecies, simple hyponymy and taxonomy), the (object-oriented) meronym-holonym-relation (with a lot of subspecies) and the instance relation (relation between a concept and an individual concept as one of its elements). KOS designers have to pay regard to the transitivity of relations.
 - There are lots of further relations, such as usefulness or harmfulness. It is possible to integrate all these relations into only one single associative relation (as in thesauri), but it is more expressive to work with the specific relations as they are necessary in a given knowledge domain.
 - We can order types of KOSs regarding to their expressiveness (quantity and quality of concepts and semantic relations): from folksonomies via nomenclatures, classification systems, thesauri up to ontologies.
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Part J
Metadata



J.1 Bibliographic Metadata

Why Metadata?

A person's intellectual or artistic achievements will be transmitted to his environment in some form. Only the transmission of these endeavors and their acknowledgement by other people gives significance to an exchange of communication. Knowledge thus does not stay hidden, but is put into motion. In knowledge representation, an author's—or artist's—potential can only be assessed once it is physically available, as a document (or, as in many libraries' rulebooks, as a “resource”). It has ever been the task of libraries to collect, arrange and display documents so that they can be retrieved. Catalog cards, which were developed and used in accordance with library rules, played the central role in the organization system. They contain compressed information *about* the document, so-called metadata, such as the author's name(s), title or publisher. An example of a catalog card is shown in Figure J.1.1.

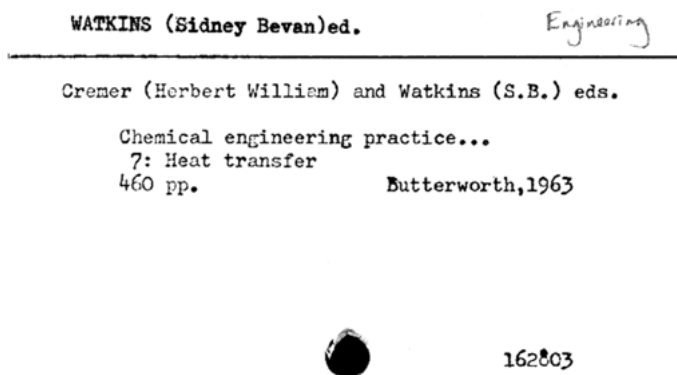


Figure J.1.1: Traditional Catalog Card. Source: University of Bristol. Card Catalogue Online.

Generally speaking, catalog cards serve to document bibliographic and—as in our example of the “engineering” entry—content information so the user can retrieve the document. If an organization schema works within a library, this does not mean that there is a unified order for all libraries. The opposite is still the case: there is a multitude of rulebooks, nationally and internationally. Ongoing digitalization only exacerbates the problem. In addition to the “palpable” library as a building containing shelves with ordered documents, there is now the virtual library. Conventional forms of gathering are no longer enough. The traditional catalog cards are replaced by data entries structured in a field schema.

This fact increases the demands placed on collecting, analyzing, indexing, retrieving and exchanging information not only in the context of libraries but in the

entire world of information. Unified access to information requires a language-independent indexing via encoding, unambiguity in rulebooks and standardization in data exchange. Public and scientific libraries in various countries employ rules for cataloging and mechanical formats for data exchange. However, the compatibility of international bibliographic data runs into problems due to differing forms of sorting. In 2010, a new rulebook called the “Resource Description & Access” (RDA, 2010) has been available. It is meant to serve as the standard form and to reach beyond the Anglo-American sphere, to be employed worldwide.

What are Metadata?

We will specify the “routine definition” of metadata, which states that they characterize data about data. As we have already seen, metadata fulfill a specific purpose: they provide aids for developing and using catalogs, bibliographies, information services and their online formats. Metadata, according to Dempsey and Heery (1998, 149), are addressed to the potential user, be that a person or a program:

(M)etadata is data associated with objects which relieves their potential users of having to have full advance knowledge of their existence or characteristics. It supports a variety of operations.

Mechanical, digital as well as human, intellectual *usage* is thus at the foreground. The task of knowledge representation is to identify and to analyze this potential usage, in order to create an organization system that provides a generalizable basis for access to metadata. Taylor (1999, 103) points out the considerable difficulties in this endeavor:

Many research studies have shown that different users do not think of the same word(s) to write about a concept; authors do not necessarily retain the same name or same form of name throughout their writing careers; corporate bodies do not necessarily use the same name in their documents nor are they known by the same form of name by everyone; and titles of works that are reproduced are not always the same in the original and the reproduction. For all these reasons and more, the library and then the archival worlds came to the realization many years ago that bibliographic records needed access points (one of which needed to be designated as the “main” one), and these access points needed to be expressed consistently from record to record when several different records used the same access point.

Formal gathering forms the basis of information processing. Without bibliographic metadata, there is no foundation to the indexing of content. In order to find out where and in what form metadata are required and used, we first have to make various preliminary considerations and decisions about how to proceed:

- Which documents should be collected, indexed, stored and made retrievable; e.g. specialist literature, music CDs, audiobooks?

- Which document types are we dealing with; e.g. patent, book, journal, magazine, newspaper?
- How deeply should the document be gathered and indexed, e.g. an anthology as an entity, or individual articles therein?
- How many and which fields are necessary, which core elements and which variant access points?
- What rulebook should be used; e.g. RAK-WB in Germany (RAK-WB, 1998), AACR in the Anglophone sphere (Anglo-American Cataloguing Rules; AACR2, 2005) or RDA as a “world standard” (Resource Description & Access; RDA, 2010)?
- Which formats of exchange should be used for an eventual transfer of external data; e.g. MAB (*Maschinelles Austauschformat für Bibliotheken* in Germany), MARC (Machine-Readable Cataloguing)?
- What methods of knowledge representation are used for indexing content; e.g. classification, thesaurus?

Just as intellectual and artistic achievements are no longer in isolation after they have been published, forming a relation to their environment, single documents form a relation to other documents through the usage of tools of knowledge representation. Uncovering these relations crystallizes the characteristics, and thus the unifying aspects, of the metadata. Correspondingly, we can put down our definition as follows: metadata are standardized data about documentary reference units, and they serve the purpose of facilitating digital and intellectual access to, as well as usage of, these documents. Metadata stand in relation to one another and provide, when combined correspondingly, an adequate surrogate for the document in the sense of knowledge representation.

Document Relations

Based on the results of the IFLA study on “Functional Requirements for Bibliographic Records”—FRBR (IFLA, 1998; Riva, 2007; Tillett, 2001), a published document is a unit comprised of two aspects: intellectual or artistic content and physical entity. This division follows the simple principle which states that the ideas of an author, initially abstract, are somehow expressed and made manifest in the form of media, which consist of individual units. Figure J.1.2 shows the four aspects of work, expression, manifestation and item. They form the primary document relations (IFLA, 1998, 12).

The entities defined as *work* (a distinct intellectual or artistic creation) and *expression* (the intellectual or artistic realization of a *work*) reflect intellectual or artistic content. The entities defined as *manifestation* (the physical embodiment of an *expression* of a *work*) and *item* (a single exemplar of a *manifestation*), on the other hand, reflect physical form.

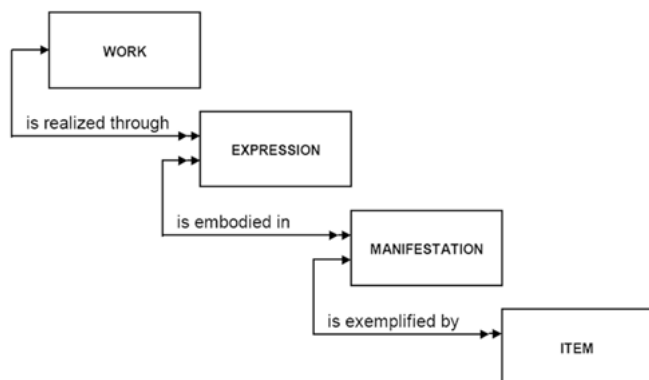


Figure J.1.2: Perspectives on Formally Published Documents. Source: IFLA, 1998, 13.

The creative achievement of the author or artist forms the foundation, the actual work underlying the document. This work is initially presented via language in the broader sense, e.g. in writing, musically or graphically. One and the same work can thus be realized via several forms of expression, but a form of expression is the realization of only one work. In the relation between work and form of expression, we have the content, the work's "aboutness". Here, content indexing attempts to gather and represent the topic of the document. However, this is impossible without the concrete form of a medium, the manifestation. Intellectual content is presented via media. It must be noted that a form of expression can be contained within several media, and that likewise, a medium can embody several forms of expression. Media are verified via items, where the single item only represents an individual within the manifestation.

A work (an author's creation) can thus have one or more forms of expression (concrete realization, including translations or illustrations), which are physically reflected in a manifestation (e.g. a printing of a book) and the single items of this manifestation (see Figure J.1.2). We will demonstrate this on an example—Neil Gaiman's "Neverwhere":

WORK₁: *Neverwhere* by Neil Gaiman

EXPRESSION₁₁: the author's text on the basis of the BBC miniseries

MANIFESTATION₁₁₁: the book "Neverwhere", published by BBC Books, London, 1996

ITEM₁₁₁₁: the book, hand-signed by the author, owned by Mary Myers

EXPRESSION₁₂: the German translation of the English text by Tina Hohl

MANIFESTATION₁₂₁: the book "Niemalsland", published by Hoffmann und Campe, Hamburg, 1997

MANIFESTATION₁₂₂: the book "Niemalsland", published by Wilhelm Heyne, München, 1998

ITEM₁₂₂₁: the copy on my bookshelf (with a note of ownership on page 5).

Depending on the goal in dealing with a document, it is important to give consideration to the four aspects. If one only wants to prove that Neil Gaiman ever wrote “Neverwhere”, it is enough to point to the *work*. However, if we want to distinguish between the English and the German edition, we are entering the level of *expressions*. If we are interested in the German editions, we must distinguish between the two *manifestations* (one in hardcover, the other in paperback). If, lastly, we are interested in a “special” copy, such as one signed by Gaiman or bearing proof of my ownership, we are dealing with *items*. Generally, documentary reference units operate on the level of manifestations; with exceptions, of course. In an antiquarian bookshop’s retrieval system, it will be of great importance whether the book has been hand-signed by the author; here, the concrete item will be displayed in the documentary unit.

How important it is to pay attention to document relations concerning bibliographic metadata can be demonstrated via the following question: what is the change that means a work is no longer the same work but a new creation? Formal gathering, and hence metadata, differ with regard to an original in relation to its heavily changed content. In order to draw a line between original and new work, further relations must be analyzed. Tillett describes the relations that act simultaneously to the primary relations as “content relationships” (Tillett, 2001, 22).

Content relationships apply across the different levels of entities and exist simultaneously with primary relationships. Content relationships can even be seen as part of a continuum of intellectual or artistic content and the farther one moves along the continuum from the original *work*, the more distant the relationship.

In Figure J.1.3, we see equivalence, derivative and descriptive relations, which are used in order to demarcate an original from a new work. The line of intersection runs through the derivative relation. The construction of the cut-off point follows the rules of the AACR.

The idea of the equivalence relation is rather precarious, according to Tillett, since here it is the perspective, or subjective interpretation, which decides whether certain characteristics are the same. On the item level, for instance, it might be of foremost importance to a book expert that the individual copy contain a detail (such as a watermark), whereas other persons are more interested in the intellectual content of a manifestation. Similar discrepancies can also appear in the relation between work and expression. Generally, equivalence relations are meant to present the most precise copies of a work’s manifestation, where possible. They are meant to do this for as long as the intellectual content of the original stays the same. This includes, in theory: microform reproductions, copies, exact reproductions, facsimiles and reprints.

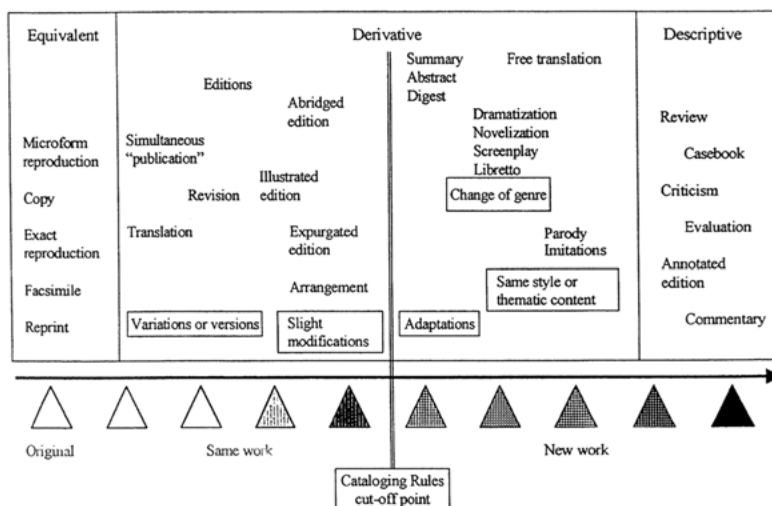


Figure J.1.3: Document Relations: The Same or Another Document? Source: Tillett, 2001, 23.

Derivative relations refer to the original work and its modification(s). The strength of the relation is of importance here, since it is used to draw the line between the same work and a new creation. Variations, such as abbreviated, illustrated or expurgated editions, revisions, translations and slight modifications (e.g. alignment with a spelling reform) are still adjudged to be the same work. Extensive reworkings of the original represent a new work. This includes: formal adaptations, such as summaries or abstracts, genre changes (such as the screenplay version of a novel) or very loose adaptations (e.g. parodies).

A special kind of derivative relations is found in gen-identical documents. Here we are dealing with the same manifestation, but ongoing changes can be made to it. Examples include loose-leaf binders (such as legal documents) or websites.

The descriptive relations concerning an earlier work and its revision always concern new creations, new works. This is the case with book reviews, criticisms or comments, for example.

Whole-Part or Part-to-Part relations play a particular role in sequences of articles spanning several issues of a journal, or electronically stored resources. On the one hand, they concern the allocation of individually separate components into a whole (e.g. images on a website, additional material for a book) or the connection of individually consecutive components, as in the case in series. If we know that a documentary reference unit is another's sequel, or that it is followed by others, we must note this within the metadata.

Documents containing shared parts must be given ample consideration. Thus, there are conference proceedings or scientific articles which are completely identi-

cal apart from their title. Since they appear in different sources, they are clearly two manifestations. However, the metadata must express that they are the same work.

Tillett (2001, 30-31) summarizes:

To summarize the relationship types operating among bibliographic entities, there are:

- Primary relationships that are implicit among the bibliographic entities of work, expression, manifestation, and item;
- Close content relationships that can be viewed as a continuum starting from an original work, including equivalence, derivative, and descriptive (or referential) relationships;
- Whole-part and part-to-part relationships, the latter including accompanying relationships and sequential relationships; and
- Shared characteristic relationships.

Document—Actors—Aboutness

FRBR are used to register document relations. Now, a work also has authors, and the manifestations a publisher. The actors associated with a document have been described via the “Functional Requirements for Authority Data”—FRAD (Patton, ed., 2009). Actors can be individual persons, families or corporate bodies. These play different roles on the different document levels: works are produced by creators (such as authors or composers), expressions have contributors (such as publishers, translators, illustrators or arrangers of music), manifestations are produced by printing presses and marketed and released by publishers, and items, lastly, are rightfully owned by their proprietors (private individuals, libraries or archives). Documents are about something—they carry aboutness. This can be described via general concepts, concrete objects, events and places. These forms of content indexing are discussed by the IFLA study on “Functional Requirements for Subject Authority Data (FRSAD)” (FRSAR, 2009). We will discuss them in chapters K through O.

Documents (i.e. works, expressions, manifestations and items) are expressed in words via a name and—particularly in publishing and libraries—clearly labeled via an identifier. This can be an ISBN (International Standard Book Number) for books, a DOI (Digital Object Identifier) for articles or a call number in libraries. Of course there are also identifiers for persons, families and corporate bodies as well as for concepts (general concepts, concrete objects, events and places).

The purposes of metadata are

- to find resources (documents) (to find resources that correspond to the user’s stated search criteria),
- to identify documents (to confirm that the resource described corresponds to the resource sought or to distinguish between two or more resources with similar characteristics),
- to select documents (to select a resource that is appropriate to the user’s need),
- to obtain documents (to acquire or access the resource described),

- to find and identify information (e.g. on names and concepts),
- to clarify information (to clarify the relationship between two such entities or more),
- to understand (why a particular name or title has been chosen as the preferred name or title for the entity) (RDA, 2010, 0-1).

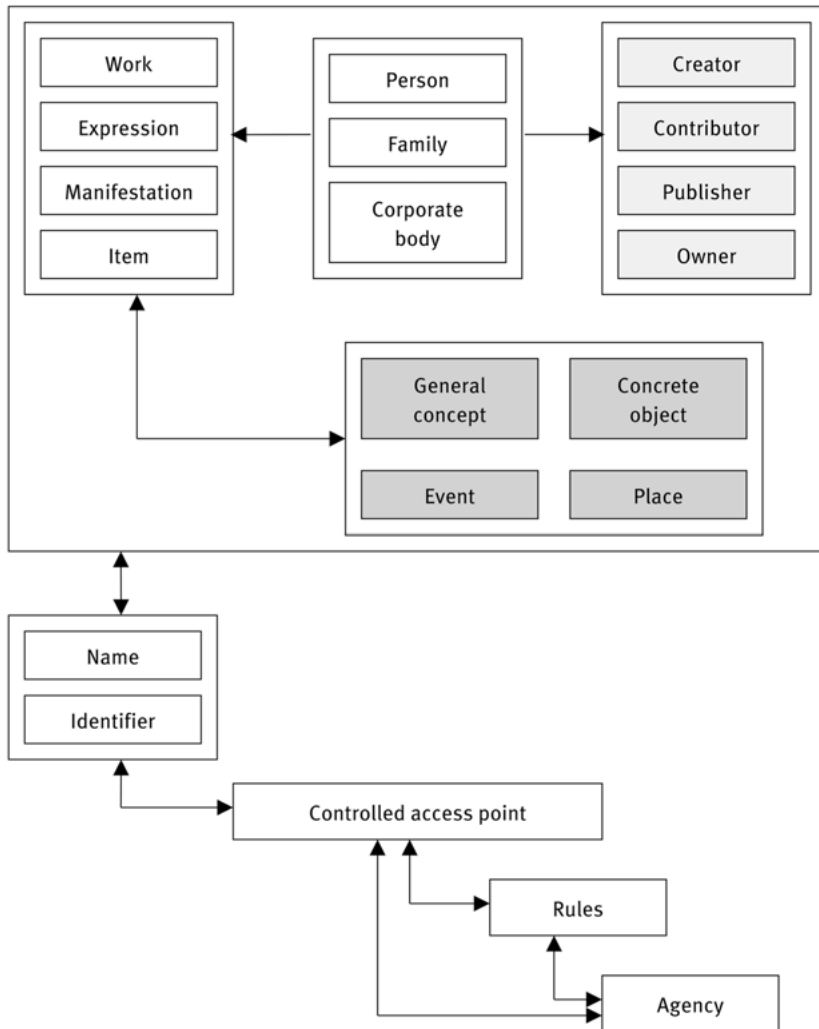


Figure J.1.4: Documents, Names and Concepts on Aboutness as Controlled Access Points to Documents and Other Information (Light Grey: Names; Dark Grey: Aboutness). Source: Modified from Patton, Ed., 2009, 23.

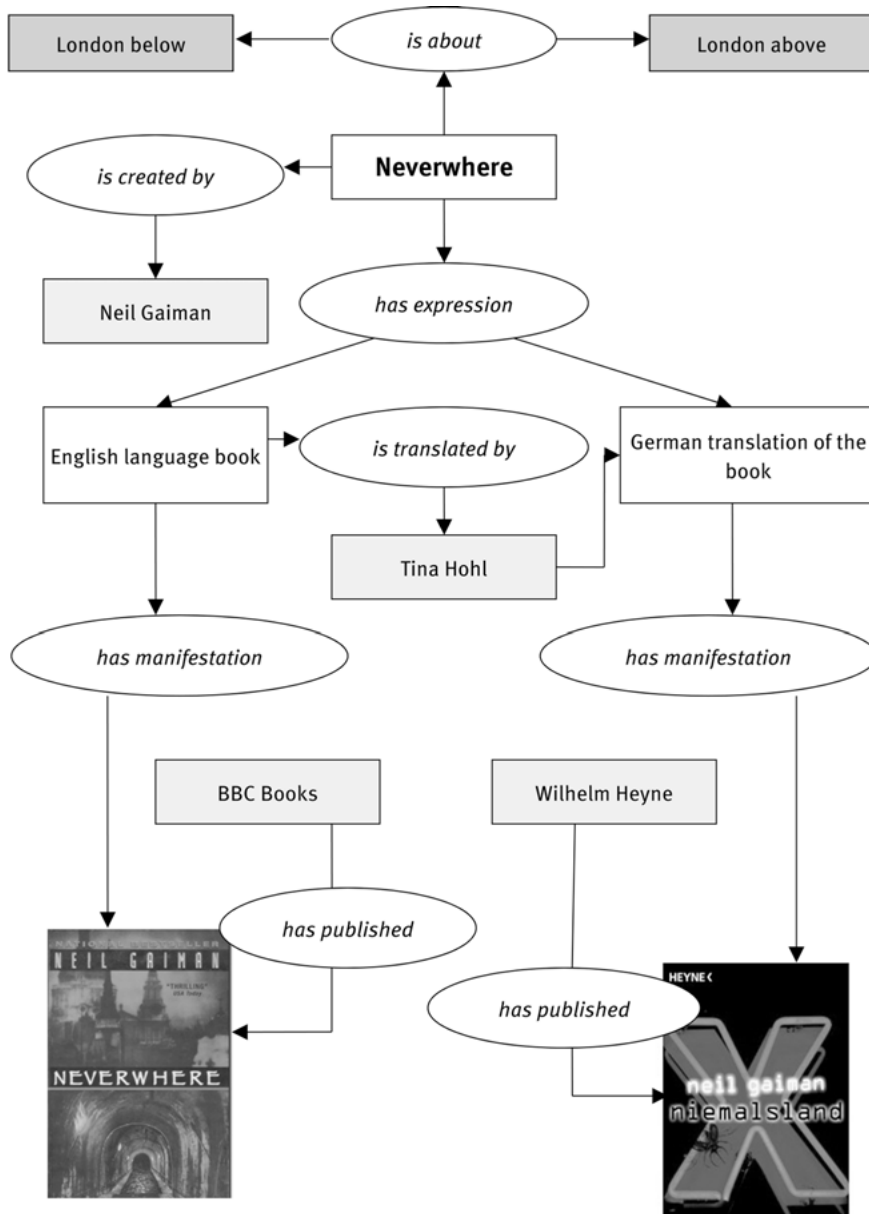


Figure J.1.5: The Interplay of Document Relations with Names and Aboutness (White: Document Relations, Light Grey: Relations with Names, Dark Grey: Aboutness).

Access to the documents and other information (names, concepts) is provided via access points. The rulebook “Resource Description & Access” (RDA) distinguishes between authorized and variant access points. An authorized access point is “the

standardized access point representing an entity” (RDA, 2010, GL-3), such as the title and author of a work. A variant access point is an alternative to the standardized access point (RDA, 2010, GL-42), such as title variants. Whereas the standardized access point for the French title is *La Chanson de Roland*, its variant titles include *Roland* or *Song of Roland* (RDA, 2010, G-15). Metadata are created in accordance with rulebooks (such as the RDA), which must be mastered by its creators and (at least in general) users. The overall context of the relations between documents, names and concepts is displayed graphically in Figure J.1.4.

We will illustrate the depictions in Figure J.1.4, which are very general, on a concrete example. In Figure J.1.5, we continue our example of ‘Neverwhere’. This work deals with two worlds in London, which are *London Below* and *London Above*. Here, content is indexed via places. The work has an author (Neil Gaiman) and a translator, for its German expression (Tina Hohl). The manifestations of ‘Neverwhere’ have appeared with different publishers (of which our graphic mentions BBC Books and Wilhelm Heyne). This always involves names that must be stated in their standardized form.

One can use the document relations, for instance, to view all expressions of one work, once it has been identified. To do so, one chooses a certain language and is thus led to the manifestations. Here, one searches for an edition and is provided with the items. If one arranges the latter list according to one’s distance from the item, one will receive a hit list with the closest library at the top. The conditions for this are that all libraries worldwide adhere to exactly one standard (or that the different standards are compatible), that catalog and ownership data of libraries are summarized digitally in one place, and that there is a user interface. Such a project is being undertaken at WorldCat (Bennett, Lavoie, & O’Neill, 2003).

Internal and External Formats

For the development, encoding and exchange of metadata, it must first be decided what data and exchange format should be taken as the basis. Internal formats state that each database can use its own format. In-house formats are mostly adjusted to the flexible and specific user demands. Exchange formats, on the other hand, are necessary in order for data entries to be swapped back and forth between institutions without any problems. This requires the simplest of standards, which restrict themselves to the bare essentials.

Documentary units (or surrogates) are encoded in order to make parts of the documentary reference unit available and searchable in the form of fields. Encoding facilitates the integration of different languages and fonts. Due to language independence, data can be transmitted from one system to the other. Programs regulate the conversion of internal data structures into an exchange format. Individual institutions, e.g. libraries, process the data transmitted in the communication format (e.g. MARC, Machine-Readable Cataloging) for display, e.g. by putting certain fields at the

very beginning or using abbreviations for given codes. The display is formatted differently in the various institutions.

Each documentary unit contains both attributes (e.g. code for author) and values (e.g. names of the authors). Encoding and metadata description go hand in hand. Taylor (1999, 57) writes, concerning the order of these two processing options:

In the minds of many in the profession, metadata content and encoding for the content are inextricably entwined. Metadata records can be created by first determining descriptive content and then encoding the content or one can start with a “shell” comprising the codes and then fill in the contents of each field. In this text encoding standards are discussed before covering creation of content. The same content can be encoded with any one of several different encoding standards and some metadata standards include both encoding and content specification.

In the following three graphics, we see how a data entry is constructed in the exchange format following MARC, and how differently the corresponding catalog entry is presented as a bibliographic unit in two different libraries using MARC.

```

LC Control No.: 85050110
000 00841nam a2200253 a 450
001 3807120
005 19850810000000.0
008 850730s1985 iaua 001 0 eng
035 __|9 (DLC) 85050110
906 __|a 7 |b cbc |c orignew |d 3 |e ocip |f 19 |g y-gencatlg
010 __|a 85050110
020 __|a 0840335296 (pbk.)
040 __|a DLC |c DLC |d DLC
050 00 |a QD251.2 |b .S74 1985
082 00 |a 547 |2 19
100 1_ |a Stacy, Gardner Wesley, |d 1921-
245 10 |a Organic chemistry : |b a background for the life sciences / |c Gardner W. Stacy, Carl C. Wamser.
250 __|a 2nd ed.
260 __|a Dubuque, Iowa : |b Kendall/Hunt Pub. Co., |c c1985.
300 __|a xiv, 465 p. : |b ill. ; |c 24 cm.
500 __|a Includes index.
650 _0 |a Chemistry, Organic.
700 1_ |a Wamser, Carl C., |d 1941-
991 __|b c-GenColl |h QD251.2 |i .S74 1985 |p 00037024776 |t Copy 1 |w BOOKS

CALL NUMBER: QD251.2 .S74 1985
Copy 1
-- Request in: Jefferson or Adams Bldg General or Area Studies Reading Rms
-- Status: Not Charged

```

Figure J.1.6: Catalog Entry in the Exchange Format (MARC). Source: Library of Congress

If a library gets a data entry in the MARC format, this entry will be reformatted for the system’s own display (e.g. by putting certain fields at the very top). Figure J.1.6 shows the formatted display of the Library of Congress’s MARC data entry. The first line contains the library’s control number. The fields beginning with “0” are further control

fields and their subfields. “020” contains, for example, the ISBN (International Standard Book Number), “050” the call number and “082” the notation under the Dewey Decimal Classification. Then follow author name with year of birth (100), title (245), edition (250), publication date and location as well as publisher (260), pages and other information (300), note on the index (500), topic (650), co-author with year of birth (700) and notes on the medium (991) as well as the signature (Call Number). The user of the Library of Congress Online Catalog can view the format and retrieve these encoded statements via “Marc Tags”.

In the standard case, the user will see the catalog entry as shown in Figure J.1.7. Here, the codes are decoded, replaced by descriptions and assembled in the form of a catalog card. The navigation field “Brief Record” contains the abridged version of the most important bibliographic data.

The Healey Library Catalog of the University of Massachusetts Boston offers a catalog entry on the same documentary reference unit, also processed from the MARC format, shown in Figure J.1.8. The individual positions of the fields as well as the field names differ from the Library of Congress’s.

◀ Previous Next ▶

Brief Record
Subjects/Content
Full Record
MARC Tags

Organic chemistry : a background for the life sciences / Gardner W. Stacy,...

LC Control No.: 85050110

Type of Material: Book (Print, Microform, Electronic, etc.)

Personal Name: Stacy, Gardner Wesley, 1921-

Main Title: *Organic chemistry : a background for the life sciences / Gardner W. Stacy, Carl C. Wamser.*

Edition Information: 2nd ed.

Published/Created: Dubuque, Iowa : Kendall/Hunt Pub. Co., c1985.

Related Names: Wamser, Carl C., 1941-

Description: xiv, 465 p. : ill. ; 24 cm.

ISBN: 0840335296 (pbk.)

Notes: Includes index.

Subjects: Chemistry, Organic.

LC Classification: QD251.2 .S74 1985

Dewey Class No.: 547 19

CALL NUMBER: QD251.2 .S74 1985

Copy 1

— Request in: Jefferson or Adams Bldg General or Area Studies Reading Rms

— Status: Not Charged

Figure J.1.7: User Interface of the Catalog Entry from Figure J.1.6 at the Library of Congress. Source: Library of Congress.

The Library of Congress Entry is a little more extensive than the other. In addition to the Healey Library entry, it also contains the fields “Type of Material” (Code 991 w), ISBN (Code 020 a) and the notation of the Dewey Decimal Classification (Code 082). Looking at these two exemplary catalog entries, one can see how useful a unified encoding of the metadata is. Apart from the respective local data (e.g. the call

number), the metadata entries are identical. The metadata of a work only have to be compiled once for each manifestation, after which they can be copied at will. This signifies a considerable lightening of the workload. It also requires an institution which possesses the authority of making these rules mandatory.



Figure J.1.8: User Interface of the Catalog Entry from Figure J.1.6 at the Healey Library of the University of Massachusetts Boston. Source: Healey Library.

The values are firmly allocated to the attributes (codes) in the MARC format. Since the codes are arranged into subsections, these subfields can be combined at will in the display. This is the case, for instance, with the Healey Library and its “Description” field (consisting of Code 250a and 300 a, b, c). The institution in question will thus only rearrange such fields or subfields for the display that are actually required for internal usage. Starting from the identical MARC data set (middle), the two exemplary libraries (left and right) produce different local surfaces:

Library of Congress	Code/Subfield	Healey Library
Personal Name	100 a, d	Main Author
Main Title	245 a, b, c	Title
Edition Information	250 a	Description
Description	300 a, b, c	Description
Published/ Created	260 a, b, c	Publisher
Related Names	700 a, d	Other Author(s)
Notes	500 a	Notes
Subjects	650 a	Subject(s)
Call Number	991 h, i	Call Number
Call Number	991 t	Number of Items.

We do not want to introduce the variants of MARC formats, or of any other encoding standards. In this context, rather, we are discussing the ways in which standards can be employed. Taylor (1999, 73) characterizes encoding standards as a sort of container, or bowl, in whose frame one need only enter the text. Metadata is the term given to the complete data entry that includes the encoding and delivers a description (instead) of a larger document.

The combination of the encoding container and its text is called *metadata* or a *metadata record*.

In a digital library, each metadata record consists of a link to the full text (e.g. in PDF format) of the entire document.

Authority Records

What entries will finally look like in their unified form will be decided by the relevant rules for authority control. This holds for titles as well as names of persons and organizations, and the goal is to circumvent the problem of multilingualism, spelling variants, transcriptions and title versions.

Let us start with authority records for titles. We assume that around 50% of bibliographical material includes derivations (Taylor, 1999, 106). Our example deals with a work that has appeared in several languages. Let our point of departure be the work “Romeo and Juliet” by William Shakespeare. Taylor (1999, 113) compares the data entries of two of the work’s derivations. An English-language manifestation bears the title “The Tragedy of Romeo and Juliet”. A Spanish translation appears as “Romeo y Julieta”. According to MARC, the surrogates have the following entries:

100	1	a Shakespeare, William d 1564-1616
240	10	t Romeo and Juliet 1 Spanish
245	10	a Romeo y Julieta ...
100	1	a Shakespeare, William d 1564-1616
240	10	t Romeo and Juliet
245	14	a The tragedy of Romeo and Juliet.

The two manifestations unequivocally refer to the work via the field entries 100 (preferred author name) and 240 10 | t (uniform title). Field 245 records the title variants.

Besides the derivative relations, there are descriptive relations. Forms of these are works about other works, e.g. treatises about a drama. A search in the “British Library” about the aforementioned work by Shakespeare leads to a surrogate with the author name Patrick M. Cunningham and the title “How to Dazzle at Romeo and Juliet”, containing the entry

600 14 | a Shakespeare, William, | d 1564-1616
 | t Romeo and Juliet.

Here the relation to the discussed work, the original, is clear. Field 600 describes “subject access entries”.

Let an example for a Part-Whole relation be one part of the work “The Lord of the Rings” by J.R.R. Tolkien, say “The Fellowship of the Ring”. Since this partial work is recorded under the concrete title, just like the other two parts, only the entry

800 1 | a Tolkien, J. R. R. | q (John Ronald Reuel),
 | d 1892-1973. | t Lord of the Rings ; | v pt. 1

brings the different parts together via | t (title of work) and | v in (volume/sequential designation) field 800 1. Field 800 serves as an access point to the series (| a refers to the preferred name for the person, | d to dates associated with the name and, finally, | q to a fuller form of the name).

Transliterations serve to unify different alphabets, providing both a target-language-neutral process and a distinct reassignment. General transcriptions cannot be used, since they are phonetically oriented and adapt to the sound of the word in the target language in question. The Russian name

Хрущев

is clearly transliterated via

Chruščev.

General transcriptions lead to language-specific variants and thus cannot be used:

Chruschtschow (German),
 Khrushchev (English),
 Kroustchev (French).

Transliterations are regulated via international norms, e.g. ISO 9:1995 for Cyrillic or ISO 233:1984 for Arabic.

Many criteria must be considered in order to bring personal names into a unified form. Different persons with the same name must be distinguishable. One and the same person, on the other hand, may be known under several names. Different names can exist for a person at different times. Names vary with regard to fullness, language and spelling, or in the special case that an author publishes his work under a pseudonym. Certain groups of people (religious dignitaries, nobles etc.) use their own respective naming conventions. RDA states that the preferred name for the person is the core element, while variant names for the persons are optional. Generally, the

name that should be chosen is the name under which the person has become best known, be it their real name, a pseudonym, a title of nobility, nickname, initials etc. Hence *Tony Blair* is deemed a preferred name, as opposed to *Anthony Charles Lynton Blair* (RDA, 2010, 9-3). Where a person is known under several names, there will correspondingly be several preferred names. *Charles L. Dodgson* used his real name in works on mathematics and logic, but resorted to the pseudonym *Lewis Carroll* for his literary endeavors (RDA, 2010, 9-11). On the other hand, there exist different forms of one and the same name with regard to its fullness and language. Where a name has changed due to a marriage or acquisition of a title of nobility, the latter name will be recorded as a preferred name. All naming variants point to the preferred name.

A list with country-specific naming authority records can be found in IFLA (1996) and, in extracts, RDA (2010). A few selected examples shall serve to illustrate this complex subject matter.

English:	Von Braun, Wernher (prefix is the first element) De Morgan, Augustus
French:	Le Rouge, Gustav (article is the first element) Musset, Alfred de (part of the name following a preposition is the first element)
German:	Braun, Wernher von (part of the name following a prefix is the first element) Zur Linde, Otto (if the prefix is an article or a contraction of an article and a preposition, then the prefix is the first element)
Icelandic:	Halldór Laxness (given name is the first element).

Metadata in the World Wide Web

The markup language HTML allows for the deposition of metadata about a webpage in a select few fields. These so-called metatags are in the head of the HTML documents. They are, among others, fields that record title, author, keywords and descriptions (abstract). One example is provided in Figure J.1.9. Every creator of a webpage is free to use metadata or not. It is a task of the webpage's creator to secure the correctness of the metadata. A misuse of metatags is principally not to be ruled out.

In the World Wide Web, we are outside of the world of libraries. Where it is already hard to agree on certain standards in the latter, the problems only grow larger when one tries to find common ground in describing and arranging the internet's data. There is no centralized instance controlling the quality and content of metadata on the Web. The entries cannot be standardized, since no webmaster can be told what to consider when creating his website. Zhang and Jastram (2006, 1099) emphasized that in the ideal case, metadata embedded in webpages can be incorporated by search engines for further data processing:

In the morass of vast Internet retrieval sets, many researchers place their hope in metadata as they work to improve search engine performance. If web pages' contents were accurately represented

in metadata fields, and if search engines used these metadata fields to influence the retrieval and ranking of pages, precision could increase and retrieval sets could be reduced to manageable levels and ranked more accurately.

```
<html>

  <head>
    <title>Fachbereich Informations- und Wissensmanagement der
Hochschule Darmstadt</title>
    <meta http-equiv="Content-Type" content="text/html;
charset=iso-8859-1">
    <meta name="description" content="Fachbereich Informations- und
Wissensmanagement der Hochschule Darmstadt">
    <meta name="author" content="Fachbereich Iuw">
    <meta name="keywords" content="Iuw, Informations- und
Wissensmanagement, IuD, Information, Dokumentation, Information und
Dokumentation, FHD, h_da, FH, Fachhochschule, Hochschule, Darmstadt, Hochschule
Darmstadt, Wirtschaftsinformation, business information, Medieninformation,
media information, Massenkommunikation, Wissensmanagement, knowledge-management,
knowledge management, Informationsmethodik, Informationswissenschaft,
Information Science, Information Broking, Information Engineering, Information
Architecture, Informationsarchitektur, Informationssystem, information system,
Informationsvermittlung, Informationsmanagement, information management,
information retrieval, informationsvisualisierung, information visualization,
informationsqualit&auml;t, information quality, Wissensrepr&auml;sentation,
Bibliotheksmanagement, Bibliothekswissenschaft, library studies, library
science, library management, digital library, Datenbanken, databases">
    <meta name="robots" content="index">
    <meta name="robots" content="follow">
    <meta name="DC.Creator" content="Fachbereich Iuw">
    <meta name="DC.Publisher" content="Hochschule Darmstadt">
    <meta name="DC.Rights" content="Fachbereich Iuw">
    <meta name="DC.Description" content="Fachbereich Informations-
und Wissensmanagement der Hochschule Darmstadt">
    <meta name="verify-v1"
content="SSB+C2IzgnuZAaIFIL7EQiOxKD09Pk5mU9dnIy1N/ZU=" />
    <script language="JavaScript">

if (top.frames.length > 0) top.location.href = self.location.href;

    </script>
  </head>
```

Figure J.1.9: A Webpage's Metatags. Source: www.iuw.fh-darmstadt.de.

How are metatags used and structured in reality? In one study, Zhang and Jastram investigate whether and how certain metadata are used in the WWW by different user groups. Around 63% of all 800 tested webpages contain metadata. Web authors often use too many or too few tags, making it hard for search engines to sift out the relevant ones. The most popular tags are those in the fields keyword, description and author (Zhang & Jastram, 2006, 1120).

The three most popular of these descriptive elements are the Keyword, Description, and Author elements while the least popular are the Date, Publisher, and Resource elements. In other words, they choose elements that they believe describe the subjective and intellectual content of the page rather than the elements that do not directly reveal subject-oriented information.

Craven (2004) compares metatag descriptions from websites in 22 different languages. He analyzes the language of the websites, the frequency of occurrence of metatags as well as the language and length of the descriptions. It was shown, for instance, that the websites in Western European languages (English, German, French and Dutch)

contained the most descriptions, as opposed to Chinese, Korean and Russian sites, only 10% of which contain metadata.

These and similar investigations are only minimal approaches to extracting any characteristics or regularities for the unified development of metadata from the internet jungle. Sifting through the internet's structure remains a very hard task for knowledge representation.

In answer to the question of what simple and wide-ranging descriptions for as many online resources as possible should be selected, the so-called Dublin Core provides a listing. In a workshop in Dublin (Ohio), the "Dublin Core Metadata Element Set" (ISO 15836:2003) was compiled. It contains 15 attributes for describing sources, where there are no exact rules for the content of the entry. The elements refer to source content (title, subject, description, type, source, relation, coverage), author (creator, publisher, contributor, rights) and formalities (date, format, identifier, language). The Dublin Core is only a suggestion which the millions of webmasters might take to heart. The precondition is honest, unmanipulated entries. Only when this ideal is adhered to can search engines exploit metatags meaningfully.

Safari (2004) sees the structuring of Web content via metadata as the future of an effective Web:

The web of today is a mass of unstructured information. To structure its contents and, consequently, to enhance its effectiveness, the metadata is a critical component and "the great web hope". The web of future, envisioned in the form of semantic web, is hoped to be more manageable and far more useful. The key enabler of this knowledgeable web is nothing but metadata.

Conclusion

- Traditionally, documents are gathered in the form of catalog cards that contain compressed information about a document, so-called metadata. In digital environments, the traditional catalog cards are replaced by structured data sets in a field schema.
- Metadata are standardized data about documentary reference units, and they serve the purpose of facilitating digital and intellectual access to and use of documents. They stand in relation to one another.
- Work, expression, manifestation and item form the primary document relations.
- Equivalence, derivative and descriptive relations can be used to distinguish an original or identical work from a new creation. Part-Whole or Part-to-Part relations are used for sequences of articles or Web documents.
- Apart from the title, names (persons', families' and corporate bodies') and concepts that express the document's aboutness are access points to the resources.
- The encoding of surrogates makes it possible to display the documentary reference unit in the form of fields, independently of language and writing, and to make it retrievable.
- Exchange formats (e.g. MARC) regulate the transmission and combination of data entries between different institutions.

- Rulebooks (e.g. RDA) prescribe the authority records of personal names, corporate bodies and titles, among others.
 - Transliterations, which provide a target-language-neutral transcription as well as a distinct reassignment, serve to unify different alphabets.
 - Metadata in websites can (but do not have to) be stated via HTML metatags. The websites' creators act independently of standards and rulebooks, and the statements' truth values are unclear. The Dublin Core Elements form a first approach to standardization.
-

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J.2 Metadata about Objects

Documents Representing Objects

Apart from textual documents (publications as well as unpublished texts) and digitally available non-textual documents (images, videos, music, spoken word etc.), there are documents which are fundamentally undigitizable (Buckland, 1991; 1997). We distinguish between five large groups of such non-text documents:

- Objects from science, technology and medicine (STM),
- Economic objects,
- Museum artifacts, or works of art,
- Real-time objects (e.g., flight tracking data).

There is a fifth group of undigitizable documents: persons. As we can find persons in other mentioned groups (as patients in the context of medicine, as employees as a special kind of economic objects, and as artists in the context of works of art), we will discuss those documents in relation to the other categories.

STM objects include, for instance, chemical elements, compounds and reactions, materials, diseases and their symptoms, patients and their medical history. Economic objects can be roughly differentiated according to sectors, industries, markets, companies, products and employees. The third group includes all exhibits in museums, galleries and private ownership, as well as objects in zoos, collections etc. Real-time objects include data from other information systems without any intellectual processing (e.g., flight tracking data).

It is obvious that such documents can never directly enter an information system. A text that is not digitally available, for example, can be scanned and prepped for further digital processing, but a company cannot be digitized. Here it is imperative that we find a surrogate. Of course it can make sense to offer aspects of an object, at least indirectly, in digital form—such as the structural formula of a chemical compound (not the compound itself) or a photograph of a work of art (where a photo of the ‘Mona Lisa’ is not the singular painting itself).

Since we never have access to the document in itself when dealing with documents about objects, the surrogate, and hence its metadata, assumes vital importance for the representation and retrieval of the knowledge in question. Metadata bundle all essential relations—in different ways for different kinds of facts—which represent the object. Depending on the type of object, the number of relations can be very large in metadata about facts. As entirely different relations must be considered for the various document types, we cannot develop a “general” field schema but have to make do with examples that show what field schemata about facts look like. As with bibliographic metadata, it is clear that the objects’ fundamental aspects must be divided into their smallest units, which are then structured into fields or subfields

in the information system. Both field schema and its values are to be regulated via standards.

Attributes and values are chosen in such a way that further processing, e.g. calculations, can be performed where possible. Here we must distinguish between two scenarios: further processing within a data record and that beyond the borders of individual surrogates. Let two examples clear up the idea of intra-surrogate processing: a database of materials will surely consider the relation “has melting point” to be of importance. When entering the values in the corresponding field, the temperature is entered in centigrade. In order to state the degrees according to the Fahrenheit or Kelvin scales, one does not have to create a new entry. Instead, the conversion is performed automatically. A company database knows the relation “has profit in the fiscal year X.” If values are available for several years, one can automatically create a time series graphic or calculate change rates for X and X-1 on the basis of the values. Inter-surrogate processing regards—to stay within the company database—comparisons between companies, e.g. a ranking by revenue of enterprises in an industry for a given year. Gaus (2004, 610) refers to statistical processing options in digital patient files:

Data of different patients can be summarized, e.g. how strongly does the number of leucocytes deteriorate on average following a certain cytotoxic therapy, and what have been the highest and lowest deteriorations so far observed?

Providers of economic time series offer both simple statistical analyses (descriptive statistics, deseasonalization etc.) and playthroughs of complex econometric models.

The database producer in question is called upon to make sure that the information about facts he offers are correct, i.e. one can make sure to consult only reliable sources.

Figure J.2.1 displays the updating process for a surrogate about an object. In this graphic, we assume that a surrogate has already been created for the document about an object. (If there is no surrogate available, all attributes for which values are known must be worked through one by one, as on the left-hand side of the graphic.) We further assume the availability of a reliable source of information (e.g. a current scientific article or patent for STM facts, a new entry in the commercial register, a current voluntary disclosure or annual report by a company for economic objects, or a current museum’s catalog).

The information source will be worked through attribute by attribute with regard to the document about an object. If the source provides information about a hitherto unallocated attribute, the value will be recorded for the first time (and the corresponding field is made searchable). Thus it is possible, for instance, that a scientific publication first reports on the boiling point of an already known chemical substance. In that case, we can allocate the published value to the attribute. If the attribute is already occupied by a value (let us say: *research leader: Karl Mayer* in a company dossier), and if the current source reports a new value (e.g. because *Karl Mayer* has left the

company, leaving his successor *Hans Schmidt* in charge of the research department), the value must be corrected. Both new entries of attributes and values and updates of values are time-critical actions that must be performed as quickly as possible in order to safeguard the accuracy of the surrogate.

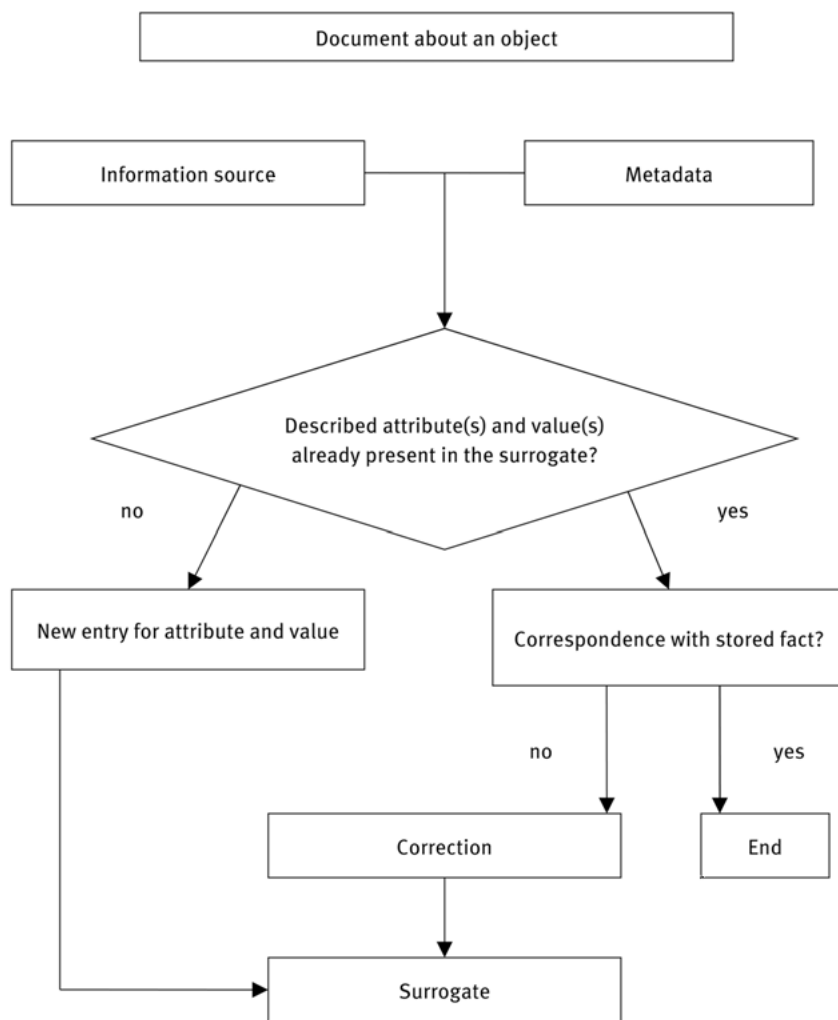


Figure J.2.1: Updating a Surrogate about an Object.

Each of the four groups of documents about objects will be presented in the following via a “textbook example”: substances from organic chemistry paradigmatically represent STM facts (Beilstein), our example for economic objects is a company database (Hoppenstedt), for museum objects we refer to the J. Paul Getty Trust’s endeavors of

describing works of art, and for real-time objects we briefly describe the flight-tracking system of Flightstats.

STM Facts: On the Example of Beilstein

In organic chemistry, knowledge of the structure of chemical elements is of the essence. Over the reference period from the year 1771 to the present, the online database “ReaxysFile” (Heller, Ed., 1990; 1998) has collected more than 35m datasets. These possess attributes and the corresponding values for over 19m organic chemical substances. The database—previously created by the Beilstein Institut zur Förderung der Chemischen Wissenschaften (Frankfurt, Germany) (Jochum, 1987; Jochum, Wittig, & Welford, 1986), now owned by Elsevier MDL—is updated on the basis of (intellectual) information extraction from articles of leading journals on organic chemistry. Beilstein’s substance database works with around 350 searchable fields (MDL, 2002, 20), each of which stands for a specific relation. Not every field in every data set is filled, i.e. there are no values available for some attributes.

Liquids (LIQ)

Boiling Point (BP)

Liquid Phase (LIQPH)

Transition Point(s) of Liquid Modification(s) (LPTP)

Gases (GAS)

Critical Temperature (CRT)

Critical Pressure (CRP)

Critical Density (CRD)

Critical Volume (CRV)

Vapour Pressure (VP)

Gas Phase (GP)

Figure J.2.2: Beilstein Database. Fields of the Attributes of Liquids and Gases. Source: MDL, 2002, 106.

The relations are structured into seven chapters, which are hierarchically fanned out from top to bottom below:

- Basic Index (across-field searches),
- Identification Data,
- Bibliographic Data,
- Concordances with other Databases,
- Chemical Data,

- Physical Data,
 - Pharmacological and Ecological Data (MDL, 2002, 27).
- We will follow the concept ladder downward via an example:

Physical Data,
 Single-Component Systems,
 Aggregate State,
 Gases.

Figure J.2.2 shows the six relations that are of significance for gases: *has critical temperature at*, *has critical pressure at* etc.

2.6.1.2.3.3 Critical Density (CRD)

Description:

The Critical Density field contains the numerical value for the density of a substance measured at its critical temperature and pressure.

Type of Indexing:

Numerical

Unit:

The default unit in the CrossFire database for this fact is $\text{g} \cdot \text{cm}^{-3}$.

Figure J.2.3: Beilstein Database. Attribute “Critical Density of Gases”. Source: MDL, 2002, 119.

In Figure J.2.3, we see the field description of the critical density of gases. The relation is described briefly; in addition, the measurement units of the values are stated. On this lowest level, facts are searched. On the hierarchy level above, searches concern whether and in what fields values are available at all.

Let us suppose that we are searching for a substance with a critical density in gaseous form of between 0.2000 and 0.2022 g/cm^3 . The attribute is expressed via the field abbreviation (here: CRD; Figure J.2.4), the value interval via the two numbers linked by “-”. Besides intervals, one can also search equality and inequality (greater than, smaller than). The correct request with the database provider STN (Buntrock & Palma, 1990; Stock & Stock, 2003a; 2003b) is as follows:

S 0.2-0.2022/CRD,

where S is the search command.

As with all metadata, it is of importance for STM facts to be able to connect several sources with each other without a problem (if possible) when these occur. There

are several databases on chemical structures; besides Beilstein, there are Chemical Abstracts, Index Chemicus and Current Chemical Reactions, among others. Problems sometimes occur for end users in in-house intranets who have to search through the different databases one by one, since they require an exact knowledge of the specifics of all sources and must be able to manipulate the systems. Zirz, Sendelbach and Zielesny (1995) report of a successful intranet endeavor in a large chemical enterprise. It used Beilstein together with other chemistry databases in the context of an “integrated chemistry system”.

State of Aggregation - Gases

Search Field Name	Default Unit	Search Code	Search Examples	Display Codes
Critical Density (1) Comment (2)	g/cm ³	/CRD	S 0.2-0.2022/CRD	CRD
Critical Pressure (1) Comment (2)	Torr	/CRP	S HANDBOOK/CRD.COM S CRP >760 MBAR	CRD CRP
Critical Temperature (1) Comment (2)	Cel	/CRT	S HANDBOOK/CRP.COM S 500-600/CRT	CRP CRT
Critical Volume (1) Comment (2)	cm ³ /mol	/CRV	S HANDBOOK/CRT.COM S 210/CRV	CRT CRV
Gas Phase Comment (2)	-	/FA	S HANDBOOK/CRV.COM S GP/FA	CRV GP
Description	-	/GP.COM	S (SATURATED (P) LIQ?)/ GP.COM	GP
Vapour Pressure (1) Comment (2)	Torr	/GP.KW	S FUGACITY/GP.KW	GP
Temperature (1)	cel	/VP	S 4-5/VP	VP
		/VP.COM	S EQUATION/VP.COM	VP
		/VP.T	S VP>80 and VP.T<5	VP

(1) Numeric search field that may be searched using numeric operators or ranges.

(2) Contains German text.

Figure J.2.4: Beilstein Database. Searching on STN. Source: STN, 2006, 18.

Metadata about Companies: Hoppenstedt

The Hoppenstedt *Firmendatenbank* (Company Database) contains profiles about roughly 300,000 large and mid-sized German enterprises. The database sets particularly great store by the points of contact on the companies' first and second levels of leadership; around 1m managers are thus listed. The database producers check the correctness of the statements via the Federal Gazette, business reports, the press and—this being by far the most important source—companies' voluntary disclosures.

Hoppenstedt has an extensive database with the full extent of fields and all entries. However, several products are derived from this database besides the full version. This full version offers around 70 attributes (Stock, 2002, 23). The relations that describe the object “enterprise” are divided into the following thematic areas:

- Company (address and the like),
- Industry,
- Operative Data (employees, revenues, etc.),
- Management (top and middle management including function and position),

- Communication (telephone, e-mail and homepage, but also SWIFT code),
- Balance Sheet Data,
- Facilities (e.g. carpool or computer systems),
- Business Operations (business type as well as import and export),
- Subsidiaries,
- Participating Interests,
- Other Matters (e.g. commercial register, legal structure, memberships in syndicates).

With its ca. 70 attributes overall, the evaluation of companies is extremely detailed compared to other company databases. However, not all fields are always filled with values. In an empirical study of company databases, it transpires that the majority of attributes are allocated a value in only 50,000 (of 225,000) data entries (Stock & Stock, 2001, 230).

LAP GmbH Laser Applikationen

Strasse :	Zeppelinstr. 23
Ort :	21337 Lüneburg
Telefon :	(04131) 95 11-95
Fax :	(04131) 95 11-96
E-Mail :	info@lap-laser.com
Internet :	http://www.LAP-LASER.com
Handelsregister :	Amtsgericht Lüneburg HRB 662



Branche	
NACE-Code :	33402; 33101
US-SIC-Code :	3827; 3845
Allgemeine Informationen	
Bank(en) :	Spark Lüneburg
Rechtsform :	GmbH
Gründung :	1984
Organe / Management / Anteilseigner	
Geschäftsführer :	Guido Jung
Technische Leitung :	Hans-Peter Wernner
Gesellschafter :	Mitarbeiter, 100%

Figure J.2.5: Hoppenstedt Firmendatenbank. Display of a Document (Extract). Source: Hoppenstedt.

We would like to address two important aspects of metadata about facts on the example of the Hoppenstedt database. Certain factual relations require their values to be entries from a knowledge organization system. For company object, for instance, we work with the relation “belongs to the industry ...” (Figure J.2.5). Hoppenstedt uses two established KOSs for naming industries: NACE and US-SIC (see chapter L.2). The entries in these fields, in the form of controlled vocabularies, are fundamentally derived from the two named industry classifications.

The second aspect regards options for further processing, which are expected of metadata about facts by the users. Hoppenstedt allows for the data to be exported in

the CSV format. CSV (Comma Separated Values) allows for a data transfer between different environments, e.g. between an online database and local office environments. (Alternatively, data exchange could be performed via XML.) A further useful application of company information is the description of addresses and information about managers. It is (almost) automatically possible to launch pinpoint mailing campaigns by using the mail merge function of text processing software.

STM facts last for a longer period of time; they only change when new discoveries make the old values obsolete. This is not the case for economic facts. Here, the values are potentially in constant flux. This regards not only operative data, which are being continuously written year in year out, but to a much larger extent almost all other attributes, such as addresses, phone numbers and information about managers. The correctness and up-to-dateness of the values must be constantly checked. What good is a marketing campaign if the people it is aimed at have already left the company, or moved house?

Metadata about Works of Art: CDWA

The “Categories for the Description of Works of Art” (CDWA) comprises a list of 507 fields and subfields for the description of works of art, architecture and other cultural goods (Baca & Harping, Eds., 2006, 1). The CDWA are developed at the “J. Paul Getty Trust & College Art Association” (previously “The Getty Information Institute”; Fink, 1999) in Los Angeles, in cooperation with other institutes.

The basic principle for the construction of such metadata is the provision of a field schema skeleton as a sort of standard for all those who are interested in representing works of art. In contrast to Beilstein and Hoppenstedt, no application is at the forefront of exactly one database. Every art historian, every museum, every gallery etc. is called upon to use the standard (for free, by the way) or, if needed, suit it to their needs. Fink (1999) reports:

(The goal of the *Categories for the Description of Works of Art* (CDWA)) was to define the categories of information about works of art that scholars use and would want to access electronically. ... The outcome ... provides a standard for documenting art objects and reproductions that serves as a structure for distributing and exchanging information via such channels as the internet.

In the CDWA, the term “category” refers to the relations, and thus to the single fields that gather the factual values. The metadata schema thus worked out is meant to help make museum information accessible to the public (Coburn & Baca, 2004).

Materials and Techniques - Watermarks - Identification**DEFINITION**

An identification of the watermark type or name.

EXAMPLES

foolscap
Briquet 6088

DISCUSSION and GUIDELINES

Optional: Record a term or reference to a watermark authority. Use lower case, but capitalize proper names.

If a watermark authority is cited, list the citation for the authority in the CITATIONS subcategory.

TERMINOLOGY/FORMAT

Controlled list: Control this subcategory with a controlled list of standard watermark descriptions based on published catalogs and watermark dictionaries.

If retrieval on watermarks is important, it is recommended to maintain a full-blown authority of watermarks rather than a controlled list. In an authority, the IDENTIFICATION and DATE subcategories described here in the CDWA could be combined with fields for geographic distribution, paper producer, measurements, and an image (beta-radiograph, photograph or negative, or tracing) of the watermark. See the data dictionary described by the *Watermark Archives Initiative*, cited below.

Figure J.2.6: Description of the Field Values for Identifying Watermarks According to the CDWA.

Source: CDWA (Online Version).

The relations are hierarchically structured; values are entered on several levels. Let us take a look at the factual hierarchy on materials and techniques (Baca & Harpring, Eds., 2006, 14).

- Materials and techniques
 - Materials and techniques—description
 - Materials and techniques—watermarks
 - Materials and techniques—watermarks—identification
 - Materials and techniques—watermarks—date
 - Materials and techniques—watermarks—date—earliest date
 - Materials and techniques—watermarks—date—latest date.

The relations on the first level have themselves relations for their values. The relations for “Materials and techniques” are, for example, “Description of the technique”, “Name of the technique”, “Name of the Material” or “Watermarks”. From the second hierarchy level onward, one can enter values. For watermarks, for instance, the value *lily blossom over two ribbons* might appear. Its identification would be—as a normalized entry—*lily blossom*, its date 1740 - 1752, with the earliest date 1740 and the latest date, correspondingly, 1752. These dates state in which years the watermark in question was in use. Figure J.2.6 shows the note on entering values into the field “Materials and techniques—watermarks—identification”.

The CDWA allow free text entries for many fields, but if a standardization is possible one must enter either numerical values in the normalized form (e.g. years) or use controlled vocabularies (KOSs).

Flight-Tracking: Flightstats

Flightstats is a system for real-time flight data. Flightstats provides information about ongoing flights and delivers them to their users via Web pages (Figure J.2.7) or via mobile applications. There are very few metadata. A flight can be identified by flight number (e.g., AA 550) and date (e.g., Dec 06, 2012). All other data about the flight (e.g., route) and the actual position (latitude, longitude, altitude, etc.) are transferred by flight dispatchers. Additionally, Flightstats calculates some statistical values (e.g., on airport delays or about on-time performance of airlines).

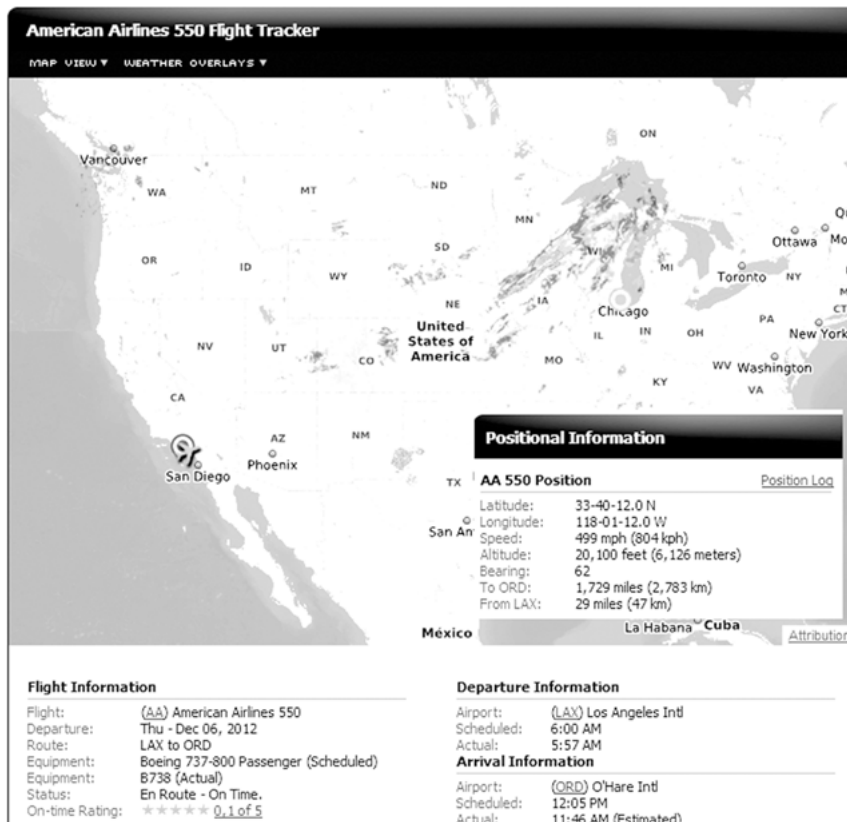


Figure J.2.7: Example of Real-Time Flight Information. Source: Flightstats.

Conclusion

-
- Non-digitizable documents about objects (STM objects, economic objects, works of art, real-time objects, and persons) are fundamentally represented via surrogates in information services.
 - There is no general schema for factual relations; rather, one must work out the specific field schema for each sort of facts. Both attributes (fields) and values (field entries) are standardized. This is set down in a (generally very detailed) rulebook for indexers and users.
 - If an attribute is located in a knowledge domain, its values are stated in the form of controlled terms from a knowledge organization system.
 - Where possible, one must provide options for further processing of field values. Intra-Surrogate relations allow for calculations within a factual surrogate (e.g. the conversion between degrees Celsius and degrees Kelvin), whereas Inter-Surrogate relations allow for processing beyond surrogate borders (e.g. comparisons of several companies' balance sheets).
 - Information about objects only fulfills its purpose when it is correct. The stored factual values must thus be continually checked for accuracy.
 - For extensive field schemata, it is useful to structure the relations hierarchically. Here it is possible to make specific field entries only on the lowest level (as in Beilstein), but also to allow for certain values on all others (as in the CDWA).
 - If there is only one standard in a knowledge domain (as is the case for the CDWA—at least in theory), and if all database producers adhere to it, there can be a smooth and frictionless data exchange. Where several standards coexist (as in economic documentation), these must be connected into a single field schema (which is usable in-house) in applications for end users.
-

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J.3 Non-Topical Information Filters

The Lasswell Formula

In communication studies, there is a classical formulation—the Lasswell Formula—which succinctly describes the entirety of a communication act. Lasswell (1948, 37) introduces the following questions:

- Who?
- Says What?
- In Which Channel?
- To Whom?
- With What Effect?

Braddock (1958) adds two questions:

- What circumstances?, leading to *context analysis*;
- What purpose?, leading to *genre analysis*.

The extended Lasswell formula reads as follows (Braddock, 1958, 88):

WHO says WHAT to WHOM under WHAT CIRCUMSTANCES through WHAT MEDIUM for WHAT PURPOSE with WHAT EFFECT?

There thus appears to be more than just the content (*What*). Knowledge representation must also take into consideration the other aspects. According to Lasswell (1948, 37), his five questions lead to five disciplines, which however can definitely cooperate.

Scholars who study the “who”, the communicator, look into the factors that initiate and guide the act of communication. We call this subdivision of the field of research *control analysis*. Specialists who focus upon the “says what” engage in *content analysis*. Those who look primarily at the radio, press, film and other channels of communication are doing *media analysis*. When the principal concern is with the persons reached by the media, we speak of *audience analysis*. If the question is the impact upon audience, the problem is *effect analysis*.

Thematic information filters are fundamentally guided by the aboutness of a document, which corresponds to Lasswell’s *What*. However, there are other aspects that nevertheless represent fundamental relations of a document. They answer questions such as these:

- On *what level* was the document created?
- *Who* has created the document (nationality, profession etc.)?
- *Which genre* does the document belong to?
- In *what medium* (reputation, circulation etc.) is the document distributed?
- For *whom* was the document created?
- *Which purpose* does the document serve?
- For *how long* is the document relevant?

As Argamon et al. (2007, 802) correctly state, one can only do justice to the complete meaning of a document when all formal and all content-related aspects are taken into consideration:

We view the full meaning of a text as much more than just the topic it describes or represents.

For textual documents, Argamon et al. (2007, 802) contrast the *What* (the aboutness) with the *How* of the text. This *How* is described as the “style” of the document:

Most text analysis and retrieval work to date has focused on the topic of a text; that is, *what* it is about. However, a text also contains much useful information in its style, or *how* it is written. This includes information about its author, its purpose, feelings it is meant to evoke, and more.

Style has several facets: on the one hand, it is about stylistic relations with regard to the knowledge contained in the document, i.e. about the manner of topic treatment. A second aspect is the target group. Finally, one can also regard a document from the perspective of time: when was it created, and for how long will it contain action-relevant knowledge? The relations of style—apart from the purely formal and purely content-related fields—are also suitable for the representation of documents and as information filters for retrieval. Crowston and Kwasnik (2004, 2) are convinced of the usefulness of non-topical information filters:

We hypothesize that enhancing document representation by incorporating non-topical characteristics of the document that signal their purpose—specifically, their genre—would enrich document (and query) representation. By incorporating genre we believe we can ameliorate several of the information-access problems ... and thereby improve all stages of the IR process: the articulation of a query, the matching or intermediation process, and the filtering and ranking of results to present documents that better represent not only the topic but also the intended purpose.

Crowston and Kwasnik (2004) explain the advantages of non-topical information filters via an example. Let a university professor search for documents about a certain topic in preparation of a lecture. “Good” results might be scripts or slide sets by colleagues. However, if the professor then searches the same topic with the aim of using his search results for a research project, the “good” results will now be scientific articles, papers in proceedings or patents. The topical search query will be the same in both cases. The two information needs can only be satisfactorily distinguished via aspects of style.

Manner of Topic Treatment: Author—Medium—Perspective—Genre

“Manner of topic treatment” summarizes all aspects that provide information about the *How* of a document with regard to content. This includes the following relations:

- Characteristics of the author,
- Characteristics of the medium,
- Perspective of the document,
- Genre.

Author-specific relations provide, very briefly, information about some crucial characteristics of the creator of a document. They include dates of birth and death, nationality and, where available, a biography. The file of personal names of the German National Library provides us with an example:

Tucholsky, Kurt (1890 – 1935)

German writer, poet and journalist (literary and theatre critic).

Emigrated to France in 1924 and to Sweden in 1929.

If we have to interpret an article on, say, Jerusalem, it will be very useful to know whether the author is from Israel or from Palestine or whether he is member of a political party. Relations concerning the medium are its circulation (number of issues and number of readers), its spatial distribution, statements concerning its editors and publisher. For academic journals, it might make sense to include an additional parameter of scientific influence (e.g. the Impact Factor). Where statements about them are available, the manner of article selection (e.g. only after a complete Peer Review) and the rejection rate of submitted contributions are of great significance, particularly for scientific journals (Schlögl & Petschnig, 2005). A medium that works with blind peer review (i.e. anonymous inspection), has a rejection rate of over 90% and also a circulation of more than 10,000 issues, is to be rated completely differently in terms of significance than a medium in which articles are selected by organs of the journal itself (without any assessment), a rejection rate of less than 10% and a circulation of 500.

The perspective of a document states from what vantage point it was created. Depending on the knowledge domain, different perspectives are possible. An information service on chemical literature, for instance, can differentiate according to discipline (*written from the perspective of medicine, of physics* etc.). The online service “Technology and Management” (TEMA) (Stock & Stock, 2004), which is one of the few databases to use such a relation in the first place, works with the following values:

- A Application-Specific Paper
- E Experimental Paper
- G Foundational Paper
- H Historical Paper
- M Aspects of Management

N Proof of Product
 T Theoretical Treatise
 U Overview
 W Economic Treatise
 Z Future Trend

Let a user, e.g. an engineer, search application-specific articles on labeling technology. He formulates his query thus:

Labeling Technology AND Perspective:A,

While his colleague from sales concentrates on the corresponding products:

Labeling Technology AND Perspective:N.

In information science literature, a lot of weight is placed on the use of genre statements as information filters. This relation is akin to the field “formal bibliographic document type”, but it is designed with far more differentiation and always deals with information content, form as well as purpose. Orlikowski and Yates (1994, 543) describe “genre” as

a distinctive type of communicative action, characterized by a socially recognized communicative purpose and common aspects of form.

As with perspective, the values for the genre attribute are domain-specific. Genre and subgenre exist for every kind of formal published and unpublished text, for the sciences (e.g. *lecture* or *review article*) as well as for company practice (e.g. *White Papers*, *Best Practices*, *problem reports*), for everyday practical areas (*cookbook*, *marriage counseling book*) as well as for private texts (*diary entry*, *love letter*). Documents in the World Wide Web also belong to different genres (e.g. *personal website*, *company press report*, *blog post*, *wiki entry*, *FAQ list*). Of course there are also different genres for pictures, movies, pieces of music, works of fine art etc.

However, analogously to perspective, very few databases include a corresponding field. Particularly for large databases, such fields are urgently needed in order to heighten the precision of search results (Crowston & Kwasnik, 2003). Beghtol (2001, 17) points out the significance of genre and subgenre for all document types:

The concept of genre has ... been extended beyond language-based texts, so that we customarily speak of genres in relation to art, music, dance, and other non-verbal methods of human communication. For example, in art we are familiar with the genres of painting, drawing, sculpture and engraving. In addition, sub-genres have developed. For painting, sub-genres might include landscape, portraiture, still life and non-representational works. Some of the recognized sub-genres of fiction include novels, short stories and novellas. Presumably, any number of sub-levels can exist for any one genre, and new sub-genres may be invented at any time.

Let us introduce the online catalog of the Harvard University Libraries as a model example of a database with a genre field (Beall, 1999). In the HOLLIS catalog, the user chooses the desired genre value from a set list of keywords (Figure J.3.1). We selected *Horror television programs—Denmark*. The surrogate is shown in Figure J.3.2. The document described is the DVD of a miniseries by Lars von Trier that ran on Danish television in 1994.

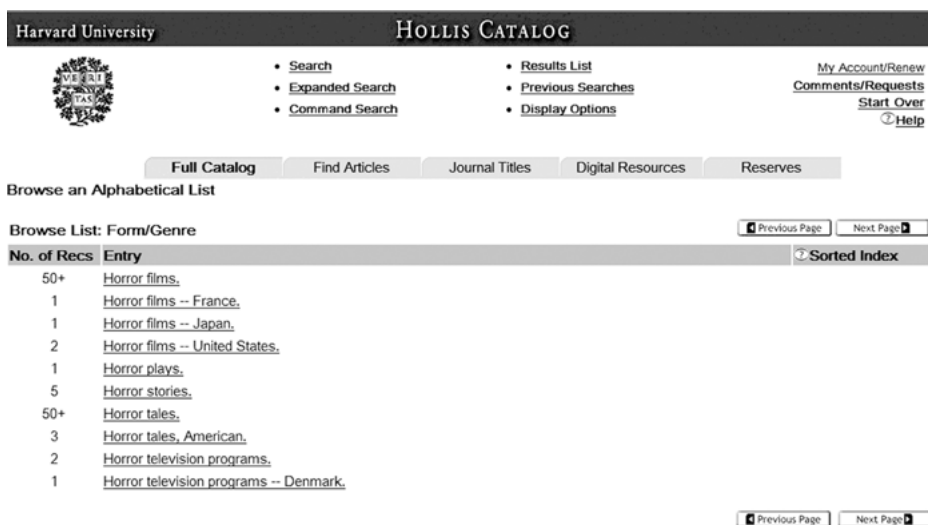


Figure J.3.1: Keyword List for Genres in the Hollis Catalog. Source: Hollis Catalog of the Harvard University Libraries.

Beall (1999, 65) tells of positive experiences representing genre information in a library catalog:

The addition of this data in bibliographic records allows library users to more easily access some materials described in the catalog, since they can execute searches that simultaneously search form/genre terms and other traditional access points (such as author, title, subject). Such searches allow for narrow and precise retrievals that match specific information requests.

Values for perspective and genre require the use of a controlled vocabulary in a knowledge organization system that must be created ad hoc for this purpose. Specific KOSs for genres can be created, for example, for the Web (Kwasnik, Crowston, Nilan, & Roussinov, 2001; Toms, 2001), for music (Abrahamsen, 2003), or for in-house company documents (Freund, Clark, & Toms, 2006). All sorts of knowledge organization systems are suitable methods, i.e. nomenclatures (as in the Hollis Catalog), thesauri or classification systems (Crowston & Kwasnik, 2004).

Title : Riget [videorecording] / Zentropa Entertainment 1 ; written by Niels Vørsel & Lars Von Trier ; producer, Ole Reim ... [et al.] ; screenplay by Tómas Gislason & Lars Von Trier ; directed by Lars Von Trier.

Published : Port Washington, NY : Koch Lorber Films, [2005]

Locations/Orders : [Availability](#)

Location : Lamont **1** Morse Music & Media DVD PN1995.9.S8 R56 1997x [Holdings](#) [Availability](#)

Music Pub. No. : KLF-DV-3044 Koch Lorber Films

Description : 2 videodiscs (272 min.) : sd., col., ; 4 3/4 in.

Participants : Ernst-Hugo Järegård, Kirsten Rolffes, Holger Juul Hansen, Ghita Nørby, Udo Kier.

Credits : Cinematography, Eric Kress ; music, Joachim Holbeck.

Summary : At The Kingdom, **Denmark's** most technologically advanced hospital, a number of strange and otherworldly events begin to occur, much to the dismay of its doctors and patients. At the urging of a spiritualist patient, the employees attempt to help the supernatural forces rest.

Technical Info : DVD; Dolby digital.

Language note : In Danish with English subtitles.

Notes : Videodisc release of the **television** mini-series Riget originally broadcast in 1994.
Bonus features: Behind the scenes footage; Selected commentary by Lars Von Trier; Misc. TV spots directed by Lars Von Trier; Trailer.

ISBN : 1417200235
9781417200238

Subject : [Hospitals -- Denmark -- Drama.](#)
[Supernatural -- Drama.](#)

Form/Genre : [Horror television programs -- Denmark.](#)
[Television programs.](#)
[Television programs -- Denmark.](#)
[Foreign language television programs -- Danish.](#)
[DVD Video-discs.](#)

Authors : [Trier, Lars von, 1956-](#)

Figure J.3.2: Entry of the Hollis Catalog. Source: Hollis Catalog of the Harvard University Libraries.

Target Group

We understand “target group” to mean the addressees of a document, i.e. all those for whom the work has been created. We wish to distinguish between two methods, each of which allow for the incorporation of target-group information in knowledge representation. The first method works with a specific relation for target groups, which is recorded in the database as a search field. The second method controls, via passwords, what documents may be read by which target group in the first place.

Public libraries sometimes arrange their stock according to so-called categories of interest, target groups among them, if a certain homogeneity of the group can reasonably be expected (e.g. *for parents, for children of preschool age*). When the categories of interest are represented by a binding KOS, one speaks of a “Reader Interest Classification” (Sapiie, 1995).

We transfer this conception to documents of all kinds. Whenever homogeneous target groups exist for a document, this fact will be displayed as a value in a special field. Thus, books about mathematics might be differentiated according to target groups:

Laymen,
Pupils (Elementary School),
Pupils (Middle School/Junior High School),
Pupils (Senior High School),
Students (Undergraduate),
Students (Graduate), etc.

Multiple allocations are always a possibility (e.g. both for *laymen* and *middle school pupils*). Target Groups can be created for professions (*scientists, lawyers*), cultural, political or social groupings (*Turks in Germany, members of the Social Democratic Party, evangelical Christians*), age groups, topic-specific interests (e.g. for this book: *information science, computer science, information systems research, knowledge management, librarianship, computational linguistics*) and other aspects. A binding KOS must be created for the target groups' values.

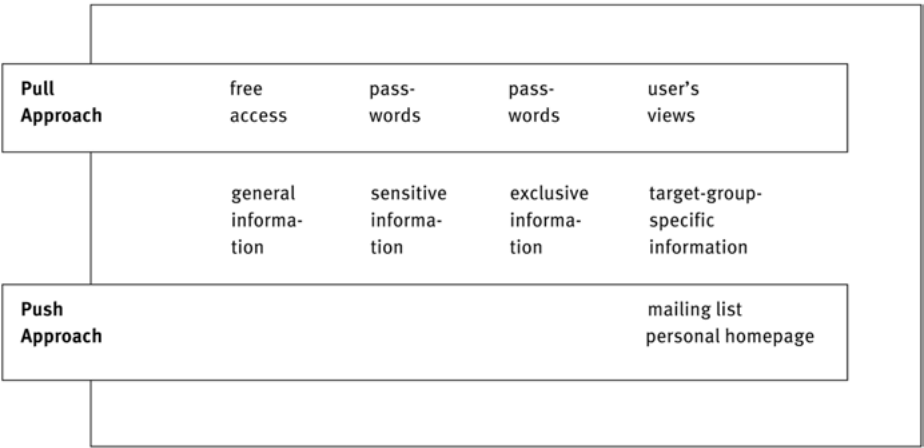


Figure J.3.3: Target-Group-Specific Access to Different Forms of Information.

In the second variant of addressing specific target groups, access to different documents is controlled via passwords as well as (password-protected) personal websites. In the Pull Approach, the user actively “pulls” down information, whereas in the Push Approach he is provided with information by the information system (from the user’s point of view: passively) (see Figure J.3.3). In the Pull model, we steer access to information, e.g. that of an enterprise, via passwords (Schütte, 1997). The class of general information (homepage, business reports and the like) is accessible to all; sensitive information is—password-protected—only available for those customers and suppliers who enjoy a certain level of trust. Exclusive information (business secrets) are, also password-protected, only for a small target group (such as the board of directors) or business units (e.g. the departments). Information needs are dependent on the type of information (e.g. scientific literature for researchers or a list with companies’ key performance indicators for controlling). In the Pull Approach, the information seeker must identify his information need and use it to perform his own research. His personal password makes sure that he will find the information that suits his purposes—and none other, if possible—in the information system.

The Push Approach is particularly suited for the distribution of target-group-specific information on the system side (Harmsen, 1998). This can mean general news

that might interest a certain user group, but also all kinds of early-warning information that must be transmitted to the decision-maker as quickly as possible. Schütte (1997, 111) reports that

Push technologies allow users to passively receive news as e-mails instead of having to actively seek them out in Web or Intranet.

In addition to mailing lists, there is also the elegant option of the person- or target-group-specific information to an employee's personal homepage. In the information service Factiva (Stock & Stock, 2003), which contains news and business facts, the user can store profiles and search queries, call up current facts (such as market prices) or digitally browse favorite sources (newspapers etc.).

Time

Documents have a time reference. They have been created at a certain time and are—at least sometimes—only valid for a certain period. The date of creation can play a significant role under certain circumstances. We will elucidate on the example of patents. Patents, when they are granted, have a 20-year term, starting from the application date. Whereas German patent law emphasizes the “First to File” principle (only the date of application is relevant for the determination of innovation), the United States use “First to Invent”: the time of creation is what is important. This time must thus be carefully documented, since it is the only way to prove the invention; the relevant documents must be time-stamped. Furthermore, US patent law uses the “grace period for patents” (35 U.S.C. 154(d)). The invention only has to be formally submitted to the patent office within twelve months of being invented.

After the legal term of 20 years (or earlier, e.g. due to non-payment of fees), the invention described in the patent is freely accessible to all. Now anyone can use and commercially exploit the ideas. Patent databases take the time aspect into consideration and publish all relevant date statements.

In the business day-to-day, many documents also have a time dimension. Memos and job instructions have a certain duration. As interesting as the cafeteria's lunch menu for today and next week might be, there is no point at all in looking at the one from yesterday or last week (unless one compiles a statistic of the serving of steaks over the course of one year). Such documents, which are only valid for a certain amount of time, are provided with a use-by date. After it has been passed, the documents are either deleted or—to use the safer path—moved to a database archive. After all, employees of the company archive might want to access these documents at some point.

Conclusion

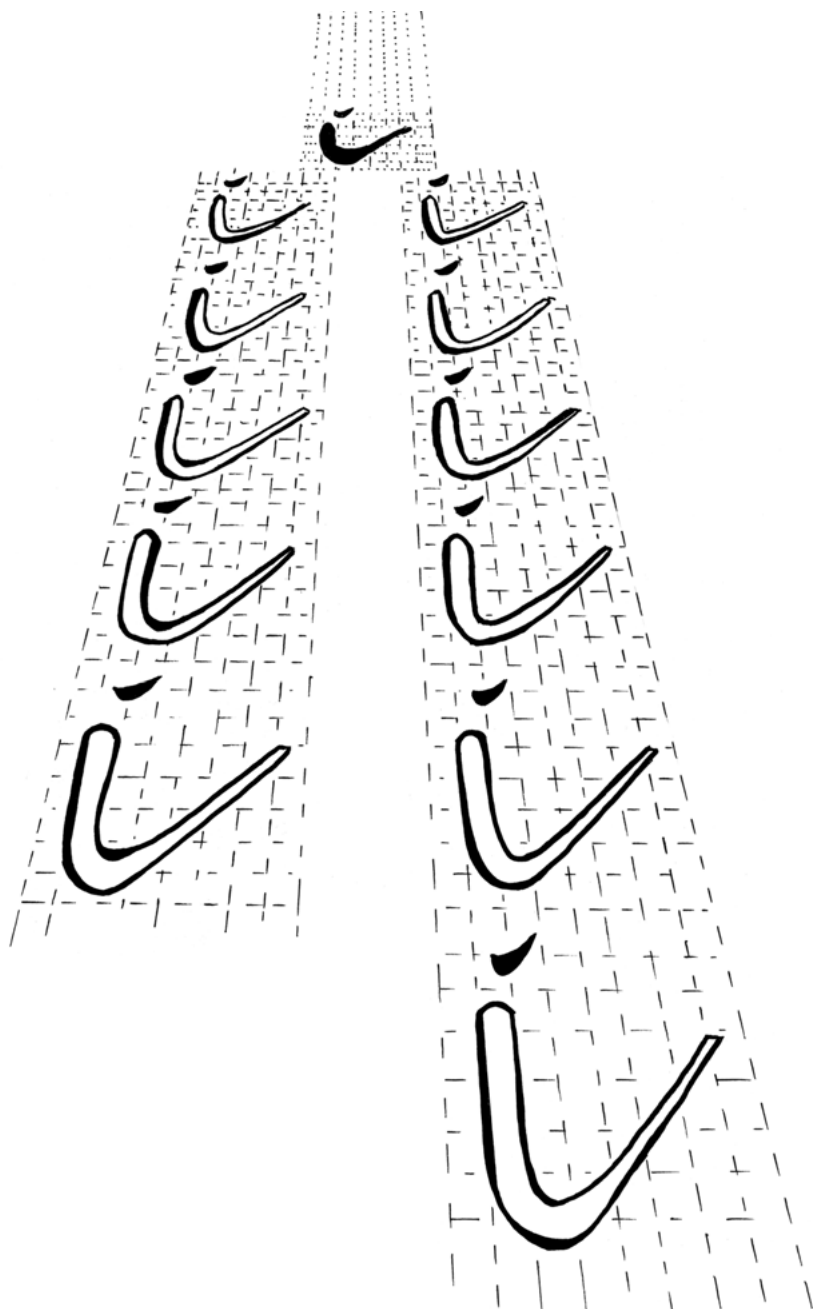
- Documents contain more than content. We summarize the aspects of *How* under the “style” of a document, using its attributes and values as non-topical information filters.
 - Search via style values mainly leads to a heightened precision in search results.
 - Depending on the type of topic treatment, we distinguish between fields for authors’ characteristics (e.g. biographical data), characteristics of the medium (e.g. circulation for print works), statements on the perspective from which a work has been created (e.g. overview) as well as genre values. Where possible, controlled vocabularies (knowledge organization systems) will be used.
 - Target groups are either designated as interested parties in a special field (e.g. laymen) or provided with the “appropriate” documents via targeted password allocation.
 - Documents have a time reference, meaning that both the date of creation and the use-by date must be noted.
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Part K
Folksonomies



K.1 Social Tagging

Content Indexing via Collective Intelligence in Web 2.0

In the early years of the World Wide Web, only a select few experts were capable of using it to distribute information; the majority of users dealt with the WWW as consumers. At the beginning of the 21st century, services began to appear that were very easy to use and which allowed users to publish content themselves. The (passive) user thus became, additionally, a(n active) Web author. The consumer of knowledge has also become its producer; a “prosumer” (Toffler, 1980). The users’ active and passive activities are united in the term “produsage” (Bruns, 2008). As authors (at least occasionally) edit, comment on, revise and “share” their documents reciprocally, in this case it is appropriate to use the term “collective intelligence” (Weiss, 2005, 16):

With content derived primarily by community contribution, popular and influential services like Flickr and Wikipedia represent the emergence of “collective intelligence” as the new driving force behind the evolution of the Internet.

“Collective intelligence” is the result of the collaboration between authors and users in “collaborative services”, which can be summarized under the term “Web 2.0” (O’Reilly, 2005). Such services are dedicated to the writing of “diaries” (weblogs), postings on microblogging services (e.g., Twitter), the compilation of an encyclopedia (e.g. Wikipedia), the arranging of bookmarks for websites (e.g. Delicious) or the sharing of images (Flickr), music (last.fm) or videos (YouTube). The contents of services, in so far as they complement each other, are sometimes summarized in “mash-ups” (e.g. housingmaps.com as a “mash-up” of real estate information derived from Craigslist and maps and satellite images from Google Maps). Co-operation does not end with the provision of content, but, in some Web 2.0 services, also includes that content’s indexing. This is done via social tagging, with the tags forming a folksonomy on the level of the information service (Peters, 2009).

Folksonomy: Knowledge Organization without Rules

Folksonomies are a sort of free allocation of keywords by anyone and everyone. The indexing terms are called “tags”. Indexing via folksonomies is thus referred to as “tagging”. The term “folksonomy”, as a portmanteau of “folk” and “taxonomy”, goes back to a blog entry on the topic of information architecture, in which Smith (2004) quotes Vander Wal:

Last week I asked the AIIA (the “Asilomar Institute for Information Architecture”; A/N) member’s list what they thought about the social classification happening at Furl, Flickr and Del.icio.us.

In each of these systems people classify their pictures/bookmarks/web pages with tags ..., and then the most popular tags float on the top ...

Thomas Vander Wal, in his reply, coined the great name for these informal social categories: *a folksonomy*.

Still, the idea of socially constructed classification schemes (with no input from an information architect) is interesting. Maybe one of these services will manage to build a social thesaurus.

Smith uses the word “classification” to describe folksonomies. This points in the wrong direction, though, as does “taxonomy”. Folksonomies are *not* classifications (Ch. L.2), as they use neither notations nor paradigmatic relations. Far more important is Smith’s suggestion that we can build on folksonomies to co-operatively compile and expand thesauri (as well as other forms of knowledge organization systems). Similar to “normal” tags are Twitter’s hashtags (e.g., #Web).

Trant (2009) defines “tags” and “folksonomy:”

User-generated keywords—tags—have been suggested as a lightweight way of enhancing descriptions of on-line information resources, and improving their access through broader indexing. “Social tagging” refers to the practice of publicly labeling or categorizing resources in a shared, on-line environment. The resulting assemblage of tags form a “folksonomy”.

For Peters (2009, 153), folksonomies “represent a certain functionality of collaborative information services.”

It must be emphasized that folksonomies and other methods of knowledge representation are not mutually exclusive in practice, but actually complement each other (Gruber, 2007). Text-oriented methods (e.g. citation indexing and the text-word method) concentrate on the author’s language; KOSs represent the content of a document in an artificial specialist language and require professional, heavily rule-bound indexing by experts or automated systems. Folksonomies bring into play the language of the user, which had hitherto been completely disregarded. Kipp (2009) found out that users use both information services with folksonomies and databases with controlled vocabularies.

Within a folksonomy, we are confronted with three different aspects (Hotho et al., 2006; Marlow et al., 2006):

- the tags (words) used to describe the document,
- the documents to be described,
- the users (prosumers) who perform such indexing tasks.

When we look at tags on the level of the information service as a whole (e.g. all tags on YouTube), we are talking about a folksonomy. If, on the other hand, we restrict our focus to the tags of exactly one document (e.g. a specific URL on Delicious), we are dealing with a docsonomy. The totality of all tags allocated by a user is called a personomy.

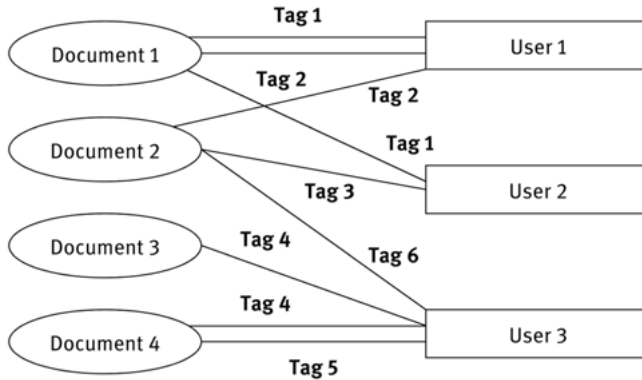


Figure K.1.1: Documents, Tags and Users in a Folksonomy.

Users and documents are connected in a social network, with the paths running along the tags in both cases. Documents are, firstly, linked with one another *thematically* if they have been indexed via the same tags. Documents 1 and 2 as well as 3 and 4 in Figure K.1.1 are thematically linked, respectively (documents 1 and 2 via tag 2; documents 3 and 4 via tag 4). Additionally, documents are also coupled *via shared users*. Thus documents 1 and 2, 3 and 4, but also 2 and 4 are connected via their users.

Users are connected if they either use the same tags or index the same documents. Users are *thematically* connected if they use the same tags (in the example: users 1 and 2 via tags 1 and 2); they are coupled *via shared documents* if they each describe the content of one or more documents (users 1, 2 and 3 via document 2). The strength of their similarity can be expressed quantitatively via similarity measurements such as Cosine, Jaccard-Sneath or Dice.

Documents are generally indexed via several tags and—depending on the kind of folksonomy used—with differing degrees of frequency. Document 1, for instance, has a total of two different tags, of which one (tag 1) has been allocated twice. Document-specific tag distributions are derived from the respective frequencies with which tags are allocated to the document in question. Analogously, it is possible to determine user-specific tag distributions.

Of course the tags are also linked among each other. When two tags co-occur in a single document, they are regarded as interlinked. On this basis we can compile tag clusters. Cattuto et al. (2007) are able to show that the resulting networks of a folksonomy represent “small worlds”, i.e. the “short cuts” between otherwise far-apart tags (and, analogously, documents and users) create predominantly short path lengths between the tags. Cattuto et al. (2007, 260)

observed that the tripartite hypergraphs of their folksonomies are highly connected and that the relative path lengths are relatively low, facilitating thus the “serendipitous discovery” of interesting contents and users.

Folksonomies are thus useful as aids for targeted searches as well as for browsing through information services.

Characteristics of Folksonomies

Following Peters (2009, 104), there are three kinds of folksonomies: systems that allow for the multiple allocation of tags (broad folksonomies), systems in which a specific tag may only be allocated once—no matter by whom—(extended narrow folksonomies), and systems in which only the document’s creator may use tags, also only once (narrow folksonomies). The distinction between broad and narrow folksonomies goes back to Vander Wal (2005). Broad folksonomies include the social bookmarking services Delicious or CiteULike; Flickr is an example of an extended narrow folksonomy; and YouTube uses a narrow folksonomy.

In a broad folksonomy, several users index a document. When a narrow folksonomy is used, both the content’s creator and (in extended narrow folksonomies) its users have the option of allocating a tag, but only once.

Consider the tag distribution for the bookmarks of two websites in a broad folksonomy (Fig. K.1.2). The upper docsonomy is dominated by a single tag. One or two further tags are still allocated relatively frequently, whereas all other used keywords appear in the little-used long tail of the distribution. In the literature, it is assumed (and justly so) (e.g. Shirky, 2005) that many documents have a docsonomy whose words, when ranked by frequency, follow a Power Law distribution.

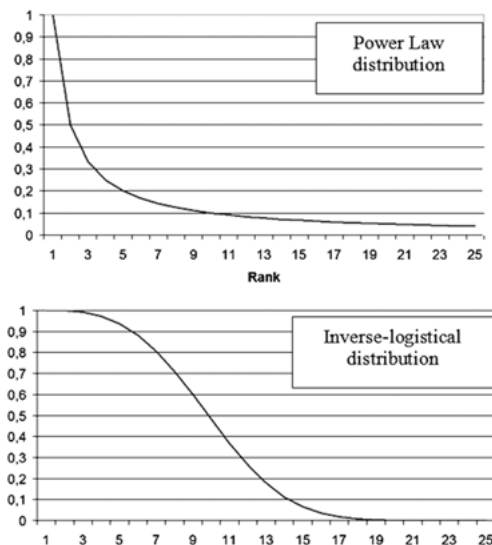


Figure K.1.2: Ideal-Typical Tag Distributions in Docsonomies.

The second example in Figure K.1.2 starts out completely differently. On the left side of the distribution, this docsonomy does not follow a Power Law distribution—rather, several tags occur at an almost identical frequency. Analogously to the “long tail”, here we can speak of a “long trunk”. On the right-hand side, the same “long tail” appears as in the Power Law curve.

The composition of distributions over the course of time follows the principle that “success breeds success” (Peters, 2009, 205 et seq.). At the beginning of a document’s tagging history, either 1st precisely one tag emerges or 2nd several (generally very few) tags are needed to describe the resource. These “successful” tags then become ever more dominant, leading, in the first scenario, to a Power Law distribution and, in the second, to an inverse-logistic distribution. From a certain number of tags upward, the relative frequency of all tags in a docsonomy consolidates itself (Terliesner & Peters, 2011; Robu, Halpin, & Shepherd, 2009).

Personomies feature person-related information regarding the tags and their indexed documents. These can be exploited in personalized recommender systems, as Shepitsen et al. (2008, 259) emphasize:

From the system point of view, perhaps the greatest advantage offered by collaborative tagging applications is the richness of the user profiles. As users annotate resources, the system is able to monitor both their interest in resources as well as their vocabulary for describing those resources. These profiles are a powerful tool for recommendation algorithms.

Folksonomy-based recommender systems (Peters, 2009, 299 et seq.) make different recommendations to their users:

- recommendations for documents,
- for users, and
- for tags (during tagging as well as retrieval).

Recommendations for documents and for other users are compiled on the basis of documents bookmarked or tagged by the current user as well as by all other users. This is a variant of collaborative filtering (Ch. G.5). We can distinguish between two paths when recommending tags during indexing (Jäschke et al., 2007). On the one hand, the recommender system orients itself on the folksonomy and suggests tags that frequently co-occur with already-entered tags. Here it is possible to exclude tags that occur with a particular frequency due to their lack of discrimination (Sigurbjörnsson & van Zwol, 2008, 331). On the other hand, the personomy is chosen as the point of reference and tags are recommended that the current user has already used (Weller, 2010, 333-336). Systems such as Weller’s tagCare make sure that the user’s vocabulary remains stable. Additionally, information services can offer recommendations for correcting entered tags.



Figure K.1.3: “Dorothy’s Ruby Slippers”. Photo by AlbinoFlea (Steve Fernie) (2006). Source: Flickr.com.

Isness, Ofness, Aboutness, and Iconology

Some Web 2.0 services store non-textual content, such as music, images and videos. In these cases, we are confronted with Panofsky’s three semantic levels (ofness, aboutness, and iconology). In addition to this, many users also tag non-topical information, e.g. the name of the museum where a photo was taken or the name of the photographer. Occasionally, authors tag technical aspects of the photograph (e.g. camera type, length of exposure, aperture). Such aspects can be called—following Ingwersen (2002, 293)—“isness” (“*is in museum X*”, “*is by Y*”, “*is taken with a Lumix*”).

Consider an example we found on Flickr (Fig. K.1.3). This is a photograph of Dorothy’s Ruby Slippers, i.e. the shoes worn by Judy Garland in “*The Wizard of Oz*”, which are exhibited in the Smithsonian National Museum of American History in Washington, DC. The photographer has labelled his image with the following tags:

<i>Tag</i>	<i>Level</i>
ruby	Ofness
slippers	Ofness
sequins	Ofness
Dorothy	Aboutness
Wizard of Oz	Aboutness
Smithsonian	Isness
Washington, DC	Isness
Museum of American History	Isness
There’s No Place Like Home	Iconology.

Panofsky’s pre-iconographical level, the world of primary objects, is described via ofness. The photographer has labelled his document with three ofness tags (*ruby*, *slippers*, *sequins*). Aboutness expresses the interpretation on the iconographical level. In the example, *Dorothy* and *Wizard of Oz* are such aboutness tags. With the tag *There’s No Place Like Home*, we are already on the iconological level. Additionally, the image

has been tagged with the name of the museum and its location. In Ingwersen's terminology, the tags addressed here display aspects of user-ness, user-aboutness, user-icology and user-iness, all jumbled up indistinguishably in a single word cloud.

Advantages and Disadvantages of Folksonomies

There are large quantities of documents on the Web, and without the use of folksonomies they would likely be impossible to index intellectually. The procedure is cheap (Wu, Zubair, & Maly, 2006), since all employees work pro bono. Tagging represents user-specific interpretations of documents, thus mirroring the users' authentic language. It allows documents to be interpreted from various different points of view (scientific, ideological or cultural) (Peterson, 2006). Neologisms are recognized almost "in real time". Mapping the users' language use provides the option of building or modifying knowledge organization systems in a user-friendly fashion. For Shirky (2005), tagging even represents a kind of quality control: the more people tag a document, the more important it appears to be. Because of the idiosyncratic and unexpected words in the "long tail", folksonomies are suited not only for searching documents, but also for browsing and exploiting serendipity, i.e. the lucky finds of relevant documents (Mathes, 2004). Users can be observed via their tagging behavior; this provides us with options for studying social networks, in so far as these use the same tags or index the same documents. The sources for recommender systems are found in the same context. The wealth of person-related information in the personomies results in an ideal basis for the construction of recommender systems (Wu, Zubair, & Maly, 2006). The fact that a broad mass of users are here for the very first time confronted with an aspect of knowledge representation appears to be of particular importance. This may lead to these users being sensitized to the aspect of indexing.

The advantages are offset by a not inconsiderable amount of disadvantages. The chief problem is the glaring lack of precision. Shirky (2004) notes:

Lack of precision is a problem, though a function of user behavior, not the tags themselves.

When using folksonomies, we find different word forms—singular nouns (*library*) and plural nouns (*libraries*), as well as abbreviations (*IA* or *IT*). Because several systems only allow one-word tags, users squash phrases into a single word (*information-science*) or link the words via underscores (*information_science*). There is no control of synonyms and homonyms. Typos are frequent (Guy & Tonkin, 2006). Since the users have different tasks, approach the documents with different motives, and are located in different cognitive contexts, they do not share a common indexing level (Golder & Huberman, 2006). Many Web 2.0 services work internationally, i.e. multilingually. Users may index documents in their own language (*London, Londres, Londra*) without

bothering to translate. Homonyms that span different languages (*Gift* in German as well as English) are not separated. In the case of non-textual documents, the various semantic levels (ofness, aboutness, iconology) are all mixed up. Users do not always distinguish between content indexing (which is what they “should” be doing) and formal description (*article*, *book*, *ebook*) or other aspects of isness. Sometimes tags contain value judgments (*stupid*), or prosumers describe a planned activity (*to_read*). Of little to no usefulness are syncategorematical tags such as the designation *me* for a photograph on Flickr. Sometimes spam tags can be found, i.e. tags that have nothing to do with the document’s content and thus knowingly mislead the users. Because of the disadvantages discussed above, the *exclusive* use of folksonomies in professional environments (e.g. in corporate knowledge management) can hardly be recommended (Peters, 2006). However, if folksonomies are combined with other methods of knowledge representation, and the users’ perspective as well as the tag distributions are taken into account, the advantages definitely outweigh the flaws. Professional databases (Stock, 2007) and online library catalogs (Spiteri, 2006) can also be significantly enriched via folksonomies.

Conclusion

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- In the so-called “Web 2.0”, prosumers (users and producers at the same time) create content in collaboration. Many Web 2.0 services use collaborative intellectual indexing carried out by the prosumers. The folksonomies used employ free keywords (tags), with no rules curtailing the freedom of indexing.
 - Documents, tags and users can be represented as nodes in a social network. Certain similarities between specific nodes can be derived from their placement in the graph (thematic proximity between documents and users, connections between documents via shared indexers or tags, connections between users via co-indexed documents or, again, via shared tags).
 - During indexing and search, users can be recommended further tags on the basis of tags they have already entered. Such recommended tags are gleaned either from the folksonomy as a whole or from the respective user’s personomy. Suggestions for correcting already entered tags are also possible.
 - Broad folksonomies allow the multiple allocation of tags, whereas in narrow folksonomies each tag is only allocated once.
 - There are two different ideal-typical tag distributions for a document. The Power Law distribution is extremely skewed to the left and has a long tail. The inverse-logistical distribution has both a long trunk and a long tail.
 - Indexing non-textual content (such as images or videos) via folksonomies leads to the mixing-together of ofness, aboutness and iconology, as well as of formal aspects (isness).
 - Advantages of folksonomies include their cheap nature, which facilitates the indexing of huge databases on the Web that otherwise could hardly be done intellectually, the application of users’ authentic language use (with the possibility of building or refining knowledge organization systems on this basis), their basis for recommender systems, as well as their search functions and (in particular) their browsing functions.

- Disadvantages of folksonomies are mainly the result of their lack of precision. There is no terminological control, and this leads to typos, value judgments, syncategoremata and descriptions of planned actions. Due to these disadvantages, folksonomies cannot effectively be used as the sole means of indexing in professional environments.
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K.2 Tag Gardening

Tag Literacy, Incentives, and Automatic Tag Processing

Folksonomies come with various (generally speaking: practical) problems. Tags used in sharing services (such as Flickr, YouTube, or Last.fm), in social bookmarking services (such as Delicious or CiteULike), or as hash-tags in microblogging systems (such as Twitter) are notoriously “unclean”.

Users who index are informational laymen and have thus not been taught to use the tools of knowledge representation correctly. While it is possible to “educate” users and raise their “tag literacy” (Guy & Tonkin, 2006), it remains likely that (at least some) disadvantages of tagging will remain. Siebenlist and Knautz (2012, 391 et seq.) emphasize the usefulness of an incentive system that motivates users to tag (well). Some workable implements include experience points, the establishment of different levels (“experienced tagger”, “pro tagger”), achievements, and leader boards (rankings of the most active taggers).

One particularly promising method is to edit the tags allocated by prosumers, thus optimizing them for the purpose of retrieval. Following Governor (2006), Peters and Weller (2008) describe this approach (metaphorically) as “tag gardening”:

(The image of tag gardening) is used to describe processes of manipulating and re-engineering folksonomy tags in order to make them more productive and effective ... To discuss the different gardening activities, we first have to imagine a document-collection indexed with a folksonomy. This folksonomy now becomes our *garden*, each tag being a different *plant*. Currently, most folksonomy-gardens are rather savaged: different types of plants all grow wildly.

Tag gardening is closely related to both knowledge representation (Peters, 2009, 235-247), specifically to the “semantic upgrades” of KOSs (Weller, 2010, 317-351), and to information retrieval (Peters, 2009, 372-388). In the following, we will study five strategies for processing tags:

- Weeding: Basic formatting,
- Seeding: Tag recommendations,
- Garden Design: Vocabulary control,
- Fertilizing: Interactions with other KOSs,
- Harvesting: Delimitation of Power Tags.

Basic Formatting (“Weeding”)

The objective in weeding is mainly to either remove “bad” tags (e.g. spam tags) or to process these in a way that improves their usefulness.

Important tasks of basic formatting include error recognition (and subsequent automatic correction) as well as the conflation of different word forms (e.g. singular and plural) into single variants. These are standard tasks of information linguistics (Ch. C.2). To identify personal names (before conflation), either a name file must be available or elaborate algorithms must be processed (Ch. C.3).

The objective is to filter out context-specific words (such as “me”) (Kipp, 2006). For the user himself (in his personomy, e.g. in photosharing services), such terms will remain searchable, but for all other users such syncategoremata, once recognized, will be replaced by their respective equivalents. “Me”, for instance, will be replaced with the document’s author’s user name.

As several Web services at present only allow using one-word terms for indexing, it becomes crucially important to deal with compounds, with their decomposition, and with phrase identification. Compounds can occur in different forms and should be conflated. In the tags *knowledgerepresentation* and *knowledge_representation*, for example, the term “knowledge representation” should be recognized. Further tasks include recognizing meaningful compound components (i.e. “knowledge” and “representation” in the tag *knowledgerepresentation*) and—in the opposite direction—forming meaningful phrases out of separate tags (e.g. deriving the phrase “knowledge representation” from the individual tags “knowledge” and “representation”). Phrase formation is also important for the recognition of personal names, as individual name components are occasionally indexed as separate tags.

Spam tags contain misinformation about a document, leading users astray. Such tags must be removed from docsonomies. In the context of their gardening metaphor, Peters and Weller (2008) describe algorithms for automatic spam removal as pesticides.

Tag Recommendations (“Seeding”)

Seeding is performed by recommending tags during the indexing process. In broad folksonomies, and when a docsonomy is available, the document’s tagger can be offered previously allocated tags for further indexing. We can distinguish between two (not mutually exclusive) variants. On the one hand, the system recommends the most frequent tags in the docsonomy, thus enhancing the effect of “success breeds success”. On the other hand, the rarer tags may lead to new perspectives and interesting suggestions. Peters and Weller (2008) discuss such “little seedlings”:

(I)t might be necessary to explicitly *seed* new, more specific tags into the tag garden. ... An inverse tag cloud (showing rarely used tags in bigger font size) can be used to display some very rarely used tags and provide an additional access point to the document collection.

Where no tags are available for a document (this is always the case when a document is first uploaded to an information service), we are confronted with a cold-start problem with regard to tag recommendation. To combat this problem, Siebenlist and Knautz (2012) discuss using methods of content-based retrieval (Ch. E.4), i.e. information that can be automatically derived from the document. For textual documents, this would mean terms gleaned via automatic indexing. For multimedia documents, the procedure is far more complex, requiring the recognition and naming of objects on the basis of pre-existing low-level features (e.g. face recognition via typical distributions of color, texture and shape).

Recommending tags on the level of the folksonomy as a whole appears to be rather implausible. Due to their low degree of discrimination tags are unsuitable for indexing a specific document, particularly when the most frequent tags are recommended. Jäschke et al. (2007, 511) report of their empirical results:

The more tags of the recommendation are regarded, the better the recall and the worse the precision will be. ... (U)sing the most popular tags as recommendation gives very poor results in both precision and recall.

Vocabulary Control (“Garden Design”)

Garden design regards the vocabulary’s terminological control. There are two challenges: to conflate synonyms (e.g. *bicycle* and *bike*) and to split up homonyms (*java* into *Java* <Island>, *Java* <Coffee> and *Java* <Programming Language>). Peters and Weller (2008) discuss two options for solving these problems. One possibility is to draw on linguistic thesauri (such as WordNet) as well as on other KOSs. For every tag, search is performed to see whether there is a synonym available. Synonyms thus recognized are incorporated into the folksonomy. The procedure is more difficult for homonyms. Even once a homonym has been identified, it is not yet clear which of the different concepts matches the document in question and which does not. Here it is necessary to additionally analyze the other terms of the docsonomy. If the co-occurrences form unambiguous and distinct term clusters, the appropriate meaning can be derived (with some insecurity left). The second option foregoes the use of KOSs and works exclusively with term co-occurrences in the docsonomies. Methods of cluster formation are used in hopes of overcoming tag ambiguity. Such a procedure is used by Flickr, for instance, to arrange documents into thematic clusters. Figure K.2.1 shows the result of clustering for images with the (ambiguous) tag *Java*. Here the separation into the two homonyms *Java* <Island> and *Java* <Coffee> has been a success.



Figure K.2.1: Tag Clusters Concerning *Java* on Flickr. Source: Flickr.com.

Interactions with Other KOSs (“Fertilizing”)

For Peters and Weller (2008), the semantic relations in KOSs serve as fertilizers for the tag garden. If a KOS entry corresponding to a tag is found via vocabulary control, this entry’s relations can then be used to recommend concepts to the users for indexing or, during retrieval, for further search arguments. Typical semantic relations in KOS (Ch. I.4) are equivalence, hierarchy and the associative relation. During tagging, all synonyms, hyperonyms and hyponyms, as well as associative concepts, are recommended as tag candidates. Only those concepts that occur in the folksonomy can be recommended to the searcher. Suppose, for instance, that someone searches for images with the tag *Milwaukee* in a filesharing service. Let this tag be deposited as a descriptor in a geographical KOS. A hyperonym of *Milwaukee* is *Wisconsin*, a meronym is *Historic Third Ward*. The system checks whether the terms *Wisconsin* and *Historic Third Ward* occur in the folksonomy (perhaps in a spelling variant, such as *WI* or *Historic_Third_Ward*). If so, the system will suggest both terms as recommendations for query expansion.

Weller (2010, 342 et seq.) encourages the use of further knowledge sources besides KOSs in order to find fertilizers for the tag garden. By way of example, she names Encyclopedia of Life, GeoNames, the Internet Movie Database and Wikipedia. Indeed, the terms *Wisconsin* and *Historic Third Ward* could have been found in the Wikipedia article for *Milwaukee*. However, this requires elaborate procedures of knowledge mining.

Delimitation of Power Tags (“Harvesting”)

Where we used KOS to enrich folksonomies in garden design and fertilizing, during harvesting we take the opposite path and use the tags to create concepts that can be used both during retrieval and in the maintenance of the KOS (Peters, 2011). On the path from the social to the semantic Web (Stock, Peters, & Weller, 2010, 148 et seq.), the so-called “Power Tags” (Peters & Stock, 2010) play a centrally important role.

To put it simply, in a broad folksonomy the Power Tags of a docsonomy are those tags that have been most frequently allocated after the tag distribution has reached a relative degree of stability. Since a tag can only be allocated once in narrow and extended narrow folksonomies, the Power Tags must here be elicited via the search tags (Peters & Stock, 2010, 86):

Every time, the user access a resource via the results list, we consider this search “successful” for this resource. ... The system stores the information with which query terms *A*, *B*, or *C* a user successfully retrieves and accesses the resource *X*. As a result, query terms are able to form a distribution of terms or tags as well.

In the docsonomies, tags (in a broad folksonomy) and search tags (in all kinds of folksonomies) are either distributed according to a Power Law (Figure K.2.2) or they form an inverse-logistic distribution (Figure K.2.3). The basic idea of Power Tags is to “harvest” the tags on the left-hand side of the distribution as particularly important document tags. When there is a Power Law distribution, relatively few Power Tags are harvested (roughly between 2 and 4). In an inverse-logistic distribution, the inflection point of the curve is regarded as the threshold value, so that in these instances more Power Tags are generally marked.

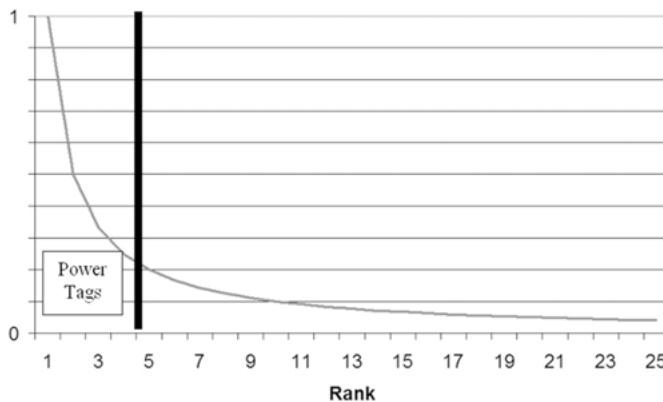


Figure K.2.2: Power Tags in a Power Law Tag Distribution. Source: Peters & Stock, 2010, 83.

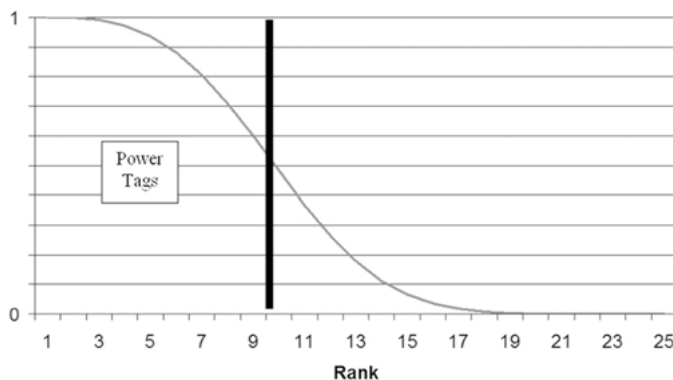


Figure K.2.3: Power Tags in an Inverse-Logistic Tag Distribution. Source: Peters & Stock, 2010, 83.

Power Tags in information retrieval play a particularly important role in such environments where documents cannot be yielded according to relevance (e.g. in some online public access catalogs of libraries). The Power Tags are administered in their own inverted file and offered to the user as a search option (Peters & Stock, 2010, 90). The user now has the option of performing targeted searches for documents that contain the search topic in a centrally important thematic role. In such systems, Power Tags thus raise the precision of the search results.

A secondary application of Power Tags aims toward the construction and maintenance of KOSs (Peters, 2011). All Power Tags on the level of docsonomies (called Power Tags I) are candidates for concepts in a KOS, i.e. descriptor candidates in a thesaurus (Ch. L.3). In KOSs, concepts' semantic relations are of fundamental importance. Candidates for further concepts that are linked to a Power Tag I are gleaned via the co-occurrences of all terms that co-occur with a Power Tag I in a document. These co-occurring tags again follow the typical tag distributions. And here, too, Power Tags (now called Power Tags II) can be partitioned. The Power Tags II are candidates for concepts that are linked to the initial tag via a semantic relation. Up to a certain point, the procedure can be performed automatically. To determine the specific type of semantic relation, however, intellectual human work is required.

We will demonstrate the procedure on an example (Stock, Peters, & Weller, 2010, 148 et seq.). In BibSonomy, the tag *Web2.0* is a Power Tag for various documents (i.e. a Power Tag I), and is thus a suitable descriptor candidate for a thesaurus. The distribution of tags co-occurring with *Web2.0* approximately follows an inverse-logistic form and yields a total of ten Power Tags II (Figure K.2.4). The inflection point of the curve lies between the tags *online* and *internet*.

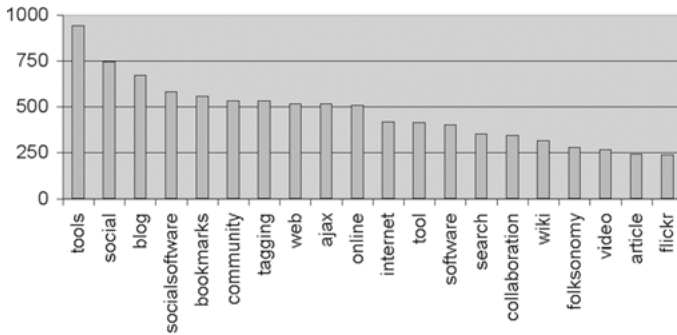


Figure K.2.4: Tag Co-Occurrences With Tag *web2.0* in BibSonomy. Source: Stock, Peters, & Weller, 2010, 151.

During intellectual revision, the tags *tools* and *social* are disregarded due to their lack of specificity. For the remaining tags, the result is the following descriptor entry (for the abbreviations, see Ch. L.3):

Web 2.0

UF	Social software	Equivalence
BT	Web	Hierarchy
NTP	Blog	Meronymy
NTP	Bookmarks	Meronymy
NTP	Tagging	Meronymy
NTP	Community	Meronymy
NTP	Ajax	Meronymy
RT	Online	Associative Relation.

Conclusion

- The fact that laymen perform indexing tasks results in practical problems when using folksonomies. Even if these users' "tag literacy" is improved, or if they are incentivized to tag "well", there remain difficulties that have a negative effect on later retrieval.
- We follow Peters and Weller in metaphorically referring to all measures relating to the processing of tags in folksonomies as Tag Gardening.
- Basic Formating ("Weeding") allows for the removal of false tags (spam tags) from the docsonomy, the replacement of syncategoremata (such as "me"), and the editing of tags via information-linguistic processes.
- Tag Recommendation ("Seeding") occurs during the indexing process and consists of recommending tags that are already extant within the docsonomy. When a document is first uploaded, there is a cold-start problem. Here, recommendable tags may be derived on the basis of content-based retrieval.

- Vocabulary control is used for “Garden Design” by conflating synonyms and separating homonyms. Here one can draw on a KOS or use cluster-analytical procedures to create meaningful tag groupings.
 - “Fertilizing” means using the semantic relations gleaned from other KOSs (or further knowledge sources, such as Wikipedia) during retrieval within a folksonomy.
 - Due to their position in the indexing and search tags, particularly frequent tags in docsonomies can be partitioned as Power Tags, i.e. “harvested”. In information services without Relevance Ranking, these Power Tags provide a further retrieval option that enhances the precision of the search results. Furthermore, Power Tags are suitable for semi-automatically creating concept entries in KOSs (e.g. descriptor entries for a thesaurus).
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K.3 Folksonomies and Relevance Ranking

When browsing and performing targeted searches for documents in information services that use folksonomies, we are confronted with the problem of Relevance Ranking (Peters, 2009, 339 et seq.; Peters, 2011). How is it possible to create a meaningful thematic ranking of tagged documents after successfully retrieving them?

Relevance Ranking Criteria of Tagged Documents

When drawing upon collective intelligence as a benchmark of ranking, there is a total of three bundles of criteria (Figure K.3.1) that determines the importance of a document (Peters & Stock, 2007):

- the tags themselves,
- aspects of collaboration,
- actions of individual users.

The Tag ranking subfactor (area 1 in Figure K.3.1) is embedded in the Vector Space Model (Ch. E.2), in which the various different tags of a database span the dimensions and the dimensions' respective values are calculated via $TF \cdot IDF$. The documents (including the queries) are modeled as vectors; the similarity between query and document vectors is calculated, in the traditional manner, via the Cosine (1a). In broad folksonomies, the relative term frequency (TF) depends upon the number of indexers who have allocated the relevant tag to the document, whereas in narrow folksonomies we work with the number of search processes that have successfully used the tag to retrieve the document.

Following Google's PageRank (Ch. F.1), Hotho, Jäschke, Schmitz and Stumme (2006, 417) introduce their folksonomy-adapted PageRank:

The basic notion is that a resource which is tagged with important tags by important users becomes important itself. The same holds, symmetrically, for tags and users, thus we have a tripartite graph in which the vertices are mutually reinforcing each other by spreading their weights.

Documents, users and tags are here linked to one another in an undirected graph. When a user indexes many documents, his node in the graph has a lot of connections; if he has tagged documents that have themselves been indexed by many other users, our user will “inherit” this importance. The same goes for documents and tags. A tag's importance thus rises when “important” users use it or when it occurs in “important” documents (1b).

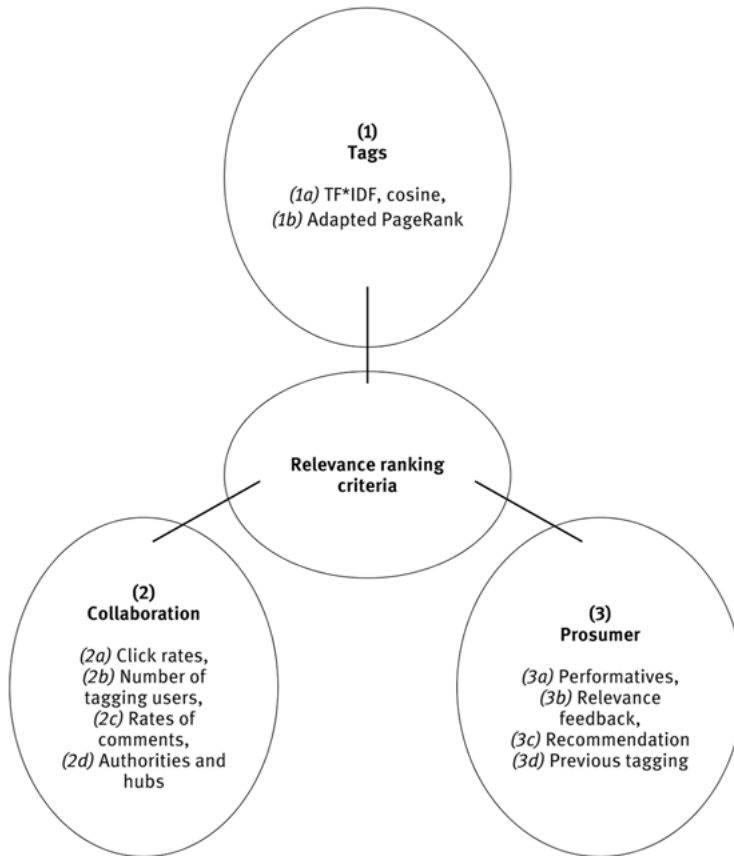


Figure K.3.1: Criteria of Relevance Ranking when Using a Folksonomy. Source: Modified Following Peters & Stock, 2007.

For the ranking subfactor Collaboration (2), as for interestingness (Ch. F.2), possible weighting factors include the number of users that have clicked or viewed a document (2a), the number of users who actively index the document (2b) and the number of comments (2c). Tagged Web pages that are linked (e.g. URLs or weblogs) can be quantitatively rated via link-topological procedures (such as PageRank or hubs and authorities) (2d).

The last ranking subfactor considers user behavior (3). When documents are indexed with performative tags such as *to_read*, the indexer expresses a certain implicit valuation that should influence the ranking (positively) (3a). In a hit list, it should be possible for users to designate certain documents as relevant. Such relevance information is analyzed in the context of Relevance Feedback in order to optimize the query at hand (e.g. following Rocchio in the Vector Space Model; Ch. E.2; or following Robertson and Sparck Jones in the probabilistic model; Ch. E.3). However,

they can also be stored separately and then function as an aspect of Relevance Ranking (3b). Some users (at the very least) are willing and able to provide explicit ratings in the context of a recommender system (Ch. G.5). This is done either via the allocation of stars (from one star: “okay”, to five stars: “excellent document”) or via a yes/no answer to the system question: “was the document useful to you?” (3c).

The individual ranking factors enter the calculation of a document’s retrieval status value with their own respective weighting. As always when Relevance Ranking is offered, the user will appreciate being able to turn this option off and instead use other criteria (e.g. author name or date) to sort the documents.

Personalized Ranking Criteria

If a specific user is known to the retrieval system (i.e. if the user is logged in to the system under a user name), and if this user has already given out tags in this system, his previous tags can be used as an additional search argument under the designation “user domain interests” (Zhou et al., 2008) (3d). Under these conditions, only those documents that match the user’s “interests” would be yielded, or, slightly less restrictively, ranked more highly. A similar personalization is used by Hotho, Jäschke, Schmitz and Stumme’s (2006, 420) “FolkRank”, which draws on “user preferences” as a ranking criterion.

Conclusion

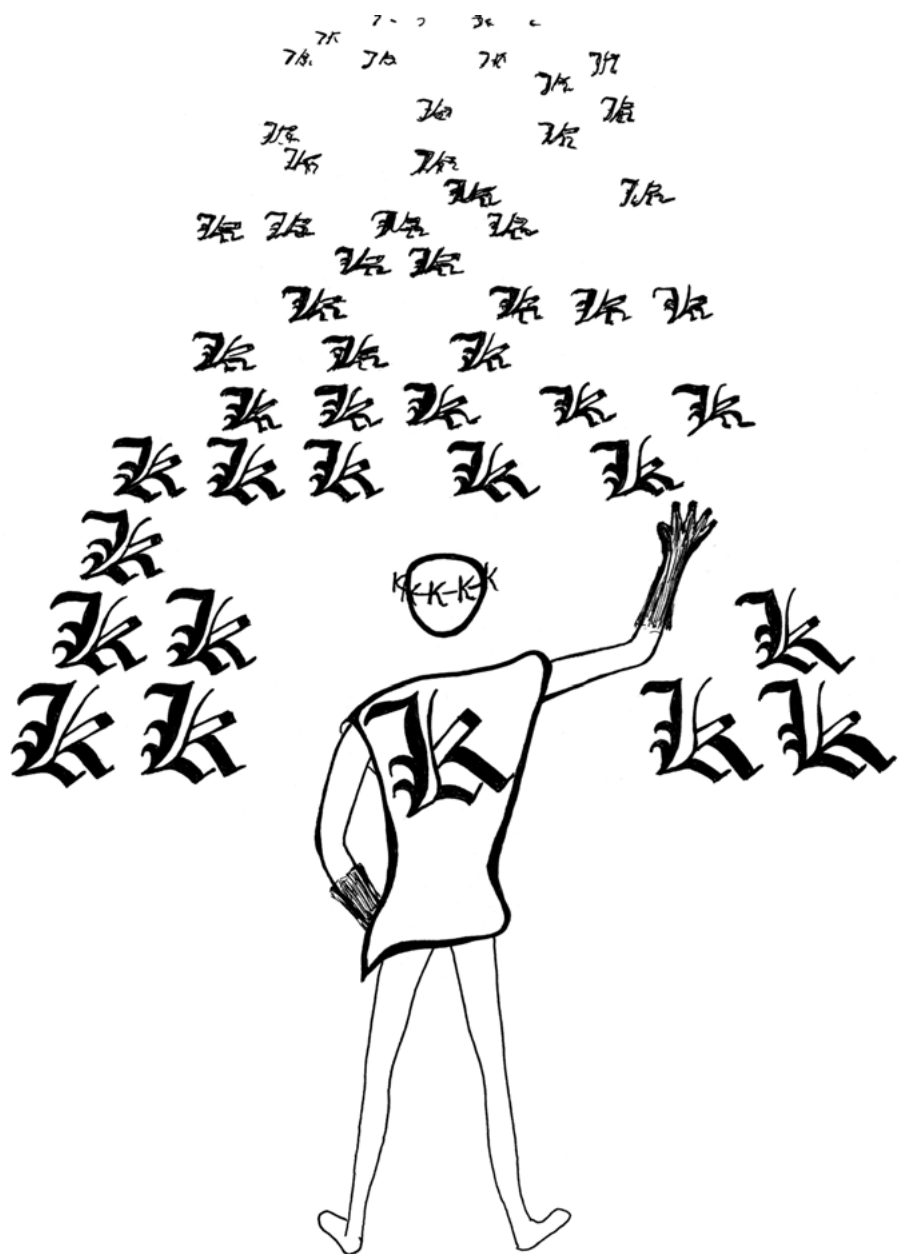
-
- Compared to other methods, folksonomies provide some completely new options for the Relevance Ranking of tagged documents.
 - An elaborate method of Relevance Ranking attempts to take into account all meaningful aspects and ranks tags via their characteristics (TF*IDF and Cosine), via the “importance” of the users that allocate them (or the “importance” of the documents containing them), via the degree of collaboration (similarly to the aspect of interestingness) as well as via the prosumer’s actions (quantitative rating of performative statements, Relevance Feedback, and explicit ratings).
 - For known and identified users, the retrieval system can perform a personalized ranking on the basis of the tags they have used.
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Part L
Knowledge Organization Systems



L.1 Nomenclature

Controlled Vocabulary

As with folksonomies, the usage of nomenclatures, or keyword systems, involves documents being allocated individual terms for indexing the knowledge contained within the documentary reference units. Analogously to folksonomies, most of these terms are derived from either natural or a specialist language; the big difference lies in folksonomies' tags being freely attributable, whereas keywords are fundamentally controlled. The only terms that may be used for indexing are those that are expressly permitted as indexing terms in the respective nomenclature—fixed in a norm form. A single norm entry (the preferred term or authority record) is a keyword; the collection of all keywords as well as all cross-references to keywords is a nomenclature. Nomenclatures distinguish themselves through a well-built synonymy relation; sometimes they also have an associative relation in the form of see-also references. Fundamentally, nomenclatures do not contain any hierarchical relations.

In the sphere of libraries, keyword systems have a long history as forms of verbal content indexing, going back to Cutter's (1904 [1876]) "dictionary catalog" from the year 1876 (Foskett, 1982, 123 et seq.).

Nomenclatures are built either generally or subject-specifically. We introduce the keyword norm file (*Schlagwortnormdatei*, SWD) of German libraries as the paradigm of a general nomenclature (Geißelmann, 1989; Gödert, 1990; Ribbert, 1992). These regulate the content indexing of libraries' stocks as a norm file of "rules for the keyword catalog" (RSWK, 1998). The keyword norm file now contains hierarchical relations, and is thus well on its way to becoming a thesaurus. In this chapter, we exclusively regard the construction of keywords. As a further case study, we will investigate a specialist nomenclature on the example of the "Chemical Registry System". This nomenclature prescribes the specialist chemical terminology of the "Chemical Abstracts Service", the worldwide leader in databases on chemical literature (Weisgerber, 1997).

Every nomenclature consists of keyword entries, which contain both the authority record and the cross-references of the concept in question (for a simple example, cf. Figure L.1.1). The synonyms do not necessarily have to be derived from a natural language, but can be formed via other languages (e.g. structural formulae or simple numbers from chemistry). They are joined by see-also references, bibliographical statements, definitions, rules of usage and management information, where available. According to the RSWK (1998, § 2), a keyword is

a terminologically controlled description which is used in indexing and retrieval for a concept contained within a document's content.

Gödert (1991, 7) adds:

The conceptual specificity of these (controlled, A/N) descriptions is here meant to correspond with the specificity of the object to be represented.

Schlagwort (4138676-0)	Keyword (4138676-0)
GKD 2081294-2	GKD 2081294-2
c Köln / Erzbischöfliche Diözesan- und Dombibliothek	Cologne / Archiepiscopal Diocese and Cathedral Library
Q GKD ; SYS 6.7 – 3.6a ; LC XA-DE-NW	Q GKD ; SYS 6.7 – 3.6a ; LC XA-DE-NW
BF Erzbischöfliche Diözesan- und Dombibliothek / Köln	Archiepiscopal Diocese and Cathedral Library / Cologne
Köln / Diözesan- und Dombibliothek	Cologne / Diocese and Cathedral Library
Diözesan- und Dombibliothek / Köln	Diocese and Cathedral Library / Cologne
Köln / Diözesanbibliothek	Cologne / Diocese Library
Diözesanbibliothek / Köln	Diocese Library / Cologne
Dom-Bibliothek / Köln	Cathedral Library / Cologne
Köln / Dom-Bibliothek	Cologne / Cathedral Library

Figure L.1.1: Example of a Keyword Entry from the Keyword Norm File (SWD). Source: Deutsche Nationalbibliothek [German National Library]. (Abbreviations: |c : Körperschaft [corporate body], Q : Quelle [source], GKD : Gemeinsame Körperschaftsdatei [common file of corporate bodies], SYS: Systematik der SWD [systematics of the SWD], LC: Ländercode [country code], BF: benutzt für [used for]).

A keyword consists of single words or compounds; individual concepts are considered in the same way as general concepts (Umlauf, 2007):

- Single Word—General Concept (*Intelligence*),
- Single Word—Individual Concept (*Oedipus*),
- Word Sequence—General Concept (*Telegu Language*)
- Word Sequence—Individual Concept (*Mozart, Wolfgang Amadeus*),
- Adjective-Noun-Compounds (*Critical Temperature*).

When building a nomenclature for libraries, it is recommended to forego highly specific keywords (Geißelmann, Ed., 1994, 51); for a chemical nomenclature, the exact opposite applies. A document dealing with, for instance, 2,7(*bis-dimethylamino*)-9,9-*dimethyantracene* must be indexed with the keyword *Anthracene Derivates* in the RSWK, whereas in a chemical database, the exact name will be used.

German-language keywords are set in the singular, except for words that occur exclusively in the plural (*Leute*, i.e. *people*), biological terms above the genus level (*Rosengewächse*, i.e. *rosaceae*, but: *Rose*), chemical group names (*Kohlenwasserstoffe*, i.e. *hydrocarbons*), names for groups of persons or countries (*Jesuiten*, i.e. *Jesuits*), groups of historical events (*Koalitionskriege*, i.e. *French Revolutionary Wars*) and joint descriptions for several sciences (*Geisteswissenschaften*, i.e. *humanities*) (Umlauf, 2007).

According to the RSWK, pleonastic terms, i.e. accumulations of elements carrying the same or a similar meaning, are taboo (RSWK, 1998, § 312):

Pleonastic terms or partial terms, which are not necessary for the full description of a concept, and generalizing forms that do not change the meaning of the root, should be avoided. In doing so, however, one should not infringe upon the specialist terminology.

Logical pleonasms (*male stallion*) are to be distinguished from factual pleonasms (*Sardinian Nuraghe culture*, viz. this culture existed exclusively on Sardinia). Both phrases will not be admitted to a nomenclature; however, for the factual pleonasms, it should be allowed—following the RSWK (1998, §324)—to index a document with *Sardinia ; Nuraghe culture*, since not every user will automatically make the connection between the island and the culture. Keyword combinations involving, for instance, *-frage* (*question*, i.e. formulate meaning, not *question of meaning*) or *-idee* (*idea*, i.e. write *equality*, not *idea of equality*) are mostly pointless and superfluous.

Aspects of time are expressed via specific time keywords (e.g. *history*, *prognosis*) (RSWK, 1998, §17). Where a specific point in time, or a certain interval of time, is addressed in a documentary reference unit, it will be represented in the surrogate as follows:

Vienna ; History 1915 – 1955
Global Economy ; Prognosis 2010 – 2015.

It must be taken care to make the individual years that are included in a period retrievable in the database. If one searches for *Vienna AND History 1949*, our sample document must be included among the search results. Historical events and epochs have a set time reference (e.g. the Battle of Nations at Leipzig in 1813), which does not have to be specially named as a keyword, but which is made retrievable as a time code (Umlauf, 2007). If one searches for *History 1813*, they will be shown a list of all keywords that are allocated to the year (explicitly or perhaps via an interval) or the time code, e.g.

Düsseldorf ; History 1800–1850
Leipzig / Battle of Nations

When a concept has several components of meaning (Kunz, 1994), there are three options of dealing with it:

- The phrase is preserved (*Kölner Dom*, i.e. *Cologne Cathedral*),
- The phrase is divided into its constituent parts, which however make up one single chain of subject headings (*Cologne / Cathedral*),
- The phrase is divided into different keywords (*Cologne ; Cathedral*).

The decision as to whether or not to split up compounds and phrases, according to Gödert (1991, 8), is made via the criteria of fidelity and the predictability of indexing

vocabularies. Geißelmann (Ed., 1994, 54) identifies customariness and proximity to natural language as the basic orientation. If one selects a chain of subject headings as a keyword (marked in the SWD via a slash between the components), the chain will be searched as a whole. In such a case, it is pertinent to incorporate the reverse order (*Cathedral / Cologne*) into the KOS as a reference (as in our example in Figure L.1.1). If one decides in favor of a split, it will be of help to the user if the compound is entered as a reference (let *Cologne Cathedral* use *Cologne ; Cathedral*). The semicolon refers to a search query via the Boolean AND or a proximity operator.

By definition, nomenclatures do not contain any hierarchies. However, chains of subject headings—when constructed skillfully—bear “hidden” hierarchical statements. This holds both for our example in Figure L.1.1 and for *Cologne / Cathedral*. The chain sometimes stretches over several hierarchical levels, as in *Cologne / Cathedral / Shrine of the Three Kings*. The concepts in the named examples form meronyms.

Homonym Disambiguation

Homonyms are identical designations for different concepts. The SWD uses qualifiers in arrow brackets to split homonymous descriptions (e.g. *Apple <IT company>*). However, in certain cases, the rulebook permits the exclusion of the qualifiers for the most common homonym (RSWK, 1998, § 10):

München (i.e. the city in Bavaria—without a qualifier, as it is far more famous than any of its homonyms)

München <Berka, Weimar> (i.e. district of a town in Thuringia)

Normally, however, all homonyms are provided a qualifier and are only displayed with it. If a user searches for *apple*, the system’s answer will consist of a list of all matching homonyms (contrary to the RSWK, since they list the first variant as the main meaning, without a qualifier):

Apple <fruit>

Apple <IT company>

Apple <music label>

It is often useful to use a scientific discipline as the qualifier (*cancer <medicine>* and *cancer <astrology>*). Whereas it can make sense in chains of subject headings to incorporate both the phrase (*Cologne / Cathedral*) and its constituent parts into the inverted file via a word index (in order to make an AND link possible), it would be entirely pointless to make homonym qualifiers searchable in their own right.

In chains of subject headings, a homonym qualifier can be omitted when clarity about the meaning of the concept has been established via combination. When, for

instance, the keyword *Münster* (*minster*) is complemented by the qualifier <Dom> (*cathedral*), this same qualifier does not have to be added to *Ulm* / *Münster*.

In certain cases, it may prove pertinent to only add a homonym to the nomenclature for the purposes of referral, in order to direct users to a non-homonymous name, e.g. (RSWK, 1998, § 306):

Class <education> USE *course*

Class <sociology> USE *social ladder*.

Synonym Conflation

Different designations are synonymous when they describe the same concept. Concepts are quasi-synonymous if their extension and intension are so similar that they are regarded as one and the same concept for the purposes of any given KOS. The different synonyms and quasi-synonyms are combined in a keyword entry, where one of the designations is set apart from the others as the preferred term or authority record. The preferred term is the keyword, all other designations are (synonymy) cross-references. In our example (Figure L.1.1), *Köln* / *Erzbischöfliche Diözesan- und Dom-Bibliothek* (Cologne / Archiepiscopal Diocese and Cathedral Library) is the keyword, while all designations named under “used for” (UF) (i.e. *Erzbischöfliche Diözesan- und Dombibliothek* / *Köln* etc.) are its cross-references. Cross-references are expressed, in the RSWK, via “use synonym” (USE), i.e.:

Erzbischöfliche Diözesan- und Dombibliothek / *Köln*

USE *Köln* / *Erzbischöfliche Diözesan- und Dombibliothek*.

All real synonyms, variants in common parlance, permutations in chains of subject headings, abbreviations, inverted arrangements of adjective-noun compounds, quasi-synonyms etc. are combined. From a practical point of view: all those designations are combined which can be viewed as the smallest semantic unit for the respective knowledge organization system. If one of these synonymous terms turns out to be more common than the rest, it will be designated as the keyword.

In chemistry, the combination of synonyms is particularly delicate, as the language of chemistry has a multitude of synonyms, molecular formulae, systematic descriptions, non-technical names, generic descriptions, trade names as well as structural formulae (Lipscomb, Lynch, & Willett, 1989). As of 2012, there are around 70m known organic and anorganic substances as well as around 64m biosequences.

The Chemical Abstracts Service’s (CAS) Registry File replaces the keyword with its CAS Registry Number, which clearly identifies every substance. The numbers themselves do not carry meaning, but, as a whole, represent a substance or biosequence. The numbers have at most nine digits, grouped into three blocks. The last entry is

a check digit. The Registry Number in Figure L.1.2 (57-88-5) identifies a substance described as *cholesterin*, *cholesterol* etc., with the molecular formula $C_{27}H_{46}O$, as well as via the depicted structure. The number of synonyms within a concept entry can be very high; 100 different designations for the same substance are nothing unusual. During search and retrieval of chemical substances, one exclusively uses the Registry Number. The synonyms as well as the molecular and structural formulae serve merely to locate the required number.

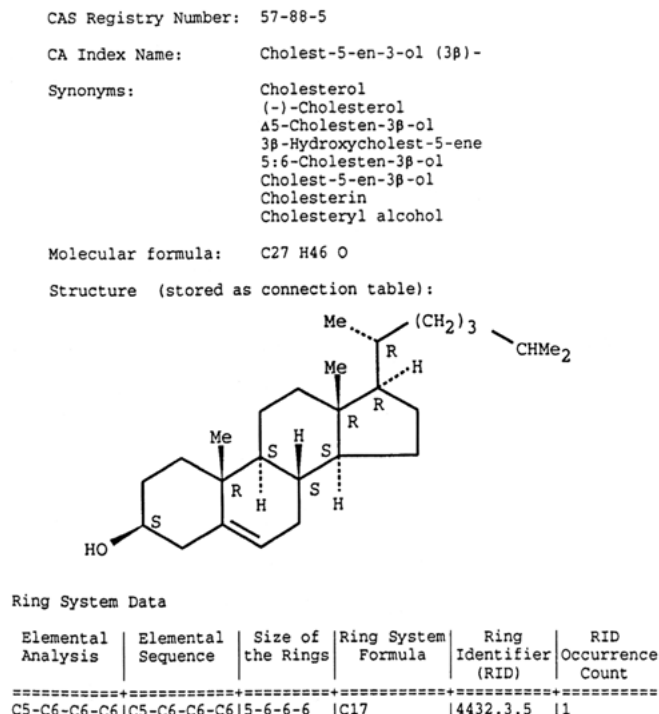


Figure L.1.2: Keyword Entry in the CAS Registry File. Source: Weisgerber, 1997, 355.

In order to provide for a graphical search, the structure is transposed from a graphical into a (searchable) tabular arrangement, called connection table (Figure L.1.3). Users can search not only for complete structures but also for substructures. Figure L.1.3 shows an example of a two-dimensional structure; processing structures in the context of stereochemistry requires more elaborate procedures.

Chemists sometimes use short descriptions, even if these “short cuts” make it clear (as in Figure L.1.4) which substance is meant. Baumgras and Rogers (1995, 628) report:

Most chemists prefer to show common or large functional groups in a shorthand notation as has been done with the 13-carbon chain. Other common shorthands include things like Ph to represent phenyl rings, COOH to represent carboxyl groups, etc. Chemists are familiar with these notations and assume the chemistry behind them, and most of the shorthand notations are interpreted by humans according to convention.

Such shorthand descriptions represent a challenge to nomenclatures.

Node no.	Atom	Connected to	Bond type
1	C	-	-
2	C	1	*1
3	C	1	*1
4	C	1	*1
5	C	2	*2
6	C	2	*1
7	C	3	*1
8	C	3	*1
9	C	4	*1
10	C	4	*2
11	C	6	*1
12	C	8	*1
13	C	1	-1
14	C	9	-1
15	C	9	-1
Ring closure	-	5-7, 9-12, 10-11	*1

1=single bond	*=cyclic bond
2=double bond	-=non-cyclic bond

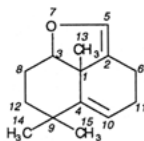


Figure L.1.3: Connection Table of a Chemical Structure in the CAS Registry File. Source: Weisgerber, 1997, 352.

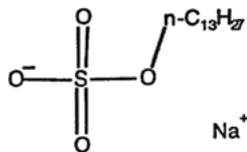


Figure L.1.4: Shorthand of a Chemical Compound. Source: Baumgras & Rogers, 1995, 628.

Problems occur with substances that only have weak bonds (Figure L.1.5). Should they be entered as *one* substance, or separated by their component parts? In Figure L.1.5, we integrated a special aspect by recording isotopes (deuterium instead of hydrogenium, as well as ^{14}C instead of ^{12}C). Are the structures with isotopes (otherwise unchanged) new substances, thus requiring their own data entry?

Another challenge for a chemical nomenclature comes in the form of Markush structures (Berks, 2001; Stock & Stock, 2005, 170). Here, instead of a specific atom, a whole group of possible elements, functional groups, classes of functional groups (e.g. esters) or groups of chemical structures (such as alkyls) is designated at a certain point within a structure (Austin, 2001, 11). This results in references to possible (“pro-

phetic”) substances in addition to actually existing substances. Markush structures are particularly popular in intellectual property protection, since they allow for ambitious definitions of patent claims. The name “Markush” goes back to a patent application by Eugene A. Markush (1923), who was the first to introduce such an unspecific description into patent literature.

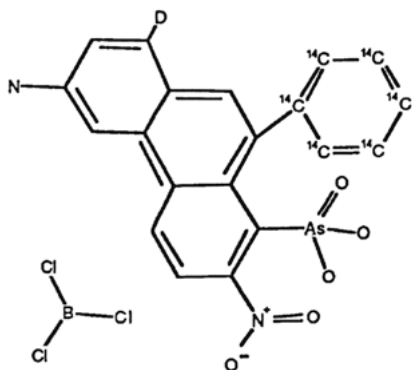


Figure L.1.5: Two Molecules (One with Isotopes) with Weak Bonds. Source: Baumgras & Rogers, 1995, 628.

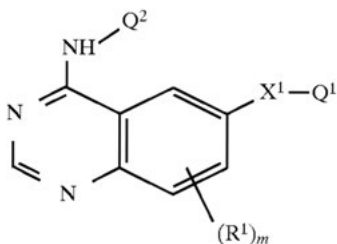


Figure L.1.6: Markush Structure. Example: Quinazoline Derivate. Source: Austin, 2001, 12.

In the example in Figure L.1.6, several such Markush elements are present: X^1 represents a direct bond, while Q^1 and Q^2 each stand for bonds (stated more explicitly within the patent) and R for organic residue.

All known Markush structures are—albeit in a special nomenclature—to be deposited in addition to the “normal” structures, and are thus available to search.

Gen Identity and the Chronological Relation

Objects change over time. If such objects are described with different designations at different times, and if the concept differs in extension or intension with regard to

other times, we speak of gen identity. The RSWK expresses gen identity via the relation of “chronological form” (CF), with its two directions “earlier” and “later”. The chronological cross-references (RSWK, 1998, § 12)

are used for geographic keywords (name changes...) as well as for corporate bodies (name change alongside a fundamental change to the nature of the body).

Name changes for people (e.g. due to a marriage) are not subject to the chronological relation, but to synonymy. After all, no new concept is created in such a case, only a new designation for the same concept.

Where only the name of a geographical fact or corporate body changes (i.e. the extent and content of the concept are preserved), personal names come within the exclusive scope of synonymy. If name changes are accompanied by modifications to the concept, the chronological relation will be used. The RSWK (1998, § 207) provide an example for a geographic keyword:

Soviet Union

CF earlier *Russia *Beginnings - 1917*

CF later *Russia *1991 - present.*

Where geographic facts are separated into several parts, or conversely, several localities are joined into a single one, this will be represented within the corresponding chronological relation (RSWK, 1998, § 209):

Garmisch-Partenkirchen

CF earlier *Garmisch*

CF earlier *Partenkirchen*

Garmisch

N used for the previously autonomous locality and current local center

CF later *Garmisch-Partenkirchen*

Partenkirchen

N used for the previously autonomous locality and current local center

CF later *Garmisch-Partenkirchen*

(*N* stands for “Note”.) The current local center (e.g. Partenkirchen) is described, following the RSWK, with the keyword of the previously autonomous locality. The alternative formulation

Garmisch-Partenkirchen-Partenkirchen

would indeed be slightly confusing.

If the areas of responsibility or self-conception of a corporate body change, or if mergers or divisions occur, the chronological relation will be used. Here, too, the RSWK (1998, § 611) provide us with an example:

German Gymnastics Association

N Founded anew in 1950, self-conceives as the successor organization to the German Turnerschaft

CF earlier *German Turnerschaft*

German Turnerschaft

N Incorporated into the German Reich's Confederation for Physical Exercise, self-dissolution in 1936, founded anew in 1950 under the name of German Gymnastics Association

CF later *German Gymnastics Association*

Nomenclature Maintenance

When new topics come up that need to be described, new norm entries must be created. Conversely, it is possible that certain keywords have hardly any appropriate documents to be allocated to, and thus need to be deleted. Nomenclature maintenance takes care of preventing both unbridled growth and shrinkage of the concept materials. Hubrich (2005, 34 et seq.) describes the work involved in creating a new keyword in library-oriented content indexing:

In order to keep the effort at a minimum, a concept is generally only introduced when it is absolutely necessary to describe an object of a document to be indexed; i.e. when the topic cannot be represented via the given means of indexing, and the concept is no "flash in the pan", and it can be assumed that it will be used for indexing and/or search in the future.

Insofar as the concept (including all synonyms and quasi-synonyms) is not yet available in the knowledge organization system, it may be considered—for compounds—whether the new concept might be expressed via a combination of old terms. Let us suppose that the possible new entry is *library statistics*. Both *library* and *statistics* are already in the file. Now, if little in the way of literature on *library statistics* is to be expected according to the current state of knowledge, and if, in addition, only sporadic queries using this compound are probable, we might make do with a cross-reference:

Library statistics USE *library* ; *statistics*.

If, however, larger quantities of literature (arbitrary point of reference: more than 25 documents) are to be expected, or it is known from user observation that the term is heavily used, a new keyword entry will be created.

The Chemical Abstracts' Registry File proceeds differently. At the moment when it becomes known that a new substance has been described in a relevant specialist publication, a keyword entry will be created in the nomenclature. Here, completeness of the KOS—updated daily, if possible—is the goal. The employees of Chemical Abstracts add roughly 15,000 new data entries to the Registry File per workday.

It is useful to provide keywords with their “life data”, i.e. their date of introduction and the point of their ceasing to be a preferred term. In this way indexers and users can see at a glance which controlled terms are currently active and, if they aren't, at which points in time they have been used in the information system.

Conclusion

-
- A record of all (general or specific) terms used in a knowledge organization system is called a “nomenclature”. Nomenclatures work with a controlled vocabulary, i.e. exactly *one* term is taken from the multitude of synonyms and quasi-synonyms and designated as the preferred term. This term is called the “keyword”. Both indexers and users employ these authority records.
 - Nomenclatures are distinguished by an elaborate form of synonymy. Further relations (e.g. unspecific see-also references) may occur; by definition, there are no hierarchical relations.
 - A (natural-language) keyword may consist of single words, word groups or adjective-noun compounds. It may refer both to individual and general concepts (such as in the Keyword Norm File SWD). However, it may also be formed via a distinct number (as in the Registry File by the Chemical Abstracts Service).
 - When creating (natural-language) keywords, attention must be paid to their genus, the prevention of pleonasms, a direct or indirect (searchable) time reference as well as (for a concept with several components of meaning) its possible division into several keywords.
 - Homonyms are always divided according to their concepts, and generally complemented by a qualifier (e.g. *bass <music>*).
 - Synonyms and quasi-synonyms are summarized into *one* class. CAS' Registry File shows that this is a formidable task, as all molecular formulae, systematic descriptions, trade names, structural formulae etc. of a substance must be located and recorded. The structural formulae (including the substructures contained therein) are made searchable via connection tables.
 - Specific problems of chemical nomenclatures include short cuts in chemists' use of language, substances with weak bonds, the occurrence of isotopes within a structure as well as (the “prophetic”) Markush structures.
 - Gen-identical concepts are interconnected via the chronological relation. The objects in question have different designations at different times and are similar in terms of extension and/or intension, but not identical. Changes merely concerning the designation (e.g. personal names after a marriage) are regarded as synonyms and not as gen identity.
 - Nomenclature maintenance's task is to prevent uncontrolled growth (and deterioration) of the KOS. It is useful to allocate to every keyword both its time of entry into the system as well as (where applicable) its date of deletion.
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L.2 Classification

Notations

Classifications as knowledge organization systems look back on a long history. This goes both for their function as shelving systems in libraries and their use in online databases (Gödert, 1987; Gödert, 1990; Markey, 2006). Standard applications have been subject to norms for years (in Germany: DIN 32705:1987). Classifications have two prominent characteristics: the concepts (classes) are described by (non-natural-language) notations, and the systems always use the hierarchy relation. Manecke (1994, 107) defines:

A classification system is a systematic compilation of concepts (*concept systematics*), which mainly represents the *hierarchical relations between concepts (superordination and subordination)* via system-representing terms (notations).

The building of classes, i.e. all efforts toward creating and maintaining classifications, is called “classifying” (this is the subject of this chapter), whereas the allocation of a specific classification system’s classes to a given document is referred to as “classing” (which is an aspect of indexing; chapter N).

In classification systems, notations have a particular significance. A few random examples to get started:

636.7 (DDC),
35550101 (Dun & Bradstreet),
DEA27 (NUTS),
Bio 970 (SfB),
A21B 1/08 (IPC).

The first notation from the Dewey Decimal Classification (DDC; Satija, 2007) is the name for the class “dogs”. In the DDC, every digit represents a hierarchy level: thus, 636.7 is a hyponym of 636, 636 in turn is hyponym to 63 and 63 to 6. This decimal principle allows for a straight-forward downward division of the KOS (hospitality in the concept ladder), but not for an enhancement in breadth exceeding ten classes. Its hospitality in the concept array (concerning the set of sister terms) is very restricted (there are only the ten digits). In Dun & Bradstreet, 35550101 stands for machines for printing envelopes. The first four digits (3555) follow the decimal principle, like the DDC, but the last two hierarchy positions are expressed by two digits each: accordingly, there are only two hierarchy levels after 3555 (355501 and 35550101). As there are now 100 options per level, we are looking at a centurial principle. (A particularly tricky aspect of Dun & Bradstreet’s notations is that inexperienced users do not know at which points in the notations the decimal and centurial principles are used.) A millennial principle (three digits) is also possible. In the bottom example from the

International Patent Classification, there is a *1* at the fourth level. In this hierarchy, 1,000 different co-hyponyms are possible (the precise notation at this point would thus actually be *001*). We could equally and just as well use letters or combinations of digits and letters. The Nomenclature des unités territoriales statistiques (NUTS) uses such a mixed system from the second hierarchy level onward. The first ten co-hyponyms receive the digits 0 through 9, after which the letters A through Z are used. This provides for hospitality options in the array of up to 36 concepts, at only one position in the notation. The first level of NUTS is formed mnemonically and stands for a country (*DE* for Germany). *A* on the level below, the second level, designates North Rhine-Westphalia, with the entire notation standing for the Rhine-Erft region.

Notations that consider mnemonics at least partly can be found in many public libraries. *Bio 970* in a German Library Systematics (SfB, 1997) stands for house cats; the concept ladder's top term is—easily comprehensibly—biology. Apart from this, classification systems put little store by the memorability of their notations (which necessarily leads to an automatic processing of user input and to a dialog between user and system).

The International Patent Classification (IPC) also uses a mixed system of letters and digits. Our notation represents steam-heated ovens. The first IPC hierarchy level (one position) expects a letter, the second, centurially, two digits (where the initial zero can be omitted), the third another letter and the fourth, millennially, three digits (again, any leading zeroes can fall away). From the fourth hierarchy level downward, the notation's makeup becomes somewhat complicated. The fourth level, the so-called “main group”, principally contains two zeroes (*/00*), and from the fifth level on, other digits are given out enumeratively.

Notations sometimes mirror the hierarchical position of their concepts, in which case we speak of hierarchical notations (DIN 32705:1987, 6). As an example, we will consider the German classification of economic sectors, which—apart from the last hierarchy level—is identical to the European NACE (Nomenclature générale des activités économiques dans les Communautés Européennes). We can see four hierarchy levels: division (centurial, e.g. *14* Manufacture of wearing apparel), group (decimal, e.g. *14.1* Manufacture of wearing apparel, except fur apparel), class (decimal, e.g. *14.14* Manufacture of underwear) as well as the lowest level, differently filled by each respective EU member country (decimal again, e.g. *14.14.3* Herstellung von Miederwaren, Manufacture of corsetry). The notations clearly show the hierarchy:

14 Manufacture of wearing apparel

14.1 Manufacture of wearing apparel, except fur apparel

14.14 Manufacture of underwear

14.14.3 Herstellung von Miederwaren (Manufacture of corsetry).

The advantage of hierarchical notations for online retrieval is obvious: the user is provided with the option for hierarchical retrieval via a single truncation symbol. If

“*” is the symbol for open truncation on the right and “?” the symbol for replacing exactly one digit,

14.1*

will retrieve all documents on manufacture of wearing apparel (except fur apparel) including its various hyponyms. Using

14.1?,

on the other hand, one searches the group including all of its classes, but not their hyponyms. If a hierarchy level works with several positions, the user will have to correspondingly use several replacement symbols; two question marks (??) in a centurial notation position, for instance.

If a classification system is not geared toward online retrieval but is meant, for example, to consolidate the shelving system of a library, one can use sequential notations for reasons of simplicity. Here the notations are simply numbered consecutively, with any neighboring notations (as it is the case in all classification systems) being closely thematically related, of course. After all, the user who has just found a relevant book on the shelf is supposed to find further appropriate offers to either side of his initial specific find.

Apart from the purely hierarchical or sequential notation systems, there are also mixtures of both. Such hierarchical-sequential notations work hierarchically on certain hierarchy levels, and sequentially on others. A good example of such a mixed notation is the IPC. The first four levels even show the hierarchical position of the concept:

G Physics
 G06 Computing, calculating, counting
 G06F Electrical digital data processing
 G06F 3 Input arrangements ...

From the group level onward, the IPC subdivides hierarchically downward on the concept ladder, but the notations do not reflect this. Only the /00 part of the notation represents the main group and thus the hyperonym of all subgroups below. In the print version of the IPC, the hierarchy is hinted at via the number of dots: *G06F 3/027* is thus the hyponym of *G06F 3/023*, and it to *G06F 3/02*, and it in turn to *G06F 3/01* (IPC, 2011).

G06F 3/01

° Input arrangements or combined input and output arrangements for interaction between user and computer (G06F 3/16 takes precedence)

G06F 3/02

- ° ° Input arrangements using manually operated switches, e.g. using keyboards or dials (keyboard switches per se H01H 13/70; electronic switches characterised by the way in which the control signals are generated H03K 17/94)

G06F 3/023

- ° ° ° Arrangements for converting discrete items of information into a coded form, e.g. arrangements for interpreting key board generated codes as alphanumeric codes, operand codes (coding in connection with key boards or like devices in general H03M 11/00)

G06F 3/027

- ° ° ° ° for insertion of the decimal point.

Hierarchical search options with truncation end on the main group level in the IPC; any classes below (and this includes the vast majority) are not searchable via truncation.

In some classification systems, related but different notations form a single class. Thus, the North American Industry Classification System (NAICS) uses the decimal principle for its subsectors (second hierarchy level). However, as there are more than ten classes in the areas of industry, retail and transportation, neighboring notations are summarized into a “chapter”. For instance, the NAICS amalgamated classes 31, 32 and 33 into a unit in the chapter on Industry. The 30 possible classes contained within the concept array apparently suffice to exhaustively categorize the industrial complex. A user who wishes to search all of industry must thus combine the notations via the Boolean OR:

31 OR 32 OR 33,

or—when incorporating all of the levels below—

31* OR 32* OR 33*.

Designating Classes

The preferred term of a class is its notation. The entire hierarchical structure of a classification system is built on these preferred terms. However, it would make little sense for users to only work with notations; they require natural-language access points to the notations. The great advantage of notations is that they are formed independently of natural languages. Users speak “their” languages, and correspondingly classes can be designated in any of the natural languages spoken by their potential users. A multinational enterprise with production sites and subsidiaries in Germany, Poland, Japan and the USA, for example, will consequently create designations in German,

Polish, Japanese and English. One of the great universal classifications—the Dewey Decimal Classification (DDC)—used in libraries in more than 135 countries, has been translated into more than 30 languages.

As shown in Figure L.2.1, the notation forms the basis, the deep structure even, of the system. Following the users' language preferences, natural-language interfaces are created. The multitude of language-specific designations is not translated literally, but adapted to the relevant linguistic, societal or cultural customs. For example: the German appellation "*Lehrling*" (or "*Azubi*") cannot be cleanly translated into English, as dual apprenticeship is scarcely heard of in Anglophone countries. The English "trainee" always carries shades of "intern", which is precisely what a *Lehrling* is not. Hence, for every natural language different numbers of designations (synonyms and quasi-synonyms) are gathered in order to allow potential users to search each respective concept.

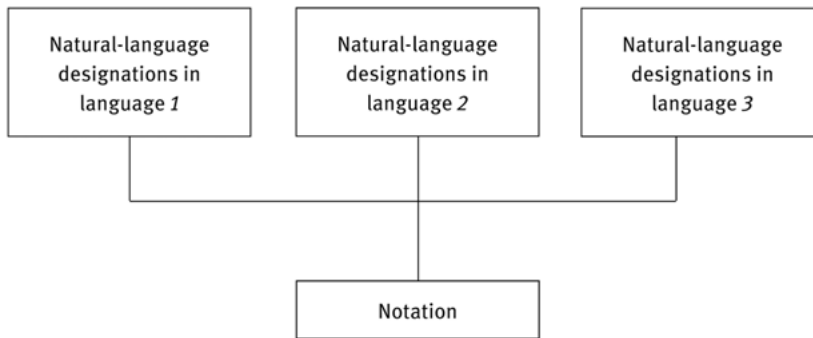


Figure L.2.1: Multiple-Language Terms of a Notation.

The natural-language terms always point to the preferred term, i.e. the notation. Since only the terms, and not the notations, have possible homonyms, the homonym problem practically solves itself in classification systems. When entering a homonym, the user is necessarily led to different notations, which represent a number of disparate concepts and thus disambiguate the homonyms. If we look up "bridge" in the DDC's index, we are shown the following list:

Bridge (Game)	795.415
Bridge circuits	621.374 2
electronics	621.381 548
Bridge engineers	624.209 2
Bridge harps	787.98
<i>see also</i> Stringed instruments	
Bridge River (B.C.)	T2-711 31
Bridge whist	795.413
Bridgend (Wales: County Borough)	T2-429 71

Bridges	388.132
architecture	725.98
construction	624.2
military engineering	623.67
public administration	354.76
transportation services	388.132
railroads	385.312
roads	388.132
Bridges (Dentistry)	617.692
Bridges (Electrical circuits)	621.374 2.

The designations, always represented as indices in printed classifications, thus fulfill the tasks of disambiguating homonyms, summarizing synonyms (in our example: the designations of class 388.132 and class 621.374 2) and—occasionally—serve as unspecific see-also references (as in *Bridge harps*). Gödert (1990, 98) thus quite rightly observes: “Index work thus always means terminological control.”

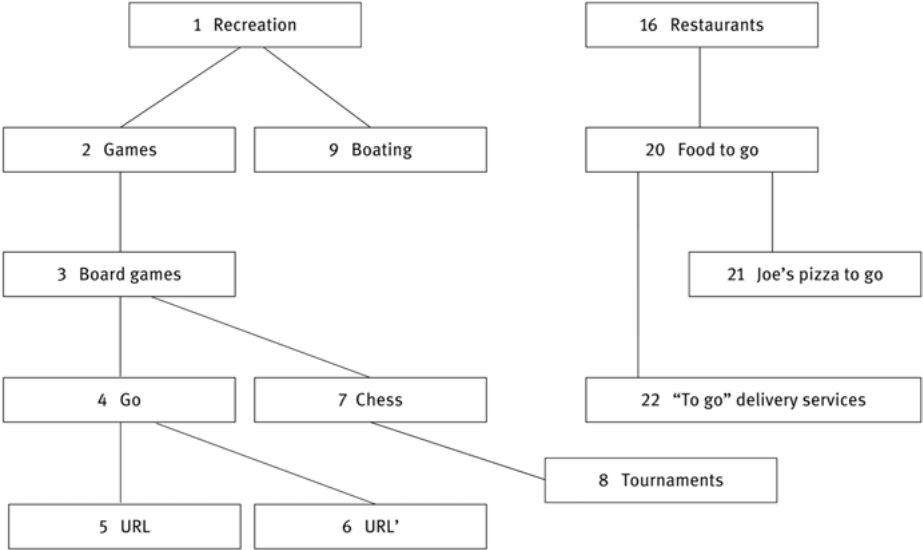


Figure L.2.2: Simulated Simple Example of a Classification System. Source: Modified from: Wu, 1997, Fig. 2.

Syncategoremata and Indirect Hits

Let us consider an example from the International Patent Classification:

A24F9/08 *Cleaning-sets*.

The designation suggests that we are merely dealing with detergents. This is entirely false, however, as we can see when looking at the next highest hierarchy level:

A24F9/04 Cleaning devices for pipes.

The entry thus exclusively concerns pipe cleaners. How will a retrieval system deal with the following search query?

“Cleaning sets” AND pipes

The task is clear: the system must lead the user to the notation A24F9/08 or directly suggest a search for this term. However, *pipes* does not occur in the notation A24F9/08, and neither does *cleaning sets* in A24F9/04. Both concepts are syncategorematic, i.e. incomplete. We thus do not receive any direct hits on the term level.

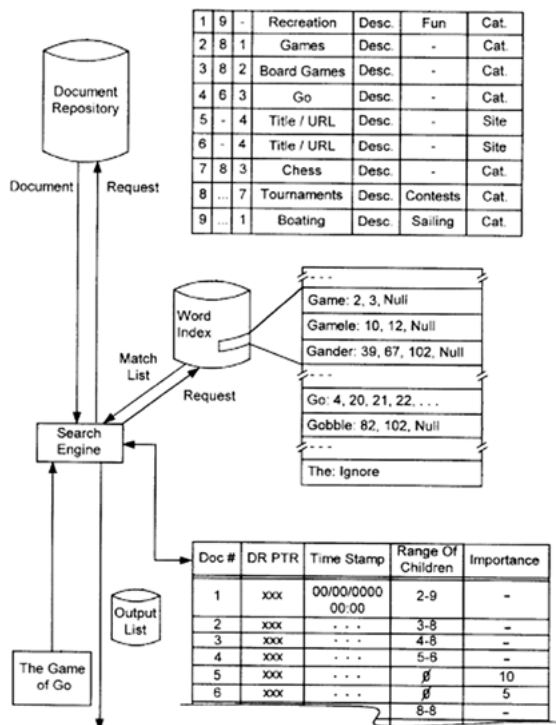


Figure L.2.3: Search Sequence with Indirect Hits for Syncategremata. Source: Modified from Wu, 1997, Fig. 3.

In such cases (where there is no direct search result in the class designation), Wu (1997) suggests, in a patent for Yahoo!, using indirect hits that incorporate the terms from the respective concept ladder into the search. Wu builds a simple classification system (Figure L.2.2). The notations are built sequentially (1, 2, 3 etc.). Let a user search for *the game of go*. There is no direct hit—which would be a class name containing *game* and *go*—but instead there are various results featuring *game* and *go*, respectively. The sequence of the search for indirect hits is displayed in Figure L.2.3.

In the document repository recording both documents and notations, including natural-language terms, in Yahoo!, each entry—where available—was allocated the notation of the next highest term as well as the notation of the bottom term on the concept ladder. Let us consider the example of notation 3 (*Board Games*), recorded in the first column of the document repository (Figure L.2.3, top). The second column records the notation of the bottom term with the highest number, which in this case is 8 (*Tournaments*). Since the notations are allocated sequentially, we know that all entries between 4 and 8 are hyponyms of 3. Finally, the third column names the hyperonym, i.e. 2 (*Games*). In the inverted file (called word index) (Figure L.2.3, middle right), all individual words are held at disposal via their document numbers (or their notations). *Game(s)* (Yahoo! used automatic lemmatization) occurs in entries 2 and 3, *Go* in entries 4, 20, 21 and 22. The intersection is empty; there is no direct hit.

The algorithm of our search for *the game of go* eliminates *the* and *of* as stop words; the remaining search arguments *game* and *go* yield no common documents in the word index. Retrieval systems that do not use the option of searching for indirect hits would yield a (false) “no search results” message. In the document repository, though, we can see that the hyponyms of *Games* (2) are the entries 3 through 8. This interval contains our second search argument (4). Hence, notation 4 is the desired indirect hit for which Yahoo! yielded not only the class but also, directly, the documents (N^os 5 and 6).

Class Building and Hierarchization

Among the crucial requirements of classification systems are the selection and hierarchical arrangement of the concepts (Batley, 2005; Bowker & Star, 2000; Foskett, 1982; Hunter, 2009; Marcella & Newton, 1994; Rowley, 2000). For Gödert (1990, 97), the

decisive question in descriptively and evaluatively dealing with classification systems ... concerns the structure of such a system. In other words, which criteria are used to define the classes, which to structure them. Aspects such as level of detail and up-to-dateness are then subsumed within these questions. The goal of such actions is to make as many users as possible accept the system on an intersubjective level.

We cannot assume that there is such a thing as a natural hierarchical order of the objects. This order cannot be *discovered*, but rather has to be *created*, with the pur-

poses of the nascent system always in mind. Spärck Jones (2005 [1970], 571) points out:

Since there is generally no natural or best classification of a set of objects as such, the evaluation of alternative classifications requires either formal criteria of goodness of fit, or, if a classification is required for a purpose, a precise statement of that purpose.

Hjørland and Pedersen (2005, 584) clarify on a simple example. Let there be three objects:



Depending on the purpose of the system, we could summarize the two black objects into one class—we could also, however, consider the two squares as one single concept. Hjørland and Pedersen (2005, 584) here observe:

(T)hree figures, namely two squares and a triangle, are presented (...). There are also two black figures and a white one. The three figures may be classified according either to form or to colour. There is no natural or best way to decide whether form or colour is the most important property to apply when classifying the figures; whether squares should form a class with triangles are excluded or whether black figures should form a class while white figures are excluded. It simply depends on the purpose of the classification.

If the purpose of the classification is general, the classes will be determined categorically—that is, logically. If the classification is needed in certain situations, though, one will classify situationally, with regard to the respective circumstances. Ingwersen (1992, 129) explains:

“Categorical” classification means that individuals sort out an abstract concept and choose the objects which can be included under this concept. “Situational” classification implies that individuals involve the objects in different concrete situations, thereby grouping objects which belong together.

The classes are formed and arranged following purpose-oriented criteria and according to their extension or intension. Those criteria and their justifications play the central role in creating and maintaining classification systems (Hjørland & Pedersen, 2005, 592):

Classification is the sorting of objects based on some criteria selected among the properties of the classified objects. The basic quality of a classification is the basis on which the criteria have been chosen, motivated, and substantiated.

Apart from technical criteria, a consideration of classifications in information science will pay particular attention to the criteria of optimal information retrieval (Hjørland & Pedersen, 2005, 593):

(W)hich criteria should be used to classify documents in order to optimise IR?

Independently of the respective criteria, which arise from the purpose of usage, there are some general criteria that apply to all kinds of classifications. Generally, classification systems are constructed monohierarchically; they do not differentiate between the abstraction relation and the part-whole relation. Instances that occur in the system are the exception (Mitchell, 2001).

Subclass A1	Subclass A2
Subclass A3	Subclass <i>A other</i>

Figure L.2.4: Extensional Identity of a Class and the Union of its Subclasses.

The extension of co-hyponyms (sister concepts in a concept array) should match the extension of their common hyperonym (Manecke, 1994, 109). In Figure L.2.4, Class A is divided into four subclasses, three of which we explicitly named (A1 through A3). In order to make sure that the principle of extensional identity between the class and the union of its subclasses is fulfilled, we use an unspecific hyponym “other”, which takes everything that is A but cannot be expressed by the other terms.

According to Manecke (1994, 110), the sister concepts of an array must be disjunct. In light of the blurriness of many general concepts, it is doubtful whether such a principle of disjunctive co-hyponyms can be kept general (one need only think of Black’s chair museum). In any case, at least the prototypes should be disjunct (Taylor, 1999, 176).

Aristotle already taught us that hierarchization must not make any jumps, i.e. we must leave out no hierarchy level.

In the case of compounds, their parts may be recorded as individual classes (including the correct notation). Here it makes sense, for practical reasons, to make the notation in question recognizable. In the DDC, for instance, France has the notation 44 and the USA the notation 73 (Batley, 2005, 43-44). The economic climates of France and the USA, respectively, are described via the notations 330.944 and 330.973,

historical events such as the reign of Louis XIV in France via 944.033 or aspects of American history via 973.8.

A last piece of advice for building classes concerns the number of classes that may be allocated to a document. Here we distinguish between two scenarios. If a document is meant to be straight-forwardly filed into exactly one class (e.g. a book on a library shelf, or an operation into a classification of surgical procedures for billing purposes), the indexer may only allocate one notation. However, if a document can be allocated to more than one class (e.g. in all electronic information services), the indexer will provide it with as many notations as are required by the different concepts it deals with.

Citation Order

A characteristic of classification systems is the principle of topical relation between adjacent documents. Such a shelving systematic is attained by determining the citation order of the classes (Buchanan, 1979, 37-38):

The purpose of a classification scheme is to show relationships by collocation—that is, to keep related classes more or less together according to the closeness of the relationship. ...

The choice of citation order, then, determines which classes are to have their documents kept together and which are to have theirs scattered—and to what extent.

The order of concepts in ladders and arrays is never arbitrary, but follows comprehensible principles (Batley, 2005, 122). These can be logical (mathematics – physics – chemistry – biology; arrangement by specialization), process-oriented (e.g. corn growing – corn crop – corn canning – can labeling – can wholesaling), chronological (Archeozoic – Paleozoic – Mesozoic – Cenozoic) or partitive (Germany – North Rhine-Westphalia – municipality of Düsseldorf).

In individual cases, particularly for compounds, arranging the classes can be a difficult decision (Batley, 2005, 18). In the area of historical sciences, for instance, the following four classes are to be ordered:

History
History – 10th century
History – Bavaria
History – Bavaria – 10th century.

Arranged thusly, the order follows the principle of Discipline – Place – Time. The problem is the third concept: following general representations of history in the tenth century, it deals with the entire history of Bavaria, in order to then turn to Bavarian history in the tenth century. Let us try the principle of Discipline – Time – Place:

History
 History – 10th century
 History – 10th century – Bavaria
 History – Bavaria.

Now Bavarian history in the tenth century follows general representations of this period, which is satisfactory. A user who starts with *History – Bavaria* and then expects to learn about its various epochs, however, will be disappointed; Bavarian history is disjointed. Here a decision is called for; a principle, once chosen, must be strictly adhered to afterward.

Thematic borders must be noted. In the DDC, for example, these three classes follow one another:

499.992 Esperanto
 499.993 Interlingua
 500 Natural sciences and mathematics.

There is a border between 499.993 and 500; the two classes are not regarded as neighbors.

A hierarchization can be pretty tricky from time to time. To exemplify, let us consider three classes:

- (1) Experiments on the behavior of primates,
- (2) Experiments on the behavior of apes,
- (3) Experiments on the playing behavior of primates.

In each case, there are several components that are formed as part of hierarchical relations (*apes* is a hyponym of *primates*, *playing behavior* is a hyponym of *behavior*). Here, following a suggestion by Buchanan (1979, 23), the compound concept follows the hierarchy of its components:

The rule is that if one or more components of one subject statement are broader than the corresponding components of another, and the remaining components are the same, then the first class is broader than the second.

Class 1 is thus the hyperonym of classes 2 and 3. To what degree the principle of extensional identity between class and subclasses can still be purposefully applied in such multi-part concepts appears to us to be an open question. It becomes entirely problematic when compound components cancel each other out. Buchanan (1979, 23-24) names the examples:

- (4) Playing behavior of primates,
- (5) Behavior of apes.

(4) would tend to be the hyperonym of (5), since *primates* is the hyperonym of *apes*; (5), on the other hand, tends to be superordinate to (4), since *behavior* is superordinate to *playing behavior*. (4) and (5) can thus be neither subordinate nor superordinate to each other.

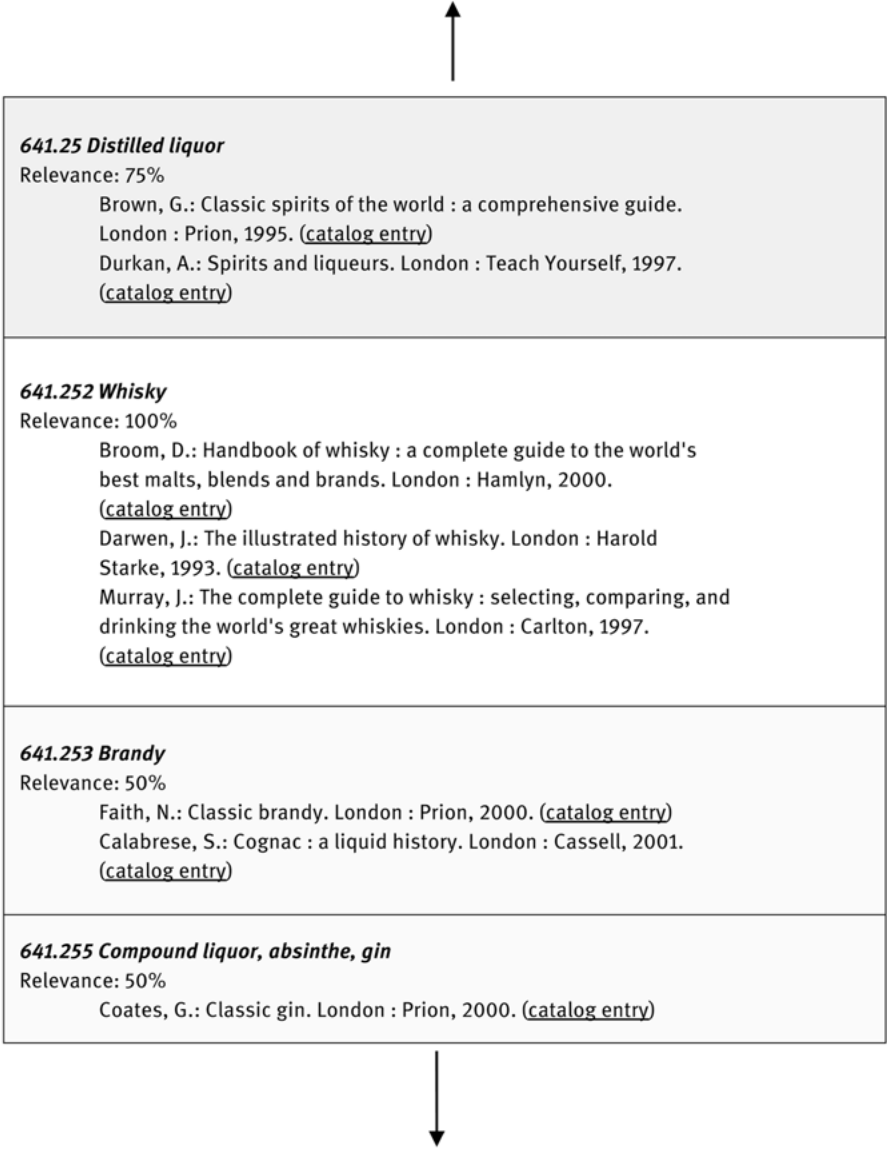


Figure L.2.5: Thematic Relevance Ranking via Citation Order. Source of the Book Titles: British Library.

The principle of the citation order is that neighboring classes have strong thematic links. If a user stands in front of a library shelf, he must be able to find further relevant documents to the right and left of his direct hit. Working in digital environments, such a shelving system must be simulated as directly as possible. An initial option involves showing the user the neighboring classes (and their notations). A second variant—which is far more user-friendly—uses the citation order as a thematic criterion for relevance ranking and directly displays the documents in their concrete environment. The difference to normal ranking (i.e. in search engines such as Google) with a single arranging direction is that ranking via citation order has two directions: thematically forward (or up) and thematically backward (or down).

In the example of Figure L.2.5, we used the DDC to search for *Whisky* (Notation 641.252). The graphic shows a “digital shelving system”, which lists the direct hits (Relevance 100%) in the center. The neighboring concept of *Whisky* in the upward direction is its hyperonym *Distilled liquor* (the weighting of which we arbitrarily set to 75% and colored correspondingly dark). The downward neighbors are its co-hyponyms *Brandy* and *Compound liquors* (whose weighting—again arbitrarily—has been set to 50%). The user can scroll both up (at which point he will reach *Beer* and *Ale*—weighted appropriately low) and down (to *Nonalcoholic beverages* in the example).

Systematic Main and Auxiliary Tables

The classes are systematically arranged in so-called “tables”. Where necessary, the following statements are recorded within the classes, among other data (Batley, 2005, 34-45; Chan & Mitchell, 2006, 42-52):

- Notes (e.g. definitions and notes on validity),
- Year (or edition) of the classification in which the class has been created,
- Year (or edition) of the class’s extinction,
- “class here!”, “class there!”, “do not class here!” instructions (lists of objects that are or are not to be classed in a certain place),
- see and see-also relation.

It is possible to unify all classes of a system in a single table. Whether this makes sense shall be determined via an example (Umlauf, 2006):

Lit 300 Literature	Art 300 Art	Hist 300 History
Lit 320 English Literature	Art 320 English Art	Hist 320 English History
Lit 340 Spanish Literature	Art 340 Spanish Art	Hist 340 Spanish History
Lit 390 German Literature	Art 390 German Art	Hist 390 German History

Such an extremely precombined system requires many different classes, and can thus become unwieldy. An elegant option is to designate the fundamental systematic

tables as “main tables”, and to shuffle some aspects, which occur multiple times in different areas, into “auxiliary tables” (also called “keys”).

In our example, we require three classes in the main tables (Lit 300, Art 300, Hist 300) and three classes in an auxiliary table for locations:

-20 England, -40 Spain, -90 Germany.

If thematic aspects from the main and auxiliary tables fall together, one must synthesize a coherent notation from the notations of the disparate tables. Here, too, a citation order applies—in this case, the set order of main and auxiliary tables. Thus it must always be determined which auxiliary table is to be used in what place.

In the DDC, there are (in auxiliary table 1) standard keys (e.g. 09 for the historical, geographical or people-related treatment of a subject) as well as specific keys, among which (in auxiliary table 2) a KOS for geographical data (where, for instance, -16336 represents the North Sea and English Channel). The citation order for our example goes: main table—auxiliary table 1—auxiliary table 2. A document about the *Channel Tunnel* will consequently be provided with the following notation:

624.194 091 633 6

624.194 *Underwater Tunnels* (from the main table)

09 (standard key for historical, geographical or people-related treatment – from auxiliary table 1)

163.36 *North Sea and English Channel* (from auxiliary table 2).

The dot after the third decimal place as well as the repeated blank after three further places are a convention of the DDC and carry no meaning.

In online retrieval, the respective individual classes should be offered to the user in addition to the synthesized notation, in order to provide him with a simple search entry. Here, the notations from main and auxiliary table are entered into different fields:

Main Table: 624.194

Table for Geographical Data: 163.36.

Only the synthesized notations containing the search arguments are displayed, or additionally—corresponding to the citation order—the thematic environment of the notation results.

Notations from auxiliary tables can either be coupled to all notations from the main tables (“general auxiliary tables” like the geographical data of the DDC) or only to a designated, thematically coherent amount of classes from the main tables (“specific auxiliary tables”, e.g. .061 *Fakes* in class 7 *Art* of the Dezimalklassifikation; Fill, 1981, 84). General auxiliary tables can be very extensive (as in the DDC and the DK), but only piecemeal in other systems. For instance, the German Modification of the International Statistical Classification of Diseases (ICD-10-GM) only has six notations

in the general auxiliary tables; three for side locations (R *right*, L *left*, B *both sides*) and a further three to describe the accuracy of diagnoses (V *suspected diagnosis*, Z *symptomless state after the respective diagnosis*, A *diagnosis ruled out*) (Gaus, 2005, 97).

The separation of main and auxiliary tables keeps a classification system easily understandable. If no such distinction is made between main and auxiliary tables, leaving all tables “equal”, we speak of a “faceted classification”.

Examples of Classification Systems

At the moment, many classification systems are being put into practice. At this point, we would like to name some of the centrally important classificatory knowledge organization systems. Without the help of these systems, it is practically impossible to search for content in some knowledge domains—for example, intellectual property rights, health care as well as industries and products—since knowledge representation proceeds (nearly) exclusively via classifications.

Table L.2.1: Universal Classifications. Sources: DDC, 2011; DK, 1953.

Notation

DDC. Dewey Decimal Classification and Relative Index

Example:

700 The arts. Fine and decorative arts
 790 Recreational and performing arts
 797 Aquatic and air sports
 797.1 Boating
 797.12 Types of vessels
 797.123 Rowing

DK. Dezimalklassifikation

Example:

7 Kunst. Kunstgewerbe. Photographie. Musik. Spiele. Sport.
 79 Unterhaltung. Spiele. Sport
 797 Wassersport. Flugsport
 797.1 Wasserfahrsport
 797.12 Wasserfahrsport ohne Segel
 797.123 Rudersport
 797.123.2 Riemenrudern

Universal Classifications (Table L.2.1) aspire to represent the entirety of human knowledge in a single system. The application goals of universal classifications are the systematic arrangement of documents in libraries as well as their (rough) content indexing in the systematic catalog and the (equally rough) filing of websites into classes (as before, in the case of Yahoo! and the Open Directory Project).

The predominant universal classification at the moment is the DDC (2011; Chan & Mitchell, 2006; Mitchell, 2001). Its European spin-offs (which are fairly similar overall), the UDC (2005; Batley, 2005, 81 et seq.; McIlwaine, 2000) and the German DK (1953; Fill, 1981) span far more classes, but were hitherto unable to widely assert themselves—particularly in large libraries. Mitchell (2000, 81) observes:

Today, the Dewey Decimal Classification is the world's most widely used library classification scheme.

Due to its subtle partitions, Batley (2005, 108) recommends that specialist libraries concentrating on particular subjects use the UDC, whereas general libraries and—we may fill in the blank here—the WWW are said to be more suited for the DDC:

The complexity of UDC's notations may not make it an ideal choice for large, general library collections: users may find notations difficult to remember and the time needed for shelving and shelf tidying would be increased.

(I)t is not suggested that UDC be used in general libraries, but rather to organise specialist collections. Here UDC has many advantages over the general schemes ... The depth of classification ... is very impressive

Both the DDC and the UDC/DK work with main tables, specific auxiliary tables as well as several general auxiliary tables. The DDC comprises around 27,000 classes in its main tables, the UDC around 65,000 terms, the DK (as of 1953) more than 150,000 classes.

Table L.2.2: Classifications in Health Care. Sources: ICD, 2007; ICF, n.d..

Notation

ICD-10. International Statistical Classification of Diseases

Example

- V01-Y98 External causes of morbidity and mortality
 - V01-V99 Transport accidents
 - V18 Pedal cyclist injured in noncollision transport accident
 - V18.3 Person injured while boarding or alighting
- (Note: 3 : special auxiliary table)

ICF. International Classification of Functioning, Disability and Health

Example:

- 1. Mental functions
 - b110–b139 Global mental functions
 - b114 Orientation functions
 - b1142 Orientation to person
 - b11420 Orientation to self

In Medicine and Health Care, classifications are widely used both for statistical purposes (i.e. causes of death) and for doctors' and hospitals' accounts (Table L.2.2). The classification for diseases (ICD; currently in its 10th edition, hence ICD-10) is overseen by the World Health Organization and put into practical use by many countries. Gaus (2005, 104) is convinced that

there is no organization system worldwide that is used as intensively as the ICD-10. This world-wide usage of the ICD-10 means that statistics on morbidity and mortality (...) are fairly comparable internationally.

The International Statistical Classification of Diseases (ICD) classifies according to disease (i.e. according to etiology, pathology or nosology), and does not follow any topological aspects (such as organs) (Gaus, 2005, 98, 100):

Should pneumonia be categorized under the locality of “lungs” or under the disease process of inflammation? When we place all diseases of a particular organ next to each other, the systematics follow the topological/organ-specific aspect (...) However, we can also place all diseases with the same process, i.e. all inflammations ..., next to each other in a systematics. This categorization is called etiological, pathological or nosological (...). The ICD-10 (...) mainly uses the etiological aspect.

We must stress the extensive index of the ICD, in which (around 50,000) terms for diagnoses refer to the respective notations.

All four areas of intellectual property rights use classification systems to index the content of documents (Table L.2.3) (Stock & Stock, 2006; Linde & Stock, 2011, 120-135):

- Patents (IPC),
- Utility Patents (IPC),
- Trademarks (Nice Classification; picture marks: Vienna Classification),
- Industrial Designs (Locarno Classification).

The technical property rights documents are always described by the International Patent Classification (IPC); for the non-technical property rights of trademarks and designs, there are specific knowledge organization systems. Regulated by an international treaty (the Strasbourg Treaty of 1971), all patent offices the world over index with reference to the IPC. Additionally, all private information providers adhere to this standard. The IPC arranges the totality of (patentable) technical knowledge into more than 60,000 classes, where—due to the heavy load of patents—thousands of documents might reside on the lowest hierarchy level of individual classes. In order to solve this unsatisfactory situation, the European Patent Office has broadened the IPC by further notation digits (decimal) downward. In this way, it has created—where necessary—around 70,000 further technical classes, comprising the European Classification (ECLA) (Dickens, 1994).

Table L.2.3: Classifications in Intellectual Property Rights. Sources: IPC, 2011; Vienna Class., 2008; Nice Agreement, 2007; Locarno Agreement, 2004.

Notation

IPC. International Patent Classification

Example:

A HUMAN NECESSITIES

A 21 BAKING; EQUIPMENT FOR MAKING OR PROCESSING DOUGHS; DOUGHS FOR BAKING

A 21 B BAKERS' OVENS; MACHINES OR EQUIPMENT FOR BAKING

A 21 B 1 / 00 Bakers' ovens

A 21 B 1 / 02 . characterised by the heating arrangements

A 21 B 1 / 06 .. Ovens heated by radiators

A 21 B 1 / 08 ... by steam-heated radiators

Vienna Classification. International Classification of the Figurative Elements of Marks

Example:

03 ANIMALS

03.01 QUADRUPEDS (SERIES I)

03.01.08 Dogs, wolves, foxes

03.01.16 Heads of animals of Series I

A 03.01.25 Animals of Series I in costume

Nice Classification. International Classification of Goods and Services for the

Registration of Trademarks

Example:

Class 1. Chemicals used in industry, science and photography, as well as in agriculture, horticulture and forestry; unprocessed artificial resins, unprocessed plastics; manures; fire extinguishing compositions; tempering and soldering preparations; chemical substances for preserving foodstuffs; tanning substances; adhesives used in industry.

A0035 Acetone

Class 35. Advertising; business management; business administration; office functions.

C0068 Commercial information agencies

Locarno Classification. International Classification for Industrial Designs and Models

Example:

04 Brushware

04-01 Brushes and brooms for cleaning

M0256 Mops

The IPC consists of systematic main tables and specific auxiliary tables ("Index Codes" in the so-called "Hybrid System"). In the IPC's print versions, one recognizes the index codes via the colon (:), which is used instead of the slash (/) normally found in main tables. The IPC does not have any general auxiliary tables.

The Nice Classification plays a significant role for trademarks, since every mark is registered in a class, or several classes, and will only gain currency in this class or these classes. The Nice Classification comprises 45 classes for trademarks and services. It is thus possible that brands with the exact same name will coexist peacefully,

registered in different Nice classes. In the case of picture marks, the representational design of the mark is attempted to be made searchable via concepts. This is done with the help of the Vienna Classification. Design Patents are arranged in product groups according to the Locarno Classification.

Working with economic classifications—industry as well as product classifications (Table L.2.4)—is difficult for users, since a multitude of official and provider-specific systems coexist. The Official Statistics of the European Union uses the NACE classification system (*Nomenclature générale des activités économiques dans les Communautés Européennes*) on the industry level; as do the EU member states, which are granted their own hierarchy level for country-specific industries. The NAICS (North American Industry Classification System) has been used in North America since 1997. Prior to that, official US statistics were based on the SIC (Standard Industrial Classification) from the years 1937 through 1939. Many commercial information providers still use the SIC, however.

The traditional arrangement of industries is based on the SIC; it follows the sectors from agriculture to services, via manufacturing. The SIC has ten main classes on the upper hierarchy level:

- A Agriculture, Forestry, Fishing
- B Mining
- C Construction
- D Manufacturing
- E Transportation, Communications, Electric, Gas, and Sanitary Services
- F Wholesale Trade
- G Retail Trade
- H Finance, Insurance, and Real Estate
- I Services
- J Public Administration.

In the 1930s, the USA were an industrial society, and the SIC was attuned to this fact. By now, many of the former industrial nations have come much closer to being service or knowledge societies. In view of this change, no mere revision of the SIC was possible; in North America, it was decided to introduce a completely new industry classification, the NAICS, which, in addition to the USA, is also used in Canada and Mexico (Pagell & Weaver, 1997). Sabroski (2000, 18) names the reasons for this “revolution” in industry classifications:

The SIC codes were developed in the 1930s when manufacturing industries were the most important component of the U.S. economy. Of the 1,004 industries recognized in the SIC, almost half (459) represent manufacturing, but today the portion of U.S. Gross Domestic Product (GDP) has shrunk to less than 20 percent.

Table L.2.4: Economic Classifications. Sources: NAICS, 2007; NACE, 2008; WZ 08, 2008; SIC, 1987; D&B-SIC, n.d.

Notation

NAICS. North American Industrial Classification System

Example:

- 31-33 Manufacturing
 - 333 Machinery manufacturing
 - 3332 Industrial machinery manufacturing
 - 33329 Other industrial machinery manufacturing
 - 333293 Printing machinery and equipment manufacturing

NACE / WZ 08. Classification of Industries, 2008 edition (WZ 08); the first three hierarchy levels correspond to the NACE, the lowest level only applies to Germany.

- 01 Crop and animal production, hunting and related service activities
 - 01.1 Growing of non-perennial crops
 - 01.19 Growing of other non-perennial crops
 - 01.19.2 Growing of flower seeds

SIC. Standard Industrial Classification (dated)

Example:

- 3000 Manufacturing industry
 - 3500 Industrial and commercial machinery and computer equipment
 - 3550 Special industry machinery (no metalworking machinery)
 - 3555 Printing trades machinery and equipment

Dun & Bradstreet

Example:

- 35550000 Printing trades machinery
 - 35550100 Printing presses
 - 35550101 Presses, envelope, printing

The first hierarchy level of the NAICS (Table L.2.4) still follows the sequence of industries, but it sets completely different emphases—particularly in the third sector. From an information science perspective, it is remarkable that information (Class 51) is now located on the highest hierarchy level (Malone & Elichirigoity, 2003), that indeed the information industry has been one of the catalysts for the revolution (Sabroski, 2000, 22):

Perhaps more than any other, the Information sector evinced the need for a new classification system. This includes those establishments that create, disseminate, or provide the means to distribute information.

Class 51 is subdivided into six industry groups:

- 511 Publishing industries (except Internet)
- 512 Motion picture and sound recording industries,
- 515 Broadcasting (except Internet),

- 517 Telecommunications,
- 518 Data processing, hosting, and related services,
- 519 Other information services.

Since class 519 represents the practical area of information science to a particularly high degree, we would like to list all NAICS classes of information and data processing services (NAICS, 2007):

- 519 Other information services
 - 5191 Other information services
 - 51911 News syndicates
 - 51912 Libraries and archives
 - 51913 Internet publishing and broadcasting and Web search portals
 - 51919 All other information services.

Industry classifications like the SIC, NAICS or the European NACE have roughly 1,000 classes, exclusively in a systematic main table. No auxiliary tables are used at all.

Product classifications can be created by adopting the basic structure of industries from one of the corresponding systems and adding further hierarchy levels for product groups or single products. This is the solution presented by the provider of company dossiers, Dun & Bradstreet. Here the SIC codes are refined by two hierarchy levels, so that D&B-SIC now has far more than 18,000 classes. However, there also exists the method of creating a classification system directly oriented on the products, which is what Kompass did with their classification. Here, too, the first level depicts industrial groups (centurially), the second product and service groups (millennially) and the third products and services (centurially again). The Kompass Classification System works with general auxiliary tables: on the product group level, the notations *I* and *E* are used to describe companies' import and export activities, and on the product level *P* (Production), *D* (Distribution) and *S* (Service) are used to describe the manner of dealing with the product. *4414901P* thus describes a manufacturer of proofing presses, *4414901D* a retailer for such products and *4414901S* a service provider, i.e. a proofing press repair service.

Finally, there are the geographic classification systems. Here, too, systems of official statistics co-exist with systems of private information providers (Table L.2.5). An example of an official geographic classification is the NUTS (Nomenclature des unités territoriales statistiques), maintained by Eurostat, which is used to authoritatively categorize the geographical units of the European Union. Widely distributed among professional information services are the Gale Group's Geographic Codes (now distributed by InSitePro), which list all of the world's countries in a unified system.

Table L.2.5: Geographic Classifications. Sources: InSitePro, n.d.; NUTS, 2007.**Notation**

InSitePro Geographic Codes

Example:

4 Europe
 4EU European Union
 4EUGE Germany

NUTS. Nomenclature de unités territoriales statistiques

Example:

DE Germany
 DEA North Rhine-Westfalia
 DEA2 Administrative District Cologne
 DEA27 Rhine-Erft County

Creation and Maintenance of Classification Systems

One cannot simply create classification systems haphazardly and then hope that they will run on their own. To the contrary: such knowledge organization systems require constant maintenance. With the exception of the fundamental classes, which are high up in the hierarchy and would thus cause great changes were they to be modified, all classes are up for reevaluation when conditions change, as Raschen (2005, 203) emphasizes:

Once the taxonomy has been implemented, it should not be regarded as a delivered, completed project. It will be added to as time goes on, although ... its initial “foundation categories” should remain as static as possible to preserve the durability of the resource.

As opposed to nomenclatures, which are merely concerned with maintaining terms (admitting and deleting them), classifications have the added task of keeping an eye on the hierarchical relations. The specific design of notations can make it very difficult indeed to incorporate a new term into a system. We need only think of a classification that is subdivided decimally at a certain point in the notation, which already has ten concepts on the hierarchy level in question and which must now add an eleventh due to terminological change.

The classification systems we sketched as examples are the result of decades of work. The use of resources in the construction of a classification (e.g. as a KOS for a company’s language) is not to be underestimated. The creation of a classification requires expertise in three areas: information science, the respective professional environment, and computer science (Raschen, 2005, 202).

The skills that the (taxonomy editorial) board should possess will include library classification expertise ..., together with more general knowledge sharing skills. It's also likely that you'll seek input and representation from IT colleagues who may well be implementing the final product to your Intranet.

We can roughly distinguish between two approaches, which perfectly complement each other. The top-down approach proceeds from systematic studies (such as textbooks) on the subject area in question and tries to derive classes and hierarchical relations from the top down. The bottom-up approach starts at the base, with the concrete documents that are to be indexed for content, as well as with the users and their queries, and thus works its way up from the bottom. It is helpful, in the bottom-up approach, to (provisionally) use other methods of knowledge representation in order to glean usable term materials in the first place. The text-word method (for the documents' terminology) as well as folksonomies (for the users' term material) are particularly suited for this purpose. It is principally advisable to consult any existing knowledge organization systems (nomenclatures, classification, thesauri) in order to build on the current state of affairs, i.e. to avoid repetitions.

Choksy (2006) suggests a program consisting of eight steps in order to develop a classification system for an organization:

- Selecting the developing team,
- Determining the role of the classification in company strategy,
- Clarifying the goal and purpose of the system,
- Acquiring “materials” from documents (concept material, existing KOS) (possibly in addition: use of the text-word method),
- Conducting empirical surveys (including interviews) in order to record the users' language (possibly in addition: use of folksonomies),
- Sighting the term material (from steps 4 and 5), consistency inspection, feedback loop to the users,
- Construction of the hierarchical structure (using the concepts that have crystallized in step 6), further consistency inspections and feedback loops with the users,
- Finishing the prototype classification (in addition: creating indices, i.e. providing for natural-language access paths to the classes).

When the classification is finished, it must be made sure that both indexers and users are worked in to the system, and to develop strategies for keeping the classification system up to date.

In those places where the system's future users will come into play (steps 5 through 7), it makes sense to embed the surveys and interviews in a concept of the analysis of cognitive work (CWA).

When constructing hierarchies, and with them—in so far as the notations are created in a structurally representational way—the notations of the classes, enough

spaces for further enhancements are to be made allowance for, according to Tennis (2005, 85):

Classification theory's concern with hospitality in classification schemes relates to how relationships between concepts—old and new concepts—in the classificatory structure are made and sustained. Well designed classificatory structures should make room for new concepts.

The “great” classification systems (like all the ones we addressed here) are maintained permanently; from time to time (about once in five years) a new edition comes out with its own authoritative terminology. Indexers and users must be made aware of innovations—the configuration of which may take a very long time (depending on the number of involved parties). In the UDC, the maintenance process seems to be relatively elaborate and time-consuming, as Manecke (2004, 133) reports:

Submitted complement suggestions were first published as P-Notes (Proposals for revision or extension), and thus submitted to professional circles for discussion. Only after an objection period of several months elapsed did they become authoritative and were made known via the yearly “Extensions and Corrections to the UDC”.

Classifications used in companies principally proceed along the same lines, except the configuration process and establishment of the innovations should take a lot less time to complete.

Conclusion

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- Classifications distinguish themselves via their use of notations as the preferred terms of classes as well as their use of the hierarchy relation.
 - Notations are derived from an artificial language (generally digits or letters); they thus grant an access which is unhindered by the limitations of natural languages.
 - When enhancing a classification system's hierarchy from the top down, we speak of hospitality in the concept ladder; when enhancing it on a hierarchical level, via co-hyponyms, we speak of hospitality in the concept array. Both aspects of hospitality must always be given, since the system cannot be adjusted to changes otherwise.
 - Numeral notations work decimally (one system digit), centurially (two digits) or millennially (three digits). Notations consisting of letters or of mixtures of digits and letters can make up codes as well.
 - In structurally representational and hierarchical notations, the systematics becomes visible in the notation itself. Such notations grant the usage of truncation symbols in hierarchical searches. Sequential notations name the classes enumeratively, so that no meaningful truncation is possible. There are mixed forms in the form of hierarchical-sequential notations.
 - Notations form the basis of a classification system (independently of natural languages). Access for the users is additionally granted via natural-language interfaces. This is done by having every class admit terms—which can vary according to the language. In this way, one solves both the

problems of synonymy (all synonyms are recorded as class designations) and homonymy (the user is referred to the different classes by the respectively homonymous words).

- Sometimes syncategorematical terms are unavoidable. Only in the context of the surrounding concepts does the meaning become obvious. In order to solve the problems posed by syncategoremata, one works with indirect hits that take into consideration the entire ladder of a concept when searching for a specific class.
 - The most important challenges posed by classification systems to their developers are the selection and the hierarchical arrangement of the concepts. We cannot assume that there are any natural classifications; such systems must be created specifically to their purposes.
 - The extension of co-hyponyms (sister concepts) should match the extension of their hyperonym. This principle of extensional identity can be secured via an unspecific co-hyponym called *other* (which records anything hitherto unrecognized). The prototypes of co-hyponyms should be disjunctive.
 - In the case of compound terms, hierarchization orients itself on the components.
 - The arrangement of classes into concept ladder and arrays, respectively, is never arbitrary but adheres to comprehensible principles. The resulting citation order, which arranges thematically related classes, and hence documents, next to each other, is one of the great advantages of classification systems. As such, it must be emulated in digital environments (“digital shelving systematics” as a criterion for topical relevance ranking).
 - When classifications systematize shelving arrangements in libraries, a user who stands in front of the text he was looking for will expect to also find relevant documents to the left and right of this direct hit. Classification systems must meet these expectations. Thematic breaks (i.e. the transition from one main class to another) must be made recognizable.
 - In digital databases, there are no physical documents arranged on a shelf. Since the principle of thematic neighborhood has proven itself, however, it must be simulated in digital environments.
 - The classes are systematically arranged in tables. Notes, year of entry, year of deletion (sometimes), instructions (such as “class here!”) and, occasionally, a see and see-also function are attached to every concept.
 - Particularly for small classifications, it is possible to work with one single table. For larger systems, it makes far more sense to divide the tables into main and auxiliary tables. The latter will record those concepts that can be meaningfully attached to different classes from the main tables.
 - Classification systems are used—have been used for decades, here and there—in libraries (particularly universal classifications), in medicine and health care, in intellectual property rights, in economic documentation (industry and product classifications) as well as in the arrangement of geographical data.
 - The construction of classifications is pursued both top-down, from a systematic perspective, and bottom-up, building on empirical material. The developing team (ideally comprising information scientists, experts in the respective field and computer scientists) must know the language of the documents and of the users. This language must then be modeled with regard to the company strategy and the goals and purposes of its application.
 - In order to clarify terms and hierarchies, it is of use to interview experts and users as well as to preliminarily use other methods of knowledge representation (mainly the text-word method and folksonomies).
 - Classifications are permanently to be adjusted to terminological change. If the knowledge in a domain changes to large degrees, the old classification can hardly be processed further. All that will be left is its deletion and the development of a new system (as in the transition from SIC to NAICS).
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L.3 Thesaurus

What Purpose does a Thesaurus Serve?

As opposed to classifications, which describe concepts via notations and sum them up into classes according to certain characteristics, thesauri, like many nomenclatures, follow natural language, linking terms into small conceptual units with or without preferred terms and setting these into certain relations with other descriptions or concepts. The term “thesaurus” is from the Greek meaning originally, according to Kiel and Rost (2002, 85), the collection and storage of treasures for the gods in small temples built specifically for this purpose. The ANSI/NISO standard Z39.19-2005 (166) describes the history of the term:

In the 16th century, it began to be used as a synonym for “Dictionary” (a treasure store of words), but later it fell into disuse. Peter Mark Roget resurrected the term in 1852 for the title of his dictionary of synonyms. The purpose of that work is to give the user a choice among similar terms when the one first thought of does not quite seem to fit. A hundred years later, in the early 1950th, the word “thesaurus” began to be employed again as the name for a word list, but one with the exactly opposite aim: to prescribe the use of only one term for a concept that may have synonyms. A similarity between Roget’s Controlled Thesaurus and thesauri for indexing and information retrieval is that both list terms that are related hierarchically or associatively to terms, in addition to synonyms.

In knowledge representation, the thesaurus represents a concept order, whose functions and characteristics are determined via rules (ISO 25964-1:2011). Following DIN 1463/1 (1987, 1), created in connection with ISO 2788:1986, “thesaurus” is defined and its field of application specified:

A thesaurus in the context of information and documentation is a structured compilation of concepts and their (preferentially natural-language) designations, which is used for indexing, storage and retrieval in a field of documentation.

A thesaurus is characterized via the following features: concepts and designations must relate unequivocally to each other (“terminological control”) by having synonyms be registered and checked as comprehensively as possible, homonyms and polysemes marked individually and especially by specifying a preferred designation for every concept (descriptor with a identification number) that conclusively represents the respective term. Hierarchical and associative relations between concepts are represented.

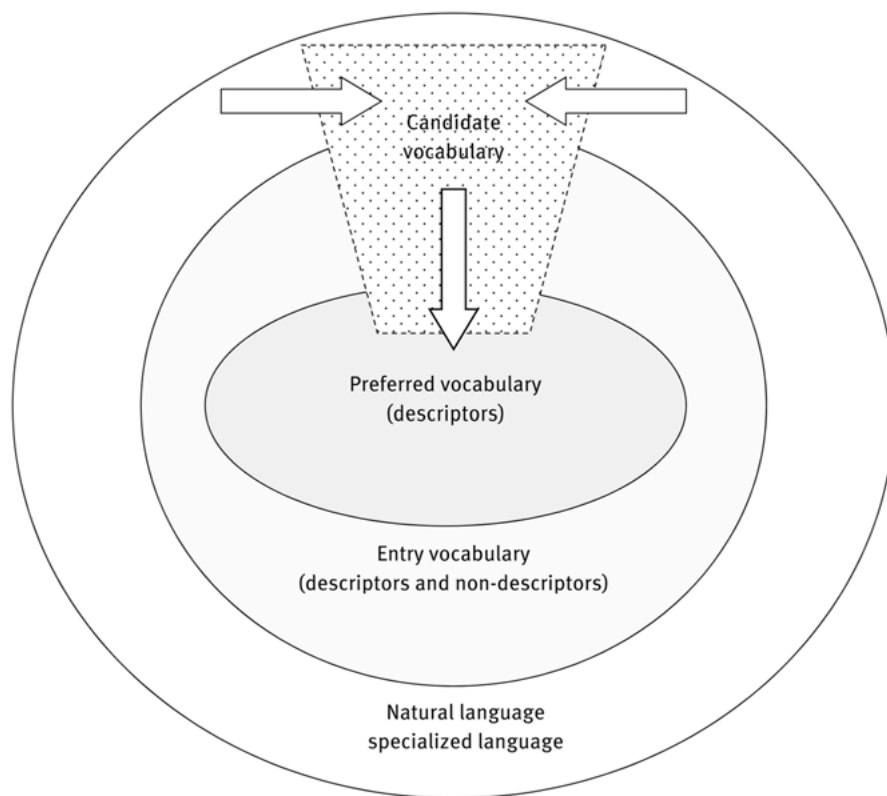


Figure L.3.1: Entry, Preferred and Candidate Vocabulary of a Thesaurus. Source: Modified from Wersig, 1985, 85.

With regard to terminological control, a distinction is made between two kinds of thesauri: in a thesaurus without preferred terms, all designations for a concept are allowed for indexing, all equivalent designations for a concept have to be given the same identification number. In a thesaurus with preferred terms, only preferred terms, called “descriptors”, are allowed for indexing. Non-descriptors are listed in the thesaurus, but they only serve the user as a tool for gaining access to the descriptor more easily. On the one hand, the thesaurus needs the controlled preferred vocabulary for indexing, storage and retrieval, and on the other hand, it needs the complementary entry vocabulary of constantly changing natural and specialized language. In a transit area, entry and preferred vocabulary are tested to see whether certain descriptor candidates are suitable for the thesaurus. Figure L.3.1 shows the interplay of the vocabularies as illustrated by Wersig (1985, 85).

Wersig (1985, 27) describes six components that make up a thesaurus. Starting from natural language and the language use of a specialist area (1), the thesaurus is used to effect a language inspection of the vocabulary (2), of the synonyms and the

homonyms. The thesaurus, as an instrument for representing meaning, also assumes a conceptual control function (3). Conventions on the part of the information service (4) play a role with regard to language use, vocabulary application and language control. Prescription (5) applies, as thesaurus terms are prescribed and interpreted for indexing and retrieval. Due to the characteristics stated thus far, the thesaurus serves as an instrument of orientation (6) between the language use and thought structures within the respective specialist area and the system in which it is applied.

Raw terms of a natural or specialist language contain ambiguities and blurry areas, and are thus edited via terminological control, i.e. the checking of synonyms, homonyms as well as morphological and semantic factoring. The vocabulary at hand is checked, compiled into conceptual units via the objects and their characteristics, and arranged into classes. The conceptual control then makes the network of relationships between the single elements visible. This procedure is meant to guarantee the unambiguity of the interrelations between concepts and their designations, respectively, and thus to create a large semantic network for the knowledge domain. Three basic types of relation here assume a predominant position: the equivalence, hierarchy and associative relations. All relations are either symmetrical (equivalence and association) or have inverse relations (hierarchy).

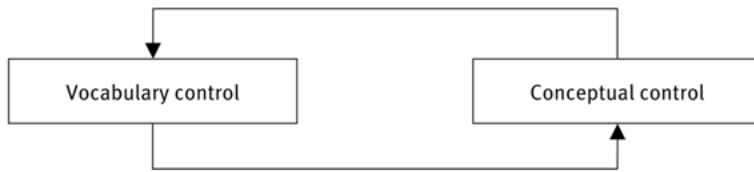


Figure L.3.2: Vocabulary and Conceptual Control.

A thesaurus is confronted with the two aspects of terminological control (vocabulary control) and conceptual control (Figure L.3.2). The results of the vocabulary control are descriptors and non-descriptors, those of conceptual control the paradigmatic relations between the concepts. Here, both aspects are coordinated. From the beginning of the vocabulary control onward, the relations must be taken into consideration, and during the conceptual control, the designations of the respective concepts are of significance. This interdependence of vocabulary and relations is what distinguishes a thesaurus from a nomenclature, since in the latter relations that exceed synonymy and association (“see also”) have no significance.

Vocabulary Relations

One of the first steps of vocabulary control is to determine what forms the designations should take. This includes grammatical forms (e.g. nouns, phrases, adjectives),

singular and plural forms (in specific or singular entities and abstract concepts), linguistic forms (e.g. British or American English), transliteration, punctuation, capitalization within one word (e.g. in proper names) as well as abbreviations (Aitchison, Gilchrist, & Bawden, 2000). When selecting a term, it must be decided whether established formulations should be adopted from other languages, as well as how neologisms, popular and vernacular expressions versus scientific expressions, brand names and place, proper or institutional names should be handled.

So far, we are merely talking about individual words, with no relation to each other. In terminological control, these are processed into conceptual units, in order to then be classified into the framework of the thesaurus. Burkart (2004, 145) writes:

The conceptual units that result are called equivalence classes, as they sum up within them all terms rated roughly the same in the thesaurus' area of influence. They form a sort of floodgate, through which all indexing results and search queries must pass.

The most commonly used description of an equivalence class receives the leading position (in a thesaurus with preferred terms) and is selected as the descriptor. Due to the terminological control, we are left with a multitude of classes that are still isolated from each other. These equivalence classes are the result of synonym control, homonym control and semantic factoring. Here we distinguish between five sorts of vocabulary relations. In this context, we will write the non-descriptors in lower-case letters and the descriptors in upper-case letters for the purposes of exemplification.

1. Designations-Concept Relation (Synonymy)

For designations with equal or similar meaning, a preferred term is set as the descriptor. Some examples for synonymy are: photograph–photo: PHOTOGRAPH; car–auto-mobile: CAR.

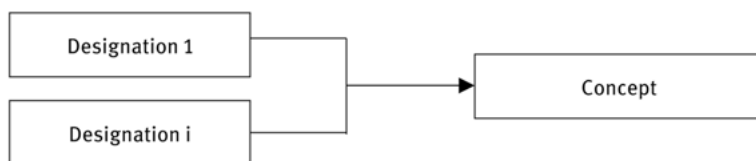


Figure L.3.3: Vocabulary Relation 1: Designations-Concept (Synonymy).

The relation between similar concepts yields identical results to synonymy. Here the terms taken to be “quasi-synonymous” in the context of the knowledge organization system (e.g. ‘search’ and ‘retrieval’ in a business thesaurus) are amalgamated into one term (e.g. SEARCH). Synonyms, or quasi-synonyms, are summarized as one individual descriptor in a thesaurus.

2. Designation-Concepts Relation (Homonymy)

In a designation with different meanings, a piece of additional information (the qualifier) is provided for the concept. Our example for homonymy is Java. Java, the Indonesian island, becomes the descriptor JAVA <ISLAND>, Java the computer language is JAVA <PROGRAMMING LANGUAGE>. Homonyms are categorically separated in the thesaurus.

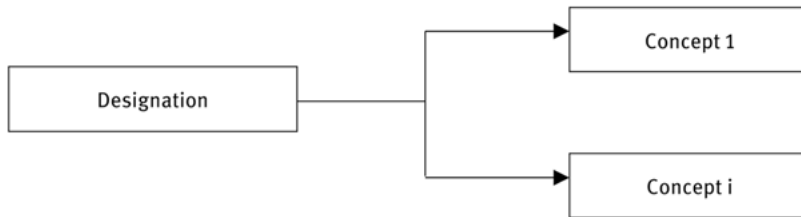


Figure L.3.4: Vocabulary Relation 2: Designation-Concepts (Homonymy).

3. Intra-Concepts Relation (Decompounding)

Expressions that consist of multiple words and express a complex concept may be changed into compound concepts via semantic splitting. This is done either via pre-combination in the thesaurus (e.g. LIBRARY STATISTICS), via precoordination by the indexer (e.g. LIBRARY / STATISTICS) or via postcoordination of the two concepts during the search process.

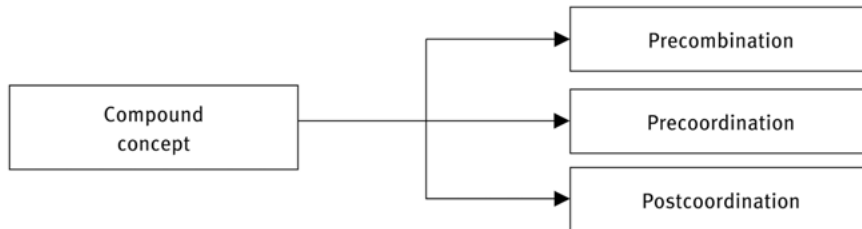


Figure L.3.5: Vocabulary Relation 3: Intra-Concepts Relation (Splitting).

4. Inter-Concepts Relation I (Bundling)

Concepts that are too specific for the thesaurus in question are summarized as a more general concept via bundling. Bundling is a hierarchy relation between non-descriptors (e.g. lemon–orange–grapefruit) and a descriptor (e.g. CITRUS FRUIT). In bundling, the non-descriptors are hyponyms of the descriptor.

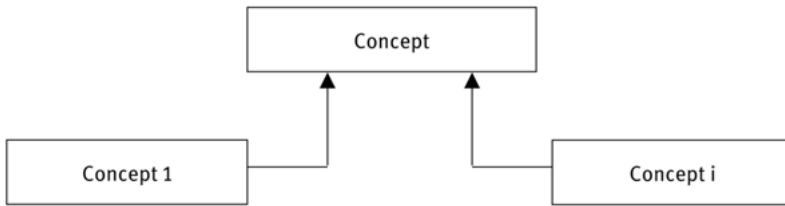


Figure L.3.6: Vocabulary Relation 4: Inter-Concepts Relation as Bundling.

5. *Inter-Concepts Relation II (Specification)*

This relation between concepts is also a hierarchy relation, this one being between a non-descriptor as the hyperonym and several descriptors. A too-general concept for the thesaurus is replaced by specific ones, e.g. natural sciences replaced by CHEMISTRY, PHYSICS and BIOLOGY.

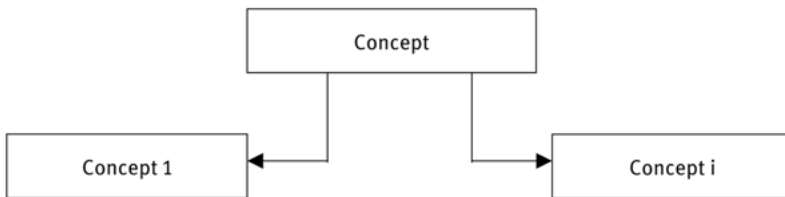


Figure L.3.7: Vocabulary Relation 5: Inter-Concepts Relation as Specification.

Descriptors and Non-Descriptors

When incorporating terms into the thesaurus, it must be decided what designation is suitable as a descriptor, or non-descriptor, and in what relations that term stands vis-à-vis other designations in the thesaurus. The selection of a possible descriptor should depend upon its usefulness: how often does it appear in the information materials? How often is its appearance to be expected in search queries? How suitable is its range of meaning? Does it conform to the current terminology of the discipline in question? Is it precise, as memorable as possible, and uncomplicated? Descriptors represent general and individual concepts.

The non-descriptor is allocated, among other tasks, to the role of indicating the range and content of the concept. It can be characterized via the following aspects: its spelling is not widely used. It is used differently in specialist language than in everyday language. It does not fit current linguistic usage. It summarizes a combination of previously available descriptors. It is a hyponym (of small importance) of a descriptor, a hyperonym (of equally small importance) of descriptors or is, finally, used as a synonym, or quasi-synonym, for another descriptor.

If a thesaurus is used for automatic indexing, the entry vocabulary will be granted a special role. Here one must anticipate, as far as possible, under which designations a concept can appear in the documents to be automatically indexed. The list of non-descriptors will, in such a case, be far larger (and possibly contain “common” typos) than in a thesaurus used in a context of intellectual indexing.

According to DIN 1463/1 (1987, 2-4), certain formal stipulations apply, concerning compound terms, concept representation via several descriptors and word forms. A descriptor is supposed to mirror the terminology common within the specialist literature in question. If clarity allows, the concept that is meant must be defined via *one* term only. Compound terms may never be separated; accordingly, “information science” and “science information” will always stay together as composites (ANSI/NISO, 2005, 39). Multi-word descriptors (as, for instance, in adjectival phrases) are rendered in their natural word order. Inverted forms can be treated as synonyms. As too many compounds will only lead to a confusing and bloated thesaurus, it makes sense to represent a concept via the combination of already available descriptors. This can happen via semantic splitting. As opposed to precombination, which is already determined during the allocation of the descriptors, postcoordination is the process of assembling the required concept via several pre-existing descriptors during the search procedure. Burkart (2004, 144) explicitly points out that splitting only ever deals with concepts and not with words, pleading for the establishment of a middle course, dependent on the respective system, between complete postcoordination, whose components of meaning are not further collapsible (uni-term procedure) and an extreme precombination:

When splitting is called for and when precombination, is thus to be decided system-specifically in the individual thesauri, and has to always stay a subjective decision, up to a certain degree. This is why it is particularly important to anchor this in the thesaurus as comprehensibly as possible.

In splitting, it is important to note that the designations at hand are only the representatives of the concepts. The purpose of splitting is to collapse the concept into conceptual components, not the word into its parts.

With regard to word form, DIN 1463/1 (1987) specifies that descriptors are, preferably, to be formed as nouns. Non-nominalized forms (the infinitive of the verb where possible) are allowed if an activity is meant. Adjectives may not stand on their own, but must be tied to a noun. Descriptors should normally be used in the noun singular, unless the use of the singular is uncommon or the plural of the term carries another meaning than its singular form.

As opposed to the German practice, in which the descriptors are generally used in the singular, English-language thesauri (with various exceptions) work with plural forms (Aitchison, Gilchrist, & Bawden, 2000, 21).

There is an equivalence relation between descriptors and non-descriptors. It relates to coextensive, or similar, designations united in an equivalence class and

regarded as synonyms or quasi-synonyms. It must be noted that the equivalence relation does not comprise terminologically “correct” terms, but—as Dextre Clarke (2001, 39) emphasizes—depends upon the genre and depth of the thesaurus, thus always requiring a subjective determination:

Terms such as “porcelain”, “bone china” and “crockery”, which might be individual descriptors in a thesaurus for the ceramics industry, could well be treated as equivalents in a thesaurus for more general use. Which relationship to apply between these terms is a subjective decision, depending on the likely scope and depth of the document collection to be indexed, as well as the background and likely interests of the users who will be searching it.

It is thus impossible to determine an unambiguous equivalence, or similarity, in meaning between the elements of the equivalence class. The following equivalence types may be contained within thesauri, among other reference works (ANSI/NISO, 2005; Dextre Clarke, 2001; Aitchison, Gilchrist, & Bawden, 2000; Burkart, 2004):

- popular–scientific names: *salt* – *sodium chloride*,
- generic–trade names: *tissues* – *Kleenex*,
- standard names–slang or jargon: *supplementary earnings* – *perks*,
- terms of different linguistic origin: *buying* – *purchasing*,
- terms from different cultures (sharing a common language): *elevators* – *lifts*,
- variant names for emerging concepts: *lap-top computers* – *notebook computers*,
- current–outdated terms: *dishwashers* – *washing-up machines*,
- lexical variants (direct–inverted order): *electric power plants* – *power plants*, *electric*,
- lexical variants (stem variants): *Moslems* – *Muslims*,
- lexical variants (irregular plurals): *mouse* – *mice*,
- lexical variants (orthographic variants): *Rumania* – *Romania*,
- full names–abbreviations, acronyms: *polyvinyl chloride* – *PVC*,
- quasi-synonyms (variant terms): *sea water* – *salt water*,
- quasi-synonyms (antonyms): *dryness* – *wetness*.

For quasi-synonyms, it is also possible to introduce different descriptors and to interlink these via an associative relation.

A particular kind of equivalence relation for an English-language thesaurus is the differentiation between British and American idioms. Dextre Clarke here talks about a dialectical form of equivalence, where the abbreviation *AF* stands for American and *BF* for the British form (Dextre Clarke, 2001, 41), as in the example of *autumn* – *AF fall* and *fall* – *BF autumn*.

Relations in the Thesaurus

The relations between concepts, or their designations, are denoted via so-called cross-references, according to the language usage of librarians. The necessity of the

relations, as well as of the corresponding cross-references, is justified thus in DIN 1463/1 (1987, 5):

In this way, the relations of a descriptor to other terms (descriptors or non-descriptors) convey a definition of the descriptor, in a way, since it shows its place in the semantic fabric.

Relations connect descriptors and non-descriptors among one another, and serve as guides through the thesaurus's network. Apart from the equivalence relation, thesauri also have hierarchy relations as well as the associative relationship (Evans, 2002).

In the hierarchy relation, the descriptors assume either a subordinate or a superordinate role vis-à-vis each other. One can also distinguish more strictly, though, between three types of hierarchy (Dextre Clarke, 2001; Aitchison, Gilchrist, & Bawden, 2000). In hyponymy (also called generic relation or abstraction relation), the narrower term possesses all characteristics of the superordinate concept, as well as at least one additional feature. This relation is used, for instance, in actions (e.g. *annealing* – *heat treatment*), properties (e.g. *flammability* – *chemical properties*), agents (e.g. *adult teachers* – *teachers*) and all cases of objects that are hierarchically coupled (e.g. *ridge tent* – *tent*), where genus and species always have to refer to the same fundamental category. In the partitive relation (meronymy), the superordinate concept corresponds to a whole, and the narrower (partial) term represents one of the components of this whole. This applies to, for instance, complexes (e.g. *ear* – *middle ear*), geographical units (e.g. *Federal Republic of Germany* – *Bavaria*), collections (e.g. *philosophy* – *hermeneutics*), social organizations (e.g. *UN* – *UNESCO*) and events (e.g. *soccer match* – *half-time*). The third kind of hierarchy relation is the instantial relation, in which the subordinate concept represents a proper name, as a one-off case (e.g. *German church* – *Frauenkirche <Munich>*).

In a hierarchy relation, it must be very carefully considered what the characteristics of broader and narrower terms are, and how many levels of subordination must be consulted. A circle is to be categorically excluded. Not too many descriptors must occur as co-hyponyms (sister concepts) in a sequence of concepts, because this would make the entire structure too confusing. Losee (2006, 959) exemplifies the problem of making a decision during the creation of a hierarchy on the example that *human* may have, on the one hand, the hyponyms *man* and *woman* or *adult* and *child* or, on the other hand—and wholly unexpectedly—*person liking broccoli* or *person disliking broccoli*, perhaps in a knowledge organization system for food marketing:

(S)hould a broad term such as *human* be broken down into *female* and *male*? Criteria are also provided for determining whether one type of class description of children is better than another type, such as whether *humans* are better defined as either *female* or *male* or whether humans should be subdivided into *adults* or *children*, or perhaps as those who like broccoli or those who dislike broccoli.

There still remains an insecurity as to whether the hierarchy has been developed to the user's satisfaction; it can, thus Losee, be lessened via feedback with the user and in the form of a dynamic thesaurus. A dynamic thesaurus is always adjusted to the requirements of the users as well as to the developments within the knowledge domain. This is the standard scenario of a thesaurus.

Descriptors that are neither equivalent to each other nor in any hierarchical relation, but are still related or sort of fit together somehow, are related associatively. The function of the associative relationship is to lead the user to further, possibly desired, descriptors. Suggestions for additional or alternative concepts for indexing or retrieval are offered. Following Aitchison, Gilchrist and Bawden (2000), Dextre Clarke (2001, 48) lists different types of associative relations:

- overlapping meanings: *ships – boats*,
- whole–part–associative: *nuclear reactors – pressure vessels*,
- discipline–object: *seismology – earthquakes*,
- process–instrument: *motor racing – racing cars*,
- occupation–person in occupation: *social work – social workers*,
- action–product of action: *roadmaking – roads*,
- action–its patient: *teaching – student*,
- concept–its property: *women – femininity*,
- concept–its origin: *water – water wells*,
- concept–causal dependence: *erosion – wear*,
- thing / action–counter agent: *corrosion – corrosion inhibitors*,
- raw material–product: *hides – leather*,
- action–associated property: *precision measurement – accuracy*,
- concept–its opposite: *tolerance – prejudice*.

Schmitz-Esser (1999; 2000, 79) knows further relations:

- usefulness: *job creation – economic development*,
- harmfulness: *overfertilization – biological diversity*.

A thesaurus in economics uses the following relations:

- customary association: *body care product – soap*,
- associated industry: *body care product – body care industry*.

Nothing can be said against foregoing the general, unspecific associative relation and instead using the relation specific to the respective knowledge domain.

Table L.3.1: Abbreviations in Thesaurus Terminology. Abbr.: ND: Non-Descriptor.

German	English	Description
Relations		
TT Top Term	TT Top Term	Topmost term
OB Oberbegriff	BT Broader Term	Hyperonym in general
OA Oberbegriff	BTG Broader Term (generic)	Hyperonym, abstraction relation
SP Verbandsbegriff	BTP Broader Term (partitive)	Holonym, meronymy
	BTI Broader Term (instantial)	Hyperonym / holonym, instancial relation
UB Unterbegriff	NT Narrower Term	Narrower term in general
UA Unterbegriff	NTG Narrower Term (generic)	Hyponym, abstraction relation
TP Teilbegriff	NTP Narrower Term (partitive)	Meronym, meronymy
	NTI Narrower Term (instantial)	Individual term, instancial relation
VB Verwandter Begriff	RT Related Term	Associative relation
BS Benutze Synonym / Quasisynonym	USE Use	Equivalence: ND–descriptor
BF Benutzt für Synonym / Quasisynonym	UF Used for	Equivalence: descriptor–ND
BK Benutze Kombination	USE Use	Partial equivalence: ND–several descriptors
KB Benutzt in Kombination	UFC Used for combination	Partial equivalence: descriptor–ND
BO Benutze Oberbegriff	USE Use	Narrower term (ND)–descriptor; bundling
FU Benutzt für Unterbegriff	UF Used for	Descriptor–narrower term (ND); bundling
BSU Benutze Unterbegriff	USE Use	Broader Term (ND)–descriptor; specification
BFO Benutzt für Oberbegriff	UF Used for	Descriptor–broader Term (ND); specification
Hinweise und Definitionen	Notes	
H Hinweis	SN Scope Note	
	XSN See Scope Note for	
	HN History Note	
D Begriffsdefinition	D Definition	

The concept relations are often represented via standard abbreviations or symbols. For multilingual thesauri, language-independent symbols may be of advantage, since the common tokens of the designations vary from language to language. Table L.3.1 lists a few abbreviations for references and relations to German and English definitions.

Descriptor Entry

All determinations that apply to a given concept are summarized in a concept entry. Since most thesauri are those with preferred terms, this terminological unit is also called a descriptor entry. This includes descriptors / non-descriptors, relations, elucidations, notes and definitions. Additionally, statements regarding concept number, notations, status statements as well as the date of entry, correction or deletion are possible (Burkart, 2004, 150). The elucidations, also called “scope notes”, provide context-dependent notes on the meaning and use of a concept and its delineation within the thesaurus in question. The definitions, on the other hand, are not meant for a specific thesaurus, but reflect the general applicability of a term within a knowledge domain. A history note indicates the development of terms over time and how a term has changed. A concept or descriptor entry generates the semantic environment. Non-descriptors accordingly receive their unambiguous relations to their preferred terms in a non-descriptor entry.

Our example for a descriptor entry has been taken from the Medical Subject Headings (MeSH, 2013). The National Library of Medicine, Bethesda, USA, develops and maintains this thesaurus. MeSH are used for sources from medicine and its fringe areas. Descriptors are termed “Main Headings” or “MeSH Headings”, non-descriptors are the “Entry Terms”. The polyhierarchically structured systematic thesaurus is called “MeSH Tree Structures” and comprises 16 main categories, which are each abbreviated via a single letter; for instance: Anatomy [A], Organisms [B], Diseases [C], Chemical Drugs [D] or Geographicals [Z].

Figure L.3.8 shows, on the example of *Tennis Elbow*, a descriptor entry (with the identification number D013716) in MeSH (Nelson, Johnston, & Humphreys, 2001).

The descriptor is located in two concept ladders of C (*Diseases*), starting, on the one hand, from C05 (*Musculoskeletal Diseases*), and from C26 (*Wounds and Injuries*) on the other. Correspondingly to the position of the descriptor in both concept ladders, *Tennis Elbow* receives two Tree Numbers. Under “Annotation”, we find notes concerning the applicability of our descriptor, e.g., that *Tennis Elbow* should not be used in conjunction with *Tennis*, under certain circumstances, unless the sport of tennis is explicitly mentioned in the document. The only non-descriptor is *Epicondylitis, Lateral Humeral*. The descriptor has been created on 03/02/1981, and is first used in the 1982 version of MeSH. The descriptor entry lists the prehistory of the concept in MeSH (*Athletic Injuries, Elbow* etc.). Apart from the number of the descriptor, MeSH uses a separate number (unique identifier UI) for the concept (here: Concept UI M0021164) and for all designations (here: UI T040293 for the preferred term and UI T040292 for the non-descriptor). For the Semantic Type of the concept, we draw on the terminology of the UMLS (Unified Medical Language System). Our term thus has to do with *Injury or Poisoning* (T037) and with *Disease or Syndrome* (T047).

MeSH Heading	Tennis Elbow
Tree Number	C05.906
Tree Number	C26.088.890
Annotation	in "housewives, artisans & violinists" as well as in tennis players; do not coord with <u>TENNIS</u> (NIM) unless the sport is particularly discussed; do not coord with <u>ATHLETIC INJURIES</u> (IM) unless tennis elbow is discussed as an athletic inj
Concept 1 (Preferred)	Tennis Elbow
Concept UI	M0021164
Scope Note	A condition characterized by pain in or near the lateral humeral epicondyle or in the forearm extensor muscle mass as a result of unusual strain. It occurs in tennis players as well as housewives, artisans, and violinists.
Semantic Type	T037 (Injury or Poisoning)
Semantic Type	T047 (Disease or Syndrome)
Term (Preferred)	Tennis Elbow
Term UI	T040293
Date	01-JAN-1999
Lexical Tag	NON
Thesaurus	NLM (1982)
Term	Epicondylitis, Lateral Humeral
Term UI	T040292
Date	02-MAR-1981
Lexical Tag	NON
Thesaurus	UNK (19XX)
Allowable Qualifiers	BL CF CI CL CN CO DH DI DT EC EH EM EN EP ET GE HI IM ME MI MO NU PA PC PP PS PX RA RH RI RT SU TH UR US VE VI
Previous Indexing	Athletic Injuries (1968-1981)
Previous Indexing	Elbow (1966-1981)
Previous Indexing	Elbow Joint (1966-1981)
Previous Indexing	Sports (1966-1975)
Previous Indexing	Sports Medicine (1966-1967)
Previous Indexing	Tendinitis (1966-1981)
Previous Indexing	Tennis (1976-1981)
History Note	\$2
Date of Entry	19810302
Unique ID	D013716

MeSH Tree Structures

Musculoskeletal Diseases [C05]

- Bone Diseases [C05.116] +
- Cartilage Diseases [C05.182] +
- Fasciitis [C05.321] +
- Foot Deformities [C05.330] +
- Foot Diseases [C05.360] +
- Hand Deformities [C05.390] +
- Jaw Diseases [C05.500] +
- Joint Diseases [C05.550] +
- Muscular Diseases [C05.651] +
- Musculoskeletal Abnormalities [C05.660] +
- Rheumatic Diseases [C05.799] +
- Tennis Elbow [C05.906]

Wounds and Injuries [C26]

Arm Injuries [C26.088]

- Forearm Injuries [C26.088.268] +
- Humeral Fractures [C26.088.390]
- Shoulder Dislocation [C26.088.666]
- Shoulder Fractures [C26.088.749]
- Tennis Elbow [C26.088.890]
- Wrist Injuries [C26.088.906]

Figure L.3.8: Descriptor Entry in MeSH. Source: Medical Subject Headings, 2013.

A particularity of MeSH is its use of qualifiers, i.e. additional information that specifies the descriptors. The descriptor and its qualifier meld into a new unit, which—in addition to the singular descriptor—can be searched as a whole. In the field of “Allowable Qualifiers”, we see a list of abbreviations of the qualifiers for *Tennis Elbow*. Let us suppose that a document describes tennis elbow surgery. Here, indexing occurs not via the two descriptors *Tennis Elbow* and *Surgery*, but via the descriptor and the qualifier as a unit, i.e.: *Tennis Elbow/Surgery* (rendered as SU in the chart). The use of qualifiers results in heightened precision during the search.

Presentation of the Thesaurus for the User

Once a thesaurus has been compiled and made accessible to users, the latter should be clearly informed as to the purpose and structure of the thesaurus via appropriate complementary information. These include elucidations of the rules that were followed when selecting descriptors and in the sorting sequence, user instructions with exemplary help, a view to tendencies for further development, statements of the number of descriptors and non-descriptors as well as the tools that were used.

The thesaurus is represented in an alphabetical and a systematic part. The systematic part can be polyhierarchically structured. Connected concept ladders can be linked with other concept ladders via cross-references. The relations, often complicated, between interconnected descriptor groups can be made clear graphically elucidated via relationship graphs (diagrams, arrows). This visually memorable form is a useful tool for the user during retrieval.

Multilingual Thesaurus

In the course of international cooperation, multilingual thesauri are being developed and implemented in order to circumvent any language barriers within a knowledge domain or a multinational corporation, as far as possible (ISO 5964:1985). Languages, however, cannot be translated with full precision, due to their dependence on culture (Hudon, 2001; Jorna & Davies, 2001). A term may exist in one language, but be missing in another due to the latter’s different tradition. There are additional problems for the multilingual thesaurus. We can distinguish between symmetrical and non-symmetrical multilingual thesauri (IFLA, 2005).

For the non-symmetrical thesauri, there exist language-dependent structures, whose terms are brought together via translations. In symmetrical multilingual thesauri, the rule is that the different national-language surfaces of the respective concepts be equivalent, since otherwise the thesaurus structure would be damaged. The independence of every single language represented in the thesaurus, however, should be taken into consideration, and hence no equivalence be *forced*. This means

that the existing demands upon single-language thesauri are to be complemented via a few aspects, if not changed. At first, it must be determined what status each of the participating languages is to be granted. One can decide between main (or source) language and secondary language. The main language here takes the position that is used for indexing and retrieval, where every concept of the system is represented via a descriptor of the main language.

Only when all languages contain equivalent descriptors for the concepts to be represented we can speak of status equality between the languages (DIN 1463/2:1988, 2). The language-independent descriptor entry is, in this case, compiled via an identification number; all natural-language equivalences are exclusively surfaces of this entry.

Designations in the main language and the identification number, respectively, are transferred into the corresponding target language. Not always are common equivalents available in the target language for translation, as the example of *teenagers* in German shows. On the other hand, there might be several terms to choose from: *mouton*, from the French, can be translated into English as either *sheep* or *mutton*.

It appears to be often necessary to coin new expressions for the target language (via new words or constructed phrases) in order to represent a concept from the first language. Thus, there may be an English translation for the German *Schlüsselkind*, *latchkey child*, but in the French language there is only the artificial, literal translation of *enfant à clé*, which requires additional information in order to be understood. Hudon (1997, 119) criticizes this solution of the problem, as a thesaurus is not meant to represent a terminological termbank that makes a language partly artificial:

The creation of neologisms is never the best solution. A thesaurus is not a terminological termbank. The role of a thesaurus is not to bring about changes in a language, it is rather to reflect the specialized use of that language in certain segments of a society.

A multilingual thesaurus, thus Hudon (2001, 69), should, where possible, consider the following problems: a language must not be overtaxed in such a way that its own speakers hardly recognize it anymore, only in order to fit a foreign conceptual structure. The entire relational structure of a cultural context need not have to be transferred to another. The translation of terms from the original language must not result in any meaningless terms in the target language.

Sometimes, a language lacks hierarchical levels. In German, we have the following concept ladder:

Wissenschaft,
Naturwissenschaft,
Physik.

In English, there is only a need for two levels:

Science,
Physics.

If we want to construct the multilingual thesaurus from a German perspective, we must—as the hierarchical structure is always to be preserved—introduce the term *Wissenschaft* as a foreign word and elucidate it (DIN 1463/2:1988, 12):

Wissenschaft (SN: loan term adopted from the German),
Science,
Physics.

Following Schmitz-Esser (1999, 14), the general structure of a multilingual thesaurus can be characterized as follows (see Figure L.3.9). A descriptor is unambiguously identified via a number (ID). Schmitz-Esser here speaks of a “Meta Language Identification Number” (MLIN). All appropriate hierarchical relations, whose goals (descriptors) are each also characterized via an identification number, are attached to the descriptor ID. The thesaurus structure matches all descriptors in the different languages. Since the individual language is determined by cultural background and linguistic usage, respectively, there are differences, in multilingual thesauri, with regard to the equivalence relation; form and quantity of the non-descriptors (Schmitz-Esser describes them as “additional access expressions”, AAE) vary in the linguistic equivalents.

Let us sketch what we stated above via an example. AGROVOC is a multilingual thesaurus that deals with the terminology of the areas of agriculture, forestry, fisheries, food and related domains. It was developed, in the 1980s, by the Food and Agriculture Organization (FAO) of the United Nations and the Commission of the European Communities. It is expressed in a formal language as a Simple Knowledge Organization System (SKOS). AGROVOC may be downloaded free of charge for educational or other non-commercial purposes. Descriptors and non-descriptors as well as their relations among one another can be searched, as of this moment, in 22 languages (2012). To exemplify, we compare the Word Tree of the descriptor with the identification number 3032 in the selected languages English (*foods*), French (*produit alimentaire*) and German (*Lebensmittel*). Behind the scope note reference, in the descriptor entry of 3032 there is the note, applicable to all languages: “Use only when a more specific descriptor is not applicable; refers to foods for human beings only; for animals use <2843>”. In English, the use of the appropriate descriptor for animals is *feeds*, in French it is *aliment pour animaux* and in German *Futter*. Differences in the relations can be found only in the non-descriptors, as they are dependent upon language, culture and even documentation, and these are provided with the note UF.

In the narrower terms, we can see that two loan terms from the English language have been adopted in the Word Tree for the German descriptor: *fast food* (in the singular) and *novel food*.

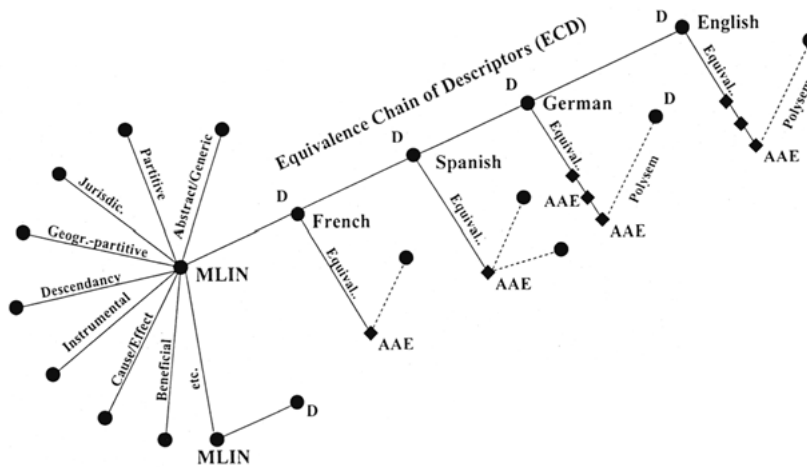


Figure L.3.9: Structure of a Multilingual Thesaurus. Source: Schmitz-Esser, 1999, 14. Abbr.: MLIN: Meta-Language Identification Number (Identification Number for Descriptor Entry); AAE: Additional Access Expressions (Language-specific Non-descriptors).

While in a single-language thesaurus it is not allowed to have one and the same term be a descriptor on the one hand and a non-descriptor on the other, this does occur in the multilingual AGROVOC thesaurus. The descriptor with the identification number 34338 is called *snack foods* in English, *snacks* in French and *Knabberartikel* in German. As opposed to the French language, where *snacks* is used as a descriptor, the same term is a non-descriptor in English, Spanish and German (ID 34956).

As not every language provides a clear translation for a descriptor, since different cultures and linguistic usages may not facilitate an equivalent translation, the AGROVOC thesaurus contains various amounts of descriptors, non-descriptors and, accordingly, various amounts of relations.

Thesaurus Construction and Maintenance

A thesaurus is a dynamic unit, which must adapt to changing requirements. Thesaurus maintenance becomes necessary when mistakes during the thesaurus construction (e.g. with regard to structure) are noticed, when new areas of research open up, when the linguistic usage of a specialist area has changed, when new forms of sources

have arisen, when user behavior has changed or the information system is no longer up to date (Wersig, 1985, 274 et seq.). Descriptors and non-descriptors may then have to be introduced anew, eliminated or re-organized. Certain descriptors may also require different relations, e.g. because the hierarchical structure is being redefined. Concept relations that are no longer in use, e.g. in related terms, require deletion.

As in nomenclature and classification, the construction and maintenance of thesauri (Broughton, 2006) is accomplished via the combined usage of top-down and bottom-up approaches. In both methods, one can proceed semi-automatically, by having lists of the frequency of individual words or phrases as well as frequent-term clusters be compiled automatically. Here, usage of informetric methods (Rees-Potter, 1989; Schneider & Borlund, 2005) results in candidates for descriptors as well as relations between descriptors that must be processed intellectually in the next step.

The top-down method is used in thematically relevant textbooks, encyclopedias, dictionaries, review articles etc. (López-Huertas, 1997, 144), extracting—automatically or intellectually—the central concepts and semantic relations, respectively. Here it is mainly those documents that define concepts which are the most useful.

In the bottom-up approach, we start with the terminological material of large quantities of literature, which are to be regarded as relevant for the knowledge domain of the thesaurus. We can—depending on the methods already put into practice—distinguish between three sources for descriptor and relation candidates (Figure L.3.10): (digitally available) full texts, tags (in folksonomies) and highlighted search entries (in the text-word method).

Descriptor candidates are mainly derived from the usage of informetric ranking methods. We thus create frequency lists of terms (words, phrases) in the full texts, tags and search entries. The terms placed at the top of the rankings provide the heuristic basis for descriptors.

We are provided with heuristic material for relation candidates, i.e. for the construction of relations between terms via a so-called “pseudo-classification” on the basis of their common occurrence in documents. According to Jackson (1970, 188), this procedure is even supposed to facilitate the automatic specification of knowledge organization systems:

Under the hypothesis that: ... “co-occurrence of terms within documents is a suitable measure of the similarity between terms” a classification may be generated automatically.

Salton (1980, 1) also holds that a co-occurrence of words in documents is at least a good indicator for a relation:

Thus, if two or more terms co-occur in many documents in a given collection, the presumption is that these terms are related in some sense, and hence can be included in common term classes.

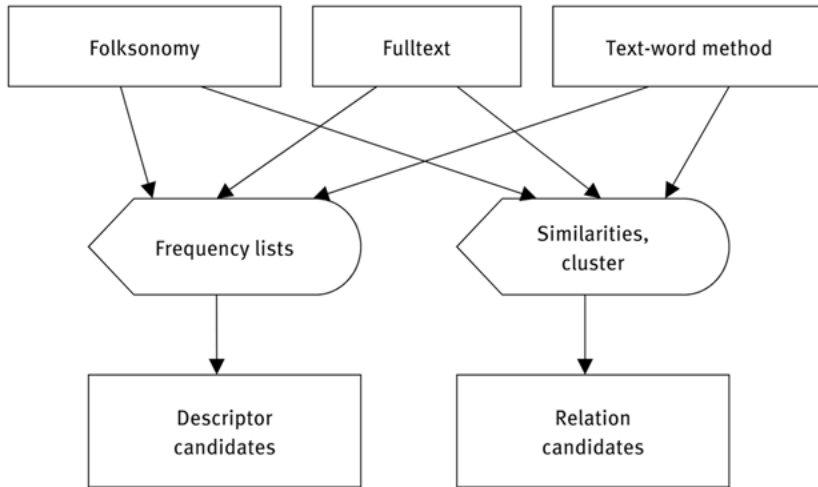


Figure L.3.10: Bottom-Up Approach of Thesaurus Construction and Maintenance.

In contrast to Jackson and Salton, we hold a purely automatic construction of KOSs to be impracticable, whereas the intellectual path supported by pseudo-classification is extremely rewarding. Here it is important that, as Salton (1980, 1) emphasizes, we apply the similarity algorithms to documents that are absolutely representative for the knowledge organization system.

When applying cluster analysis to the syntagmatic relations in the documents or the surrogates (in folksonomy and text-word method, respectively), the results will be semantic networks between the terms. Similarities can be calculated via the Jaccard-Sneath or Dice coefficients, vectors or via relative frequencies of co-occurrence (Park & Choi, 1996). Furthermore, the use of methods from cluster analysis, such as k-nearest neighbors, single links or complete links, is a valid option. The result is always a quantity of terms, while in hierarchical clustering there is, in addition, the (automatically calculated) pseudo-hierarchy. We know what term is connected with what other terms, and we also know the extent of the correlation. However, we find out nothing about the form of the underlying relation. This interpretation of the respective correlation, and thus the transition from the syntagmatic to the paradigmatic relation, requires the intellectual work of experts. They decide, whether there is an equivalence, hierarchical or associative relation.

In the analysis of large amounts of specialist full texts (e.g. of scientific articles on biology), procedures of information extraction can be used to glean suggestions for hierarchical relations. If patterns used by authors to thematize hierarchies are known, candidates for relations will be derivable. If, for example, “belongs to the genus” (or even the mere occurrence of “genus”) is a pattern of an abstraction relation, the phrase *ruminant artiodactyl* in the sentence “The goat belongs to the genus of ruminant artiodactyls” will be identified as the hyperonym to *goat*.

It thus appears to make sense to check and perhaps to take into consideration any reasonable suggestions of the users as thesaurus maintenance support (in the same way that AGROVOC, for instance, asks the users for assistance, or MeSH provides an electronic mailbox for vocabulary suggestions).

Conclusion

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- Thesauri (like nomenclatures) work with expressions of natural language. In a thesaurus without preferred terms, all designations of a concept are allowed for indexing and search; in thesauri with preferred terms, one designation is excelled as the descriptor. The other designations of the concepts are non-descriptors.
 - Thesauri generally refer to the vocabulary of a specific knowledge domain. We distinguish between entry vocabulary, preferred vocabulary and candidate vocabulary.
 - In the thesaurus, one must distinguish between vocabulary control (with regard to exactly one concept) and conceptual control (with regard to the semantic relations).
 - Vocabulary control is performed via (1) the conflation of synonyms and quasi-synonyms, (2) the separation of homonyms, (3) the fragmentation, or retaining, of multi-word expressions (via pre-combination, precoordination or postcoordination), (4) the bundling of specific hyponyms as well as (5) the specification of too-general hyperonyms.
 - There is an equivalence relation between a descriptor and all of its non-descriptors.
 - The relations between descriptors are expressed via hierarchical relations and the (generally unspecific) associative relation. Thesauri work with hyponymy (abstraction relation), meronymy (partitive relation) and—sometimes additionally—with the instantial relation.
 - All determinations regarding a concept are summarized in the descriptor entry. This contains, among other things, statements relating to all non-descriptors, all neighboring concepts in the equivalence and associative relations, “life data” of the concept, elucidations and (where possible) a definition.
 - Descriptors can be specified via qualifiers (as in MeSH), and thus result in heightened precision during search.
 - The user presentation of a thesaurus contains a systematic as well as an alphabetical entry. A retrieval system allows for searching the KOS; links between the entries allow for browsing the system. Additionally, a graphical representation of the concepts, including their semantic environment, would make a lot of sense.
 - Multilingual thesauri work with either a main language (from the perspective of which the terms are translated) or with a language-independent term number (with corresponding natural-language surfaces).
 - The hierarchical structure stays fundamentally the same, even in the case of multiple languages; the non-descriptors, however, are adjusted to the respective environment of every language.
 - In constructing and maintaining thesauri, top-down and bottom-up approaches complement each other, with the former using relevant documents (e.g. textbooks or review articles) and the latter dealing with larger quantities of thematically relevant literature, which are then processed statistically as well as cluster-analytically. In this way, one derives (intellectually or automatically) candidates for descriptors and for semantic relations as heuristic material for further intellectual processing.
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L.4 Ontology

Heavyweight Knowledge Organization System

Whereas the previously discussed methods of knowledge representation can be observed independently of their technical implementation, the case is different for ontologies: here one must always pay attention to the standardized technical realization in a specific ontology language, since ontologies are created both for man-machine interaction and for collaboration between different computer systems. In their demarcation of “ontology”, which is closely oriented on Gruber’s (1993, 199) classical definition, Studer, Benjamins and Frensel (1998, 185) emphasize the aspects of a terminology’s formal, explicit specification for the purpose of using concepts from a knowledge domain jointly.

However, this definition is so broad that it includes all knowledge organization systems (nomenclature, classification, thesaurus and ontology in the narrow sense). If “formal” is defined in the sense of formal logic or formal semantics, the subject area is narrowed down (Staab & Studer, 2004, VII):

(A)n ontology has to be specified in a language that comes with a formal semantics. Only by using such a formal approach ontologies provide the machine interpretable meaning of concepts and relations that is expected when using an ontology-based approach.

Like all other KOS, an ontology always orients itself on its users’ language use (Staab & Studer, 2004, VII-VIII):

On the other hand, ontologies rely on a social process that heads for an agreement among a group of people with respect to the concepts and relations that are part of an ontology. As a consequence, domain ontologies will always be constrained to a limited domain and a limited group of people.

The concept of “ontology” is derived from philosophy, where it is generally understood as “the study of being” (in substance: ever since Aristotle’s metaphysics; under the label “ontologia”: from the 17th century onward). In contemporary analytical philosophy, ontology is discussed in the context of formal semantics and formal logic. This relationship between the philosophical and the information science conceptions of ontology is discussed by Smith (2003):

The methods used in the construction of ontologies (in computer and information science; A/N) thus conceived are derived on the one hand from earlier initiatives in database management systems. But they also include methods similar to those employed in philosophy (...), including the methods used by logicians when developing formal semantic theories.

Ontology as a method of knowledge representation also distinguishes itself by enabling automated reasoning via the use of formal logic. A further characteristic of ontologies is the continuous consideration not only of general concepts, but also of individual ones (instances).

Corcho, Fernández-López and Gómez-Pérez (2003, 44) roughly distinguish between “lightweight” and “heavyweight” ontologies:

The ontology community distinguishes ontologies that are mainly taxonomies from ontologies that model the domain in a deeper way and provide *more restrictions* on domain semantics. The community calls them *lightweight* and *heavyweight ontologies* respectively. On the one hand, lightweight include concepts, concept taxonomies, relationships between concepts and properties that describe concepts. On the other hand, heavyweight ontologies add axioms and constraints to lightweight ontologies.

“Lightweight ontologies” correspond to our KOSs nomenclature, classification and thesaurus. To prevent any confusion in the following, we will only speak of “ontology” when “heavyweight ontologies” are discussed. To wit: a KOS is only an ontology (in the narrow sense) when all of these four aspects are fulfilled:

- Use of freely selectable specific relations (apart from hierarchical relations),
- Use of a standardized ontology language,
- Option for automated reasoning,
- Occurrence of general concepts and instances.

Freely Selectable Relations

One of the knowledge domains that has been the focus of particular attention in the context of ontological knowledge representation is biology. “Gene Ontology” (GO) in particular has attained almost exemplary significance (Ashburner et al., 2000). However, in the early years, GO was not an ontology in the sense defined above but a thesaurus (specifically: three partial thesauri for biological processes, molecular functions and cellular components, respectively), since this KOS only used the relations *part_of* and *is_a*, i.e. only meronymy and hyponymy. Gene Ontology represents a good starting point for demonstrating which relations, apart from the hierarchical one, are even necessary in biomedicine in the first place. For us, Smith et al.’s (2005) approach is an example for the way that the previously unspecific associative relation can be specified into various different concrete relations. In ontologies, too, hierarchical relations provide a crucial framework (Smith et al., 2005):

Is_a and *part_of* have established themselves as foundational to current ontologies. They have a central role in almost all domain ontologies ...

For the other relations, we find such links between those concepts that are characteristic for the respective knowledge domain. In the area of genetics, Smith et al. (2005) distinguish between the components C (“continuant”, as a generalization of the original GO’s “cellular components”) and processes P (as a generalization of “biological processes”). For Smith et al. (2005), the following eight relations are essential for bio-domains in addition to hierarchical relations:

<i>Relation</i>	<i>Example</i>
<i>C located_in C_i</i>	66s pre-ribosome <i>located_in</i> nucleolus chlorophyll <i>located_in</i> thylakoid
<i>C contained_in C_i</i>	cytosol <i>contained_in</i> cell compartment space synaptic vesicle <i>contained_in</i> neuron
<i>C adjacent_to C_i</i>	intron <i>adjacent_to</i> exon cell wall <i>adjacent_to</i> cytoplasm
<i>C transformation_of C_i</i>	fetus <i>transformation_of</i> embryo mature mRNA <i>transformation_of</i> pre-mRNA
<i>C derives_from C_i</i>	plasma cells <i>derives_from</i> lymphocyte mammal <i>derives_from</i> gamete
<i>P preceded_by P_i</i>	translation <i>preceded_by</i> transcription digestion <i>preceded_by</i> ingestion
<i>P has_participant P_i</i>	photosynthesis <i>has_participant</i> chlorophyll cell division <i>has_participant</i> chromosome
<i>P has_agent C</i>	transcription <i>has_agent</i> RNA polymerase translation <i>has_agent</i> ribosome.

Here in the terminological field of genetics, we can see that the relation *derives_from* embodies the chronological relation (which we have otherwise defined in a general sense) of gen-identity (Ch. L.1).

When defining specific relations, one must always state which characteristics these relations bear:

- Transitivity,
- Symmetry,
- Functionality,
- Inversion.

Transitive relations also allow for conclusions to be made concerning semantic distances greater than one, i.e. they are not confined to direct term neighbors. Symmetrical relations are distinguished by the fact that the relation between *x* and *y* also holds in the opposite direction between *y* and *x*. A symmetrical relation is e.g. *has_neighbor*. A relation is called inverse when the initial relation $x \rho y$ has a counterrelation $y \rho' x$, as it is the case for hyponymy and hyperonymy. Here, conclusions can be drawn in both directions:

Pomaceous fruit has_hyponym *Apple* → *Apple* has_hyperonym *Pomaceous fruit*
Apple has_hyperonym *Pomaceous fruit* → *Pomaceous fruit* has_hyponym *Apple*.

Finally, a relation is functional if it leads to precisely one value. As an example, we name the relation *has_birthday*. Inverse relations exist for functional relations (e.g. *is_birthday_of*).

Web Ontology Language

OWL is an ontology language for the Web (Antoniou & van Harmelen, 2004; Horrocks 2005). In its variant of description logic (Gómez-Pérez, Fernández-López, & Corcho, 2004, 17-21), it builds on terminological logic. An alternative method works with frames (Gómez-Pérez, Fernández-López, & Corcho, 2004, 11-16). The expected acronym WOL (Web Ontology Language) does not work, however; Hendler (2004) explains:

Actually, OWL is not a real acronym. The language started out as the “Web Ontology Language” but the Working Group disliked the acronym “WOL”. We decided to call it OWL. The Working Group became more comfortable with this decision when one of the members pointed out the following justification for this decision from the noted ontologist A.A. Milne who, in his influential book “Winnie the Pooh” stated of the wise character OWL: “He could spell his own name WOL ...”.

What makes OWL a language for the Web is its option of using URI (Uniform resource identifiers; being the unification of URL [Uniform resource locator] and URN [Uniform resource name]) as values for objects in order to be able to refer to Web documents and to take into account the knowledge contained in documents (if it follows OWL), respectively.

OWL always defines two classes: *owl:Thing* being the top term of the KOS and *owl:Nothing* being an empty quantity. To exemplify, we introduce two simple general concepts (“classes”), *Wine* and *Region*:

```
<owl:Class rdf:ID="Wine"/>
<owl:Class rdf:ID="Region"/>.
```

Within a given ontology, the above style of representation is enough; to work cooperatively in the Web, however, one must state the correct URI for every class. We now wish to define *Wine* hierarchically as a hyponym of *Alcoholic Beverages*, and to create further vernacular access points in languages other than English:

```
<owl:Class rdf:ID="Wine">
  <rdfs:subClassOf rdf:Resource="#Alcoholic Beverages"/>
  <rdfs:label xml:lang="en">Wine</rdfs:label>
  <rdfs:label xml:lang="ge">Wein</rdfs:label>
  <rdfs:label xml:lang="fr">Vin</rdfs:label>
</owl:Class>.
```

rdf stands for Resource Description Framework, *rdfs* for an RDF Schema (McBride, 2004). Individual concepts (“things”) are introduced analogously:

```
<owl:Thing rdf : ID="Cedar Creek Platinum"/>
<owl:Thing rdf : about="#Cedar Creek Platinum">
  <rdf : type rdf:resource="#Wine"/>
</owl:Thing>.
```

Let us now turn to the properties. In order to express the fact that wine is made from grapes, we create the relation *made_from* and the value *Grape*:

```
<owl:Object Property rdf : ID="MadeFrom">
  <rdfs : domain rdf : resource="#Wine"/>
  <rdfs : range rdf : resource="#Grape"/>
</owl:Object Property>.
```

Quantifiers in OWL are described via *allValuesFrom* (universal quantifier) and *someValuesFrom* (existential quantifier), while the number is expressed via *Cardinality*. Values of relations must be stated via *hasValue*.

As it cannot be asked of any user to apply such a complicated syntax, ontology editors have been developed that allow users to enter the respective data via forms while they manage them in the background in OWL. The editor Protégé is particularly popular at present (Noy, Fergerson, & Musen, 2000; Noy et al., 2001).

General and Individual Concepts

A concrete ontology rises and falls with the respective knowledge base registered by the terminology. Terminologies for ontologies are either deposited in the TBox (“terminology box”), for general concepts, or in the ABox (“assertional box”), for individual ones.

In the TBox, general concepts are defined on the basis of already introduced concepts. Supposing that the KOS already has the terms *Person* and *Female*, the term *Woman* can be introduced in this basis (let $\sqcap$ be the sign for AND):

$$\text{Woman} \equiv \text{Person} \sqcap \text{Female}.$$

The TBox has a hierarchical structure; in the final analysis, it is a classification system (Nardi & Brachman, 2003, 14):

In particular, the basic task in construction a terminology is classification, which amounts to placing a new concept expression in the proper place in a ... hierarchy of concepts. Classification

can be accomplished by verifying the subsumption relation between each defined concept in the hierarchy and the new concept expression.

Hyperonyms are contained in their hyponyms. Let C be a concept (e.g. *bald eagle*) and D its hyperonym (*eagle*). It is then true that:

$$C \sqsubseteq D$$

(where $\sqsubseteq$ is the sign for “being contained in”). All characteristics (i.e. all relations) that hold for D thus also hold for C . (All characteristics and relations that an eagle has can thus be found in the case of bald eagles.) On such a basis, automated reasoning is performed. For instance, if it has been determined that eagles have feathers, we then deduce that bald eagles have feathers as well.

The ABox takes statements about individual concepts, concerning both the individual’s characteristics (“concept assertions”) and relations (“role assertions”). If we wish to express e.g. that Anna is a female person, the following entry will be entered into the ABox:

$$\text{Female} \sqcap \text{Person}(\text{ANNA}).$$

Female and *Person* must, of course, have been previously defined in the TBox. If we suppose that Anna has a child named Jacopo, the entry, which now contains a relation, will read:

$$\text{has_Child}(\text{ANNA}, \text{JACOPO}).$$

On the Way to the Semantic Web?

Ontologies are ill suited for the search and retrieval of documents. They are far too complex in their construction, in their maintenance, and in the methods required to support user search to be of use in large information services (e.g. for all of chemistry, for patent documents, for images). Here it is a far better option to employ lightweight KOSs such as nomenclatures, classifications or thesauri. However, with a view to data exchange and computer-computer communication it makes sense to make these lightweight KOSs available in a formal language. Ontologies show their advantages in applications that do not aim for the search and retrieval of documents, but for the provision of the knowledge contained in the documents.

When the complete knowledge inside a document is formally represented via ontologies, we speak of the “Semantic Web”, and if we restrict ourselves to the structured data contained in the document, we are dealing with “linked data” or—if the information is freely accessible—“linked open data” (Bizer, Heath, & Berners-Lee,

2009). For instance, DBpedia extracts structured information from Wikipedia (Bizer, Lehmann et al., 2009). The structured information is gleaned from the infoboxes of certain Wikipedia articles, represented as RDF in DBpedia, and made available as linked data. The more providers join in the “Linking Open Data”-project and produce formalized data, the richer the “data cloud” will be, as data that used to be stored separately in different sources are brought together.

Via ontologies, we have arrived at the core of the Semantic Web (Berners-Lee, Hendler, & Lassila, 2001; Shadbolt, Hall, & Berners-Lee, 2006). Shadbolt, Hall and Berners-Lee formulate the claims of the Semantic Web as follows (2006, 96):

The Semantic Web is a Web of actionable information—information derived from data through a semantic theory for interpreting the symbols. The semantic theory provides an account of “meaning” in which the logical connection of terms establishes interoperability between systems.

At first, discussions on the Semantic Web are highly technical in nature. They concern the resource description framework, universal resource identifiers, the most suitable ontology language (such as OWL, the Web ontology language), the rules of automated reasoning sketched above, and ontology editors such as Protégé (Noy, Fergerson, & Musen, 2000; Noy et al., 2001). Both the background (in the sense of concept theory) and the methods for creating suitable KOSs have sometimes been left unaddressed. Current attempts at finding a solution are discussed by Shadbolt, Hall and Berners-Lee in the form of two approaches based on the cooperation of participating experts. Ontologies (as described here) that are separately constructed and maintained are suited for well-structured knowledge domains (Shadbolt, Hall, & Berners-Lee, 2006, 99):

In some areas, the costs—no matter how large—will be easy to recoup. For example, an ontology will be a powerful and essential tool in well-structured areas such as scientific applications ... In fact, given the Web’s fractal nature, those costs might decrease as an ontology’s user base increase. If we assume that ontology building costs are spread across user communities, the number of ontology engineers required increases as the log of the community’ size.

This approach can only be used a) if the knowledge domain is small and overseeable and b) if the members of the respective community of scientists are willing to contribute to the construction and maintenance of the ontology. Such an approach seems not to work at all across discipline borders. The second approach, consolidating the Semantic Web, proceeds via tagging and folksonomies (Shadbolt, Hall, & Berners-Lee, 2006, 100):

Tagging on a Web scale is certainly an interesting development. It provides a potential source of metadata. The folksonomies that emerge are a variant on keyword searches. They’re an interesting emergent attempt at information retrieval.

Folksonomies, however, only have syntagmatic relations. If one wants to render such an approach usable for the Semantic Web (now in the sense of the “Social Semantic Web”; Weller, 2010), focused work in “Tag Gardening” would seem to be required (Peters & Weller, 2008) (Ch. K.2). Equally open, in our view, is the question of indexing documents in the Semantic Web. Who will perform this work—the author, the users or—as automatic indexing—a system?

For Weller (2010), using only ontologies for the Semantic Web is an excessive approach. Ontologies are far too complicated for John Q. Web User in terms of structure and use. Weller’s suggestion boils down to a simplification of the concept system. Not all aspects of ontologies have to be realized, as the desired effects might also be achieved via a less expressive method (e.g. a thesaurus). The easier the concept systems are to use for the individual, the greater will be the probability of many users participating in the collaborative construction of the Semantic Web.

Semantically rich KOSs, notably thesauri and ontologies, only work well in small knowledge domains. Therefore we have to consider the problem of semantic interoperability between different KOSs—i.e. the formulation of relations of (quasi-) synonymy, hierarchy etc. (as means of semantic crosswalks)—as well as between singular concepts and compounds beyond the borders of individual KOSs in order to form the kind of “universal” KOS that is needed for the Semantic Web.

The vision of a universal Semantic Web on the basis of ontologies is given very narrow boundaries. The ruins of a similar vision of summarizing world knowledge—Ottlet’s and La Fontaine’s “Mundaneum”—can today be admired in a museum in Mons, Belgium. We do not propose that a Semantic Web is impossible in principle; we merely wish to stress that the solution of theoretical and practical problems, besides technological ones, will be very useful on the way to the Semantic Web.

Conclusion

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- In the narrow sense, we understand an ontology to be a knowledge organization system that is available in a standardized language, allows for automated reasoning, always disposes of general and individual concepts, and, in addition to the hierarchy relation, uses further specific relations. Hyponymy forms the supporting framework of an ontology.
 - In the further relations, it must be noted that the fundamental relations of the respective knowledge domain are incorporated into the KOS. The relations have characteristics, each of which must be determined. We distinguish between symmetrical, inverse, and functional relations. The range of inferences (only one step or several ones) depends upon whether the relation is transitive or not.
 - Concepts and relations form the ontology’s knowledge base. General concepts are generally deposited as a classification system in the TBox, individual ones in the ABox.
 - There exist standardized ontology languages (such as the Web Ontology Language OWL) and ontology editors (such as Protégé).

- All types of KOSs—i.e., not only ontologies—are capable of forming the terminological backbone of the Semantic Web. Constructing the Semantic Web is not merely a technical task, but calls for such efforts as the construction of KOSs and the (automated or manual) indexing of documents.
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L.5 Faceted Knowledge Organization Systems

Category and Facet

A faceted knowledge organization system does not work with one set of concepts only, but instead uses several. The crucial factor for the construction of faceted systems is whether several fundamental categories occur in the knowledge domain. If so, the construction of a faceted KOS must be taken into consideration as a matter of principle (Vickery, 1968). Broughton (2006, 52) demonstrates the advantages of faceted systems on a simple example. The objective here is to arrange terms around the concept *socks*. A classification with a single table would certainly yield a concept ladder of the following dimensions:

- Grey socks
 - Grey wool socks
 - Grey wool work socks
 - Grey wool hiking socks
 - Grey wool ankle socks for hiking
 - Grey wool knee socks for hiking
 - Grey spotted wool knee socks for hiking

Even a relatively small knowledge domain such as “socks” would lead to a comprehensive classification system in this way. The first step in the faceted approach is to detect the fundamental categories of the subject area. Here, these would be color, pattern, material, function and length. The second step consists of collecting the respective subject-specific concepts for each facet. Broughton (2004, 262) makes the following suggestion:

Color	Pattern	Material	Function	Length
Black	Plain	Wool	Work	Ankle
Grey	Striped	Polyester	Evening	Calf
Brown	Spotted	Cotton	Football	Knee
Green	Hooped	Silk	Hiking	
Blue	Checkered	Nylon	Protective	
Red	Novelty	Latex		

Of course, this is only a hypothetical example; however, it demonstrates the workings of faceted KOSs (Broughton, 2006, 52):

Such an arrangement is often presented as an example of a faceted classification, and it does give quite a good sense of how a faceted classification is structured. A faceted bibliographic classification has to do a great deal more than this, and a proper faceted classification will have many more facets, covering a much wider range of terminology.

The expressiveness of a faceted knowledge organization system is the result of its possible combinations: in the end, any concept from any facet can be connected to any of the other concepts from the other facets. This increases the amount of synthesizable concepts in comparison with non-faceted knowledge organization systems, even though the former generally contain far less concept material in their facets. For the users, the results are comprehensible search options as well as entirely different search entries, since the facets all emphasize different dimensions. Uddin & Janecek (2007, 220) stress the flexibility of such systems:

In short, faceted classification is a method of multidimensional description and arrangement of information resources by their concept, attributes or “aboutness”. It addresses the fact that users may look for a document resource from any number of angles corresponding to its rich attributes. By encapsulating these distinct attributes or dimensions as “facets”, the classification system may provide multiple facets, or main categories of information, to allow users to search or browse with greater flexibility (...).

We already met a similar kind of KOS in the classifications (Chapter L.2), except these distinguished between main and auxiliary tables. In faceted systems, all tables have the same status.

The categories, which subsequently become facets, form homogeneous knowledge organization systems, in which the facets are disjunct vis-à-vis each other. One concept is thus allocated exactly one facet. The concepts are generally simple classes, also called foci (Buchanan, 1979), i.e. they do not form any compounds. When defining the foci, one must perform terminological control (homonymy, synonymy). Within their facets, the foci use all the known relations—particularly hierarchy.

The principle of faceting works independently of the method of knowledge representation. This provides for faceted classification systems—the most frequent variant—as well as faceted nomenclatures, faceted thesauri and faceted folksonomies. The historical point of departure for faceted knowledge organization systems, however, is that of classifications. The “Colon Classification” by Ranganathan (1987[1933]) is the first approach of a faceted knowledge organization system. The field of application of current faceted KOSs ranges from libraries to search tools on the World Wide Web (Gnoli & Mei, 2006; Ellis & Vasconcelos, 2000; Uddin & Janecek, 2007), and further on to the documentation of software (Prieto-Díaz, 1991).

Faceted Classification

A faceted classification (Foskett, 2000) merges two bundles of construction principles: those of classification (notation, hierarchy, citation order) and those of faceting. Each focus of every facet is named by a notation, there is a hierarchical order within the facets—as far as this is warranted—and the facets are worked through in a certain order (citation order). As opposed to the Porphyrian tree, which Ranganathan

(1965, 30) explicitly names as a counterexample, his network of concepts is far more branched out and works in many more dimensions. The “trick” of creating a multitude of compounds from a select few (simple) basic concepts lies in combinatorics (we need only recall Llull). Any focus of the second facet may be attached to any focus of the first facet (which Ranganathan calls the ‘Personality’). To this combination, in turn, one can attach foci of the third facet etc. It is possible to use several foci of a facet for content description. The foci are created specifically for a discipline, i.e. the energy facet under the heading of Medicine will look completely different to that of Agriculture. The Colon Classification uses its own “facet formula” for every discipline, i.e. a facet-specific citation order (Satija, 2001). Ranganathan (1965, 32-33) writes the following about this “true tree of knowledge”:

For in the true Tree of Knowledge, one branch is grafted to another at many points. Twigs too get grafted in a similar way among themselves. Any branch and any twig are grafted similarly with one another. The trunks too become grafted among themselves. Even then the picture of the Tree of Knowledge is not complete. For the Tree of Knowledge grows into more than three dimensions. A two dimensional picture of it is not easily produced. There are classes studded all along all the twigs, all the branches, and all the trunks.

A compound is only created in the classification system when a document discusses this term for the first time. As is well known, there are syntagmatic relations that apply to the terms of a document. When using a facet classification, these syntagmatic relations are translated into paradigmatic relations via the act of content indexing. Such a paradigmaticization of the syntagmatic is typical for this method of knowledge representation (Maniez, 1999, 251-253).

Let us discuss this on an example by Ranganathan (1965, 45). Let a document be the first to report on fungus diseases of rice stems, which first occurred in Madras in the year 1950. The task is now to synthesize this compound terminologically, from pre-existing foci. The disciplinary allocation is clear: we are talking about *agriculture* (J). The personality (who?) is rice (notation hierarchy: 3 *Food Crop*—38 *Cereal*—381 *Rice Crop*), respectively its stem (4). The material facet (what?) remains unoccupied in this example. The energy facet (how?) expresses the fungus disease (notation hierarchy: 4 *Disease*—43 *Parasitic Disease*—433 *Fungus Disease*). The geographical aspect is Madras (notation hierarchy: 44 *In India*—441 *In Madras*), the chronological aspect is 1950 (N5). We remember the facet-specific truncation symbols—personality (.), material (:), energy (:), space (.) and time (')—and synthesize the syntagmatic relations in the document into the following paradigmatic notation:

J381.4:433.441'N5.

In order to allow a user to not only search the synthesized notation but also to be able to act synthetically during retrieval (Ingwersen & Wormell, 1992, 194), the notation components must additionally be recorded separately, in different fields

Scientific Discipline: J,
 Personality/Agriculture: 381,
 Personality/Agriculture: 4,
 Energy/Agriculture: 433,
 Space: 441,
 Time: N5.

From the first document onward, the finished synthesized notation will be available to all documents with the same or similar topics. The citation order makes sure that thematically related documents are located next to each other.

As with the non-faceted classifications, it must be noted that the synthesized notation is to be saved as a whole. Only the synthesized notation can provide for a citation order, and thus—on library shelves and in purely digital environments—a shelving system. Gnoli and Mei (2006, 79) emphasize, regarding the use of faceted classifications in the World Wide Web:

However, the basic function of notation is not just to work as a record identifier to retrieve items sharing a given subject; rather, it is designed primarily to produce (in Ranganathan's terms) *helpful sequences* of documents, that is, to present selected information items sorted in meaningful ways to be browsed by users. As on traditional library shelves some systematic arrangement is usually preferred to the alphabetic arrangement, website menus, browsable schemes, and search results can benefit from a classified display, especially where the items to be examined are numerous.

When dealing with synthesized notations, the system must recognize where a new facet begins, and then save this facet separately (Gödert, 1991, 98).

Following Ranganathan's Colon Classification, the British "Classification Research Group" elaborates the approach of faceted classification (CRG, 1955), leading to a further faceted universal classification, the "Bliss Bibliographic Classification" (Mill & Broughton, 1977). Here we can find the following 13 standard categories (Broughton, 2001, 79):

Thing / entity, Kind, Part,
 Property, Material, Process, Operation,
 Patient, Product, By-product,
 Agent, Space, Time.

Broughton (2001, 79-80) observes, on the subject of this selection:

These fundamental thirteen categories have been found to be sufficient for the analysis of vocabulary in almost all areas of knowledge. It is however quite likely that other general categories exist ...

The facets prescribe the citation order (Broughton, 2006, 55): 1. Thing / entity; 2. Kind; 3. Part, etc.

Depending on the knowledge domain, other categories might very well make sense (Broughton & Slavic, 2007). Working out which facets are suited to certain uses is the task of facet analysis. Once created, the facets provide for the preservation of the status quo of the KOS (expressed negatively: they provide for rigidity). After all, any exchange or modification of the facets during the lifetime of a classification system is highly unlikely.

Since it can hardly be asked of users to deal practically with the notation monstrosities of synthesized terms and also with the single notations, verbal access is assigned a special meaning.

Prieto-Díaz (1991) demonstratively sums up the elements of a faceted classification in the retrieval system. First, the focus is on the facets, which contain, the notations (here called “descriptors”, with reference to thesauri). Second, the concepts, expressed by the notations, stand in certain (mostly hierarchical) relations within their facets. Third, vernacular designations are created for all notations, taking into consideration all languages of the potential users.

Somewhat surprisingly, a thesaurus appears in Prieto-Díaz’ classification system. In a pure classification system, the vernacular designations are synonyms and quasi-synonyms, which exclusively refer to the preferred term (i.e. the notation). However, it is possible to insert further relations between the designations, such as the associative relation. In a multilingual environment, this will create a multilingual thesaurus. Such a hermaphroditic creation, consisting of thesaurus and (faceted) classification, has been implemented as early as 1969, in the form of “Thesaurofacet”, by Aitchison, Gomershall and Ireland.

Faceted Thesaurus

Delimiting the other faceted KOSs from the “prototype”, the faceted classification, is not an easy thing to do, since a lot of mixed forms exist in practice. We understand a faceted thesaurus to be a method of knowledge representation that adheres to the principles of faceting and works with vernacular descriptors as well as (at the very least) the hierarchy relation. Although theoretically possible, the principle of concept synthesis hardly comes to bear on faceted thesauri. In light of this fact, we will use it as an additional criterion of demarcation from classifications. Spiteri (1999, 44-45) views the post-coordinate proceedings of thesauri as a useful companion to synthesis:

Strictly speaking, can a faceted thesaurus that does not use synthesis be called faceted? One wonders, however, about the ability to apply fully the measure of synthesis to a post-coordinate thesaurus. Since most post-coordinate thesauri consist of single-concept indexing terms, the assumption is that these terms can be combined by indexer and searcher alike. ... In a post-

coordinate thesaurus, it is not necessary to create strings of indexing terms. It could therefore be argued that synthesis is inherent to these types of thesauri, and hence mention of this principle in the thesaurus could be redundant.

Without synthesis, there is of course no citation order and hence no direct virtual shelving systematic.

	A	B	C	D
1	Factiva Intelligent Indexing™ - Industries			
2				
3	Factiva Code	Factiva Descriptor	Factiva Description	Direct Parents
	i3302	Computers/Electronics	All computing and electronics news. Stories about computer hardware and software, semiconductors, networking, computer services and the manufacture, sale and rental of computer equipment or electronic goods, components, instruments and appliances.	
	247 icomp	Computing	All computing news. Stories about computer hardware and software, semiconductors, networking, computer services and the retailing of computer equipment.	i3302
	248 icph	Computer Hardware	Stories about the physical aspects of computers. Computer hardware refers to the "box", the electronic circuitry and components, and also includes the cables, connectors, power supply units, and peripheral devices.	icomp
	249 icircb	Circuit Boards	Stories about printed circuit boards or motherboards. The physical arrangement in	icph

Figure L.5.1: Thesaurus Facet of Industries in Dow Jones Factiva (Excerpt). Source: Stock, 2002, 34.

A faceted thesaurus thus contains several (at least two) coequal subthesauri, each of which displaying characteristics of a normal thesaurus. Different fields are available for the facets. The indexer will draw on all facets to add as many descriptors to a document as necessary in order to express aboutness. The user will search through the thesaurus facets one by one and select descriptors from them (or he will mark the always-available option “all”). Either the user or the system will connect the search arguments via the Boolean AND. A menu-assisted query, with search and entry fields for all facets, is particularly suited to this task, as no facet can be overlooked by the user in such a display.

The commercial news provider Dow Jones Factiva uses a faceted thesaurus in its system “Factiva Intelligent Indexing” as the basis for automatic indexing and as a search tool (Stock, 2002, 32-34). Factiva works with four facets:

- Companies (ca. 400,000 names),
- Industries (ca. 900 descriptors),
- Geographic Data (ca. 600 descriptors),
- “Topics” (around 400 descriptors).

Such a method of faceting according to industry (e.g. *computer hardware*), geographic data (*Germany*) and (economic) topic (*revenue*), linked with a company facet, has asserted itself in economic databases. The company facet, too, works with the

hierarchy relation: in the concept ladder, the parent company and its subsidiaries are displayed (ideally including the percentage of interest). Factiva only knows the hierarchy relation, which is always displayed polyhierarchically. In Figure L.5.1, we see an excerpt from the industry facet. The hierarchies are indicated by the number of vertical bars on the left. The preferred term for the concepts is a code (e.g. *icph* for *computer hardware*), which is only used for internal editing, however. The user works with one of the more than 20 interfaces, which cover all of the world's major languages.

Faceted Nomenclature

In nomenclatures, the concepts of the knowledge organization system are entered into the single facets without any hierarchization. If one wants to synthesize the concepts from the different facets, a citation order will be required (exactly as it is for classifications), i.e. the order of the facets will be determined. However, it is also possible, as with the faceted thesauri, to work without keyword synthesis.

The Keyword Norm File (SWD) is such a faceted nomenclature with synthesizable keywords (Ch. L.1). Using a KOS for libraries, the following five categories emerge:

- Persons (p),
- Geographic and ethnographic concepts (g),
- “Things” (s),
- Time (z),
- Form (f) (RSWK, 1998, §11).

The category of thing keywords is a “residual” facet, which is filled with everything that is not covered under the other four facets.

The entries from the facets are arranged, one after the other, according to the order stated above (p–g–s–z–f), i.e. the persons first, then geographic data etc. If several concepts could be used as content-descriptive terms within a facet, these will be noted and brought into a meaningful order. An alphabetical arrangement may also be a possibility (RSWK, 1998, §13). The rulebook recommends not admitting more than six (up to ten in special situations) keywords into such a chain. Apart from the basic chain thus created, the rulebook recommends forging further chains that are formed via permutations of the chain links. However, this is only of significance for printed indices, in order to provide access points for the respective places in the alphabet. For online retrieval, permutations may be waived.

recipe categories
(about recipe categories)

☐ Chef Recipes	☐ Kid-Friendly	☐ One Dish Meal	☐ Quick
☐ Epicurious TV	☐ Meatless	☒ Part of Menu	☐ Wine Pairing

healthy options
(about healthy options)

☐ High Fiber	☐ Low Carb	☐ Low Sodium	☐ Wheat/Gluten Free
☒ Low Cal	☐ Low Fat	☐ Low/No Sugar	

course cuisine season / occasion

Desserts German Christmas

type of dish preparation method source

Cookies all methods all sources

main ingredients

☐ Beans	☐ Duck	☐ Grains	☐ Olives	☐ Rice
☐ Beef	☐ Eggs	☐ Greens	☐ Onions	☐ Shellfish
☐ Berries	☐ Fish	☐ Herbs	☐ Pasta	☐ Soy
☐ Cheese	☒ Fruits	☐ Lamb	☐ Peppers	☐ Tomatoes
☐ Chicken	☐ Game	☐ Mushrooms	☐ Pork	☐ Turkey
☐ Chocolate	☐ Garlic	☐ Mustard	☐ Potatoes	☐ Vegetables
☐ Citrus	☐ Ginger	☐ Nuts	☐ Poultry	☐ Yogurt
☐ Dairy				

Figure L.5.2: Faceted Nomenclature for Searching Recipes. Source: www.epicurious.com.

As a further example of a faceted nomenclature, we will leave the world of libraries and turn to recipes on the World Wide Web. The information service Epicurious offers a search tool for recipes (without the option of keyword synthesis); here one works with keywords that are divided into nine facets (Figure L.5.2). In the top two facets as well as the bottom one (recipe categories, healthy options, main ingredients), all keywords available in the respective facet can be ticked off, whereas in the six facets in the middle (course, cuisine, season/occasion, type of dish, preparation method, source), the keywords are displayed in a drop-down menu. All facets only contain very few concepts (the ingredients facet is the most terminologically rich, with 36

keywords). This shows very clearly how faceted knowledge organization systems with a minimum of terminological effort can be used to achieve ideal retrieval results.

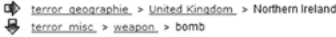
Faceted Folksonomy

Folksonomies, which are normally without any structure at all, are provided by faceting with a certain amount of terminological organization. The basic idea of faceted folksonomies (Spiteri, 2010) is to present the tagging user with a field schema in which the fields correspond to the facets. Thus it would be possible, for example, to preset fields for locations (*Vancouver, BC*), people (*Aunt Anne*) or events (*70th birthday*) in sharing services such as Flickr and YouTube (Bar-Ilan et al., 2006). Within the fields, there are no rules for assigning the individual tags—as is the normal procedure in folksonomies. The preset fields, according to the studies conducted by Bar-Ilan et al. (2006), entice the users to fill the fields with the appropriate tags, as warranted by the content. “This results in higher-quality image description” (Peters, 2009, 193).

Online Retrieval when Using Faceted KOSs

When retrieving documents online that have been indexed via faceted knowledge organization systems, several options arise which are not available for non-faceted KOSs. The user has to look for and enter (or mark) the desired concepts from the individual facets. Therefore, after the first search argument has been transmitted the option arises of only showing the user those terms in the remaining facets that are syntagmatically related to the first one. The number of documents no longer refers to all data entries, but only to those that contain the first search argument as well as the respective further term. Such a context-specific search term selection also allows for active browsing, since when opening a facet the user already knows which further search arguments (and which hit lists) are available.

Furthermore, there are the options of digital shelving systematics. Here we must distinguish between two variants. Where a system works with concept synthesis, we will enter the system at exactly the point where the desired synthesized term is located and display the exact result as well as its neighbors. Apart from this direct shelving systematic, we can create an indirect one—when a system does not offer synthesis—by calculating thematic similarities between documents. For this purpose, algorithms such as Jaccard-Sneath, Dice or Cosine are consulted and the common classes are set in relation to all assigned classes. Proceeding from a direct hit, all documents with which it shares certain classes, descriptors or keywords will be shown in descending order according to their similarity value.





	Stormont (35/19)	Fermanagh (22/10)	Tyrone (28/29)	Bogside (5/3)	Down (61/31)	Co. Antrim (70/28)	Ballykelly (7/3)	Co. Derry (65/23)	Creggan (7/5)
pipe bomb (81/14)	 (7)	 (3)	 (12)	 (2)	 (13)	 (14)	 (2)	 (13)	 (2)
time bomb (6/1)	 (1)		 (1)		 (1)	 (1)		 (1)	
semtex (95/17)	 (10)	 (5)	 (16)		 (13)	 (15)	 (1)	 (16)	 (4)
mortar bomb (44/9)	 (3)	 (4)	 (6)		 (7)	 (9)	 (1)	 (7)	
car bomb (112/22)	 (9)	 (7)	 (14)	 (1)	 (19)	 (21)	 (2)	 (18)	 (1)
petrol bomb (42/7)	 (4)	 (2)	 (6)	 (2)	 (6)	 (7)	 (1)	 (7)	
incendiary bomb (16/3)	 (1)	 (1)	 (3)		 (2)	 (3)		 (3)	

Figure L.5.3: Dynamic Classing as a Chart of Two Facets. Source: Experimental IRA Database of the Department of Information Science at the University of Düsseldorf (Retrieval Software: Convera).

Finally, we can class documents dynamically via the terms contained in facets. Here, a search result will not be displayed as a one-dimensional sorted list, but two-dimensionally, as a chart, in which the two axes each correspond to a facet. When using a faceted classification or a faceted thesaurus, concepts that have hyponyms can be further refined, with reference to their respective search result. Figure L.5.3 shows a search on the Northern Irish terrorist organization PIRA, represented as a chart with the two facets of geographic data and topics. In the figure, we can see the number of documents that have been indexed with *PIRA* on the one hand and with the corresponding region and corresponding subject on the other hand. In the case of underlined names (such as *Co. Derry*), further hyponyms are available. The literature on *PIRA* and *Co. Derry* frequently discusses *car bomb* and *semtex* (18 and 16 documents), as well as *mortar bomb* (far less frequently, with 7 documents), this latter weapon appearing to be more connected to the Northern Irish region of *Antrim* (9 documents). In the dynamic classing of the facets' foci, the user receives certain pieces of heuristic information about the thematic environment of the retrieved documents even before the results are yielded. In cases of information needs that exceed the retrieval of precisely one document, and which might be meant to locate trends, such display options are extremely useful.

Conclusion

- Faceted knowledge organization systems work not only with *one* system of concepts, but use several, where the fundamental categories of the respective KOS form the facets.
- The facets are disjunct vis-à-vis each other and form homogeneous systems within themselves. Each concept of the KOS is assigned to precisely one facet. The concepts in the facets are called “foci”; generally, they are (terminologically controlled) simple concepts. The foci can span relations (particularly hierarchies) within their facets.
- Faceted KOSs often require far less term material than comparable precombined systems, since they consistently use combinatorics.

- There are faceted KOSs with concept synthesis and there are those without this option. When a system uses synthesis, a citation order will be predetermined, which one can use to arrange the terms from the individual facets.
 - Faceted classification systems bring the principles of the construction of classifications (notation, hierarchy and citation order) together with the principle of faceting. The historical point of departure of this type of knowledge organization system is the Colon Classification by Ranganathan. Since notations (and hence concepts) can only be synthesized when a document initially discusses the object in question, the paradigmaticization of the syntagmatic is characteristic for faceted classification systems. Because notations (and, to an even greater extent, synthesized notations) are very problematic for end users, verbal class designations are extremely significant.
 - A faceted thesaurus works with descriptors and (at least) the hierarchy relation, as well as, additionally, the principle of faceting, but not with concept synthesis.
 - In faceted nomenclatures, the concepts of the KOS are listed in the facets without any hierarchization. Here, systems with and without concept synthesis are possible.
 - A faceted folksonomy makes use of different fields related to the categories (facets).
 - In online retrieval, faceted knowledge organization systems provide for elaborate search options: context-specific search term selection, digital shelving systematics as well as dynamic classing in the form of charts.
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L.6 Crosswalks between Knowledge Organization Systems

Retrieval of Heterogeneous Bodies of Knowledge and the Shell Model

In the preceding chapters, we introduced different methods of building and maintaining knowledge organization systems. There are single tools, based on these methods of representation, which are used for the respective practical tasks of working with content-indexed repositories. In other words, we can use completely different tools for every database. If the objective is to scan thematically related databases together in one search, we will be faced with a multitude of heterogeneously indexed bodies of knowledge. The user is confronted by professionally indexed specialist databases (each of which may very well be using different methods and tools), publishers' databases (again using their own indexing procedures), library catalogs (with library-oriented subject and formal indexing), Web 2.0 services (with freely tagged documents in the sense of folksonomies) as well as by search tools in the World Wide Web (which generally index automatically).

If we wish to come closer to the utopia that is the Semantic Web, we must take care to make the respective ontologies being used compatible with one another. If we want to provide users with a (somewhat) unified contentual access to the entirety of the respectively relevant bodies of knowledge, we must start thinking about the desired standardization from the perspective of heterogeneity (Krause, 2006).

There is a tendency to claim that the heterogeneity of content indexing results exclusively from the fact that different systems, or working groups, each implement their own ideas—without any knowledge of the respective others'—which, in total, cannot help but be incompatible with one another. However, this is not entirely true. Even in “protected areas”, e.g. within an enterprise, it proves necessary to knowingly bank on heterogeneity. Not all documents are equally important in enterprises. The objective is to find a decentralized structure, which provides as much consistency as possible while still creating a—controlled—freedom for heterogeneity. As one possible solution, Krause discusses the Shell Model (Figure L.6.1). This model unites various different levels of document relevance, i.e. stages of worthiness of documentation, as well as different stages of content indexing that correspond to the respective levels.

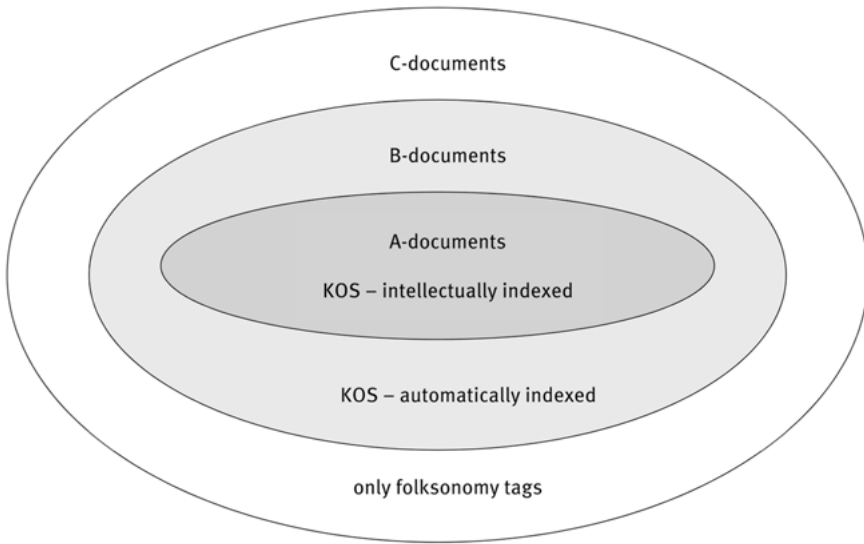


Figure L.6.1: Shell Model of Documents and Preferred Models of Indexing. Source: Stock, Peters, & Weller, 2010, 144.

Let there be a corporate environment, in which there exist some A-documents, or core documents. These are of extreme importance for the institution—say, a strategy paper by the CEO or a fundamental patent describing the company’s main invention. Every A-document has to be searchable at all times—there must be a maximum degree of retrievability. B-documents are not as important as the core documents, but they also have to be findable in certain situations. Finally, C-documents contain very little important information, but they, too, are worth being stored in a knowledge management system—perhaps some day a need for retrieving them will arise.

All users of the corporate information service are allowed to tag all documents. Since they will tag any given document with their own specific (even implicit) concepts, the corporate folksonomy will directly reflect the language of the institution. The less important C-documents are indexed only with tags, if at all. All other documents are represented in terms of the corporate KOS. Professional indexing is an elaborate, time-consuming and therefore costly task. Hence only the A-documents are indexed intellectually by professional indexers, and the B-documents are indexed automatically. If the full texts of the documents are stored as well, there will be many additional (but uncontrolled) access points to the documents (full text search).

It must be attempted, whether by using the Shell Model or via the uncoordinated work of several information producers, to minimize the problems of heterogeneity for the user. This is addressed by Crosswalks between KOSs.

Forms of Semantic Crosswalks

Semantic crosswalks refer to connections between knowledge organization systems, their concepts as well as their relations. The problem of heterogeneity also exists in other metadata areas, e.g. the desired unification of several rulebooks or practices when using authority records (e.g., of personal names or of book titles). Semantic crosswalks serve two purposes:

- they provide for a unified access to heterogeneously indexed bodies of knowledge,
- they facilitate the reuse of already introduced KOSs in different contexts (Weller, 2010, 265).

The objective is to achieve both comparability (compatibility) and the option of collaboration (interoperability) (Zeng & Chan, 2004). We distinguish between the following five forms of semantic crosswalks:

- “Multiple Views” (different KOSs, untreated, together in one application),
- “Upgrading” (upgrading a KOS, e.g. developing a thesaurus into an ontology),
- “Pruning” (selecting and “cropping” a subset of a KOS),
- “Mapping” (building concordances between the concepts of different KOSs),
- “Merging” and “Integration” (amalgamating different KOSs into a new whole).

Parallel Usage: “Multiple Views”

An extremely simple form of crosswalks is the joint admittance of several knowledge organization systems into one application (Figure L.6.2). Likely candidates include different KOSs sharing the same method (e.g. two thesauri) as well as tools of different methods (e.g. a thesaurus and a classification system). One can also regard the single KOSs in faceted systems as different knowledge orders. The parallel usage of several KOSs, or of several facets, provides the user with different perspectives on the same database. If the retrieval system allows dynamic classing, two KOSs may be used concurrently in one and the same chart.

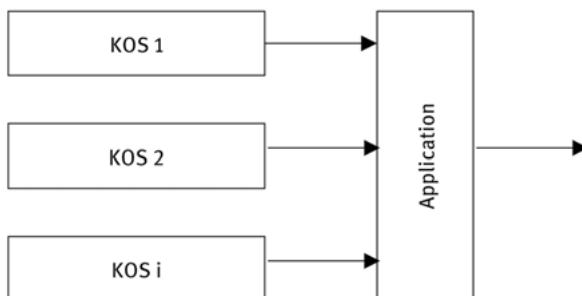


Figure L.6.2: Different Semantic Perspectives on Documents.

Upgrading KOSs

Nomenclatures contain no relations apart from synonymy, classifications mostly have none but the (general) hierarchy relation as well as thesauri use hyponymy and meronymy (in some cases they, too, only use the general hierarchy relation) as well as the unspecific associative relation. We describe the respective transitions of nomenclatures, classifications and thesauri toward ontologies as “upgrading”. The amount of concepts can generally remain the same in these cases, but the relations between them will change—sometimes massively. It is always worth thinking about refining the previously used hierarchy relation into hyponymy and meronymy, and going further, refining the latter into the individual (now transitive) part-whole relations. Additionally, the associative relation must be specified. If not yet available, important specific relations of the knowledge domain must be introduced (indicated in Figure L.6.3 via the bolder lines). Of course the other characteristics of an ontology (such as the usage of an ontology language as well as the consideration of instances) can also be introduced into the KOS.

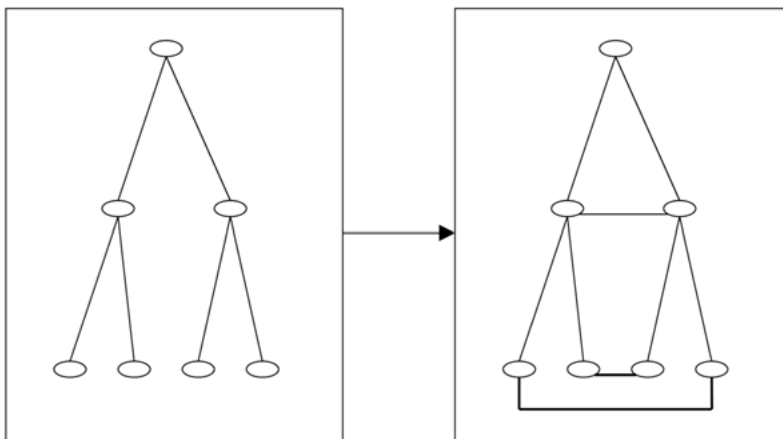


Figure L.6.3: Upgrading a KOS to a More Expressive Method.

Soergel et al. (2004) demonstrate the procedure via examples of thesauri on their way toward ontologies. In an educational science thesaurus, we find the following descriptor entry and its hypothetical refinement into an ontology:

reading instruction

BT instruction

RT reading

RT learning standards

reading instruction

is_a instruction

hasDomain reading

governedBy learning standards.

Here the hierarchy has been left alone, but the associative relation was tampered with. In an agricultural thesaurus, the hierarchy relation is being refined (Soergel et al., 2004):

milk	milk
<i>NT</i> cow milk	<i>includesSpecific</i> cow milk
<i>NT</i> milk fat	<i>containsSubstance</i> milk fat.

When the new relations are deftly formulated, one can exploit the reasoning rules in order to automatically create relations between two concepts. Let us proceed (again following Soergel et al., 2004) from two refinements:

animal	animal
<i>NT</i> milk	<i>hasComponent</i> milk
cow	cow
<i>NT</i> cow milk	<i>hasComponent</i> cow milk

The respectively specific relation (on the right-hand side) can then be automatically derived for all other animals that the KOS tells us about (and which produce milk):

goat	goat
<i>NT</i> goat milk	<i>hasComponent</i> goat milk.

Excerpts: “Pruning”

There already exist high-quality knowledge organization systems that do indeed cover large knowledge domains. The spectrum reaches from (general) world knowledge (DDC), via technology (IPC), industries (NAICS or NACE), medicine (MeSH), genetics (Gene Ontology) up to recipes (Epicurious Nomenclature). It is prudent when creating a new KOS—e.g. within a company—to draw on preexisting systems. If part of a KOS is relevant, one should use this excerpt for further processing. Conesa, de Palol and Olivé (2003) describe such an adaptation as “pruning”, the cutting out of certain concepts and their neighbors from the relations of the original knowledge organization system.

A cohesive (via the relations) part is taken out of the more general KOS (indicated in Figure L.6.4 via the circle) and pruned further in the next step, as needed, by removing less useful concepts and their relations. If new terminologies must be introduced, the respective concepts will be worked into the new system, thus broadening the knowledge domain to the desired extent. Since there is a particular danger during pruning that relations (especially hierarchies) can be destroyed, it is necessary to perform consistency checks in the developing new KOSs.

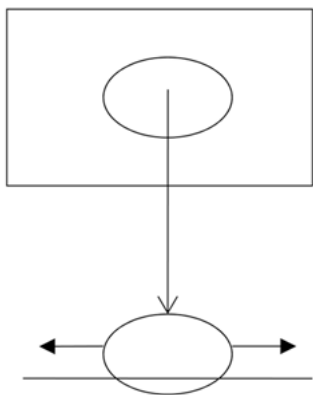


Figure L.6.4: Cropping a Subset of a KOS.

Concordances: “Mapping”

Concordances are probably the most frequently encountered variant of semantic crosswalks. Here the concepts of the different knowledge organization systems are set in relation to one another (Kalfoglou & Schorlemmer, 2003). The work of mapping is performed either purely intellectually or (semi-)automatically, for example on the basis of statistically derived values of co-occurrence. The statistical method requires the existence of parallel corpora in which the same documents are indexed via the respective KOSs.

When more than two KOSs are due for concordances, two methods suggest themselves (Nöther, 1998, 220). In the first method (Figure L.6.5), the total amount of these KOSs is drawn upon to create pairs, in which the concepts must always be set in relation to each other. The alternative works with a “Master KOS” (Figure L.6.6), which must then be used to create concordances in a radial shape. Whereas the direct concordance is used to create a multitude of individual concordances, the second method raises the task of developing a new, preferably neutral knowledge organization system (the Master). It might be possible, under the right circumstances, to raise a preexisting KOS to the “rank” of Master.

The ideal scenario in linking two concepts is when their extension and their intension can be unambiguously represented in a double-ended way. However, we must also take into consideration those cases where *one* initial concept corresponds with several target concepts. The Standard-Thesaurus Wirtschaft (STW), for instance, has a concordance with the NACE. The STW descriptor *Information Industry* refers to the NACE notations 72.3 (data processing services), 72.4 (databases) and 72.6 (other activities involving data processing) (top of Figure L.6.7) (in the sense of one-manyness). In the reverse scenario NACE—STW, we would clearly have many-oneness. Cases of many-manyness are also to be expected.

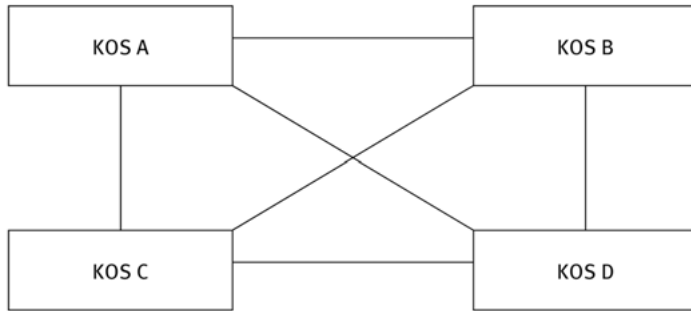


Figure L.6.5: Direct Concordances. Source: Nöther, 1998, 221.

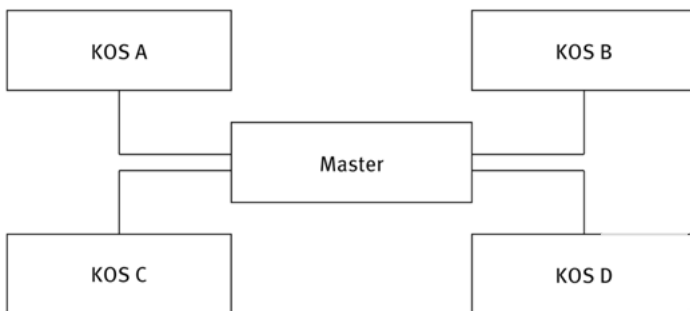


Figure L.6.6: Concordances with Master. Source: Nöther, 1998, 221.

If there is no quasi-synonymy, but there is still a best possible counterpart, we will speak of a “non-exact match” relation—in the sense of: there is a (significant) intersection (Doerr, 2001). If it can be clearly determined that the initial and the target concept are both in a hierarchical relation, this must be noted separately. In this case, one of the discussed KOSs produces far more detailed formulations in the thematic environment of the concept. A remarkable special case is stated in Figure L.6.8. There is a non-exact match, but it is such that we only know the respective next closest hyponyms and hyperonyms. There are thus only borders available for one step upward and downward, respectively, which we will state while mapping (under “lies between [next closest hyperonyms] and [next closest hyponyms]”). In the inverse relations, we have hierarchy relations.

In the one-manyness case discussed above, the target KOS has several concepts that the user must connect in the sense of a Boolean OR in order to receive an equivalent of the initial concept. One must note the special case of *one* concept of the initial KOS (e.g. *library statistics*) having to be expressed via a combination of several concepts from the target KOS (*library; statistics*) (bottom of Figure L.6.7). Here the user must formulate using AND or a proximity operator during his search. In the reverse

case (concordance of *library* and *statistics*, respectively), the relation “used in combination” as *library statistics* is in evidence.

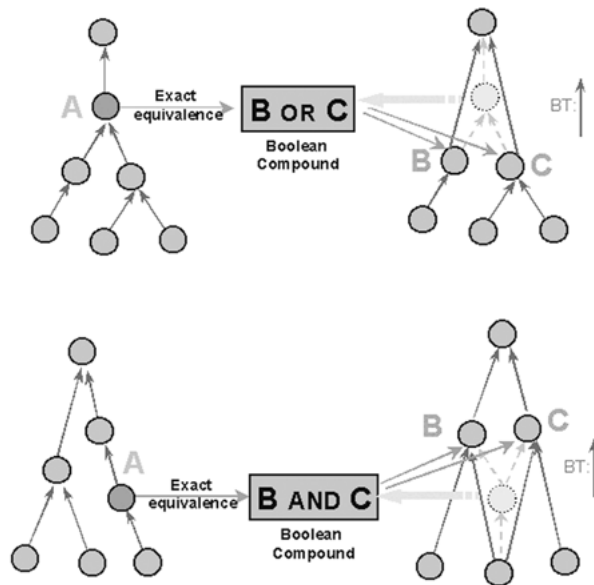


Figure L.6.7: Two Cases of One-Manyess in Concordances. Source: Doerr, 2001, Fig. 3 and 4.

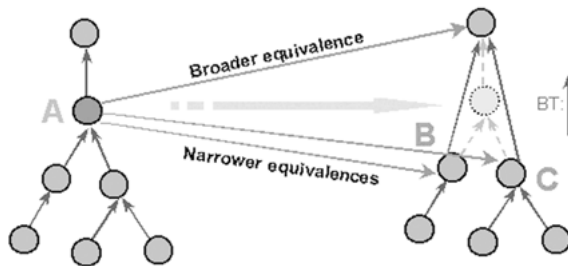


Figure L.6.8: Non-Exact Intersections of Two Concepts. Source: Doerr, 2001, Fig. 5.

A concordance must take into consideration the following relations between the concepts of the KOSs that are to be linked (Bodenreider & Bean, 2001):

- one-to-one (quasi-)synonymy,
- one-to-many (quasi-)synonymy; inverse: many-to-one (quasi-)synonymy,
- many-to-many (quasi-)synonymy,
- non-exact match,

- broader term; inverse: narrower term (perhaps specified into hyponymy and meronymies),
- “lies between ... and ...”; inverse: hyperonym, hyponym,
- “use combination”; inverse: “used in combination”.

Statistical procedures help during the intellectual tasks of creating concordances. Here, parallel corpora must be on hand, more specifically documentary units, which have been indexed in a first database with KOS A and in a second database with KOS B (Figure L.6.9). The documentary unit A2, for instance, has been indexed with the concept *c* from KOS A, B2 with *z* from B. Since we know that A2 and B2 represent the same documentary reference unit, we have now found an indicator for *c* from A being used analogously to *z* from B. If this relation is confirmed in further documents, it will clearly indicate that there is a concordance relation between *c* and *z*.

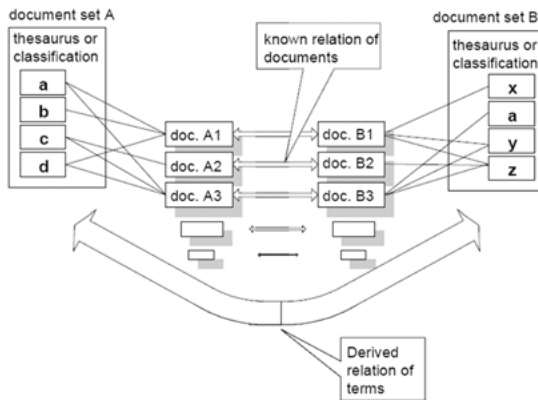


Figure L.6.9: Statistical Derivation of Concept Pairs from Parallel Corpora. Source: Hellweg et al., 2001, 20.

The similarity between *c* and *z* can be calculated via the relevant algorithms Jaccard-Sneath, Dice and Cosine, respectively. We would like to demonstrate this via the Jaccard-Sneath formula. Let our corpus contain only documents that occur in both databases (indexed differently). Let *g* be the amount of documents (from both databases) that have been indexed by *c* in A and by *z* in B, let *a* be the amount of documents in A that name *c* and *b* the amount of documents in B that name *z*. Then, the similarity between *c* and *z* is calculated via Jaccard-Sneath:

$$\text{SIM}(c-z) = g / (a + b - g).$$

The resulting SIM values are indicators for similarity, but they are not a legitimate expression of a semantic relation (of whatever nature it may be). In cases of a shortage in means, it can be justifiable to create a (statistical) mapping only via similarity

algorithms. Here, two concepts are accepted to be linked if their SIM value exceeds a threshold value (to be determined). However, it is preferable to proceed via the intellectual creation of concordances (while taking into consideration the different concordance relations), for which statistics provide a useful heuristic basis.

If concordances are present, these can be automatically used for cross-database searches (e.g. for searches that span all shells of Krause's model). If the user has formulated a search request via a KOS, the system will translate the search arguments into the language of the respective database using the concordance relation and search in every database using the appropriate terminology.

Unification: “Merging” and “Integration”

We understand the unification of knowledge organization systems to mean the creation of a new KOS on the basis of at least two sources, where the new work normally replaces the old ones (Predoiu et al., 2006, 7). Here, the objective is to amalgamate both the concepts and the relations (Udrea, Getoor, & Miller, 2007; Wang et al., 2006) into a new unit. One must distinguish between two cases: the unification of thematically identical KOSs (merging) and the unification of thematically complementary KOSs (integration).

One definition of the merging of KOSs is provided by Pinto, Gómez-Pérez and Martins (1999, 7-7):

In the merge process we have, on one hand, a set of ontologies (at least two) that are going to be *merged* ($O_1, O_2, \dots, O_n$), and on the other hand, the *resulting* ontology (O). The goal is to make a more general ontology about a subject by gathering into a coherent bulk, knowledge from several other ontologies in that same subject. The subject of both the merged and the resulting ontologies are the same (S) although some ontologies are more general than others, that is, the level of generality of the merged ontologies may not be the same.

During integration, the aspect of the same subject shared by all participating KOSs falls away (Pinto & Martins, 2001). Here, the sources complement each other (Pinto, Gómez-Pérez, & Martins, 7-4):

The ontology resulting from the integration process is what we want to build and although it is referenced as one ontology it can be composed of several “modules”, that are (sub)ontologies. ... In integration one can identify regions in the resulting ontology that were taken from the integrated ontologies. Knowledge in those regions was left more or less unchanged.

Integrations are far easier to create than KOSs unified via mergers, as the latter must decide for every concept that is not identical in terms of extension, intension and designation, as well as for every different specific relation, which variant should enter the target KOS.

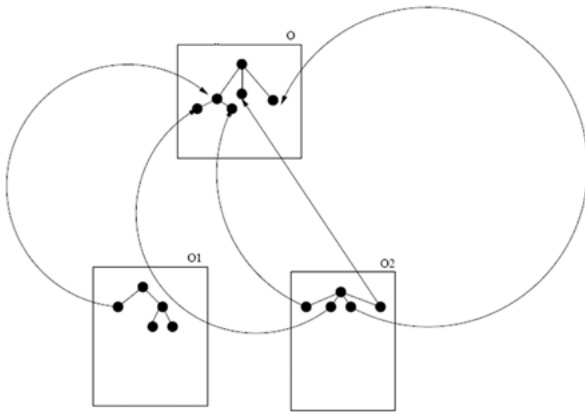


Figure L.6.10: The Process of Unifying KOSs via Merging. Source: Pinto, Gómez-Pérez, & Martins, 1999, 7-8.

A hypothetical example (Figure L.6.10) shall illustrate the process of merging. Let there be two initial KOSs O_1 and O_2 , which only have the hierarchy relation. The two top terms are identical, as are some concepts on the lower hierarchy levels. However, the individual hyponymy and hyperonymy relations are not the same. The resulting ontology O is a compromise, which attempts to retain the strengths of the individual initial KOS where possible, and to eliminate their weaknesses. Can such a process be automatized? Pinto, Gómez-Pérez and Martins (1999, 7-7) are skeptical:

The way the merge process is performed is still very unclear. So far, it is more of an art.

Conclusion

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- Both heterogeneously indexed, distributed bodies of knowledge and the use of the Shell Model lead to the user being confronted by a multitude of different methods and tools of knowledge representation.
 - Semantic crosswalks facilitate a unified access to such heterogeneous databases, and, further, they facilitate the reuse of previously introduced KOSs in other contexts.
 - The parallel application of different KOSs (not processed further) within a system provides perspectives on one and the same body of documents from different perspectives (polyrepresentation).
 - The upgrading of KOSs means the transition from a relatively unexpressive method to a more expressive one (following the path: folksonomy—nomenclature—classification—thesaurus—ontology).
 - The cropping of subject-specific conceptual subsets from a more comprehensive KOS (with possible complements and preservation of consistency) is a simple and inexpensive variant of the reuse of KOSs.

- Concordances relate the concepts of different KOSs to one another. We distinguish between direct concordances and those with a Master. The semantic concordance relations take into consideration (quasi-)synonymy (in the variants one-to-one, one-to-many, many-to-one and many-to-many), hyponymy and hyperonymy, combinations, non-exact matches and accordance with hierarchical borders.
 - Statistical procedures that calculate similarities between the concept materials of two KOSs are of use for the creation of concordances. The precondition is the existence of a parallel corpus.
 - The unification of KOSs occurs in two variants. The first (and far more easily performable) variant, integration, amalgamates sources that complement each other, whereas merging unifies thematically identical sources. In merging, consistency control, particularly of the relations, is assigned a particular role.
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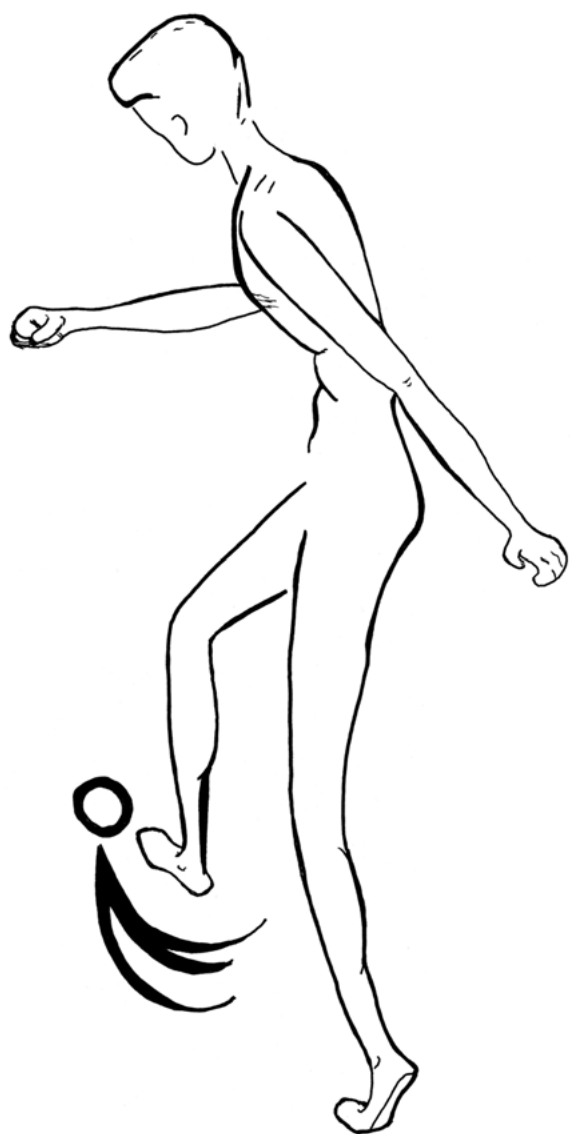
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Part M

Text-Oriented Knowledge Organization Methods



M.1 Text-Word Method

Restriction to Text Terms

The basic idea of developing an electronic database exclusively dedicated to philosophical literature dates back to the 1960s and was formulated by the philosopher Diemer (1967). The preparation, development and implementation of the project *Philosophische Dokumentation* (documentation in philosophy) were undertaken, in 1967, by Henrichs, who developed the Text-Word Method as a method of knowledge representation for documents in the arts and humanities in particular (Henrichs 1970a; 1970b; 1975a; 1975b; 1980; 1992; Stock & Stock, 1991; Stock, 1981; 1984; 1988; Werba & Stock, 1989). Why did Henrichs not draw on previously established methods, like classification or thesaurus? The variety and individuality of philosophical terminology on the one hand and the ideological background of the single publications on the other, according to Henrichs, require a specific method of processing. Henrichs rejects any standardization or narrowing down, respectively, of a humanistic author's language via documentation. According to him, philosophy in particular does not have any set language, meaning that standardization could lead to users being provided with misinterpretations. Each respective scientific community creates its own terminology. Henrichs (1977, 10) even describes the usage of thesaurus and classification as discrimination:

Today's so-called documentary languages (meaning thesauri, A/N)—not to mention yesterday's classification systems—are frequently crude, interpreting and often prematurely standardizing instruments of information indexing and transmission.

The philosophical, and particularly the hermeneutical orientation on the text gives rise to a method of knowledge representation that operates exclusively on the available term material of the specific texts. It does not matter whether the text expresses knowledge, suppositions or assertions; the focus is purely on the object under discussion. The text-word method is not a knowledge organization system; it is a text-oriented method of knowledge representation. Only those terms that occur in the individual text are allowed for content indexing. This also demarcates the limited field of application of the text-word method: movies, images or acoustic music are disregarded as documentary reference units, since they do not have any text.

Where can knowledge organization systems be used and where can't they be applied? Why are the existing KOSs unsuited for those disciplines that do not have a widely prevalent, established and consistent specialist language? Let us turn to Henrichs's argumentation!

First, we will draw on Kuhn's deliberations on normal science knowledge. Wherever there is a consensus between scientists over a certain period of time about their discipline's system of concepts, and—following Kuhn (1962)—about a paradigm, a

knowledge organization system may be used as an apt tool or as an aid for the paradigm. This only applies in the context of a normal science, in the context of a holistically conceived theoretical tradition. Here, several specific KOSs can be worked out for the respective sciences. Areas outside of normal science, or areas that touch upon several adjacent or subsequent paradigms, are unsuited to knowledge organization systems. According to Henrichs, philosophy is one such non-paradigmatic discipline. In philosophy, but also in many humanistic and social disciplines, the terminology depends upon various different systems, theories, ideologies and its authors' personal linguistic preferences.

Classification as well as thesaurus do not represent the evolution and history of a scientific discipline (Henrichs, 1977), but only a synchronous view of it. Arguments against a classification, according to Henrichs (1970a, 136), include:

On the one hand, classifications are always oriented on the science's temporary state of development, and any permanent adjustment to changing states of facts, even though within the realm of possibility, can only ever refer to the respective material to be processed anew, while disregarding that which has already been stored. On the other hand classifications always bear the mark of scholastic orientation, and are hardly ever free of ideology.

Since philosophy does not have its own specialist language, Henrichs (1970a, 136-137) speaks out against the use of a thesaurus for this discipline:

As concerns the list of concepts relevant to this area ..., there is hardly any philosophical specialist language. The history of philosophical literature teaches us that practically every word of every respective ordinary language has been the subject of discussion at some point.

Would a full-text storage not be pertinent, then, since it would, ideally, completely store the philosophical documents? Henrichs denies this, too (1969, 123). The indexing of full texts by machine

may lead to usable results in stylistic comparisons or other statistical textual examinations, but it is of no use for a targeted literary search because it would inevitably lead to a gargantuan information overload. The answer to a search query would of course yield an unbroken catalog of instances of the desired terms, but the mere occurrence of a term at any point in the text does not in itself mean that it is actually being discussed, which is the sole point of interest for the user of the documentation.

Henrichs is thus not interested in the mere occurrence and listing of some words in a text, but in the recording of the textual context, whose "significant" text-words are selected and thematically linked by the indexer.

Here we encounter hermeneutical problems. The indexer must at the very least have a rough understanding of what the text is about, which text-word is meant to be extracted and into which context it should be placed with other text-words or names.

The text-word method cannot be used without any preunderstanding and interpretation. There is, as of yet, no automatized text-word method.

The text-word method does not deal with concepts but only with words that authors use in their texts. Synonyms are not summarized and homonyms are not disambiguated.

Low-Interpretative Indexing

According to this underlying pluralistic worldview, the indexer should analyze the documents as neutrally as possible. It must be stressed, regarding the selection of text-words, that these are not non-interpretative in the hermeneutical sense, but are at best low-interpretative. After all, it is left to the indexer's discretion which specific text-word he selects for indexing and which one he leaves out.

As a selection method, the text-word method indexes literature via those text-words that occur either frequently or at key points in the text (e.g. in the title, in the subheadings or in summarizing passages). The selected text-words mark "search entries" into the text. In his work process, the indexer assesses whether a user should be able to retrieve the text via a certain text-word or not. It is a balancing act between information loss and information overload. When the indexer marks a text-word, he must consider whether the user who is working on the topic being described by the text-word would be (a) disappointed if he were not led to the text, as he would have deemed it important had he known of its existence, or (b) disappointed if he were led to it, as he would deem it irrelevant to his work. This balancing act becomes even more precarious when we imagine the different groups of users of a database. A specialist needs other information for an essay to be published than a student does for a seminar paper.

Usage of the text-word method is absolutely reliant on the very elaborate intellectual work of the indexer, which—as opposed to the other methods of knowledge representation—requires a considerable amount of time and money.

In indexing, the text at hand is scoured for thematized text-words and personal names. Apart from linguistic standardizations toward a preferred grammatical form for topics and toward a preferred form for personal names, the selected indexing terms are always identical to the text terms.

Meinong, Alexius: Über Gegenstandstheorie, in Untersuchungen zur Gegenstandstheorie und Psychologie, ed. by Alexius Meinong. Leipzig: Johann Ambrosius Barth, 1904, pp. 1-50.

Thematic Context:

Topics: Gegenstandstheorie (1-18); Etwas (1); Gegenstand (1-15); Wirkliche, das (2-3); Erkenntnis (2,10); Objektiv (3,10); Sein (4,6-8); Existenz (4-5); Bestand (4); Sosein (5-6); Nichtsein (5); Unabhängigkeit (6); Gegenstand, reiner (7-8); Außersein (7-8); Quasisein (7); Psychologie (9); Erkenntnisgegenstand (10); Objekt (10); Logik, reine (11); Psychologismus (11-12); Erkenntnistheorie (12);

Mathematik (13,18); Wissenschaft (14,18); Gegenstandstheorie, allgemeine (15); Gegenstandstheorie, spezielle (15,18); Philosophie (17); Metaphysik (17); Gegebene, das (17); Empirie (17); Apriorische, das (17); Gesamtheit-der-Wissenschaften (18)

Names: Mally, Ernst (6); Husserl, Edmund (11); Höfler, Alois (16)

Figure M.1.1: Surrogate According to the Text-Word Method. Source: Stock & Stock, 1990, 515.

The text is always indexed in its original language. In indexing practice, an indexing depth of around 0.5 to 2 text-words per page has proven to be effective. The thematic relationships are expressed via digits, the so-called “chain numbers”. Identical chain numbers following different text-words show (independently of their numerical value) the thematic connectivity of the text-words in the text at hand. The text-word method is a means of syntactical indexing via chain formation. The user recognizes, via the allocation of chain numbers behind the text-words, in which relation names and topics stand vis-à-vis each other and how relevant a given text-word is in the discussed text.

Our example of a surrogate in Figure M.1.1 shows 18 groups of topics. The text-word *Gegenstandstheorie* (theory of objects) is at the center here, being the core theme of the text and occurring in all 18 groups. *Bestand* (subsistence; chain number 4) is thematically connected with *Gegenstand* (object), *Sein* (being) and *Existenz* (existence). *Psychologie* (psychology; chain number 9) is only linked to *Gegenstandstheorie*.

In retrieval, syntactical indexing is needed to refine the search results. Let us suppose someone is searching for “*Existenz* and *Mathematik*” (existence and mathematics). The formulation of this search *without* syntax, i.e.

Existenz AND Mathematik

will find our example (Figure M.1.1), as both terms occur within it. This attestation would be superfluous, though, as our text discusses the two subjects at completely different places and never together. An analogous search *including* syntax, such as

Existenz SAME Mathematik

will, correctly, fail to unearth our example, as the chain numbers for *Existenz* (4-5) and *Mathematik* (13, 18) are different.

Syntactical indexing via chain formation facilitates a weighted retrieval. In our example, it is obvious that the text-word *Gegenstandstheorie* is much more important than, for instance, *Bestand*, as the latter only occurs in one chain. A weighting value is calculated for every text-word via the frequency of occurrence in the chains as well as the structure of the chains. This value lies between greater than zero and one hundred (Henrichs, 1980, 164 et seq.). We now turn to the central literature on this topic via a search request

Bestand [Weighting > 60]

and are, consequently, not led to our example. (The retrieval software should be capable of freely choosing the weighting value.)

Text-Word Method with Translation Relation

Hermeneutically speaking, authors who speak different languages belong to different communities of interpretation, which potentially have different horizons of understanding. If we are to do justice to this phenomenon in the translation of text-words, we cannot insist upon a one-to-one translation but must accept ambiguities that stem from the respective authors' language usages.

A bibliographical reference database built to the specifications of the text-word method contains indexed text-words in as many languages as there are publication languages for the given object of investigation. Our sample database on the philosophy and psychology of the Graz School shows that a combined multilingual bibliographical and terminological database can be built and managed via the text-word method (Stock, 1989; Stock & Stock, 1991). As our unified language, we chose English for demonstration (In the original research, it was German.). For non-English literature, the topics are listed both in the documentary reference unit's original language and, in parallel and with thematic chains, its translations into the unified language. The principal objective is to maintain the linguistic nuances both on the basis of the text-word method and in the context of the translations. The terms in the original language are thus enhanced via "doubles" in the unified language. Ambivalences in the linguistic variants are taken into consideration.

Ambiguous Translation Relations are the result of different usages of a syntactically equivalent term in different contexts. One-manyness is present when there are several unified-language translations with different meanings for a foreign-language term. In the Graz School's philosophy, there is a distinction between *Gegenstand* (object in the more general meaning of "a thing or a proposition") and *Objekt* (single object), where objects represent a certain class of *Gegenstände* alongside others. *Gegenstand* is the hyperonym of *Objekt*; the two terms' meaning is thus completely different. In English, however, some authors use *object* to mean both terms. We speak of many-oneness in the reverse case scenario, when there are several foreign-language terms that can only be translated into a unified language via one single term. Thus the two English variants *so-being* and *being-so* can only be rendered into German as *Sosein*.

Meinong, Alexius: Über Gegenstandstheorie, in: Untersuchungen zur Gegenstandstheorie und Psychologie, ed. by Alexius Meinong. Leipzig: Johann Ambrosius Barth, 1904, pp. 1-50.

Thematic Context:

Topics (Original Language):

Gegenstandstheorie (1-18)
 Etwas (1)
 Gegenstand (1-15)
 Wirkliche, das (2-3)
 Erkenntnis (2,10)
 Objektiv (3,10)
 Sein (4,6-8)
 Existenz (4-5)
 Bestand (4)
 Sosein (5-6)
 Nichtsein (5)
 Unabhängigkeit (6)
 Gegenstand, reiner (7-8)
 Außersein (7-8)
 Quasisein (7)
 Psychologie (9)
 Erkenntnisgegenstand (10)
 Objekt (10)
 Logik, reine (11)
 Psychologismus (11-12)
 Erkenntnistheorie (12)
 Mathematik (13,18)
 Wissenschaft (14,18)
 Gegenstandstheorie, allgemeine (15)
 Gegenstandstheorie, spezielle (15,18)
 Philosophie (17)
 Metaphysik (17)
 Gegebene, das (17)
 Empirie (17)
 Apriorische, das (17)
 Gesamtheit-der-Wissenschaften (18)

Topics (Unified Language)

Theory-of-objects (1-18)
 Something (1)
 Object (1-15)
 Real (2-3)
 Knowledge (2,10)
 Objective (3,10)
 Being (4,6-8)
 Existence (4-5)
 Subsistence (4)
 Being-so (5-6)
 Nonbeing (5)
 Independence (6)
 Object, pure (7-8)
 Being, external (7-8)
 Quasi-being (7)
 Psychology (9)
 Object-of-knowledge (10)
 Object, single (10)
 Logic, pure (11)
 Psychologism (11-12)
 Epistemology (12)
 Mathematics (13,18)
 Science (14,18)
 Theory-of-objects, general (15)
 Theory-of-objects, special (15,18)
 Philosophy (17)
 Metaphysics (17)
 Given (17)
 Empiricism (17)
 Apriori (17)
 Totality-of-sciences (18)

Names: Mally, Ernst (6); Husserl, Edmund (11); Höfler, Alois (16)

Figure M.1.2: Surrogate Following the Text-Word Method with Translation Relation.

A precarious problem is posed by many-manyness. This sort of relation occurs when one-manyness and many-oneness coincide. The English *idea* can be translated both as *Vorstellung* (perception) and *Repräsentation* (representation). These two German terms, however, also serve as fitting equivalents for *representation*. In the case of such crisscrossing, it must be decided during indexing which translation is the appropriate one in the individual documents.

The input into the combined bibliographical and terminological database ensues in the same cycle. As the bibliographical database is being constructed, the termino-

logical database is still empty; however, as the former becomes ever more productive, it will itself become more complex and complete. For every foreign-language term, it is always checked whether it is already present in the terminological database. If so, it is additionally checked whether the available translation does justice to the terms at hand. If it does not, a new translation variant is created. Following this procedure, the dictionary of the specialist language is built up one step at a time and will always be up to date on the terminology. Figure M.1.2, continuing the example from Figure M.1.1, demonstrates a surrogate which has been created following the text-word method with translation relation.

Fields of Application

The text-word method has the disadvantage of not relating the documents' topics to concepts but remaining on the basis of the words. Synonyms, quasi-synonyms and homonyms are not paid any attention, and neither are there any paradigmatic relations (as these exist solely between concepts, not between words). The user will not find any controlled vocabulary. The text-word method has the advantage, however, of representing the authors' languages authentically. It is thus of excellent usefulness for historical studies of languages, language development and disciplines. For instance, it was possible to describe the history of the Graz School on the basis of the text-word method (Stock & Stock, 1990, 1223 et seq.).

In information practice, the text-word method can be used in two places. On the one hand, it is suited to such disciplines that have no firm term material, e.g. philosophy or literary studies. If the respective discipline is to be evaluated from a certain perspective, though (e.g., philosophy from a Marxist-Leninist viewpoint), one might prefer to use a knowledge organization system. The second area of usage is related to knowledge domains in which no KOS is available yet. Such a case is at hand, for instance, when a company begins indexing its internal documents in a corporate information service. The goal here is to represent the language of the company in such a way that it can be used for indexing and search in an in-company knowledge organization system. The text-word method helps (in a similar manner to a folksonomy)—in the sense of an emergent semantics—during the creation and expansion of KOSs (Stock, 2000, 31-32).

Conclusion

-
- The text-word method does not work with KOSs but represents a text-oriented method of knowledge representation. The text-word method works with words, not with concepts.
 - For content indexing, only those words are admitted that do in fact occur in the text to be indexed. The method is thus relatively low-interpretative.

- The text-word method does not only represent single topics, but also groups of them. These thematic relations are established via syntactic indexing and chain formation.
 - Since the text-word method always works with the original language, an additional translation of the text-words is necessary for multilingual access. This is accomplished by using the text-word method with translation relation.
 - The text-word method is used in the information practice of non-normal scientific disciplines as well as in the creation and expansion of KOSs.
-

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M.2 Citation Indexing

Citation Indexing in knowledge representation focuses on bibliographical data in publications, either as footnotes, end notes or in a bibliography. It thus claims to be a text-oriented method of content indexing. Inspecting the text that cites and the document being cited, we can reconstruct the transmissions of information that have contributed toward the success of the citing text (Stock, 1985). At the same time, we can see in the reverse direction how reputation is awarded to the cited document. From the citer's perspective, this is a "reference", from the citee's perspective, a "citation". There is always a timeline in the model: the cited document is dated at an earlier time than the one doing the citing. The cited and the citing document are directly linked to each other via a directed graph (in the direction cited—citing: information transmission, in the opposite direction: reputation). References and citations are viewed as concepts that represent the content of the citing and the cited document, respectively. Citation indexing is used wherever formal citing occurs, i.e.

- in law (citations of and references in court rulings),
- in academic science (citations of and references in scientific documents),
- in technology (citations of and references in patents).

Data gathering in citation indexing occurs either manually (with minimal intellectual effort) or automatically.

Citation Indexing of Court Rulings: Shepardizing

In the United States, court decisions have a special relevance. In verdicts and their history, we are being told what is regarded as "good law" at any given time. The idea of evaluating legal citations was coined by Shepard. Since the year 1873, references to verdicts have been collected and intellectually analyzed. Young (1998, 209) observes:

In the United States, legal researchers and librarians are weaned on the citator services offered by *Shepard's Citations*. Begun in 1873, *Shepard's* has grown over the past century into a legal institution. It is a tool used by virtually everyone to determine the history and treatment of a case, a legislative enactment, a court rule, or even a law review article.

By gathering all references in published rulings, it is possible to elicit where a previous verdict (or law) has been cited. The bases of the legal citation index are individual judges' and courts' interpretations of the law, as well as those of commentators and legal scholars, who record them in their published verdicts, comments or articles. As opposed to the "simple" indexing of references and citations (as implemented for citations in the technological and academic spheres), Shepard's intellectually elicits and indexes the types of citation. The question that guides the indexer during this process is always (Spriggs II & Hansford, 2000, 329):

What effect, if any, does the citing case have on the cited case?

View: [KWIC](#) | [Full](#) | [Display Options](#) | [Unrestricted](#) | [All Neg](#) | [All Pos](#) | [Custom](#) | [EQUIS™](#) | [FAST Print](#) | [Print](#) | [Download](#) | [Fax](#) | [Email](#) | [Text Only](#)

1 - 50 of 790 Total Cites [NEXT](#)

Shepard's® [United States v. Dunn, 480 U.S. 294 \(TOA\)](#)

Signal: [△](#) Caution: Possible negative treatment
Trail: [Unrestricted](#)

United States v. Dunn, 480 U.S. 294, 94 L. Ed. 2d 326, 107 S. Ct. 1134, 1987 U.S. LEXIS 1057, 55 U.S.L.W. 4251 (1987)

SHEPARD'S SUMMARY • [Hide Summary](#)

Shepard's FULL Summary: [△](#) Caution - Possible negative treatment
Subsequent appellate history contains possible negative analysis.

Citing References:

Cautionary Analyses:	Criticized (2) , Distinguished (7)
Positive Analyses:	Followed (131) , Concurring Opinion (15)
Neutral Analyses:	Dissenting Op. (18) , Explained (7)
Other Sources:	Law Reviews (176) , Statutes (1) , Treatises (60) , American Law Rpts/Lawyers' Edition Annos (4)

PRIOR HISTORY (11 citing references) • [Hide Prior History](#)

[Select for Delivery](#)

- [1. United States v. Dunn](#), 674 F.2d 1093, 1982 U.S. App. LEXIS 19446 (5th Cir. Tex. 1982)
- [2. Appeal after remand at:](#)
[United States v. Dunn](#), 706 F.2d 153, 1983 U.S. App. LEXIS 27219 (5th Cir. Tex. 1983)

Figure M.2.1: Shepardizing on LexisNexis. Source: Stock & Stock, 2005, 61.

Looking at Figure M.2.1, we can clearly see a(n originally yellow) triangle in front of the citation details, signaling “attention”. The citations are structured into the following groups (where they are subdivided even further):

- caution: negative reference (signaling color: red),
- questionable—validity of a verdict is unclear (orange),
- attention: possible negative interpretation (yellow),
- positive—case is discussed favorably (green),
- neutral—neither negative nor positive (blue “A”),
- citation information available in other sources (blue “I”).

Shepard's draws on around 400 legal journals as well as on all US rulings available on LexisNexis. Negative citations, which are extremely important in order to recognize whether a verdict still applies, are in the database after two to three days, according to LexisNexis (Stock & Stock, 2005, 61).

The indexers may follow a detailed rulebook (Spriggs II & Hansford, 2000, 329-332), yet a certain room for interpretation in assessing a reference cannot be entirely ruled out. However, a study on the reliability of the assessments, conducted by Spriggs and Hansford (2000, 338), yields very good results:

Our analysis indicates that *Shepard's* provide a reliable indicator of how citing cases legally treat cited cases. We are particularly sanguine about the reliability of the stronger negative treatment codes (Overruled, Questioned, Limited, and Criticized), while the neutral treatment codes (Harmonized and Explained) appear to be the least reliable.

In 1997, the publishing house Reed Elsevier bought Shepard's Citation Index from McGraw-Hill and shortly thereafter, understandably, withdrew it from the host of its

competitor Westlaw. Westlaw has by now introduced its own tool for citation indexing, called KeyCite (Linde & Stock, 2011, 212-214), so that there are currently two legal citation services for American law. According to their critics, both products differ only marginally (Teshima, 1999, 87):

Because of the substantive similarities, it is hard for researchers to make a bad choice. Which one is right for a particular firm probably depends more on personal preferences and which online giant offers the better deal. In any event, the consequence of this war of citators has resulted in substantial improvement to both products, making legal researchers the ultimate victor.

Citation Indexing for Academic Literature: Web of Science

In the 1950s, building on Shepard's Citations (Wouters, 1999, 128-129), Garfield introduced the idea of an academic citation index that—carried by the idea of unified science—would cover all scientific disciplines (Garfield (2006 [1955], 1123):

This paper considers the possible utility of a citation index that offers a new approach to subject control of the literature of science. By virtue of its different construction, it tends to bring together material that would never be collated by the usual subject indexing. It is best described as an association-of-ideas index, and it gives the reader as much leeway as he requires.

In contrast to Shepard's, Garfield's "Science Citation Index" puts no store by the citation's qualifications and instead exclusively notes the respective facts of information transmission and reputation. This is justified by the difficulty of such an assessment in the academic sphere as well as by the sheer amount of information, which in this case makes intellectual qualification practically impossible (Garfield & Stock, 2002, 23).

Garfield's academic citation indices, as well as his Institute for Scientific Information (ISI)—now belonging to Thomson Reuters—are published in the four series "Science Citation Index" for the natural sciences, "Social Sciences Citation Index" for the social sciences, "Arts & Humanities Citation Index" for the humanities and "ISI Proceedings" for conference publications. All of these are available in a unified presentation in "Web of Science" (WoS). WoS forms the product "Web of Knowledge" (Stock, 1999; Stock & Stock, 2003) in combination with further databases. Outside the scope of knowledge representation, WoS and products derived from it, such as the "Journal Citation Reports" (Stock, 2001b) and "Essential Science Indicators" (Stock, 2002) are developing into a tool of scientometrics as well as of the bibliometrics of academic journals (Garfield, 1972; Haustein, 2012). "Web of Science" and "Web of Knowledge" are thus becoming central sources of informetric analyses (Cronin & Atkins, Eds., 2000).

Since it is impossible, for practical reasons, to analyze all academic journals (Linde & Stock, 2011, ch. 9), Garfield attempts to mark and register those periodicals that have the greatest weight within their respective field. It turns out that journals—

arranged in descending order by number of citations received—follow a Power Law (Garfield, 1979, 21):

One study of the *SCI* data base (...) shows that 75% of the references identify fewer than 1000 journals, and that 84% of them are to just 2000 journals.

Additionally, it turns out that the journals in the “long tail” of a given discipline’s distribution often belong to the core journals of other disciplines, i.e. that there is a strong overlap (Garfield, 1979, 23):

This type of evidence makes it possible to move ... to Garfield’s law of concentration (...), which states that the tail of the literature of one discipline consists, in a large part, of the cores of the literature of other disciplines.

With a current (as of 2012) amount of more than 12,000 analyzed journals, a representative (if, inevitably, incomplete) volume of the most important academic periodicals can thus be made available in the sense of a general-science database (Testa, 2006). The journals analyzed will be the most important ones for their respective disciplines (Garfield, 1979, 25).

How can a simple footnote be a bearer of knowledge? Is citation indexing really a method of content indexing? Garfield (1979, 3) is certain of it:

Citations, used as indexing statements, provide ... measures of search simplicity, productivity, and efficiency by avoiding the semantics problem. For example, suppose you want information on the physics of simple fluid. The simple citation “Fisher, M.E., *Math. Phys.*, 5, 944, 1964” would lead the searcher directly to a list of papers that have cited this important paper on the subject.

The literature cited in a document as well as that citing the document are both closely related to that document’s topic. The precondition for a thematic search is that the user must already have received a direct hit (such as the article by Fisher in the example), which he can use as his point of departure. He will then search this result in another database or use WoS with the search via titles, abstracts and keywords offered there. The search via citations is independent of the documents’ language of publication as well as of the language of any knowledge organization systems otherwise implemented.

In citation databases, the user is provided with two simple retrieval options, which allow him to pursue the relations of intertextuality:

- Search for and navigation to the references of a source document (search “backward”),
 - Search for and navigation to the citations of a source document (search “forward”).
- In the documentary unit, the link “Cited References” leads to the sources cited in the article, while the link “Times Cited” leads to the documents that name the initial doc-

ument in their bibliography. Additionally, the user can search for documents related to the source via an analysis of the bibliographic couplings.

“Web of Science” provides the search mode “Cited References”. Here it is possible to launch a targeted search for the citations of a document, and going further, or all citations of an author (including statements of the year they date from). Every hit list on “Web of Science” allows the documents to be ranked according to their number of citations.

For decades the databases of the Institute for Scientific Information had a monopoly in the area of citation indexing. With Scopus and Google Scholar, there now exist some further general-science information services that analyze footnotes.

The indexing of citations in an academic environment is not entirely unproblematic, as the footnotes and bibliographies in the scientific articles can be highly fraught to begin with (Smith, 1981, 86-93; Cronin, 1984; MacRoberts & MacRoberts, 1989; Stock, 2001a, 29-36). Sometimes authors conceal thematically relevant literature, citing instead less pertinent sources that have the advantage of supporting their arguments. Self-citations or reciprocal citations within a citation cartel make it even harder to assess the correctness of bibliographical data. But there are also practical problems in retrieval, such as the homonymy of names or typing errors made by authors when citing. Concerning multiple-author papers, the mere statement of authorship tells us nothing about the actual role played by this individual during the production of the article (e.g. as idea generator, statistician, technical aid or scientist in charge).

In many scientific publications, the acknowledgements occupy a “floating” intermediate position between co-authorship and reference. Here one points to the knowledge of other people which informed the document, which is not mentioned anywhere else but here (Cronin, 1995). Neither citation databases nor any other scientific information services analyze acknowledgements.

Citation Indexing for Patents

References in patents are distinct from references in academic publications in one crucial aspect (Meyer, 2000). They are constructed in the patent office, on the basis of the division of labor between the patent applicant and the examiner, where most of the cited documents are generally introduced by the latter. Garfield (1966, 63) observes:

The majority of references ... are provided by the examiner and constitute the prior art which the examiner used to disallow one or more claims. ... How relevant are these references to the subject matter of any given search? Obviously the examiner considers them relevant “enough” to disallow claims.

On the basis of the citations that touch upon the object of invention, the patent examiner decides whether a patent is granted. Hence, it can be assumed that there will be as high a degree of completeness as possible in the references to these citations of prior art. The citations are printed on page 1 in patents (or, in case of extensive lists, on the first couple of pages); the applicants' references are harder to find, incorporated into the continuous text of the description of the invention.

If a user finds a patent specification that perfectly fits the topic he is searching for, he can assume that the references in this document represent a small special bibliography on the subject. References in patents refer both to other patents and to all other document types, such as scientific literature or press reports from scientifically oriented enterprises.

Apart from the usefulness of patent references for searches on the state of technology, they also prove useful for the surveillance of one's own patents. Performing a search for citations of his own patents, the patent holder will glean information about who has used his inventions for their own technological developments, and how often his inventions have been cited in other patent specifications. Such searches form the basis of informetric patent analyses (Narin, 1994); they also represent indicators for an invention's degree of innovation, and—in the case of aggregation on the company or industry level—indicators for the technological performance of the company or respective industry. Albert et al. (1991, 258) can show that a patent's number of citations correlated positively with its degree of innovation:

It can be quite directly concluded from this study that highly cited patents are of significantly greater technological importance than patents that are not cited at all, or only infrequently cited.

Automatic Citation Indexing: CiteSeer

The references in legal, academic and technological citation databases are recorded intellectually by human indexers. Can this process of indexing be automatized for digitally available documents? The system CiteSeer (Giles, Bollacker, & Lawrence, 1998; Lawrence, Giles, & Bollacker, 1999) pursues an automatized approach of citation indexing.

The first task is to recognize the references in the document, a task which is controlled by the retrieval of certain markers, such as (*Meyer 2000*), [*5*], or *MEY00*. Additionally, CiteSeer elicits, for every reference in the bibliography, that point in the text which refers to the reference. Thus in a search for citations one can state not only that the document is being named at all, but the user can also be shown the precise context of the way the foreign knowledge is being used.

The second task consists of dividing the references into their component parts. These are the citation marker, the authors' names, title, source and year. Here an attempt is made to recognize and exploit patterns in the indexing. For instance, many

bibliographical statements begin with the marker, followed by the authors' names. Lists of prominent author names and journal titles facilitate the localization of the respective subfield.

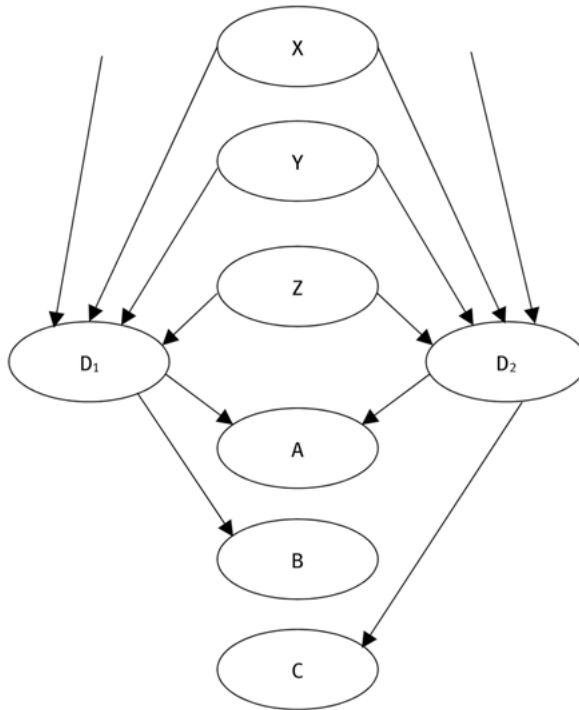


Figure M.2.2: Bibliographic Coupling and Co-Citation. The Arrows Represent Reputation (References), e.g. D_1 Contains a Reference to A.

In the third step, the references concerning one and the same work—which have been formulated and formatted in very different ways—are summarized. This is not a trivial matter (Lee et al., 2007), as is shown by the following three bibliographical statements from different documents, each “meaning” the same work (Lawrence, Giles, & Bollacker, 1999, 69):

Aha, D. W. (1991), Instance-based learning algorithms, *Machine Learning* 6(1), 37-66.

D. W. Aha, D. Kibler and M. K. Albert, *Instance-Based Learning Algorithms*. Machine Learning 6 37-66. Kluwer Academic Publisher, 1991.

Aha, D. W., Kibler, D. & Albert, M. K. (1990). Instance-based learning algorithms. Draft submission to *Machine Learning*.

CiteSeer deletes the markers, hyphens, certain special characters (such as &, (,), [,], or :) and some words (e.g. pp., pages, in press, accepted for publication, vol., volume,

no., number, et al., ISBN) in the references. The remaining words are changed to all lower-case letters. It is now calculated whether these normalized bibliographical statements already fit into the system of pre-existing references. This is accomplished, among other methods, via the personal names and journal titles already recognized in Step 2, as well as via the words' degree of co-occurrence.

Bibliographic Coupling and Co-Citations

There are two principal options of creating connections between two works, building on indexed references:

- directed connections of the information transmissions,
- undirected connections in the two variants
 - bibliographic coupling (analyzing the documents' references)
 - co-citations (analyzing the documents' citations).

Directed connections trace the information streams between cited and citing documents. Undirected connections between documents can be created via bibliographic coupling and via co-citations. Two documents are bibliographically coupled when they make references to the same documents. In the words of Kessler (1963a, 10), the developer of this marker:

[It is described] a new method for grouping technical and scientific papers on the basis of bibliographic coupling units. A single item of reference used by two papers was defined as a unit of coupling between them.

or—expressed more succinctly (Kessler, 1963b, 169):

One item of reference used by two papers.

In Figure M.2.2, the two documents D_1 and D_2 are bibliographically coupled, since they have a shared reference, namely A. The extent of bibliographic coupling depends upon how many common references occur in both. The respective value is time-independent, in contrast to the co-citations, since nothing in the citation apparatus of the citing documents can still change. In a simple version (used by Kessler himself), the coupling strength is expressed by the absolute number of common references. All documents linked to a source document and whose coupling strengths exceed a threshold value, thus spanning a common (undirected) graph, represent a (scientific, technical or legal) topic from the perspective of the citing documents and their authors, respectively (Kessler, 1965).

Co-Citations do not use common references, but common citations. Two documents (both cited) are deemed co-cited when they co-occur in the bibliographical apparatus of citing documents. Our sample documents D_1 and D_2 from Figure M.2.2

are co-cited, since they are named in the citation apparatus of X, Y and Z. A co-citation relation can change at any time, in so far as new publications may appear that cite D_1 or D_2 . The marker for co-citations was introduced by Small (1973, 265):

(C)o-citation is the frequency with which two items of earlier literature are cited together by the later literature.

The relations in the co-citation are created exclusively by the citing authors and their documents. If we look at the co-citations in a scientific specialist field over a longer period of time, we can trace its scientific development (Small, 1973, 266):

Changes in the co-citations pattern, when viewed over a period of years, may provide clues to understanding the mechanism of specialty development.

One either starts with a model document and follows the co-citations that exceed a threshold value (to be defined), or one proceeds cluster-analytically from the pair with the highest co-citation value and completes the cluster, e.g. following the single-link or the complete-link procedure. The documents that are heavily co-cited form the “core” of the respective academic knowledge domain’s research front (Small & Griffith, 1974; Griffith, Small, Stonehill, & Dey, 1974). Small (1977) succeeded in citation-analytically proving a scientific revolution (in the sense of Kuhn) on an example (of Collagen Research).

The simple version of the expression of bibliographic couplings and co-citations counts the absolute frequency of common references and citations, respectively. When calculating the relative similarity between two documents via their bibliographic couplings or co-citations, one uses the Cosine, the Dice Index as well as the Jaccard-Sneath procedure. Small (1973, 269) prefers the Jaccard index:

If A is the set of papers which cites document a and B is the set which cites b , then $A \cap B$ is the set which cites both a and b . The number of elements in $A \cap B$, that is $n(A \cap B)$, is the co-citation frequency. The relative co-citation frequency could be defined as $n(A \cap B) / n(A \cup B)$.

An elaborate variant of bibliographic couplings is submitted by Giles, Bollacker and Lawrence (1998, 95). Analogously to the calculation of TF * IDF in text statistics, they use the formula

$$CC * IDF.$$

CC are the references co-occurring in the two citation apparatuses (“common citations”), IDF the inverse document frequency of the references. The IDF of a reference i is calculated via

$$IDF(i) = [\log(N/n)] + 1,$$

where N counts the total number of data sets in a database and n counts those documents that contain i as a reference. IDF grows larger in proportion to how seldomly documents cite the document in question. We assume that the IDF value of every reference has been calculated. For an initial document A , we now elicit all documents B_1 through B_j that have at least one reference in common with A . The IDF values of the references in common with A are added for all B . In the last step, the B are ranked in descending order via the sum of the IDF values of the “shared references”. The intuitive justification for the combination of the bibliographic couplings with the IDF value is that the co-occurrence of very rare documents is to be weighted more highly than the common citation of already frequently cited documents.

Bibliographic couplings and co-citations pursue two goals:

- proceeding from a model document, the user can find “related” documents via its references and citations,
- documents that are heavily coupled or co-cited form classes of similar works. Our citation-analytical procedures are thus based on methods of automatic quasi-classification (Garfield, Malin, & Small, 1975).

Since both methods have different reference frameworks—stable references for bibliographic couplings, changing citation sets in the co-citations—a combined approach in knowledge representation and information retrieval is particularly promising (Bichteler & Eaton III, 1980).

Conclusion

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- Citation indexing is a text-oriented method of knowledge representation, which analyzes bibliographical statements in publications—in the sense of concepts. It can be used wherever formal citing is being practiced (law, academic science, technology).
 - A bibliographical statement is a reference from the perspective of the citing document, and a citation is its counterpart from the perspective of the cited document.
 - Citation indexing in the legal sphere (Shepardizing) assesses the citation by recording the influence of the cited on the citing (i.e. negative, questionable or positive). In countries where case law is very highly regarded (as in the U.S.A.), the citations can be used to glean what was deemed “good law” at any given time.
 - In the citation indexing of academic literature, all that is noted is the fact of information transmission; there is no assessment. The citation indices (now called “Web of Science”) created by Garfield have attained central importance for scientific literature. Especially in the academic sphere, citation indexing is problematic, since the allocation of bibliographical statements is not always complete, sometimes even containing citation overload.
 - Patent examiners compile little bibliographies for every patent, which, as citations of prior art, decide whether an invention can be regarded as an innovation or not.
 - The automatic indexing of bibliographical statements in digitally available documents faces the tasks of recognizing the references in the first place, of subdividing each reference into its component parts (like author, title, source, year) and of summarizing all statements (which are often formulated differently) into one identical work.

- Citation analysis makes it possible to record connections between documents. We distinguish between the directed paths of information transmission and the undirected options of bibliographic couplings and co-citations. These connections can always be represented in the form of graphs.
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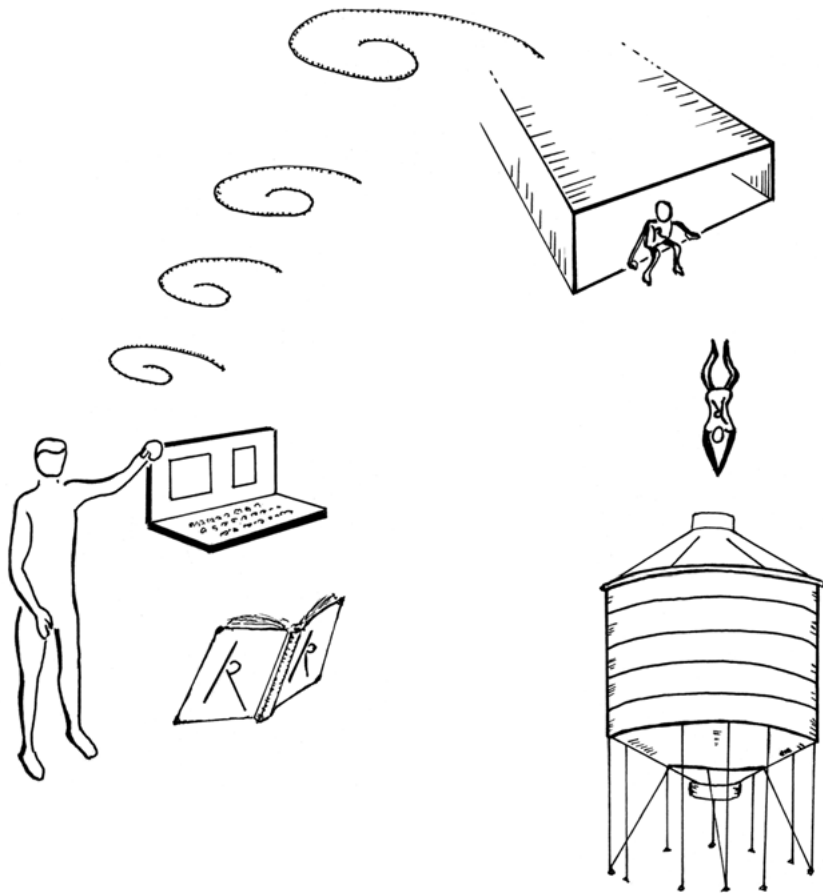
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Part N

Indexing



N.1 Intellectual Indexing

What is Indexing, and what is its Significance?

Information activities help produce informational added value via intermediation. Intermediation is divided into the three phases (1) Information Indexing (Borko, 1977; Cleveland & Cleveland, 2001; Lancaster, 2003), (2) Information Retrieval and (3) the further processing of retrieved information by the user. In information practice, we are thus confronted with three great sources of errors:

- Error Type 1: Representation Error,
- Error Type 2: Retrieval Error,
- Error Type 3: Processing Error.

Documentary reference units are represented in a database via the documentary units (surrogates). The structuring and representation are guaranteed by the meta-data—both the formal and the content-oriented kinds. Information may get lost on the way from a document to its placeholder in a database, particularly if the document is not available digitally. This is Error Source 1, leading to representation errors. Since formal statements (e.g. about author or year of publication) are far less prone to error than the representation of knowledge, particular attention must be paid to content errors, i.e. errors during the indexing process. Error source 2, the retrieval error, lurks in the process of query formulation and in dealing with the search results. Suboptimally formulated search requests, missing system-user dialogs or confusing hit lists further bear the danger of losing information. Error Type 3 consists of the user assessing a retrieved result incorrectly, not understanding it (due to a lack of necessary foreknowledge) or simply ignoring it. Error Type 1 gives rise to the other two: one can only find what has been properly entered into the system. Mai (2000, 270) writes on this subject:

Retrieval of documents relies heavily on the quality of their representation. If the documents are represented poorly or inadequately, the quality of the searching will likewise be poor. This reminds one of the trivial but all too true phrase “garbage in, garbage out”. The chief task for a theory of indexing and classification is to explain the problems related to representation and suggest improvements for practice.

The significance of indexing can scarcely be overstated: here, each individual case decides under which content-oriented aspects a document will be retrieved in an information service. The best knowledge organization system will fail in practice if the right concepts are not found and used during the indexing process. But what does “indexing” mean? The ISO standard 5963:1985 defines “indexing” as

(t)he act of describing or identifying a document in terms of its subject content.

Indexing is thus the practical usage of a method of knowledge representation (additionally, in KOSs, the use of a concrete tool) on the content of a document. As a hyponym of indexing, we can use “classing” when talking about indexing via a classification system.

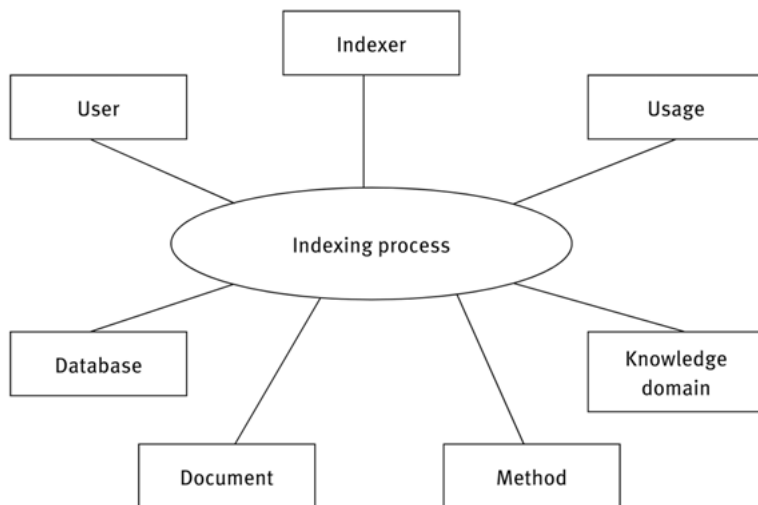


Figure N.1.1: Fixed Points of the Indexing Process.

The process of indexing depends upon several fixed points (Figure N.1.1). First, there is the documentary reference unit that is to be indexed. It is confronted by the indexer with his expertise (firstly, in the knowledge domain; secondly, in information science and, more specifically, in the practice of indexing as well as thirdly, where necessary, in foreign languages). In small organizations, it is possible for the indexer to be personally acquainted with “his” users, and be able to think: “this would be a job for Ms X,” but in most cases, this spot is occupied by an imaginary ideal-typical user. In contrast to the user, who represents an anthropocentric role, usage involves a position in an organization—independently of who occupies said position. When orienting himself on user and usage, the indexer tries to anticipate information needs that can be satisfied by the respective right document. The fixed point of the database summarizes all the documentary units that are available in the database so far. Let us assume that a document discusses a topic T on half a page. If hundreds of search results are already available for T, the indexer may forego the representation of the topic; if T has never been previously discussed, however, the indexer will probably keep this aspect in mind. Always at the back of the indexer’s mind are the methods of knowledge representation to be used: using the text-word method will make for an entirely different indexing than using a thesaurus. Finally, the knowledge domain plays an important role. This involves the context in which the database is set; according to Mai (2005,

606), this is a group of people with shared goals. In the scientific arena, we would be talking about a community of scientists in a “normal science” in the sense of Kuhn. Thus humanistic literature must be indexed differently than a natural-science document or a patent, since in the humanities concepts are inconsistently used. Moreover, there are a lot of longer treatises (monographs in the form of books) being published in this sphere (Tibbo, 1994).

The goal of indexing is to find the most appropriate concepts to serve as representatives and surrogates for the document in the database. During an initial preliminary approach, the indexer is guided by questions such as:

- Document: Which concepts are of importance in the document? Are they useful access points to the document?
- Indexer: Am I sufficiently acquainted with the subject matter?
- User: Are there concrete or ideal-typical users for which the document would be a helping hand on the job?
- Usage: Does someone need to find this document when searching for a particular concept? Does anyone even use this concept to search for this document?
- Database: Are there already “better” documents for this concept? Does the document contribute something new to it?
- Method of Knowledge Representation: Can I even adequately represent the concept in the context of the methods used?
- Knowledge Domain: Is the concept relevant for the scientific debate? How is it to be classified, historically and systematically?

Phases of Indexing

The process of indexing proceeds over several phases, from an analytical point of view. In practice however, these phases are strongly interwoven and can easily be performed simultaneously (Figure N.1.2). The starting point for indexing is the relevant documentary reference unit (Element 1). Document Analysis (Phase 1) provides the indexer with an understanding of the document’s content (Element 2). In Phase 2, the indexer puts down his understanding of the aboutness via concepts (Element 3). When using a KOS, the last step of processing (Phase 3) is a translation of the concepts into the controlled vocabulary of the KOS being used. Mai (2000, 277) describes the three steps that connect the four elements:

The first step, the *document analysis process*, is the analysis of the document for its subjects. The second step, the *subject description process*, is the formulation of an indexing phrase or subject description. The third step, the *subject analysis process*, is the translation of the subject description into an indexing language.

The three steps link four elements of the process. The first element is the *document* under examination. The second element is the *subject* of the document. This element is only present in the mind of the indexer in a rather informal way. The third element is a *formal written description of*

the subject. The fourth is the *subject* entry, which has been constructed in the indexing language and represents the formal description of the subject.

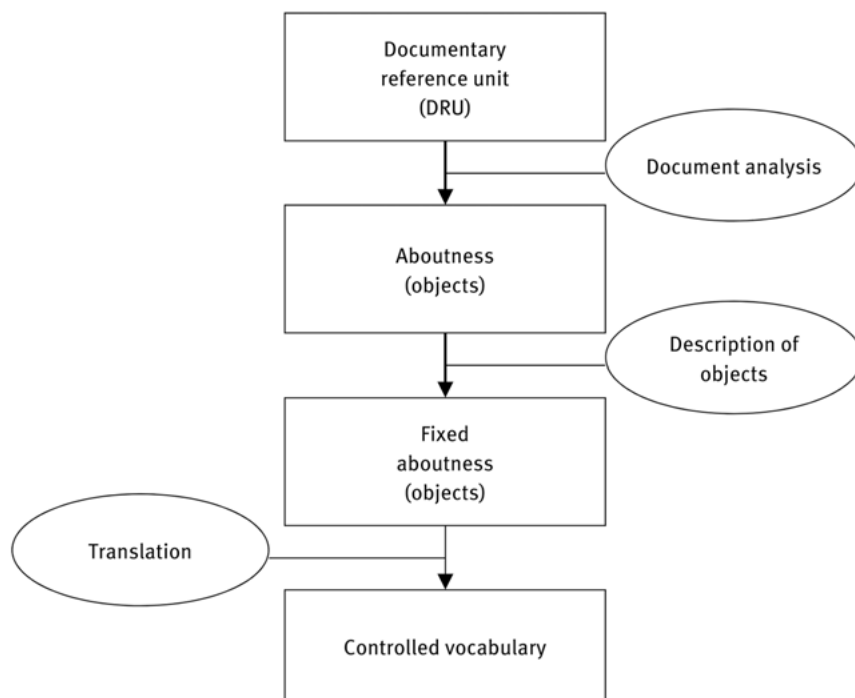


Figure N.1.2: Elements and Phases of Indexing.

Indexing is cognitive work (Milstead, 1994, 578) and can thus be object of cognitive work analysis (CWA). At the center stands the indexer with his characteristics. Analysis of activity leads to the working steps detailed in Figure N.1.2, each of which require strategies (how does an indexer read a document?) and decisions (which are the relevant objects in the document?). The activities are embedded in an organizational framework. Governed by economic necessities, this framework might stipulate that an indexer must submit his indexed terms (including an abstract) within 15 to 20 minutes—which is not an unrealistic value in the context of commercial database production. The analysis of the work domain uses the “Means End” theory, which analytically divides activities into levels of abstraction (goals, priorities, functions, processes, resources), which finally lead the way to the desired goal. The all-important question, from top to bottom, is *How?* (e.g. how can we implement the process?), whereas the question from the bottom up is *Why?* (why are we using this resource to implement the process?) (Mai, 2004, 208). The goal of any indexing is to facilitate thematic, concept-oriented retrieval in databases. How does this work? The priorities are set by the seven fixed points (Figure N.1.1). How does this work, in turn? The functions

of indexing are the three indexing phases (Figure N.1.2). Here, specific processes such as reading and understanding the document, as well as browsing through knowledge organization systems, are required. In order to perform these, resources such as KOSs and indexing rules must be available (Mai, 2004, 209).

How must a document analysis be conducted? Standards such as DIN 31.623:1988 (parts 1 to 3) or ISO 5963:1985 suggest paying particular attention to titles, summaries, subheadings, images and captions, first and last paragraphs as well as typographically distinguished areas. But this only tells us *where* to start, not *how*. On this subject, DIN 31.623/2 (1988, 2) succinctly states:

Indexing begins with the reading and understanding of the document's content.

The concept of understanding has led us to the center of hermeneutics. The indexer needs a “key” in order to be granted thematic access to the document in the first place. Such a key will only be given to a person who knows his way around the knowledge domain and who will be able to locate the document within it. When concentrating on the document (and not on what the author “meant”, for instance), one must reconstruct the question to which the document is the answer. This can only succeed when the horizon of the given document is amalgamated with that of the indexer. The indexer's horizon provides a preunderstanding of the document. This preunderstanding can, however, be revised over the course of reading the text, since now the hermeneutic circle enters the fray: understanding single components allows one to understand the whole document—but to adequately understand the single parts, one must have understood the entire document. This interplay of part-whole and the appropriately adjusted preunderstanding gives rise (in the best-case scenario) to understanding.

There are documents that are easily indexed, and those that permanently refuse to be objectively and correctly described. In an empirical study of document analysis, Chu and O'Brien (1993, 452-453) were able to name five aspects that benefit indexing. (1.) The main topic can be identified without a problem. This is generally a given for articles in the natural and social sciences, but not in the humanities. (2.) The document restricts itself to objective descriptions and keeps out any subjective impressions the author may have had. This is not the case in feuilleton articles, for instance. (3.) Complex subject matters described clearly lead to a good indexing performance, whereas a simple concept (expressed via exactly one atomic term) can lead to problems if it cannot be clearly delineated from secondary terms. (4.) A positive influence on the indexing tasks is exerted by an ongoing structuring of the document (clear title and subheading, presence of section headings, author's abstract and introductory paragraph). (5.) The clarity of the text—in terms of both content and layout—benefits indexing (Chu & O'Brien, 1993, 453):

Clarity of the text was associated with a high degree of ease in determining the general subject of texts. The body of the text was helpful in the identification of primary and secondary topics while the layout was cited as more helpful in analysing the general subject.

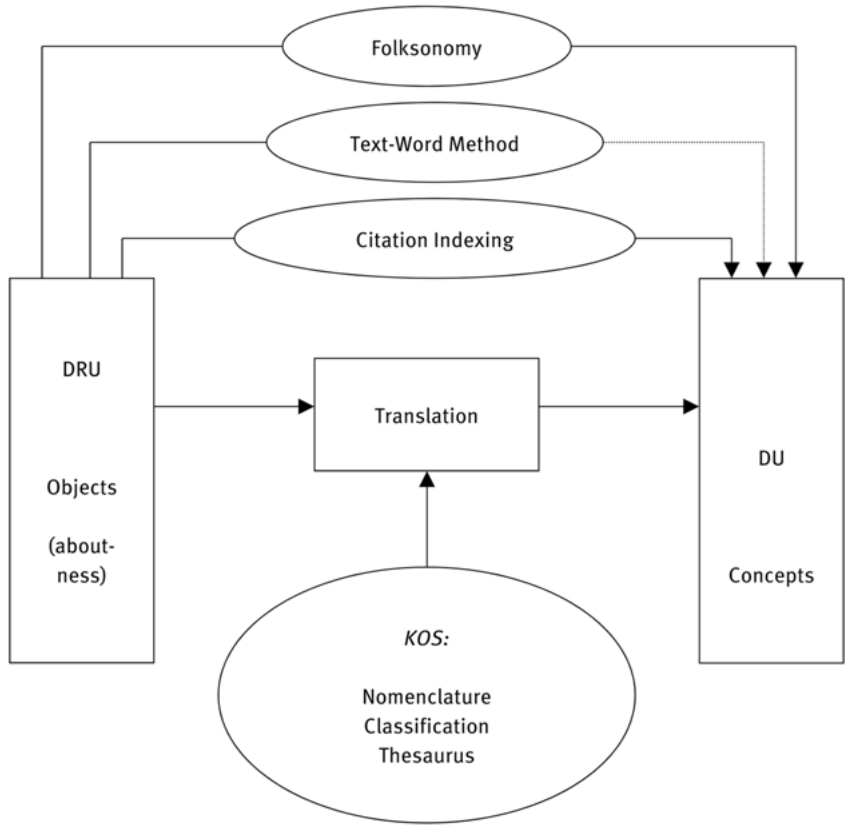


Figure N.1.3: Allocation of Concepts to the Objects of the Aboutness of a Documentary Reference Unit (Note: Text-Word Method only Works with Words, not with Concepts).

The goal of Phase 2 is to describe the aboutness of the comprehended document via concepts. (The recorded propositions are put down in the abstract.) The concepts can be contained in the document, but they do not have to be. If the concepts are contained within the text, we speak of the extraction method. A document can also address topics, however, that are not expressed via exactly one concept. DIN 31.623/2 (1988, 6) notes, on the subject of “implicitly contained” concepts:

To mark the content, it may be necessary to add [concepts; A/N] that are not verbally contained in the text of the document. Thus, we must not concentrate on the indexing of a document’s special contents to such a degree that we forget to allocate these [concepts; A/N] that mediate the integration of the document’s content into larger coherencies.

If, for instance, an author reports of events in a country, but the fact of the country's identity is so obvious for the author that he does not bother to explicitly name it, the indexer must additionally index the name of the country. Such an addition method is not allowed in some cases, though (such as during the text-word method), since it does not preclude misunderstandings and misinterpretations on the part of the indexer.

In the last working step, the objective is to translate the objects of aboutness into the language of the method of knowledge representation that is being used (Figure N.1.3). This process is dependent upon the method and on the tool, and is guided by indexing rules—some of them very specific. There are no firm rules for the use of folkonomies; here, the user can tag at his discretion. The text-oriented methods of citation indexing and the text-word method fundamentally follow the extraction method and only allow linguistic standardizations of the terms.

As soon as a knowledge organization system is being used, the indexer must translate the extracted or added term into the terminology of the KOS, i.e. replace it with the most appropriate keyword (in a nomenclature), the most adequate notation (in a classification system) or the ideal descriptor (in a thesaurus). It is thus useful to look at the semantic environment of the retrieved entry, in order to estimate whether different, or further, concepts might be suitable for indexing.

When concepts are candidates for indexing terms that stand in a hierarchical relation to one another, the indexer will principally choose the most specific term that can be found in the KOS, as Lancaster (2003, 32) emphasizes:

The single most important principle of subject indexing ... is that a topic should be indexed under the most specific term that entirely covers it. Thus, an article discussing the cultivation of oranges should be indexed under *oranges* rather than under *citrus fruits* or *fruit*.

However, if a document discusses several hierarchically connected concepts (to continue Lancaster's example: oranges in one paragraph, and citrus fruits in general in another), both terms will be used for indexing.

Co-Ordinating and Syntactical Indexing

Co-ordinating indexing (DIN 31.623/2:1988) notes those terms that could be used as a search entry to a document, e.g. for an economic article about Finland:

*Finland, Production, Industrial Production, National Product,
Fiscal Policy, Monetary Policy, Foreign Exchange Policy, Inflation,
Trade Balance, Unemployment.*

If we index all concepts independently of their interrelations within the document, we risk a loss of precision. Let us suppose that a user searches for information on the

connection between *inflation* AND *unemployment*. Since both descriptors occur in the sample document, the sample document would be yielded as a search result. This is misleading, though, since the article deals, among other topics, with the groups of themes *inflation and Finland* as well as *unemployment and Finland*, but not with *inflation and unemployment*. A solution, and thus a heightened precision in information retrieval, is provided by syntactical indexing (DIN 31.623/3:1988, 1).

Syntactical indexing is the indexing method in which the document-specific interlinking and/or role of the descriptors or notations (or, more generally, of the concepts; A/N) is made recognizable, according to all rules of syntax, in the corresponding documentary reference unit.

One method of syntactical indexing is the formation of thematic subsets of concepts, in which several chains of concepts are created (DIN 31.623/3:1988, 4). We already got to know this procedure in the text-word method (Chapter M.1). Every concept used for indexing will be allocated (e.g. via digits) the thematic chains to which it belongs. Hence, a (simulated) syntactical indexing of our example would go like this:

Finland (1-4), *Production* (1), *Industrial Production* (1), *National Product* (1,3), *Fiscal Policy* (2), *Monetary Policy* (2), *Foreign Exchange Policy* (2), *Inflation* (2), *Trade Balance* (3), *Unemployment* (4).

Inflation belongs to the thematic chain 2, *unemployment* to 4. If a person searches for *Inflation* SAME *Unemployment* under these circumstances (i.e. using a proximity operator that can be applied to thematic chains), the document will not be retrieved, and rightly so.

Weighted Indexing

Syntactical indexing not only provides for more precise searches via the thematic chains, but, furthermore, it facilitates weighted indexing. In relation to its occurrence in the thematic chains, the weightiness of the respective thematic chains as well as the complexity of the document, one can calculate a numerical value for every concept in order to express the importance of the concept within each respective document. We use the Henrichs algorithm (Henrichs, 1980; Stock, 1984) and receive the following values:

Finland (1-4) <100>, *Production* (1) <29>, *Industrial Production* (1) <29>, *National Product* (1,3) <54>, *Fiscal Policy* (2) <29>, *Monetary Policy* (2) <29>, *Foreign Exchange Policy* (2) <29>, *Inflation* (2) <29>, *Trade Balance* (3) <25> *Unemployment* (4) <18>.

A value of 100 means that the concept occurs in every chain; smaller values refer to a less-pronounced degree of importance. In retrieval, these weighting values are exploited in order to operate via threshold values during searches or to consolidate a relevance ranking of the search results.

Conceivably, an indexer could intellectually allocate each indexing term its document-specific weighting during co-ordinating indexing (in which, after all, the basis for automatic weighting calculations is not given) (Lancaster, 2003, 186). This procedure is very elaborate and error-prone, though, and is not used in the everyday practice of professional information services.

A variant of intellectual weighting is the allocation of only two weighting values: important (“major”) and less important (“minor”). Such an assessment of important concepts is used, for instance, via medical information services such as Medline. The major descriptors can be recognized by the stars. It is possible for searchers to implement the major-minor distinction during retrieval. Lancaster (2003, 188) emphasizes, for all procedures of weighted indexing:

Note that weighted indexing, in effect, gives the searcher the ability to vary the exhaustivity of the indexing.

Indexing Non-Text Documents

For non-text documents (images, videos, music) (Neal, 2012), the task of indexing is fundamentally to allocate concepts to the documentary reference units. It is true, of course, that “a picture says more than a thousand words.” If, however, one requires “more than a thousand words” for the textual description of an image, this is hardly cost-effective and even less useful for retrieval (Shatford, 1984). Although the indexing of images in particular is frequently discussed in information science (Rasmussen, 1997), we are even farther from any satisfactory results in this area than we are for indexing texts. (The situation is similar concerning the intellectual indexing of pieces of music; Kelly, 2010). The translation of non-text information into textual information seems to be very difficult. Svenonius (1994, 600) explored the problem of “lack of translatability between different media on subject indexing.” Neal (2012, 5) notes:

It can be difficult for librarians to describe non-text materials without any accompanying textual metadata: a photo of children standing in a field, for example, tells us a few things on its own, but it does not tell us the location the photo was taken, the names of the children, the name of the photographer, when it was taken, what is it “about”, and so on.

In the indexing practice for images, we can observe a high degree of indexing inconsistency (Markkula & Sormunen, 2000, 273). Ornager (1995, 214) regards the following minimal criteria as important for images that are used in newspapers:

(N)amed person (who), background information about the photo (when, where), specific events (what), moods and emotions (shown or expressed), size of photo.

Indexing via individual concepts is prevalent; general concepts are used less frequently, according to the results obtained by Markkula and Sormunen (2000, 270-271):

The most often used index terms referred to specifics, i.e. to individual objects, places, events and linear time, and to the theme of the photo.

One theoretical approach for the indexing of non-text documents is the distinction of the pre-iconographical, iconographical and iconological levels, following Panofsky. Indexing (no matter which document is up for content description) presupposes that there is a semantic level of the document that can be indexed terminologically. Svenonius (1994, 605) claims:

Subject indexing presupposed a referential or propositional use of language. It presupposes the aboutness model ... which postulates a thing or concept being depicting, about which propositions are made. It presupposes as well that what is depicted can be named. In short, it presupposes a terminology.

Shatford(-Layne) (1986, Layne, 2002) takes up Panofsky's approach of the semantic levels and reformulates them in information science. The objects of the pre-iconographical level of meaning comprise the ofness of the non-textual document, whereas the iconographical level contains its aboutness. Since the indexing of the iconological level requires expert knowledge (e.g. of art history), this level is disregarded in information practice. The indexing of non-textual documents is thus performed with the help of concepts belonging to two levels, entered into two different fields. These fields record the terms of (factual) ofness and of (interpretative) aboutness, respectively. A photo of a Parisian building could thus be indexed in the following manner (Lancaster, 2003, 219):

Ofness: Tower, River, Tree,
Aboutness: Eiffel Tower, Seine.

The Book Index

A special form of indexing is the compilation of book indices. Here, the objective is not to describe the content of documents in such a way that they can be found within an information service; the goal is to provide the reader of a book with comfortable entry points (generally placed at the end of the book) into the text. Book indices do not exist in the digital world, but in the world of printed works. Mulvany (2005, 8) defines "indices" as follows:

An index is a structured sequence—resulting from a thorough and complete analysis of text—of synthesized access points to all the information contained in the text. The structured arrangement of the index enables users to locate information efficiently.

There is an international standard (ISO 999:1996) for creating indices. Headings are created, in the sense of a controlled vocabulary, which are based on places in the text (reference locators) and refer mostly to pages—or, e.g. in legal texts, to paragraphs. Since the number of reference locators must remain manageable, groups of topics are sometimes named (in the form of subheadings). Synonyms (without reference locators) refer to the headings (“see”), related terms are incorporated via cross-references (“see also”) (Figure N.1.4). Depending on the work being discussed, it may make sense to create several indices—e.g. divided by topics and people.

duration	<i>Main heading</i>
copyright, 39-41	<i>Subheadings</i>
moral rights, 85-86	
noncompetition agreement, 96	
nondisclosure agreement, 97-99	
patent, 45-47	
trade secret, 142-145	<i>Numbers: Reference locators</i>
see also expiration	<i>Cross-reference</i>

Figure N.1.4: Typical Index Entry. Source: Mulvany, 2005, 18.

Optimization of Indexing Quality

How can the quality of indexing be guaranteed during the production of information services? The precondition for optimization efforts is the continuous performance of test runs in the indexing team in the sense of internal quality audits (Stubbs, Mangia-terra, & Martinez, 1999). The indexers receive the same documents for indexing; the surrogates are discussed by the team. Only via the continuous feedback between the indexer and his work and the rest of this team members will the indexing terms be adjusted—and, when concentrating on “hits”, it will probably be improved.

A pointer on possible problems with the knowledge organization system being used, or with the indexing rules, is represented by low indexing consistency. In pairs of surrogates with low consistency values, the respective concepts are collected and the elicited terms entered into a ranking based on frequency of occurrence. For concepts that are at the very top of this list, it must be worked out whether they are ambiguously defined and thus give rise to misinterpretations (Lancaster, 2003, 93).

Conclusion

- Indexing is the practical usage of a method of knowledge representation with the purpose of representing the content of a document as optimally as possible, using concepts. Intellectual indexing means that humans perform this task.
 - During indexing, it is decided under which aspects a document can be retrieved in the first place. Optimal indexing is thus the precondition of the possibility of optimal information retrieval.
 - The indexing process depends upon (at least) seven factors: document, indexer, user, usage, database, method (and—as far as it is used—tool) of knowledge representation, knowledge domain.
 - The indexing runs through several phases: in Phase 1 (document analysis), the indexer attempts to understand the respective document; Phase 2 (object description) leads to a list of concepts that describe the aboutness of the document.
 - In methods of knowledge representation that do not work with KOSs (folksonomies and the two text-oriented methods), the keywords, text-words or references thus derived enter the surrogate. When using a KOS (nomenclature, classification or thesaurus), Phase 3 follows, in which the derived concepts are translated into the language of the respective tool being used.
 - Document analysis requires the usage of hermeneutical methods: selecting the correct key, striving for amalgamating the horizons between the document and the indexer, passing the hermeneutic circle as well as taking into consideration the positive role of preunderstanding.
 - The description of the comprehended aboutness via concepts occurs either via the extraction method (when the term is available within the document) or via the addition method (when the indexer allocates a concept that is not available in the document).
 - In co-ordinating indexing, the concepts are admitted to the document independently of their relationships. This method can lead to information ballast, and thus to a lowered precision during search. Syntactical indexing, on the other hand, takes into consideration the thematic relationships of concepts in a document.
 - In weighted indexing, the concepts are allocated numerical values that express their importance in the document. If one works with syntactical indexing, the calculation of importance can be automatized, whereas in the case of co-ordinated indexing the indexer must allocate the values intellectually.
 - In the indexing of non-text documents, we distinguish between ofness (facts without any context information) and aboutness (interpretation).
 - The compilation of a book index represents a special case of intellectual indexing. A book index provides thematic entry points to a work.
 - Indexing is to be organized in such a way that indexing teams periodically check both their surrogates and the underlying KOS.
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N.2 Automatic Indexing

Fields of Application for Automatic Procedures

In which areas can automatic procedures be used during indexing? The fundamental decision when building and operating a retrieval system is whether to work with terminological control or without. When deciding to use this feature, we are be called upon to create and use a knowledge organization system. Here, we will be working with concepts throughout (hence: “concept-based information indexing”). In intellectual indexing, the concrete task of indexing is performed by human indexers, whereas in automatic indexing this task is delegated to the information system.

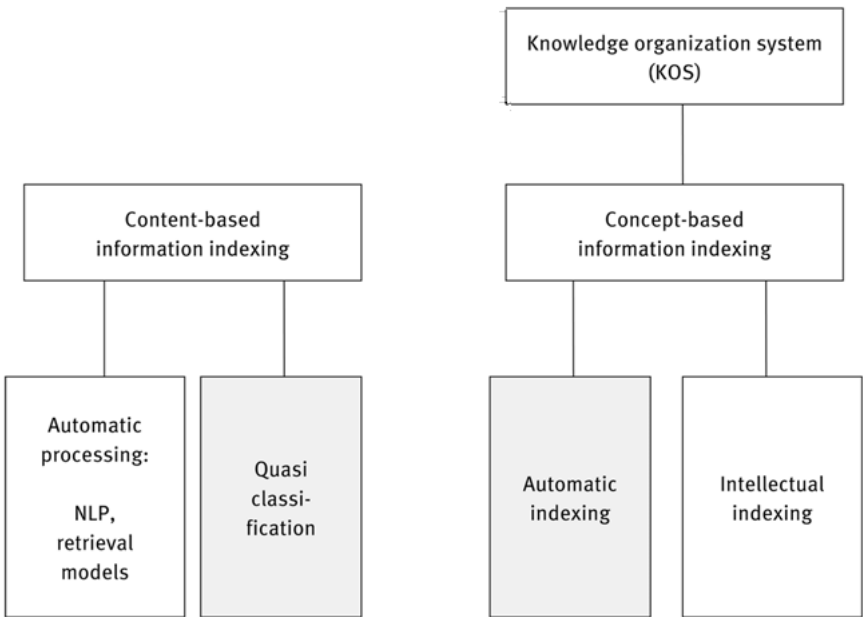


Figure N.2.1: Fields of Application for Automatic Procedures during Indexing.

There are two theoretical approaches that attempt to achieve this task. The probabilistic indexing model inquires about the probability of a certain concept to be allocated to a given document. Rule-based indexing procedures create if-then clauses: If words in the text as well as adjacent words in a text window follow certain rules, then a certain concept will be designated as the indexing term. The rule base may contain text-statistical characteristics (e.g. absolute frequency, position, WDF, $TF \cdot IDF$ of words). In information practice, we also see mixed forms of the probabilistic and the rule-based model.

If we waive terminological control, we will start with the concrete document in front of us; with its script (in texts), its colors and forms (for images) or its pitch and rhythm (in music). Apart from the text-word method, this area is completely devoid of human indexers. The so-called “content-based information indexing” requires multiple processing steps; for text documents, the entire panoply of the methods of information linguistics (“natural language processing”, NLP) must be drawn upon, and a retrieval model must be used. In one version of automatic processing, the indexing system arranges documents into classes—without a prescribed classification system, purely on the basis of the words in the documents. We will call this procedure (classification without classification system) “quasi-classification”. The two fields colored gray in Figure N.2.1—automatic indexing via a KOS and quasi-classification—form the subject of this chapter.

Probabilistic Indexing

The core of probabilistic indexing is the availability of probability information concerning the relevance to a document of a certain concept from a KOS, under the condition that a certain word or phrase occur in the document. Here, two conditions must be met: the document (or a part thereof—such as an abstract) is available digitally and a certain amount of documents (a “training corpus”) has already been intellectually indexed. According to Fangmeyer and Lustig (1969), we can now calculate the association factor z between a document term t and a descriptor (or a notation, depending on what kind of KOS is being used) s . Let $f(t)$ be the amount of documents containing the term t , and $h(t,s)$ the number of documents that contain t and have been intellectually indexed via s . Afterwards, the association factor z is calculated following

$$z(t,s) = h(t,s) / f(t).$$

The result is a relative frequency; z is given the value 0 when s does not co-occur with t even once; z becomes 1 when s is always indexed following the occurrence of t in the text. Accordingly, z can be interpreted as the probability of whether the concept s being relevant for a document d or not.

A probabilistic indexing system that builds on this association factor is AIR/PHYS, developed in Darmstadt, Germany. It is an indexing system for a physics database, also used in practice, and which works with English-language abstracts on the document side and with a thesaurus (and, in addition, a classification system) on the side of the KOS (Biebricher, Fuhr, Knorz, Lustig, & Schwantner, 1988; Biebricher, Fuhr, Lustig, Schwantner, & Knorz, 1988a, 1988b). The fact that the physics database had already been in practice long before automatization began means that several hundreds of thousands of indexed documents are available from which the association factors can be gleaned. For the terms, single words and phrases that are reduced to

their basic forms are both equally drawn upon. All term-descriptor pairs that exceed the threshold values $z(t,s) \geq 0.3$ and $h(t,s) \geq 3$ are entered into a dictionary. We will introduce association factor and dictionary on the example of the descriptor *stellar winds*:

term t	descriptor s	h(t,s)	f(t)	z(t,s)
stellar wind	stellar wind	359	479	0.74
molecular outflow	stellar wind	11	19	0.57
hot star wind	stellar wind	13	17	0.76
terminal stellar wind velocity	stellar wind	12	13	0.92.

If the phrase “terminal stellar wind velocity” will be in the text of an abstract, the probability of the descriptor *stellar wind* being relevant lies at 92%. If the text contains “hot star wind”, the probability of the document’s relevance will be 76%. The respective z values control, together with other criteria (e.g. the frequency of occurrence of t or the occurrence of t in the title), whether a certain descriptor is allocated or not. At this point, two options present themselves. In a binary approach, the objective is to define threshold values which, when they are exceeded, activate the use of the descriptor. The goal is an average indexing breadth of twelve descriptors per document. The previously gleaned relevance probabilities only control this binary decision and are then discarded. This was the Darmstadt solution for the physics database. In the weighted approach, however, the probability values are retained (Fuhr, 1989) and form the basis of the calculation of retrieval status values—and hence the relevance ranking of search results. Biebricher, Fuhr, Lustig, Schwanter and Knorz (1988a, 322) describe the working steps:

The decision ... as to whether a descriptor s should be allocated to a document d , can be represented as the mapping g of the document-descriptor pair (s,d) onto one of the value 0 or 1:

$$g(s,d) = \begin{cases} 1, & \text{if } s \text{ is allocated} \\ 0, & \text{if } s \text{ is not allocated.} \end{cases}$$

In automatic indexing, this mapping g can be divided into a *description step* and a *decision step*. In the description step, all information that should feed into the decision whether or not to allocate the descriptor s is drawn from the text d . The decision-making basis for the second step formed via this information is described as the relevance description of s with regard to d . In the decision step, each relevance description x is mapped onto one of the values 0 or 1 via an indexing function $a(x)$ that is analogous to $g(s,d)$, or in the case of a weighted indexing, onto an indexing weight.

The Darmstadt approach to indexing amounts to a semi-automatic process. Subsequently to the automatic indexing, indexers control the workings of the machine. Of the automatically allocated twelve descriptors, the human experts delete four, on average, and in turn add four new descriptors (Biebricher, Fuhr, Knorz, Lustig, & Schwantner, 1988, 141). This feedback loop creates the option (not applied in Darm-

stadt, however) of constructing a learning indexing system. Since some association factors change after each (controlled) indexing scenario, the z values are recalculated in each case. The system should thus optimize itself over time. In contrast to probabilistic retrieval, which works with relevance feedback in the context of precisely one search, the path via the association values of the term-descriptor pairs paves the way for long-term mechanical learning. This is emphasized by Fuhr and Buckley (1991, 246):

Unlike many other probabilistic IR models, the probabilistic parameters do not relate to a specific document or query. This feature overcomes the restriction of limited relevance information that is inherent to other models, e.g., by regarding only relevance judgments with respect to the current request. Our approach can be regarded as a long-term learning method ...

Rule-Based Indexing

In the rule-based process, we equally presuppose the availability of both digital documents and a knowledge organization system, also in digital form. For every concept from the KOS being used, we must construct rules according to which the concept will be used for indexing purposes or not. The efforts of preparatory work and system maintenance rise in tandem with the number of concepts in the KOS. Hence, it is extremely sensible in this case not to use *one* extensive KOS, but to divide it into several small facets. One such approach is pursued by the database provider Factiva, for example, which represents the language of economic news via a faceted KOS.

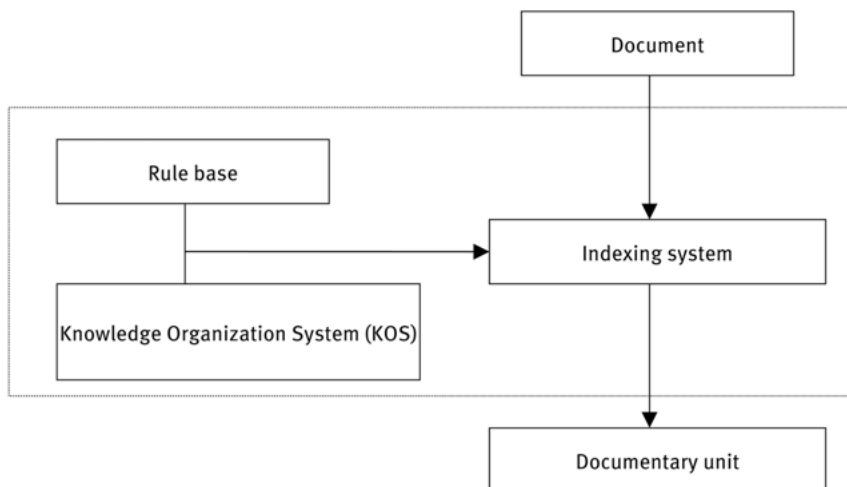


Figure N.2.2: Rule-Based Automatic Indexing.

Indexing in Factiva occurs in real time, i.e. directly after an entry has been registered by the system (Figure N.2.2). The indexing system processes the respective text and applies both the rules and the knowledge basis. The result is a documentary unit containing the document (thus making the full text searchable) and adding a controlled vocabulary from the thesaurus facets. Used practically in Factiva is the system Construe-TIS (Categorization of news stories, rapidly, uniformly, and extensible / Topic Identification System), originally developed for Reuters Ltd., which is applicable to news documents in the economic sector (agency releases, newspapers articles, text versions of broadcasts etc.) (Hayes & Weinstein, 1991).

For every concept that occurs in the facets of the KOS, rules are defined. These rules control two processing steps: firstly, the recognition of a concept in the text and secondly, the allocation of a descriptor to the document. Concepts are recognized via patterns of words and phrases that occur in the text respectively in a certain text window. Let us assume that we wish to find out whether a certain document thematizes *gold* as a tradable economic good. The simple occurrence of the word *gold* is insufficient, since it could mean *gold medals* or *goldsmiths*. One option is to proceed negatively by excluding semantically inappropriate meanings. For the English language, the following rule applies to the recognition of the term *gold (commodity)*:

gold NOT (reserve OR medal OR jewelry)

(Hayes & Weinstein, 1991, 58). The NOT-functor signals that the term will not be allocated if one of the following words occurs in the text window. The phrase *gold medal* is thus rejected, whereas the occurrence of, for instance, *gold mine* or *gold production* are not excluded by the term *gold (commodity)*. The second step decides, on the basis of recognized concepts, which descriptors are to be allocated to the document. Hayes and Weinstein (1991, 59) describe the procedure:

Categorization decisions are controlled by procedures written in the Construe rule language, which is organized around if-then rules. These rules permit application developers to base categorization decisions on Boolean combinations of concepts that appear in a story, the strength of the appearance of these concepts, and the location of a concept in a story.

The employed rules draw on Boolean functionality (AND, OR, NOT), the term's frequency of occurrence (i.e. following absolute values or, more elaborately, TF*IDF) as well as the term's position in the text (e.g. in the title, introductory paragraph or body). A concrete rule (here: for the descriptor *gold*) looks as follows (Hayes & Weinstein, 1991, 60):

```

if
test:      (or      (and  [gold-concept:scope headline 1]
                        [gold-concept:scope body 1])
                        [gold-concept:scope body 4])
action:    (assign gold-category).
```

If the word *gold* occurs (at least) once in the title and the body, or if it occurs (at least) four times in the body of a document, the descriptor *gold* will be allocated. Factiva uses a rule editor to catalog new rules and to perform maintenance on existing ones.

Factiva Intelligent Indexing works semi-automatically, i.e. indexers are needed to check the results. According to statements by the company itself, around 80% of all automatically compiled surrogates are correct and require no further intellectual processing (Golden, 2004).

Quasi-Classification

Quasi-classification is the arrangement of similar documents into a class based purely on characteristics within the text, without the help of a KOS. Automatic classing is done in two steps: first, similarity values between documents are calculated. This is done in order to bring together, in the second step, similar documents in one class, where possible. Conversely, dissimilar documents are separated into different classes. This is achieved by using methods of numerical classification (Anderberg, 1973; Rasmussen, 1992; Sneath & Sokal, 1973).

When estimating the similarity of textual documents on the basis of the words, it is advisable to remove stop words from the documents as well as to merge the different forms of a word that occur in the text into the stem form or the lexeme. Instead of only using single words to describe documents, it is also possible—and, very probably, successful—to draw on “higher” term forms such as phrases or the designation of “named entities”. Since we work with words (and not with concepts), problems such as synonymy and homonymy remain unheeded. Additionally, quasi-classification can only be applied to texts that are written in the same language.

An alternative procedure works with references and citations, or to be more precise: with bibliographic coupling and co-citations. This method is independent of the language of the text, but it can only be used in domains where formal citation is used.

In practice, quasi-classification begins at the end of a search and calculates the similarity values for all search results (e.g. via the algorithms by Dice, Jaccard-Sneath or the Cosine). The similarity is calculated via the number of shared words in two texts. If a document A consists of, say, 100 different words (a), and document B of 200 (b) and there are 75 words (g) which occur in both documents, there will be a similarity (for example, Dice: $2g / [a + b]$) of $150/300 = 0.5$. Now, a similarity matrix is created:

D_1	D_2	D_3	D_4	...	D_n
D_1	1				
D_2	S_{21}	1			
D_3	S_{31}	S_{32}	1		
D_4	S_{41}	S_{42}	S_{43}	1	
..					
D_n	S_{n1}	S_{n2}	S_{n3}	S_{n4}	1.

It makes sense in practical applications to apply a hierarchical quasi-classification (Voorhees, 1986; Willett, 1988), as too many search results would clutter up individual classes otherwise—particularly in Web search engines. It is possible, via appropriate modifications to the threshold values or the k value, to predetermine the number of classes for each hierarchy level. For reasons of clarity and the graphical representability of the classes in search tools, the number of classes per level should not exceed about 15 or 20.

A primitive procedure of quasi-classification involves starting with a document D and then admitting as many documents into a quasi-class as are similar to the initial document until a count value k has been reached. In this k -nearest-neighbors procedure, one calculates the similarity of all other documents to a document D_i . Then, the values thus gleaned are ranked in descending order of similarity. The first k documents in this list make up the class.

We will now introduce three “classical” methods of forming classes from similarity matrices via cluster analysis (Rasmussen, 1992, 426): the Single-Link procedure, the Complete-Link procedure and the Group-Average procedure. The cluster algorithm always begins with the document pair with the highest S_{ij} value. In the Single-Link procedure, those documents are added that have a similarity value to one of the two initial documents which lies above the preset threshold value. This is repeated for all documents from the matrix, until no more documents are found that have an above-the-threshold similarity value to one of those already in the cluster. The documents retrieved up to that point form a class (Figure N.2.3). This procedure is then repeated until all documents in the matrix have been checked off.

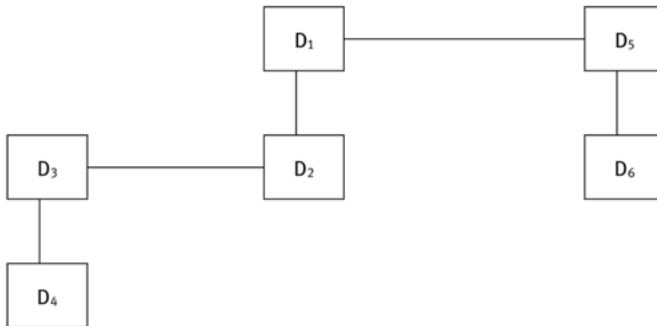


Figure N.2.3: Cluster Formation via Single Linkage.

If we wish to form disjunct classes, we must make sure that the documents already entered into a class are not selected again after the first readout of the similarity value in the respective iteration loop. This is achieved, for instance, by setting the similarity value to zero (for this particular iteration). If we fail to do so, overlapping classes may be created as a result.

The Single-Link procedure orients itself on its closest neighbor (Sneath & Sokal, 1973, 216), whereas the Complete-Link procedure has its fixed point in the furthest-away neighbor (Sneath & Sokal, 1973, 222). Complete Linkage requires all documents that form a class to be interconnected with a similarity value that must be predefined (Figure N.2.4). This procedure leads to rather small clusters, whereas the Single-Link procedure may form extensive classes.

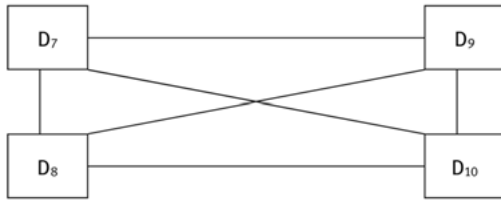


Figure N.2.4: Cluster Formation via Complete Linkage.

A sort of compromise between both methods is formed by the Group-Average Link procedure. It initially proceeds in the same way as Single Linkage, but calculates the arithmetic mean of all similarity values for the resulting class. This mean now serves as a threshold value, so that all documents will be removed which are connected to documents that remain in the class merely via a value below the threshold.

For hierarchical classification, the cluster analysis is repeated within the respective classes. This can lead to several hierarchy levels in case of extensive search results. The procedure should be aborted when only around 20 documents are left for further cluster analysis. To present the quasi-classes of the search results in the user interface, it makes sense to use information visualization.

How do we call those classes that have just been automatically generated? Once all stop words have been removed from the documents, the vocabulary of the respective centroid vector (mean vector of all documents in the class) offers its services. For the class centroid, the terms are ranked in descending order of frequency (or, alternatively, according to TF*IDF). The first two or three words should provide a fairly clear description of the class.

Conclusion

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- In information indexing, there are two points at which automatic procedures may be used: when indexing via a knowledge organization system and when classing documents according to words or citations and references (quasi-classification, respectively).
 - Automatic indexing (via KOSs) is either guided by probabilistic indexing or by rule-based indexing.
 - Probabilistic indexing (as practiced by AIR/PHYS in the Darmstadt approach, for instance) requires the availability of probability information about whether a descriptor (a notation etc.) will be

of relevance for the document if the document contains a certain term. If human indexers correct the machine, it becomes possible to initiate long-term mechanical learning.

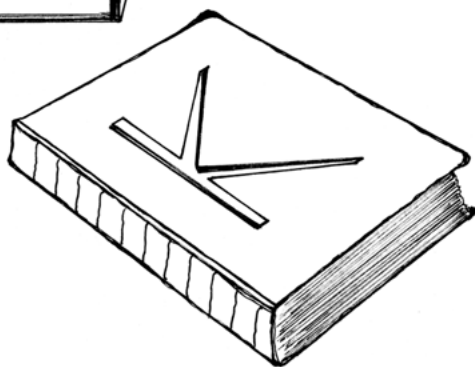
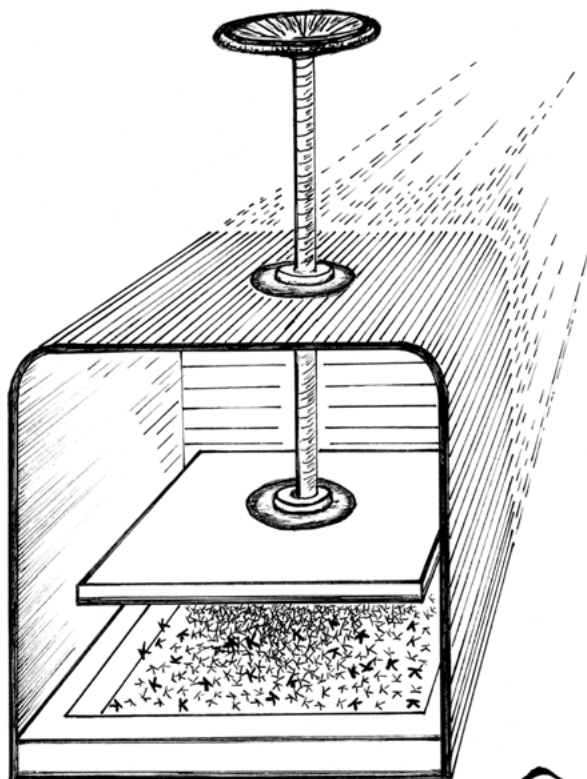
- In the rule-based indexing procedure, a set of rules is defined for every concept in the KOS. These govern the allocation of the descriptor (the notation etc.) as an indexing term. In order to keep the efforts of development and maintenance at a tolerable level, the system should contain as few concepts as possible, as is the case in faceted KOSs. An example of such a system used routinely is Factiva Intelligent Indexing.
 - Quasi-classification means the allocation of similar documents to a class based solely on textual characteristics (i.e. without recourse to a knowledge organization system), by applying methods of numerical classification.
 - Similarity coefficients are used for a list of search results and lead to a similarity matrix of the retrieved documents. This forms the basis of cluster analysis, which follows either the Single-Link procedure, the Complete-Link procedure or the Group-Average Link procedure. Hierarchical classing is a sensible option.
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Part 0

Summarization



O.1 Abstracts

Abstracting as a Method of Summarization

Methods of summarization are used to “condense” documents—even longer ones—down to an overview of their main content. Information condensation ignores all that is not fundamentally important in the document (thus abstracting from its content) and, as a result, provides an “abstract” of several lines. In this chapter we say goodbye to the world of concepts and turn to that of sentences. These latter do not represent object classes (as concepts do) but propositions. Abstracts thus consist of sentences representing propositions. Abstracts are defined via national (e.g. DIN 1426:1988) and international standards (ISO 214:1976).

What is an abstract? The German standard (DIN 1426:1988, 2) defines as follows:

The abstract briefly and precisely reflects the document’s content. The abstract should be informative, offering neither interpretation nor judgment (...) and should also be understandable without reference to the original. The title should not be repeated—rather, if necessary, it should be complemented or elucidated. Not all components of the document’s content must be represented; but those of particular importance can be chosen.

Pinto (2006, 215) succinctly formulates the main characteristics of abstracts:

Abstracts are reduced, autonomous and purposeful textual representations of original texts, above all, representations of the essential content of the represented original texts.

The starting point is always a document, the content of which is described in a condensed manner. For Borko and Bernier (1975, 69),

a well-written abstract must convey both the significant content and the character of the original work.

Cremmins (1996) views abstracting as an art—which points to potentially serious problems when using automatized methods. In talking about summarization, we must distinguish between extracts and abstracts. Whereas the former only select sentences contained within a document, the latter are a specific kind of text in themselves: the result of creatively editing the original document. Lancaster (2003, 100) emphasizes:

An extract is an abbreviated version of a document created by drawing sentences from the document itself. For example, two or three sentences from the introduction, followed by two or three from the conclusions or summaries, might provide a good indication of what a certain journal article is about. A true abstract, while it may include words occurring in the document, is a piece of text created by the abstractor rather than a direct quotation from the author.

Are summaries even necessary when the full text itself is available digitally? At this point we need to ask which full texts we are talking about. A short Web page surely does not need its own intellectually produced abstract, but a scientific article most certainly does. Especially in the latter case, full texts are generally available digitally today, but they are often located in the (fee-based) Deep Web. In their user study, Nicholas, Huntington and Jamali (2007, 453) arrive at the following result:

It is clear that abstracts have a key role in the information seeking behavior of scholars and are important in helping them deal with the digital flood.

Even if the document is accessible free of charge, researchers express a desire for abstracts (Nicholas, Huntington, & Jamali, 2007, 446):

In a digital world, rich in full text documents, abstracts remain ever-popular, even in cases where the user has complete full-text access.

This is because a summary guides the user's decision as to whether or not he will acquire the full text (or just follow the link leading to it). The user sees at first glance whether the content of the complete document will satisfy his information need. Abstracts thus fulfil a marketing function: they influence demand for the complete document. Following Koblitz (1975, 16-17), information condensation serves two main purposes:

(The abstract), in connection with the indices (i.e. the controlled vocabulary from knowledge organization systems, A/N), must facilitate or simplify the user's assessment as to whether the abstracted information source corresponds (is relevant) to his information need or not.

It must facilitate or simplify the decision whether the information value of the information source deemed relevant justifies its perusal or not.

A third aspect arises for the case of foreign-language literature (particularly in languages the user does not speak). The user gets a brief overview of a document, possibly serving as a substitute for the full text (or as the basis for deciding whether to have the document translated).

Even though information condensation occurs throughout all forms of human communication (Cleveland & Cleveland, 2001, 108; Endres-Niggemeyer, 1998, 45 et seq.), information science is most concerned with the summaries of documents from the scientific system. The author of such a summary is either the publishing author himself or an information service employing professional abstractors. According to Endres-Niggemeyer (1998, 98-99), quality concerns dictate that the creation of abstracts is left to information specialists who know how to write the best possible summaries:

(P)rofessional summarizers can summarize the same information with greater competence, speed, and quality than non-professionals.

However, the services of such experts are so expensive that even professional information services often use abstracts written by the authors themselves.

Characteristics of Abstracts

Following the German Abstract Standard (DIN 1426:1988, 2-3), abstracts should display the following characteristics:

- a) Completeness. The abstract must be understandable to experts in the respective area without reference to the original document. All fundamental subject areas must be explicitly addressed in the abstract—also with regard to mechanical search. ...
- b) Accuracy. The abstract must accurately reflect the content and opinions contained in the original work. ...
- c) Objectivity. The abstract must abstain from judgments of any kind. ...
- d) Brevity. The abstract must be as short as possible. ...
- e) Intelligibility. The abstract must be understandable.

The standard itself admits (DIN 1426:1988, 3) that some of the listed characteristics partly contradict each other. Kühlen (2004, 196) names the example of completeness clashing with brevity.

In an empirical study, Pinto (2006) was able to work out characteristics that are of particular interest to the users: exactitude (in the sense of accuracy), representativity (similarity between source and abstract), quality as discerned by the user (a very subjective criterion, compiled on the basis of the difference between the degrees of expectation and perception), usefulness and exhaustiveness (the amount of important topics reported from the original).

Lancaster (2003, 113) brings the number of decisive characteristics down to three: brevity, accuracy and clarity:

The characteristics of a good abstract can be summarized as brevity, accuracy, and clarity. The abstractor should avoid redundancy. ... The abstractor should also omit other information that readers would be likely to know or that may not be of direct interest of them. ... The shorter the abstract the better, as long as the meaning remains clear and there is no sacrifice of accuracy.

Occasionally the discussion revolves around the consistency of abstracts (e.g. in Pinto & Lancaster, 1998). It is difficult enough to safeguard inter-indexer consistency when using knowledge organization systems; hence, inter-abstractor- as well as intra-abstractor consistency would appear to be even more difficult to achieve. Lancaster (2003, 123) describes this as follows:

No two abstracts for a document will be identical when written by different individuals or by the same individual at different times: what is described may be the same but how it is described

will differ. Quality and consistency are a bit more vague when applied to abstracts than when applied to indexing.

Homomorphous and Paramorphous Information Condensation

The first step of abstracting is to decide whether the documentary reference unit needs an abstract in the first place. Analogously to documentary reference units' (DRU) worthiness of documentation, we here speak of “worthiness of abstracting”. This is determined via a bundle of criteria, which include: degree of innovation, manner of representation, document type or the publishing journal’s reputation. The abstractor then performs a content analysis of the work that is to be abstracted in order to work out the most important content components (the aboutness). Each of these pieces of aboutness are represented (a) via its topics, and (b) via propositions. The topics are represented by using concepts from knowledge organization systems, the propositions via sentences.

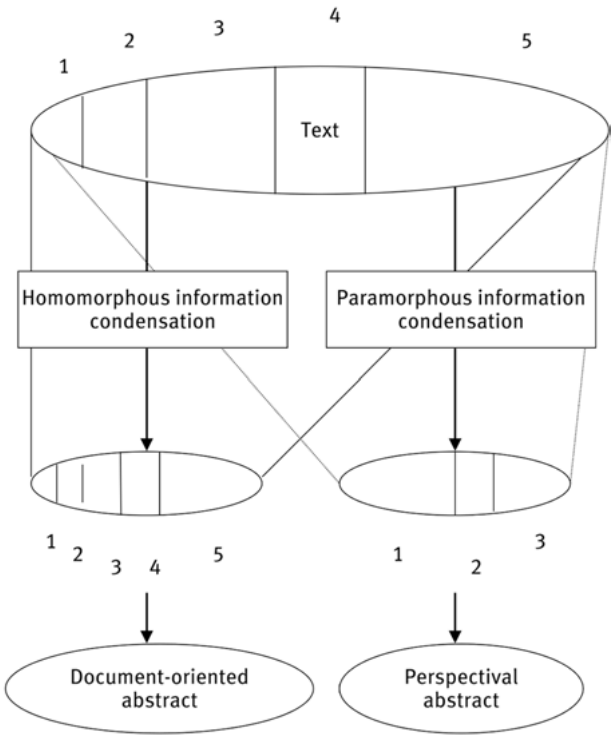


Figure O.1.1: Homomorphous and Paramorphous Information Condensation.

Each textual document can be condensed to its fundamental components. For van Dijk and Kintsch (1983, 52-53), this is the “macrostructure” of the text:

Whereas the textbase represents the meaning of a text in all its detail, the macrostructure is concerned only with the essential points of a text. But it, too, is a coherent whole, just like the textbase itself, and not simply a list of key words or of the most important points. Indeed, in our model the macrostructure consists of a network of interrelated propositions which is formally identical to the microstructure. A text can be reduced to its essential components in successive steps, resulting in a hierarchical macrostructure, with each higher level more condensed than the previous one.

Starting from the original, the aboutness can be condensed almost steplessly. Such an isomorphous representation appears to be unattainable via the practical work of abstracting. What can be achieved, however, is a similar, homomorphous reduction (Heilprin, 1985). Tibbo (1993, 34) describes it as follows:

Thus, if one topic comprises 60% of an original document’s text and another topic only 10% (and they are both of potential interest to the audience), an abstract should represent the former more prominently than the latter.

In Figure O.1.1 we see a documentary reference unit addressing five different topics, each with a different degree of importance. Topic 5, for instance, is processed far more intensively than Topic 1. Homomorphous information condensation more or less retains the relative proportions of the topics in the abstract. The result is an abstract that orients itself unequivocally on the original document.

One can also proceed in a different way, however. Heilprin (1985) calls this “paramorphism” (*para* being the Ancient Greek word for *next to*). Here we must distinguish between a negative point of view (“the abstract is off target”) and a positive one. This latter explicitly demands that the text is abstracted from a certain perspective (Lancaster, 2003, 103). The perspectival abstract (in Figure O.1.1, right) condenses paramorphously, by abstracting certain parts of the documentary reference unit’s aboutness and leaving others out if they are irrelevant for the target group. Suppose that only subjects 1 through 3 of the documentary reference unit in Figure O.1.1 can be assigned to discipline A, while the others belong to another discipline. An information service for A will consequently only be interested in subjects 1 through 3. A paramorphously representing perspectival abstract will concentrate on those subjects that interest the given user group while ignoring all others.

The Abstracting Process

How is an abstract (intellectually) created? As a precondition, the acting abstractor must be proficient in three areas: he must know the subject area of the document, he

must speak foreign languages (for literature from other countries) and he must have methodological experience in creating abstracts. According to Pinto Molina (1995, 230), the working process splits up into the three steps of reading, interpreting and writing. Abstractors read a text in a way that provides them with a comprehension of the text at hand, a knowledge of what it is about in the first place. Pinto Molina (1995, 232) sketches this first step:

This stage ... is an interactive process between text and abstractor (...), strongly conditioned by the reader's base knowledge, and a minimum of both scientific and documentary knowledge is needed. Reading concludes with comprehension, that is to say, textual meaning interpretation.

The step of interpretation, which is split off from step 1 on a purely analytical basis, creates a comprehension of the abstractable components of the aboutness. For this, the abstractor must always bear in mind his quantitative guidelines for the abstract. The standard (DIN 1426:1988, 5) speaks of less than 100 words for short articles; even for comprehensive monographs, the abstract should not be more than 250 words. Interpretation is situated in the hermeneutic circle's area of tension (Ch. A.3). On the one hand, there is the text from which a bottom-up interpretation is attempted; on the other hand, there is the reader's pre-understanding (including his ideas for interpretation keys and his expectation of meaning), from which he attempts to derive a top-down understanding. Pinto Molina sees the circle as an interplay between inductive inference (bottom-up) and deductive conclusion (top-down). Since not all subjects discussed in the original can be entered into the abstract, the objective is to designate certain aspects of the aboutness as positive (and to mark them for the abstract) while eliminating others as negative. Abstractors skip repetitions of subjects ("contraction"), reduce the representation of subject areas that are of little relevance ("reduction") and eliminate irrelevant aspects ("elimination").

The last step of writing an abstract depends both on the document's interpretation and on the aimed-for abstract type (indicative, informative, etc.; see below). Abstracts are regarded as autonomous (secondary) documents that meet the following conditions (Pinto Molina, 1995, 233):

Any kind of synthesis to be done must be entropic, coherent, and balanced, retaining the schematic (rhetoric) structure of the document.

According to Endres-Niggemeyer (1998, 140), there are several strategies for writing abstracts. At this point, we will discuss the method we deem to be the most important: the phase-oriented approach. After reading and initial understanding, there follows a relevance assessment of the subjects which results in a written specification of notes (or alternatively marked text passages). From this material the abstractor compiles a draft for the abstract, either in a single working step or via the intermediary stage of notes. In the last working steps, the abstract is written. There should be multiple drafts: this goes back to the hermeneutic aspects of circularity and to the

maximum word count that must not be exceeded. As the subjects are processed in several working steps, mistakes are minimized, as Endres-Niggemeyer (1998, 141-142) emphasizes:

As the information items are accessed several times and in context (first read and understood, then assessed for their relevance, later reintegrated into the target text plan, etc.), the summarizer is more certain to bring her or his work to a good conclusion, to learn enough about the document, and to notice errors. At the moment of relevance decision and writing, many knowledge items from the source document are known and can be considered together.

What does a professional abstractor's work look like in practice? In an empirical study, Craven (1990) has detected three trends. (1.) Apparently, counter to the standards' encouragement, there is no correlation between the length of the abstract and that of the full text. Rather, abstractors orient themselves more on the density of the subject matters in the source document than on any rigid length guidelines. (2.) The abstractors do not adopt phrases from the original, not even longer sentence fragments, but they do adopt individual words (Craven, 1990, 356):

(T)he abstractors extracted longer word sequences relatively rarely, preferring to rearrange and condense. They generally took the larger part of the abstract's vocabulary of individual words from the original. But they also added many words of their own.

(3.) The words adopted from the original often stem from its opening paragraph (the first 200 words); the abstractors thus particularly orient themselves on the terminology of a document's first paragraphs. (4.) Following studies by Dronberger and Kowitz (1975), as well as by King (1976), abstracts are more difficult to read when compared to full texts. Furthermore, demanding full texts generally lead to even more demanding abstracts. Abstracts by scientific authors themselves are frequently deemed extremely difficult to read (Gazni, 2011).

Indicative and Informative Abstract

Document-oriented and perspectival abstracts have three "classical" subforms: the indicative, the informative, and the mixed form of the informative-indicative abstract. To clarify this differentiation, we separate the mere listing of subject matters discussed from the description of all abstractable results regarding the subject matters contained within the document. From this, Lancaster (2003, 101) derives the two basic forms:

The indicative abstract simply describes (indicates) what the document is about, whereas the informative abstract attempts to summarize the substance of the document, including the results.

The indicative abstract (Figure O.1.2) refers to the subject matters, but yields no results. The abstract standard (DIN 1426:1988, 3) defines:

The *indicative* abstract merely states what the document is about. It points the reader to the subjects discussed in the document and mentions the manner of discussion, but it does not state specific results of the deliberations contained in the document or of the studies undertaken by it.

Russ, H. (1993). Einzelhandel (Ost): Optimistische Geschäftserwartungen.
ifo Wirtschaftskonjunktur, 45(3), T3.

There is a description of the business situation in the East German retail industry as of January, 1993. The business trend to be expected over the following six months is sketched. Particular objects of discussion are the areas of durables and consumables, respectively.

Figure O.1.2: Indicative Abstract.

In contrast to the indicative abstract, the informative abstract (Figure O.1.3) also discusses the concrete results posited by the documentary reference unit (DIN 1426:1988, 3):

The *informative* abstract provides as much information as permitted by the document's type and style. In particular, there is mention of the subject area discussed as well as of the goals, hypotheses, methods, results and conclusions of the deliberations and representations contained in the original document, including the facts and data.

Russ, H. (1993). Einzelhandel (Ost): Optimistische Geschäftserwartungen.
ifo Wirtschaftskonjunktur, 45(3), T3.

In January of 1993, the business situation in East German retail has deteriorated significantly from its situation of the month before. However, the participants of the ifo Business Climate Test were optimistic as to their prospects over the following six months. The business situation in the area of durables is, on average, satisfying; in the area of consumables, assessments are largely negative.

Figure O.1.3: Informative Abstract.

It is preferable to produce an informative abstract, as it offers the user more information about the source document. However, to do so is more elaborate and thus more expensive. For more extensive documents in particular, there is no point in trying to abstract every important result; in such cases, abstractors tend to use the indicative form or a mixed version with both informative and indicative components.

A marginal form of abstract used in knowledge representation is the judging abstract. Here the content of the document is evaluated critically and a (positive or negative) recommendation may be tendered. Cleveland and Cleveland (2001, 57) regard such abstracts as useful, but point out:

The key ... is that the abstractor is sufficiently knowledgeable of the subject and the methodologies in the paper so that they can make quality judgments. This kind of abstract is generally used on general papers with broad overviews, on reviews, and on monographs.

For laymen in particular, judging abstracts can have a useful orientating function. However, in our opinion they are more closely related to the text forms of literary notes and reviews (DIN 1426:1988, 2 and 4) than to abstracts, as they explicitly require a critical statement on the part of the abstractor. The judging abstract is in violation of the objectivity guideline posited by the DIN standard 1426.

Structured Abstract

Indicative and informative abstracts have no subheadings; they are “seamless”. Not so structured abstracts (Hartley, 2004; Zhang & Liu, 2011) (also called “More Informative Abstracts”; Haynes et al. 1990). These are mainly used in scientific journals, written by the authors themselves. Starting from journals in the areas of medicine and biosciences, many academic periodicals currently use structured abstracts. Such abstracts are always organized in sections, which are characterized by subheadings (see Figure O.1.4). The authors are thus obliged to organize their texts systematically. According to Hartley (2002, 417), structured abstracts have the following characteristics:

(T)he texts are clearly divided into their component parts; the information is sequenced in a consistent manner; and nothing essential is omitted.

Russ, H. (1993). Einzelhandel (Ost): Optimistische Geschäftserwartungen.
ifo Wirtschaftskonjunktur, 45(3), T3.

Introduction. Every month the ifo Institute for Economic Research (Munich, Germany) analyzes the economic cycle in the German retail industry. *Method.* To do so, it uses the ifo Business Climate Test, which is based upon survey results by representatives of the industry. *Results.* In January, 1993, the business climate in East German retail deteriorated significantly from the previous month. However, the participants of the ifo Business Climate Test voiced optimism with regard to expected developments over the following six months. The business situation in the area of durables is, on average, satisfying; in the area of consumables, assessments are largely negative. *Discussion.* Despite a short-term drop in the business climate of East German retail in January, 1993, the outlook on economic development is generally positive.

Figure O.1.4: Structured Abstract.

Structured abstracts are generally more information-rich than informative abstracts and far more informative than indicative ones; they are also easier to read (Hartley, 2004, 368), and users in online retrieval can orient themselves more quickly, which

leads to fewer mistakes when assessing the document's relevance (Hartley, Sydes, & Blurton, 1996, 353):

The overall results ... indicate that the readers are able to search structured abstracts more quickly and more accurately than they are able to search traditional versions of these abstracts in an electronic database.

However, they take up (marginally) more space in the journals (Hartley, 2002).

Currently, articles in scientific and medical journals often use the IMRaD structure (Introduction / Background, Methods, Results, and Discussion / Conclusion) (Sollaci & Pereira, 2004). As an obvious consequence, abstracts in these disciplines also follow the same IMRaD structure. Alternatives are the eight-chapter format (Objective, Design, Setting, Participants, Interventions, Measurement, Results, and Conclusions) (Haynes et al., 1990) as well as free structures. A study of abstracts in medical journals from the year 2001 showed that around 62% of abstracts are structured abstracts, of which two thirds adhere to the IMRaD structure and one third to the eight-chapter format (Nakayama et al., 2005).

It is important for searching abstracts in discipline-specific information services (e.g. for MEDLINE as a source of medical bibliographical information) to mark their individual chapters as specific search fields, allowing users to perform concrete searches in a certain chapter (e.g. the Method section) (O'Rourke, 1997, 19).

Discipline-Specific Abstracts

There are various different types of text. Some works narrate, others explain. Within the scientific disciplines, too, text types have developed that are specific to certain fields; a scientific article is generally structured completely differently from an article in the humanities, and both distinguish themselves fundamentally from technical patent documents. Tibbo (1993, 31) emphasizes:

Scholarly and scientific research articles are highly complex, information-rich documents. While they may all serve the same purpose of reporting research results irrespective of their disciplinary context, it is this context that shapes both their content and form. Authors may build these papers around conventionalized structures, such as introductions, statements of methodology, and discussion sections, but a great variation can exist. This format is, of course, the classic model for the scientific, and more recently, the social scientific research article ...

This is not, however, the form humanists typically use. They seldom include sections labeled "findings", "results", or even "methodology".

Discipline-oriented abstracting adapts itself to the customs of the respective discipline and uses these to structure its abstracts. Since both authors and readers of a certain scientific discipline have been scientifically socialized to a similar degree, the

structure of the documents as well as of the abstracts reflects their perceptions and standards with regard to formal scientific communication (Tibbo, 1993, 34).

Patent abstracts are characterized by the fact that in many cases they only become understandable once they are partnered with a related graphic. Images and text are interdependent; one cannot be grasped without the other. The links between text and images are created via numbers. However, not all numbers are always represented in both areas (which makes it more difficult to assess the relevance of a document via the abstract, thus requiring perusal of the original document).

Multi-Document Abstracts

With the multi-document abstract we are turning away from an individual document as the unit being summarized and concentrating instead on a certain amount of sources published around a certain topic. Here, too, the possible forms are informative, indicative and structured abstracts. According to Bredemeier (2004, 10), multi-document abstracts combine documentary and journalistic endeavors, which assume three functions:

- Identification of subject areas currently under discussion (...);
- Status of the discussion as evident from current publications and
- Provision of the corresponding full texts, if the user wants to acquaint himself with subareas or with the subject as a whole.

Multi-document abstracts are distinguished from purely journalistic works by the absence of value judgments. However, there is always an implicit valuation in the selection of documents that enter into such an abstract in the first place. Depending on the number of sources summarized, a multi-document abstract can run up to several pages in length. The “Knowledge Summaries” by Genios (Bredemeier, 2004) prefix a so-called “Quickinfo” to the abstract proper in order to present the user with an overview of the subject’s fundamental aspects. The subsequent multi-document abstract is divided into chapters; the referenced sources are summarized in a bibliography at the end. Since Genios has access to the sources’ full texts, the bibliographical entries always come with a link to the original document.

Conclusion

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- Abstracts are autonomous (secondary) documents that briefly, precisely and clearly reflect the subject matter of a source document in the form of sentences.
 - The abstract has a marketing function in that it guides the user’s decision whether or not to acquire the full text (or click a link leading to it). In the case of literature in a foreign language that the user does not understand, he will at least get an overview of the document.

- Homomorphous information condensation leads to a document-oriented abstract, paramorphous information condensation leads to a perspectival abstract.
 - The process of abstracting is divided into three working steps: reading, interpreting, and writing. Reading and interpreting are hermeneutic processes with the goal of working out the central propositions of the source document's aboutness. Writing an abstract depends on the (understood) aboutness, the abstract form and length guidelines.
 - Indicative abstracts exclusively report the existence of discussed subject matter, whereas informative abstracts—as a standard form of professional abstracts—also describe the fundamental conclusions of the documents.
 - Structured abstracts are divided into individual sections, marked by subheadings. Such abstracts are used in many medical and scientific journals; they are written by the full texts' authors themselves. Typical structures are IMRaD and the eight-chapter format.
 - Abstracts in different scientific disciplines follow different customs. Thus IMRaD is fairly typical for medicine, whereas combined abstracts with graphics and text are the rule for patent documents.
 - Multi-document abstracts condense the subject matter of several documents into one topic. They are to be viewed as a mixed form of journalistic and information science activity.
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O.2 Extracts

Extracting Important Sentences

When understood as a specific variety of text that foregoes all extraneous elements in a source and condenses its fundamentals into sentences, abstracting is a form of creative scientific work that depends on an understanding of the source and the abstractor's writing skills. There is, at the moment, no indication that this process can be satisfactorily executed on a purely automatic basis. However, the path of extracting information from a source can be treated mechanically, as Salton, Singhal, Mitra and Buckley (1997, 198) emphasize:

(T)he process of automatic summary generation reduces to the task of *extraction*, i.e., we use heuristics based upon a detailed statistical analysis of word occurrence to identify the text-pieces (sentences, paragraphs, etc.) that are likely to be most important to convey the content of a text, and concatenate the selected pieces together to form the final extract.

After decades of attempts to perform automatic abstracting (while unduly ignoring the hermeneutic problems) that have not been vindicated by satisfactory results, there is currently a noticeable renaissance of extracting. The automatic creation of extracts is based on statistical methods that operate under the objective of retrieving the most important sentences in a digitally available source, which then—adhering to length guidelines and after several processing steps—serve as an extract (and thus as substitute abstracts) (Brandow, Mitze, & Rau, 1995; Lloret & Palomar, 2012; Saggion, 2008; Spärck Jones, 2007). Here we distinguish between static extracts (which are created exactly once and then stored in the surrogate) and dynamic summaries that are only worked out on the basis of a concrete query. While the static extracts are in competition with the (mostly) higher-quality, intellectually created abstracts, dynamic extracts are exclusively automatically processed and serve as perspectival summaries that lead from the respective user's search aspects. Dynamic extracts are able to replace the information-poor snippets which Web search engines offer on their results pages (Spärck Jones, 2007, 1457).

Sentence Weighting

The task of mechanical extracting is to retrieve a document's important sentences and to link them to each other in the extract. The extent of a sentence's importance in a text is guided by several factors (Hahn & Mani, 2000):

- the weighting values of its words (the sum of the TF*IDF values of all words in the sentence)—the most important factor,

- its position in the text (in the introduction, square in the middle, or at the end, in the context of a discussion of the text’s conclusions),
- after cue phrases (bonus words, rated positively, as well as negative stigma words),
- after indicator phrases (such as “in conclusion”),
- after terms that also occur in the document’s title (or in subheadings).

The idea of statistical information extraction goes back to Luhn (1958), who gave us the first weighting factor for sentences. Spärck Jones (2007, 1469) describes the fundamentals of the statistical strategy of automatically creating extracts:

The simplest strategy follows from Luhn, scoring source sentences for their component word values as determined by *tf*idf*-type weights, ranking the sentences by score and selecting from the top until some summary length threshold is reached, and delivering the selected sentences in original source order as the summary.

After a text has been divided into its sentences, the objective is to find a statistical score, which Luhn (1958) calls the “significance factor”, on the basis of the words that occur therein (Edmundson, 1964, 261). One option is to use the established TF*IDF procedure. After marking and skipping stop words, we calculate the statistical weighting value $w(t,d)$ for each term t of the document d following

$$w(t,d) = \text{TF}(t,d) * \text{IDF}(t).$$

A sentence s from d receives its statistical weight $ws(s)$ as the sum of the weights of the terms t_1 through t_i contained within it:

$$ws(s) = w(t_1,d) + \dots + w(t_i,d).$$

In a variant of this procedure, Barzilay and Elhadad (1999) exclusively work with nouns or with recognized phrases that contain at least one noun. This presupposes that the system has a part-of-speech tagger that can reliably identify the different word forms.

It can be empirically shown that sentences at certain positions of a text are, on average, more useful for the purposes of summary than the rest. Myaeng and Jang (1999, 65) observe:

Based on our observation, sentences in the final part of an introduction section or in the first part of a conclusion section are more likely to be included in a summary than those in other parts of the sections.

Sentences at salient positions in introduction and conclusion receive a positional weight $wp(s)$ of greater than 1, all others get a wp value of 1.

Edmundson remarked that the occurrence of certain reference words in a sentence can raise or lower its probability of appearing in an extract. His “cue method” works with three dictionaries (Edmundson, 1969, 271):

The Cue dictionary comprises three subdictionaries: Bonus words, that are positively relevant; Stigma words, that are negatively relevant; and Null words, that are irrelevant.

Bonus terms (such as “important”, “significant” or “definitely”) express the particular importance of a sentence and are allocated with weighting values greater than 1, whereas stigma words (such as “unclear”, “incidentally” or “perhaps”; Paice et al., 1994, 2) lead to values smaller than 1 (but greater than zero). Null words are weighted at 1 and play no particular role in this factor. The cue weight $w_c(s)$ of a sentence s follows the weighting value of the reference word it contains. If more than one such term occurs within a sentence, we work with the product of these cue words’ values.

Indicator phrases play a role analogous to that of reference words. These former are words or sequences of words that are often found in professionally created abstracts (Paice et al., 1994, 84-85). Kupiec, Pedersen and Chen (1995, 69) emphasize:

Sentences containing any of a list of fixed phrases, mostly two words long (e.g., “this letter ...”, “In conclusion ...” etc.), or occurring immediately after a section heading containing a keyword such as “conclusions”, “results”, “summary”, and “discussion” are more likely to be in summaries.

Sentences that contain an indicator phrase receive an indicator weight $w_i(s)$ of greater than 1, whereas all other sentences at this position get a w_i of 1.

Assuming that the document titles or subheadings contain important terms, those sentences that repeat words from titles and headings should receive a higher sentence weight than sentences without title terms (Ko, Park, & Seo, 2002). The effectiveness of this method depends on the quality of the titles that authors give their documents. For instance, high-quality titles can be expected for scientific articles, but not in the case of e-mails. An e-mail may very well have the title “question”, but there is no guarantee that this term will have any relation to the document’s content. If the document type is suitable for this method, sentences with title terms will be assigned a title weight $w_t(s)$ of greater than 1.

In individual cases, authors use further factors of sentence weighting. Kupiec, Pedersen and Chen (1995) suggest excluding very short sentences (e.g. with fewer than five words) from extracting as a matter of principle, and granting higher weights to sentences that contain upper-case abbreviations (such as *ASIST*, *ALA*, *ACM* or *IEEE*).

To be able to express the importance of a sentence in a document, we must accumulate the individual weighting values. Here we use (as a very simple procedure) the product of the individual weighting factors. The weight of a sentence s in a document d is thus calculated following:

$$w(s,d) = ws(s) * wp(s) * wc(s) * wi(s) * wt(s).$$

Alternatively, we could work with the sum of the individual weighting values (Hahn & Mani, 2000, 30). The specific values of the weighting factors w_p , w_c , w_i and w_t depend upon the database and the type of documents contained within them. Here it is likely necessary to play through different values and to evaluate the results in order to identify the ideal settings.

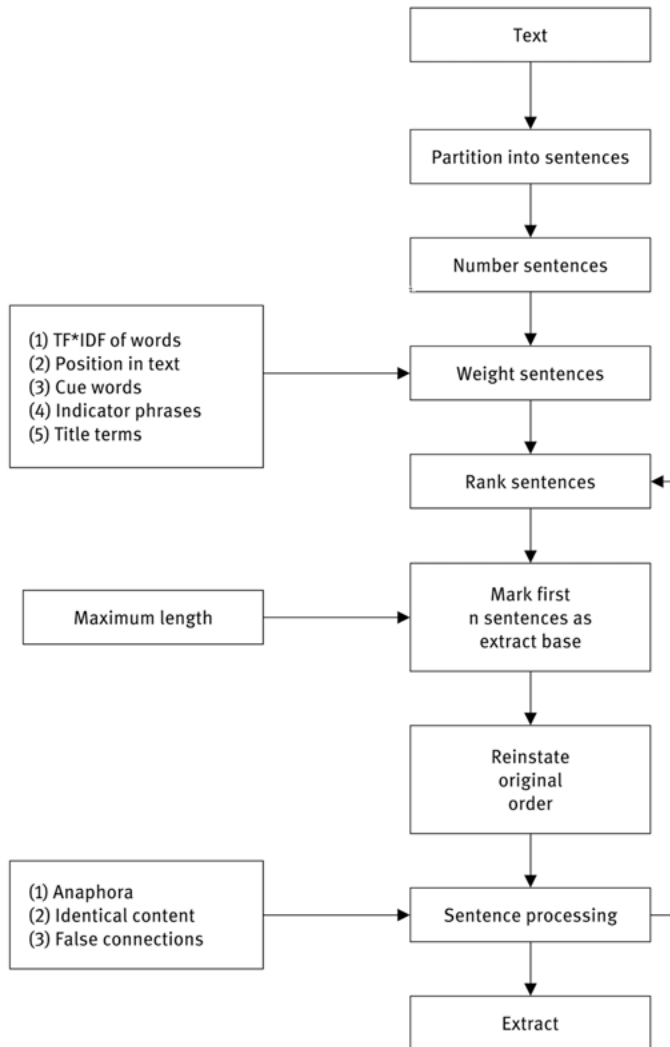


Figure 0.2.1: Working Steps in Automatic Extracting.

The sentences of a document are initially numbered and then arranged in descending order of their document-specific importance $w(s,d)$. This involves counting through the characters. If any given sentence reaches or surpasses the predefined maximum length of an extract, this sentence as well as all sentences ranked before it will be tagged as the basis for the extract. We presuppose that the number of sentences in the extract base is n . Using their serial number, the original order of these n sentences in the text is then reinstated.

Sentence Processing

The sentences in the extract base have been torn from their textual context and rearranged. In many cases, this can lead to faulty or incomplete information.

A first problem arises in the form of anaphora; these are expressions (such as the personal pronouns *he*, *she*, *it*) that refer to an antecedent introduced elsewhere in the text. If a sentence that contains anaphora is featured in the extract base without its antecedent, the reader will not know what the antecedent is and thus not understand the sentence. The first step of processing is to identify the anaphor. Not every expression that looks like an anaphor actually is one. In the sentence “it is raining”, *it* is not an anaphor and is thus unproblematic in the extract. “True anaphora” mostly have their antecedent either in the same sentence or in a neighboring one. If anaphor and antecedent are in the same sentence (“the solution is salty because *it* contains NaCl”), the sentence will be understandable in the extract. The situation is different for anaphora whose antecedents are located in other sentences. Consider the two sentences

Miranda Otto is Australian.

According to director Peter Jackson, *she* was the perfect choice for the role of Eowyn in *The Lord of the Rings*.

The second sentence is featured in the extract base due to its high weight, but the very short first sentence is not. In this case, anaphora resolution must make sure that *she* is replaced with *Miranda Otto*. However, anaphora resolution is very elaborate and not always satisfactory (Ch. C.4).

As the sentences are entered into the extract base independently of each other and following purely statistical methods, it is not impossible for them to contain repetitive content. In such cases, the most informative sentence will be retained and the others deleted (Carbonell & Goldstein, 1998). Similar sentences are recognized by the fact that they contain many of the same words. A similarity calculation (following Jaccard-Sneath, Dice or Cosine) must thus be performed for all sentences in the extract base. Sentences with a similarity value that surpasses a threshold value (to be defined) are deemed thematically related. Of these, only the sentence with the highest

weighting value is retained in the extract. Since this has made space in the summary, further sentences can be admitted to the extract base.

A final problem is that of false sentence connections. Extracted sentences that begin with “On the other hand ...” or “In contrast to ...”, for instance, lead to no usable information in the extract. Such logical or rhetorical connections tend to create confusion, as they are now situated next to another sentence than they were in the original. It makes sense to work out a list of typical connections and to remove the corresponding words from the sentence. Paice discusses a variant of false connections: “meta-textual references to distant parts of the text”. Examples for such distant connections are “... the method summarized earlier ...” or “... is discussed more fully in the next section ...” (Paice, 1990, 178). Here, too, the objective is to use lists of corresponding terms in order to discover the false connections and then correct them in the next step via adherence to the rules. The examples named lead to the following abbreviations: “... the method ...” and “... is discussed ...”.

It should be possible to derive some useful extracts via sentence weighting, the selection of central sentences, as well as the processing of sentences in the extract base. These extracts would then be firmly allocated to the document’s surrogate in the database.

Perspectival Extracts

A variant of extracting is the creation of summaries in information retrieval via reference to the user’s specific query. This is always an instance of paramorphous information condensation in the sense of perspectival extracts (in analogy to perspectival abstracts), where the perspective is predefined via the query. The procedure is analogous to the production of general extracts, the only difference being that only those sentences are selected that contain at least one query term (or—if a KOS is available—semantically similar terms such as synonyms, quasi-synonyms or any hyponyms that are contained in the sentence). There are two ways to select the corresponding sentences. In the primitive variant, the general ranking of the sentences remains the same and the system deletes all sentences that do not contain the search terms. In a more elaborate version, the statistical weight of the sentences is calculated not via all document terms but for the words from the (possibly expanded) query exclusively.

Multi-Document Extracts

Topic Detection and Tracking (TDT) (Ch. F.4) summarizes several documents with similar content into a single topic. This task often arises for news search engines (such as Google News). Currently Google News uses the first lines of the highest-weighted news source as the “abstract”, standing in for all identical or similar documents. In

actual fact, an extract of the topic is required here—not of a single document but of the totality of texts available for the topic.

Radev, Jing, Styś and Tam (2004) propose working with the topic's centroid. A previously recognized topic is expressed via the mean vector, called the centroid, of its stories. Topic tracking always works out whether the vector of a new document is close to the centroid of a known topic. If so, the document is allocated to the topic (and the centroid adjusted correspondingly); otherwise, it is assumed that the document discusses a new, previously unknown topic. TDT systems thus already have centroids. We exploit this fact in order to produce the extract (Radev et al., 2004, 920):

From a TDT system, an *event cluster* can be produced. An event cluster consists of chronologically ordered news articles from multiple sources. These articles describe an event as it develops over time. ... It is from these documents that summaries can be produced.

We developed a new technique for multi-document summarization, called *centroid-based summarization* (CBS). CBS uses the centroids of the clusters ... to identify sentences central to the topic of the entire cluster.

In statistical sentence weighting, $TF \cdot IDF$ is not calculated via the values of exactly one document, but via the values of the centroid. As a result, we get sentences from different documents in the cluster (Wang & Li, 2012). Since it is very probable that thematically related sentences will enter the extract base, we must take particular care to identify sentences discussing the same topic as well as to select the most information-rich sentence.

Fact Extraction

Systems with passage-retrieval components and question-answering systems never yield whole documents but only those text passages that most closely correspond to the query (Ch. G.6). The procedures used here can be refined in such a way that only the subject matter mentioned in the query is yielded. Suppose a user asks a system “What is the population of London, Ontario?” A current up-to-date search engine will produce a list of Web pages, in which the user must search for the desired facts by himself. A system with fact extraction gives the direct answer:

Population London, ON: 366,151 (2012),

which facts the system might have extracted from the Wikipedia article on London, ON, for instance. Fact extraction is only at the beginning of its development, which means that producers of fact databases, e.g. in chemistry or material engineering, currently rely on the intellectual work of experts. However, it is possible to support these activities heuristically via automatic fact extraction. If, for instance, fact extraction is meant to support the expansion of a company database, the system can peri-

odically visit pages on the WWW and align the data found there with those contained in the company dossier. In this way a fact extraction system can signal, for example, that a certain statement is not (or no longer) true.

Fact extraction is the translation of text passages into the attribute-value pairs of a predefined field schema. Moens (2006, 225) defines:

Information extraction is the identification, and consequent or concurrent classification and structuring into semantic classes, such as natural language text, providing additional aids to access and interpret the unstructured data by information systems.

For the purposes of illustration, picture the following field schema:

Person (leaves position)
 Person (fills position)
 Position
 Organization.

Now let us assume that a local news portal on the Web is reporting the following (imaginary) bulletin:

Gary Young, prior CEO of LAP Laser Applications, Ltd., yesterday announced his retirement. His successor is Ernest Smith.

After successful fact extraction, the field schema can be filled with values:

Person (leaves position)	Gary Young
Person (fills position)	Ernest Smith
Position	CEO
Organization	LAP Laser Applications, Ltd.

The extraction process requires templates that describe the relation between the field description (as in the example *Person / leaves position*) and the respective value (here: *Gary Young*). Possible templates in the example might be the formulations *prior* or *announced his retirement*. Such templates are either created intellectually or automatically via examples (Brin, 1998).

Conclusion

-
- Automatic extracting selects important sentences from a digitally available text, processing them in a second step.
 - The selection of sentences follows statistical aspects. Five criteria have proven to be of crucial value: the importance of the words contained in the sentence (calculated via TF*IDF), the position

of the sentence in the text, the occurrence of cue words, the occurrence of indicator phrases, and the repetition of title terms.

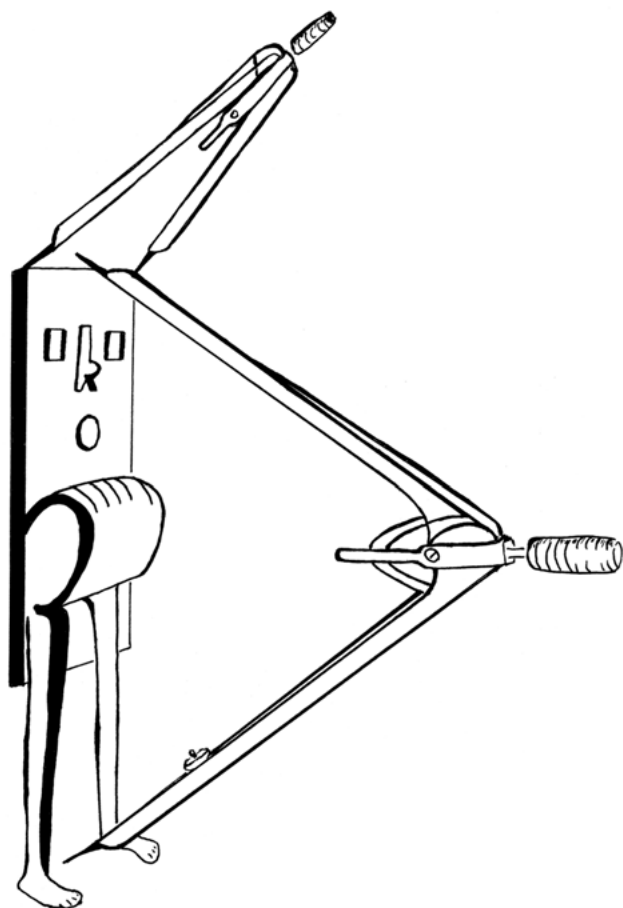
- The top-ranked sentences (up to a pre-set length) form the basis of the extract.
 - When processing the sentences in the extract base, anaphora across sentences must be resolved, content-identical sentences must be removed (except for the most informative one) and false sentence connections must be corrected.
 - Perspectival extracting takes into account user queries. Only those sentences that contain query terms are adopted in the extract.
 - In Topic Detection and Tracking (TDT), the extract is compiled from sentences of all documents in the thematic cluster. The statistical guidelines for the calculation of TF*IDF stem from the cluster's centroid vector.
 - Fact extraction is the transposition of textual content into the attribute-value pairs of a predefined field schema.
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Part P
Empirical Investigations on Knowledge
Representation



P.1 Evaluation of Knowledge Organization Systems

When evaluating knowledge organization systems, the objective is to glean data concerning the quality of a KOS (nomenclature, classification system, thesaurus, ontology) via quantitative measurements. To do so, we need measurements for describing and analyzing KOSs. Following Gómez-Pérez (2004) and Yu, Thorn and Tam (2009), we introduce these dimensions of KOS evaluation criteria:

- Structure of a KOS,
- Completeness of a KOS,
- Consistency of a KOS,
- Overlaps of multiple KOSs.

Structure of a KOS

Several simple parameters can be used to analyze the structure of a KOS (Gangemi, Catenacci, Ciaramita, & Lehmann, 2006; Soergel, 2001). These parameters relate both to the concepts and to the semantic relations. An initial base value is the number of concepts in the KOS. Here the very opposite of the dictum “the more the better” applies. Rather, the objective is to arrive at an optimal value of the number of terms that adequately represent the knowledge domain and the documents contained therein, respectively. If there are too few terms, not all aspects of the knowledge domain can be selectively described. If a user does not even find “his” search term, this will have negative consequences for the recall, and if he does find a suitable hyperonym, the precision of the search results will suffer. If too many concepts have been admitted into the KOS, there is a danger that users will lose focus and that only very few documents will be retrieved for each concept. When documents are indexed via the KOS (which—excepting ontologies—is the rule), the average number of documents per concept is a good estimate for the optimal number of terms in the KOS. Of further interest is the number of designations (synonyms and quasi-synonyms) per concept. The average number of designations (e.g. non-descriptors) of a concept (e.g. of a descriptor) is a good indicator for the use of designations in the KOS.

Analogously to the concepts, the number of different semantic relations used provides an indicator for the structure of a KOS (semantic expressiveness). The total number of relations in the KOS is of particular interest. Regarded as a network, the KOS’s concepts represent the nodes and their relations the lines. The size of relations is the total number of all lines in the KOS (without the connections to the designations, since these form their own indicator). A useful derived parameter is the average number of semantic relations per concept, i.e. the term’s mean degree. The indicators for size of concepts and size of relations can be summarized as the “granularity of a KOS” (Keet, 2006).

Information concerning the number of hierarchy levels as well as the distribution of terms throughout these individual levels are of particular interest. Also important are data concerning the number of top terms (and thus the different facets) and bottom terms (concepts on the lowest hierarchy level), each in relation to the total number of all terms in the KOS. The relation of the number of top terms to the number of all terms is called the “fan-out factor”, while the analogous relation to the bottom terms can be referred to as “groundedness”. “Tangledness” in turn measures the degree of polyhierarchy in the KOS. It refers to the average number of hyperonyms for the individual concepts. By counting the number of hyponyms for all concepts that have hyponyms (minus one), we glean a value for each concept’s average number of siblings.

Soergel (2001) proposes measuring the degree of a term’s precombination. A KOS’s degree of precombination is the average number of partial terms per concept. The degree of precombination for *Garden* is 1, for *Garden Party* it is 2, for *Garden Party Dinner* 3, etc.

Completeness of a KOS

Completeness refers to the degree of terminological coverage of a knowledge domain. If the knowledge domain is not very small and easily grasped, this value will be very difficult to determine. Yu, Thorn and Tam (2009, 775) define completeness via the question: Does the KOS “have concepts missing with regards to the relevant frames of reference?”

Portaluppi (2007) demonstrates that the completeness of thematic areas of the KOS can be estimated via samples from indexed documents. In the case study, articles on chronobiology were searched in Medline. The original documents, i.e. the documentary reference units, were acquired, and the allocated MeSH concepts were analyzed in the documentary units. Portaluppi (2007, 1213) reports:

By reading each article, it was (...) possible to identify common chronobiologic concepts not yet associated with specific MeSH headings.

The missing concepts thus identified might be present in MeSH and may have been erroneously overlooked by the indexer (in which case it would be an indexing error; Ch. P.2), or they are simply not featured in the KOS. In the case study, some common chronobiologic concepts are “not to be associated with any specific MeSH heading” (Portaluppi, 2007, 1213), so that MeSH must be deemed incomplete from a chronobiologic perspective.

If one counts the concepts in the KOS’s thematic subset and determines the number of terms that are missing from a thematic point of view, the quotient of the number of missing terms and the total number of terms (i.e. those featured in the KOS plus those missing) results in an estimated value of completeness in the corresponding knowledge subdomain.

Consistency of a KOS

The consistency of a KOS relates to three aspects:

- Semantic inconsistency,
- Circularity error,
- Skipping hierarchical levels.

Inconsistencies may particularly occur when several KOSs (that are consistent in themselves) are unified into a large KOS.

In the case of semantic inconsistency, terms have been wrongly arranged in the semantic network of all concepts. Consider the following descriptor entry:

Fishes

- BT Marine animals
- NT Salt-water fishes
- NT Freshwater fishes.

BT (broader term) and NT (narrower term) span the semantic relation of hyponymy in the example. In this hierarchical relation, the hyponyms inherit all characteristics of their hyperonyms (represented graphically in Fig. P.1.1). The term *Marine animals*, for instance, contains the characteristic “lives in the ocean.” This characteristic is passed on to the hyponym *Fishes* and onward to its hyponyms *Salt-water fishes* and *Freshwater fishes*. The semantic inconsistency arises in the case of *Freshwater fishes*, as these do not live in the ocean.

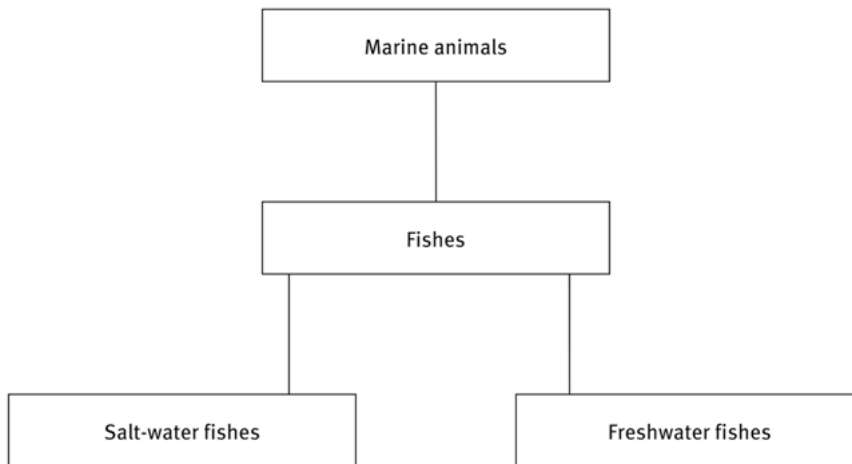


Figure P.1.1: An Example of Semantic Inconsistency.

Circularity errors occur in the hierarchical relation when one and the same concept appears more than once in a concept ladder (Gómez-Pérez, 2004, 261):

Circularity errors ... occur when a class is defined as a specialization or generalization of itself.

Suppose that two KOSs are merged. Let KOS 1 contain the following set of concepts:

Persons

NT Travelers,

whereas KOS 2 formulates:

Travelers

NT Persons.

When both KOS are merged, the result is the kind of circle displayed in Figure P.1.2 (example taken from Cross & Pal, 2008).

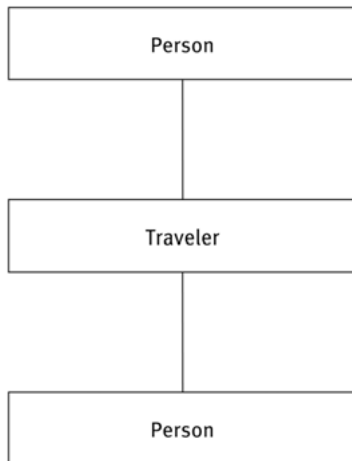


Figure P.1.2: An Example of a Circularity Error.

Skipping errors are the result of hierarchy levels being left out. Here, too, we can provide an example:

Capra

NT Wild goat

NT Domestic Goat

Wild goat

NT Domestic goat.

In the biological hierarchy, *Capra* is the broader term for *Wild goat* (*Capra aegagrus*). *Wild goat*, in turn, is the broader term for *Domestic goat* (*Capra hircus*). By establish-

ing a direct relation between *Capra* and *Domestic goat*, our KOS (Figure P.1.3) skips a hierarchy level. The cause of the skipping error is the erroneous subsumption of NT *Domestic goat* within the concept *Capra*.

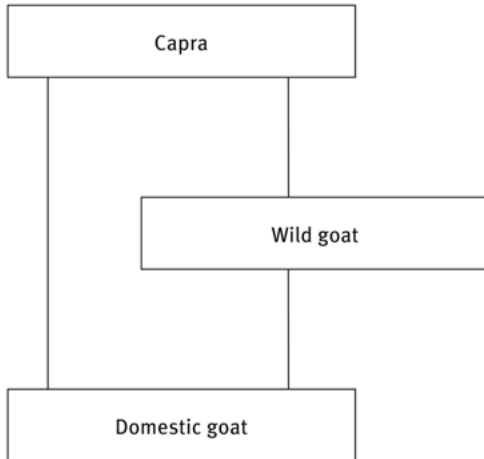


Figure P.1.3: An Example of a Skipping Error.

Overlaps of Multiple KOSs

In the case of polyrepresentation, different methods of knowledge representation as well as different KOSs are used to index the same documents. Haustein and Peters (2012) compare the tags (i.e., in the sense of folksonomies, the readers' perspective), subject headings of Inspec (the indexers' perspective), KeyWords Plus (as a method of automatic indexing) as well as author keywords and the words from title and abstract (the authors' perspective) of over 700 journal articles. Haustein and Peters use three parameters to determine overlap. The value g represents the number of identical tags and other denotations (author keywords, Inspec subject headings, etc.) per document, a is the number of unique tags per document, and b the number of unique terms (author keywords, etc.) per document. Mean overlap tag ratio means the overlap between tags and other terms (g) relative to the number of tags (a); mean overlap term ratio calculates the overlap from g relative to the number of respective terms (b). Thirdly, Haustein and Peters work with the Cosine. The authors are particularly interested in the overlap between folksonomy-based tags and other methods of knowledge representation (displayed in Table P.1.1). Of course one can also compare several KOS with one another, as long as they have been used to index the same documents.

Table P.1.1: Mean Similarity Measures Comparing Different Methods of Knowledge Representation.
Source: Haustein & Peters, 2012.

	Author key- words	Inspec subject headings	KeyWords Plus	Title terms	Abstract terms
Mean overlap tag ratio	11.8%	13.3%	2.9%	36.5%	50.3%
Mean overlap term ratio	10.4%	3.4%	3.0%	24.5%	4.8%
Mean cosine similarity	0.103	0.062	0.026	0.279	0.143

In Table P.1.1 it is shown, for instance, that roughly one in two tags (50.3%) also occurs as a word in the respective document's abstract. The other way around, though, only every twentieth word from the abstract (4.8%) is also a tag. All in all, there is only a minor overlap between social tagging and the other methods of knowledge representation, leading Haustein and Peters (2012) to emphasize:

(S)ocial tagging represents a user-generated indexing method and provides a reader-specific perspective on article content, which differs greatly from conventional indexing methods.

The Haustein-Peters method can also be used to comparatively evaluate different KOSs in the context of polyrepresentation. When the similarity measurements between two KOSs are relatively low, this points to vocabularies that complement each other—which is of great value to the users, as it provides additional access points to the document. If similarities are high, on the other hand, one of the two KOSs will probably be surplus to requirements in practice.

In Table P.1.2 we show an overview of all dimensions and indicators of the evaluation of knowledge organization systems.

Table P.1.2: Dimensions and Indicators of the Evaluation of KOSs.

Dimension	Indicator	Calculation
Structure of a KOS	Size of concepts (granularity, factor I)	Number of concepts (nodes in the network)
	Size of relations (granularity, factor II)	Number of relations (lines in the network)
	Degree of concepts	Average number of relations per concept
	Semantic expressiveness	Number of different semantic relations
	Use of denotations	Average number of denotations per concept
	Depth of hierarchy	Number of levels
	Hierarchical distribution of concepts	Number of concepts on the different levels
	Fan-out factor	Quotient of the number of top terms and the number of all concepts
	Groundedness factor	Quotient of the number of bottom terms and the number of all concepts
	Tangledness factor	Average number of hyperonyms per concept
	Siblinghood factor	Average number of co-hyponyms per concept
	Degree of precombination	Average number of partial concepts per concept
Completeness	Completeness of knowledge subdomain	Quotient of the number of missing concepts and the number of all concepts (in the KOS and the missing ones) regarding the subdomain
Consistency	Semantic inconsistency	Number of semantic inconsistency errors
	Circulation	Number of circulation errors
	Skipping hierarchical levels	Number of skipping errors
Multiple KOSs	Degree of polyrepresentation	Overlap

Conclusion

- On the four dimensions Structure of a KOS, Completeness, Consistency, and Multiple KOSs, there are indicators that serve as quantitative parameters pointing to the quality of a KOS.
- The structure of a KOS is calculated via the number of concepts (nodes in a network) and their semantic relations (lines). Further important indicators include fan-out (the relative frequency of the top terms), groundedness (relative frequency of bottom terms), tangledness (average number of broader terms per concept) and siblinghood (average number of co-hyponyms per concept).

- Completeness is a theoretical measurement for the degree of terminological coverage of a knowledge domain. For small subdomains, an analysis of the original documents can be used to determine how many concepts would be necessary for an exhaustive description of the content. The quotient of this number and that of the sum of concepts actually contained in the domain and those missing provides an indicator of the completeness of the KOS relative to the subdomain.
 - Consistency comprises semantic inconsistency (the erroneous insertion of a concept into the semantic network), circle errors in the hierarchy relation as well as the skipping of hierarchical levels.
 - When comparing different KOSs, and KOSs with other methods of knowledge representation (e.g. folksonomy-based tags or author keywords), the overlap of the respective terms can be calculated in the case of polyrepresentation (i.e. when different methods or KOSs are applied to a single document).
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P.2 Evaluation of Indexing and Summarization

Criteria of Indexing Quality

Sub-par indexing leads to problems with recall and precision in information retrieval, as Lancaster (2003, 85) vividly demonstrates:

If an indexer fails to assign *X* when it should be assigned, it is obvious that recall failures will occur. If, on the other hand, *Y* is assigned when *X* should be, both recall and precision failures can occur. That is, the item will not be retrieved in searches for *X*, although it should be, and will be retrieved for *Y*, when it should not be.

Indexing quality means the accurate representation of the content of a document via the surrogate (Rolling, 1981; White & Griffith, 1987). It can be fixed via the following criteria, which refer both to the terms used for indexing and to the surrogates as a series of wholes:

- Indexing Depth of a Surrogate,
 - Indexing Exhaustivity of a Surrogate,
 - Indexing Specificity of a Concept,
- Indexing Effectivity of a Concept,
- Indexing Consistency of Surrogates,
 - Inter-indexer and intra-indexer consistency,
 - Meeting user interests,
 - Ideal Type,
 - Thematic Clusters.

What is required are the parameters for information science research into information services and for the internal controlling of information producers as well as, for libraries, for the subscription decisions concerning information services (particularly those that are charged) (DeLong & Su, 2007). In the business scenario, the information science parameters are regarded as relative to the costs. Which degree of indexing quality corresponds to the respective costs needed for creating the services?

Indexing Depth

Indexing depth has two aspects: indexing exhaustivity and specificity (DIN 31.623/1:1988, 4):

Indexing exhaustivity states the degree of indexing with regard to the specialist content of a document; a first approximation is expressed in the number of those (concepts; A/N) to be allocated. Indexing specificity states how general or how specific they (the concepts; A/N) are in

relation to the document's content; a first approximation is expressed via the hierarchical level of the (concepts; A/N).

Indexing exhaustivity is the number of concepts B_1 through B_n that are allocated to a document during indexing (Maron, 1979; Wolfram, 2003, 100-102). The simple counting of concepts does not always lead to satisfactory results (Sparck Jones, 1973). This is why we combine indexing depth and specificity into a single parameter.

We understand indexing specificity to be the hierarchical level HL_i ($1 \leq i \leq m$), where we can find a concept in a knowledge organization system. In the polyhierarchical scenario, we consider the shortest path to the respective top term. A top term of a concept ladder is always on level 1. The indexing depth is made up of indexing exhaustivity and indexing specificity. Each allocated concept thus enters, weighted relative to its specificity, the following formula measuring the indexing depth of a documentary unit DU:

$$\text{Indexing Depth (DU)} = \{ld[HL(B_1)+1] + \dots + ld[HL(B_n)+1]\} / \#S.$$

$\#S$ counts the number of standard pages (e.g. translated into A4 from a given format). We work with logarithmic values instead of absolute ones, since it is hardly plausible that a concept from the second level should be exactly twice as specific as one from the first. Since $ld1 = 0$, we will add 1 in each respective case so as to receive a value that is not equal to 0 when the concept is a top term. For non-text documents, which contain no pages, the value $\#S$ in the formula is discarded.

How exhaustively should a document be described? It is not the case at all that a greater indexing depth is always of advantage to the user. Rather, a certain value represents the apex; not enough indexing depth can lead to a loss of information, too much indexing depth to information overload (Seely, 1972). Cleveland and Cleveland (2001, 105) point out:

Exhaustivity is related to how well a retrieval system pulls out all documents that are possibly related to the subject. Total exhaustivity will retrieve a high proportion of the relevant documents in a collection, but as more and more documents are retrieved, the risk of getting extraneous material rises. Therefore, when indexers are aiming for exhaustiveness they must keep in mind that at some point they may be negatively affecting the efficiency of the system.

The ideal system will give users all documents useful to them and no more.

If not enough terms are allocated, information loss may occur; choose too many, information overload looms. Ideal indexing amounts to walking a thin line between information loss and overload.

Defining an ideal indexing depth becomes particularly problematic when the database's user group is not coherent (Soergel, 1994, 596). To illustrate, we distinguish between two types of users: scientists (e.g. medical doctors) looking for specialized literature on the one hand, and laymen (e.g. patients) looking for an easy entry to a specialist topic on the other. Now there are two approaches, one with heavily

exhaustive indexing (deep indexing) and one without deep indexing (surface indexing). The following overview is created:

User	great indexing depth	meager indexing depth
Scientist	good recall, good precision, satisfactory	weak recall, weak precision, too few exact hits
Layman	recall too large, precision too large, too many specific hits	good recall, good precision, satisfactory.

A layman is served ideally while the expert finds too few fitting documents; in return, the scientist will find satisfactory working conditions while the laymen is presented with too many specific (and thus, for him, useless) documents. The dilemma can only be resolved when the database has implemented the non-topical information filter for the respective target group (Chapter J.3).

We will come back once more to our indexing example from Chapter N.1, comparing it now to a surrogate of the same documentary reference unit—which comes from another information service this time. Both information services use the same KOS, a scientific thesaurus.

Information Service 1:

Finland, Manufacturing, Industrial Manufacturing, National Product, Financial Policy, Monetary Policy, Foreign Exchange Policy, Inflation, Trade Balance, Unemployment;

Information Service 2:

Finland, Economy, Economic Forecast.

We calculate the indexing depth of both surrogates. The descriptors of information service 2 are located on the hierarchy levels 2 (Economy and Economic Forecast) and 3 (Finland), and the number of pages in the article is 11. The surrogate's indexing depth is:

$$(ld3 + ld3 + ld4) / 11 = 0.47.$$

The surrogate for information service 1 contains ten descriptors on the hierarchical levels 1 through 4; its indexing depth is calculated via:

$$(ld4 + ld2 + ld3 + ld4 + ld4 + ld5 + ld2 + ld2 + ld3 + ld5) / 11 = 1.53.$$

The indexing in information service 1 thus stands for great indexing depth, its counterpart in information service 2 for rather shallow indexing depth.

Indexing Effectivity

A further criterion for indexing quality is the effectivity of the concepts used for indexing (White & Griffith, 1987). By this we mean the discriminatory power of the respective terms. If a user searches for a concept X, he will be presented with a hit list, great or small, representing the topic X. Borko (1977, 365) derives a measurement from this:

The clustering effect of each index term, and by implication, the retrieval effectiveness of the term, can be measured by a signal-noise ration based on the frequency of term use in the collection.

Instead of the absolute frequency of occurrence of an indexing term in documentary units, we propose the usage of inverse document frequency IDF (similarly, Ajiferuke & Chu, 1988):

$$\text{Effectivity}(B) = \text{IDF}(B) = [\text{ld}(N/n)] + 1.$$

N is the total number of documentary units in the database, n the number of those surrogates in which B has been allocated as an indexing term. The effectivity of B becomes smaller the oftener B occurs in documentary units; it reaches its maximum when only a single document is indexed with the concept.

A surrogate's indexing effectivity is calculated as the arithmetic mean of the effectivity of its concepts.

Indexing Consistency

Looking back on the two surrogates of information services 1 and 2 (which, after all, index the same document using the same tool), what leaps to the eye is the considerable discrepancy between the allocated concepts. Only a single descriptor (*Finland*) co-occurs in both surrogates. The degree of coincidence between different surrogates from the same template is an indicator of indexing consistency. For indexing consistency (Zunde & Dexter, 1969), we distinguish between inter-indexer consistency (comparison between different indexers) and intra-indexer consistency (comparison of the work of one and the same indexer at different times). The indexing consistency between two surrogates is calculated via the known similarity measurements, e.g. following Jaccard-Sneath:

$$\text{Indexing Consistency}(DU_1, DU_2) = g / (a + b - g).$$

DU_1 and DU_2 are the two documentary units being compared, g is the number of those concepts that occur in both surrogates, a the number of the concepts in DU_1 and b the

number of concepts in DU_2 . We calculate the indexing consistency of the two exemplary surrogates:

$$\text{Indexing Consistency}(\text{Service}_1, \text{Service}_2) = 1 / (10 + 3 - 1) = 0.083.$$

A value of 0.083 points to an extremely low indexing consistency. Does this mean that one surrogate or the other is better? Looking at Figure P.2.1, we see four indexers (each with their own surrogate of the same document). Indexers A, B and C work consistently (indicated by the arrows pointing to one and the same circle), whereas indexer D uses completely different concepts. The inter-indexer consistency of A, B and C is thus very high, that between D and his colleagues very low. But D—and only D—“hits” user interests precisely, which A, B and C fail to do. The blind inference according to which indexing consistency leads to indexing quality is thus disproved (Cooper, 1969). Lancaster (2003, 91) observes:

Quality and consistency are not the same: one can be consistently bad as well as consistently good!

For Fugmann, a new perspective on consistency relations opens up. He considers indexer-user consistency to be more important than inter-indexer consistency (Fugmann, 1992).

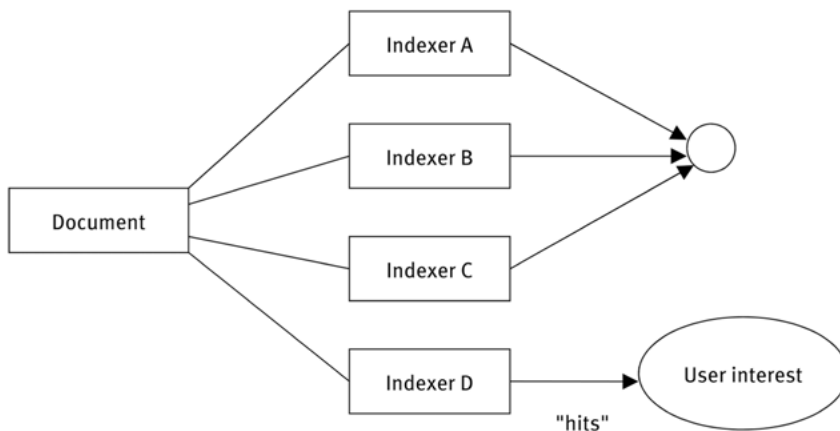


Figure P.2.1: Indexing Consistency and the “Meeting” of User Interests.

How can such a content-oriented (i.e. grounded in “hits”) indexing consistency be measured? To do so, we require the precondition of ideal-typically correct templates. The consistency of the actually available surrogates is then calculated in relation to these templates. Lancaster discusses two variants of this procedure. On the one hand,

it is possible to construct centrally important queries for a document under which it must be retrievable in every case (Lancaster, 2003, 87). The measure of question-specific consistency is the relative frequency with which a given surrogate is retrieved under the model queries. The alternative procedure works out an ideally correct surrogate as the “standard” (i.e. as a compromise surrogate between several experienced indexers) and compares the actual surrogates with the standard (Lancaster, 2003, 96). The measure can be the above-mentioned formula for indexing consistency. As a possible alternative to these procedures, or also to complement them, Braam and Bruil (1992) suggest asking the authors of the indexed documents.

A further procedure of analyzing indexing consistency works comparatively (Stock, 1994, 150-151). Using preliminary investigations, e.g. co-citation analysis or expert interviews, clusters of documents that unambiguously belong to a topic and should thus be indexed at least in a similar fashion are marked in thematically related information services. Afterward, one can consider filtering out from the surrogates all those concepts whose indexing effectivity is too low. Now all concepts that occur in at least 50% of the surrogates in the respective cluster are identified. (Experience has shown that to consider *all* effective terms leads to too small and hence unusable hit sets.) In the last step, the effective 50%-level concepts are counted for every information service. Chu and Ajiferuke (1989, 19) name an example: the indexing quality of three information science databases (ISA: Information Science Abstracts; LISA: Library and Information Science Abstracts; LL: Library Literature) for the topic *Cataloguing Microcomputer Software*:

<i>Information Service</i>	<i>Number of Effective Concepts</i>
LISA	8
ISA	4
LL	2.

The interpretation of the respective values of the indexing quality criteria always depends on the fixed points of the indexing process. What counts as a good indexing depth for a knowledge domain, a user, a method etc. (indexing effectivity, indexing consistency) can prove to be inadequate for another domain, another user etc. Thus, the measurement 2 for LL is an acceptable value for an information service that predominantly addresses laymen, whereas a value like 8, as in LISA, appears adequate for a specialist database.

Indexing consistency depends upon the type of documents to be indexed. As early as 1984, Markey was able to demonstrate that images lead to a lower indexing consistency than textual documents. The different indexers use completely different indexing depths when working with graphical material (Hughes & Rafferty, 2011), which necessarily leads to lower consistency values.

Quality of Tagging

Information services in Web 2.0 draw on the collaboration of their clients by letting them tag the documents. In contrast to professional information services (such as Medline or Chemical Abstracts Service), it is not information professionals who tag in Flickr, YouTube, CiteULike, Delicious etc., but amateurs. The parameters of indexing quality can also be used here, of course. Typical research questions are: What does indexing exhaustiveness look like in such services? How many terms are allocated per document in narrow folksonomies? What is the extent of inter-tagger consistency in broad folksonomies (Wolfram, Olson, & Bloom, 2009)?

In broad folksonomies, new options for analysis open up. For instance, the document-specific tagging behavior of users can be observed over time. How do such tag distributions come about for each document? Do certain “power tags”, which are used consistently often, assert themselves for the tagged document? How long does it take for the tag distributions to become stable (Ch. K.1)?

Summarization Evaluation

Research topics for the evaluation of summaries are found in intellectually created abstracts, in automatically compiled extracts as well as in snippets from search engine results pages (SERPs). Abstracts, extracts and snippets are evaluated either intrinsically or extrinsically (Lloret & Palomar, 2012). While extrinsic evaluation relates to the use of summaries in other applications (such as retrieval systems), intrinsic evaluation attempts to register the quality and informativeness of the text itself (Spärck Jones, 2007, 1452). Both forms of summarization evaluation are problematic, as Lloret and Palomar (2012, 32) emphasize:

The evaluation of a summary, either automatic or human-written, is a delicate issue, due to the inherent subjectivity of the process.

The informativeness of a summary has its fixed point in its relative utility for a user and concentrates on the text’s content. Quality evaluation, on the other hand, registers formal elements such as the coherence or non-redundancy of the summary (Mani, 2001). A possible indicator for a summary’s lack of quality is the occurrence in the text of an anaphor without its antecedent.

Some possible methods of summarization evaluation are the comparison of a summary with a “gold standard” (i.e. an “ideal” predefined summary) (Jing, McKeown, Barzilay, & Elhadad, 1998) as well as user surveys (Mani et al., 1999). In TIPSTER (Mani et al., 1999), for example, the user was presented with documents (indicative summaries as well as full texts) that exactly matched a certain topic. The test subjects then had to decide whether the text was relevant to the topic or not. Then

the relevance judgments' respective matches with full texts and their summaries were analyzed (Mani et al., 1999, 78):

Thus, an indicative summary would be "accurate" if it accurately reflected the relevance or irrelevance of the corresponding source.

The quotient of the number of matching relevance judgments (between full text and corresponding summary) and the number of all relevance judgments can be calculated in order to arrive at a parameter for the accuracy (as an aspect of informativeness) of a summary.

Conclusion

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- Indexing quality can be represented via the parameters of indexing depth (indicator consisting of indexing exhaustiveness and specificity), the indexing effectivity of the concepts and indexing consistency. The latter indicator has the dimensions of inter-indexer consistency, intra-indexer consistency and user-indexer consistency.
 - Indexing depth unifies the number of allocated indexing terms with their specificity (placement in the hierarchy of a KOS) and relativizes (for textual documents) the value via the number of the documentary reference unit's pages.
 - The indexing effectivity of a surrogate is the average effectivity of its indexing terms, which is calculated via the terms' inverse document frequency (IDF).
 - Indexing consistency means the consensus between different surrogates regarding one and the same documentary reference unit. The parameter initially has nothing to do with indexing quality (surrogates may very well be consistently false). Only consensus between a surrogate and user interest indicates information quality.
 - In the case of folksonomies in Web 2.0, the indexing is performed by laymen, and not—as in professional information services—by trained indexers. Here, too, the parameters of indexing quality can be usefully applied. Additional parameters are gleaned via analysis of document-specific tag distributions and their development in broad folksonomies.
 - Abstracts, extracts and snippets are evaluated either extrinsically (the utility of summaries for other systems, e.g. information retrieval systems) or intrinsically (by evaluation of the quality and the informativeness of the text).
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Part Q
Glossary and Indexes



Q.1 Glossary

Aboutness means “what it’s about”. It describes the topic that is to be represented in any given document. In multimedia documents, aboutness expresses the interpretation on the iconographical level, described by Panofsky. *See also* Ofness.

Abstract, being the result of summarization, ignores all that is not of fundamental importance in the document. It is an autonomous (secondary) document that briefly, precisely and clearly reflects the subject matter of the source document in the form of sentences.

Abstracting is the process of condensing the content of a document by summarizing the thematically significant subject matters.

Abstraction Relation (also called Hyponym-Hyperonym Relation) is a hierarchy relation that is subdivided from a logical perspective; it represents the logical perspective on concepts.

Actors in a social information network are documents, authors (as well as any further actors derived from them, such as institutions) and subjects. The prominence of an actor in a network is described via its centrality (degree, closeness, and betweenness), its exposed position (bridge, cutpoint) as well as through a prominent position in a “small worlds” network. The measurements are criteria for relevance ranking. In CWA, actors are referred to as carriers of activities.

Ambiguity. *See* Homonym and Disambiguation.

Anaphor denotes an expression (e.g. pronoun) that refers to an antecedent located elsewhere in the text. The term referred to is called antecedent (e.g. the name phrase *Miranda Otto*), while the referring term is the anaphor (*she*). Such referring expressions cause problems in information retrieval when using proximity operators and the counting basis of information statistics. Anaphor resolution tries to combine anaphora with their respective antecedents. In knowledge representation, anaphor resolution is important for automatic extracting.

Anchor appears above a link in the browser. It briefly describes the content of the linked page.

Antonym is an opposite concept. Contradictory antonyms only have extremes (mutually exclusive examples include *pregnant–non-pregnant* or *dead–live*), while contrary antonyms allow for other values to lie between the extremes (between *love* and *hate* lies *indifference*).

Associative Relation is a relation between two concepts that are neither synonymously nor hierarchically connected to each other. Generally, a *see-also* connection is used, but this can be specified depending on the application (e.g. usefulness respectively harmfulness).

Authorities are Web pages that are linked to by many other pages (i.e. documents with many inlinks). Authorities play an important role both in PageRank and the Kleinberg algorithm.

Authority Record determines the unified form of entries in bibliographic metadata or KOSs. This holds for titles, names of persons, organizations, as well as transliteration, etc.

Automatic Indexing is divided into concept-based and content-based procedures. The concept-based approach is performed either by using a KOS (probabilistic or rule-based indexing) or by classing documents according to terms or citations and references (quasi-classification), respectively. Content-based indexing goes through the two phases of information linguistics (or, for multimedia documents, of feature identification) and document ranking by relevance (using suitable retrieval models).

Automatic Reasoning is based upon both a KOS and description logic. In ontologies, such reasoning is a (content-based) strict implication.

Berrypicking is a strategy for searching through diverse information services. The user works his way from one database to another until his information need is satisfied.

B-E-S-T. In DIALOG's command terminology, the search for relevant documents in selected databases requires going through the basic structure BEGIN (call up database), EXPAND (browse dictionary), SELECT (search) as well as TYPE (output).

Bibliographic Coupling. Two documents are bibliographically coupled when both include references to the same documents.

Bibliographic Metadata are standardized data about documentary reference units, serving to facilitate digital and intellectual access to, as well as usage of, these documents. They stand in relation to one another. Metadata on websites can be expressed via HTML meta tags or Dublin Core elements.

Bibliographic Relation formally describes a document. *See also* Document Relations.

Bluesheets are lists of special characteristics of any given database (e.g. field schema, thesaurus, output options, and fees) that are listed in the database descriptions.

Boolean Operator joins together search atoms into complex search arguments. There are four Boolean operators in information retrieval: AND (in set theory: intersection—logically: conjunction); (inclusive) OR (set union—disjunction); NOT (exclusion set—postsection); XOR in the sense of an exclusive OR (symmetric difference—contravariance).

Boolean Retrieval Model dates back to the mathematician and logician George Boole and allows a binary perspective on truth values (0 and 1) in connection with the functions AND, OR, NOT and XOR. Relevance ranking is principally impossible.

Building Blocks Strategy is a form of intellectual query modification where the user divides his information problem into different facets. Within the facets, the search atoms are connected via the Boolean OR. The facets are interconnected either via AND (or NOT) or a proximity operator.

Category is a special form of general concepts within a knowledge domain. A category is the highest level of abstraction of concepts and includes a minimum amount of properties (e.g. space, time). For the knowledge domain, the forming of even more general concepts would create empty or useless concepts. Categories substantiate facets.

Centroid is a mean vector.

Character Set is a standard for representing the characters of natural languages in a binary manner. There are several character sets, including ASCII and Unicode.

Chronological Relation expresses the respective temporal direction in cases of gen-identity. Gen-identical objects, i.e. objects that are described at different times via different designations, are temporally connected to one another.

Citation transmits information from one (cited) document to another (citing) document, and is indicated in the latter as a reference.

Citation Indexing in knowledge representation is a text-oriented method and focuses on bibliographical statements in publications (as foot notes, end notes or in a bibliography). It is used wherever formal citing occurs (in law, academic science or technology). References and citations are viewed as concepts that represent the content of the citing and the cited documents.

Citation Pearl Growing Strategy is a form of intellectual query modification as an intermediate goal, in which the user tries to retrieve an ideally appropriate document via an initial search. New search terms are then gleaned from the retrieved document.

Citation Order arranges thematically related classes (i.e. documents sitting next to each other), in the sense of a shelving systematics while using classification systems.

Classification describes concepts (classes) via non-natural-language notations. This knowledge organization system is characterized by its use of notations as the preferred terms of classes, and by utilizing the hierarchy relation.

Classifying means the building of classes, i.e. all efforts toward creating and maintaining classifications.

Classing refers to the allocation of classes of a specific classification system to a given document.

Co-Citation. Two documents are co-cited when both co-occur in the bibliographical apparatus of other documents.

Co-Hyponym. See Sister Term.

Co-Ordinating Indexing gathers the concepts independently of their relationships within the respective document (as opposed to syntactical indexing).

Colon Classification, by Ranganathan, is the first approach toward a faceted KOS (with concept synthesis) and uses a facet-specific citation order for each discipline.

Cognitive Model combines a user's background knowledge, his language skills and his socio-economic environment. Both the user's expertise and his degree of information literacy are deemed to be of interest.

Cognitive Work Analysis (CWA) deals with the human work that requires decision-making. The actor and his activity are at the center of, and dependent on, his environment. The procedure following CWA helps reveal the right conditions for knowledge representation and information retrieval to aid an actor's specific tasks (e.g. constructing KOSs, indexing, abstracting).

Compound. Several individual concepts are merged into a concept unit (e.g. *housemaid*).

Compound Decomposition concerns the meaningful partition of singular terms or parts from their respective multi-word terms. When decomposing individual characters of a compound (from left to right and vice versa), the longest term that can be found in a dictionary will be marked. Ambiguities must be noted.

Concept is an abstraction (as a semantic unit) that bears meaning and refers to certain objects. It is verbalized, i.e. expressed via words. Concepts are defined by their extension (objects) and their intension (properties). They are interlinked with other concepts via relations. There is a distinction between categories, general concepts and individual concepts.

Concept Array is formed by sister terms that share the same hyperonym or holonym in a hierarchical relation.

Concept Explanation assumes that concepts are composed of partial concepts. According to Aristotle, a term is explained by specifying its genus (being a concept from the directly superordinate genus) and its differentia specifica (the fundamental difference to its sister concepts, i.e. those concepts that belong to the same genus). In a concept ladder, the properties are always passed down from top to bottom. These specifications necessarily embed terms in a hierarchical structure.

Concept Ladder is formed by concepts within a hierarchical relation. A broader term (hyperonym or holonym) is that term in the concept ladder that sits precisely one level higher than an initial term. A narrower term (hyponym or meronym) is a term that is located on the next lowest level of the hierarchy.

Concordance sets concepts of different KOSs in relation to one another.

Content Condensation is accomplished via summarizing (abstracting or extracting).

Conflation. Variant word forms of one and the same word are brought into a single form of expression. There are two forms of word form conflation: Lemmatization leads to basic forms, and stemming liberates word forms from their suffixes.

Cosine Coefficient is a measurement for calculating the similarity between two items A and B, following the formula $SIM(A-B) = g / (a \ b)^{1/2}$ (*a*: number of occurrences related to A; *b*: number of occurrences related to B; *g*: number of common occurrences related to A and B).

Crawling. A software package, called a “robot” or a “crawler”, makes sure that new documents admitted into a database are automatically retrieved and copied into a search engine’s database. Crawlers are the search engines’ “information suppliers”, performing the task of automatically discovering or updating documents on the WWW. The process starts with an initial quantity of known Web documents (“seed list”), then processes the links contained within these Web pages, and then the links contained in the newly retrieved pages.

Cross-Language Information Retrieval (CLIR) uses only the search queries, but not the documents that are to be translated into various other languages. In contrast to this, see Multi-Lingual Information Retrieval (MLIR).

Damerau Method performs approximate string matching in order to achieve fault-tolerant retrieval. It uses letter-by-letter comparison to recognize and correct individual input errors. The basis of this comparison is a dictionary.

Decimal Principle. A concept will be divided into a maximum of ten narrower terms, which are represented by decimal digits.

Deep Web. Digital documents (partly surrogates) are stored in databases of (partly fee-based) information services; only their entry pages are available on the World Wide Web.

Definition describes or explains a concept according to certain criteria of correctness. Definitions should be accurate, orient themselves on prototypes and—above all—be useful for the respective knowledge domain. Besides other sorts of definitions (incl. definition as abbreviation, explication, or nominal and real definition), knowledge representation mainly works with concept explanation and family resemblance.

Description Logic (also called terminological logic) allows the option of automatic reasoning to be introduced to a concept system, in the sense of ontologies.

Descriptor is the preferred term in a thesaurus. An equivalence relation connects a descriptor with all of its non-descriptors.

Descriptor Entry in a thesaurus summarizes all specifications that apply to a given preferred term. Non-descriptors accordingly receive their unambiguous assignment to preferred terms in a non-descriptor entry.

Designation is a natural-language word or a combination of terms from artificial languages (e.g. numbers, character strings) that expresses concepts in a KOS.

Dice Coefficient is a measurement for the calculation of the similarity between two items A and B, following the formula $SIM(A-B) = 2g / (a + b)$ (*a*: number of occurrences related to A; *b*: number of occurrences related to B; *g*: number of common occurrences related to A and B).

Differentia Specifica means the fundamental difference of the sister terms (or co-hyponyms) with regard to their superordinate genus concept (or hyperonym).

Dimension. The Vector Space Model regards terms as dimensions of an n-dimensional space. Documents and user queries are located in said space as vectors. The respective value in a dimension is calculated via text statistics.

Disambiguation. The ambiguity of homonyms is solved by recognizing the respective “right” semantic unit.

Document. An object is a document if it is physically available—including its digital form—, carries meaning, is created and is perceived to be a document. Apart from texts, there are non-textual (factual)

documents that are either digital or at least digitizable (movies, images, music etc.) or they are principally non-digital (for instance facts from science, museum artifacts).

Document Representing Objects applies to objects which are fundamentally undigitizable (STM facts, business and economic facts, facts about museum artifacts and works of art, persons, real time facts). Such a document can never directly enter an information system. It is thus replaced in information services by a surrogate.

Document-Term Matrix. According to the Vector Space Model, a retrieval system disposes of a large matrix. The documents are deposited in the rows of this matrix and the terms in the columns. The weight value of an individual term in the respective document is registered in the cells.

Documentary Reference Unit (DRU). In information indexing, documents are analytically divided into DRUs that form the smallest units of representation, and which always stay the same (e.g. an article in a scientific literature information service or video sequences in a movie database).

Documentary Unit (DU). Documentary reference units are represented in a database via surrogates that have an informational added value (i.e. that are created via precise bibliographical description, via concepts serving as information filters as well as via a summary as a kind of information condensation). Those surrogates are the DUs.

Document File is the central repository in a database storing documentary units as data entries.

Document Relations play an important role regarding the use of bibliographic metadata. A published document is formed by the four primary document relations: work, expression, manifestation and item. Content relations (such as the equivalence, derivative and descriptive relations) act simultaneously to the primary relations and are used to draw a line between original and new work. Part-Whole or Part-to-Part relations are significant for sequences of articles or Web documents.

Document-specific Term Weight. See Term Frequency (TF) and Within-Document Frequency (WDF).

Ellipsis means that terms have been left out, since the text's context makes it clear which antecedent is meant. Ellipses lead to related problems such as anaphora.

Emotional Information Retrieval (EmIR) aims to search and find documents that express feelings or that provoke feelings in the viewer. In addition to their aboutness, users can also inquire into documents' emotiveness.

E-Measurement unites the two effectiveness values of Recall and Precision into a single value. It is used for the evaluation of retrieval system quality.

Equivalence Relation (Document) considers identical or similar documents (e.g. original, facsimile, copy).

Equivalence Relation (KOS) refers to designations that are similar in meaning and to similar concepts, and unites them in an equivalence class. There is an equivalence relation between descriptors and non-descriptors in a thesaurus.

Evaluation is an empirical method for calculating the quality of technical systems and tools.

Evaluation of Indexing refers both to the concepts used for indexing and to the surrogates. It takes into consideration indexing depth, indexing effectiveness and indexing consistency.

Evaluation of Knowledge Organization Systems includes the characteristics of a KOS's structure, its completeness, its consistency and overlaps between multiple KOSs. The objective is to use quantitative measurements in order to glean indicators concerning the quality of a KOS.

Evaluation of Retrieval Systems comprises the four main dimensions of IT service quality, knowledge quality, IT system quality as well as retrieval system quality. Each dimension makes a contribution to the usage or non-usage—or, rather, the success or failure—of retrieval systems.

Evaluation of Summarization applies to abstracts, extracts and snippets. Intrinsic evaluation attempts to register the quality and informativeness of the text itself, while extrinsic evaluation regards the use of summaries in other systems, such as information retrieval systems.

Exchange Format regulates the transmission and combination of data entries between different institutions.

Explicit Knowledge is knowledge fixed in documents.

Extract is an automated selection of important sentences from a digitally available document.

Facet is a partial knowledge organization system derived from a more extensive KOS. It comes about as the result of an analysis performed by specifying a category.

Faceted Classification combines two bundles of construction principles: those of classification (notation, hierarchy and citation order) and those of faceting. Any focus of a given facet is named via a notation, a hierarchical order applies within the facets and the facets are processed in a specific order. One example of a faceted classification is the Colon Classification by Ranganathan.

Faceted Folksonomy makes use of different fields related to the categories (facets).

Faceted Knowledge Organization System works with several sub-systems of concepts, where the fundamental categories of the respective KOS form the facets. The respective subject-specific terms are collected for each facet. Any term from any facet can be connected to any of the other terms from the facets. Faceted KOSs consistently use combinatorics in order to require far less term material. *See also* Faceted Classification, Faceted Thesaurus, Faceted Nomenclature and Faceted Folksonomy.

Faceted Nomenclature lists the concepts of the KOS into the individual facets without any hierarchization. There can be systems with and without concept synthesis.

Faceted Thesaurus contains at least two coequal subthesauri, each of which has the characteristics of a normal thesaurus. It combines the construction principles of descriptors and of the hierarchy relation—in the latter, at least those that involve the principle of faceting (but not concept synthesis).

Factual Document. *See* Documents Representing Objects.

Family Resemblance, according to Wittgenstein, is a sort of definition that works via a disjunction of properties (e.g. *game*). The different concepts do not pass along all of their properties to their narrower terms.

Fault-Tolerant Retrieval is a form of information retrieval that identifies input errors (in documents as well as in user queries) on a limited scale and then corrects those mistakes.

Field is the smallest unit for processing uniform information. There are administrative fields (e.g. the date of a documentary unit's being recorded), formal-bibliographic fields (e.g. author and source) and content-depicting fields (e.g. descriptors and abstract).

Focus is a terminologically controlled simple term in a facet (of a KOS).

Folksonomy is a portmanteau of “folk” and “taxonomy”. It allows users to describe documents via freely selectable keywords (tags), according to their preferences. This sort of free tagging system is not bound by rules. A folksonomy evolves through the assemblage of user tags. There are three kinds of folksonomies: broad folksonomy (the multiple allocation of tags is allowed), narrow folksonomy (the document's author may use tags only once) and extended narrow folksonomy (a specific tag may only be allocated once, no matter by whom).

Frame, according to Barsalou, regards concepts in the context of sets of attributes and values, structural invariants (relations) and rule-bound connections. Properties are allocated to a concept, and values to the properties, where both properties and values are expressed via concepts. The frame

approach leads to a multitude of relations between concepts, and, furthermore, to rule-bound connections.

Freshness can be consulted as a weighting factor for Web documents (the more current, the more important). Particularly on the Web, pages are constantly changed or even deleted by Webmasters and thus become out-of-date. Crawlers use certain strategies of refreshing, on the behalf of search engines, in order to update the pages retrieved thus far. A crawled page is considered to be “fresh” and will continue to be so until it is modified.

Fuzzy Operator. ANDOR dilutes the strict forms of Boolean AND and OR as a combination of both.

Fuzzy Retrieval Model uses many-valued logic and fuzzy logic in weighted Boolean retrieval.

Gen-Identity is a weak form of identity which disregards certain temporal aspects. It describes an object over the course of time. The object is always the same, while its intension or extension changes. Gen-identical concepts are interconnected via the chronological relation.

General Concept is a concept whose extension contains more than one element. General concepts are situated between categories and individual concepts.

Genre, used as non-topical information filter, depends less on the information’s content than on its communicative purpose and form. There are genres and subgenres for all formally published and unpublished texts in various specific domains (e.g. lecture, best practices, cookbook, love letter, FAQ list).

Genus Proximus is a concept from the directly superordinate hierarchical level. When defining, it exists as the superordinate genus concept and as the nearest hyperonym, respectively.

Geographical Information Retrieval (GIR) allows the ranking of documents with location-dependent content according to their distance from the user’s stated location.

Graph consists of nodes (actors) and lines. Directed graphs of interest to information retrieval depict information flows and link relations on the World Wide Web. In undirected graphs there are co-authorship, co-subjects, co-citations and bibliographic coupling.

Hierarchy is the most important relation in concept systems. The knowledge organization system is arranged either monohierarchically or polyhierarchically. There are three different variants of hierarchy: hyponymy, meronymy and instance. *See also:* Abstraction Relation (or Hyponym-Hyperonym Relation), Part-Whole Relation (or Meronym-Holonym Relation) and Instance Relation.

Hierarchical Retrieval allows for the incorporation of narrower terms and broader terms into a search request.

HITS (Hyperlink-Induced Topic Search) is a link-topological algorithm for use in relevance ranking. *See* Kleinberg algorithm.

Holonym is a concept of wholeness in a part-whole relation.

Homograph is a word where the spelling is the same but the meanings are different (e.g. *lead* the verb and *lead* the metal).

Homomorphous Information Condensation is the process of producing a document-oriented abstract; it more or less retains the relative proportions of the topics discussed in the document.

Homonym is an ambiguous word that designates different concepts (e.g. *Java*). Homonymous words must be disambiguated in retrieval.

Homophone is a word that sounds the same but represents different concepts (e.g. *see—sea*). It causes problems both for retrieval of spoken language as well as for spoken queries.

Homonymy (Designation-Concepts Relation) starts from designations and is described as the relation between one and the same designation for different concepts (e.g. *Java* <Island>, *Java* <Programming language>).

Hospitality in the Concept Ladder means the expansion of a KOS in the downward direction of the hierarchy.

Hospitality in the Concept Array means the expansion of a KOS to include sister terms (co-hyponyms) within a hierarchical level.

Hubs are denoted as prominent Web pages (focal points that link to many other pages, i.e. documents with many outlinks). They play an important role in the Kleinberg algorithm.

Hyperlink is a relation between singular Web pages. On the one hand, there is a linking page (with an outgoing link) and on the other hand, a linked page (with an incoming link). Link topology regards outlinks as analogs to references, and inlinks as analogs to citations.

Hyperonym is the broader term in the abstraction relation.

Hyponym is the narrower term in the abstraction relation.

Hyponym-Hyperonym Relation (also called abstraction relation) represents the logical perspective on concepts. In a concept ladder, from the top term down to the bottom terms, the path runs from the general to the specific. Each hyponym (defined via concept explanation) inherits all properties of its respective hyperonym; additionally, it will have at least one further fundamental property that sets it apart from its sister terms (co-hyponyms).

Iconography, according to Panofsky, represents the middle semantic level in the description of non-textual documents. For this purpose, social and cultural foreknowledge on the subject of the document is necessary.

Iconology, according to Panofsky, represents the highest semantic level in the description of non-textual documents. Here expert knowledge (e.g. from art history) is required.

Image Retrieval is a non-textual form of retrieval that allows for the searching of images on the basis of their content. The basic dimensions are color, texture and form.

Implicit Knowledge, according to Polanyi, is tacit subjective knowledge that is physically embedded in a person. It is an articulated (but not via forms of expressions) intimacy that can scarcely be objectified (externalized).

IMRaD is a method of structuring scientific articles and abstracts (Introduction, Methods, Results and Discussion).

Indexing means the practical work of representing the individual objects thematized in a documentary reference unit in the documentary unit (surrogate), with the help of concepts. The content of a document should be represented as ideally as possible. Indexing is the representation of the aboutness of objects via concepts. Indexing consolidates information filtering.

Indexing Consistency means the consensus between different surrogates regarding one and the same documentary reference unit.

Indexing Depth has the two aspects of a surrogate's exhaustiveness and a concept's specificity.

Indexing Effectiveness relates to the terms that are used for indexing. By this we mean the discriminatory power of the given terms.

Indicative Abstract exclusively reports on subject matters discussed in the source document.

Individual Concept is a concept whose extension comprises exactly one element (such as the proper name of a person, organizations etc.). It has a maximum amount of properties, i.e. the extension will stay the same even after the introduction of more properties.

Information is knowledge put in motion. The concept of information unites the two aspects of physical signal and knowledge in itself (content). Information does not (in contrast to knowledge) have a truth claim.

Information Barrier is an obstacle for free flow of information to the user, diminishing recall. Examples include access, time and terminology barrier.

Information Behavior is the most general level of human behavior with regard to information and encompasses all types of human information processing. At the next level, there is information-seeking behavior, which regards all manner of information search habits. Information search behavior, at the lowest level, exclusively refers to information seeking in digital systems. A distinction is made between human information behavior and corporate information behavior.

Information Filter serves as a tool for the targeted searching and finding of information and is created in the context of knowledge organization. Apart from knowledge organization systems, one works with text-oriented methods as well as user-fixated approaches. A thematic information filter is generally oriented towards the aboutness of a document, while a Non-Topical Information Filter regards style.

Information Flow Analysis uses references and citations in order to retrace past information flow paths between documents. Such reporting systems have been established in academic research, technological development and legal practice. It is one of the basic forms of informetric analyses.

Information Hermeneutics is the study of understanding information. Pre-understanding, background knowledge, interpretation, tradition, language, hermeneutic circle, and horizon are fundamental aspects concerning the field of information and thus influence information retrieval and knowledge representation.

Information Linguistics (Natural Language Processing, NLP) applies linguistic methods and results to information retrieval, analyzing words, concepts, anaphora and input errors. Particular problems are compound formation (recognition of phrases), decompounding and the recognition of personal names.

Information Literacy empowers the user to apply information science results in everyday life, in the workplace and at school. A distinction is drawn between information retrieval literacy and literacy in creating and representing information.

Information Need. The (potential) user experiences a lack of relevant information and thus a need for action-relevant knowledge. A concrete information need (CIN), generally a factual question, is satisfied via the transmission of exactly one piece of information. Problem-oriented information need (POIN) has no clearly definable thematic borders and regards a certain number of bibliographical references. Only the transmission of multiple documents can (however rudimentarily) satisfy the information need.

Information Profile describes a user's information need that remains constant over a longer period of time. The user deposits his request as a profile, so that the system may act as a push service and inform the user about current retrieval results.

Information Retrieval is the science, engineering and practice of searching and finding information. Written text, images and sound, as well as video (as a mixed form of moving images and sound) are the basic media in information retrieval. Text-based surrogates are searched via written queries. The search for non-textual documents is performed in multimedia retrieval.

Information Science studies the representation, storage and supply as well as the search for and retrieval of relevant (predominantly digital) documents and knowledge, including the environment of information. It comprises a spectrum of five sub-disciplines: (1) information retrieval, (2) knowledge representation, (3) knowledge management and information literacy, (4) research into the information society and information markets, and (5) informetrics (including Web science).

Information Service is a combination of a database (stored documentary units in a document file and an inverted file) and a retrieval system.

Informative Abstract (as a standard form of professional abstracts) reports the subject matters discussed in a source document and additionally describes the fundamental conclusions of this document (e.g. goals, methods, results).

Informetric Analysis creates new information that has never been explicitly entered into a database and which only arises out of the informetric search and analysis procedure itself. Suitable processing methods are rankings, time series, semantic networks and information flow analyses.

Informetrics includes all empirically oriented information science research activities. Its subjects are information itself, information systems as well as the users and usage of information. In contrast to nomothetic informetrics, which tends to discover laws, there is descriptive informetrics that describes concrete objects.

Inlink (on the World Wide Web) is a link that is directed towards the respective Web page.

Input Error. There are three forms: Errors involving misplaced blanks; errors in words that are recognized in isolation (typographical, orthographical and phonetic mistakes) as well as errors in words that are only recognized via their context.

Instance Relation is a hierarchical relation in which the narrower term is always an individual concept.

Intellectual Indexing is the practical application of a method of knowledge representation to the content of a document; it is performed via human intellectual brainwork.

Inverse Document Frequency (IDF) is a database-specific weighting factor used in text statistics. It expresses the degree of discrimination of a term within an entire database's set of documents. The rarer a term, the more discriminatory it is.

Inverted file is a second file in addition to the document file in a database. It is—depending on the field—either a word or a phrase index that is sorted alphabetically (or according to another sorting criterion). The entries are combined with the document file via the allocation of an address.

IT Service Quality. The service quality of a retrieval system relates, on the one hand, to the processes the user has to implement in order to obtain results, and on the other hand, to the attributes of the offered services and the way in which these are perceived by the user.

IT System Quality is evaluated on the basis of the technology acceptance model (TAM), which consists of these four dimensions: perceived usefulness, perceived ease of use, trust and fun.

Jaccard-Sneath Coefficient is a measurement for calculating the similarity between two items A and B, following the formula $SIM(A-B) = g / (a + b - g)$ (a : number of occurrences related to A; b : number of occurrences related to B; g : number of common occurrences related to A and B).

Keyword is the denotation for a single norm entry (the preferred term, or authority record) in the nomenclature. Natural-language terms are normalized, or, as in the Registry File of the Chemical Abstracts Service, a unique number clearly identifying each of the chemical substances is used instead of the keyword.

Keyword Entry in a nomenclature summarizes both the authority records and the cross-references of the respective normalized concept (keyword).

Kleinberg Algorithm (HITS) uses link-topological procedures to rank Web pages via pseudo-relevance feedback. A (pruned) initial hit list is enhanced by outlinks of the documents in the hit set and by parts of their inlinks. The objective is to calculate hubs and authorities, which, when taken together, form a community.

Knowledge is the content of information. It is subjective when a user disposes of it (know-that and know-how). It is objective in so far as content is stored user-independently. It is thus fixed either in a human consciousness (as subjective knowledge) or in another store (as objective knowledge). Since knowledge as such cannot be transmitted, a physical form is required, i.e. an inFoRMation.

Knowledge about. In addition to knowing how (i.e. correctly dealing with an object) and knowing that (i.e. correctly comprehending an object), the informational added value is knowing about. Information activities lead to knowledge about documents as well as to knowledge about the knowledge that is fixed in the documents.

Knowledge Management means the way knowledge is dealt with in organizations. In the context of knowledge representation it involves, above all, the tasks of optimally distributing and utilizing knowledge.

Knowledge Organization safeguards (“organizes”) the accessibility and availability, respectively, of knowledge contained within documents. It comprises all types of KOSs as well as further user- and text-oriented procedures. A folksonomy represents knowledge organization without rules.

Knowledge Organization System (KOS), also called “documentary language”, is an order of concepts used to represent documents (and their knowledge), including their ofness, aboutness and style. Basic forms of KOSs are nomenclature, classification system, thesaurus and ontology.

Knowledge Quality of a retrieval system involves the evaluation of the content of documentary reference units and of documentary units.

Knowledge Representation is the science, technology and application of the methods and tools for representing knowledge in such a way that it can be optimally searched and retrieved in digital databases. The knowledge that has been found in documents is represented in an information system, namely via surrogates. Knowledge representation includes the subareas of information condensation and information filters.

Latent Semantic Indexing (LSI) is a variant of the Vector Space Model in which the number of dimensions is obtained via factor analysis, thus being greatly reduced. Terms summarized into pseudo-concepts are factors that correlate with word forms and with documents.

Lemmatization is a linguistic method that conflates the variant word forms into a basic form (lemma). There are rule-based approaches as well as dictionary-based procedures.

Levenshtein Distance performs approximate string matching in fault-tolerant retrieval. It uses a proximity measurement between two sequences of letters in order to recognize and correct input errors. This measurement counts the editing steps between the words that are to be compared. The basis of comparison is a dictionary.

Line is the connection between two nodes within a social network.

Link. See Hyperlink.

Link Topology locates Web documents by analyzing the structure of their hyperlinks, and is used for the relevance ranking of Web pages. Important variations of link topology are the Kleinberg algorithm and the PageRank.

Lovins Stemmer is a longest-match stemmer that recognizes the longest ending of word forms and removes it.

Luhn's Thesis states that a word's frequency of occurrence in a text is a criterion for its significance. “Good” text words are those with a medium-level frequency of occurrence in a document. Frequently occurring words are usually stop words.

Merging means the unification of thematically identical KOSs, where both the concepts and their relations have to be amalgamated into a new unit.

Meronym is the concept of a part in a part-whole relation.

Meronym-Holonym Relation (also called Part-Whole Relation, Part-of Relation or Partitive Relation) is an object-oriented hierarchy relation. Concepts of wholeness (holonyms) are divided into the concepts of their parts (meronyms). Meronymy is not just one relation, but a bundle of different part-whole relations.

Metadata are standardized data consisting of attributes (fields) and values (field entries). They are defined in a rulebook for indexers and users. Metadata provide information about documents.

Metadata about Objects require a standardized representation of all significant relationships that characterize a non-digitizable object.

Min and Max Model (MM Model) is a simple variant of weighted Boolean (fuzzy) retrieval that exclusively uses the Minimum of the values for conjunction, and on the other hand, their Maximum for disjunction.

Mirror, on the Web, is a duplicate that is hosted in a different location than the original. Since the pages must not be indexed twice, duplicates need to be recognized.

Mixed Min and Max Model (MMM Model) is the elaborate variant of weighted Boolean (fuzzy) retrieval that introduces softness coefficients combining the Minimum and Maximum values.

Model Document. Applied to the context of relevance feedback, this denotes a positive (or, where appropriate, negative) designated document within a preliminary hit list resulting from a search request. The original query is modified via terms from documents defined as both relevant and non-relevant (via the Robertson-Sparck formula or the Rocchio algorithm). It is also possible for a user to select a certain document as a model in order to search for “more like this”.

Multi-Document Abstract condenses the subject matter of several documents on one and the same topic into one summary.

Multi-Lingual Information Retrieval (MLIR) is retrieval across language barriers, in which the documents themselves are available in translation, i.e. in which both query and documents are in the same language. In contrast to this, see Cross-Language Information Retrieval (CLIR).

Multimedia Retrieval means the search for non-textual documents. The points of interest here are both images and sound in spoken texts, music and other audio documents, as well as videos and movies.

Music Information Retrieval (MIR), performed as a form of multimedia retrieval, works with the content dimensions of music itself. This is done either via an excerpt from a piece of music (model document) or via a sung query (“query by humming”).

***n*-Gram** is a formal word that brings character sequences in texts to a length of *n* (where *n* is a number between 1 and about 6).

Named Entities are concepts that refer to a class containing precisely one element (e.g. personal names, place names, names of regions, companies etc., but also names of scientific laws).

Name Recognition. Named entities are identified automatically via characteristics inherent to the name (indicator terms, e.g. the first name, and rules) as well as via external specifics (of the given text environment). The disambiguation of homonymous names is performed via external specifics.

Natural Language Processing (NLP). See Information linguistics.

Network Model is a retrieval model that uses the position of an actor in a social network as a ranking criterion.

Node represents an actor in a social network.

Nomenclature is a collection of controlled terms (keywords), as well as of cross-references to keywords from a given natural or specialist language. By definition, there are no hierarchical relations. Nomenclatures have a well-developed synonymy relation, and they may also have an associative relation (see-also references).

Non-Descriptor, listed in a thesaurus, is a designation or a concept without priority. It only serves users as a tool for gaining access to the preferred term more easily.

Non-Text Document. *See* Document Representing Objects.

Non-Topical Information Filter is geared to aspects of a non-formal nature, which nevertheless represent fundamental relations of a document. It concerns the “how” of a text, or more precisely, the “style” of a document. The attributes and values are used as a filter. Depending on the type of topic treatment, a distinction is made between characteristics of the author, characteristics of the medium, the perspective of the document as well as its genre. Search via style values mainly leads to a heightened precision in search results. *See also* Information Filter.

Normal Science, according to Kuhn, is a research field that is not affected by any great scientific breakdowns over a longer period of time. There is a consensus between scientists concerning the terminology of their special language. The scientific community adheres to its paradigm.

Notation has a particular significance in classification systems. It is a (non-natural-language) preferred term of a class, generally constituted via numbers or letters.

Objective Information Need is the information need without regard to the person concerned, i.e. the information need “as such”.

Ofness describes the pre-iconographical semantic level of images, videos, pieces of music etc. (discussed by Panofsky). *See also* Aboutness.

Ontology, as a knowledge organization system, is available in a standardized language, permits automatic reasoning, always disposes of general and individual concepts and uses further specific relations in addition to the hierarchical relation. Ontologies are the fundamental KOSs of the Semantic Web.

Outlink (on the World Wide Web) is a link going out from a certain Web page.

PageRank calculates the probability of a random surfer’s visiting a certain Web page. This calculation considers the number of the page’s inlinks as well as the PageRank of the linking pages.

Paradigmatic Relation is one of the semantic relations. From the perspective of knowledge representation, these relations form “tight” relations, which have been established or laid down in a certain KOS. They are valid even without the presence of specific documents. We distinguish three groups of paradigmatic relations: equivalence, hierarchy and further specific relations.

Parallel Corpora are documents of identical content in different languages. They are used in cross-language retrieval.

Paramorphous Information Condensation is a process of producing an abstract according to certain perspectival criteria.

Parsing means the breaking down of a text into its smallest units (words).

Passage is a sub-unit of a documentary reference unit for text documents. Distinctions are made between discourse locations (roughly speaking: paragraphs), semantic locations (the area of a topic) and text windows (excerpts of fixed or variable length).

Passage Retrieval assigns the retrieval status value of a text's most suitable passage to the entire document, thus orienting the text's relevance ranking on the most appropriate given passage. It is also used for the within-text retrieval of long documents, as well as in question-answering systems.

Part-Whole Relation represents an objective perspective on concepts. See Meronym-Holonym Relation.

Path Length indicates the number of hierarchy levels in a file directory tree that lie above a Web page. It may serve as a weighting factor for Web pages (the higher up a page is in the folder hierarchy, the more important it is).

Personalized Retrieval always takes into account user-specific requirements, both in modifying search arguments as well as during relevance ranking.

Perspectival Extract is an extract with reference to the user's specific query. Only those sentences that contain query terms are adopted in the extract.

Pertinence is the relation between the subjective information need and a document (or the knowledge contained therein). The latter is pertinent if the subjective information need has been satisfied. Pertinence incorporates the concrete user into the observation, alongside his cognitive model.

Phonix provides fault-tolerant retrieval for input error correction, searching for words based on their sound. It enhances Soundex by phonetic replacement.

Phrase is a semantic unit that consists of several individual words (e.g. "soft ice", "information retrieval").

Phrase Building. Two methods attempt to identify the relations of concepts that consist of several words (i.e. phrases), and to make the phrases thus recognized searchable as a whole. On the one hand, there is the statistical method, which counts joint occurrences of the phrase's components, and on the other hand there is a procedure for "text chunking", which eliminates text components that cannot be phrases, thus forming "large chunks".

Porter-Stemmer is an iterative procedure for stemming that processes the suffixes in several go-throughs.

Postcoordination combines individual terms during the search process using Boolean operators. This method of coordination provides the user with a lot of freedom in formulating his search arguments.

Pre-Iconography, according to Panofsky, represents the lowest semantic level in the description of non-textual documents. It concerns the world of primary objects. There exists only some practical experience with the thematized objects, but there is no available knowledge of their social or cultural background. Pre-Iconography describes the document's oneness.

Precision is a basic measurement of the quality of search results, describing the information's appropriateness (in the sense of freedom from ballast). It is calculated as the quotient of the number of relevant documentary units retrieved and the total number of retrieved data entries.

Precombination occurs during indexing, i.e. while searching in a KOS. A concept comprised of several components is formed into a fixed connected unit (e.g. *library statistics*). It is exactly this combination which must then be used for the respective concept. Precombination allows for a high conceptual specification in a KOS.

Precoordination means the syntactical combination of concepts during the indexing process (e.g. *library / statistics*).

Preferred Term is a natural-language concept, or an artificial designation (with digits or letters shown) that has primacy over similar designations. In a thesaurus it is called the descriptor, in a nomenclature the keyword, in a classification the notation, and in an ontology the concept.

Probabilistic Indexing requires the availability of probability information about whether a descriptor (notation etc.) will be of relevance for the document if the document includes a certain term. It is used for automatic indexing via KOSs.

Probabilistic Retrieval Model asks for the probability of a given document's matching a search query (the conditional probability of a document's relevance under a query). It requires relevance information about the documents, which is done via relevance feedback or pseudo-relevance feedback. The documents are ranked in descending order of attained probability.

Prosumer means, according to Toffler, that the consumer of knowledge has also become its producer.

Prototype helps in defining general concepts that have fuzzy borders. It can be treated as the best example for a basic-level concept.

Proximity Operator intensifies the Boolean AND; as adjacency operator, incorporating the intervals between search atoms, or as syntactic operators, incorporating the limits of sentences or of paragraphs as well as syntactic chains, respectively.

Pseudo-Relevance Feedback is an automatically performed relevance feedback that defines the top-ranked documents of the initial search as relevant, and uses the term material thus obtained as positive model documents for the following search step.

Pull Service is an information service in which a user actively searches for information within a retrospective search. Pull services meet ad-hoc information needs.

Push Service is an information service that satisfies fairly long-term information needs. A user deposits a profile in a retrieval system. Then, the system supplies the user with ever-new information pertaining to the stored information profile.

Qualifier is a supplementary piece of information that specifies a descriptor in a thesaurus (e.g. MeSH).

Quasi-Classification means the allocation of similar documents to a class by applying methods of numerical classification. It is automatic indexing without the help of a KOS.

Quasi-Synonymy is a synonymy of partial nature and stands for more or less closely related concepts (as opposed to the absolute synonymy relation, which is a relation between designations and a concept).

Query is a search argument, consisting of one or more search atoms that a user directs toward a retrieval system.

Query by Humming is a search request in music information retrieval where the user sings, whistles or hums an excerpt from a piece of music in order to perform his search.

Query Expansion becomes necessary when an initial query has not yielded the desired search results. There are three methods of optimizing the initial query formulation: intellectually, via appropriate search strategies by the user; automatically, through the system; and semi-automatically, via interaction between system and user.

Question-Answering System serves to satisfy a specific information need by gathering factual information. The user is not presented with an entire document as a search result, but only with the most appropriate passage from a relevant document. *See* Passage Retrieval.

Rankings sort entries according to their frequency of occurrence in an isolated document set. This is one of the basic forms of informetric analyses. For Ranking of Search Results *see* Relevance Ranking.

Recall is a basic measurement of the quality of search results, describing the information's completeness. It is calculated as the quotient of the number of relevant documentary units retrieved and

the total number of relevant documents. The Recall measurement is a purely theoretical construct, since the total number of DUs that were not retrieved cannot be directly obtained.

Recommender System presents a user with personalized document recommendations which are new and might be of interest to him. The system works either user-specifically (by matching a user profile with the content of documents) or collaboratively (by comparing the user with similar users). Hybrid systems consider both the behavior of the current user and that of the others.

Reference, in a (citing) document, is a bibliographic indication showing where the author has obtained certain information (from a cited document). It is indicated in the latter as a citation.

Relevance is established between the poles of user and retrieval system. It is the relation between an objective information need and a document (or the knowledge contained therein). The latter is relevant if the objective information need is satisfied. Relevance aims for user-independent, objective observations.

Relevance Feedback is a query modification method with the goal of removing the query from the areas of irrelevant documents and moving it closer to the relevant ones. It is the result of feedback loops between system and user that contain relevance information about documents. It is performed either via an explicit relevance assessment of documents by the user or via pseudo-relevance feedback. Relevance feedback plays an important role both in the probabilistic retrieval model (Robertson-Sparck Jones-formula) and in the Vector Space Model (Rocchio algorithm).

Relevance Ranking sorts a hit list according to the documents' retrieval status values relative to the initial search argument. Relevance Ranking is performed differently in accordance with the retrieval model being used.

Relevance Ranking of Tagged Documents is applied to folksonomies and is mainly based upon aspects of collective intelligence. It takes into account the tags themselves, aspects of collaboration and actions by individual prosumers.

Retrieval Model deals with the modeling of documents and their retrieval processes. Apart from NLP and the Boolean model, all approaches fulfill the task of arranging the documents (retrieved through information-linguistic working steps) into a relevance-based ranking. Important retrieval models are: Boolean model, weighted Boolean model, vector space model, probabilistic model, link-topological model, network model and user/usage model.

Retrospective Search satisfies ad-hoc information needs of a user in the manner of a pull service.

Retrieval Status Value (RSV) is a numerical value to be assigned to a document, based on search arguments and operators. The calculation of the retrieval status value depends on the retrieval model used. On the search engine results page, the documents are ranked in descending order of their retrieval status value, thus yielding the relevance ranking.

Retrieval System provides functions for searching and retrieving information. Depending on the operational purpose of the respective retrieval system, the scope of the functions offered can vary significantly.

Retrieval System Quality. Evaluating the quality of a retrieval system aims to analyze its functionality (quality of the search functions) and usability, as well as to calculate the traditional parameters of Recall and Precision.

Rocchio Algorithm works for relevance feedback in the context of the vector space model. It modifies a query vector via the sum of the original query vector, the centroid of those document vectors that the user has marked as relevant, and (with a minus sign) the centroid of the documents estimated to be non-relevant.

Rule-based Indexing requires the formulation of rules according to which a concept will be used for indexing purposes or not. It is a method for automatic indexing via KOSs.

Rulebook, in the context of metadata, prescribes the authority records of personal names, corporate bodies, and titles, among other things. In the context of information filter and information condensation, it is used as a collection of rules for indexing and summarizing.

Scientometrics is the quantitative study of science. Its main subjects are scientific documents, authors, scientific institutions, academic journals, topics and regional aspects of science.

Search Atom is the smallest unit of a complex search argument, e.g. a singular term.

Search Strategy is characterized by the plan according to which a user executes his search. This applies to the search for and retrieval of appropriate databases, as well as of documentary units contained therein. Over the course of the search, sometimes search arguments are modified via measures that heighten Recall or Precision. Basic search strategies include the building blocks strategy, the citation pearl growing strategy as well as berrypicking. The three phases of search are: searching for and retrieving the relevant databases, searching for and retrieving the relevant documentary units, and modifying the search arguments.

Search Tool is a retrieval system on the WWW. There are algorithmic search engines, Web catalogs and meta search engines.

Selective Dissemination of Information (SDI). Deposited user-specific information profiles are worked through one by one as new documents enter the database. The user is notified about the new information either via e-mail or via his website.

Semantic Crosswalk facilitates unified access to heterogeneous databases, and, moreover, the reuse of previously established KOSs in other contexts. It aims to provide for both comparability (compatibility) and cooperation (interoperability) between KOSs. There are five forms: Parallel usage of different KOSs together in one application, upgrading of a KOS, selecting and cutting out a subset from a KOS, forming of concordances between the concepts of different KOSs with the final goal of merging and integrating these different KOSs into a whole.

Semantic Network is spanned by concepts and their relations to each other. We distinguish between paradigmatic networks (relations in a KOS) and syntagmatic networks (relations between concepts, developed from frequent co-occurrences). Syntagmatic semantic networks are one of the basic forms of informetric analyses.

Semantic Relation means a relation between concepts. There are either syntagmatic relations (co-occurrences of terms in documents) or paradigmatic relations (relations in KOSs).

Semiotic Triangle. In information science, it describes the connection between designation (e.g. word), concept and both properties and objects.

Sentiment Analysis and Retrieval uncovers positive or negative opinions, emotions and evaluations (about people, companies, products etc.) that may be contained in documents. Objects of analysis are, generally, press reports (newspapers, magazines, agency articles) as well as social media in Web 2.0 (blogs, message boards, rating services, product pages of e-commerce services). Applications of sentiment analysis and retrieval are press reviews, media resonance analyses, Web 2.0 monitoring services, as well as Web page rating.

Shell Model, according to Krause, demonstrates the structure of a (controlled) heterogeneity in indexed bodies of knowledge. Differing levels of document relevance (as stages of worthiness of documentation) are united with their respectively corresponding quality of content indexing.

Shepardizing was introduced by Shepard and relates to the citation indexing of published court rulings.

Signal is the physical basis (e.g. printing ink, sound waves or electromagnetic waves) of information. Signals transmit signs.

Similarity between Concepts within a KOS is the measurement of the semantic proximity derived from the position/distance of two concepts (e.g. via the hierarchy or associative relation). It can be quantitatively described by counting the paths along the shortest route between two concepts.

Similarity between Documents is calculated as the distance between two documents via the co-occurrences of terms or—for non-text documents—via similar attributes concerning document-specific dimensions (e.g. the distribution of colors in two image documents). Similarity metrics include the cosine, Jaccard-Sneath and Dice coefficients.

Similarity Thesaurus (also called statistical thesaurus) is used for query expansion when the retrieval system is not supported by a KOS, or if the full text is also required for query modification. Words that frequently co-occur in documents from a specific database are merged into one concept. The objective is to offer the user words similar to his original search atoms.

Sister Term (first degree parataxis in a hierarchical relation) shares the same hyperonym with other terms in the concept array. A sister term is also called a co-hyponym.

Small Worlds (in the theory of social networks) are networks with high graph density and small graph diameters. In a “small world” network there are shortcuts between actors who are far apart.

Snippet is a low-quality form of an extract in search engines results pages.

Social Network focuses on the structured representation of actors and their connections. From the perspective of information retrieval, actors can be documents, authors (as well as any further actors derived from them) or topics. Actors (described as nodes) are interconnected via lines. In a folksonomy, for instance, documents, tags and users can be represented as nodes in a social network.

Sound Retrieval is a form of non-textual retrieval that searches for audio documents (music, spoken language, noise).

Soundex is a method of fault-tolerant retrieval for input error correction which searches words based on their sound (*see also* Phonix).

Spam page on the WWW has no relevance for a certain topic at all, but merely simulates relevance. Spammers use methods such as the fraudulent heightening of relevance, as well as techniques of concealing spam information.

Spoken Query means a spoken search argument (e.g. via telephone) directed toward a retrieval system. Particular problems in recognizing spoken language are posed by words that sound the same but represent different concepts (i.e. homophones).

Sponsored Links are texts and links with advertising content. Their ranking position is dependent on the price offered per click (in combination with other factors).

Statistical Language Model describes and compares the distribution of terms in texts, in entire databases, or in the general everyday use of language. It can be theoretically linked with text statistics and probabilistic retrieval.

Stemming is a method that processes the suffixes of the word forms and unifies them in a common stem. The stems do not have to be linguistically valid word forms. There are two different procedures of processing suffixes: the longest-match approach (e.g. by Lovins) and the iterative approach (as in the Porter stemmer).

Stop Word (e.g. “the” or “if”) is a word in a document or query that bears no knowledge. It is content-free. It must be marked and excluded from “normal” retrieval. Stop words are collected in negative lists.

Structured Abstract is divided into individual sections and marked by sub-headings (e.g. in medical and scientific journals). Typical structures are IMRaD and the eight-chapter format.

Subjective Information Need is the information need of a concrete person in a concrete situation.

Suffix (such as *-ing* in the English language) has to be deleted by means of lemmatization or stemming.

Summarization represents the important subject matter of a document in the form of sentences. It should be done briefly but as comprehensively as possible. Short representations are abstracts or extracts. Summarization consolidates content condensation.

Surface Web is the entirety of digital documents that are firmly linked and stored *on* the Web, resulting in no additional cost to the user.

Surrogate is a documentary unit (DU) that represents a documentary reference unit (DRU) in a database.

Synkategoremata are incomplete concepts that only achieve meaning in relation to other terms. They are to be completed via additions (e.g. *with filter*).

Synonym is one of several distinct words that each designates the same concept (e.g. *autumn, fall*). Two designations are synonymous if they denote the same concept.

Synonymy (Designations-Concept Relation) is described as the relation between designations that each refer to the same concept. Here we have absolute synonymy (e.g. *photograph, photo*). *See also* Quasi-Synonymy.

Synset (set of synonyms) summarizes words that express exactly one concept, and forms one of the basic building blocks of a semantic network. WordNet's synsets are separated by the word classes of nouns, verbs, adjectives and adverbs.

Syntagmatic Relation is one of the semantic relations, existing between concepts within an actual document. It deals with the co-occurrences of terms in specific documents.

Syntactical Indexing incorporates the thematic relations between concepts in a document (in contrast to co-ordinating indexing). It allows for precise searches via topic chains, and facilitates weighted indexing.

Table. In a classification system, classes are systematically arranged in main tables as well as auxiliary tables. The basic aspects are included in the main tables. The aspects that occur multiple times in different areas are accommodated in auxiliary tables. The latter will record those concepts that can be meaningfully attached to the classes of main tables.

Tag refers to the keyword (indexing term) that is freely generated by users of folksonomies.

Tag Cloud can also be described as a “visual summary” of the contents of a database, hit set or document. They are offered by many Web 2.0 services.

Tag Gardening is a strategy for editing folksonomy tags in such a way as to make them more effective. Tags can be processed via basic formatting (weeding), tag recommendations (seeding), vocabulary control (garden design), interactions with other KOS (fertilizing) and the delimitation of power tags (harvesting).

Tagging stands for indexing via folksonomies.

Taxonomy is a form of the hyponym-hyperonym relation (or abstraction relation), where the *IS-A* relation can be strengthened into *IS-A-KIND-OF*.

Term is the hyperonym of: unprocessed (inflected) word form, basic form (lexeme), stem, phrase, compound, named entity, concept as well as anaphor.

Term Cloud is the generic designation for the visual presentation of terms in search tools and results. The font size of the processed terms shows their frequency within the cloud. Term clouds are used for different sets of documents: in entire information services, in search results as well as in documents. A term cloud shows no relations between terms. (These are only achieved via the construction of term clusters). *See also* Tag Clouds.

Term Frequency (TF) is the quantitative expression of a term's importance in the context of a given document. The document-specific term weight works with the relative frequency of a term's occurrence in a text and is used in text-statistical methods. *See also* Within Document Frequency (WDF).

Terminological Control refers designations and concepts to each other in an unequivocal fashion. Words are processed into conceptual units in order to then be arranged into the framework of the thesaurus. Homonyms must be disambiguated and Synonyms must be summarized. Expressions that consist of multiple words must be adequately decomposed. Concepts that are too general are replaced by specific ones, and too-specific concepts are bundled. Terminological control is used in nomenclature, classification system, thesaurus and ontology.

Text Retrieval Conferences (TRC) provide experimental databases for the evaluation of retrieval systems. The test collection comprises documents, queries and relevance judgments.

Text Statistics works with characteristics of term distributions, both in individual texts and in entire databases. It calculates weight values for each term in a document or in a query, and uses variants of term frequency (TF; in particular WDF) as well as IDF. The purpose is to arrange documents into a relevance ranking.

Text Window is a text excerpt of fixed or variable length.

Text-Word Method operates exclusively via the available words of the respective text. It is a text-oriented method of knowledge representation, but not a KOS. This method limits itself to text words, for the purpose of low-interpretative indexing. The text-word method deals with syntactical indexing via chain formation.

Thesaurus draws on natural language. The KOS connects terms into small conceptual units, with or without preferred terms, and sets these into certain relations with other concepts. We distinguish between specialist thesauri (e.g. MeSH) and natural-language thesauri (e.g. WordNet).

Time Series show the development of subjects over the course of time. This is one of the basic forms of informetric analyses.

Topic Detection and Tracking (TDT) analyzes the stream of news from the WWW, the Deep Web, and broadcasts, with the objective of identifying new events and allocating stories to previously known topics.

Truncation fragments a search atom via wildcards (right, left, and center).

Ubiquitous Retrieval uses other types of context-specific information besides geographical information, such as real-time information. It is a form of context-aware retrieval, with the context changing continuously.

Usage Model is a retrieval model that uses the usage frequency of documents as a ranking criterion.

Usage Statistics can be used as ranking criteria for a Web page. Usage frequencies are gleaned either by observing the clicked links in the hit lists of a search engine or by evaluating all page views by users via toolbar.

User and Usage Research is a sub-discipline of empirical information science (informetrics) that describes and explains the information behavior of users. Objects of examination are information professionals, professional end users and end users (information laymen). Methods of user research may be: observation of users in a laboratory situation, user survey and analysis of log files of search tools.

Vagueness of Concepts often reveals itself in general terms. The corresponding term has fuzzy boundaries, i.e. there are objects that do not fall exactly within that definition (e.g. chair or not-chair in Black's imaginary chair exhibition). *See also* Prototype.

Vector Space Model interprets terms from the texts, as well from the queries, as dimensions of an n -dimensional space and locates both the documents and the user queries as vectors in said space. n counts the number of different terms. The vector's position is determined by the terms' dimensions. The values in the dimensions result from text-statistical weighting (generally $WDF \cdot IDF$).

Video Retrieval is a form on non-textual retrieval in which the documentary reference units are shots and scenes. In addition to the basic dimensions of image retrieval, it also has the dimension of motion. Video retrieval combines methods of image and sound retrieval.

Visual Retrieval Tools are alternatives to the usual (alphabetically sorted) list rankings, and attempt to process search tools and results visually. Search terms can be visually processed in term clouds (e.g. tag clouds in folksonomies). Search results can be represented via graphs or sets. When results of informetric analyses are visualized, the document sets are described in their entirety.

Vocabulary Relation is a relation between designations and concept(s) in a thesaurus. Such relations include synonymy, homonymy, splitting, bundling and specification.

Web 2.0 is a section of the World Wide Web in which prosumers (users and producers at the same time) create, edit, comment on, revise and share their documents reciprocally.

Web Information Retrieval is a special case of general retrieval in which additional web-specific subject matter (e.g. hyperlinks between Web documents) can be used for relevance ranking.

Webometrics applies informetric approaches to the World Wide Web. Web documents, their content, their users and their links are analyzed quantitatively.

Web Science is a blend of science and engineering; its subject is the World Wide Web.

Weighted Boolean Retrieval uses weight values for search terms, for terms in the documents and for entire documents. This, in contrast to the classical Boolean model, facilitates a relevance-ranked output.

Weighted Indexing assigns numerical values to concepts, expressing their importance in the document.

Weighting. A numerical value is allocated to a given term or sentence, expressing its importance in documents as well as in queries.

Weighting Factors guide the ranking algorithms of Web retrieval systems, with the goal of arriving at a relevance-ranked hit list. Ranking factors used by search engines include, for instance, PageRank, TF and IDF of the search terms, anchor texts, Web page path length, freshness or usage statistics. Social media, on the other hand, require different criteria, such as weighting via interestingness in sharing services, ranking of bookmarks by affinity, etc.

Within Document Frequency (WDF) is the quantitative expression of a term's importance in the context of a given document. The document-specific term weight works with a logarithmic measurement that sets the term frequency in relation to the total length of a text. It is used in text-statistical methods. *See also* Term Frequency (TF).

Word is a linguistic expression of a concept. Words are sometimes ambiguous (homonymy) and must then be disambiguated. A natural-language word in a text is recognized via blanks and punctuation marks. A formal word (n -gram) consists of n characters.

Word-Concept Matrix. The dictionary of a language is regarded as a matrix, in which the columns are filled with all of a language's words, and the rows contain the different concepts. It identifies homonyms (words that refer to several concepts) and synonyms (concepts that are described via several words).

Word Form is a flectional word.

WordNet includes an Anglophone online-dictionary.

Worthiness of Documentation. The selection of which specific documentary reference units should be admitted into a database and which should not is based upon a bundle of criteria (e.g. only documents in a specific language, only scientific documents, number of documentary reference units to be processed in a time unit, user needs).

Q.2 List of Abbreviations

ASIST	Association for Information Science and Technology
ASK	Anomalous State of Knowledge
CAS	Chemical Abstracts Service
CBIR	Content-Based Image Retrieval
CBMR	Content-Based Music Retrieval
CDU	<i>Classification Décimale Universelle</i>
CDWA	Categories for the Description of Works of Art
CIN	Concrete Information Need
CIR	Collaborative Information Retrieval
CIS	Computer and Information Science
CLIR	Cross-Language Information Retrieval
COUNTER	Counting Online Usage of Networked Electronic Resources
CSV	Comma-Separated Values
CWA	Cognitive Work Analysis
DDC	Dewey Decimal Classification
DK	<i>Dezimalklassifikation</i>
DNS	Domain Name Server
DOI	Digital Object Identifier
DRU	Documentary Reference Unit
DU	Documentary Unit
EmIR	Emotional Information Retrieval
FID	International Federation for Information and Documentation
FIFO	First in, First out
FILO	First in, Last out
FREQ	Frequency
GIR	Geographical Information Retrieval
GPS	Global Positioning System
HITS	Hypertext-Induced Topic Search
HTML	Hypertext Markup Language
ICD	International Classification of Diseases
ICF	International Classification of Functioning, Disability and Health
ICT	Information and Communication Technology
IDF	Inverse Document Frequency
IF	Impact Factor
IFLA	International Federation of Library Associations and Institutions
II	Immediacy Index
IP	Internet Protocol
IPC	International Patent Classification
IR	Information Retrieval
ISBN	International Standard Book Number
ISSN	International Standard Serial Number
IT	Information Technology
J	Journal

KOS	Knowledge Organization System
KWIC	Keyword in Context
LIS	Library and Information Science
LSA	Latent Semantic Analysis
LSI	Latent Semantic Indexing
MARC	Machine-Readable Cataloging
MARS	Mitkov's Anaphora Resolution System
MeSH	Medical Subject Headings
MIDI	Musical Instrument Digital Interface
MIR	Music Information Retrieval
MPEG	Moving Picture Experts Group
MLIR	Multi-Lingual Information Retrieval
NACE	<i>Nomenclature Générale des Activités Économiques dans les Communautés Européennes</i>
NAICS	North American Industry Classification System
NUTS	<i>Nomenclature des Unités Territoriales Statistiques</i>
OWL	Web Ontology Language
P	Precision
PDF	Portable Document Format
PL	Path Length
POIN	Problem-Oriented Information Need
QA	Question Answering
R	Recall
RDF	Resource Description Framework
RDFS	RDF Schema
RDR	Reversed Duplicate Removal
RSV	Retrieval Status Value
SCC	Strongly Connected Component
SDI	Selection Dissemination of Information
SDR	Spoken Document Retrieval
SERP	Search Engine Results Page
SIC	Standard Industrial Classification
SIM	Similarity
SMART	System for the Mechanical Analysis and Retrieval of Text
STM	Science, Technology, Medicine
SWD	<i>Schlagwortnormdatei</i> (Keyword Norm File)
TAM	Technology Acceptance Model
TDT	Topic Detection and Tracking
TF	Term Frequency
TReC	Text Retrieval Conference
UDC	Universal Decimal Classification
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
WDF	Within Document Frequency

WoS	Web of Science
WWW	World Wide Web
XML	Extensible Markup Language

List of Variables

d	Document
e	Retrieval Status Value (RSV)
n	Counting Variable
p	Web Page
s	Sentence
t	Term
w	Weighting Value

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